QUANTITATIVE APTITUDE SIMPLIFIED

(Useful for all Competitive Examinations like Banks, IT, CAT, CSAT, CLAT, Defence, G.I.C. GMAT, GRE, IBPS ICET, KPSC, L.I.C, MAT, Railways, SSC, SNAP, TNPSC, UPSC, UGC, XAT and all Government & other Competition Examination etc.)

JAGGAN SANEJA

QUANTITATIVE APTITUDE SIMPLIFIED

(Useful for all Competitive Examinations like Banks, IT, CAT, CSAT, CLAT, Defence, G.I.C. GMAT, GRE, IBPS ICET, KPSC, L.I.C, MAT, Railways, SSC, SNAP, TNPSC, UPSC, UGC, XAT and all Government & other Competition Examination etc.)

By
JAGGAN SANEJA

notionpress.com

Notion Press

Old No. 38, New No. 6
McNichols Road, Chetpet
Chennai - 600 031

First Published by Notion Press 2017
Copyright © Jaggan Saneja 2017
All Rights Reserved.

eISBN 978-1-947949-84-3

This book has been published with all reasonable efforts taken to make the material error-free after the consent of the author. No part of this book shall be used, reproduced in any manner whatsoever without written permission from the author, except in the case of brief quotations embodied in critical articles and reviews.

The Author of this book is solely responsible and liable for its content including but not limited to the views, representations, descriptions, statements, information, opinions and references [“Content”]. The Content of this book shall not constitute or be construed or deemed to reflect the opinion or expression of the Publisher or Editor. Neither the Publisher nor Editor endorse or approve the Content of this book or guarantee the reliability, accuracy or completeness of the Content published herein and do not make any representations or warranties of any kind, express or implied, including but not limited to the implied warranties of merchantability, fitness for a particular purpose. The Publisher and Editor shall not be liable whatsoever for any errors, omissions, whether such errors or omissions result from negligence, accident, or any other cause or claims for loss or damages of any kind, including without limitation, indirect or consequential loss or damage arising out of use, inability to use, or about the reliability, accuracy or sufficiency of the information contained in this book.

Dedicated to My Parents

Late (Master) Chetan Dass Saneja & Late Smt. Bishan Devi

PREFACE

To understand and resolve any problem, it requires a clear understanding of the concept – may it pertain to area concerning Mathematics or Data Interpretation or Quantitative Aptitude. No questions can be dealt with by guessing until you have a good and perfect hand over your steps. A clear concept of the subject will immensely help the students in having an in-depth knowledge to solve any problem. The present work is an attempt to simplify the same. After successful launch of my two books **(i) MATHEMATICS Simplified and (ii) DATA INTERPRETATION Simplified –** which have been widely accepted by the aspiring candidates for various competition examinations with warm welcome – it is a matter of pride for me to publish my another book titled **"QUANTITATIVE APTITUDE Simplified by JAGGAN SANEJA"** through **M/s Notion Press Media Private Limited, Chennai - 600005.**

With the growing competition in different examinations and Quant occupying major part in most of the examinations/question papers – the author felt a strong need to present the basics in a lucid manner. Having an experience of more than 35 years in imparting teaching and guiding thousands of successful students for different competitive examinations, the author has tried to present the subject matter in an easy and understandable manner. Every aspect of each chapter has been thoroughly dealt with. At the end of each chapter, Practice Exercises, with varied time schedules for each Test Papers with MCQs, given in an increasing order of toughness, have been presented for recapitulation and to have a better grip over the subject. Students are advised to try to finish the question set in the given stipulated time and mark the answers with utmost care. As there is a negative marking system in most of the examinations – wide guessing is not advised.

Every effort has been made to present the subject matter error-free, from both technical and typographical aspect. However, constructive and helpful suggestions for the improvement of the book are welcome
(at mail **jnsaneja@yahoo.co.in**) which will be acknowledged with thanks.

It is my pleasant duty to convey my sincere thanks to **Mrs. Sarla Saneja** and all the members of her team – **Mr. Tanush Kohli, Mr. Navansh Saneja and Ms. Vriddhi** – **Smriddhi –** who have always been a constant source of motivational support in the preparation of the book without hindrance.

Thanks must also be conveyed, from my heart, to my guides and mentors **Astro Yash Pal Malhotra** and **CA Manav Saneja**, without whose help and guidance, at multiple stages, this work could not have been presented.

JAGGAN SANEJA

M.Com, CAIIB

jnsaneja@yahoo.co.in

ABOUT THE AUTHOR

JAGGAN SANEJA, has already authored and launched successfully his two books **(i) MATHEMATICS Simplified** and **(ii) DATA INTERPRETATION Simplified** for various competition examinations. Academically, he holds a Master Degree in Commerce and is a Certified Associate of Indian Institute of Bankers.

JAGGAN SANEJA started guiding students for Mathematics in Chandigarh Tricity way back in 1980. He has also been associated with **M/s SANEJA TUTORIALS**. With an extensive and rich experience to his credit, he has got an expertise in identification, understanding and cracking the mathematics problems by quicker and shorter methods, in the shortest possible time or sometimes even by glance of the questions which has helped thousands of his students come successful in various examinations.

For a good number of years, his question papers for various competition examinations had been occupying prime space in famous magazines and periodicals.

Simplified series…by JAGGAN SANEJA

(i) MATHEMATICS Simplified	…	already with the readers
(ii) DATA INTERPRETATION Simplified	…	already with the readers
(iii) QUANTITATIVE APTITUDE Simplified	…	Now with the readers

*All these books are available on all leading e-commerce stores like **Amazon.com**, **Bookadda.com**. **Flipkart.com**, **Infabeam.com** and **Notion Press.com***

Readers can reach the author by e-mail at jnsaneja@yahoo.co.in

ABOUT THE BOOK

In **QUANTITATIVE APTITUDE Simplified** you will find numerous aptitude questions with answers and explanation in the shape of **MCQs** (**Multiple Choice Questions)**. These questions are extremely helpful for competitive examinations. All the answers have been explained in detail, on need basis. The collection of questions contains a good number of aptitude questions asked in various placement examinations and competitive examinations and shall help students prepare for **any type of competitive examinations**.

> **Questions dealt with in "QUANTITATIVE APTITUDE Simplified" are extremely useful for all kind of Competitive Examinations like Bank Competitive Examinations, Career Aptitude Tests (IT Companies), Common Aptitude Test (CAT), CLAT, CSAT, Defence Competitive Examinations, G.I.C. Competitive Examinations, GMAT, GRE, IBPS Examinations, ICET, KPSC, L.I.C, MAT, Railway Competitive Examinations SSC Competitive Examinations, SNAP Test, TNPSC, UPSC Competitive Examinations, University Grants Commission (UGC), XAT and all Government and any other Competition Examination etc.**

"QUANTITATIVE APTITUDE Simplified" deals with almost all chapters which are prominently asked in various examinations. Some of these are like : Age related, Areas and volumes, Average (Mean), Banker's Discount, Boats and Streams, BODMAS – Simplifications, Calendar, Chain Rule or Unitary methods, Clocks, Data Interpretation, Decimal Fractions, H.C.F. and L.C.M, Heights and Distance, Interest – Simple and Compound, Logarithms, Missing Numbers Series, Mixtures and Alligations, Numbers Systems, Odd Man out Series, Partnerships, Percentage, Permutations and Combinations, Pipes and Cistern, Probability, Profit & Loss, Races and Games, Ratio & Proportion, Square Root and Cube Root, Stocks and Shares, Surds and Indices, Time and Distance, Time and Work, Trains etc.

Normally competitive examinations consist of **MCQs** which need to be completed within the stipulated period of time. Some students prepare very hard but give little consideration to the importance to time. Time provided for competitive examinations is tightly tagged and one can solve all the questions only if the questions are solved faster BUT accurately. Out of numerous methods to solve a problem, some might help solve the problem faster. Selecting the right method is, therefore, a key factor for success in competitive examinations. All this calls for a thorough but solid practice on these questions for which the present book **"QUANTITATIVE APTITUDE Simplified"** has been prepared.

ARITHMETIC APTITUDE QUESTIONS are important for Bank POs, CAT, CDS, Clerks, CBI, CGL, CLAT, CTET, CPO, GMAT, IIFT, IGNOU, IMA, MBA, MAT, NDA, RBI, SBI-PO, SSC, Specialist Officers and other competitive examinations.

AREA AND VOLUMES QUESTIONS include questions of Area of Rectangle, Square, Triangle, Rhombus, Trapezium, Circle , Cuboids, Cube, Cylinder, Cone, Sphere, Hemisphere etc which are important for competitive examinations like SSC, BANK PO, Clerk, IBPS, SBI-PO, RBI, MBA, MAT, CAT, IIFT, IGNOU, SSC CGL, CBI, CPO, CLAT, CTET, NDA, CDS, Specialist Officers.

PERMUTATION AND COMBINATION QUESTIONS include questions on calculating combinations, permutations by different tricks which are important for many competitive examinations.

PROBABILITY QUESTIONS include questions of determining chances of happening an event in different type of problems using different methods which are important for many competitive examinations.

HEIGHT AND DISTANCE QUESTIONS include questions of calculating distances, angles etc. using short trick methods which are important for many competitive examinations.

DATA INTERPRETATION QUESTIONS (MCQs) deal with various types of questions pertaining to Line, Bar, Pie, Tabulation, Sets, Statement type questions.

CONTENTS

NUMBER SYSTEMS

1

A measurement carried out in the physical world leads directly to a number or some meaningful value which is known as **Real Number**.

Real Numbers can be divided into two main groups:

(A) **Rational Numbers:** A rational number can always be represented by a fraction of the form (p/q) where p and q are integers and q is not = 0. Every integer and all the fractions are rational numbers including finite decimal numbers.

(B) **Irrational Numbers:** An irrational number cannot be expressed in the form p/q where q is not = O. e.g., under root of 7 gives an approximate answer in the form of a fraction or decimal but the digits after the decimal point are non-ending, hence $\sqrt{7}$ is an irrational number. Similarly, $\pi = 3.14....$ is irrational. In other words, an infinite non-recurring decimal is an irrational number which is non-terminating.

Note: No number can be both rational and irrational.

Prime Numbers: A prime number is a number which has no other factors except itself and unity

e.g., 2, 3, 5, 7, 11 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97, 101, 103, 107, …. 173, 179, 181, 191, 193, 197, 199, etc., 2 is the only even number which is prime.

- *There are 25 prime numbers between 1 & 100.*
- *To check if a particular number ‘k’ is prime or not, we should apply divisibility tests of all primes up to $\sqrt{k}$*
- *If a number has no factor equal to or less than its square root, then that number must be prime.*
- *‘1’ is neither Prime nor composite.*
- *Largest prime no. known so far is (22281 – 1) which is of about 700 digits.*

Co-Prime Numbers: Two numbers are relatively prime to each other when their highest common factor is 1 or we can say that they have no common factor except 1.

For being co-prime the numbers themselves need not be prime

- *E.g. 35 has factors 1, 5, 7, 35 i.e., it is not a prime number*
- *And 49 has factors 1, 7, 49 i.e., again it is not a prime number*

BUT these are Co-prime numbers as no common factor except 1.

Composite Numbers: A composite number is a number which has more than 2 factors i.e., itself and unity, e.g., or in other words we can say any number that has atleast 3 factors. e.g., 14, 36, 345 etc., i.e., it must have 3 factors – (1, itself and one More)

Even numbers are divisible by 2 and end up with 0, 2, 4, 6 or 8

Odd numbers are not divisible by 2 and end up with 1, 3, 5, 7 or 9.

Some important results on Numbers:

- *If n is odd, $n(n^2 - 1)$ is divisible by 24 and if n is an odd prime greater than 3, $n^2 - 1$ is divisible by 24 e.g., if $n = 11$ then $11^2 - 1 = 120$ is divisible by 24.*
- *If n is odd, $2^n + 1$ is divisible by 3, and if n is even, $2^n - 1$ is divisible by 3.*
- *If n is prime to 5, then $\{n(n^4 - 1\})$ is divisible by 30.*
- *If n is odd, $\{2^{2n} + 1\}$ is divisible by 5, and if n is even, $\{2^{2n} - 1\}$ is divisible by 5.*
- *If n is odd $\{5^{2n} + 1\}$ is divisible by 13 and if n is even, $\{5^{2n} - 1\}$ is divisible by 13.*
- *If $a > b$, then $a - b = 1 - b$ and if $a < b$, then $a - b = b - a$.*

There are an infinite number of rational numbers between any two rational numbers.

Test for Divisibility:

(1) A number is divisible by 2, if it ends with 0, 2, 4, 6 or 8 e.g., 4172, 790, 714, etc.

(2) A number is divisible by 3, if the sum of its digits is divisible by 3. e.g., 6072, 4701063.

(3) A number is divisible by 4, the number formed by the last two right hand digits is divisible by 4, or if the last two digits are 00 e.g., 362164, 725500.

(4) A number is divisible by 5, when its unit's digit is 5 or 0. e.g., 2745, 5760.

(5) A number is divisible by 6, when it is divisible by 2 and 3 **both**, e.g., 654, 984.

(6) A number is divisible by 8, if the number formed by the last three right hand digits is divisible by 8, or when the last three digits are 000. e.g., 80800, 8844832.

(7) A number is divisible by 9, when the sum of its digits is divisible by 9. e.g., 4263567.

(8) A number is divisible by 10, when its units digit is 0. e.g., 120, 4350.

(9) A number is divisible by 11, when the difference between the sum of the digits in the odd and even places is 0 or a multiple of 11. e.g., 6172859. Here $(9 + 7 + 8 + 6) = 30$ and $1 + 2 + 5 = 8$, $30 - 8 = 22$ which is a multiple of 11.

Note: When any number with an even number of digits is added to its reverse, the sum is always a multiple of 11. e.g., abcd + dcba = pqrs…. Where "pqrs" is divisible by 11.

(10) A number is divisible by 12, when it is divisible by 3 and 4 both. e.g., 81004572.

(11) A number is divisible by 25, when the number formed by the last two right hands digits is divisible by 25, e.g., 783425.

(12) A number is divisible by 125, when the number formed by the last three right hand digits is divisible by 125. e.g., 824625.

TABLES AND MULTIPLES – please remember and write down for better memory

1	2	3	4	5	6	7	8	9	10
2	4	6	8	10	12	14	16	18	20
3	6	9	12	15	18	21	24	27	30
4	8	12	16	20	24	28	32	36	40
5	10	15	20	25	30	35	40	45	50
6	12	18	24	30	36	42	48	54	60
7	14	21	28	35	42	49	56	63	70
8	16	24	32	40	48	56	64	72	80
9	18	27	36	45	54	63	72	81	90
10	20	30	40	50	60	70	80	90	100
11	22	33	44	55	66	77	88	99	110
12	24	36	48	60	72	84	96	108	120
13	26	39	52	65	78	91	104	117	130
14	28	42	56	70	84	98	112	126	140
15	30	45	60	75	90	105	120	135	150
16	32	48	64	80	96	112	128	144	160
17	34	51	68	85	102	119	136	153	170
18	36	54	72	90	108	126	144	162	180
19	38	57	76	95	114	133	152	171	190
20	40	60	80	100	120	140	160	180	200

SQUARES AND CUBES – please remember and write down for better grip

X	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
X^2	1	4	9	16	25	36	49	64	81	100	121	144	169	196	225
X^3	1	8	27	64	125	216	343	512	729	1000	1331	1728	2197	2744	3125

PRIME NUMBERS - please remember and write down for better grip

2	3	5	7	11	13	17	19	23	29
31	37	41	43	47	53	59	61	67	71
73	79	83	89	97	101	103	107	109	113
127	131	137	139	149	151	157	163	167	173
181	191	193	197	199					

Practice Exercise - 1

Number of Questions: 20 **Suggested time: 10 minutes**

1. The numbers which are not divisible by 2 are called as
(a) prime numbers
(b) rational numbers
(c) whole numbers
(d) odd numbers

2. The numbers which are divisible by 2 are known as
(a) even numbers
(b) odd numbers
(c) natural numbers
(d) prime numbers

3. "N" numbers are called
(a) prime numbers
(b) even numbers
(c) natural numbers
(d) odd numbers

4. The numbers which are not only divisible by themselves and 1 are known as
(a) odd numbers
(b) composite numbers
(c) even numbers
(d) integers

5. 4, 6, 8, 10, 12, 14, 16, 18, ……. are
(a) even numbers
(b) odd numbers
(c) prime numbers
(d) none of the above

6. 1, 2, 3, ..., 99, 100 are called
(a) natural numbers
(b) even numbers

(c) odd numbers
(d) prime numbers

7. 2, 3, 5, 7, 11, 13, 17, 19, ………. are
(a) even numbers
(b) prime numbers
(c) odd numbers
(d) composite numbers

8. The sum of first seven even numbers is
(a) 30
(b) 56
(c) 24
(d) 36

9. The sum of first six odd numbers is
(a) 30
(b) 36
(c) 24
(d) 42

10. The least prime number is
(a) 0
(b) –1
(c) 2
(d) 1

11. Which of the following numbers is prime
(a) 79
(b) 204
(c) 822
(d) 644

12. A number divisible by 8 can never end with
(a) 20
(b) 54
(c) 48
(d) 12

13. Which one is divisible by 15

(a) 39825
(b) 575450
(c) 57235
(d) 614255

14. A prime number nearest to 68 is
(a) 69
(b) 70
(c) 67
(d) 63

15. How many prime numbers are there in the following numbers 3, 18, 28, 37, 54, 67, 79, 91, 113, 171, 253, 346
(a) 2
(b) 3
(c) 4
(d) 5

16. Two numbers differ by 168. If the greater one is 350. The smaller one is
(a) 400
(b) 700
(c) 182
(d) 180

17. The least number of 3 digits subtracted from greatest number of 6 digits yields.
(a) 999899
(b) 99999
(c) 10000
(d) 10999

18. The number which when added to itself 17 times gives 180 is
(a) 2
(b) 5
(c) 10
(d) 8

19. Find the number which when multiplied by 13 is increased by 240 is
(a) 40
(b) 30

(c) 10
(d) 20

20. The number which is neither prime nor composite is
(a) 2
(b) –1
(c) 1
(d) 0

Practice Exercise - 1 Answers

1 (d)	2 (a)	3 (c)	4 (b)	5 (a)
6 (a)	7 (d)	8 (b)	9 (b)	10 (c)
11 (a)	12 (b)	13 (a)	14 (e)	15 (d)
16 (c)	17 (a)	18 (c)	19 (d)	20 (d)

Practice Exercise - 2

Number of Questions: 20 **Suggested time: 10 minutes**

- *You are advised to prefer a hit on the choices.*
- *Don't ever try to solve such questions fully or actually until the situation so warrants.*
- *Try to do the questions as much orally as you can.*

1. In a two-digit number, the digit in the unit's place is more than twice the digit in ten's place by 1. If the digits in the unit's place and the ten's place are interchanged, difference between the newly formed number and the original number is less than the original number by 1. What is the original number?
(a) 35
(b) 36
(c) 37
(d) 39

2. In a two-digit number, if it is known that its unit's digit exceeds its ten's digit by 2 and that the product of the given number and the sum of its digits is equal to 144, then the number is:
(a) 12
(b) 24
(c) 36
(d) 48

3. Three numbers are in the ratio 4: 5: 6 and their average is 25. The largest number is:
(a) 30
(b) 40
(c) 50
(d) 60

4. The product of two numbers is 9375 and the quotient, when the larger one is divided by the smaller, is 15. The sum of the numbers is:

(a) 100
(b) 200
(c) 300
(d) 400

5. The denominator of a fraction is 3 more than the numerator. If the numerator as well as the denominator is increased by 4, the fraction becomes 4/5. What was the original fraction?
(a) 7/11
(b) 8/11
(c) 9/11
(d) 10/11

6. The sum of three numbers is 136. If the ratio between first and second be 2: 3 and that between second and third is 5: 3, then the second number is:
(a) 30
(b) 40
(c) 50
(d) 60

7. Three times the first of three consecutive odd integers is 3 more than twice the third. The third integer is:
(a) 12
(b) 137
(c) 14
(d) 15

8. If two painters can complete two rooms in two hours, how many painters would it take to do 18 rooms in 6 hours?
(a) 2
(b) 6
(c) 4
(d) 9

9. Fifty-four is to be divided into two parts such that the sum of 10 times the first and 22 times the second is 780. The bigger part is?
(a) 35

(b) 36
(c) 37
(d) 39

10. A number consists of 3 digits whose sum is 10. The middle digit is equal to the sum of the other two and the number will be increased by 99 if its digits are reversed. The number is:
(a) 253
(b) 263
(c) 273
(d) 283

11. 243 has been divided into three parts such that half of the first part, one-third of the second part and one-fourth of the third part are equal. The largest part is?
(a) 27
(b) 81
(c) 123
(d) None

12. There are two numbers such that the sum of twice the first and thrice the second is 39, while the sum of thrice the first, and twice the second is 36. The larger of the two is:
(a) 3
(b) 6
(c) 9
(d) 12

13. The difference between a two-digit number and the number obtained by interchanging the two digits is 63. Which is the smaller of the two numbers?
(a) 12
(b) 15
(c) 17
(d) None

14. The sum of the squares of two numbers is 3341 and the difference of their squares is 891. The numbers are:

(a) 35, 46
(b) 35, 50
(c) 40, 55
(d) 45, 60

15. The product of three consecutive even numbers when divided by 8 is 720. The product of their square roots is:
(a) $12\sqrt{10}$
(b) $24\sqrt{10}$
(c) $2\sqrt{10}$
(d) $4\sqrt{10}$

16. The difference between two numbers is 1365. When larger number is divided by the smaller one, the quotient is 6 and the remainder is 15. The smaller number is?
(a) 240
(b) 225
(c) 255
(d) None

17. If the product of 6 integers is negative, at most how many of the integers can be negative?
(a) 2
(b) 3
(c) 4
(d) 5

18. A number is doubled and 9 is added. If the resultant is trebled, it becomes 75. What is that number?
(a) 2
(b) 4
(c) 6
(d) 8

19. The total weight of a tin and the cookies it contains is 2 pounds. After 3/4 of the cookies are eaten, the tin and the remaining cookies weigh 0.8 pounds. What is the weight of the empty tin in pounds?

(a) 0.2
(b) 0.3
(c) 0.4
(d) 0.5

20. What approximate value will come in place of the question mark(?) in the below question? (47% of 1442 – 36% of 1412) + 63 =?
(a) 4
(b) 5
(c) 3
(d) 6

21. If 50 is subtracted from two-third of a number, the result is equal to sum of 40 and one-fourth of that number. What is the number?
(a) 156
(b) 179
(c) 198
(d) 216

22. If the sum and difference of two numbers are 20 and 8 respectively, then the difference of their squares is:
(a) 120
(b) 140
(c) 160
(d) 180

23. Half the people on a bus get off at each stop after the first, and no one gets on after the first stop. If only one person gets off at stop number 7, how many people got on at the first stop?
(a) 128
(b) 64
(c) 32
(d) 16

24. $(3x + 2)(2x - 5) = (ax^2 + kx + n)$ then find is the value of $(a - n + k)$?
(a) 5
(b) 8
(c) 9

(d) 10

25. Find the number which when multiplied by 15 is increased by 196.

(a) 10

(b) 12

(c) 14

(d) 16

Practice Exercise - 2 Answers with Hints

1 (c)	2 (b)	3 (a)	4 (d)	5 (b)
6 (d)	7 (d)	8 (b)	9 (b)	10 (a)
11 (d)	12 (c)	13 (d)	14 (a)	15(b)
16 (d)	17 (d)	18 (d)	19 (c)	20 (c)
21 (d)	22 (d)	23 (b)	24 (a)	25 (c)

4. Let the numbers be x and y; Then, $xy = 9375$ and $x/y = 15$; $\frac{xy}{\left(\frac{x}{y}\right)} = \frac{9375}{15}$

$\Rightarrow y^2 = 625 \Rightarrow y = 25 \Rightarrow x = 15y = 15 \times 25 = 375.$

11. No given choices can be divided by 4.

15. Let the numbers be x, $x + 2$ and X + 4.

Then, $x(x + 2)(x + 4)/8 = 720 \Rightarrow x(x + 2)(x + 4) = 5760.$

$\sqrt{x} * \sqrt{x+2} * \sqrt{x+2} = \sqrt{x(x+2)(x+4)} = \sqrt{5760} = 24\sqrt{10}$

19. Let the weight of the empty tin = w

(2 – w) = weight of the cookies; and one quarter of the cookies weigh (2 – w)/4

One quarter of the cookies + tin = 0.8 = w + (2 – w)/4

Multiply through by 4; 3.2 = 4w + 2 – w

1.2 = 3w; w = 0.4

21. Let the number be x; Then, $\frac{2}{3}x - 50 = \frac{1}{4}x + 40 \Rightarrow \frac{5}{12}x = 90$; $x = 216$

24. This is a quadratic equation.

To find ‘n’ compare constant on each side; ‘n’ = –10

To find ‘k’ we compare co-efficient of ‘x’; ‘k’ = –11

To find ‘a’, we compare the co-efficient of x^2 on each side. So a = 6

a – n + k = 6 – (–10) + (–11) = 5

Quick Test - I

(5 minutes job)

1. $\left(\frac{13}{12}\right)^2 - \left(\frac{1}{2}\right)^2 - \left(\frac{1}{3}\right)^2 - \left(\frac{1}{4}\right)^2 = ?$

(a) $\frac{5}{6}$ (b) $\frac{5}{6}$ (c) $\frac{5}{6}$ (d) $\frac{5}{6}$

2. Find the number whose double is greater than its half by 45.
(a) 33
(b) 21
(c) 27
(d) 30

3. If you add to a number $\frac{5}{6}$ of the same number; the sum would be twice the number less by 27.
What is the number?
(a) 63
(b) 49
(c) 56
(d) 52

4. B is now 1/3 as old as A, but 8 years ago A was 7 times as old as B. How old is B now.
(a) 8
(b) 16
(c) 12
(d) 9

5. The difference between two numbers is 3 and the difference of their squares is 51. Find the numbers.
(a) 6, 9
(b) 8, 11
(c) 7, 10
(d) 9, 12

Quick Test - I Answers with Hints

1 (b)	2 (d)	3 (a)	4 (c)	5 (c)

1. Apply $X^2 - Y^2 = (x - y)(x + y)$ and solve.

2. $2x - \frac{5}{6}x = 45$.

3. Prefer a hit on the choices OR conclude that 3/7 of the number = 27(?).

Quick Test - II

This test is exclusively to check your divisibility tests perfection **(Not more than 5 minutes)**

1. Which one is divisible by 3?
(a) 35432
(b) 38792
(c) 318412
(d) 51782
(e) none

2. Which one is divisible by 12?
(a) 56238
(b) 39848
(c) 38468
(d) 56324
(e) none

3. What should be in the place of A so that the number 58743A8 is divisible by 8?
(a) 1
(b) 3
(c) 4
(d) 5
(e) 6

4. To make 75 × 368 divisible by 9, x should be?
(a) 3
(b) 4
(c) 5
(d) 7
(e) none

5. A number divisible by 11 is?
(a) 3824521

(b) 398321
(c) 563281
(d) 368285
(e) none

6. Which one is divisible by 15?
(a) 38532
(b) 57545
(c) 39825
(d) 57235
(e) none

7. A number divisible by 8 may have _____ in the end.
(a) 2
(b) 4
(c) 6
(d) All
(e) None

8. A number divisible by 25 is?
(a) 583630
(b) 58560
(c) 395425
(d) 58005
(e) none

9. Which one is divisible by 6?
(a) 35432
(b) 38351
(c) 85254
(d) 53823
(e) none

10. Which one is divisible by 45?
(a) 38565
(b) 38543
(c) 58555

(d) 85555

(e) 98655

Quick Test - II Answers

1 (c)	2 (c)	3 (e)	4 (d)	5 (b)
6 (c)	7 (d)	8 (c)	9 (c)	10 (a)

Quick Test - III

(only 6 minutes)

1. How many two's are contained in 48!?
(a) 45
(b) 46
(c) 48
(d) 50

2. How many three's are involved in 55!?
(a) 22
(b) 24
(c) 26
(d) 28

3. How many zeros does 72! Contain?
(a) 15
(b) 14
(c) 16
(d) 7

4. How many factors will 144 have?
(a) 15
(b) 10
(c) 6
(d) 8

5. No. of prime factors of 4410 is?
(a) 4
(b) 15
(c) 18
(d) 21

6. 1, 2, 3, … written without break what will be the 2448^{th} digit?
(a) 7
(b) 8
(c) 9
(d) none

7. $27^{45} + 45^{45}$ can be divided by:
(a) 8
(b) 24
(c) 36
(d) 72

8. $58^{72} – 42^{72}$ is divisible by:
(a) 400
(b) 1600
(c) 200
(d) 2400

9. $(12^{72} – 1)$ when divided by 11 leaves:
(a) 4
(b) 2
(c) 0
(d) 9

10. How many months will have 28 days in a 400–year calendar?
(a) 40
(b) 100
(c) 103
(d) 400

Quick Test - III Answers with Hints

1 (a)	2 (c)	3 (c)	4 (c)	5 (a)
6 (d)	7 (d)	8 (b)	9 (c)	10 (d)

Q 1

2	48
2	24
2	12
2	6
2	3
2	1
	0

There are 24+12+6+3+1+0 = 46 two's in 48!

Q 3 Check 72! for 5 & 2 both and select lower of the two powers of 5 and 2 involved.

i.e., If it contains 5^x and 2^y select x if $x < y$

Q 7, 8 & 9 $(X^n - Y^n)$ is divisible by

- $(X - Y)$ if n is odd
- $(X - Y)$ and $(X + Y)$ both if n is Even

  $(X^n + Y^n)$ is divisible by
- $(X + Y)$ if n is odd
- Neither by $(X - Y)$ nor by $(X + Y)$ both if n is Even

SIMPLIFICATION

BODMAS rule defines the correct sequence in which operations are to be performed in a given mathematical expression to find its value.

B = Bracket – O = Order (Powers, Square Roots, etc.)

D M = Division and Multiplication (left-to-right) – AS = Addition and Subtraction (left-to-right)

While simplifying an expression, the following order must be followed.

- *Do operations in brackets first, strictly in the order [{ ( ) }]*
- *Evaluate exponents (powers, roots, etc.)*
- *Perform division and multiplication (division and multiplication rank equally and are solved left to right).*
- *Perform addition and subtraction (addition and subtraction rank equally and are done left to right).*

Examples

(a) $17 + 22 \div 11 \times (18 \div 3)^2 - 13$

$= 17 + 22 \div 11 \times 6^2 - 13$ ($\because$ brackets first)

$= 17 + 22 \div 11 \times 36 - 13$ ($\because$ exponents)

$= 17 + 2 \times 36 - 13$ ($\because$ division and multiplication)

$= 17 + 72 - 13$ ($\because$ division and multiplication, left to right)

$= 89 - 13$ ($\because$ addition and subtraction, left to right)

$= 76$

(b) $9 + 14 - 3 \times 6/3 + 2$

$= 9 + 14 - 18/3 + 2$ ($\because$ division and multiplication, left to right)

$= 9 + 14 - 6 + 2$ ($\because$ division and multiplication, left to right)

$= 23 - 6 + 2$ ($\because$ addition and subtraction, left to right)

$= 17 + 2$ ($\because$ addition and subtraction, left to right)

$= 19$

(c) $7 + 8/2 \times 2$

$= 7 + 4 \times 2$ ($\because$division and multiplication, left to right)

$= 7 + 8$ ($\because$division and multiplication, left to right)

$= 15$

Modulus of a Real Number

Absolute value or modulus x is denoted by $|x|$ and it is the non-negative value of x
regard less of its sign. $|x|$ is defined as

$|x| = x$; if $x \geq 0$

$= -x$; if $x < 0$

For example $|48| = |-48| = 48$

SIMPLIFICATIONS

Practice Exercise - 1

Number of Questions: 20 **Suggested time: 15 minutes**

1. The price of 80 apples is equal to that of 120 oranges. The price of 60 apples and 75 oranges together is Rs. 1320. The total price of 25 apples and 40 oranges is Rs.
(a) 660
(b) 620
(c) 820
(d) 780

2. The price of 24 apples is equal to that of 28 oranges. The price of 45 apples and 60 oranges together is Rs. 1350. The total price of 30 apples and 40 oranges is Rs.
(a) 920
(b) 940
(c) 880
(d) 900

3. There are two buildings P and Q. If 15 persons are sent from P to Q, then the number of persons in each building is the same. If 20 persons are sent from Q to P, then the number of persons in P is double the number of persons in Q. How many persons are there in building P?
(a) 80
(b) 140
(c) 120
(d) 90

4. The price of 3 tables and 4 chairs is Rs. 3300. With the same money one can buy 2 tables and 10 chairs. If one wants to buy 1 table and 1 chair, how much does he need to pay?
(a) 80
(b) 140
(c) 120

(d) 90

5. There are 6 working days in a regular week and for each day, the working hours are 10. A man earns Rs 2.10 per hour for regular work and Rs. 4.20 per hour for overtime. If he earns Rs. 525 in 4 weeks, how many hours did he work?

(a) 245
(b) 285
(c) 275
(d) 255

6. A man has some hens and cows. If the number of heads be 48 and the number of feet equals 140 then the number of hens will be:

(a) 22
(b) 24
(c) 26
(d) 20

7. A sum of Rs. 2200 has been divided among A, B and C such that A gets 1/4 of what B gets and B gets 1/5 of what C gets. What is B's share?

(a) 341
(b) 364
(c) 372
(d) 352

8. A fires 5 shots to B's 3 but A kills only once in 3 shots while B kills once in 2 shots. When B has missed 27 times, A has killed:

(a) 30
(b) 22
(c) 18
(d) 38

9. If $p - q = 6$ and $p^2 + q^2 = 116$, what is the value of pq?

(a) 30
(b) 40
(c) 20
(d) 50

10. To fill a tank, 25 buckets of water is required. How many buckets of water will be required to fill the same tank if the capacity of the bucket

is reduced to two-fifth of its present?
(a) 63
(b) 64.5
(c) 62.5
(d) 60.5

11. John gets on the elevator at the 14^{th} floor of a building and rides up at the rate of 84 floors per minute. At the same time, Vinod gets on an elevator at the 58^{th} floor of the same building and rides down at the rate of 92 floors per minute. If they continue travelling at these rates, then at which floor will their paths cross?
(a) 38
(b) 36
(c) 32
(d) 35

12. A man has Rs. 312 in the denominations of one-rupee notes, five-rupee notes and twenty-rupee notes. The number of notes of each denomination is equal. What is the total number of notes that he has?
(a) 36
(b) 24
(c) 28
(d) 32

13. Free notebooks were distributed equally among children of a class. The number of notebooks each child got was one-eighth of the number of children. Had the number of children been half, each child would have got 16 notebooks. Total how many notebooks were distributed?
(a) 602
(b) 528
(c) 423
(d) 512

14. Eight people are planning to share equally the cost of a rental car. If one person withdraws from the arrangement and the others share equally the entire cost of the car, then the share of each of the remaining persons increases by:

(a) 2/5

(b) 1/8

(c) 2/7

(d) 1/7

15. $(723 + 1992)^2 – (723 – 1992)^2 – 2912 \times 1992$ =?

(a) 5

(b) 18

(c) 128974

(d) None

16. One-third of Rahul's savings in National Savings Certificate is equal to one-half of his savings in Public Provident Fund. If he has Rs. 1, 80, 000 as total savings, how much has he saved in Public Provident Fund?

(a) 72000

(b) 44000

(c) 58000

(d) 92000

17. 8/4/2 =?

(a) 4

(b) 2

(c) 1

(d) 0

18. $20 + 20 \times 2$ =?

(a) 40

(b) 50

(c) 60

(d) 70

19. $25/5 \times 5$ =?

(a) 25

(b) 15

(c) 20

(d) 30

20. $5 \times 5/5$ =?

(a) 5
(b) 1
(c) 10
(d) 25

SIMPLIFICATIONS

Practice Exercise - 1 Answers with Hints - need based

1 (b)	2 (d)	3 (c)	4 (b)	5 (a)
6 (c)	7 (d)	8 (a)	9 (b)	10 (a)
11 (d)	12 (a)	13 (d)	14 (d)	15 (a)
16 (a)	17 (b)	18 (c)	19 (a)	20 (a)

1. Price of 80 apples = Price of 120 oranges ⇒ $80a = 120b \Rightarrow 2a = 3b$ (1)
and also $60a + 75b = 1320 \Rightarrow 4a + 5b = 88 \Rightarrow (6b) + 5b = 88$ or $b = 8$, $a = 12$

3. $\Rightarrow p - 15 = q + 15 \Rightarrow p - q = 30$ (1)
$\Rightarrow 2(q - 20) = p + 20 \Rightarrow 2q - p = 60$ (2)
(1) + (2) ⇒ $q = 90$ or $p = 30 + q = 30 + 90 = 120$ i.e., Building P has 120 persons

4. $3t + 4c = 3300$ (1)
$2t + 10c = 3300 \Rightarrow t + 5c = 1650$ (2)
From the above we can find the cost of 1 table and 1 chair.
Cost of 1 table and 1 chair
$= c + t = 150 + 900 = 1050$

5. Regular working hours in 4 weeks = $4 \times 6 \times 10 = 240$ hours
Amount earned by working in these regular working hours = 240×2.10 = Rs. 504
Additional amount earned = $525 - 504$ = Rs. 21
Hours he worked overtime = $21/4.2 = 210/42 = 5$ hours
Total hours he worked = $240 + 5 = 245$ hours

6. Let hens be ‘h’ and cows are ’48 – h’;
Therefore, legs are $2h + 4(48 - h) = 140$ or $h = 26$

7. Let C get = $100x$, B gets = $20x$, A gets = $5x$;

$100x + 20x + 5x = 125x = 2200$;

$x = 88/5$ or B gets 352

8. 2B missed 27 times. Hence, B fired $27 \times 2 = 54$ shots

Therefore, number of shots fired by A = $54/3 \times 5 = 90$

Therefore, number of birds killed by A = $90 \times 1/3 = 30$

9. $(p - q)^2 = p^2 - 2pq + q^2$

$\Rightarrow 6^2 = 116 - 2pq$

$\Rightarrow 36 = 116 - 2pq$

$\Rightarrow 2pq = 80$

$\Rightarrow pq = 40$

10. Suppose their paths cross after x minutes. Then,

$14 + 84x = 58 - 92x \Rightarrow 176x = 44 \Rightarrow x = 44/176 = 1/4$

Therefore, required floor = $14 + 84x = 14 + 84 \times 1/4 = 35$

11. Let each note be 'x' in number.

Therefore, $x + 5x + 20x = 312$

12. Given condition is that iIf number of children was half, each child would have got 16 books. Therefore, actually each child got $16/2 = 8$ books and the number of children is $8 \times 8=64$. Hence, total number of books distributed = $64 \times 8 = 512$.

13. Suppose each of the eight persons spends Rs.7 initially. Then the total cost is $8 \times 7 = 56$ If seven persons share the cost equally, each of them has to spend 567 = Rs. 8, leading to an increase of Rs. 1; Required fraction =1/7.

SIMPLIFICATIONS

Practice Exercise - 2

Number of Questions: 20 **Suggested time: 12 minutes**

1. $b - [b - (a + b) - \{b - (b - a + b)\} + 2a] =?$
(a) 0
(b) 4a
(c) a
(d) –2 a

2. $213 + 312 + 414 =? - 63$
(a) 1012
(b) 1016
(c) 1014
(d) 939

3. If $a * b = 2a - 4b + 2ab$, then $2 * 3 + 3 * 2 =?$
(a) 2
(b) 0
(c) 14
(d) 12

4. $14 - 14 \times 14/14=?$
(a) 2
(b) 0
(c) 14
(d) 12

5. If the number of boys in a class is 8 times the number of girls, which value can never be the total number of students?
(a) 27
(b) 45
(c) 81
(d) 42

6. What fraction of 3/5 needs to be added to itself to become 21/4

(a) 4/5
(b) 23/4
(c) 31/4
(d) 3/4

7. An organization decided to raise Rs. 6 lakh by collecting equal contribution from each of its employees. If each of them had contributed Rs. 60 extra, the contribution would have been Rs. 6.24 lakhs. How many employees are there in that organization?
(a) 400
(b) 300
(c) 200
(d) 100

8. In a group of ducks and cows, the total number of legs is 28 more than twice the number of heads. Find the total number of cows.
(a) 14
(b) 12
(c) 16
(d) 8

9. If $a - b = 6$ and $a^2 + b^2 = 116$, then what is the value of ab?
(a) 20
(b) 40
(c) 60
(d) 80

10. A room has equal number of men and women. Eight women left the room, leaving twice as many men as women in the room. What was the total number of men and women present in the room initially?
(a) 32
(b) 34
(c) 28
(d) 30

11. From a group of boys and girls, 15 girls leave. There are then left 2 boys for each girl. After this, 45 boys leave. There are then 5 girls for each boy. Find the number of girls in the beginning
(a) 40

(b) 20
(c) 32
(d) 60

12. How many pieces of 85 cm length can be cut from a rod of 42.5 metres long?
(a) 15
(b) 20
(c) 2
(d) 50

13. In a garden, there are 10 rows and 12 columns of mango trees. The distance between two trees is 2 metres and a distance of one metre is left from all sides of the boundary of the garden. What is the length of the garden?
(a) 30
(b) 28
(c) 26
(d) 24

14. In a garden, 26 trees are planted at equal distances along a yard 300 metres long, one tree being at each end of the yard. What is the distance between two consecutive trees?
(a) 10
(b) 20
(c) 14
(d) 12

15. A boy was asked to multiply a number by 22. He instead multiplied the number by 44 and got the answer 308 more than the correct answer. What was the number to be multiplied?
(a) 16
(b) 10
(c) 14
(d) 12

16. In an examination, a student scores 4 marks for every correct answer and loses 1 mark for every wrong answer. If he attempts all 80

questions and secures 120 marks, How many questions does he answer correctly?
(a) 30
(b) 60
(c) 50
(d) 40

17. In an examination, a student scores 4 marks for every correct answer and loses 1 mark for every wrong answer. If he attempts all 60 questions and secures 130 marks, the no of questions he attempts correctly is:
(a) 35
(b) 38
(c) 40
(d) 42

18. In a regular week, there are 5 working days and for each day, the working hours are 8. A man gets Rs. 2.40 per hour for regular work and Rs. 3.20 per hours for overtime. If he earns Rs. 432 in 4 weeks, then how many hours does he work for?
(a) 160
(b) 175
(c) 180
(d) 195

19. If 2 tables and 3 chairs cost Rs, 3500 and 3 tables and 2 chairs cost Rs. 4000, then how much does a table cost?
(a) 500
(b) 1000
(c) 1500
(d) 2000

20. If $a * b = 2a - 3b + ab$, then $3 * 5 + 5 * 3$ is equal to:
(a) 22
(b) 23
(c) 24
(d) 25

SIMPLIFICATIONS

Practice Exercise - 2 Answers with Hints - need based

1 (a)	2 (a)	3 (c)	4 (c)	5 (d)
6 (b)	7 (a)	8 (a)	9 (b)	10 (a)
11 (a)	12 (d)	13 (d)	14 (d)	15 (c)
16 (d)	17 (b)	18 (b)	19 (b)	20 (a)

3. $2 * 3 = 2(2) - 4(3) + 2(2 \times 3) = 4 - 12 + 12 = 4$

$3 * 2 = 2(3) - 4(2) + 2(3 \times 2) = 6 - 8 + 12 = 10$

Therefore, $2 * 3 + 3 * 2 = 4 + 10 = 14$

5. Let number of Girls be 'G', then Boys are = '8G'; total is '9G' means the answer must be multiple of 9.

6. K $(3/5) + 3/5 = 21/4$

7. Extra 24, 000 is the collection by additional contribution of Rs. 60 per employee

10. Let initial number of men = initial number of women = x

$2(x - 8) = x \Rightarrow 2x - 16 = x \Rightarrow x = 16$

11. 15 girls leave and then there are 2 boys for each girl.

$\Rightarrow 2(g - 15) = b \Rightarrow 2g - b = 30°$ (1)

After this, 45 boys leave and then there are 5 girls for each boy

$5(b - 45) = (g - 15) \Rightarrow 5b - g = 210°$ (2)

$\Rightarrow 10g - 5b = 150°$ (3)

(1) multiplied by 5

$\Rightarrow 9g = 360 \Rightarrow g = 40$ i.e., number of girls in the beginning = 40

14. Between 26 trees, there are 25 gaps. Length of each gap =300/25 = 12

i.e., distance between two consecutive trees = 12

15. Let required number = x

Therefore, $22x + 308 = 44x \Rightarrow 22x = 308 \Rightarrow x = 308/22 = 14$

16. Let number of correct answers = x

Then, number of wrong answers = $(80 - x)$

Therefore, $4x - (80 - x) = 120 \Rightarrow x = 40$

SIMPLIFICATIONS

Practice Exercise - 3

Number of Questions: 20 **Suggested time: 10 minutes**

1. How many $\left(\frac{1}{4}\right)$ are there in $\left(\frac{1}{4}\right)$?

(a) 2
(b) 3
(c) 4
(d) 1/3

2. How many $\frac{5}{6}$s are there in $37\frac{1}{2}$?

(a) 300
(b) 400
(c) 500
(d) 600

3. If $a + b + c = 13$, $a^2 + b^2 + c^2 = 69$ then find the $ab + bc + ca$:

(a) 10
(b) 30
(c) 50
(d) 70

4. The difference of $1\frac{3}{16}$ and its reciprocal is equal to:

(a) 1/8
(b) 4/3
(c) 15/16
(d) 105/304

5. In a group of Cows and hens, the number of legs are 24 more than twice the number of heads. What is the number of cows in the group?

(a) 6
(b) 10
(c) 12

(d) Data inadequate

6. A tank is 2/5 full. If 16 litres of water is added to the tank, it becomes 6/7 full. The capacity of the tank is:

(a) 35
(b) 26
(c) 48
(d) 42

7. The highest score in an inning was 3/11 of the total and the next highest was 3/11 of the reminder. If the score differ by 9, the total score was:

(a) 110
(b) 121
(c) 132
(d) 143

8. Simplify the following equation.

$$\frac{19}{43}+\cfrac{1}{2+\cfrac{1}{3+\cfrac{1}{1-\cfrac{1}{4}}}}$$

(a) 43/19
(b) 1
(c) 38/17
(d) None

9. 38% of 7500 +? % of 375 = 50% of 6000.

(a) 50
(b) 40
(c) 60
(d) 45

10. If $3x + 7 = x^2 + p = 7x + 5$, what is the value of P?

(a) 17/2
(b) 25/3
(c) 33/4
(d) 41/5

11. By how much is three-fifth of 350 greater than four-seventh of 210?
(a) 90
(b) 100
(c) 110
(d) 120

12. $48\sqrt{?}+32\sqrt{?}=320$
(a) 4
(b) 8
(c) 16
(d) 32

13. If $\frac{a}{b}=\frac{4}{5}$ and $\frac{b}{c}=\frac{15}{16}$, then $\frac{c^2-a^2}{c^2+a^2}$ is?
(a) 7/5
(b) 7/25
(c) 7/125
(d) 7

14. 14.43% of 616.6 + 37% of 217 =?
(a) 345.428
(b) 354.428
(c) 344.428
(d) 364.428

15. $17\frac{4}{7}+13\frac{2}{12}-?=11\frac{2}{3}$
(a) 16
(b) 17
(c) 18
(d) 19

16. If $a-b=3$, $a^2+b^2=29$, find the value of ab.
(a) 10
(b) 12
(c) 15
(d) 18

17. If $x=1-q$ and $y=2q+1$, then for what value of q, x is equal to y?
(a) 0

(b) 1
(c) –1
(d) –2

18. Water boils at 212°F or 100°C and melts at 32°F or 0°C. If the temperature of the particular day is 35°C, it is equal to:
(a) 85°F
(b) 95°F
(c) 96°F
(d) 97°F

19. $[2 - 3\ \{4 - (5 \times 6 - 7 + 8)\}] = ?$
(a) 84
(b) 83
(c) 95
(d) 108

20. 48 of $24 \times 12\ [16 + 8 - 3$ of $(5 + 3)]$ is =
(a) 248
(b) 120
(c) 72
(d) None

SIMPLIFICATIONS

Practice Exercise - 3 Answers with Hints - need based

1 (b)	2 (a)	3 (c)	4 (d)	5 (c)
6 (a)	7 (b)	8 (d)	9 (a)	10 (a)
11 (a)	12 (c)	13 (b)	14 (a)	15 (d)
16 (a)	17 (a)	18 (b)	19 (b)	20 (d)

18. Let F and C denotes the temperature in Fahrenheit and Celsius respectively.
Then, $(F - 32)/(212 - 32) = (C - 0)/(100 - 0)$, if $c = 35$, then $F = 95$.

DECIMAL FRACTIONS & BODMAS

3

A *fraction* is a quantity which expresses a part of the whole.

e.g., $\frac{5}{6}$ means four-fifth of the whole.

$\frac{5}{6}$ mean seven – eighth of the whole.

$\frac{13}{24}$ means thirteen – twenty fourth of the whole.

Numerator and Denominator:

The lower number, which shows the fractional division of the number into which it has been distributed is called the ***denominator***. The number, that is being divided by this denominator or we can call it antecedent which is to be distributed and shows how many of these equal parts are taken to from the fraction is called the ***numerator*.**

BODMAS Rule:

In signifying fractions, remember the order indicated by each letter of the word B O D M A S, where

B = bracket; O = of; D = division; M = multiplication; A = addition; S = subtraction.

Example (a) Write down $44\frac{5}{6}$ in the form of an improper fraction.

(b) Write down $\frac{174}{45}$ in the form of a mixed fraction.

(c) Write down the descending order of the fraction $\frac{3}{4}, \frac{5}{6}, \frac{5}{8}$

Solution

(a) $44\frac{1}{5} = 44\frac{1}{5} = \frac{220+1}{5} = \frac{221}{5}$

(b) $\frac{174}{25} = \frac{150+24}{25} = \frac{150}{25} + \frac{24}{25} = 6 + \frac{24}{25} = 6\frac{24}{25}$

(c) The L.C.M. of 4, 6, 8 is 24.

$$\therefore \frac{3}{4} = \frac{3}{4} \times \frac{6}{6} = \frac{18}{24} \qquad \text{(i)}$$

$$\therefore \frac{5}{6} = \frac{5}{6} \times \frac{4}{4} = \frac{20}{24} \qquad \text{(ii)}$$

$$\therefore \frac{5}{8} = \frac{5}{8} \times \frac{3}{3} = \frac{15}{24} \qquad \text{(iii)}$$

Clearly, from (i), (ii) and (iii) the descending order is

$\frac{15}{24}, \frac{18}{24}, \frac{20}{24}$

Thus $\frac{3}{4}, \frac{5}{6}, \frac{5}{8}$ is the required order.

Example Find the sum of $6\frac{5}{6} + 7\frac{5}{6} + 8\frac{5}{6}$

Solution

First we add all the integers 6, 7 and 8 i.e., $6 + 7 + 8 = 21$

Now we add the proper fractions $\frac{5}{6}, \frac{5}{6}$ and $\frac{5}{6}$

Hence, $6\frac{3}{4} + 7\frac{1}{5} + 8\frac{3}{8} = 21\frac{(3\times10)(1\times8)(3\times5)}{40}$

$$= 21\frac{30+8+15}{40}$$

$$= 21\frac{53}{40} \times [21 \times 1\frac{13}{40}] = 22\frac{13}{40}$$

Example Solve $0.15 \times 0.025 \times 0.00045$.

Solution

Since, $15 \times 25 \times 45 = 16875$

Hence, $0.15 \times 0.025 \times 0.00045 = 0.0000016875$ $\because$Sum of decimal places = 2 + 3+5 =10]

To multiply a decimal by 10, 100, 1000, ……., shift the decimal point 1, 2, 3, …… places to the right.

e.g., $3.986 \times 10 = 39.86$

$3.986 \times 100 = 398.6$

DIVISION OF DECIMALS

Note the following:

(1) ***To divide a decimal number by an integer, put decimal point just after dividing the unit's digit.***

e.g., $\frac{35.3}{3} = 11.8$

(2) ***To divide one decimal by another***

(i) ***Find difference between the number of decimal in numerator and the denominator***

(ii) ***Simply divide as if decimals were not existing***

(iii) ***Put decimal symbol before as many as digits as was the answer in (i)***

(3) ***To divide a decimal by 10, 100, 1000, ….., shift the point 1, 2, 3, …, places to the left.***

e.g., $\frac{74425.1}{1000} = 744.251$

Practice Exercise - 1

Evaluate the following:

BLOCK 'A'

Q 1 – 15 Try finishing this block of questions in 10 minutes

1. (i) $2.123 + 0.0048 + 2.002$
(ii) $4.986 + 12.76 + 3.00005$
(iii) $186.34 + 8.91 + 1000.57 + 47.83$

2. (i) $48.02 – 37.895$
(ii) $229.432 – 218.543$
(iii) $114.05 – 58.04 – 52.035$

3. (i) $0.0124 \times .00325$
(ii) $17.6 \times .0125$

4.(i) $0.35209 \div 25.7$
(ii) $.00405594 \div .555$
(iii) $7.02 \div 39$

5. $(0.75)^2 - (0.25)^2$

6. $(2.31)^2 - (1.69)^2$

7. $\frac{7.0128 \times 6.025}{42.25212}$

8. $[(12.3)^2 \times 0.2 \times 0.12^2]$ divided by $[2.46 \times 36.9 \times 0.492]$

9. $\frac{(0.51)^2 - (0.49)^3}{(0.51)^2 + 0.51 \times 0.49 + (0.49)^2}$

10. $\frac{(0.51)^2 - (0.49)^3}{(0.51)^2 + 0.51 \times 0.49 + (0.49)^2}$

11. Evaluate $\frac{1.63 \times 1.63 - 4.74 \times 1.63 + 2.37 \times 2.37}{2.37 \times 2.37 - 1.63 \times 1.63}$ correct to two decimal places.

12. Evaluate $\frac{18.45 \times 0.184 \times 0.091}{2.32 + 4.58} \div 1.23$ correct to two significant figures.

13. $1.09 + (0.05 \times 3.62)$ and also give the value correct to one decimal place.

14. (i) Write .125 as a vulgar fraction.

(ii) Write $20 + 3\ \frac{174}{45}$ as one number using a decimal point.

(iii) Write $2 \times 10\frac{3}{10} + \frac{174}{45}$ as one number using a decimal point.

15. State for each of the following whether it is true or false.

(a) (i) $0.04 \times 0.04 = 0.16$ (ii) $0.1 \div 0.01 = 10$

(b) $\frac{5}{6} = 75$

(c) $\frac{13}{24} = 2.5\ \%$

BLOCK 'B'

Q 16 – 30 ***Finish this block in 10 minutes***

16. Express the following fraction in the form of improper fraction:

$5\frac{5}{9}, 7\frac{8}{15}, 9\frac{20}{27}, 10\frac{120}{121}$

17. Express the following fraction in the form of mixed fraction.

$\frac{75}{14}, \frac{123}{101}, \frac{3456}{789}$

18. Simplify the following:

$\frac{142}{22}, \frac{264}{192}, \frac{1450}{600}, \frac{375}{625}, \frac{625}{225}, \frac{278}{417}$

19. Write down the following fractions with same least denominators:

(i) $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}$

(ii) $\frac{74425.1}{1000}$

20. Arrange in ascending order:

(i) $\frac{16}{19}, \frac{26}{36}, \frac{76}{80}$

(ii) $\frac{16}{19}, \frac{26}{36}, \frac{76}{80}$

21. Add the following fractions:

(i) $\frac{15}{24}, \frac{18}{24}, \frac{20}{24}$

(ii) $3\frac{1}{5}, 2\frac{1}{15}, \frac{1}{18}, 1\frac{8}{45}$

22. Find the difference between the fractions:

(i) $41\frac{5}{32}, 37\frac{16}{33}$

(ii) $420\frac{7}{60}, 309\frac{43}{220}$

23. Find the value of:

(i) $13\frac{1}{3}+4\frac{1}{4}-12\frac{1}{2}$

(ii) $14\frac{2}{15}+23\frac{11}{20}-12\frac{1}{12}-23\frac{7}{18}$

24. State whether the following are true or

(i) $\frac{36\times60\times32\times126\times38}{19\times24\times36\times25\times84\times72}=\frac{2}{15}$

(ii) $\frac{1}{3}\times48=\frac{3}{4}\times24$

(iii) $92\frac{2}{3}+128\frac{1}{6}-322\frac{5}{6}=0$

(iv) $92\frac{2}{3}+128\frac{1}{6}-322\frac{5}{6}=0$

25. State whether the following statements are true or

(a) The value of $\frac{1}{2}+\frac{2}{13}\div\frac{4}{13}$ is $\frac{5}{8}$

(b) $\frac{5}{3}-\frac{6}{50}\div\frac{4}{25}=\frac{4}{3}$

26. Find out whether the following statements are true or

In the group of fractions $\frac{3}{8}, \frac{1}{3}, \frac{2}{7}, \frac{5}{15}, \frac{3}{10}$

(i) The largest is $\frac{3}{8}$

(ii) All the fractions have different values.

27. Simplify: $\dfrac{2\frac{3}{4}+\frac{1}{6}-1\frac{5}{9}}{11\frac{3}{4}-2\frac{4}{9}}$

28. Simplify. $\frac{(3\frac{1}{2})^2}{2\frac{1}{4}} - \frac{(2\frac{1}{3})^2}{2\frac{1}{9}}$

29. In the group of fractions $\frac{4}{24}, \frac{5}{31}, \frac{6}{37}, \frac{7}{43}, \frac{8}{50}$ find

(a) The least and the greatest fraction.
(b) The difference between the least and the greatest.
(c) The least is what fraction of the greatest?
(d) Whether all the fractions have different values?
(e) Ascending order of their magnitudes.

30. Show that

$$2\frac{4}{15} \div [1\frac{4}{15} \div 1\frac{20}{36} \div 6\frac{1}{2} \text{ of } (3\frac{1}{3} - 1\frac{1}{2} - 1\frac{5}{9})] - \frac{27}{415} = 1$$

BLOCK 'C'

Q 31 – 46 ***Time is 10 minutes only***

31. Divide the sum of $33\frac{1}{2}$ and $33\frac{1}{2}$ by the difference of $33\frac{1}{2}$ and $33\frac{1}{2}$.

32. A man spends one quarter of his income on food, $1/5^{th}$ of it on rent and the remaining, which is Rs. 231, on other commodities. Find his total income.

33. My son's age is 1/3 of my wife's age, my wife's age is 4/5 of my age, and my age is 3/5 of my father's age. Find the age of my son, if my father is 50 years old.

34. The highest score in a cricket test match is an innings was 5/21 of the total, and the next highest was 5/24 of the reminder. Difference between the score was 25 runs. What was the total score.

35. A candidate in an examination was asked to find 5/16 of a certain number. He made the mistake of finding 5/6 of it. His answer was 125 more than the correct answer. Find the number.

36. Write down three rational numbers P, Q and R between 1 and 2, such that PQ = QR. Is the solution unique?

37. From the adjoining number line, write down three rational numbers.
P, Q, R between – 1 and – 2, such That AP = PQ = QR =RB.
Is the solution unique?

38. Write down the following rational numbers with the same least denominator:

(i) $\frac{3}{4}, \frac{4}{5}, \frac{5}{6}$

(ii) $\frac{5}{6}, \frac{8}{9}, \frac{10}{11}, \frac{7}{18}$

39. Arrange the following rational numbers in ascending order:

(i) $\frac{18}{20}, \frac{15}{18}, \frac{12}{17}$

(ii) $\frac{7}{11}, \frac{21}{22}, \frac{16}{20}$

40. Evaluate the following:

(i) $\frac{3}{8}, \frac{1}{3}, \frac{2}{7}, \frac{5}{15}, \frac{3}{10}$

(ii) $\frac{5}{3} - \frac{6}{50} \div \frac{4}{25} = \frac{4}{3}$

41. Simplify

(i) $\{(5\frac{3}{4})^2 - (4\frac{1}{4})^2\}$ divided by $\{5\frac{3}{4} - 4\frac{1}{4}\}$

(ii) $\{(6\frac{4}{5})^2 - (3\frac{1}{5})^2\}$ divided by $\{5\frac{3}{4} - 4\frac{1}{4}\}$

42. State whether the following statements are true or false:

(i) The value of $\frac{1}{3} + \frac{3}{4} \div \frac{13}{12}$ is 1

(ii) The value of $\frac{2}{5} + \frac{5}{2} \div \frac{10}{3}$ is $\frac{23}{20}$

43. Find out whether the following statements are true or false:
In the group of fractions $\frac{2}{3}, \frac{3}{4}, \frac{5}{7}, \frac{6}{7}, \frac{3}{5}, \frac{5}{8}$

(i) The largest is $\frac{2}{3}$.

(ii) All the fractions have different values.

44. Square the following irrational numbers:

(i) $\sqrt{7}$
(ii) $\sqrt{11}$
(iii) $\left(\sqrt{5}\right)^2$

45. Square the following irrational numbers.
(i) $3\sqrt{3}$
(ii) $2\sqrt{2}$
(iii) $\frac{2}{\sqrt{2}}$

46. Insert 4 irrational numbers between two rational number 5 and 6.

Practice Exercise - 1 Answers

Block A

1. (i) 4.13 (ii) 20.74605 (iii) 343.65
2. (i) 10.125 (ii) 10.889 (iii) 3.975
3. (i) 0.0000403 (ii) 22
4. (i) 0.0137 (ii) 0.007308 (iii) 0.18
5. 0.5
6. 2.48
7. 1
8. 1
9. 0.02
10. 0.09
11. 0.19
12. 0.0336
13. 1.2710; 1.3
14. (i) 1/8 (ii) 23.03 (iii) 20.34
15. (a) (i) F (b) T (iii) F

Block B

16. $\frac{4}{24},\frac{5}{31},\frac{6}{37},\frac{7}{43},\frac{8}{50}$

17. $\frac{2}{5}+\frac{5}{2}\div\frac{10}{3}$ is $\frac{23}{20}$

18. $\frac{71}{11},\frac{11}{8},\frac{29}{12},\frac{3}{5},\frac{25}{9},\frac{2}{3}$

19. (i) $\frac{5}{3}-\frac{6}{50}\div\frac{4}{25}=\frac{4}{3}$ (ii) $\frac{32}{40},\frac{35}{40},\frac{36}{40}$

20. (i) $\frac{13}{18},\frac{16}{19},\frac{19}{20}$ (ii) $\frac{13}{18},\frac{16}{19},\frac{19}{20}$

21. (i) $\frac{c^2-a^2}{c^2+a^2}$ (ii) $110\frac{152}{165}$

22. (i) $33\frac{1}{2}$ (ii) $110\frac{152}{165}$

23. (i) $5\frac{2}{12}$ (ii) $2\frac{19}{90}$

24. (i) T (ii) F (iii) F (iv) T

25. (i) F (ii) F

26. (i) T (ii) F

27. $33\frac{1}{2}$

28. $33\frac{1}{2}$

29. (a) $\frac{8}{50},\frac{4}{240}$ (b) $5\frac{2}{12}$ (c) $\frac{24}{25}$ (d) Yes

30. $\frac{8}{50};\frac{5}{31};\frac{6}{37};\frac{4}{24}$

Block C

31. $33\frac{1}{2}$

32. Rs. 420

33. 8 Years

34. 15

35. 240

36. 1.25, 1.5, 1.75, No

37. −1.25, −1.50, −1.75, Yes

38. (i) $\frac{7}{11}, \frac{21}{22}, \frac{16}{20}$ (ii) $\frac{165}{198}, \frac{176}{198}, \frac{180}{198}, \frac{77}{198}$

39. (i) $\frac{13}{18}, \frac{16}{19}, \frac{19}{20}$ (ii) $\frac{7}{11}, \frac{21}{22}, \frac{16}{20}$

40. (i) 1.6 (ii) 2.983

41. (i) 10 (ii) 10

42. (i) True (ii) False

43. (i) False (ii) True

44. (i) 7 (ii) 11 (iii) $\sqrt{5}$

45. (i) 27 (ii) 8 (iii) 2

46. $\sqrt{26}$ $\sqrt{26}$ $3\sqrt{3}$ $3\sqrt{3}$

Practice Exercise - 2

Number of Questions: 20 **Suggested time: 12 minutes**

1. $55 \div 165 \div 1.5 =$
(a) 5
(b) .5
(c) 4.84
(d) .125

2. (?) $\times$ 9.99 = 999.999
(a) 100.1
(b) 1001
(c) 1.001
(d) 10.01

3. $2.075 \times .24 =$
(a) .49
(b) .498
(c) .488
(d) .48880

4. $\frac{18}{20}, \frac{15}{18}, \frac{12}{17}$
(a) 82
(b) 164
(c) .8
(d) 205

5. $2.75 \times 4 + 3.3 \times 5 - 1.8 \times 15 =$
(a) 27.5
(b) 1.5
(c) .5
(d) 27

6. $\frac{96.8}{8.81} =$
(a) 12

(b) 12.1
(c) 11
(d) 11.8

7. $\frac{0.74+0.074}{0.82-0.006}=$
(a) 1.1
(b) .1
(c) 1
(d) 10

8. $\frac{2.89-0.64}{0.25}=$
(a) .9
(b) .09
(c) 9
(d) 90

9. .1 of .1 ÷ .1 × .1 + .1 =
(a) .1
(b) .01
(c) .011
(d) .11

10. .2 + .2 − .2 + .2 −.2 + .2 =
(a) .2
(b) .22
(c) .04
(d) .4

11. A boy instead of multiplying 48 by 0.5 divided it by 0.5. He differs from the correct answer by:
(a) 48
(b) 12
(c) 72
(d) 192

12. To get .05 of a number it should be divided by ______
(a) 20
(b) .05

(c) 1/.5
(d) 1/20

13. A Number is 0.25 time of the other which itself is 0.3 time of a third number. Third number is _____ time of the first number?
(a) .75
(b) 5/6
(c) 6/5
(d) .075

14. √1257.68 =
(a) 34.4
(b) 35.4
(c) 36.4
(d) 35.8

15. √125.768 =
(a) 10.2
(b) 10.8
(c) 11
(d) 11.2

16. 5.56 is 2/5th of _____?
(a) 24.6
(b) 22.24
(c) 27.8
(d) 13.9

17. 1.08 × 12 × 03 =
(a) 12.96
(b) 3.88
(c) 1.29
(d) 38.8

18. 75 ÷175 ÷.75 =
(a) .057
(b) .57
(c) .58
(d) .59

19. $\frac{8}{50}, \frac{4}{240}$

(a) 9.6

(b) 4.8

(c) 6.4

(d) 7.2

20. (?) × (.99) = 999

(a) 1009

(b) 1008

(c) 100.9

(d) 1010

Practice Exercise - 2 Answers

1 (d)	2 (d)	3 (b)	4 (d)	5 (c)
6 (c)	7 (c)	8 (c)	9 (d)	10 (a)
11 (c)	12 (d)	13 (d)	14 (b)	15 (c)
16 (d)	17 (b)	18 (b)	19 (d)	20 (a)

Practice Exercise - 3

Number of Questions: 10 **Suggested time: 8 minutes**

1. When 0.36 is written in the simplest form, the sum of the numerator and the denominator is:
(a) 35
(b) 47
(c) 34
(d) 19

2. Which of the following fractions is greater than 3/4 and less than 5/6?
(a) 4/5
(b) 8/5
(c) 12/19
(d) 4/5

3. If 1.5x = 0.04y, then the value of $((y-x)/(y+x))$ is:
(a) 730/777
(b) 7.03/77
(c) 73/77
(d) None

4. If 47. 2506 = 4 ∗ A + 7/B + 2 ∗ C + 5/D + 6 ∗ E, then
Find the value of 5 ∗ A + 3 ∗ B + 6 ∗ C + D + 3 ∗ E is:
(a) 53.603
(b) 153.6003
(c) 55.6003
(d) 213.003

5. A tailor has 37.5 metres of cloth and he has to make 8 pieces out of a centimetre of cloth. How many pieces can he make out this cloth?
(a) 450
(b) 470
(c) 468
(d) 467

6. Aryan Chawla, a future mathematician, never replies direct but moves all replies in the shape of equations. Once asked to state his age in years, his reply was, “Take my age 3 years hence, multiply it by 3 and reduce 3 times my age 3 years ago and you will know how old I am.” What is his age in years?
   (a) 20
   (b) 24
   (c) 18
   (d) None
7. Shruti converted (0.232323....) into a fraction, her result is:
   (a) 1/5
   (b) 3/9
   (c) 23/999
   (d) None
8. What decimal of an hour is a second?
   (a) 0.0028
   (b) 0.0027
   (c) 0.0026
   (d) None
9. The number 0.127 is how much greater than 1/8?
   (a) 1/9
   (b) 1/15
   (c) 1/2
   (d) None
10. When (4.036) is divided by (0.04) the result is:
   (a) 100.9
   (b) 100.09
   (c) 1.0009
   (d) 10.09

Practice Exercise - 3 Answers with Explanations

1. (c)
2. (d)

3. (c)

4. (b)

5. (c)

6. (c)

7. (d) Explanation: $92\frac{2}{3}+128\frac{1}{6}-322\frac{5}{6}=0$

8. (b)

9. (d) 1/500; 0.127 expressed as a fraction = 127/1000; 1/8 as (1 × 125)/(8 × 125) = 125/1000

The difference is 2/1000, which is 1/500.

10. (d)

Practice Exercise - 4

Number of Questions: 15 **Suggested time: 10 minutes**

1. To make the expression $(11.98 * 11.98 + 11.98 * (k) + 0.02 * 0.02)$ a perfect square the value of **k** equals to:
(a) 0.010
(b) 0.030
(c) 0.04
(d) None

2. When (52416) is divided by 312, the quotient is 168. What will be the quotient when (52.416) is divided by (0.0168)?
(a) 3210
(b) 3.120
(c) 3120
(d) 32.10

3. How many digits will be to the right of the decimal point in the product of 95.75 and 0.2554?
(a) 5
(b) 6
(c) 7
(d) 4

4. The price of commodity P increases by 40 paise every year, while the price of commodity Q increases by 15 paise every year. If in 2016, the price of commodity P was Rs. 4.20 and that of Q was Rs. 6.30, in which year commodity P will cost 40 paise more than the commodity Q?
(a) None
(b) 2022
(c) 2025
(d) Can't say

5. $(3 \times 0.3 \times 0.03 \times 0.003 \times 30) =?$
(a) 0.0000243
(b) 0.000243

(c) 00.0024300
(d) 000.024300000

6. The rational number for the recurring decimal 0.125125...... is:
(a) 125/999
(b) 1250/999
(c) 125/99
(d) None

7. What will be the value of $(25.732)^2 - (15.732)^2$
(a) 4414.64
(b) 414.64
(c) 314.64
(d) 514.64

8. What will come in place of "k" in the following equation?
543 + 5.43 + 54.k3 = 603.26
(a) 8
(b) 6
(c) 1
(d) 4

9. $(0.002 \times 0.5) =?$
(a) 0.001
(b) 0.01
(c) 0.1
(d) 0.0001

10. $(54.327 \times 357.2 \times 0.0057)$ is equal to?
(a) $5.4327 \times 3.572 \times 0.57$
(b) $5.4327 \times 3.572 \times 5.7$
(c) $54327 \times 3572 \times 0.0000057$
(d) None of these

11. When 5.2416 is divided by 31.2, the quotient is 0.168 What will be the quotient when 524.16 is divided by 0.0168?
(a) 312
(b) 3120
(c) 31200

(d) 312000

12. Correct expression of 6.46 is:

(a) 64/99

(b) 640/9

(c) 64000/99

(d) 640/99

13. (138.009 + 341.981 – 146.305 = 123.6 + K), then K is:

(a) 210.085

(b) 295.05

(c) 120.085

(d) None

14. (10.3 ∗ 10.3 ∗ 10.3 + 1)/(10.3 ∗ 10.3 − 10.3 + 1) is:

(a) 12.3

(b) 14.3

(c) 11.3

(d) None

15. When (0.32) is written in simplest form, the product of the numerator and the denominator is:

(a) 32

(b) 200

(c) 34

(d) 27

Practice Exercise - 4 Answers with Explanations

1. (d) 0.04; Given = (11.98)^2 + (0.02)^2 + 11.98 ∗ k; For the given expression to be a perfect square, we must have 11.98 ∗ k = 2 ∗ 11.98 ∗ 0.02 ⇒ k = 0.04

2. (c)

3. (b) Sum of the decimal places = 7; Since the last digit to the extreme right will be zero (5 ∗ 4 = 20), so there will be 6 significant digits to the right of the decimal point.

4. (a) 2026; Suppose commodity P will cost 40 paise more than Q after R years.

Then, $(4.20 + 0.40R) - (6.30 + 0.15R) = 0.40 \Leftrightarrow 0.25Z = 0.40 + 2.10 \Leftrightarrow R = \frac{2.50}{0.25} = \frac{250}{25} = 10$

$\because$ P will cost 40 paise more than Q 10 years after 2016 . i.e in 2026.

5. (c)

6. (a)

7. (b)

8. (a)

9. (a)

10. (b) Number of decimal places in the given expression = 8.

Number of decimal places in choice (A) = 9; in (B) = 8 ; in (C)= 7; Clearly, the expression in (B) is the same as the given Expression.

11. (c)

12. (d) 640/99; $6.\overline{46} = 6 + 0.\overline{46} = 6 + \frac{46}{99} = \frac{594 + 46}{99} = \frac{640}{99}$

13. (a)

14. (c)

15. (b)

SQUARE ROOTS & CUBE ROOTS

4

The square root of a number 'k' is a number denoted by a number "$\sqrt{7}$" such that the number i.e., "$\sqrt{5}$" when multiplied by itself is equal to the given number i.e. "k"

e.g., $\sqrt{169}$ = 13, since 13 × 13 = 169

$\sqrt{1681}$ = 41, since 41 × 41 = 1681

Square Roots by Factors:

Steps:

(1) ***Find out prime factors of the given number.***

(2) ***Arrange the factors in pairs.***

(3) ***Take one factor from each pair.***

(4) ***Find the product of these factors, which represents the square root of the given number:***

$(a + b)^2 = a^2 + 2ab + b^2 \quad \therefore a^2 = (a+b)^2 - 2ab - b^2$

$(a - b)^2 = a^2 - 2ab + b^2 \quad \therefore a^2 = (a - b)^2 + 2ab - b^2$

e.g., $96^2 = (96 + 4)^2 - 2 \times 96 \times 4 - 4^2 = 100^2 - 768 - 16$

$= 9216$

Square of a number ending in 5:

Multiply the ten's digit 'n' by the next higher integer (n + 1) and write 25 to the right of the product.

e.g., 125^2 : 12 × 13 = 156 Hence square is 15625

e.g., 145^2 : 14 × 15 = 210 Hence square is 21025

Square of a number ending in 25:

Multiply the hundred's digit by a number consisting of the hundred's digit with a 5 to its right and annex 625 to the right of the product.

e.g., $625^2 : 6 \times 65 = 390$ Hence square is 390625.

e.g. $755^2 : 7 \times 75 = 525$ Hence square is 525625

Squares of $4\frac{1}{2}$, $7\frac{1}{2}$, $8\frac{1}{2}$ etc.

Multiply the integral portion by the next higher integer and add $\frac{1}{4}$.

e.g., $\left(9\frac{1}{2}\right)^2 = 9 \times 10 + \frac{1}{4} = 90\ \frac{1}{4}$

To write down by inspection the square, cube and higher powers of a number consisting wholly of 9's say 'k' nines.

Rule 1: The square is found by; writing down (k–1) nines followed by 8 and then (k–1) zeros followed by 1 (where n is the total number of 0s in the number).

$999^2 = 998001$

$99999^2 = 9999800001$

Rule 2: The cube of a number having 'k' nines is found by writing down (k–1) nines followed by 7, then (k–1) zeros followed by 2 and finally 'k' nines

9992 = 997002999

Rule 3: The fourth power is found by writing down (k–1) nines followed by 6, then (k–1) zeros
followed by 5, then (k–1) nines followed by 6, and lastly (k–1) zeros followed by 1.

Important Formulas - Square Root and Cube Root

$\sqrt{m} \times \sqrt{m} = m$

$\sqrt{m} \times \sqrt{m} = mn$

$p\sqrt{m} \times p\sqrt{n} = p\sqrt{mn}$

$\sqrt{m}\sqrt{n} = \sqrt{mn}$

$p\sqrt{mp}\sqrt{n} = p\sqrt{mn}$

$\sqrt{a}.\sqrt{a}.\sqrt{a}\cdots n \text{ times} = a\,1-12\,n$

$\sqrt{a}.\sqrt{a}.\sqrt{a}\cdots\infty = a$

If $\sqrt{a} + \sqrt{a} + \sqrt{a}\cdots\infty = s$, then s (s – 1) = a

Some Properties of Square Numbers:

(1) ***A square cannot end with an odd number of zeros.***

(2) ***A square no. cannot end with 2, 3, 7 or 8.***

(3) ***The square of an odd no. is odd.***

(4) ***The square of an even no. is given.***

(5) ***Every square no. is a multiple of 3, or exceeds a multiple of 3 by unity.***

(6) ***Every square no. is a multiple of 4, or exceeds a multiple of 4 by unity.***

(7) ***If a square number ends in 9, the preceding digit is even.***

Square Root:

The square root of a given number is that number whose square is equal to the given number.

(1) To find the square root of a number by breaking it into prime factors.

$$\sqrt{36072036} = \sqrt{3\times3\times4\times4\times5\times5\times7\times7\times11\times11\times13\times13}$$

$$= 3 \times 4 \times 5 \times 7 \times 11\times 13 = 6006$$

Also remember......

- ***A two digit number has a single digit square root***
- ***A three digit number has a two digit square root.***
- ***For a fraction square root is found by separate roots of numerator as well as of denominator provided the denominator is a perfect square.***
- ***If denominator is not a perfect square then rationalize in such a way that the denominator becomes a perfect square.***

Example 1 What is the least square number which is divisible by 12, 15, 16 & 9?

Here we have to find the least number which is divisible by 12, 15, 16 and 9 and is also a perfect square.

The LCM = 720 = (36 × 2 × 2 × 5). Hence the required number is 36 × 4 × 25 = 3600. (the LCM has to be multiplied by the unpaired factor i.e. 5 in this case)

Example 2 By what smallest number must 151200 be multiplied or divided in order to make it a perfect square?

$151200 = 2 \times 2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 3 \times 5 \times 5 \times 7$

Hence the number should be multiplied or divided by $2 \times 3 \times 7 = 42$ to make it a perfect square.

To have a number a perfect square it should be multiplied or divided by its unpaired factor.

Cube Root:

For finding Cube Roots, we remember that the fact that the 1000 and 1000000 have easy Cube Roots, 10 and 100 respectively.

Example Find the square root of 1936

To find the prime factors of 1936 we shall use successive divisions method.

2 ! 193600

2 ! 96800

2 ! 48400 $\therefore 193600 = 2 \times 2 \times 2 \times 2 \times 11 \times 11 \times 100$

2 ! 24200 $= 2^2 \times 2^2 \times 10^2 \times 11^2$

11! 12100 $\therefore \sqrt{193600} = 2 \times 2 \times 10 \times 11$

11! 1100 $= 440$ **(Answer)**

100 (Or we could divide 193600 by 100, 4 , 4 and 121 (the last perfect square factor). The idea is to have divisions by 4 , 9 , 16 , 25 , 100 or any other visible perfect square – as it will save a lot of time)

Square Root of decimal numbered

Steps:

(a) Pairing after decimal is a must.

(b) If there are odd number of digits after decimal add zero to make full pairs.

(c) Find square root of the number without consideration of decimals and put decimals before as many as digits as many pairs are.

Example Evaluate: $\sqrt{246.8}$

$246.8 = 246.80$ (one pair after decimals)

$$\therefore \sqrt{24680} = 157.09$$
$$= 15.7$$ **(Answer)**

Practice Exercise -1

Number of Questions: 10 **Suggested time: 5 minutes**

1. Which of the following statements is/are false.

i. $(a + b)^2 = a^2 + b^2 + 2ab$

ii. $(a + b)^2 - (a - b)^2 = 4ab$

iii. $(a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$

iv. $(a - b)^2 = a^2 - b^2 + 2ab$

v. $(a - b)^2 = a^2 + b^2$

(a) iv and v only

(b) v only

(c) ii and iv only

(d) iii and i only

2. Which of the following statements is/are true?

i. $28^2 = 227^2 + 227 + 228$

ii. $71^2 = 72^2 - 71 - 72$

iii. $61^2 = 60^2 + 61 + 62$

iv. $(x + 1)^2 = x^2 + x + 1$

v. $242 = 252 - 24 - 23$

(a) i only

(b) i and ii only

(c) i and iv only

(d) iii and v only

3. The last two digits of 37^2 is.

(a) 69

(b) 39

(c) 49

(d) 79

4. What is the value of 47^2?

(a) 2249

(b) 2349

(c) 2209
(d) 2149

5. What is the value of 53^2?
(a) 2509
(b) 2849
(c) 2809
(d) 2909

6. What is the value of 23×29?
(a) 687
(b) 677
(c) 657
(d) 667

7. What is the value of $1.27 \times 1.27 + 2.54 \times 1.73 + 1.73 \times 1.73$.
(a) 9.0146
(b) 9.0219
(c) 9.1349
(d) None of these

8. What is the value of $\frac{(4.216+6.929)^2+(4.216-6.929)^2}{4.216\times 4.216+6.929}$.
(a) 2.2172
(b) 4.9167
(c) 2.1916
(d) None of these

9. What is the minimum number you should multiply 648 with to get a perfect square?
(a) 2
(b) 3
(c) 5
(d) 7

10. What is the minimum number you should divide 23958 by to get a perfect cube?
(a) 6
(b) 12

(c) 18
(d) 24

Practice Exercise 1 - Answers

1 (b)	2 (b)	3 (a)	4 (c)	5 (c)
6 (d)	7 (d)	8 (d)	9 (a)	10 (c)

Practice Exercise - 2

Find out the square root by factors:

1. 1681, 1225, 324 , 484

2. 676, 9216, 577600

3. $R=\frac{2.50}{0.25}=\frac{250}{25}=10$

4. $2\frac{7}{9}, 1\frac{25}{144}$

5. 7.84, 156.25

6. What should be multiplied or divided to make the following factors a perfect square:
(i) $2 \times 3 \times 4 \times 5 \times 6 \times 8$ (ii) $3 \times 21 \times 14$.

7. 288, 980,

8. Find out the square root using tables. 990, 2456, 12345

9. 153.76 0.001296, 341.14

10. 235.6 correct to three significant figures.

Practice Exercise 2 - Answers

1. 41; 35; 18; 22

2. 26; 96; 760

3. $\frac{4}{7}, \frac{25}{13}, \frac{28}{29}, \frac{41}{42}$

4. $\frac{5}{3}, \frac{13}{12}$

5. 2, 8; 12.5

6. 8; 2

7. 2; 5; 2

8. 8.775; 49.56; 111.1

9. 12.4; 0.036; 18.47

10. 15.4

Practice Exercise - 3

Number of Questions: 22 **Suggested time: 15 minutes**

1. $\frac{\sqrt{72}}{\sqrt{24}}\times\frac{\sqrt{343}}{\sqrt{245}}\times\frac{\sqrt{126}}{\sqrt{84}}=?$

(a) 39.06
(b) 25
(c) 2.5
(d) 2.5

2. $\sqrt{x+2}$

(a) 8.41
(b) 841
(c) 5.38
(d) 29

3. 91/169 = $\sqrt{9}/x$ then x =?

(a) 5
(b) 5.6
(c) 62
(d) 31

4. $\frac{\sqrt{192}\times\sqrt{729}\times\sqrt{45}}{\sqrt{125}\times\sqrt{567}}=$

(a) 8.4
(b) 9.2
(c) 81
(d) 9.42

5. $\sqrt{\mathrm{a}}.\sqrt{\mathrm{a}}.\sqrt{\mathrm{a}}\cdots\infty=\mathrm{a}$

(a) 2.2
(b) 4.4
(c) 3.3
(d) 1.1

6. $\sqrt{1681}=35\times3-(?)=$

(a) 54
(b) 41
(c) 64
(d) 105

7. $\sqrt{\frac{578}{250}} =$
(a) 1.3
(b) 1.4
(c) 1.52
(d) 1.6

8. 5^{th} root of (161051) is:
(a) 11
(b) 13
(c) 19
(d) 21

9. The square root of $(8-2\sqrt{15})$ is:
(a) $\sqrt{2}$ - $\sqrt{5}$
(b) $\sqrt{5}$ - $\sqrt{5}$
(c) 2$\sqrt{5}$ - 5$\sqrt{5}$
(d) 2 + $\sqrt{5}$

10. $\sqrt{484} \div \sqrt{121} = ?$
(a) 8
(b) 9
(c) 16
(d) 2

11. $\left\{\sqrt{175} \times \sqrt{45}\right\}$ **divided by** $\left\{\sqrt{245} \times \sqrt{7}\right\}$
(a) 1.92
(b) 2.14
(c) 2.25
(d) 2.5

12. Fifth root of (64 × 36 × 216 × 16) is =
(a) 12

(b) 18
(c) 24
(d) 32

13. $\sqrt{0.01+\sqrt{0.0064}} = ?$
(a) 0.003
(b) 0.00003
(c) 0.018
(d) 0.3

14. $\dfrac{\sqrt{9.5\times0.0085\times18.9}}{\sqrt{0.0017\times1.9\times2.1}} =$
(a) .015
(b) 0.59
(c) .15
(d) none

15. ${}_{8}23^{2} =$
(a) 8^{1024}
(b) 8^{512}
(c) 8^{256}
(d) 2^{2048}

16. The smallest perfect square number which is divisible by 5, 54 and 25, is:
(a) 225
(b) 900
(c) 1225
(d) 625

17. The least square number, exactly divisible by 6, 12, 15, 18 is:
(a) 8100
(b) 7500
(c) 676
(d) none

18. $\dfrac{\sqrt{02.04\times4.2}}{\sqrt{0.0007\times0.034}} = ?$

(a) .6

(b) 0.06

(c) 0.006

(d) (1/6)

19. Find the value of $\sqrt{6+\sqrt{6+\sqrt{6+\sqrt{6+........}}}}$

(a) 1

(b) 2

(c) 3

(d) 4.

20. Find the square root of $7+2\sqrt{10}$.

(a) $\sqrt{5}+\sqrt{2}$

(b) $\sqrt{5}-\sqrt{2}$

(c) $\sqrt{5}+\sqrt{2}$

(d) $\sqrt{5}-\sqrt{2}$

21. Rationalise the denominator of $\frac{1}{\sqrt{41}-6}$.

(a) $\frac{\sqrt{41}-6}{3}$

(b) $\frac{5}{\sqrt{41}+6}$

(c) $\frac{\sqrt{41}+6}{5}$

(d) $\frac{\sqrt{41}+6}{5}$

22. Rationalise the denominator of $\frac{1}{2\sqrt{3}+\sqrt{5}}$.

(a) $5 + 3\sqrt{5}$

(b) $\sqrt{11}+3$

(c) $\sqrt{5} + \sqrt{5}$

(d) $\frac{2\sqrt{3}-\sqrt{5}}{7}$

- ***Remember to factorise bigger numbers into smallest factors, preferably which are perfect squares (Q 1, 4, 5, 10, 11..)***
- ***Try to have pairs after decimals. (Q 13, 14, 18..).***
- ***Factorise the number into two consecutive factors and select bigger if "+" and select smaller if "-" is involved.(Q 19).***
- ***Try to have denominator in a^2 - b^2 shape and rationalize.***

Practice Exercise 3 - Answers

1 (c)	2 (d)	3 (b)	4 (d)	5 (c)
6 (c)	7 (c)	8 (a)	9 (b)	10 (d)
11 (b)	12 (c)	13 (d)	14 (d)	15 (b)
16 (b)	17 (d)	18 (b)	19 (c)	20 (a)
21 (c)	22 (d)			

Practice Exercise - 4

Number of Questions: 25 **Suggested time: 12 minutes**

1. If $\frac{x}{\sqrt{512}} = \frac{\sqrt{648}}{x}$, find the value of x?
(a) 24
(b) 12
(c) 48
(d) 36

2. $\sqrt{5.4756} = ?$
(a) 2.24
(b) 1.24
(c) 1.34
(d) 2.34

3. If $3\sqrt{5} + \sqrt{125} = 17.88$, then what will be the value of $\sqrt{80} + 16\sqrt{5} = ?$
(a) 21.66
(b) 13.41
(c) 22.35
(d) 44.7

4. The cube root of 0.000729 is:
(a) 0.09
(b) 0.9
(c) 0.21
(d) 0.11

5. What is the least perfect square which is divisible by each of 21, 36 and 66?
(a) 213444
(b) 214434
(c) 231444
(d) None

6. $\sqrt{144} / 11 \times 11 / \sqrt{225} \times 15 / \sqrt{196} = ?$

(a) 0.85
(b) 0.72
(c) 2.8
(d) 0.4

7. $\sqrt{7}-1/\sqrt{7}/2=?$

(a) $36\sqrt{7}$
(b) $36\sqrt{7}$
(c) $\sqrt{26}$
(d) $\sqrt{26}$

8. The square root of 16641 is?
(a) 129
(b) 121
(c) 211
(d) 229

9. $3\sqrt{5}+\sqrt{125}=17.88$
(a) 10
(b) 0.1
(c) 1.0
(d) None

10. $\sqrt{0.000256}\times ?=1.6$
(a) 0.1
(b) 10
(c) 10000
(d) 1000

11. How many two-digit numbers satisfy this property: The last digit (units digit) of the square of the two-digit number is 8?
(a) 1
(b) 2
(c) 3
(d) Not possible

12. if a = 0.2917, then the value of $\sqrt{4a^2-4a+1+3a}$ is:
(a) 0.5834

(b) 0.2917
(c) 1.2917
(d) 2.2917

13. A group of students decided to collect as many paise from each member of group as is the number of members. If the total collection amounts to Rs. 98.01 , the number of member in the group is?
(a) 101
(b) 98
(c) 99
(d) 88

14. If $\sqrt{5.4756} = ?$, then find the value of $3\sqrt{5} + \sqrt{125} = 17.88$
(a) 7.22
(b) 8.92
(c) 6.72
(d) 10.58

15. If $x = \sqrt{3+1}/\sqrt{3-1}$; and $y = \sqrt{3-1}/\sqrt{3+1}$, what is the value of $(x^2 + y^2)$.
(a) 15
(b) 14
(c) 13
(d) 10

16. The square root of $(14+2\sqrt{13})/(14-2\sqrt{13})$ is:
(a) 8
(b) 9
(c) 10
(d) 12

17. $\sqrt{24.8} + \sqrt{6.4} = ?$
(a) 21
(b) 14
(c) 12
(d) 16

18. $\sqrt{191.6} = ?$
(a) 11.6
(b) 11.3

(c) 11.2
(d) 11.4

19. $\sqrt{41} - \sqrt{21} + \sqrt{19} - \sqrt{9} = ?$
(a) 3
(b) 4
(c) 5
(d) 6

20. $3\sqrt{412125} = ?$
(a) 1
(b) 125
(c) 135
(d) 145

21. $3\sqrt{0.000064} = ?$
(a) 0.2
(b) 0.4
(c) 0.1
(d) 0.3

22. What is the smallest number by which 3600 be divided to make it a perfect cube?
(a) 110
(b) 210
(c) 420
(d) 450

23. $140\sqrt{?} + 315 = 1015.$
(a) 25
(b) 15
(c) 5
(d) 50

24. $\sqrt{0.2890}\sqrt{0.00121} + \sqrt{9} = ?$
(a) 4.205
(b) 3.205
(c) 2.205
(d) 1.205

25. A man plants 49284 apple trees in his garden and arranges them so that there are as many rows as there are apples trees in each row. The number of rows is:

(a) 182
(b) 202
(c) 222
(d) 122

Practice Exercise 4 - Answers

1 (a)	2 (d)	3 (d)	4 (a)	5 (a)
6 (a)	7 (d)	8 (a)	9 (c)	10 (c)
11 (d)	12 (c)	13 (c)	14 (d)	15 (b)
16 (d)	17 (d)	18 (d)	19 (d)	20 (c)
21 (c)	22 (b)	23 (a)	24 (b)	25 (c)

Practice Exercise - 5

Number of Questions: 25 **Suggested time: 10 minutes**

1. What is the difference between $(\sqrt{18}+\sqrt{3})$ and $(\sqrt{2}+\sqrt{12})$?

(a) $(3\sqrt{2}-2\sqrt{3})$

(b) $(\sqrt{18}+\sqrt{3})$

(c) $3\sqrt{2}-\sqrt{3}$

(d) $(\sqrt{2}+\sqrt{12})$

2. If a × b = a + b$-\sqrt{ab}$, then find the value of 16 × 9.

(a) 14

(b) 13

(c) 12

(d) 11

3. Find square root of (14 × 12/28 × 2/3)

(a) 1

(b) 2

(c) 3

d) 4

4. What is the square root of $8+2\sqrt{15}$?

(a) $2\sqrt{5}+2\sqrt{3}$

(b) $\sqrt{5}+\sqrt{3}$

(c) $3\sqrt{2}-\sqrt{3}$

(d) $2\sqrt{2}+2\sqrt{6}$

5. $\sqrt{9}+\sqrt{8}-\sqrt{8}-\sqrt{7}+\sqrt{7}+\sqrt{6}-\sqrt{6}-\sqrt{5}+\sqrt{5}-\sqrt{4}=?$

(a) 1.5

(b) 0.25

(c) 0.5

(d) 1

6. The cube root of .000216 is:

(a) 0.6
(b) 0.06
(c) . 77
(d) 0.006

7. A group of students decided to collect as many paise from each member of group as is the number of members. If the total collection amounts to Rs. 59.29, the number of the member is the group is:
(a) 57
(b) 67
(c) 77
(d) 87

8. If $\sqrt{0.00000676} = 0.0026,$ the square root of 67, 60, 000 is:
(a) 2.6
(b) 26
(c) 260
(d) 2600

9. If $x = (7 - 4\sqrt{3}),$ then the value of $\left(x + \frac{1}{x}\right)$ is:
(a) $\sqrt{ab}$
(b) $\sqrt{ab}$
(c) 14
(d) $14 + 8\sqrt{3}$

10. What should come in place of both the question marks in the equation $\frac{?}{\sqrt{128}} = \frac{\sqrt{162}}{?}.$
(a) 12
(b) 14
(c) 144
(d) 196

11. If $\frac{5 + 2\sqrt{3}}{7 + 4\sqrt{3}} = a + b\sqrt{3},$ then:
(a) a = -11, b = -6
(b) a = -11, b = 6

(c) a = 11, b = -6
(d) a = 11, b = 6

12. If $x \times y = x + y + \sqrt{xy}$, the value of 6 × 24 is:
(a) 41
(b) 42
(c) 43
(d) 44

13. If $\sqrt{18225} = 135$, then the value of $\left(\sqrt{182.25} + \sqrt{1.8225} + \sqrt{0.018225} + \sqrt{0.00018225}\right)$ is:
(a) 1.49985
(b) 14.9985
(c) 149.985
(d) 1499.85

14. $\sqrt{\frac{0.204 \times 42}{0.07 \times 3.4}}$ **is equal to:**
(a) 0.16
(b) 0.06
(c) 0.6
(d) 6

15. The value of $\frac{\sqrt{80} - \sqrt{112}}{\sqrt{45} - \sqrt{63}}$ is:
(a) 3/4
(b) 4/3
(c) 4/7
(d) 7/4

16. The value of $\sqrt{10 + \sqrt{25 + \sqrt{180 + \sqrt{154 + \sqrt{225}}}}}$ is:
(a) 4
(b) 7
(c) 3
(d) 1

17. If 3a = 4b = 6c and a + b + c = $27\sqrt{29}$, then $\sqrt{a^2 + b^2 + c^2}$ is:

(a) 84
(b) 80
(c) 116
(d) 88

18. The square root of $\frac{(0.75)^3}{1-0.75}+\left[0.75+(0.75)^2+1\right]$ is:

(a) 5
(b) 2
(c) 7
(d) 0.25

19. $1.5^2 \times \sqrt{0.0225} = ?$

(a) 0.00375
(b) 0.3375
(c) 3.2705
(d) 32.0075

20. $\sqrt{0.01+\sqrt{0.0064}} = ?$

(a) 0.035
(b) 0.3
(c) 0.428
(d) 0.325

21. $28\sqrt{?}+1426=\frac{3}{4}$ of 2872

(a) 276
(b) 676
(c) 296
(d) 444

22. If $3\sqrt{5}+\sqrt{125}=17.88,$ then what will be the value of $\sqrt{5.4756}=?$

(a) 3.41
(b) 28.46
(c) 21.6
(d) 22.35

23. The square root of $\frac{(0.75)^3}{1-0.75}+\left[0.75+(0.75)^2+1\right]$ is:

(a) 25
(b) 2
(c) 35
(d) 0.45

24. $1.5^2 \times \sqrt{0.0225} = ?$
(a) 0.03755
(b) 0.3375
(c) 3.175
(d) 33.75

25. If $x = 2 + 2^{\frac{2}{3}} + 2^{\frac{1}{3}}$ then find the value of $x^3 - 6x^2 + 6x$?
(a) 17
(b) 2
(c) 6
(d) 21

Practice Exercise 5 - Answers

1 (c)	2 (b)	3 (b)	4 (b)	5 (d)
6 (c)	7 (d)	8 (c)	9 (d)	10 (a)
11 (c)	12 (b)	13 (b)	14 (d)	15 (b)
16 (a)	17 (c)	18 (b)	19 (b)	20 (b)
21 (b)	22 (d)	23 (b)	24 (b)	25 (b)

HCF & LCM

5

LCM (Least Common Multiple) & HCF (Highest Common Factor)

The concept of LCM would find a wide application in a lot of problems in the subsequent chapters. This is an important concept which can be used to simplify many problems.

LCM or Lowest Common Multiple i.e., is the smallest positive number which is a multiple of each of the given numbers. That means which can be divided by all the given numbers. Though 6, 12, 18, 24 can divided 72, 144, 288, 576... but amongst all these numbers 72 is the smallest i.e., **LCM is 72**.

HCF or GCD (Greatest Common Divisor) is the largest factor of the given set of numbers. Similarly 24, 48, 72 can be divided by each of 2, 3, 4, 6, 12… but 12 is the greatest common divisor of all these numbers hence **HCF or GCD is 12**.
LCM/HCF can be found for integers, fraction polynomials, or decimals.

Example Find the LCM and HCF of 6 and 8

Solution

Multiples of 6 are 6, 12, 18, 24, 30, 42, 48… so on. The factors are 1, 2, 3, 6.

Multiples of 8 are 8, 16, 24, 32, 40, 48…. so on. The factors are 1, 2, 4, 8.

Common multiples are: 24, 48…. etc., Least of these is 24. **Thus, 24 is the LCM of 6 and 8**.

The common factors are 1, 2. The largest of these is 2. **So the HCF is 2**.

But method is tedious for determining the LCM/HCF of large numbers or more than 2 numbers.

A better method is the ‘Factorisation Method’.

In this method, each of the numbers is factorised as:

$6 = 2 \times 3 = 2^1 \times 3^1$;

$8 = 2 \times 2 \times 2 = 23 \times 30$

HCF is 2^1 **or HCF is COMMON FACTOR/S with MINIMUM power appearing anywhere**.

LCM is $2^3 \times 3^1$ **or LCM is ALL FACTOR/S with MAXIMUM power appearing anywhere.**

Example Find the LCM and HCF of 12 and 18.

Solution

Factorising the two numbers we get,

$12 = 2 \times 2 \times 3 = 2^2 \times 3^1$; $18 = 2 \times 3 \times 3 = 2^1 \times 3^2$

Hence, LCM = $2^2 \times 3^2$ and the HCF = $2^1 \times 3^1$

Alternate method of finding the LCM:

When there are 2 large numbers of more than 2 numbers, use the methods/process as shown in the following examples.

Example Find the LCM of 12, 15, 27 and 30.

Solution

Put the number in a row as shown alongside and start 2! 12, 15, 27, 39 dividing by the smallest number that divides all of them 2! 6, 15, 27, 39.

Stop the process when the common divisor is 1. 3! 3, 15, 27, 39.

The LCM is the product of all the divisors and ! 1, 5, 9, 13 the dividends left.

Then, LCM of 12, 15, 27 and 39 = $2 \times 2 \times 3 \times 1 \times 5 \times 9 \times 13 = 7020$.

This method is most appropriate for more than 2 large numbers.

Note: LCM is always greater than or equal to the greatest of the given numbers.

Example Find the HCF of 24 and 34.

Solution

The "Synthetic method" for finding the HCF of any 2 numbers say 24 and 34 is as follows:

(1) Divide the large number by the smaller.

(2) In the next step divide the previous divisor by the previous remainder.

Continue with the process till you end up with a remainder 0.

The **LAST divisor** when the remainder becomes 0 is the HCF of the 2 numbers. Applying it to the given problem we get.

If two numbers a and b are given, if their LCM and HCF are L and H respectively, then $L \times H = a \times b$.

i.e., product of LCM and HCF of two numbers = product of numbers themselves.

LCM and HCF of Fractions: $\text{LCM of fractions} = \dfrac{\text{LCM of Numerators}}{\text{HCF of Denominators}}$

$$\text{HCF of fractions} = \frac{\text{HCF of Numerators}}{\text{LCM of Denominators}}$$

Example Find the LCM and HCF of $\frac{25}{12}$ and $\frac{25}{12}$ LCM = $\dfrac{\text{LCM of 25 and 35}}{\text{HCF of 12 and 18}} = \dfrac{175}{6}$

Solution

$$\text{HCF} = \frac{\text{HCF of 25 and 35}}{\text{LCM of 12 and 18}} = \frac{5}{36}$$

Example Find the LCM and HCF of $\frac{100}{24}$ and $\frac{60}{16}$.

Solution

Simplify both the fractions to their lowest forms i.e., $\frac{100}{24} + \frac{25}{6}$ and $\frac{60}{16} = \frac{15}{4}$

Now the LCM $\left(\frac{25}{6}, \frac{15}{4}\right) = \dfrac{\text{LCM }(25,25)}{\text{HCF }(6,4)} = \dfrac{75}{2}$

And the HCF $\left(\frac{25}{6}, \frac{15}{4}\right) = \frac{\text{HCF }(25,25)}{\text{LCM }(6,4)} = \frac{5}{12}$

Note: Do not directly apply the formula if the fractions are not in their simplest form. The applications of LCM and HCF galore. You would come across it in a practically the entire arithmetic.

SOME FUNDAMENTAL APPLICATIONS

Example A heap of coconuts is divided into groups of 2, 3 and 5 and each time no coconut is left over. Find the least number of coconuts in the heap.

Solution

Let X be the number of coconuts desired. X must be a common multiple of 2, 3 and 5, otherwise some coconuts will be left over. Further, the number X must be the least possible number. So X has to be the LCM of 2, 3, 5 which is 30. So the least number of coconuts in the heap is 30.

Practice Exercise - 1

Number of Questions: 12 **Suggested time: 10 minutes**

1. LCM of $\frac{3}{5}, \frac{60}{16}, \frac{3}{5}, \frac{12}{15}$

(a) $\frac{12}{15}$

(b) $\frac{60}{16}$

(c) $\frac{60}{16}$

(d) $\frac{60}{16}$

2. HCF of $\frac{3}{5}, \frac{60}{16}, \frac{12}{15}, \frac{12}{15}$

(a) $\frac{60}{16}$

(b) $\frac{12}{15}$

(c) $\frac{60}{16}$

(d) $\frac{60}{16}$

3. HCF of 105, 90, 195, 215 =

(a) 5
(b) 10
(c) 15
(d) 25

4. The greatest no which shall leave equal reminders on dividing 60, 132, 180:

(a) 30
(b) 36
(c) 12
(d) 24

5. A number when multiplied by 5 becomes divisible by 12, 18, 24, 32 & 40 is.
(a) 7200
(b) 288
(c) 480
(d) 460

6. A number when divided by 6, 7, 15, 20, 21 leaves remainder '1' in each case but is completely divisible by 29. The number is given by.
(a) 441
(b) 841
(c) 6561
(d) 2401

7. Kalpana puts 1P, 2P, 3P in her cash box during February 2016. She will have saved a total sum of Rs. ______ during the month.
(a) 4
(b) 4.35
(c) 4.06
(d) 6.12

8. LCM of 121, 1331, 1221
(a) 1331
(b) 1221
(c) 2662
(d) none

9. LCM of 2.5, .075, .0015 =
(a) .15
(b) .075
(c) .75
(d) 7.5

10. HCF of two numbers is 15, while their product is 2025. Numbers are ______
(a) 15, 175
(b) 45, 75
(c) 75, 75

(d) 45, 45

11. Two numbers have HCF as 15. Their sum is 75. The numbers are:
(a) 40, 35
(b) 30, 40
(c) 60, 15
(d) 250, 25

12. The greatest number that leaves 6 on division on 5193 or 5856 is.
(a) 21
(b) 29
(c) 39
(d) 37

Practice Exercise 1 - Answers with Hints

1 (d)	2 (d)	3 (a)	4 (d)	5 (b)
6 (b)	7 (b)	8 (d)	9 (d)	10 (d)
11 (c)	12 (c)			

4. Find HCF of difference of the numbers i.e., 72, 48, 120

5. As the number required should be only **after it is multiplied by 5**. Hence it is LCM of the numbers divided by 5.

6. The required number is (LCM × N + 1) divisible by 29, for varied values of N = 1, 2, 3……

11. Let the numbers of 15A, 15B; therefore 15A + 15B = 75 for A + B = 5
Here A = 1, 2 and B = 4, 3 respectively.

13. Let the numbers be $7x$, $9x$; Remember **Product of two numbers** = **LCM × HCF.**

Practice Exercise - 2

Number of Questions: 10 **Suggested time: 5 minutes**

1. What is the LCM of 24, 39 and 54.
(a) 936
(b) 2808
(c) 2106
(d) 1872

2. What is the LCM of 14, 36, 45 and 64.
(a) 20160
(b) 40320
(c) 22680
(d) 21820

3. What is the HCF 36, 64 and 100.
(a) 1
(b) 2
(c) 4
(d) 8

4. Find the HCF of 108, 63 and 981.
(a) 3
(b) 7
(c) 11
(d) 9

5. What is the LCM of $\frac{16}{19}, \frac{26}{36}, \frac{76}{80}$.
(a) 42
(b) 33
(c) $\frac{1}{12}$
(d) $\frac{1}{12}$

6. What is the HCF of the following fractions: $\frac{12}{25}, \frac{4}{5}$ and $\frac{16}{15}$.

(a) $\frac{12}{15}$

(b) $\frac{60}{16}$

(c) $\frac{60}{16}$

(d) $\frac{60}{16}$

7. Which of the following statements is/are definitely false?

(i) LCM of a set of numbers is necessarily greater than the largest number in that set.

(ii) HCF of a set of numbers can be equal than the smallest number in that set.

(iii) The HCF of a, b and c is H, a, b and c are therefore multiples of H.

(iv) The LCM of a, b and c is L, a, b and c are therefore factors of L.

(a) (i) only

(b) (i) and (iii) only

(c) (i) and (ii) only

(d) (iii) and (iv) only

8. What is the number which when increased by 7 is divisible by 27, 21, 35 & 77.

(a) 10402

(b) 10388

(c) 10397

(d) 20783

9. The greatest number that will divide 964, 1238 and 1400 leaving remainders 41, 31, 51 respectively is given by.

(a) 64

(b) 71

(c) 69

(d) 72

10. Which is divisible by 33 and 36 both.

(a) 1854724

(b) 745350
(c) 875380
(d) none

Practice Exercise 2 - Answers

1 (b)	2 (a)	3 (c)	4 (d)	5 (a)
6 (b)	7 (d)	8 (b)	9 (b)	10 (d)

8. The required number is (LCM – 7).

9. The required number will be HCF of (964 – 41), (1238 – 31), (1400 – 51).

10. Instead of actual methods, it is better to apply divisible tests of 33 i.e., 3 and 11
and 36 i.e., 4 & 9.

Practice Exercise - 3

Number of Questions: 10 **Suggested time: 10 minutes**

1. The greatest number which on dividing 1657 and 2037 leaves remainders 6 and 5 respectively, is:
(a) 129
(b) 245
(c) 257
(d) 127

2. The product of two numbers is 2028 and their H.C.F. is 13. The number of such pairs is:
(a) 3
(b) 2
(c) 6
(d) 7

3. A rectangular floor 3.78 meters long 5.25 meters wide is to be paved with square tiles, all of the same size. What is the largest size of the tile which could be used for the purpose?
(a) 14
(b) 35
(c) 54
(d) 21

4. Product of two co-prime numbers is 117. Their L.C.M should be?
(a) 13
(b) 9
(c) 117
(d) HCF

5. The sum of two numbers is 528 and their H.C.F is 33. The number of pairs of numbers satisfying the above condition is?
(a) 6
(b) 12
(c) 8

(d) 4

6. The L.C.M of two numbers is 495 and their H.C.F is 5. If the sum of the numbers is 100, then their difference is:
(a) 70
(b) 90
(c) 10
(d) 127

7. Three number are in the ratio of 3 : 4 : 5 and their L.C.M. is 2400. Their H.C.F. is:
(a) 120
(b) 200
(c) 80
(d) 40

8. The H.C.F and L.C.M of two numbers are 11 and 385 respectively. If one number lies between 75 and 125, then that number is:
(a) 99
(b) 88
(c) 77
(d) 110

9. The least multiple of 7, which leaves a remainder of 4, when divided by 6, 9, 15 and 18 is:
(a) 184
(b) 364
(c) 74
(d) 94

10. The smallest number which when diminished by 7, is divisible by 12, 16, 18, 21 and 28 is
(a) 1032
(b) 1022
(c) 1015
(d) 1008

Practice Exercise 3 - Answers with Explanations

1. (d)

Required number = H.C.F. of (1657 – 6) and (2037 – 5) = 127.

2. (b)

Let the numbers 13a and 13b; Then, 13a × 13b = 2028 ⇒ ab = 12. Now, the co-primes with product 12 are (1, 12) and (3, 4). [Two integrals 'a' and 'b' are said to be co-prime or relatively prime if they have no common positive factor other than 1 or, equivalently, if their greatest common divisor is 1].

So, the required numbers are (13 × 1, 13 × 12) and (13 × 3, 13 × 4). i.e., 2 such pairs.

3. (d)

4. (c)

H.C.F of co-prime numbers is 1. So, L.C.M = $\frac{100}{24}$ =117.

5. (d)

6. (c)

Let the numbers be x and $(100 - x)$; Then, $x(100 - x) = 5 \times 495$.

$\Rightarrow x^2 - 100x + 2475 = 0$

$\Rightarrow (x - 55)(x - 45) = 0$

$\Rightarrow x = 55$ or $x = 45$;

⇒ Difference = 10

7. (d)

8. (c)

9. (b)

10. (c)

Practice Exercise - 4

Number of Questions: 10 **Suggested time: 7 minutes**

1. The least number which when divided by 5, 6, 7 and 8 leaves a remainder 3, but when divided by 9 leaves no remainder, is:
(a) 1677
(b) 2523
(c) 3363
(d) 1683

2. L.C.M of two prime numbers ‘m’ and ‘n’ ($m > n$) is 161. The value of ($3n - m$) is:
(a) 2
(b) 1
(c) –1
(d) –2

3. The G.C.D of 1.08, 0.36 and 0.9 is:
(a) 0.0018
(b) 0.108
(c) 0.9
(d) 0.18

4. If the sum of two numbers is 55 and the H.C.F. and L.C.M. of these numbers are 5 and 120 respectively, then the sum of the reciprocals of the numbers is equal to:
(a) $\frac{100}{24}$
(b) $\frac{100}{24}$
(c) $\frac{100}{24}$
(d) $\frac{100}{24}$

5. The maximum number of students among them 1001 pens and 910 pencils can be distributed in such a way that each student gets the same

number of pens and same number of pencils is:
(a) 91
(b) 21
(c) 22
(d) 31

6. Three numbers are in the ratio 1 : 2 : 3 and their H.C.F is 12. The difference between the greatest and the smallest number is:
(a) 8
(b) 10
(c) 20
(d) 24

7. The L.C.M of 22, 54, 108, 135 and 198 is:
(a) 11880
(b) 5940
(c) 1980
(d) 660

8. The H.C.F of two numbers is 11 and their L.C.M is 7700. If one of the numbers is 275, then the other is:.
(a) 308
(b) 318
(c) 279
(d) 283

9. The H.C.F and L.C.M of two numbers are 84 and 21 respectively. If the ratio of the two numbers is 1 : 4, then the larger of the two numbers is:
(a) 48
(b) 12
(c) 84
(d) 108

10. Which of the following has the most number of divisors?
(a) 176
(b) 101
(c) 99

(d) 182

Practice Exercise 4 - Answers with Explanations

1. (d)

2. (d)

HCF of two prime numbers is 1. Product of numbers = 1 × 161 = 161;
Let the numbers be 'm' and 'n'. Then, mn = 161.
Now, co-primes with product 161 are (1, 161) and (7, 23).
Since 'm' and 'n' are prime numbers and m > n, we have 'm = 23 and n = 7.
∴ 3n – m = (3 × 7) – 23 = –2

3. (d)

GCD is nothing but HCF.

4. (c)

$\frac{100}{24}$; Let the numbers be a and b; Then, a + b = 55 and ab = 5 × 120 = 600.

The required sum = $\frac{1}{a}+\frac{1}{b}=\frac{a+b}{ab}=\frac{55}{600}=\frac{11}{120}$.

5. (b)

6. (d)

Let the required numbers be x, $2x$, $3x$. Then, their H.C.F = x or = 12
∴ The numbers are 12, 24, 36.

7. (b)

8. (a)

other number = $\frac{11\times 7700}{275}$ = 308.

9. (c)

10. (a)

Practice Exercise - 5

Number of Questions: 10 **Suggested time: 6 minutes**

1. $343^3 - 216^3$ must be ÷ by.
(a) 13, 127 both
(b) 13 only
(c) 127 only
(d) 13 or 127

2. $4096^2 + 256^3$ is ÷ by.
(a) 2^{12}
(b) 2^7
(c) 8^3
(d) all of these

3. $(19^{26} + 18^{17} - 7^{17} - 8^{26})$ is ÷ by.
(a) 11
(b) 22
(c) 33
(d) 44

4. HCF of $8x^2y^3z^6$, $16y^4z^8k^8$, $20x^6z^8y^5$ is:
(a) $4x^2yz^4$
(b) $4x^0y^3z^2$
(c) $4y^3z^6$
(d) $4z^2$

5. The HCF of $2x^2 - 4x$, $3x^4 - 12x^2$ and $2x^5 - 2x^4 - 4x^3$ is:
(a) $2x\ (x + 2)$
(b) $2x\ (2 - x)$
(c) $2x\ (x - 2)$
(d) $x\ (x - 2)$

6. The factors of $x^4 + 4$ are:
(a) $(x^2 + 2)^2$

(b) $(x^2 + 2)(x^2 - 2)$
(c) $(x^2 + 2x + 2)(x^2 - 2x + 2)$
(d) none of the above

7. The H.C.F. of the expressions $(x^3 + x^2 + x + 1)$ and $(x^4 - 1)$ is given by.
(a) $(x^2 - 1)(x^2 + 1)$
(b) $(x + 1)(x^2 - 1)$
(c) $(x + 1)(x^2 + 1)$
(d) $(x^2 + 1)(x + 1)(x^3 + 1)$.

8. LCM of 10, 15, 25, 30 & 35 is how times their HCF.
(a) 21
(b) 1050
(c) 5
(d) 210

9. HCF of two Nos. is 15, their product is 5400. How many pairs are there? Difference of the Greatest & the smallest such no.
(a) 2, 345
(b) 3, 345
(c) 2, 315
(d) 3, 375

10. LCM of 12, 34 and 48 is:
(a) 408
(b) 816
(c) 576
(d) 1632

Practice Exercise 5 - Answers

1 (c)	2 (d)	3 (a)	4 (c)	5 (d)
6 (d)	7 (b)	8 (d)	9 (a)	10 (c)

1. $X^n - Y^n$ is divisible by $X - Y$ if n is odd.

2. It can be factorised as $256^2(16^2 - 1)$.

6. $x^4 + 4 + 4x^2 - 4x^2 = (x^2 + 2)^2 - (2x)^2$.

7. Let the numbers be 15X, 15Y for all co-prime set of values of X & Y.

PERCENTAGE

6

Percentage represents a base of 100 in calculations.
When we say what% of Y is X, we mean as under:

Let "k"% of Y = X or kY = 100 X.

Any percentage question can be presented in one of the three methods only:

- *X is what % of Y?*
- *Y is what % of X?*
- *X % of Y = ?*

Whatever shape of the question may be the relationship will always be better explained as

- *X = k% of Y*
- *Y = k% of X*
- *X · Y/100 = k respectively for different values of k*

The following illustrations shall further explain the concept:

Example 1

(a) What is the value of $8\frac{1}{4}$% of Rs. 400?

Solution

$8\frac{1}{4}$ of Rs. 400 = $\frac{17}{2} \times \frac{1}{100} \times 400 = 34$. Ans

(b) What percentage is Rs. 25 of Rs. 500?

Solution

$\frac{25}{500} \times 100 = 5\%$. Ans

(c) A boy spends 40% of his pocket money, and has Rs. 3 left. How much had he at first?

Solution

The boy spends 40% of his pocket money, therefore he has 60% of the whole left. Hence we can conclude on what is left with him, rather than what he has spent. i.e., he is left with 60%.

Hence, 60% of the whole = Rs. 3 *(Always consider what is left with you)*

$\therefore$100% of the whole = $\frac{3}{60} \times 100$ = Rs. 5 Ans

Example 2 A man receives an annual salary of Rs. 8800/- in the year 1977. This is 10% more than his salary in 1976. What was his salary in 1976?

Solution

Let the salary in 1976 be Rs. 100.

From the question, the salary in 1977 was 10% more than in 1976, thus we have:

That if salary was Rs. 110 in 1977; then it was Rs. 100 in 1976

Rs. 8800 in 1977; then Rs.? in 1976 = $100 \times 8800/110$

= Rs. 8000

Example 3 In a certain examination there are three papers and it is necessary for a candidate to get 35% of the total to pass. In one paper the candidate obtained 62 out of 120, and in the second paper he obtained 35 out 150. How many marks must he obtain out of 180 in the third paper to just qualify for pass?

Solution

Let the marks obtained in the third paper be 'x'

From the question, pass marks are 35% of the total.

Or

$62 + 35 + x = 35\% \,[120 + 150 + 180]$

$$\therefore\ 97 + x = \frac{35}{100}[450] = 157.5$$

$\therefore\ x = 157.5 = 60.5$ **Ans**

Example 4 If A's income be 20 percent more than B's, how much percent is B's income less than A's?

Solution

All such like questions must belong to One of the four categories given below:

- ***P gets k% more than Q; Q gets what % less than P***
- ***P gets k% less than Q; Q gets what % more than P***
- ***P gets k% more than Q; Q gets what % of P***
- ***P gets k% less than Q; Q gets what % of P***

In the given example the problem can be dealt as under:
Less the income of B = Rs. 100
Since A's income is 20% more than B's income,
$\therefore$ A's income = 100 + 20 = Rs. 120
If A's income is Rs. 120, then B's income = Rs. 100
If A's income is Rs. 100, then B's income = Rs. 100 × 100/120

$$= \text{Rs. } \frac{250}{3}$$

$\therefore$ B's income is less than A's income by

$$= 100 - \frac{250}{3} = \frac{50}{3} = 16\frac{2}{3}\% \textbf{ Ans}$$

Example 5 A man's net income, after paying income tax at the rate of 5 paise in a rupee, is Rs. 760. Find his gross income.

1st method

Since Income Tax on Rs. 1 is 5p = Rs. $\frac{1}{20}$

$$\therefore \text{Net income} = 1 - \frac{1}{20} = \text{Rs. } \frac{19}{20}$$

If the net income is Rs. $\frac{19}{20}$, then gross income = Rs. 1

If the net income is Rs. 760, then gross income = $\frac{1 \times 760 \times 20}{19}$ = Rs. 800/- **Ans**

Or

2nd method

After paying 5p in a Re as Income tax he is left with 95p i.e., 95% of his income.

Hence 95% of Income = Rs.760

Therefore Income = Rs.800/-

Practice Exercise - 1

Number of Questions: 15 **Suggested time: 15 minutes**

1. (a) Write down 15% of Rs. 34.
(b) What percentage is 55 Paise of Rs. 11?
(c) Express 2/3 as a percentage.
(d) Express 6 1/4% as a fraction.
(e) Express the statement in the form of a percentage: "In a class of 45, 36 students passed'
(f) What is 80% of 25?

2. (i) Find the value of 6 2/3% of 300 kg.
(ii) What is the commission at 6 1/4 percent on the sales of goods for Rs. 1,234.40?

3. (i) Express 1/4 as a percentage of 25/9.
(ii) In an examination a girl got 507 marks out of 650. Find out what percentage of the total marks she got.

4. (i) What is the number, 11 1/2 percent of which is Rs. 69?
(ii) 96% population of a village is 7680. Find the total population of the village.

5. Three candidates in an election received 1136; 7636 and 11,628 votes. What percentage of the total votes did the winning candidates get.
(a) 43%
(b) 57%
(c) 52%
(d) 58%

6. The population of a village is 4500, 11/18th of them are males and the rest are females, 40% of the males are married. Find the number of married males.
(a) 900
(b) 1000
(c) 1100
(d) 1200

7. Out of a class of 38 girls, three were absent, 20% of the remainder failed to do the home work. How many did the home work?

(a) 18

(b) 20

(c) 24

(d) 28

8. If 60% of the students in a school are the boys and the girls number is 812, How many boys are there?

(a) 1118

(b) 1218

(c) 1318

(d) 1418

9. You are to pay a bill of Rs. 120 by cheque and are required to add bank commission to this amount @ 1%. Write, in words, the total amount to be paid including bank commission.

10. (a) Find the gross value of an Estate which yield Rs. 3800 after a commission of 5% has been deducted

(a) 3900

(b) 4000

(b) 4100

(d) 3610

(b) A man's income is Rs. 5000. What income remains after paying an income tax of 5 p per rupee?

(a) 4950

(b) 4650

(c) 4800

(d) None of these

11. The lead ore from a certain mine yields 60% of metal. Of this metal 3/4% is silver. The rest of the metal is lead. How much of silver in Kg would be obtained from 8,000 Kg of the ore.

(a) 30

(b) 36

(c) 4764

(d) 4770

12. If A's income be 25% less than B's, how much percent B's income is more than that of A?
(a) 20
(b) 25
(c) 33
(d) 40

13. A clerk receives an annual salary of Rs. 3660 in the year 1975. This is 20% more than his salary in 1974. What was his salary in 1974?
(a) 2850
(b) 3050
(c) 3250
(d) 3400

14. A liters of water is evaporated from 6 liters of sugar solution containing 4% of sugar. Find the percentage of sugar in the remaining solution.
(a) 3.33%
(b) 4.5%
(c) 4.8%
(d) 6.57%

15. Peter can do a piece of work in 12 days. Mohan is 50% more efficient that Peter. If Mohan is given that piece of work, how long will he take to do it?
(a) 18
(b) 9
(c) 6
(d) 8

Practice Exercise 1 - Answers

1. (a) 5.1
(b) 5%
(c) 66 2/3%
(d) 1/16
(e) 80% passed
(f) 20

2. (i) 20
(ii) Rs. 77.15

3. (i) 9%
(ii) 78%

4. (i) 600
(ii) 8000

5. 57%

6. 1100

7. 28

8. 1218

9. Rupees one hundred twenty one and paise twenty only.

10. (a) Rs. 4000 (b) Rs.4750

11. Silver 36 Kg

12. 33 1/3%

13. 3050

14. 4.8%

15. 8 days

Hints

5. 11628 = k% of (1136 + 7636 + 11628).

6. We are to find 40% of 11/18 of 4500 = ?

7. Home work is done by 80%, hence we should find 80% of (38 − 3) only.

8. 40% girls are 812, then what is 60% OR (using ratios it is 40% : 812 :: 60% : X).

11. The silver obtained is = 3/4% of 60% of 8000.

14. Out of 6 litre solution, 1 litre evaporates i.e., 4% of 6 = k% of 5 to find k.

15. Peter does 1/12 job in a day. Mohan does 150% of (1/12) of the job in a day.

Practice Exercise - 2

Number of Questions: 15 **Suggested time: 10 minutes**

1. (i) What percent of 60 is 54?

(a) 90
(b) 90%
(c) 9
(d) 9%

(ii) What percent of 160 is 56?

(a) 30
(b) 35%
(c) 32
(d) 32%

2. (i) 12 is 15% of a number. What is the number?

(a) 60
(b) 80
(c) 75
(d) 125

(ii) 35% of x is 140, find the value of x?

(a) 40
(b) 40000
(c) 4
(d) None

3. Amit secures 49% marks in an examination. His score is 294 and his sister's 372, Express her score as a percentage.

(a) 58
(b) 60
(c) 62
(d) 64

4. Find the gross output of a coal mine, if after 25% wastage, the net output is 60,000 tonnes.

(a) 40000

(b) 60000

(c) 70000

(d) 80000 tonnes

5. Price of sugar is increased by 20%. What percent of consumption is to be; decreased so that there would be no increase in expenditure.

(a) $12\frac{1}{4}$

(b) $14\ \frac{1}{4}$

(c) $11\ \frac{1}{9}$

(d) $16\ \frac{1}{4}$

6. A’s salary is 50% above B’s How much percent is B’s salary below A’s?

(a) 25%

(b) 30%

(c) $33\frac{1}{3}$

(d) $37\ \frac{1}{4}$

7. Kant secured 20% marks in an examination and failed by 10 marks. Rika secured 42% and got 1 mark more than the bare minimum to pass. Determine the percent necessary to pass.

(a) 30

(b) 35

(c) 37

(d) None

8. If x is 25% less than y, then by what percent does y exceed x?

(a) 20

(b) 25

(c) 33

(d) 40

9. A transistor is sold for Rs. 510 at a loss of 15%. Find the selling price in order to gain 15%.

(a) 690

(b) 720

(c) 750

(d) 810

10. A producer blends two qualities of tea from two different tea gardens, one costing Rs. 25 per kg and the other Rs. 20 per kg in the ratio of 8 : 12 He sells the blended variety at Rs. 24 per kg. Determine his profit.

(a) Rs. 30

(b) 40

(c) 50

(d) No profit

11. A grocer makes a profit of 20% by selling loose tea at Rs. 17.40 per kg.

(i) What did 40 kg of loose tea cost him?

(a) 460

(b) 500

(c) 520

(d) 580

(ii) If he sells half of the quantity at Rs.17.40 per kg and the remaining half at Rs. 16.00 per kg; what is his profit percent on the whole?

(a) 13.2%

(b) 14.2%

(c) 15.2%

(d) 16.2%

12. A sells an electric press to B at a profit of 5% and B sells it to C at a loss of 5%. If C pays Rs. 23.94 for it, what did it cost to A?

(a) 22

(b) 24

(c) 26

(d) 28

13. A man buys a plot of agricultural land for Rs. 36,000. He sells one-third of this piece of land at a loss of 20% and two-fifth at a gain of 25%. At what price must he sell the remaining land so as to make an overall profit of 10%.

(a) 10000
(b) 12000
(c) 14000
(d) 16000

14. A fruit seller purchases 6 oranges for a rupee and sells them at the rate of Rs. 2.50 per dozen. Find his gain or loss percent.

(a) 20
(b) 25
(c) 30
(d) 33

15. A merchant sells tea at the rate of Rs. 7.50 per kg and loses 25% in the transaction. At what price must he sell the tea per kg in order that he may gain 10%.

(a) 9
(b) 10
(c) 11
(d) 12.25

Practice Exercise 2 - Answers with Hints

1. (i) 905 (ii) 35%

2. (i) 80 (ii) *400*

3. 62%

4. 80000 tonnes

5. $16\frac{1}{4}\%$

6. $33\frac{1}{9}\%$

7. Maximum marks = 50; 40%

8. $33\frac{1}{9}\%$

9. Rs. 690

10. Profit = Rs. 40; $9\frac{1}{11}\%$

11. (i) (d) *(ii) (c)*

12. Rs 24

13. Rs. 12000

14. (b)

15. (c) 75% of Cost = Rs 7.50; 110% of CP = ?

Practice Exercise - 3

Number of Questions: 20 **Suggested time: 15 minutes**

1. In a class with 30 students, 18 like Base ball, 5 like both base ball and foot ball, and 8 do not like either sport. How many students like Football but not baseball?
(a) 4
(b) 5
(c) 7
(d) 9

2. Jagjit and Surjit have three children. The sum of the weights of the two smallest children is 71 kg. The sum of the weights of two largest children is 96 kg and the sum of the weights of the smallest and largest child is 87 kg. What is the sum of the weights of all three children.
(a) 125
(b) 127
(c) 128
(d) 130

3. If the Nos. 17/24, 1/2, 3/8, 3/4 and 9/16 were ordered from the greatest to least, the middle number of the resulting sequence would be:
(a) 17/54
(b) 1/2
(c) 9/16
(d) 3/4

4. If a 10% deposit that has been paid towards the purchase of a certain product is Rs. 110, how much more remains to be paid?
(a) 880
(b) 990
(c) 1000
(d) 1100

5. The original price of an article was increased by 15% and then again by 15%. New price now is equivalent to increasing the original price by:

(a) 32.25%
(b) 31%
(c) 30.25%
(d) 30%

6. In a discount store all paperbacks are discounted 20%. In a promotion sales the stores offers a further discount of 10%. What is the effective discount on a paperback bought at the promotion sales?
(a) 20%
(b) 22%
(c) 25%
(d) 28%

7. An Engineer had her monthly wages increased by 8% and then by 10%. If she now receives Rs. 12,376 find her wages in Rs. before the first increase?
(a) 10200
(b) 1900
(c) 19125
(d) 11625

8. The owners of a business divided up the profits as follows: The share of the second owner was 3 times that of the third and the share of the first owner was 4 times that of the second. The first received Rs. 1.210 more than the third. How many Rs did the first owner receive?
(a) 13200
(b) 14000
(c) 25500
(d) 23000

9. If k is an integer and P = (0.0010101)(k) where P > 1,000, what is the least possible value of 10 raised to the power k?
(a) 2
(b) 3
(c) 4
(d) 5

10. Sherry wishes to leave a 15% tip for her waiter. If she paid a total of Rs. 24.15 for her meal and the tip how much was the cost in Rs. of the meal?

(a) 20

(b) 21

(c) 22

(d) 523

11. Goldy purchased brand X pen for Rs.4 a brand Y pens for Rs. 2.80 . If Goldy purchased a total of 12 of those pens for Rs. 42 how many brand X pens did he purchases?

(a) 4

(b) 5

(c) 6

(d) 7

12. A VCR was advertised for sale with successive discounts of 10% followed by another 10%. How much in Rs will it be sold if its original price was Rs. 300?

(a) 240

(b) 243

(c) 253

(d) 207

13. A 9 mg shaving lotion contains 30% spirit. How much water added to shall convert it into a resulting lotion of 5% spirit.

(a) 30

(b) 40

(c) 45

(d) 63

14. A perfect number is a natural number with the property that the sun of the natural number divisors of that number except for the number itself, is that number what is the first perfect number?

(a) 4

(b) 6

(c) 8

(d) 9

15. A worker is paid @ Rs. 60.75 per hour. The minimum number of hours that Jolly needs to work to earn Rs.1,500 is –

(a) 35
(b) 22
(c) 30
(d) 38

16. In 1985, 45 percent of a document storage capacity was availed by 630 banks. But in 1987, when the capacity was increased, number of banks rose by 82 using overall 25 percent of its new capacity. What was the minimum percent increase in storage capacity.
(a) 6
(b) 12
(c) 9
(d) 8

17. The original price of an item was increased by 30%. Then the new price was reduced by 30% to get the final selling price. The final selling price is what percent of the original price?
(a) 91
(b) 95
(c) 98
(d) 100

18. The amounts of time that three secretaries work on a special project are in the ratio of 1 : 2 : 5. If they worked a combined total of 112 hours, how many hours did the secretary who worked the longest spend on the project more than the slowest?
(a) 80
(b) 70
(c) 56
(d) 16

19. If "basis points" are defined so that 1 percent is equal to 100 basis points, then 82.5 percent is how many basis points greater than 62.5 percent?
(a) .02
(b) 0.2
(c) 20.00
(d) 200

20. A union contract specifies a 6 percent salary increase plus a Rs. 450 bonus for each employee. For a certain employee, this is equivalent to an 8 percent salary increase. What was this employee's salary in Rs. before the new contract?

(a) 21500

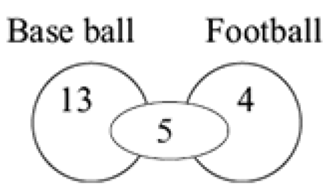
(b) 22500

(c) 23500

(d) 24300

Practice Exercise 3 - Answers with Hints

1 (a)	2 (b)	3 (c)	4 (b)	5 (a)
6 (d)	7 (a)	8 (a)	9 (d)	10 (b)
11 (d)	12 (b)	13 (d)	14 (b)	15 (c)
16 (d)	17 (a)	18 (c)	19 (c)	20 (b)

1. 

Total players – Not playing any game = 30 − 8 = 22

2. Let the children be A>B>C

B+C = 71, A+B = 96, A+C = 87, Add and find the solution.

5. 115% of 115% of 100 – 100 = Net change%.

7. Let original wages be W Rs. therefore, 108% of 110% of W = 12,376/-

8. Let the owners be A, B and C.

Let their shares be $12x$, $3x$, x; therefore $11x$ = 1,210

11. Let X pens be purchased @ Rs.4/- and (12 − X) pens @ Rs.2.80/-

Therefore, 4(X) + (12 − X)(2.80) = Rs. 42.00

13. Let x mg water is added to it.

Therefore 30% of 9 = 5% of $(9 + x)$

14. Refer to the Number system chapter

15. He can earn this required amount in 23 hrs, why should he work extra than the choices given?

16. In 1985, 45% of Capacity = 630 i.e., Capacity was 1400 documents

Let the rise in capacity be "x" documents

Therefore 25% of $(1400 + x) = 630 + 82 = 712$; $x = 1488$

Rise in capacity 88 which is 6.3% i.e., 8 **(the only possible answer amongst the choices given)**.

20. Here (8% − 6%) = 2% rise = 450 i.e., bonus

Number of Questions: 25 **Suggested time: 25 minutes**

1. Two students, Preeti and Radhika, appeared in an examination. One secured 9 marks more than the other and her marks were 56% of the sum of their marks. Marks obtained by them?

(a) 42, 33
(b) 42, 36
(c) 44, 33
(d) 44, 36

2. If A = x% of y and B = y% of x, then which of the following is true?

(a) 42, 33
(b) 42, 36
(c) 44, 33
(d) 44, 36

3. If 20% of a = b, then b% of 20 is the same as:

(a) None of these
(b) 10% of a
(c) 4% of a
(d) 20% of a

4. Two numbers P and Q are such that the sum of 5% of P and 4% of Q is two-third of the sum of 6% of P and 8% of Q. Find the ratio of P : Q.

(a) 2 : 1
(b) 1 : 2
(c) 1 : 1
(d) 4 : 3

5. Two employees Rawat and Puri are paid a total of Rs. 550 per week by their employer. If Rawat is paid 120 percent of the sum paid to Puri, how much is Puri paid per week?

(a) Rs. 150
(b) Rs. 300
(c) Rs. 250

(d) Rs. 200

6. Samant went to a shop and bought things worth Rs. 25, out of which 30 paise went on sales tax on taxable purchases. If the tax rate was 6%, then what was the cost of the tax free items?
   (a) Rs. 15
   (b) Rs. 12.10
   (c) Rs. 19.70
   (d) Rs. 16.80

7. The population of Sirsa town increased from 1,75,000 to 2,62,500 in a census decade of 2001−2011. What is the average percent increase of population of Sirsa per year?
   (a) 4%
   (b) 6%
   (c) 5%
   (d) 50%

8. Three candidates contested an election and received 1136, 7636 and 11628 votes respectively. What percentage of the total votes did the winning candidate get?
   (a) 57%
   (b) 50%
   (c) 52%
   (d) 60%

9. A fruit seller had some oranges. He sells 40% oranges and still has 420 oranges. How many oranges he had originally?
   (a) 420%
   (b) 700%
   (c) 220%
   (d) 400%

10. A batsman scored 110 runs which included 3 boundaries and 8 sixes. What percent of his total score did he make by running between the wickets?
   (a) 42, 33
   (b) 42, 36
   (c) 44, 33

(d) 44, 36

11. What percentage of numbers from 1 to 70 have 1 or 9 in the unit's digit?
(a) 21
(b) 22
(c) 23
(d) Data Inadequate

12. In an election between two candidates, Mr Kamal Awasthi got 55% of the total valid votes. 20% of the votes were invalid. If the total number of votes was 7,500, what was the number of valid votes that the other candidate got?
(a) 2800
(b) 2700
(c) 2100
(d) 2500

13. In a competitive examination in State of Haryana, 6% candidates got selected from the total appeared candidates. State of Punjab had an equal number of candidates appeared and 7% candidates got selected with 80 more candidates got selected than Haryana. What was the number of candidates appeared from each state?
(a) 8200
(b) 7500
(c) 7000
(d) 8000

14. What percent of a day is 6 hours?
(a) 6.25%
(b) 20%
(c) 25%
(d) 12.5%

15. A student has to obtain 33% of the total marks to pass. He got 125 marks and failed by 40 marks. The maximum mark is:
(a) 600
(b) 500
(c) 400
(d) 300

16. In a certain school, 20% of students are below 8 years of age. The number of students above 8 years of age is 23 of the number of students of 8 years of age which is 48. What is the total number of students in the school?

(a) 100

(b) 102

(c) 110

(d) 90

17. In an examination, 5% of the applicants were found ineligible and 85% of the eligible candidates belonged to the general category. If 4275 eligible candidates belonged to other categories, then how many candidates applied for the examination?

(a) 28000

(b) 30000

(c) 32000

(d) 33000

18. A student multiplied a number by 35 instead of 53. What is the percentage error in the calculation?

(a) 64%

(b) 32%

(c) 34%

(d) 42%

19. 270 students appeared for an examination, of which 252 passed. What is the pass percentage?

(a) 93.13

(b) 93.23

(c) 92.23

(d) 92

20. Jagmohan's salary was decreased by 50% and subsequently increased by 50%. How much percent does he stand to gain or loss?

(a) 35%

(b) 25%

(c) 32%

(d) 28%

21. How many litres of pure acid are there in 8 litres of a 20% solution?
(a) 2 litres
(b) 1.4 litres
(c) 1 litres
(d) 1.6 litres

22. The price of a car is Rs. 3,25,000. It was insured to 85% of its price. The car was damaged completely in an accident and the insurance company paid 90% of the insurance. What was the difference between the price of the car and the amount received?
(a) Rs. 76,375
(b) Rs. 34,000
(c) Rs. 82,150
(d) Rs. 70,000

23. If number x is 10% less than another number y and y is 10% more than 125, then find out the value of x.
(a) 123
(b) 122.00
(c) 122.25
(d) 123.75

24. A housewife saved Rs. 2.50 in buying an item on sale. If she spent Rs. 25 for the item, approximately how much percent she saved in the transaction?
(a) 9%
(b) 10%
(c) 7%
(d) 6%

25. A pipe X is 30 meters and 45% longer than another pipe Y. Find the length of pipe Y.
(a) 20.12
(b) 20.68
(c) 20
(d) 20.5

Practice Exercise 4 - Answers

1 (a)	2 (a)	3 (c)	4 (d)	5 (c)
6 (c)	7 (c)	8 (a)	9 (b)	10 (c)
11 (b)	12 (b)	13 (d)	14 (c)	15 (b)
16 (a)	17 (b)	18 (a)	19 (a)	20 (b)
21 (d)	22 (a)	23 (d)	24 (a)	25 (b)

Practice Exercise - 5

Number of Questions: 30 **Suggested time: 20 minutes**

1. On my sister's 15**th** birthday, her height was 159 cm, having grown 6% since the year before. How tall was she in the previous year?
(a) 150 cm
(b) 140 cm
(c) 142 cm
(d) 154 cm

2. Q as a percentage of P is equal to P as a percentage of (P + Q). Find Q as a percentage of P.
(a) 62%
(b) 50%
(c) 75%
(d) 66%

3. If the price of petrol increases by 25% and Benson intends to spend only an additional 15% on petrol, by how much percent will he reduce the quantity of petrol purchased?
(a) 8%
(b) 7%
(c) 10%
(d) 6%

4. Arun got 30% of the maximum marks in an examination and failed by 10 marks. However, Sujith who took the same examination got 40% of the total marks and got 15 marks more than the passing marks. What were the passing marks in the examination?
(a) 90
(b) 250
(c) 75
(d) 85

5. 30% of the men are more than 25 years old and 80% of the men are less than or equal to 50 years old. 20% of all men play football. If 20% of the men above the age of 50 play football, what percentage of the football players are less than or equal to 50 years?

(a) 60%

(b) 70%

(c) 80%

(d) 90%

6. In an election between two candidates, one got 55% of the total valid votes, 20% of the votes were
invalid. If the total number of votes was 7500, the number of valid votes that the other candidate got, was:

(a) 60%

(b) 70%

(c) 80%

(d) 90%

7. If 20% of a = b, then b% of 20 is the same as _____ % of 'a'

(a) 4%

(b) 6%

(c) 8%

(d) 10%

8. Fresh fruit contains 68% water and dry fruit contains 20% water. How much dry fruit can be obtained from 100 kg of fresh fruits?

(a) 20

(b) 30

(c) 40

(d) 50

9. A candidate scoring 25% in an examination fails by 30 marks, while another candidate scores 50% mark, gets 20 marks more than the minimum pass marks. Find the minimum pass marks. Find the minimum pass percentage.

(a) 40%

(b) 46%

(c) 38%

(d) 50%

10. The value of a machine depreciates at the rate of 10% every year. It was purchased 3 years ago. If its present value is Rs. 8748, its purchase price was:
(a) 10,000
(b) 12,000
(c) 14,000
(d) 16,000

11. A student multiplied a number by 3/5 instead of 5/3, What is the percentage error in the calculation?
(a) 54%
(b) 64%
(c) 74%
(d) 84%

12. A student has to obtain 33% of the total marks to pass. He got 125 marks and failed by 40 marks. The maximum marks are:
(a) 500
(b) 600
(c) 800
(d) 1000

13. If the price of a book is first decreased by 25% and then increased by 20%, then the net change in the price will be:
(a) 10
(b) 20
(c) 30
(d) 40

14. Ghanshyam spends 30% of his monthly income on food articles, 40% of the remaining on conveyance and clothes and saves 50% of the remaining. If his monthly salary is Rs. 18,400, how much money does he save every month?
(a) 3864
(b) 4903
(c) 5849
(d) 6789

15. Hari Parshad spends 35% of his income on food, 25% on children's education and 80% of the remaining on house rent. What percent of his income he is left with?
(a) 6
(b) 8
(c) 10
(d) 12

16. If Vipul's height is 40% less than that of Alok, how much percent Alok's height is more than that of Vipul?
(a) 67
(b) 77
(c) 97
(d) 28

17. Chocolates (405 in numbers) were distributed equally among children in such a way that the number of chocolates received by each child is 20% of the total number of children. How many chocolates did each child receive?
(a) 25
(b) 20
(c) 16
(d) 18

18. In an election only two candidates contested. 20% of the voters did not vote and 120 votes were declared as invalid. The winner got 200 votes more than his lone opponent and he secured 41% votes of the total vote on the voter list. Find the percentage of votes of the defeated candidate out of the total votes cast?
(a) 47.5
(b) 41
(c) 38
(d) 45

19. If 75% of a number is added to 75, results as the number itself. The number is:
(a) 100
(b) 200

(c) 300
(d) 400

20. A number was decreased by 10% and then was increased by 10%. The result so obtained is 10 less than the original number. Find the original number?
(a) 1000
(b) 2000
(c) 3000
(d) 4000

21. What percent of 7.2 kg is 18 grams?
(a) 0.25
(b) 0.5
(c) 0.75
(d) 1.0

22. The ratio 5 : 4 expressed as a percent equals:
(a) 12.5
(b) 40.0
(c) 80.0
(d) 125

23. Anubhav's salary is 40% of Bhambri's salary which is 25% of Chelsy's salary. What percentage of Chelsy's salary is Anubhav's salary?
(a) 10
(b) 20
(c) 30
(d) 40

24. The population of Fatehabad was 1,60,000 in 2014. If it increased by 3%, 2.5% and 5% respectively in the last three years, then the present population in the year 2017 is?
(a) 1,55,679
(b) 1,67,890
(c) 1,79,890
(d) 1,77,366

25. In Chemistry examination, the average marks for the entire class were 80. If 10% of the students scored 35 marks and 20% scored 90 marks, find the average marks of the remaining students of the class?

(a) 25

(b) 50

(c) 75

(d) 100

26. If 252 candidates passed out of a total of 270 candidates appeared then the nearest pass percentage is:

(a) 91

(b) 93

(c) 97

(d) 98

27. The price of sugar increased from Rs. 36 per kg to Rs. 45 per kg. Mr Mehta does want to have any increase in his expenditure on sugar. He will reduce his consumption of sugar by%:

(a) 15

(b) 20

(c) 25

(d) 30

28. If ‘x’ is 80% of ‘y’, then what percent of ‘2x’ is ‘y’?

(a) $65\frac{1}{2}$ (b) $64\frac{1}{2}$ (c) $63\frac{1}{2}$ (d) $62\frac{1}{2}$

29. In a School examinations, 1100 boys and 700 girls took a test. If 42% of the boys and 30% of the girls passed the examinations, find the percentage of the strength which failed:

(a) 58

(b) 62.66

(c) 64.64

(d) 66.66

30. In building an Airliner in 1978, the materials cost 45% of the total, wages 42% and other expenses 13%. By 1979 the cost of materials had increased by 8%, wages had gone up by one-seventh, but other expenses were reduced by 10%. Find the percentages increases in the cost of the Airliner.

(a) 7.2%

(b) 8.1%
(c) 8.3%
(d) 8.7%

Practice Exercise 5 - Answers with Hints

1 (a)	2 (a)	3 (a)	4 (d)	5 (c)
6 (b)	7 (a)	8 (c)	9 (a)	10 (b)
11 (b)	12 (a)	13 (a)	14 (a)	15 (b)
16 (a)	17 (a)	18 (d)	19 (c)	20 (a)
21 (a)	22 (d)	23 (a)	24 (d)	25 (c)
26 (b)	27 (b)	28 (d)	29 (b)	30 (c)

5. Let total number of men = 100; Then, 20 men play football.

80 men are < or = to 50 years old.

Remaining 20 men > 50 years old.

Number of football players above 50 years old = 20 × 20/100 = 4

Number of football players less than or equal to 50 years old = 20 − 4 = 16

Required percentage = 1620 × 100 = 80%

6. Total number of votes = 7500 (20% votes invalid) ⇒ Valid votes = 80%

Total valid votes = $7500 \times \frac{80}{100}$

1st candidate got 55% of the total valid votes means the 2nd candidate got 45% of votes.

⇒ Valid votes that 2nd candidate got = total valid votes × $\frac{45}{100}$

$= 7500 \times \frac{80}{100} \times \frac{45}{100} = 2700$

8. The fruit content in both the fresh fruit and dry fruit is the same. Given, fresh fruit has 68% water, so remaining 32% is fruit content, weight of fresh fruits is 100kg.

Dry fruit has 20% water, so remaining 80% is fruit content.

Let weight of dry fruit be = Y kg.

fruit% in fresh fruit = fruit% in dry fruit

$\therefore$ (32/100) * 100 = (80/100)* Y,

Here we get, Y = 40 kg

18. Let there be x voters and k votes goes to loser then

$0.8x - 120 = k + (k + 200)$ or $k + 200 = 0.41x$

$\Rightarrow k = 1440$ or $(k + 200) = 1640$

Therefore $\frac{1440}{3200} \times 100 = 45\%$

30. Net change = 8% of 45 + 1/7 of 42 – 10% of 13.

AVERAGE

7

The **Average** or **Mean** or **Arithmetic Mean** of a number of quantities of the same kind is their sum divided by their number.

$$\text{Mean of 'n' quantities} = \frac{\text{Sum of 'n' quantities}}{\text{'n'}}$$

Example 1 The population of 4 towns is 35560, 30000, 27500 and 25600, respectively. What is the average population of a town?

$$\text{Average} = \frac{35560 + 30000 + 27500 + 25600}{4} = \frac{188660}{4} = 29665$$

Example 2 The average weight of 8 men in a boat is increased by 1 kg, when one of the crew, who weights 60 kg is replaced by a new man. What is the weight of the new man?

The average weight of 8 men is increased by 1 kg. Therefore, the total weight is increased by $8 \times 1 = 8$ kg.
Hence the weight of the new man is $60 + 8 = 68$ kg.

Example 3 The average marks of three batches of students containing 70, 50 and 30 students respectively, are 50, 55 and 45. Find the average marks of all the 150 students.

Total of I batch $= 70 \times 55 = 3500$

Total of II batch $= 50 \times 55 = 2700$

Total of III batch $= 30 \times 45 = 1350$

Total $= 7600$

$$\text{Hence average is } \frac{7600}{150} = 50.67$$

Example 4 A student secured 70%, 80%, 50% and 48% in four papers. It was decided to give double weight age to the 1^{st} and the 3^{rd}

paper and half to the 2^{nd} and 4^{th} paper.

(a) find the changed average marks.

(b) by what % does it change from the average before the decision was taken?

(c) what would have been the average had the weightage system been inter changed?

Solutions

(a) New average would be $70 \times 2 + 50 \times 2 + 80 \times \frac{1}{2} + 48 \times \frac{1}{2} =$ $304/500 = 60.8\%$

(b) Original average $70 + 80 + 50 + 48 = 248/4 = 62\%$
Therefore changed % is $1.2/62 \times 100 = - 2\%$.

(c) New average would have been $70 \times \frac{1}{2} + 50 \times \frac{1}{2} + 80 \times 2 + 48 \times 2 = 316/500 = 63.2\%$

Remember

If a car covers a certain distance at x kmph and an equal distance at y kmph. Then, the average speed of the whole journey = 2xy/x + y kmph. (Here the Distance is same, but speed varies)

Practice Exercise - 1

Number of Questions: 11 **Suggested time: 10 minutes**

1. The average salary of 50 workers in a factory was Rs. 350. 5 new workers with salaries 250, 300, 320, 350 and 400 were employed. What would be the new average salary?

(a) 346.64
(b) 347.64
(c) 348.64
(d) 349.36

2. Typist A can type a sheet in 5 minutes, typist B in 6 minutes and typist C in 8 minutes. The average number of sheets typed per hour per typist is……sheets.

(a) 8.83
(b) 7.83
(c) 7
(d) None

3. A car runs its first 5000 miles at 40 miles to the gallon of petrol and its next 500 miles at 36 miles to the gallon. What is the average petrol consumption for the whole 10000, miles (gallon/mile)?

(a) .0263
(b) .0264
(c) .0264
(d) .0266

4. Find out the average rate of motion in the case of a person who rides the first mile @ 10 m.p.h., the next mile @ 8 m.p.h and the third mile @ 6 m.p.h.

(a) 6
(b) 7.33
(c) 7.66
(d) 8.33

5. The population of a town increased from 100000 to 150000 in a decade. What is the average % increase per year

(a) 2

(b) 3

(c) 4

(d) 5

6. The average monthly sales for the first eleven months of the year in respect of a certain salesman were Rs. 12,000. Due to his illness during the last month, the average monthly sales for the whole year came down to Rs. 11,375. What was the value of sales during the last month?

(a) 4100

(b) 3900

(c) 4300

(d) 4500

7. My income for 15 days averaged Rs. 7, the average for the first 5 days was Rs. 6 and the average for the last 9 days was Rs.8. What was the income on the 6^{th} day?

(a) Rs. 2

(b) Rs. 2.25

(c) Rs.2.50

(d) None

8. If a man receives on a fourth of his capital 3 p.c., on two-third 5 p.c. and on the remainder 11 p.c., what percentage does he receives on the whole?

(a) 2%

(b) 3%

(c) 5%

(d) 7%

9. The average weight of 10 men is increased by $1\frac{1}{2}$ kg when one of the men who weights 68 kg is replaced by a new man. Find the weight of the new man.

(a) 73 Kg

(b) 83 Kg

(c) 88 Kg
(d) 78 Kg

10. The average age of a committee of seven trustees is the same as it was five years ago, a younger man having been substituted for one of them. How much younger was he than the trustee whose place he took?
(a) 25
(b) 32
(c) 35
(d) 42

11. Six men A, B, C, D, E, F agree with a seventh man G to provide a sum of money for some project and share among them. A, B, C, D, E, F are to subscribe Rs. 10 each, and G is to pay Rs. 3 more than the average of the seven. What is the whole sum to be provided?
(a) Rs. 67.50
(b) Rs. 71.50
(c) Rs. 73.50
(d) Rs. 78.50

Practice Exercise - 1 Answers with Hints - need based

1. Rs. 347.64
2. 9.83
3. 0.026388 gal/mile.
4. 7.66 m.p.h.
5. 4.1%
6. Rs. 4500
7. Rs. 3
8. 5%
9. 83 kg
10. 35 years
11. Rs. 73.50

Hints

2. Let in 120 minutes (LCM) A types 24, B types 20 and C types 15 sheet.

4. If distances are same Average speed is $\frac{n}{1/S1+1/S2+1/S3}$

11. Let the average be 'A'
Therefore (6 × 10 + (A + 3)) = 7A.

Practice Exercise - 2

Number of Questions: 30 **Suggested time: 20 minutes**

1. In the first 10 overs of a cricket game, the run rate was only 3.2. What should be the run rate in the remaining 40 overs to reach the target of 282 runs?
(a) 6.25
(b) 5.5
(c) 7.4
(d) 5

2. A grocer has a sale of Rs. 6435, Rs. 6927, Rs. 6855, Rs. 7230 and Rs. 6562 for 5 consecutive months. How much sale must he have in the sixth month so that he gets an average sale of Rs. 6500?
(a) 4800
(b) 4991
(c) 5004
(d) 5000

3. The average of 20 Integers is zero. Of them, How many of them may be greater than zero, at the most?
(a) 1
(b) 20
(c) 0
(d) 19

4. The captain of a cricket team of 11 members is 26 years old and the wicket keeper is 3 years older. If the ages of these two are excluded, the average age of the remaining players is one year less than the average age of the whole team. Find out the average age of the team.
(a) 23 years
(b) 20 years
(c) 24 years
(d) 21 years

5. The average monthly income of A and B is Rs. 5050. The average monthly income of B and C is Rs. 6250 and the average monthly income of A and C is Rs. 5200. What is the monthly income of A?
(a) 2000
(b) 3000
(c) 4000
(d) 5000

6. A car owner buys diesel at Rs.7.50, Rs. 8 and Rs. 8.50 per litre for three successive years. What approximately is the average cost per litre of diesel if he spends Rs. 4000 each year?
(a) Rs. 8
(b) Rs. 7.98
(c) Rs. 6.2
(d) Rs. 8.1

7. In Kiran's opinion, his weight is greater than 65 kg but less than 72 kg. His brother does not agree with Kiran and he thinks that Kiran's weight is greater than 60 kg but less than 70 kg. His mother's view is that his weight cannot be greater than 68 kg. If all are them are correct in their estimation, what is the average of different probable weights of Kiran?
(a) 70 kg
(b) 69 kg
(c) 61 kg
(d) 67 kg

8. The average weight of 16 boys in a class is 50.25 kg and that of the remaining 8 boys is 45.15 kg. Find the average weights of all the boys in the class.
(a) 48.55
(b) 42.25
(c) 50
(d) 51.25

9. A library has an average of 510 visitors on Sundays and 240 on other days. What is the nearest average number of visitors per day in a month of 30 days beginning with a Sunday?
(a) 290

(b) 304
(c) 285
(d) 270

10. A student's mark was wrongly entered as 83 instead of 63. Due to that his average marks got increased by 2. What is the number of subjects taken by him?
(a) 5
(b) 10
(c) 4
(d) 2

11. A family consists of two grandparents, two parents and three grandchildren. The average age of the grandparents is 67 years, that of the parents is 35 years and that of the grandchildren is 6 years. The average age of the family is
(a) 32.27
(b) 31.57
(c) 28.17
(d) 30.57

12. The average weight of A, B and C is 45 kg. If the average weight of A and B be 40 kg and that of B and C be 43 kg, what is the weight of B?
(a) 31
(b) 28.12
(c) 32
(d) 30.12

13. Average of 15, 26, 28, 32, 35 and X is 29.45. Find the value of X.
(a) 45.7
(b) 40.7
(c) 39.7
(d) 43.7

14. The average age of husband, wife and their child 3 years ago was 27 years and that of wife and the child 5 years ago was 20 years. What is the present age of the husband?
(a) 40
(b) 32

(c) 28

(d) 30

15. The average weight of 8 person's increases by 2.5 kg when a new person comes in place of one of them weighing 65 kg left the group. What is the weight of the new person?

(a) 75 Kg

(b) 50 Kg

(c) 85 Kg

(d) 80 Kg

16. There are two divisions A and B of a class, consisting of 36 and 44 students respectively. If the average weight of divisions A is 40 kg and that of division B is 35 kg. What is the average weight of the whole class?

(a) 38.25

(b) 37.25

(c) 38.5

(d) 37

17. A batsman makes a score of 87 runs in the 17th inning and thus increases his averages by 3. What is his average after 17th inning?

(a) 39

(b) 35

(c) 42

(d) 40.5

18. A student needed to find the arithmetic mean of the numbers 3, 11, 7, 9, 15, 13, 8, 19, 17, 21, 14 and 'x'. He found the mean to be 12. What is the value of 'x'?

(a) 12

(b) 5

(c) 7

(d) 9

19. Anamica obtained 76, 65, 82, 67 and 85 marks (out in 100) in English, Mathematics, Chemistry, Biology and Physics. What is her average mark?

(a) 53

(b) 54
(c) 72
(d) 75

20. Distance between Jammu and Agra is 778 km. A train covers the journey from Jammu to Agra at 84 km per hour and returns back to Jammu with a uniform speed of 56 km per hour. Find the average speed of the train during the whole journey?
(a) 69.0 km /hr
(b) 69.2 km /hr
(c) 67.2 km /hr
(d) 67.0 km /hr

21. The average age of Girls in a class is 16 years and that of the Boys is 15 years. There are 3 trans genders in the class with average age of 14 years. What is the average age for the whole class?
(a) 15
(b) 16
(c) 15.5
(d) Insufficient Data

22. The average age of 36 students in Class IX is 14 years. When teacher's age is included to it, the average increases by one. Find out the teacher's age in years?
(a) 51 years
(b) 49 years
(c) 53 years
(d) 50 years

23. The average of five numbers is 27. If one particular number "x" is excluded, the average becomes 25. What is the excluded number "x"?
(a) 30
(b) 40
(c) 32.5
(d) 35

24. The batting average for 40 innings of a cricket player is 50 runs. His highest score exceeds his lowest score by 172 runs. If these two innings are excluded, the average of the remaining 38 innings is 48 runs. Find out the highest score of the player.

(a) 150

(b) 174

(c) 180

(d) 166

25. The average score by Nimrata for ten matches is 38.9 runs. If the average for the first six matches is 42, what is her average score for the last four matches?

(a) 34.25

(b) 36.4

(c) 40.2

(d) 32.25

26. The average marks of six students is 'x' and the average marks of three of these students is 'y'. If the average marks of the remaining three students is 'z', then the relationship of x,y and z can be expressed as

(a) $3x = 2y - 5z$

(b) $x = y + z$

(c) $2x = y + z$

(d) $x = 2y + 2z$

27. Samunder drives his car to a place 150 km away at an average speed of 50 km/hr but drives back at 30 km/hr. What is his average speed for the entire journey?

(a) 32.5 km/hr.

(b) 35 km/hr.

(c) 37.5 km/hr

(d) 40 km/hr

28. The average age of Mandip and his wife was 23 years at the time of their marriage. After five years they have a one year old chil(d) What is the new average age of the family?

(a) down by 2 years

(b) up by 1 year

(c) down by 1 years

(d) down by 4 years

29. In an examination, a student's average marks were 63. If he had obtained 20 more marks for his Geometry and 2 more marks for his Hindi, his average would have been 65. How many subjects did he appear in the examination?

(a) 12

(b) 11

(c) 13

(d) 14

30. The average salary of all the workers in a factory is Rs.8,000. The average salary of 7 Technical mechanics is Rs.12,000 and the average salary of the remaining work force is Rs.6,000. How many workers are there in the workshop?

(a) 21

(b) 22

(c) 23

(d) 24

Practice Exercise - 2 Answers with Hints

1 (a)	2 (b)	3 (a)	4 (a)	5 (a)
6 (b)	7 (d)	8 (a)	9 (c)	10 (b)
11 (b)	12 (a)	13 (b)	14 (a)	15 (d)
16 (d)	17 (a)	18 (c)	19 (d)	20 (c)
21 (d)	22 (a)	23 (d)	24 (b)	25 (a)
26 (c)	27 (c)	28 (d)	29 (b)	30 (a)

1. $10 \times 3.2 + 40\,A = 282$

4. Let the average of the whole team be 'x' years
Therefore $11\,x - 55 = 9\,(x - 1)$

9. 4 Sundays and 26 other days i.e., Average: $(4 \times 510 + 26 \times 240)/30$

20. As the distance is same, average speed = $(2 \times 56 \times 84)/(56 + 84)$

30. If the Number of workers be 'x'; then $8000\,x = 7\,(12000) + (x - 7)\,6000$

AGE PROBLEMS

8

Remember

- ***Ages are generally not in fraction.***
- ***Start and assume the age of present status else from a stage where a direct relationship exists.***
- ***Try to do the questions orally as much as possible.***

Practice Exercise - 1

Number of Questions: 10 **Suggested time: 10 minutes**

1. Ten years ago, P was half of Q's age. If the ratio of their present ages is 3 : 4, what will be the total of their present ages?
(a) 45
(b) 40
(c) 35
(d) 30

2. Father is aged three times more than his son Sunil. After 8 years he would be two and a half times of Sunil's age. After further eight years, how many times would he be of Sunil's age?
(a) 4
(b) 5
(c) 2
(d) 3

3. A man's age is 125% of what it was 10 years ago. It is $5/6^{th}$ of what it will be after 10 years. What is his present age?
(a) 70
(b) 60
(c) 50
(d) 40

4. A man is 24 years older than his son. In two years, his age will be twice the age of his son. What is the present age of his son?
(a) 23
(b) 22
(c) 21
(d) 20

5. Present ages of Karan and Sohan are in the ratio of 5 : 4 respectively. Three years from now, their ages will be in the ratio 11 : 9 respectively. What is the present age (years) of Sohan?
(a) 28

(b) 27
(c) 26
(d) 24

6. The sum of ages of 5 children born at the intervals of 3 years each is 50 years. Find out the age of the youngest child in the family.
(a) 6
(b) 5
(c) 4
(d) 3

7. A is two years older than B who is twice as old as C. The total of the ages of A, B and C is 27. How old is B?
(a) 10
(b) 9
(c) 8
(d) 7

8. The average age of a class of 22 students is 21 years. The average increased by 1 when the teacher's age also included. What is the age of the teacher?
(a) 48
(b) 45
(c) 43
(d) 44

9. Mr Manav said to his son- Navansh, "I was - as old as you are at present - at the time of your birth". If Manav's age is 38 years now. What was the age of the Navansh three years back?
(a) 20
(b) 18
(c) 16
(d) 22

10. Alisha's age is $1/6^{th}$ of her father's age. Alisha's father's age will be twice the Himanshu's age after 10 years period. If Himanshu's 8^{th} birthday was celebrated two years before, then what is Alisha's present age.
(a) 10

(b) 12
(c) 8
(d) 5

11. The sum of the present ages of a son and his father is 60 years. Six years ago, father's age was five times the age of the son. After 6 years, what will be son's age?
(a) 23
(b) 22
(c) 21
(d) 20

12. Kalpana is younger than Kiran by 7 years and their ages are in the respective ratio of 7:9. How old is Kalpana?
(a) 25
(b) 24.5
(c) 24
(d) 23.5

13. Six years ago, the ratio of the ages of Vimal and Saroj was 6 : 5 Four years hence the ratio of their ages will be 11 : 10. What is Saroj's age at present?
(a) 18
(b) 17
(c) 16
(d) 15

14. At present, the ratio between the ages of Shekhar and Shobha is 4 : 3. After six years, Shekhar's age will be 26 years. Find out the age of Shobha at present?
(a) 15
(b) 14
(c) 13
(d) 12

15. A is as much younger than B as he is older than C. If the sum of the ages of B and C is 50 years, what is the difference between the ages of A and B?
(a) 3

(b) 2
(c) 5
(d) Data inadequate

16. The age of Gurjeet is thrice the total ages of his daughters Ria and Amani. (0.5) decades hence, his age will be twice of the total ages of his daughters. Then what is the Gurjeet's current age? [0.5 Decades = 5 Years]
(a) 35
(b) 60
(c) 45
(d) 48

17. A man's current age is $\frac{1}{2}$ of the age of his father's age. After 8 years, he will be $\frac{1}{2}$ of the age of his father. What is the age (years) of father now?
(a) 50
(b) 40
(c) 33
(d) 30

18. The total age of A and B is 12 years more than the total age of B and C. C is how many years younger than A?
(a) 12
(b) 16
(c) 18
(d) 6

19. Sobha's father was 38 years of age when she was born while her mother was 36 years old when her brother four years younger to her was born. What is the difference between the ages of her parents?
(a) 6
(b) 5
(c) 4
(d) 3

20. The age of father 10 years ago was thrice the age of his son. Ten years hence, father's age will be twice that of his son. What is the ratio of their present ages?

(a) 7 : 3

(b) 3 : 7

(c) 9 : 4

(d) 4 : 9

21. The ages of two persons differ by 16 years. 6 years ago, the elder one was 3 times as old as the younger one. What are their present ages of the elder person?

(a) 10

(b) 20

(c) 30

(d) 40

22. Present age of a father is 3 years more than three times the age of his son. Three years hence, father's age will be 10 years more than twice the age of the son. What is father's present age?

(a) 30

(b) 31

(c) 32

(d) 33

23. Kamal was 4 times as old as his son 8 years ago. After 8 years, Kamal will be twice as old as his son. Find out the present age of Kamal.

(a) 40

(b) 38

(c) 42

(d) 36

24. If six years are *subtracted from the present age of Ajay and the remainder is divided by 18, then the* present age of Rahul is obtained. If Rahul is 2 years younger to Denis whose age is 5 years, then what is Ajay's present age?

(a) 50

(b) 60

(c) 55

(d) 62

25. The ratio of the age of a man and his wife is 4 : 3. At the time of marriage the ratio was 5 : 3 and After 4 years this ratio will become 9 : 7. How many years ago were they married?

(a) 8

(b) 10

(c) 11

(d) 12

Practice Exercise - 1 Answers with Hints - need based

1 (c)	2 (c)	3 (c)	4 (b)	5 (d)
6 (c)	7 (a)	8 (d)	9 (c)	10 (d)
11 (d)	12 (b)	13 (c)	14 (a)	15 (d)
16 (c)	17 (b)	18 (a)	19 (a)	20 (a)
21 (c)	22 (d)	23 (a)	24 (b)	25 (d)

14. Let the son's present age be x years. Then, $(38 - x) = x \Leftrightarrow x = 19$.
Son's age 5 years back $= (19 - 5) = 14$ years

Practice Exercise - 2

Number of Questions: 25 **Suggested time: 15 minutes**

1. The product of the ages of Syam and Sunil is 240 If twice the age of Sunil is more than Syam's age by 4 years, what is Sunil's age?
(a) 16
(b) 14
(c) 12
(d) 10

2. One year ago, the ratio of Sartaj's and Vaani's age was 6:7 respectively. Four years hence, this ratio would become 7 : 8. How old is Vaani?
(a) 32
(b) 34
(c) 36
(d) 38

3. The total age of P and Q is 12 years more than the total age of Q and R. R is how many year younger than P?
(a) 10
(b) 11
(c) 12
(d) 13

4. SUNDER's age after 15 years will be 5 times his age 5 years back. Find out the present age of SUNDER?
(a) 15
(b) 11
(c) 12
(d) 10

5. Sandeep's age after six years will be three-seventh of his father's age at that time. Ten years ago the ratio of their ages was 1 : 5. What is Sandeep's father's age at present?
(a) 30
(b) 40

(c) 50

(d) 60

6. Kartar said to his son, "Bitte, I was as old as you are at present at the time of your birth. " If Kartar age is 38 now, the son's age 5 years hence will be —— years?

(a) 24

(b) 19

(c) 33

(d) 38

7. A person's present age is two-fifth of the age of his mother Parsanni Devi. After 8 years, he will be one-half of the age of his mother. How old is the mother at present?

(a) 38

(b) 40

(c) 42

(d) 44

8. Six years ago Anu was 'P' times as old as Penny was. If Anu is now 17 years old, how old is Penny now in terms of 'P'?

(a) $11/P + 6$

(b) $P/11 + 6$

(c) $17 - P/6$

(d) $17/P$

9. Five years ago, the average age of A, B, C and D was 45 years. With E Joining them now, the average of all the five is 49 years. The age of E is?

(a) 48

(b) 50

(c) 52

(d) None of these

10. Shamsher is younger than Ranjan by 7 years. If the ratio of their ages is 7 : 9, find the age of Shamsher.

(a) 24.5

(b) 25.5

(c) 26.5

(d) 27.5

11. In 10 years, Amrit will be twice as old as Brahmin was 10 years ago. If Amrit is now 9 years older than Brahmin, the present age of Brahmin is:

(a) 19
(b) 29
(c) 39
(d) 49

12. The sum of the ages of 5 children born at the intervals of 3 years each is 50 years. What is the age of the youngest child?

(a) 1 4
(b) 18
(c) 4
(d) 13

13. The age of Namisha is 4 times of her daughter Asmi. Five years ago, she was nine times old as her daugther was at that time. The present age of Namisha is?

(a) 32
(b) 40
(c) 56
(d) 23

14. The ratio of Laxmi's age to the age of her mother is 3 : 11. The difference of their ages is 24 years. The ratio of their ages after 3 years will be?

(a) 3 : 8
(b) 4 : 9
(c) 5 : 7
(d) 3 : 1

15. The sum of the present ages of Hemant and his son Dhairya is 60 years. Five years ago, Hemant's age was four times the age of Dhairya. So now the son's age is —— years:

(a) 5
(b) 10
(c) 15
(d) 20

16. The average age of students of Mr Gambhir's class is 15.8 years. The average age of boys in the class is 16.4 years and that of the girls is 15.4 years. The ratio of number of boys to the number of girls in the class is?

(a) 4 : 7
(b) 3 : 5
(c) 3 : 2
(d) None

17. If two times of the Amani's age in years is included to the Nina's age, the total is 70 and if two times of the Nina's age is included to the Amani's age, the total is 95. So Nina's age is —— years?

(a) 30
(b) 38
(c) 40
(d) 41

18. The ratio of the ages of Meenu and Kalu is 4 : 3. The total of their ages is 2.8 decades. The proportion of their ages after 0.8 decades will be [1 Decade = 10 years]?

(a) 4 : 3
(b) 12 : 11
(c) 7 : 4
(d) 6 : 5

19. The ages of Soumaya and Krishan are in the proportion of 3 : 5. After 9 years, the proportion of their ages will be 3 : 4. Then the current age of Krishan is:

(a) 10
(b) 13
(c) 15
(d) 25

20. If Raj was one-third as old as Rakesh 5 years back and Raj is 17 years old now, How old is Rakesh now?

(a) 40
(b) 41
(c) 53

(d) 50

21. The average age of a group of 10 students is 15 years. When 5 more students join the group, the average age increase by 1 year. The average age of the new students is?

(a) 18
(b) 10
(c) 12
(d) None

22. It was (0.1) decade ago, that Nitin was quadrice as old as his daughter Vriddhi. (0.6) decades hence, Nitin's age will beat Vriddhi's age by (0.9) decades. The proportion of their current ages is? [0.1 Decades = 1 Year; quadrice = 4 times].

(a) 8:11
(b) 13:3
(c) 11:5
(d) 13:4

23. The sum of the present ages of two persons Rajesh and Kanchan is 60. If the age of Rajesh is twice that of Kanchan, find the sum of their ages 5 years hence.

(a) 36
(b) 70
(c) 65
(d) 60

24. The average age of 12 men is decreased by a year when two of them having ages 28 years and 32 years are replaced by two women of same age. The age of replaced woman is?

(a) 30
(b) 34
(c) 32
(d) None

25. The ratio of the present ages of Ishan and Parteek is 3 : 4. Five years ago, the ratio of their ages was 5 : 7. Find sum of their present ages.

(a) 28
(b) 60

(c) 42
(d) None

Practice Exercise - 2 Answers with Hints - need based

1 (c)	2 (c)	3 (c)	4 (d)	5 (c)
6 (a)	7 (b)	8 (a)	9 (d)	10 (a)
11 (c)	12 (c)	13 (a)	14 (d)	15 (c)
16 (d)	17 (c)	18 (d)	19 (c)	20 (b)
21 (a)	22 (d)	23 (b)	24 (d)	25 (d)

7. Let the mother's present age be x years. Then, his present age = $\frac{2}{5}x$ years.

$\therefore \left(\frac{2}{5}x+8\right)=\frac{1}{2}(x+8) \Rightarrow 2(2x + 40) = 5(x + 8) \Rightarrow x = 40$

9. $\therefore$ 5 years ago total age of A, B, C, D = 45 × 4 = 180 years
$\therefore$ Total present age of A, B, C, D = 180 + 5 × 4 = 200 years
E's present age is ^{x} years $\Rightarrow \frac{200+x}{5}=49 \; \therefore x = \mathbf{5}$ years

10. If Ranjan age is x, then Shamsher age is $x - 7$,
so, $\frac{(x-7)}{x}=\frac{7}{9} \Rightarrow 9\,^{x} - 63 = 7\,^{x} \Rightarrow 2\,^{x} = 63 \Rightarrow {}^{x} = 31.5$
So Shamsher's age is 31.5 − 7 = 24.5 (***Normally ages are Integral, exceptions ignored)***

11. Let B's present age = x years. Then, A's present age = $(x + 9)$ years.
$\therefore (x + 9) + 10 = 2(x - 10) \Rightarrow x + 19 = 2x - 20 \Rightarrow x = 39$.

12. Let x = the youngest child. Then other four children are $x + 3$, $x + 6$, $x + 9$, $x + 12$.
Given that, $x + (x + 3) + (x + 6) + (x + 9) + (x + 12) = 50 \Rightarrow x = 4$
$\therefore$ The youngest child is 4 years old.

14. The difference of their ages $11x - 3x = 24 \Leftrightarrow x = 3$
$\therefore$ Ratio of their ages after 3 years = $3\xi + 3 : 11^{x} + 3 = 3 \times 3 + 3 : 11 \times 3 + 3 = 12 : 36 = 1 : 3$

16. Suppose the number of boys and girls are x and y respectively

$\Rightarrow \frac{x \times 16.4 + y \times 15.4}{x+y} = 15.8 \Rightarrow 16.4^x + 15.4y = 15.8x + 15.8y$

$\Rightarrow 0.6x = 0.4y \therefore \frac{x}{y} = \frac{0.4}{0.6} = \frac{2}{3} = 2:3$

17. 2 A + N = 70; A + 2 N = 95; Therefore N = 40 years

12 Men go down by One year each i.e., by 12 years when a total of 60 year is replaced by two new women. i.e., the new women bring 60 − 12 = 48 years means 24 years each.

Practice Exercise - 3

Number of Questions: 10 **Suggested time: 7 minutes**

(try to hit as many questions as you can – but carefully)

1. Ashok's age is ($1/6^{th}$) of his father's age. The father's age will be twice of Kamal's age after 10 years. If Kamal's eighth birthday was celebrated 2 years before, what is Ashok's present age?
(a) 24 years
(b) 30 yrs
(c) $6\frac{1}{4}$ yrs
(d) None

2. The sum of the ages of a father and son is 45 years. Five years ago the product of their ages was 4 times the father's age at that time. The present ages of the father and son respectively are :
(a) 25 years, 10 yrs
(b) 36 yrs, 9 yrs.
(c) 39 yrs, 6 yrs
(d) None

3. 10 years ago, Mona's mother was 4 times older than her daughter. After 10 years, the mother will be twice older than the daughter. Mona's present age is :
(a) 5 years
(b) 10 yrs
(c) 20 yrs
(d) 30 yrs

4. Three years ago, the average age of A and AB was 18 years. With C joining them, the average becomes 22 years. How old is C now?
(a) 24 years
(b) 27 yrs
(c) 28 yrs
(d) 30 yrs

5. Kamla got married 6 years ago. Today her age is $1\frac{1}{4}$ times her age at the time of marriage. Her son's age is $\frac{17}{2}$ times her age. Her son's age is:

(a) 2 years
(b) 3 yrs
(c) 4 yrs
(d) 5 yrs

6. One year ago the ratio of Varun and Veena's age was 4 : 5. One year hence the ratio of their ages will be 5 : 6. The present age of Veena is:

(a) 9 years
(b) 10 yrs
(c) 11 yrs
(d) 6 yrs

7. A is as much younger to B as he is older to C. If the sum of the ages of B and C is 48 years, what is the present age of A?

(a) 20 years
(b) 24 yrs
(c) 30 yrs
(d) 36 yrs

8. Ten years ago A was half of B in age. If the ratio of their present ages is 3 : 4, what is the sum of their present ages?

(a) 8 years
(b) 20 yrs
(c) 35 yrs
(d) 45 yrs

9. The ratio of Laxmi's age to the age of her mother is 3 : 11. The difference of their ages is 24 years. The ratio of their ages after 3 years will be :

(a) 1 : 3
(b) 2 : 3
(c) 3 : 5
(d) 2 : 5

10. The age of a man is 4 times that of his son. Five years ago, the man was nine times as old as his son was at that time. The present age of the man is:

(a) 28 years

(b) 32 yrs

(c) 40 yrs

(d) 44 yrs

Practice Exercise - 3 Answers with Hints and Solutions

1 (d)	2 (b)	3 (c)	4 (a)	5 (b)
6 (c)	7 (b)	8 (c)	9 (a)	10 (b)

1. Kamal's present age = 10 years.

Kamal's age after 10 years = 20 years.

Father's age after 10 years = 40 years.

Father's age now = 30 years.

Ashok's age now = $\left(\frac{1}{6} \times 30\right)$ years = 5 years.

2. Let son's age = x. Then, father's age = $(45 - x)$

$(x - 5)(45 - x - 5) = 4(45 - x - 5) \Leftrightarrow x - 5 = x \Leftrightarrow x = 9$

∴ Father's age now = 36 years and son's age = 9 years.

3. Let Mona's age 10 years ago = x. Her mother's age then = $4x$

$2 (x + 20)= 4x + 20 \Leftrightarrow 2x = 20 \Leftrightarrow x = 10$

∴ Mona's present age = (10 + 10) years = 20 years.

4. (A + B)'s age 3 years ago = (18 × 2)= 36 years.

(A + B) now = (36 + 3 × 2) years = 42 years.

(A + B + C) now = (3 × 22) years = 66 years.

C's present age = (66 – 42) years = 24 years.

5. Let kamla's age 6 years ago be x.

$\therefore (x + 6) = \frac{1}{4}x \Leftrightarrow 4 (x + 6) = 5x \Leftrightarrow x = 24.$

∴ Kamla's present age = 30 years.

$\therefore$ Kamla's son's present age $= \left(\frac{1}{10} \times 30\right)$ years = 3 years.

6. Let their ages 1 year ago be $4x$ and $5x$

$\frac{4x+2}{5x+2} = \frac{5}{6} \Leftrightarrow 6(4x+2) = 5(5x+2) \Leftrightarrow x = 2$

$\therefore$ Veena's present age = $(5x + 2)$= 12 years.

7. B –A = A– C $\Leftrightarrow$ 24A = B + C $\Leftrightarrow$ 2A = 48 $\Leftrightarrow$ A = 24

8. Let their present ages be $3x$ and $4x$. Then,

2 $(3x – 10)$= $4x – 10 \Leftrightarrow 2x = 10 \Leftrightarrow x = 5$.

$\therefore$Sum of their present ages = $7x$ = 35 years.

9. Let their ages be $3x$ and $11x$

$11x – 3x = 24 \Leftrightarrow x = 3$. Their present ages are 9 years and 33 years.

Ratio of their ages after 3 years = 12 : 36 = 1 : 3

10. Let the son's age be x. Then, father's age = $4x$.

$\therefore (4x - 5) = 9 (x - 5) \Leftrightarrow 5x = 40 \Leftrightarrow x = 8$.

BANKER'S DISCOUNT

9

Assume that a merchant Mr Ashok Kumar purchases goods worth, say Rs. 1,000 from another merchant Mr. Suresh Chawla at a credit of say 4 months.

Then Mr Chawla prepares a bill called bill of exchange (also called Hundi). On receipts of goods, Mr Kumar gives an agreement by signing on the bill allowing Mr Chawla to withdraw the money from Mr Kumar's bank exactly after 4 months of the date of the bill.

The date exactly after 4 months is known as **NOMINALLY DUE DATE**. Three more days (called grace days) are added to this date to get a date known as **legally due date.**

The amount given on the bill is called the **FACE VALUE (F)** which is Rs. 1,000 in this case.

Assume that Mr Chawla needs this money before the legally due date. He can approach a banker or broker who pays him the money against the bill, but somewhat less than the face value. The banker deducts the simple interest on the face value for the unexpired time. This deduction is known as **BANKERS DISCOUNT (BD).** In another words, Bank Discount (BD) is the simple interest on the face value for the period from the date on which the bill was discounted and the legally due date.

The **PRESENT VALUE** is the amount which, if placed at a particular rate for a specified period will amount to that sum of money at the end of the specified period. The interest on the present value is called the **TRUE DISCOUNT (TD).** If the banker deducts the true discount on the face value for the unexpired time, he will not gain anything.

Banker's Gain (BG) is the difference between banker's discount and the true discount for the unexpired time.

Note: When the date of bill is not given, grace days are not to be added.

Important Formulas - Banker's Discount

Let F = Face Value of the Bill, TD = True Discount,
BD = Bankers Discount, BG = Banker's Gain,
R = Rate of Interest,
PW = True Present Worth and T = Time in Years

BD *= Simple Interest on the face value of the bill for unexpired time = FTR/100*

TD *= Simple Interest on the present value for unexpired time = PW × TR/100*

TD *= BD × 100 (100 + TR)* ***PW*** *= F – T/D*

F *= BD × TD/BD – TD* ***BG*** *= BD – TD = Simple Interest on TD = (TD)/2PW*

TD *= √PW × BG* ***TD*** *= BG × 100/TR*

Practice Exercise - 1

Number of Questions: 15 **Suggested time: 15 minutes**

1. The banker's discount on a bill due 4 months hence at 15% is Rs. 420. What is the true dicount?
(a) Rs. 410
(b) Rs. 400
(c) Rs. 390
(d) Rs. 380

2. The banker's discount on a certain amount due 2 years hence is 1110 of the true discount. What is the rate percent?
(a) 1%
(b) 5%
(c) 10%
(d) 12%

3. The present worth of a sum due sometimes hence is Rs. 5760 and the banker's gain is Rs. 10. What is the true discount?
(a) Rs. 480
(b) Rs. 420
(c) Rs. 120
(d) Rs. 240

4. What is the banker's discount if the true discount on a bill of Rs. 540 is Rs. 90?
(a) Rs. 108
(b) Rs. 120
(c) Rs. 102
(d) Rs. 106

5. A bill for Rs. 3000 is drawn on 14^{th} July at 5 months. It is discounted on 5^{th} October at 10%. What is the Banker's Discount?
(a) Rs. 60
(b) Rs. 82
(c) Rs. 90

(d) Rs. 120

6. The bankers discount and the true discount of a sum at 10% per annum simple interest for the same time are Rs. 100 and Rs. 80 respectively. What is the sum and the time?

(a) Sum = Rs. 400 and Time = 5 years
(b) Sum = Rs. 200 and Time = 2.5 years
(c) Sum = Rs. 400 and Time = 2.5 years
(d) Sum = Rs. 200 and Time = 5 years

7. The banker's gain on a sum due 6 years hence at 12% per annum is Rs. 540. What is the banker's discount?

(a) 1240
(b) 1120
(c) 1190
(d) 1290

8. The present worth of a certain bill due sometime hence is Rs. 1296 and the true discount is Rs. 72. What is the banker's discount?

(a) Rs. 76
(b) Rs. 72
(c) Rs. 74
(d) Rs. 4

9. The banker's discount of a certain sum of money is Rs. 36 and the true discount on the same sum for the same time is Rs. 30. What is the sum due?

(a) Rs. 180
(b) Rs. 120
(c) Rs. 220
(d) Rs. 200

10. The banker's gain on a bill due 1 year hence at 10% per annum is Rs. 20. What is the true discount?

(a) Rs. 200
(b) Rs. 100
(c) Rs. 150
(d) Rs. 250

11. The banker's gain of a certain sum due 3 years hence at 10% per annum is Rs. 36. What is the present worth?

(a) Rs. 400
(b) Rs. 300
(c) Rs. 500
(d) Rs. 350

12. The present worth of a certain sum due sometime hence is Rs. 3400 and the true discount is Rs. 340. The banker's gain is:

(a) Rs. 21
(b) Rs. 17
(c) Rs. 18
(d) Rs. 34

13. The banker's discount on Rs. 1600 at 15% per annum is the same as true discount on Rs. 1680 for the same time and at the same rate. What is the time?

(a) 3 months
(b) 4 months
(c) 5 months
(d) 6 months

14. The banker's gain on a certain sum due 2 1/2 years hence is 9/25 times of the banker's discount. What is the rate percent?

(a) 18.13%
(b) 18.12%
(c) 24.13%
(d) 22.22%

15. The banker's gain on a sum due 3 years hence at 12% per annum is Rs. 360. The banker's discount is:

(a) Rs. 1360
(b) Rs. 1000
(c) *Rs. 360*
(d) *Rs*. 640

Practice Exercise - 1 Answers with Hints - need based

1 (d)	2 (b)	3 (d)	4 (a)	5 (a)
6 (c)	7 (d)	8 (a)	9 (a)	10 (a)
11 (a)	12 (d)	13 (b)	14 (d)	15 (a)

1. TD = BD × 100 (100 + TR) = 420 × 100 (100 + 4/12 × 15) = 420 × 100 (100 + 5) = 400

2. BD = FTR/100 ⇒ 1110 = 11 × 2 × R/100 ⇒ R = 5%

3. TD = √PW × BG = √5760 × 10 = √57600 = Rs. 240

Present Worth, PW = F − TD = 540 − 90 = Rs. 450

Hence Simple Interest on 450 = 90 −−−−−−−−−−−(Equation 1)

Simple Interest on the face value = Bankers Discount ⇒ Simple Interest on 540

Hence, Simple Interest on 540 = 90/450 × 540 = Rs. 108 = Bankers Discount

4. Date on which the bill is drawn = 14^{th} July at 5 months

i.e., Nominally Due Date = 14^{th} December

Legally Due Date = 14^{th} December + 3 days = 17^{th} December

Date on which the bill is discounted = 5^{th} October

Unexpired Time = 73 Days = 1/5 year

BD = Simple Interest on the face value of the bill for unexpired time

= FTR/100 = 3000 × 1/5 × 10/100 = Rs. 60

8. BG = $(TD)^2$/PW = 72 × 72/1296 = Rs. 4 ⇒ BD = 72 + 4 = Rs. 76

11. TD = BG × 100/TR = 36 × 100/3 × 10 = Rs. 120

TD = PW × TR/100 ⇒ 120 = PW × 3 × 10/100 ⇒ = Rs. 400

14. Let the banker's discount, BD = Rs. 1

Then, banker's gain, BG = 9/25 × 1 = Rs. 9/25

BG = BD − TD ⇒ 9/25 = 1 − TD ⇒ TD = 1 − 9/25 = 16/25

F = BD × TD/BD − TD = 1 × 16/25 (1 − 16/25) = (16/25)/(9/25) = Rs. 16/9

BD = Simple Interest on the face value of the bill for unexpired time = FTR/100

$\Rightarrow 1 = (16/9) \times (5/2) \times R/100 \Rightarrow R = 100 \times 9/40 = 22\ 1/2\%$

Practice Exercise - 2

Number of Questions: 15 **Suggested time: 15 minutes**

1. The true discount on a certain sum due 6 months hence at 15% is Rs. 240. What is the banker's discount on the same sum for the same time at the same rate?
(a) Rs. 248
(b) Rs. 278
(c) Rs. 228
(d) Rs. 258

2. A bill is discounted at 10% per annum. If banker's discount is allowed, at what rate percent should the proceeds be invested so that nothing will be lost?
(a) 10%
(b) 11 − 1/9%
(c) 11
(d) 9 − 1/9%

3. A banker paid Rs. 5767.20 for a bill of Rs. 5840, drawn of Apr 4 at 6 months. If the rate of interest was 7%, what was the day on which the bill was discounted?
(a) 3^{rd} March
(b) 3^{rd} September
(c) 3^{rd} October
(d) 3^{rd} August

4. The banker's discount on a sum of money for 3 years is Rs. 1116. The true discount on the same sum for 4 years is Rs. 1200. What is the rate percent?
(a) 8%
(b) 12%
(c) 10%
(d) 6%

5. The true discount on a bill of Rs. 2160 is Rs. 360. What is the banker's discount?
(a) Rs. 432
(b) Rs. 422
(c) Rs. 412
(d) Rs. 442

6. The banker's gain of a certain sum due 2 years hence at 10% per annum is Rs. 24. What is the present worth?
(a) Rs. 600
(b) Rs. 500
(c) Rs. 400
(d) Rs. 300

7. The true discount on a bill for Rs. 2520 due 6 months hence at 10% per annum is:
(a) Rs. 180
(b) Rs. 140
(c) Rs. 80
(d) Rs. 120

8. What is the present worth of a bill of Rs. 1764 due 2 years hence at 5% compound interest is:
(a) Rs. 1600
(b) Rs. 1200
(c) Rs. 1800
(d) Rs. 1400

9. What is the difference between the banker's discount and the true discount on Rs. 8100 for 3 months at 5%?
(a) Rs. 2
(b) Rs. 1.25
(c) Rs. 2.25
(d) Rs. 0.50

10. The B.G. on a certain sum 4 years hence at 5% is Rs. 200. What is the present worth?
(a) Rs. 4,500
(b) Rs. 6,000

(c) Rs. 5,000
(d) Rs. 4,000

11. The B.D and T.D on a certain sum is Rs. 200 and Rs. 100 respectively. Find out the sum.
(a) Rs. 400
(b) Rs. 300
(c) Rs. 100
(d) Rs. 200

12. The banker's discount on a bill due 6 months hence at 6% is Rs. 18.54. What is the true discount?
(a) Rs. 24
(b) Rs. 12
(c) Rs. 36
(d) Rs. 18

13. The banker's discount on a sum of money for 112 years is Rs. 120. The true discount on the same sum for 2 years is Rs. 150. What is the rate percent?
(a) 313%
(b) 413%
(c) 323%
(d) 423%

14. The present worth of a certain bill due sometime hence is Rs. 400 and the true discount is Rs. 20. What is the banker's discount?
(a) Rs. 19
(b) Rs. 22
(c) Rs. 20
(d) Rs. 21

15. If the discount on Rs. 498 at 5% simple interest is Rs. 18, when is the sum due?
(a) 8 months
(b) 11 months
(c) 10 months
(d) 9 months

Practice Exercise - 2 Answers with Hints - need based

1 (d)	2 (b)	3 (d)	4 (d)	5 (a)
6 (a)	7 (d)	8 (a)	9 (b)	10 (c)
11 (d)	12 (d)	13 (a)	14 (d)	15 (d)

1. TD = BG × 100/TR ⇒ 240 = BG × 100/(1/2 × 15) ⇒ BG = Rs. 18
BG = BD − TD ⇒ 18 = BD − 240 ⇒ BD = Rs. 258

2. Let the amount = Rs. 100

Then BD = Rs. 10 (∵ banker's discount, BD is the simple Interest on the face value of the bill for
unexpired time and bill, in this case, is discounted at 10% per annum)

Proceeds = Rs. 100 − Rs. 10 = Rs. 90

Hence we should get Rs. 10 as the interest of Rs. 90 for 1 year so that nothing will be lost

⇒ 10 = 90 × 1 × R/100 ⇒ R = 11 1/9%

3. BD = FTR/100 ⇒ 72.8 = 5840 × T × 7/100 ⇒ T = 13 × 365/73 days = 65 days

⇒ Unexpired Time = 65 days

Given that Date of Draw of the bill = 4^{th} April at 6 months ⇒ 4^{th} October

⇒ Legally Due Date = (4^{th} October + 3 days) = 7^{th} October

Hence, The date on which the bill was discounted

= (7^{th} October − 65 days) = 3^{rd} August

5. True Discount is the Simple Interest on the present value for unexpired time

⇒ Simple Interest on Rs. 1800 for unexpired time = Rs. 360

Banker's Discount is the Simple Interest on the face value of the bill for unexpired time

= Simple Interest on Rs. 2160 for unexpired time

= 360/1800 × 2160 = 1/5 × 2160 = Rs. 432

7. $TD = F/TR\ (100 + TR) = 2520\ (1/2 \times 10)/(100 + (1/2 \times 10)) = 2520/21 =$ Rs. 12

9. $TD = PW \times TR/100 \Rightarrow 18 = 480 \times T \times 5/100 \Rightarrow T = 18/24 = 3/4$ years = 9 months

10. $BD = FTR/100 = 8100 \times 1/4 \times 5/100 =$ Rs. 101.25
$TD = FTR/(100 + TR) = 8100 \times 1/4 \times 5/(100 + (1/4 \times 5)) = 2025 \times 5 \times 4/405 =$ Rs. 100
$BD - TD =$ Rs. $101.25 -$ Rs. $100 =$ Rs. 1.25

11. $TD = BG \times 100/TR = 200 \times 100/(4 \times 5) =$ Rs. 1000

12. $F = BD \times TD\ (BD - TD) = 200 \times 100\ (200-100) = 200 \times 100/100 =$ Rs. 200

13. $TD = BD \times 100/(100 + TR) = 18.54 \times 100/(100 + (1/2 \times 6)) =$ Rs. 18

15. $BG = (TD)^2/PW = 20^2/400 =$ Rs. 1
$BG = BD - TD \Rightarrow 1 = BD - 20 \Rightarrow BD = 1 + 20 =$ Rs. 21

Practice Exercise - 3

Number of Questions: 10 **Suggested time: 10 minutes**

1. The difference between Simple Interest and True Discount on a certain sum of money, paid by Mr Manohar, for 6 months at $12\frac{1}{2}$% per annum is Rs. 25. Find the sum (Rs.).
(a) 6,800
(b) 6,500
(c) 6,000
(d) 6,200

2. Harkaran buys a watch for Rs. 1,950 in cash and sells it for Rs. 2,200 at a credit of 1 year. If the rate of interest is 10% per annum, he _____:
(a) gains Rs. 55
(b) gains Rs. 50
(c) loses Rs. 30
(d) gains Rs. 30

3. Vivek Bhatia wants to sell his scooter. Before him, there are two offers (i) at Rs. 12,000 cash and (ii) at a credit of Rs. 12,880 to be paid after 8 months, money being at 18% per annum which is better offer?
(a) 12,800
(b) 12,000
(c) Both equal
(d) Can't say

4. The profit earned by Mrs Malhotra in selling an article for Rs. 900 is double the loss incurred by her when the same article is sold for Rs. 490. At what price (Rs.) should the article be sold to make 25% profit?
(a) 715
(b) 469
(c) 400
(d) 750

5. Mr Jaspal purchased a cow for Rs. 3,000 and sold it to Mr Dharampal, on the same day, for Rs. 3,600, allowing Dharampal a credit of 2 years. If the rate of interest be 10% per annum, then the Jaspal has a gain of:
(a) Nil
(b) 5%
(c) 750
(d) 10%

6. The true discount on Rs. 2,562 due 4 months hence is Rs. 122. The rate percent is?
(a) 12
(b) 13
(c) 15
(d) 14

7. Find the present worth of Rs. 9300, due 3 years hence at 8% per annum. Also find the discount?
(a) 1,860
(b) 1,800
(c) 1,850
(d) 1,890

8. The present worth of Rs. 702 due in two equal half-yearly installments at 8% per annum calculated at simple interest is Rs.
(a) 600
(b) 1,500
(c) 1,325
(d) 1,000

9. If Rs. 40 be allowed as true discount on a bill of Rs. 440 at the end of a certain time, then the discount allowed on the same sum due at the end of double the time is Rs.?
(a) 72.80
(b) 73.32
(c) 72.00
(d) 73.20

10. A bill, drawn on 23rd Jun 2017 falls due in 1 year. The creditor agrees to accept immediate payment of the half and to defer the payment of the other half payable on 22nd Jun 2019. By this arrangement interest charged is Rs. 60. What is the amount (Rs.) of the bill, if the money be worth 12.5%?

(a) 1,800
(b) 4,500
(c) 5,400
(d) 1,950

Practice Exercise - 3 Answers with Explanations

1 (a)	2 (b)	3 (b)	4 (d)	5 (a)
6 (c)	7 (b)	8 (c)	9 (b)	10 (c)

1. (a) Rs. 68,000

Let amount be Rs. x; T.D $= \frac{17}{2}$; S.I $= \frac{x \times \frac{25}{2} \times \frac{1}{2}}{100} = \frac{17}{2}$;

$\therefore \frac{x}{16} - \frac{x}{17} = 25 \Rightarrow 17x - 16x = 25 \times 16 \times 17 \Rightarrow x = 6800$

2. (b) gains Rs. 50

S.P. = P.W. of Rs. 2200 due 1 year hence

$= \text{Rs.} \left[\frac{2200 \times 100}{100 + (10 \times 1)}\right]$

= Rs. 2000.

$\therefore$ Gain = Rs. (200 = 0 − 1950) = Rs. 50.

3. (b) Rs. 12000

PW of Rs. 12,880 due 8 months hence = Rs. 11500; Clearly 12000 in cash is a better offer.

4. (d) 750

Let C.P be Rs. x; $900 - x = 2(x - 450) \Rightarrow x =$ Rs. 600

C.P = 600 gain required is 25%; S.P = [(100 + 25)*600]/100 = Rs. 750

5. (a) 0%

C.P = Rs. 3000; S.P = $\text{Rs.}\left[\dfrac{3600\times100}{100+(10\times2)}\right]=\text{Rs}3000$; Gain = 0%

6. (c) 15%

P.W = 2562 − 122 = Rs. 2440; $\text{Rate}=\dfrac{100\times122}{2440\times\frac{1}{3}}$; = 15%

7. (b) 1800

T.D = Amount − P.W = 9300 − 7500 = Rs. 1800

8. (c) 1325

Required Sum = PW of Rs. 702 due 6 months hence + PW of Rs. 702 due 1 year hence = Rs. 1325

9. (c) 73.32

Present worth = Amount − True Discount = 110 − 10 = Rs. 100

SI on Rs. 100 for a certain time = Rs. 10

SI on Rs. 100 for double the time = Rs. 20

True Discount on Rs. 120 = 120 − 100 = Rs. 20

True Discount on Rs. 440 = Rs. 73.32

10. (c) 5,400

Let the sum be Rs. x.

Then $x/2 + 2x/5 - 8x/9 = 60$, $x = 5{,}400$; Therefore amount of bill = Rs. 5,400.

Practice Exercise - 4

Number of Questions: 15 **Suggested time: 10 minutes**

1. The present worth of Rs. 2,310, which is due $2\frac{1}{2}$ years hence, the rate of interest being 15% per annum, is Rs.:
(a) 1,440
(b) 1,680
(c) 1,840
(d) 1,750

2. The true discount on a bill due 9 months hence at 16% per annum is Rs. 189. The amount of the bill is Rs. ———
(a) 1,500
(b) 1,764
(c) 2,000
(d) 2,820

3. The True discount on Rs. 17,600 due after a certain time at 12% per annum is Rs. 1,600. The time (in months) after which it is due is?
(a) 15
(b) 5
(c) 12
(d) 10

4. The interest on Rs. 7,500 for 2 years is the same as the True Discount on Rs. 9,600 due 2 years hence. If the rate of interest is same in both cases, then the rate being charged is ——— %.
(a) 14
(b) 15
(c) 16
(d) 12

5. The true discount on a bill due 9 months hence at 12% per annum is Rs. 135. Find the amount of the bill and its present worth.
(a) 1,500

(b) 1,600
(c) 6,200
(d) 6,000

6. What is the true discount on a bill of Rs. 2,916 due in 3 years hence at 8% compound rate of interest?
(a) Rs. 600
(b) Rs. 601
(c) Rs. 602
(d) Rs. 603

7. The present value of a bill, drawn on 01 Jan 2016 due on 31 Dec 2017 is Rs. 1,250. If the bill were due on 30 Nov 2017, its present worth would be Rs. 1,200. Find the rate of interest and the sum.
(a) 1,175
(b) 1,375
(c) 1,475
(d) 1,575

8. If Rs. 20 is the True Discount on Rs. 260 due after a certain time. What will be true discount on the same sum due after half of the former time, the rate of interest being the same?
(a) 9.40
(b) 10.14
(c) 10.40
(d) 9.14

9. In what time a debt of Rs. 7,920 due may be cleared by immediate cash down payment of Rs. 3,600 at $\frac{1}{2}$% per month.
(a) 10
(b) 30
(c) 20
(d) 40

10. The true discount on a bill due 9 months hence at 12% per annum is Rs. 540. Find the amount of the bill and its present worth (Rs.)?
(a) 1,600
(b) 7,000

(c) 6,000
(d) 6,500

11. Anand has to pay Rs. 220 to Babita after 1 year. Babita asks Anand to pay Rs. 110 in cash and defer the payment of Rs. 110 for 2 years to which Anand agrees. If the rate of interest be 10% per annum in this mode of payment, then Anand stands to a:
(a) Gain Rs. 11
(b) Loss Rs. 7.43
(c) At par
(d) Gain Rs. 7.34

12. The true-discount charged by Mr Brahmputra on Rs. 2,480 - due after a certain period at 5% - is Rs. 80. Find the due period in months?
(a) 6
(b) 5
(c)8
(d) 3

13. If a Transistor is purchased for Rs. 490 and sold for Rs. 465.50 find the loss percentage?
(a) 10
(b) 5
(c) 15
(d) 20

14. Which is a better offer out of (i) a cash payment now of Rs. 81,000 or (ii) a credit of Rs. 82,500 after 6 months when the rate of interest is simple at 6.5% p.a.
(a) (i)
(b) (ii)
(c) Either
(d) Data inadequate

15. The true discount on a certain sum of money due after 3 year is Rs. 1,000 and the\simple interest is Rs. 1,200 when Sum, Rate and Time are unchanged. Find the rate percent.
(a) $6 - 2/3$
(b) $5 - 2/3$

(c) 2 − 2/3
(d) 4 − 2/3

Practice Exercise - 4 Answers with Explanations

1 (c)	2 (b)	3 (d)	4 (a)	5 (d)
6 (c)	7 (b)	8 (c)	9 (c)	10 (c)
11 (d)	12 (c)	13 (b)	14 (a)	15 (a)

1. (c) $\text{P.W} = \text{Rs.}\left[\dfrac{100\times 2310}{100+\left(15\times\dfrac{5}{2}\right)}\right]$ = Rs. 1680

2. (b) Let P.W. be Rs. x.

Then, S.I. on Rs. x at 16% for 9 months = Rs. 189.

$\therefore x \times 16 \times \dfrac{9}{12}\times\dfrac{1}{100}$ = **189 or x = 1575.**

∴ **P.W. = Rs. 1575.**

∴ **Sum due = P.W. + T.D. = Rs. (1575 + 189) = Rs. 1764.**

3. (d) P.W = Rs. (1760 − 160) = Rs. 1600; ∴ S.I on Rs. 1600 at 12% is Rs. 160

∴ **Time** = $\left(\dfrac{100\times 160}{1600\times 12}\right) = \dfrac{5}{6}$ **years** = $\left(\dfrac{5}{6}\times 12\right)$ **months = 10 months**

4. (a) S.I on Rs. 750 = T.D on Rs. 960; This means P.W of Rs. 960 due 2 years hence is Rs. 750

Therefore, T.D = Rs. (960 − 750) = Rs. 210; Thus, S.I on Rs. 750 for 2 years is Rs. 210

$\therefore$ Rate $= \left[\dfrac{100 \times 210}{750 \times 2}\right]$**% = 14%**

5. (d) Let amount be Rs. X; Then, $\dfrac{X \times R \times T}{100 + (R \times T)} = TD \Rightarrow \dfrac{X \times 12 \times \frac{3}{4}}{100 + \left(12 \times \frac{3}{4}\right)} = 540$

$X =$ Rs. 6540

Amount = Rs. 6540; P.W. = Rs. (6540 − 540) = Rs. 6000.

6. (c) $PW = \dfrac{Amount}{\left(1 + \dfrac{R}{100}\right)^T}$

7. (b) $PW = \dfrac{100 \times Amount}{100 + (R \times T)}$

8. (c) S.I. on Rs. (260 − 20) for a given time = Rs. 20; for a half time = Rs. 10

True Discount on Rs. 250 = Rs. 10 or on Rs. 260 = Rs.$\left(\dfrac{10}{250} \times 260\right)$ = **Rs. 10.40**

9. (c) $PW = \dfrac{100 \times Amount}{100 + (R \times T)}$

10. (c) Let amount be Rs. x then; $\Rightarrow x =$ Pσ. 6540 $\therefore$ Amount = Rs. 6540

P.W = Rs.(6540 − 540) = Rs. 6000

11. (d) A has to pay = Present worth of Rs. 220 due 1 year hence = Rs. $\left[\dfrac{220 \times 100}{100 + (10 \times 1)}\right]$ = Rs. 200

A actually he pays = Rs. 110 + PW of Rs. 110 due 2 years hence

$= \left[110 + \frac{110 \times 100}{100 + (10 \times 2)}\right]$ **= Rs. 192.66; So, A gains Rs. (200 − 192.66) = Rs. 7.34**

12. (c) 8 months or P.W = Amount − (T.D) i.e., Time = $\frac{100 \times TD}{PW \times R}$

13. (b) Loss = C.P − S.P = 490 − 465.50 = 24.50 or loss% = (loss*100)/C.P = (24.50*100)/490 = 5%

14. (a) (i) $PW = \frac{100 \times Amount}{100 + (R \times T)}$

15. (a) (6 + 2/3)% or Sum ⇒ Amount i.e., $\frac{x}{16} - \frac{x}{17} = 25$

RACES AND GAMES

10

In a 100 meter race A can give B 10 meters or 10 meters "start" means that while A runs 100 meters B runs (100 − 10) or 90 meters. The same idea is expressed by saying that in a 100 meter race A beats B by 10 meters.

Example A runs $1\frac{5}{6}$ times as fast a B. If A gives B a start of 120 meters, how far must the winning post be so that A and B might reach it at the same time?

Solution

The rates of A and B are as 11 : 8 (1 3/8: 1)

$\therefore$ A gains (11 − 8) or 3 m in a race of 11 m.

$\therefore$ A gains 120 m, in a race of $\frac{5}{6} \times 120$ or 440 m.

Example A can give B 40 meters start and C 70 meters start in a km race. How many meters start can B give C in a km race?

Solution

A runs 1000 m while B runs (1000 − 40) or 960 m.

A runs 1000 m while C runs (1000 − 70) or 930 m.

$\therefore$ B runs 960 in while C runs 930 m.

$\therefore$ B runs 1000 m while C runs $\frac{1000 \times 930}{960} = 968\frac{3}{4}$ m.

$\therefore$ B can give C $(1000 - 968\,{}^{3}\!/_{4})$ or 31 3/4 m start.

Example In a km, race A beats B by 40 meters, or 7 seconds. Find A's time over the course.

Solution

Here B runs 40 meters in 7 second.

$\therefore$ B runs 1000 m in $\frac{9}{12} \times \frac{1}{100}$ = 175 seconds.

Hence A's time over the course = (175 − 7) = 168 seconds.

Example In a km. race, if A gives B 40 m start, A wins by 19 second, but if A gives B 30 seconds start B wins by 40 m. Find the time that each takes to run a km?

Solution

Suppose A takes x seconds and B takes y seconds to run 1000 m.

Then $x + 19 = \frac{960}{1000}$ y and $\frac{960x}{1000} + 30 = y$

Solve for x and y.

A takes 125 seconds and B takes 150 seconds.

Example Two men A and B walk around a circle 1200 m in circumference. A walks at the rate of 150 meters and B at the rate of 80 m/minute. If they both start at the same time from the same point, and walk in the same direction.

1. When will they first be together again at the starting point?
2. When will they be together again?

Solution

(a) A makes one complete round of the circle in $\frac{960}{1000}$ = 8 minutes.

B in $\frac{1200}{80}$ = 15 minutes.

That is, after every 8 minutes A is at the starting point and after every 15 minutes B is at the starting point.

Now A and B will be together again at the starting point at the end of the time during which each can make an exact number of rounds. Hence the required time is the LCM of 8 and 15 minutes

i.e., 120 minutes or 2 hours.

(b) A and B will be together again for the first time when A has gained one complete round over B

Now A gains (150 − 80) or 70 meters on B in 1 minute

$\therefore$ A will gain 1200 meters in $\frac{1200}{80}$ or 17 $\frac{1}{4}$ minutes.

$\therefore$ A and B will be together in 17 $\frac{1}{4}$ minutes.

Example Three man A, B and C walk around a circle 1760 meters in circumference at the rate of 160, 120 and 105 m/minute respectively. If they all start together and walk in the same direction, when will they first be together again?

Solution

A, the quickest man gains one complete round on C, the slowest man in $\frac{1760}{160-105}$ or 32 min.

A gains one complete round on B the next slowest man in $\frac{1760}{160-105}$ or 44 minutes.

Thus A and C are together after every 32 minutes and A and B are together after every 44 minutes. Hence A, B and C will be together in the time which is LCM of 32 and 44, i.e., 352 minutes or 5 hours 52 minutes.

Practice Exercise - 1

Number of Questions: 20 **Suggested time: 20 minutes**

1. Rabbit is $2\frac{5}{6}$ times as fast as Tortoise. If Rabbit gives Tortoise a start of 80 m, how long should the race course be so that both of them reach at the same time?
(a) 170 metre
(b) 150 metre
(c) 140 metre
(d) 160 metre

2. Mohan can run 224 metre in 28 seconds and Makhan in 32 seconds. By what distance Mohan beat Makhan?
(a) 36 metre
(b) 24 metre
(c) 32 metre
(d) 28 metre

3. In a 100 m race, Naresh can give Suresh 10 m and Ramesh 28 m. In the same race Suresh can give Ramesh:
(a) 20 m
(b) 15 m
(c) 10 m
(d) 25 m

4. At a game of billiards, Manav can give Manuj 15 points in 60 and Manav can give Harkaran to 20 points in 60. How many points can Manuj give Harkaran in a game of 90?
(a) 22 points
(b) 20 points
(c) 12 points
(d) 10 points

5. In a 500 m race, the ratio of the speeds of two contestants Divya and Rancy is 3 : 4. Divya has a start of 140 m. Then, Divya wins by:

(a) 60 m
(b) 20 m
(c) 40 m
(d) 10 m

6. In a race of 200 m, A can beat B by 31 m and C by 18 m. In a race of 350 m, C will beat B by:
(a) 20.25 m
(b) 21.5 m
(c) 22.75 m
(d) 25 m

7. Hans can run 22.5 m while Pruthi runs 25 m. In a kilometre race Pruthi beats Hans by:
(a) 102 m
(b) 100 m
(c) 75 m
(d) 112.5 m

8. In a 100 m race, A can beat B by 25 m and B can beat C by 4 m. In the same race, A can beat C by:
(a) 21 m
(b) 28 m
(c) 26 m
(d) 29 m

9. In a 300 m race Pinky beats Geeta by 22.5 m or 6 seconds. Geeta's time over the course is:
(a) 86 sec
(b) 80 sec
(c) 76 sec
(d) None of these

10. A and B take part in 100 m race. A runs at 5 kmph. A gives B a start of 8 m and still beats him by 8 seconds. The speed of B is:
(a) 4.4 kmph
(b) 4.25 kmph
(c) 4.14 kmph
(d) 5.15 kmph

11. A runs $1\ ^{2/3}$ times as fast as B. If A gives B a start of 80 m, how far must the winning post be so that A and B might reach it at the same time?
(a) 200 m
(b) 270 m
(c) 300 m
(d) 160 m

12. In a game of 100 points, Atul can give Ajay 20 points and Sarita 28 points. Then, Ajay can give Sarita:
(a) 40 points
(b) 10 points
(c) 14 points
(d) 8 points

13. In 100 m race, Roshan covers the distance in 36 seconds and Kuldip in 45 seconds. In this race Roshan beats Kuldip by:
(a) 9 m
(b) 25 m
(c) 22.5 m
(d) 20 m

14. In a 100 m race, A beats B by 10 m and C by 13 m. In a race of 180 m, B will beat C by:
(a) 5.4 m
(b) 5 m
(c) 4.5 m
(d) 6 m

15. In a 200 metres race Jaspal beats Bittu by 35 m or 7 seconds. Jaspal's time over the course is:
(a) 33 sec
(b) 40 sec
(c) 47 sec
(d) None of these

16. A, B and C are the three contestants in one km race. If A can give B a start of 40 metres and A can give C a start of 64 metres. How many metres start can B give C?
(a) None of these

(b) 20 m
(c) 25 m
(d) 35 m

17. In a game of 90 points P can give Q 15 points and R 30 points. How many points can Q give R in a game of 100 points?
(a) 140
(b) 20
(c) 300
(d) 50

18. In a 100 metres race. A runs at a speed of 2 metres per seconds. If A gives B a start of 4 metres and still beats him by 10 seconds, find the speed of B.
(a) 1.6 m/sec.
(b) 4 m/sec.
(c) 2.6 m/sec.
(d) 1 m/sec.

19. P runs 1 km in 3 minutes and Q in 4 minutes 10 seconds. How many metres start can P give Q in 1 km race, so that the race may end in a dead heat?
(a) 210 metre
(b) 180 metre
(c) 220 metre
(d) 280 metre

20. A runs $1\frac{5}{6}$ times As fast as B. If A gives B a start of 90 m and they reach the goal at the same time. The goal is at a distance of:
(a) 330 m
(b) 440 m
(c) 120 m
(d) 280 m

Practice Exercise - 1 Answers with Hints - need based

1 (c)	2 (d)	3 (a)	4 (d)	5 (b)
6 (d)	7 (b)	8 (b)	9 (b)	10 (c)

11 (a)	12 (b)	13 (d)	14 (d)	15 (a)
16 (c)	17 (b)	18 (a)	19 (d)	20 (a)

1. Speed of Rabbit : Speed of Tortoise = 7/3 : 1 = 7 : 3
It means, in a race of 7 m, Rabbit gains (7 − 3) = 4 metre
If Rabit needs to gain 80 metre, race should be of 74 × 80 = 140 metre

3. "A can give B 10 m" ⇒ In a 100 m race, while A starts from the starting point, B starts 10 meters ahead from the same starting point at the same time.
"A can give C 28 m" ⇒ In a 100 m race, while A starts from the starting point, C starts 28 meters ahead from the same starting point at the same time.
i.e., in a 100 − 10 = 90 m race, B can give C 28 − 10 = 18 m
⇒ In a 100 m race, B can give C 1890 · 100 = 20 m

6. It means, when A covers 200 m, B covers (200 − 31) = 169 m; and C (200 − 18) = 182 m
⇒ When C covers 182 m, B covers 169 m
⇒ When C covers 350 m, B covers 169182 × 350 = 325 m
Hence, C beats B by 350 − 325 = 25 metre

8. When A covers 100 m, B covers (100 − 25) = 75 m
When B covers 100 m, C covers (100 − 4) = 96 m
⇒ When B covers 75 metre, C covers 96100 × 75 = 72 m
i.e., When A covers 100 m, B covers 75 m and C covers 72 m
⇒ In a 100 m race, A beat C by (100 − 72) = 28 m

14. While A runs 100 m, B runs (100 − 10) = 90 m and C runs (100 − 13) = 87 m
i.e., when B runs 90 m, C runs 87 m
⇒ when B runs 180 m, C runs 8790 × 180 = 174 m
Hence, in a 180 m race, B will beat C by (180 − 174) = 6 m

17. While A scores 90 points, B scores (90 − 15) = 75 and C scores (90 − 30) = 60 points
i.e., when B scores 75 points, C scores 60 points
⇒ When B scores 100 points, C scores 6075 × 100 = 80 points
i.e., in a game of 100 points, B can give C (100 − 80) = 20 points

Practice Exercise - 2

Number of Questions: 15 **Suggested time: 10 minutes**

1. Two boys A and B run at $4\frac{5}{6}$ and 8 km an hour respectively. A having 150 m start and the course being 1 km, B wins by a distance of:
(a) 325 m
(b) 60 m
(c) 120 m
(d) 275 m

2. In a 100 metres race, A can beat B by 10 metres and B can beat C by 5 metres. In the same race, A can beat C by:
(a) 14.5 metre
(b) 14 metre
(c) 15.5 metre
(d) 15 metre

3. X, Y and Z are the three contestants in one km race. If X can give Y a start of 52 metres and X can also give Z a start of 83 metres, how many metres start Y can give Z?
(a) 33.3 m
(b) 33 m
(c) 32 m
(d) 32.7 m

4. In a race of 600 m. A can beat B by 60 m and in a race of 500 m. B can beat C by 50 m. By how many metres will A beat C in a race of 400 m?
(a) 76 metre
(b) 74 metre
(c) 72 metre
(d) 78 metre

5. A can run 220 metres in 41 seconds and B in 44 seconds. By how many seconds will B win if he has 30 metres start?
(a) 8 sec

(b) 4 sec
(c) 2.5 sc
(d) 3 sec

6. In one km race Dilawar beats Hashim by 4 seconds or 40 metres. How long does Hashim take to run the kilo meter?
(a) 112 sec
(b) 110 sec
(c) 101 sec
(d) 100 sec

7. In a game, A can give B 20 points, A can give C 32 points and B can give C 15 points. How many points make the game?
(a) 120 points
(b) 90 points
(c) 80 points
(d) 100 points

8. In a game A can give B 20 points in 60 and C 18 points in 90. How many points can C give B in a game of 120?
(a) 20 points
(b) 22 points
(c) 18 points
(d) 40 points

9. In a km race A can beat B by 100 m and B can beat C by 60 m. In the same race A can beat C by:
(a) 144 m
(b) 164 m
(c) 144 m
(d) 154 m

10. In a flat race, A beats B by 15 metres and C by 29 metres. When B and C run over the course together, B wins by 15 metres. Find the length of the course?
(a) 225 m
(b) 125 m
(c) 220 m
(d) 256 m

11. In 100 m race, A covers the distance in 36 seconds and B in 45 seconds. In this race A beats B by:

(a) 20 m

(b) 25 m

(c) 22.5 m

(d) 9 m 12.

12. In a 100 m race, A beats B by 10 m and C by 13 m. In a race of 180 m, B will beat C by:

(a) 5.4 m

(b) 4.5 m

(c) 5 m

(d) 6 m

13. In a 1 km race, Parteek beats Ishan by 28 meters in 7sec. Find Parteek's time over the course?

(a) 5 min 4 sec

(b) 4 min 3 sec

(c) 2 min 3 sec

(d) 3 min 4 sec

14. In a race of One kilometer, A can beat B by 155 m and C by 90 m. In a race of 350 m, C will beat B by how many meter?

(a) 21.69

(b) 17.25

(c) 13.5

(d) 25.15.

15. In a race of 100 meters, A can give B 20 meters and C 28 meters. Then, B can give C is:

(a) 10 m

(b) 12 m

(c) 17 m

(d) 35 m

Practice Exercise - 2 Answers with Hints - need based

1 (a)	2 (a)	3 (d)	4 (a)	5 (d)

6 (d)	7 (d)	8 (a)	9 (d)	10 (a)
11 (a)	12 (d)	13 (b)	14 (d)	15 (a)

1. A has a start of 150 m. So A has to run 1000 - 150 = 850 metre while B runs 1000 metre

Speed of A = 21/5 km/hr =21/5 × 5/18 = 21/18 = 76 m/s

Speed of B = 8 km/hour = 8 × 5/18 = 20/9 m/s

Time taken by B to travel 1000 metre = distance x

Speed of B = 1000/(20/9) = 450 second.

Distance travelled by A by this time = time × speed of A = 450 × 7/6 = 525 metre

Hence, A wins by 850 - 525 = 325 metre

CHAIN RULE – UNITARY METHODS 11

1. Direct Proportion

Two quantities are said to be directly proportional, if on the increase or decrease of one, the other increases or decreases the same extent.

Example ***(a) Sales of the goods is directly proportional to the number of goods sold. (More goods sold, More sale).***

(b) Amount of work done i.e., output is directly proportional to the number of work force who were placed to do the work. (More work force, More Work output).

2. Indirect Proportion (inverse proportion)

Two quantities are said to be indirectly proportional (inversely proportional) if on the increase of one, the other decreases to the same extent and vice-versa.

Example ***(a) Number of days needed to complete a work is indirectly proportional (inversely proportional) to the number of work force employed to do the work (More work force, Lesser Days needed).***

(b) The time taken to travel a distance is indirectly proportional (inversely proportional) to the speed of travel (More Speed, Less Time).

Practice Exercise - 1

Number of Questions: 25 **Suggested time: 15 minutes**

1. If the cost of x meters of wire is 'd' rupees, that find the cost (in Rupees) of 'y' meters of wire at the same rate.
(a) x/dy
(b) x − d
(c) ydx
(d) y + d

2. In a dairy farm, 40 cows eat 40 bags of husk in 40 days. In how many days one cow will eat one bag of husk?
(a) 15
(b) 40
(c) 38
(d) 32

3. If 7 spiders make 7 webs in 7 days, then how many days are needed for 1 spider to make 1 web?
(a) 10
(b) 7
(c) 3
(d) 4

4. 3 pumps, working 8 hours a day, can empty a tank in 2 days. How many hours a day should 4 pumps work in order to empty the tank in 1 day?
(a) 10
(b) 12
(c) 8
(d) 15

5. 39 persons can repair a road in 12 days, working 5 hours a day. In how many days will 30 persons, working 6 hours a day, complete the work?
(a) 9
(b) 14
(c) 12

(d) 13

6. A certain industrial loom weaves 0.128 meters of cloth every second. Approximately how many seconds will it take for the loom to weave 25 meters of cloth?

(a) 220

(b) 200

(c) 170

(d) 195

7. A contract is to be completed in 56 days if 104 persons work, each working at 8 hours a day. After 30 days, 25 of the work is completed. How many additional persons should be deployed so that the work will be completed in the scheduled time, each person now working 9 hours a day.

(a) 60

(b) 58

(c) 12

(d) 56

8. 'x' men working 'x' hours per day can do 'x' units of a work in 'x' days. How much work can be completed by 'y' men working 'y' hours per day in 'y' days?

(a) x^2y^2 units

(b) y^3x^2 units

(c) x^2y

(d) y^2x^2 *units*

9. 21 goats eat as much as 15 cows. How many goats eat as much as 35 cows?

(a) 38

(b) 39

(c) 37

(d) 41

10. A flagstaff 17.5 m high casts a shadow of length 40.25 m. What will be the height of a building, which casts a shadow of length 28.75 m under similar conditions?

(a) 12.5 m
(b) 11.5 m
(c) 14.8
(d) 12.9

11. 39 persons can repair a road in 12 days, working 5 hours a day. In how many days will 30 persons, working 6 hours a day, complete the work?
(a) 10
(b) 13
(c) 18
(d) 136

12. Four mat-weavers can weave 4 mats in 4 days. At the same rate, how many mats would be woven by 8 mat-weavers in 8 days?
(a) 16
(b) 12
(c) 18
(d) 14

13. If a quarter Kg of potato costs 60 paise, How many paise will 200 gm cost?
(a) 48
(b) 45
(c) 59
(d) 72

14. In a dairy farm, 40 cows eat 40 bags of husk in 40 days. In how many days one cow will eat one bag of husk?
(a) 1
(b) 32
(c) 40
(d) 76

15. If 18 binders bind 900 books in 10 days, How many binders will be required to bind 660 books in 12 days?
(a) 21
(b) 14
(c) 18
(d) 11

16. An industrial loom weaves 0.128 metres of cloth every second. Approximately, how many seconds will it take for the loom to weave 25 metre of cloth?

(a) 184

(b) 195

(c) 190

(d) 197

17. Running at the same constant rate, 6 identical machines can produce a total of 270 bottles per minute. At this rate, how many bottles could 10 such machines produce in 4 minutes?

(a) 648

(b) 1800

(c) 1700

(d) 1080

18. If 8 men can reap 80 hectares in 24 days, then how many hectares can 36 men reap in 30 days?

(a) 350

(b) 460

(c) 425

(d) 450

19. A fort had provision of food for 150 men for 45 days. After 10 days, 25 men left the fort. The number of days for which the remaining food will last, is:

(a) 39

(b) 37

(c) 42

(d) 56

20. If the cost of x metres of wire is d rupees, then what is the cost of y metres of wire at the same rate?

(a) xy/d

(b) xd

(c) yd

(d) yd/x

21. In a camp, there is a meal for 120 men or 200 children. If 150 children have taken the meal, how many men will be catered to with remaining meal?
(a) 56
(b) 45
(c) 30
(d) 27

22. If the price of 23 toys is Rs. 276, then what will the price (Rs.) of 12 toys?
(a) 144
(b) 158
(c) 159
(d) 166

23. Ten men, working 6 hours a day can complete a work in 18 days. How many hours a day must 15 men work to complete the work in 12 days?
(a) 4
(b) 3
(c) 6
(d) 8

24. If the Price of 6 toys is Rs. 264.37, What will be the approximate price (Rs) of 5 toys?
(a) 230
(b) 120
(c) 220
(d) 220

25. If 16 men working 7 hours day can plough a fieid in 41 days, inhow many days will 14 men working 12 hours a clay plough the same field?
(a) 56
(b) 32
(c) 38
(d) 30

Practice Exercise - 1 Answers with Hints - need based

1 (c)	2 (b)	3 (b)	4 (b)	5 (d)

6 (d)	7 (d)	8 (b)	9 (b)	10 (a)
11 (b)	12 (a)	13 (a)	14 (c)	15 (d)
16 (b)	17 (b)	18 (d)	19 (c)	20 (d)
21 (c)	22 (a)	23 (c)	24 (d)	25 (b)

11. Let the required number of days be x.

Less persons, More days (Indirect Proportion)

More working hours per day, Less days (Indirect Proportion)

$$\left.\begin{array}{ll} \textit{persons} & 30:39 \\ \textit{working hours/day} & 6:5 \end{array}\right\} :: 12:x$$

$\therefore 30 \times 6 \times x = 39 \times 5 \times 12$

$\Rightarrow x = \dfrac{39\times5\times12}{30\times6} \Rightarrow x = 13$

12. Let the required number of bottles be x.

More weavers, More mats (Direct Proportion)

More days, More mats (Direct Proportion)

$$\left.\begin{array}{ll} \textit{weavers} & 4:8 \\ \textit{days} & 4:8 \end{array}\right\} :: 4:x$$

$\therefore 4 \times 4 \times x = 8 \times 8 \times 4$

$\Rightarrow x = \dfrac{8\times8\times4}{4\times4}$

$\Rightarrow x = 16.$

13. Let the Required cost be x paise. Less Weight, Less cost (Direct Proportion)

$\therefore 250 : 200 :: 60\ x \Leftrightarrow 250 \times x = (200 \times 60) \Leftrightarrow x = \dfrac{(200\times60)}{250} \Leftrightarrow x = 48$

14. ***Less cows, More days (Indirect Proportion)***

Less bags, Less days (Direct Proportion)

$$\left.\begin{matrix} Cows & 1:40 \\ Bags & 40:1 \end{matrix}\right\} :: \ 40:x$$

$\therefore 1 \times 40 \times x = 40 \times 1 \times 40$

$\Rightarrow x = 40.$

15. Let the required no of binders be x.

Less books, Less binders (Direct Proportion)

More days, Less binders (Indirect proportion)

$$\left.\begin{matrix} Books & 900:600 \\ Days & 12:10 \end{matrix}\right\} ::18:x$$

$\therefore (900 \times 12 \times x) = (600 \times 10 \times 18) \Leftrightarrow x = \frac{x}{16} - \frac{x}{17} = 25 = 11$

16. Let the time required by x seconds.

Then, More cloth means More time (Direct Proportion)

So, $0.128 : 1 :: 25 : x \Rightarrow x = \frac{25 \times 1}{0.128}$

$\Rightarrow x = 195.31$

So time will be approx 195 seconds

17. Let the required number of bottles be x.

More machines, More bottles (Direct Proportion)

More minutes, More bottles (Direct Proportion)

$$\left.\begin{matrix} machines & 6:10 \\ time(in\ minutes) & 1:4 \end{matrix}\right\} ::270:x$$

$\therefore 6 \times 1 \times x = 10 \times 4 \times 270$

$\Rightarrow x = \frac{10 \times 4 \times 270}{6} \Rightarrow x = 1800$

18. Let the required no of hectares be x. Then

More men, More hectares (Direct proportion)

More days, More hectares (Direct proportion)

$$\left.\begin{array}{ll} \textit{Men} & 8:36 \\ \textit{Days} & 24:30 \end{array}\right\}:80:x$$

$\therefore$ 8 × 24 × x = 36 × 30 × 80 ⇔ $x = \frac{36\times30\times80}{8\times24}$ ⇔ x = 450

18. After 10 days : 150 men had food for 35 days.

Suppose 125 men had food for x days.

Now, *Less men, More days (Indirect Proportion)*

$\therefore$125 : 150 **::** 35 : x ⇔ 125 × x = 150 × 35

$\Rightarrow x = \frac{150\times35}{125} \Rightarrow x = 42$

19. Cost of x metres = Rs. d.

Cost of 1 metre = Rs.(d/x)

Cost of y metres = Rs.[(d/x).y] = Rs.(yd/x)

20. There is a meal for 200 children. 150 children have taken the meal.

Remaining meal is to be catered to 50 children.

Now, 200 children = 120 men.

50 children = $\left(\frac{120}{200}\right)\times50$ = 30 men

Practice Exercise - 2

Number of Questions: 25 **Suggested time: 12 minutes**

1. Running at the same constant rate, 6 identical machines can produce a total of 270 bottles per minute. At this rate, how many bottles could 10 such machines produce in 4 minutes?

(a) 1800
(b) 1600
(c) 2500
(d) 1700

2. A person works on a project and completes 5/8 of the job in 10 days. At this rate, how many more days will he it take to finish the job?

(a) 12
(b) 6
(c) 10
(d) 9

3. A fort had provision of food for 150 men for 45 days. After 10 days, 25 men left the fort. Find out the number of days for which the remaining food will last.

(a) 45
(b) 42
(c) 49
(d) 46

4. If the price of 357 apples is Rs. 1517.25, what will be the approximate price of 49 dozens of such apples?

(a) Rs. 2500
(b) Rs. 2800
(c) Rs. 2900
(d) Rs. 2400

5. On a scale of a map 0.6 cm represents 6.6 km. If the distance between two points on the map is 80.5 cm, what is the actual distance between these points?

(a) 885.5 km
(b) 870 km
(c) 872.5 km
(d) 895 km

6. A rope can make 70 rounds of the circumference of a cylinder whose radius of the base is 14 cm. how many times can it go round a cylinder having radius 20 cm?
(a) 49 rounds
(b) 54 rounds
(c) 56 rounds
(d) 77 rounds

7. Eight persons can build a wall 140 m long in 42 days. In how many days can 30 persons complete a similar wall 100 m long?
(a) 11
(b) 13
(c) 8
(d) 19

8. A certain number of persons can finish a piece of work in 100 days. If there were 10 persons less, it would take 10 more days finish the work. How many persons were there originally?
(a) 190
(b) 180
(c) 110
(d) 196

9. Nine examiners can examine a certain number of answer books in 12 days by working 5 hours a day. How many hours in a day should 4 examiners work to examine twice the number of answer books in 30 days?
(a) 9
(b) 11
(c) 13
(d) 17

10. Nine engines consume 24 metric tonnes of coal, when each is working 8 hours day. How much coal is required for 8 engines, each running 13 hours a day, if 3 engines of former type consume as much as 4 engines of latter type?

(a) 20 metric tonnes
(b) 39 metric tonnes
(c) 23 metric tonnes
(d) 26 metric tonnes

11. A garrison had provisions for a certain number of days. After 10 days, 15 of the men desert and it is found that the provisions will now last just as long as before. How long was that?

(a) 50 days
(b) 70 days
(c) 45 days
(d) 48 days

12. A garrison of 500 persons had provisions for 27 days. After 3 days a reinforcement of 300 persons arrived. For how many more days will the remaining food last now?

(a) 27 days
(b) 28 days
(c) 26 days
(d) 15 days

13. A hostel had provisions for 250 men for 40 days. If 50 men left the hostel, how long will the food last at the same rate?

(a) 58 days
(b) 50 days
(c) 45 days
(d) 67 days

14. In a camp, food was sufficient for 2000 people for 54 days. After 15 days, more people came and the food last only for 20 more days. How many people came?

(a) 1900
(b) 2400
(c) 1940
(d) 2220

15. If 40 men can make 30 boxes in 20 days, How many more men are needed to make 60 boxes in 25 days?
(a) 20
(b) 24
(c) 27
(d) 28

16. 2 men and 7 boys can do a piece of work in 14 days; 3 men and 8 boys can do the same in 11 days. Then, 8 men and 6 boys can do three times the amount of this work in.
(a) 18
(b) 21
(c) 34
(d) 39

17. A certain number of men can finish a piece of work in 100 days. If there were 10 men less, it would take 10 days more for the work to be finished. How many men were there originally?
(a) 75
(b) 92
(c) 120
(d) 110

18. Three pumps working 8 hours a day, can empty a tank in 2 days. How many hours a day must 4 pumps work to empty the tank in 1 day?
(a) 19
(b) 14
(c) 11
(d) 12

19. If 20 men can build a wall 56 meters long in 6 days, what length of a similar wall can be built by 35 men in 3 days?
(a) 40
(b) 43
(c) 48
(d) 49

20. If 7 spiders make 7 webs in 7 days, then 1 spider will make 1 web in how many days?

(a) 1
(b) 8
(c) 7
(d) None

21. A wheel that has 6 cogs is meshed with a larger wheel of 14 cogs. When the smaller wheel has made 21 revolutions, then the number of revolutions mad by the larger wheel is:
(a) 4
(b) 9
(c) 14
(d) 7

22. A contractor undertakes to do a piece of work in 40 days. He engages 100 men at the begining and 100 more after 35 days and completes the work in stipulated time. If he had not engaged the additional men, how many days behind schedule would it be finished?
(a) 7
(b) 18
(c) 12
(d) None

23. 36 men can complete a piece of work in 18 days. In how many days will 27 men complete the same work?
(a) 24
(b) 23
(c) 34
(d) 39

24. Some persons can do a piece of work in 12 days. Two times the number of such persons will do half of that work in:
(a) 16
(b) 12
(c) 8
(d) 3

25. If 36 men can do a piece of work in 25 hours, in how many hours will 15 men do it?
(a) 45

(b) 55
(c) 60
(d) 72

Practice Exercise - 2 Answers with Hints - need based

1 (a)	2 (b)	3 (b)	4 (a)	5 (a)
6 (a)	7 (c)	8 (c)	9 (a)	10 (d)
11 (a)	12 (d)	13 (b)	14 (a)	15 (b)
16 (b)	17 (d)	18 (d)	19 (d)	20 (c)
21 (b)	22 (d)	23 (a)	24 (d)	25 (c)

19. $\therefore (20 \times 6 \times x) = (35 \times 3 \times 56) \Leftrightarrow x = \frac{35 \times 3 \times 56}{120} = 49$

20. *Less spiders, More days (Indirect Proportion)*

Less webs, Less days (Direct Proportion)

$\frac{\sqrt{9.5 \times 0.0085 \times 18.9}}{\sqrt{0.0017 \times 1.9 \times 2.1}} =$

$\therefore 1 \times 7 \times x = 7 \times 1 \times 7$

$\Rightarrow x = 7$

21. Then, *More cogs, Less revolutions (Indirect Proportion)*

$\therefore$ 14 : 6 **::** 21 : $x \Rightarrow 14 \times x = 6 \times 21$

$x = \frac{25 \times 1}{0.128} \Rightarrow x = 9$

22. [(100 × 35) + (200 × 5)] men can finish the work in 1 day

$\therefore$ 4500 men can finish the work in 1 day. 100 men can finish it in $\frac{4500}{100} =$ 45 days.

This is 5 days behind Schedule

23. Let the number of days be x then, 27 : 36 :: 18 : x

24. Let x men can do the in 12 days and the required number of days be z

More men, Less days [Indirect Proportion]

Less work, Less days [Direct Proportion]

$$\left.\begin{array}{ll} \textit{Men} & 2x : x \\ \textit{Work} & 1 : \dfrac{1}{2} \end{array}\right\} :: 12 : z$$

$$\therefore (2x \times 1 \times z) = \left(x \times \frac{1}{2} \times 12 \right) \Rightarrow 2xz = 6x \Rightarrow z = 3$$

25. Let the required no of hours be x. Then

Less men, More hours (Indirct Proportion)

$\therefore$ 15 : 36 :: 25 : x $\Leftrightarrow$ $(15 \times x) = (36 \times 25)$ $\Leftrightarrow$ $x = \dfrac{36 \times 25}{15} = 60$

Hence, 15 men can do it in 60 hours.

CLOCKS

12

The face of a clock or a watch is a circle which is divided into 60 minutes spaces.

- ***The minute hand passes over 60 minute spaces whilst the hour hand goes over 5 minute spaces.***
- ***In 60 minutes the minute hand gains 55 minutes on the hour hand.***
- ***Hands are straight 22 times in a day.***
- ***Hands coincide 22 times in a day.***
- ***Hands are at right angle twice per hour.***

Example At what time between 4 and 5 will the hands of a watch.

(a) Point in opposite directions?
(b) Coincide
(c) Be at right angle

Solution

(a) They will be opposite each other *(i.e., angle between them is 180 degrees)* when there is a space of 30 minutes between them. This will happen when minute hand gains (30 20) minutes, i.e.,

$\frac{52 \times 60}{55} = 54\frac{6}{11}$ mts. Or $54\frac{6}{11}$ minutes past 4.

(b) At 4 O'clock the hands are 20 minutes apart. Clearly the minute hand must gain 20 minutes before the hands can be coincident (*I,e angle between them ZERO)*. But the minute hand gains 55 minutes in 60 minutes. Hence to find out in what time minute hand will gain 20 minutes, we have the following proportion:

55 : 20 60 : x

$\therefore x \frac{20 \times 60}{55} = \frac{240}{11} = 21\frac{9}{11}$ minutes

$\therefore$Hands will be coincident at $21\frac{1}{11}$ minutes past 4.

(c) At 4 O'clock the hands are 20 minutes apart. They will be at right angles when there is a space of 15 minutes between them. This will happen twice (i) when the minute hand has gained (20 – 15) or 5 minutes; (ii) when the minute hand has gained (20 + 15) or 35 minutes.

Minute hand gains 5 minutes in $\frac{x}{16} - \frac{x}{17} = 25$ minutes.

i.e., $5\frac{1}{11}$ minutes past 4.

Or minute hand gains 35 minutes in $\frac{35 \times 60}{55} = 38\frac{2}{11}$ minutes

i.e., $38\frac{1}{11}$ minutes past 4.

Example Two clocks begin to strike 12 together. One strikes its stroke in 33 seconds and the other in 22 seconds. What is the interval between the 6th stroke of the first and the 8th stroke of the second?

Solution

Since the first clock strikes (12 – 1) or 11 strokes in 33 seconds the interval between its two successive strokes is $\frac{3}{60}$ or 3 seconds.

Similarly, the interval between two successive strokes of the other clock is $\frac{22}{11}$ or 2 seconds.

Now the 6th stroke of the first clock will come after 5 × 3 or 15 seconds and the 8th stroke of the second clock will come after 7 × 2 or 14 seconds.

Example My watch which gains uniformly is 2 min. slow at noon on Sunday, and is 4 minutes 48 seconds fast at 2 p.m. on the following Sunday. When was it correct?

Solution

From Sunday noon to the following Sunday at 2 p.m. 7 days 2 hrs 170 hrs.

The watch gains $2 + 4 + \frac{3}{60} = 6 + \frac{3}{60}$ minutes of $6\ \frac{1}{4}$ minutes in 170 hrs.

$\therefore$The watch gains 2 minutes in $\frac{2}{6\frac{4}{5}} \times 170$ or 50 hrs.

Now 50 hrs = 2 days 2 hrs.

2 days 2 hrs from Sunday noon = 2 p.m. on Tuesday.

Practice Exercise - 1

1. Find the time between 3 and 4 O'clock when the angle between the hands of a watch is one-third of a right angle.

(a) $3 - 10\frac{17}{2}$

(b) $3 - 21\frac{1}{11}$

(c) both

(d) None

2. A watch which is set correctly at noon indicates 10 minutes after 9 the same evening when the true time is 10. What is the true time when the watch indicates 11 on the same evening?

(a) 11.30 p.m.

(b) 11.52 p.m.

(c) 11.46 p.m.

(d) None

3. A watch which usually gains 5 seconds in every 3 minutes is set right at 6 a.m. What is the true time in the afternoon of the same day when the watch indicated a quarter past 3 O'clock.

(a) 2.45

(b) 3.00

(c) 3.20

(d) 3.15

4. Does a watch gain or lose per day if its hands coincide every 66 minutes?

(a) Gain

(b) Loss

(c) Can't say

(d) None

5. A man who went out between five and six and returned between six and seven, found that the hands of the watch had exactly changed places. When did he go out?

(a) 5.29
(b) 5.31
(c) 5.32
(d) 5.38

Practice Exercise - 1 Answers

1. $10\frac{17}{2}$ or $21\frac{1}{11}$ min. past 3.

2. mid night

3. 3 p.m.

4. Loses 11 min. $54\frac{3}{60}$ sec.

5. $32\frac{17}{2}$ min. past 5.

CALENDAR

13

A Solar year consists of 365 days 5 hours 48 minutes and 48 seconds. It was due to this reason that the calendar system prevailing now-a-days, an year is taken for $365\frac{1}{2}$ days. In order to get rid of this $\frac{1}{2}$ day an extra day is added in every 4^{th} year which is called a '*leap year*'. This calendar is known as Julian Calendar, arranged 47 BC by Julius Caesar.

In India, *Vikrami* and a number of other calendars were in use till 1952. On the basis of report of a committee formed, Govt of India adopted the National Calendar based on *Saka* with *Chaitra* as the first month. The days of this Vikrami calendar have permanent correspondence with those of Gregorian calendar – Chaitra beginning on March 22 and March 21 in ordinary and leap years respectively.

Remember:

(a) an ordinary year contains 365 days i.e., 52 weeks and 1 odd day

(b) a leap year contains 366 days i.e., 52 weeks and 2 odd days

(c) a leap year must be divisible by 4 but if complete century then by 400

(d) 100 years contain 5 odd days

(e) 01.01.0001 was Monday - hence count Sunday = 0, Monday = 1… odd days

(f) 400 years contain NIL odd days.

(g) An ordinary year leaves one odd day while a leap year leaves 2 odd days

(1) **Odd Days**:

Number of days more than the complete weeks are called odd days in a given period.

(2) **Leap Year**:

A leap year has 366 days.
In a leap year, the month of February has 29 days.
Every year divisible by 4 is a leap year, if it is not a century.
Examples: 1952, 2008, 1680 etc., are leap years.
1991, 2003 etc., are not leap years
Every 4^{th} century is a leap year and no other century is a leap year.
Examples: 400, 800, 1200 etc., are leap years.
100, 200, 1900 etc., are not leap years

(3) **Ordinary Year**:

The year which is not a leap year is an ordinary year.
An ordinary year has 365 days

(4) **Counting odd days and calculating day of any particular date**

1 ordinary year ≡ 365 days ≡ (52 weeks + 1 day)
Hence number of odd days in 1 ordinary year = 1.
1 leap year ≡ 366 days ≡ (52 weeks + 2 days)
Hence number of odd days in 1 leap year = 2.
years ≡ (76 ordinary years + 24 leap years)
≡ (76 × 1 + 24 × 2) odd days ≡ 124 odd days.
≡ (17 weeks + 5 days) ≡ 5 odd days.
Hence number of odd days in 100 years = 5.
Number of odd days in 200 years = (5 × 2) = 10 ≡ 3 odd days
Number of odd days in 300 years = (5 × 3) = 15 ≡ 1 odd days
Number of odd days in 400 years = (5 × 4 + 1) = 21 ≡ 0 odd days
Similarly, the number of odd days in all 4^{th} centuries (400, 800, 1200 etc.,) = 0

Mapping of the number of odd day to the day of the week

No. of odd days	**Week day**

0	Sunday
1	Monday
2	Tuesday
3	Wednesday
4	Thursday
5	Friday
6	Saturday

(5) **Additional Notes**

- ***Last day of a century cannot be Tuesday or Thursday or Saturday.***
- ***For the calendars of two different years to be the same, the following conditions must be satisfied:***

  (a) Both years must be of the same type. i.e., both years must be ordinary years or both years must be leap years;

  (b) 1^{st} Day of the year i.e., 1^{st} January of both the years must be the same day of the week.

Example What day was 15^{th} August 1947

Solution

1947 means 1946 complete number of years
1946 = 1600 + 300 + 46 + leap years
therefore # of odd days = 0 + 15 + 46 + 11 = 72 days
# of odd days in January = 31
February = 28
March = 31
April = 30
May = 31
June = 30
July = 31
August = 15

Total number of odd days up to 15.08.1947 = 299 days = 42 weeks 5 days

i.e., 5th day is Friday; hence it was Friday on 15th August 1947

Example How many times does the 29th day of the month occur in 400 consecutive years?

Solution:

In 400 years February occurs 97 times having 29 days while remaining 11 months also have 29th day of the month hence $400 \times 11 + 97 = 4497$ time shall the day 29th will appear in a leap year.

Example Calendar of 1993 can be used for which of the following years

1997 1998 2001 2010

1997 there is a run of 4 years i.e., 4 odd days + one (leap year i.e., 1996) = 5 odd days which does not make a complete week

1998 there is a run of 5 years i.e., 5 odd days + 1 leap year = 6 days – not a complete week.

2001 run of 8 years i.e., 8 days + leap years of 1996, 2000 i.e., total run 10 days does not make a week

2010 a run of 17 years + leap years are 1996, 2000, 2004, 2008 i.e., a total run of 21 days which makes complete weeks- hence can be used

Practice Exercise - 1

1. Mrs Sunita Bhatia was born on Monday 23rd March 1912. On which date she shall be 8 years 2 months and 5 days of age?

2. Find day of the week on

(a) 2nd January 1921 (b) 26th January 1950 (c) 01st June 1955

(d) 24th August 1985 (e) 23rd March 1923 (f) 10th May 1857

3. If the first day of the year 1910 was Monday, what day of the week shall be the 1st day of 20th Century?

4. If 3rd November 1984 was Monday. What day shall it be on this day in 1987?

5. Calendar for the year 1983 can be used for earliest of which years?

6. How many are not the leap years of 1956,1972,1984,1674,1702,1700,684,600

7. Calendar of 1912 can be used for 1928 1920 1917 1914

8. If 7th May is 3 days after Friday. 8th June will be 2 days before

9. The day of the week on 18th July 1937 was

10. If Tuesday falls on the 4th of the Month, what day will dawn three days after the 24th

11. If Saturday falls four days after today which is 5th January, on what day did the 1st December of the previous year fall?

12. How many time the date 28 shall come in a period of 4 years?

Remember :

- ***As the calendar requires actual working it is of no use hitting the Choices. Therefore, practice the questions and their methods to solve***

  For solution:

- ***Always observe how many complete years have gone***
- ***The total number of odd days till***

- ***Total number of leap years***
- ***Odd days for each month***

Practice Exercise - 1 Answers

1. 28^{th} May 1920 – Friday

2. a) Thursday b) Thursday c) Wednesday d) Saturday e) Sunday f) Sunday

3. Monday

4. Thursday

5. 1994

6. 4 (1674, 1702, 1700, 600)

7. 1928

8. Sunday

9. Tuesday

10. Thursday

11. Monday

12. 48 times (it will appear in every month) eithic ware

INTEREST – SIMPLE & COMPOUND

14

SIMPLE & COMPOUND INTEREST

Interest or Simple Interest is the money/consideration paid by the borrower to the lender for the use of money lent for a specified period. The sum lent is called the Principal. Interest is calculated at a rate % for a certain period of time.

Simple interest : SI = $\frac{1200}{80}$

where SI is the Simple Interest, P the principal amount, T the time of year, R the rate %

The sum of the principal and the interest is called the **Amount** A = P + I

Important Points on Simple Interest

(1) Initial money that has been borrowed is called Principal or Sum.

(2) Interest is the reward for risk taking and is dividend on the Principal lent.

(3) When the interest is computed uniformly, it is called **Simple Interest,** or **S.I.**

(4) Amount = Principal + Interest.

(5) **Formulae**:

Let Principal = P; Rate = R% and Time = T years. Then,

S.I. = $\frac{P \times R \times T}{100}$; **P** = $\frac{100 \times S.I.}{R \times T}$;

T = $\frac{100 \times S.I.}{P \times R}$ **R** = $\frac{100 \times S.I.}{R \times T}$

Compound Interest: When the interest is earned on the already earned interest this is called a system of **Compound Interest** when interest is added to the sum lent, and the amount thus obtained becomes the principal for the next year or period. The system of adding interest to the principle thus calculated after addition of previous interest is repeated until the amount for the last period has been found. The difference between the final amount and the original principal is the **Compound Interest** (CI)

Formulae – Compound Interest:

Let, Principal = P, Time = n years & Rate = R % per annum.

(1) When **Interest** is compounded **Annually**: **Amount = P** $\left(1+\frac{R}{100}\right)^n$

(2) When **Interest** is compounded **Half-Yearly** Rate will be halved and time (years) converted into half years. e.g. @8% for 1.5 years. It means now the rate will be 4% per half years for 3 half years.

(3) When **Interest** is compounded **Quarterly** : **Amount = P** $\left(1+\frac{R/4}{100}\right)^{4n}$

Remember

Rate	1st Year	2nd Year	3rd Year
4%	$\left(\frac{26}{25}\right)$	$\left(\frac{676}{625}\right)$	$\left(\frac{17576}{15625}\right)$
5%	$\left(\frac{21}{20}\right)$	$\left(\frac{441}{400}\right)$	$\left(\frac{9261}{8000}\right)$
10%	$\left(\frac{11}{10}\right)$	$\left(\frac{121}{100}\right)$	$\left(\frac{1331}{1000}\right)$

These are paired quantities. One figure appearing anywhere will have an involvement of the remaining somewhere connected.

e.g. From the above table, 8000 given at 5% in 3 years will become 9261 or shall earn 1261 and so on.

Example The simple interest on a sum of money is 1/9 of the principal, and the number of years is equal to the rate per cent per annum. Find the rate of per cent.

Solution

If the principal = P, Time = n years, Rate p.c. = n then I = $\frac{Pnn}{100} = \frac{P}{9}$ ∴ n2 = $\frac{1}{100}$ ∴ = $\frac{17}{2} = 3\ \frac{1}{9}$ ∴Rate p.c. is $3\ \frac{1}{9}$

Example If Rs. 5600 amount to Rs. 6678 in $3\ \frac{1}{4}$ years, what will Rs. 9400 amount to in $5\ \frac{1}{4}$ years at the same rate per cent per annum, simple interest?

Solution

We first find the rate p.c.

Interest on Rs. 5600 = Rs. 6678 – Rs. 5600 = Rs. 1078

$$\therefore \text{rate } \% = \frac{100 \times 1078}{5600 \times 3\frac{1}{2}} = \frac{2 \times 100 \times 1078}{5600 \times 7} = \times\ 5\ \frac{1}{4}$$

Interest on Rs. 9400 = Rs. $\frac{9400 \times 21 \times 11}{100 \times 4 \times 2} = \frac{10857}{4}$ = Rs. 2714.25

∴ The required amount = Rs. 9400 + Rs. 2714.25 = Rs. 12114.25

SIMPLE INTEREST

Practice Exercise - 1

Number of Questions: 20 **Suggested time: 20 minutes**

1. How much time will it take for an amount of Rs. 900 to yield Rs. 81 as interest at 4.5% per annum of simple interest?
(a) 2 years
(b) 3 years
(c) 1 year
(d) 4 years

2. Acharya took a loan of Rs. 1,40,000 with simple interest for as many years as the rate of interest. If he paid Rs. 68,600 as interest at the end of the loan period, what was the rate of interest?
(a) 8%
(b) 6%
(c) 4%
(d) 7%

3. A sum of money at simple interest amounts to Rs. 8,150 in 3 years and to Rs. 8,540 in 4 The sum is:
(a) Rs. 7,000
(b) Rs. 6,900
(c) Rs. 6,500
(d) Rs. 6,980

4. A sum fetched a total simple interest of Rs. 1,858.40 at the rate of 8% p.(a) in 5 years. What is the sum?
(a) Rs. 4,646
(b) Rs. 2,446
(c) Rs. 5,112
(d) Rs. 2,706

5. Mr. Thakur ji invested a total amount of Rs. 1,39,000 in two different schemes “Dhan Vridhi” and “Dhan Varsha” at the simple interest rate of 14% p.(a) and 11% p.(a) respectively. If the total amount of simple interest earned in 2 years be Rs. 35,080, what was the amount invested in Scheme Dhan Varsha?

(a) Rs. 6400

(b) Rs. 7200

(c) Rs. 6500

(d) Rs. 7500

6. Mr Garib Dass borrows Rs. 50,000 for 2 years at 4% p.(a) simple interest. He immediately lends it to Mr Zaroorat Mand at $6\frac{1}{4}$% p.a for 2 years. Find his gain in the transaction per year

(a) Rs. 1,675

(b) Rs. 1,500

(c) Rs.2,250

(d) Rs. 1,125

7. What will be the ratio of simple interest earned by Mr Kanzoos on a certain amount at the same rate of interest for 5 years and that for 15 years?

(a) 3 : 2

(b) 1 : 3

(c) 2 : 3

(d) Data in adequate

8. A sum of money amounts to Rs. 98,000 after 5 years and Rs. 1,20,050 after 8 years at the same rate of simple interest. The rate of interest per annum is.

(a) 15%

(b) 12%

(c) 8%

(d) 5%

9. A certain amount earns simple interest of Rs. 1200 after 10 years. Had the interest been 2% more, how much more interest would it have earned?

(a) Rs. 25

(b) None of these

(c) Rs. 120

(d) Cannot be determined

10. Mr Faqir Chand took a loan from a bank at the rate of 8% p.(a) simple interest. After 4 years he had to pay Rs. 6,200 interest only for the period. The principal amount borrowed by him was:

(a) Rs.17,322

(b) Rs.20,245

(c) Rs.18,230

(d) Rs.19,375

11. A sum of Rs. 14,000 amounts to Rs. 22,400 in 12 years at the rate of simple interest. What is the rate of interest?

(a) 7%

(b) 6%

(c) 5%

(d) 4%

12. A sum of Rs. 725 is lent in the beginning of a year at a certain rate of interest. After 8 months, a sum of Rs. 362.50 more is lent but at the rate twice the former. At the end of the year, Rs. 33.50 is earned as interest from both the loans. What was the original rate of interest?

(a) 3.46%

(b) None of these

(c) 4.5%

(d) 5%

13. A leading financer claims to be lending money at simple interest, but he includes the interest every six months for calculating the principal. If he is charging an interest of 10%, the effective rate of interest after one year becomes:

(a) 10%

(b) 10.25%

(c) 10.5%

(d) None of these

14. Mr Shahukaar lent Rs. 50,000 to Mr Needy for 2 years and Rs. 30,000 to Mr Mauzee for 4 years on simple interest at the same rate of interest and received Rs. 22,000 in all from both of them as interest. The rate of interest per annum is:

(a) 5%

(b) 10%

(c) 7%

(d) 8%

15. What annual payment will discharge a debt of Rs. 6,450 due in 4 years at 5% per annum?

(a) Rs.1500

(b) Rs.1400

(c) Rs.1800

(d) Rs.1600

16. Mr Dhani Ram lends Rs. 1500 to Mr Garib Dass and a certain sum to Mr Majhdaar at the same time at 8% per annum simple interest. If after 4 years, Dhani Ram altogether receives Rs. 1400 as interest from Mr Dass and Mr Majhdaar, then the sum lent to Mr Majhdaar is

(a) Rs.2875

(b) Rs.1885

(c) Rs.2245

(d) Rs.2615

17. A sum of Rs. 10,000 is given as a loan to be returned in 6 monthly instalments at Rs. 3,000. What is the rate of interest?

(a) 820%

(b) 620%

(c) 780%

(d) 640%

18. If the simple interest on a certain sum of money after $3\frac{1}{8}$ years is $\frac{1}{4}$ of the principal, what is the rate of interest per annum?

(a) 6%

(b) 4%

(c) 8%

(d) 12%

19. If a sum of Rs. 9,00,000 is lent to be paid back in 10 equal monthly instalments of Re. 1,00,000 each, then the rate of interest is?

(a) 11.33%

(b) 11%

(c) 266.67%

(d) 26.67%

20. Divide Rs. 2379 into 3 parts so that their amount after 2,3 and 4 years respectively may be equal, the rate of interest being 5% per annum at simple interest. The first part is?

(a) Rs. 828

(b) Rs. 746

(c) Rs. 248

(d) Rs. 1024

Practice Exercise - 1 Answers with Hints

1 (a)	2 (d)	3 (d)	4 (a)	5 (a)
6 (d)	7 (b)	8 (b)	9 (d)	10 (d)
11 (c)	12 (a)			

13. (b) Let the automobile financier lends Rs. 100

Simple Interest for first 6 months = PRT/100 = 100 × 10 × 1/2/100 = Rs. 5

After 6 months, he adds the simple interest to principal

i.e., after 6 months, principal becomes Rs. 100 + Rs. 5 = Rs. 105

Simple Interest for next 6 months = PRT/100 = 105 × 10 × 1/2/100 = Rs. 5.25

Amount at the end of 1 year = Rs. 105 + Rs. 5.25 = Rs. 110.25

i.e., Effective Simple Interest he gets for Rs. 100 for 1 year = 110.25 - 100 = 10.25

i.e, the Effective Rate of Interest = 10.25%

14. (b) Let the rate of interest per annum be R%

Simple Interest for Rs. 5000 for 2 years at rate R% per annum + Simple Interest for Rs. 3000 for

4 years at rate R% per annum = Rs. 2200

$\Rightarrow$ 5000 × R × 2/100 + 3000 × R × 4/100 = 2200 $\Rightarrow$ 100R + 120R = 2200 $\Rightarrow$ 220R = 2200 $\Rightarrow$ R = 10%

15. (a) The amount needs to be repaid in 4 years = Rs. 6450

Suppose Rs. x is paid annually to repay this debt

Then, amount paid after 1^{st} year = Rs. X

Interest for this Rs. x for the remaining 3 years = $x \times 5 \times 3/100 = 15x/100$

Then, amount paid after 2nd year = Rs. x

Interest for this Rs. x for the remaining 2 years = $x \times 5 \times 2/100 = 10x/100$

Amount paid after 3rd year = Rs. x

Interest for this Rs. x for the remaining 1 year = $x \times 5 \times 1/100 = 5x/100$

Amount paid after 4th year = Rs. x and this closes the entire debt

$\Rightarrow x + 15x/100 + x + 10x/100 + x + 5x/100 + x = 6450 \Rightarrow x = 1500$

16. (a)

17. (d) Amount borrowed = Rs. 10

Let rate of interest = R%

Simple Interest for Rs. 10 for 6 months at R% = 10 × R × 1/200 = R20

i.e, 10 + R20 is due in 6 months

Payment after 1st month = Rs. 3

Interest for this Rs. 3 for the remaining 5 months = $3 \times R \times 5/1200$

Payment after 2nd month = Rs. 3

Interest for this Rs. 3 for the remaining 4 months = $3 \times R \times 4/1200$

Payment after 5th month = Rs. 3 similarly...

Interest for this Rs. 3 for the remaining 1 month = $3 \times R \times 1/1200$

Payment after 6th month = Rs. 3 and this closes the loan

$\Rightarrow (3 + 3 + 3 + 3 + 3 + 3) + [3 \times R \times 5/1200 + 3 \times R \times 4/1200 + ... + 3 \times R \times 1/1200\ ] = 10$ R = 640%

18. (c)

19. (d)

20. (a) Let the parts be x, y and z

R = 5%

x + interest on x for 2 years = y + interest on y for 3 years = z + interest on z for 4 years

SIMPLE INTEREST

Practice Exercise - 2

Number of Questions: 20 **Suggested time: 20 minutes**

1. In how many years, Rs. 7,500 will produce the same interest at 6% as Rs. 40,000 produce in 2 years at $4\frac{1}{2}$%?

(a) 4 years
(b) 6 years
(c) 8 years
(d) 9 years

2. A sum of Rs. 7,500 amounts to Rs. 11,625 in 4 years at the rate of simple interest. What is the rate of interest?

(a) 12.25%
(b) 12%
(c) 6%
(d) 13.75%

3. What is the interest due after 40 days for Rs. 32,000 at 10% per annum.

(a) 35.07
(b) 36.21
(c) 35.52
(d) 34

4. What is the rate of interest at which Rs. 15,000 becomes Rs. 22,000 in 10 years.?

(a) 113%
(b) 143%
(c) 12%
(d) 14%

5. Mr Suresh Chawla invested in all Rs. 26,00,000 at 4%, 6% and 8% per annum simple interest. At the end of the year, he got the same interest in all three cases. The money invested at 4% is:

(a) Rs. 22,00,000

(b) Rs. 8,00,000
(c) Rs.16,00,000
(d) Rs. 12,00,000

6. Mr Dasson invested certain amount in three different schemes "Slow", "Medium" and "Fast" with the rate of interest 10% p.a., 12% p.a. and 15% p.a. respectively. If the total interest accrued in one year was Rs. 32,000 and the amount invested in Scheme "Fast" was 150% of the amount invested in Scheme "Slow" and 240% of the amount invested in Scheme "Medium" , what was the amount invested in Scheme "Medium"?
(a) Rs. 50,000
(b) Rs. 20,000
(c) Rs.60,000
(d) Rs. 30,000

7. A sum of Rs. 15,500 was lent partly at 5% and partly at 8% p.a. at simple interest. The total interest received after 3 years was Rs. 3,000. The ratio of the money lent at 5% to that at 8% was:
(a) 16 : 15
(b) 15 : 16
(c) 15 : 8
(d) 8 : 15

8. A sum of money doubles in 12 years. In how many years, it will treble at S.I.
(a) 12 years
(b) 8 years
(c) 6 years
(d) 24 years

9. Mr Navansh invests a certain sum of money at 6% per annum simple interest and another sum at 7% per annum simple interest. His income from interest after 2 years was Rs. 35,400. One-fourth of the first sum is equal to one-fifth of the second sum. The total sum invested was:
(a) Rs. 3,10,000
(b) Rs. 2,70,000
(c) Rs.2,20,000

(d) Rs. 1,80,000

10. The interest on a certain deposit at 5% per annum is Rs. 101.20 in one year. How much will the additional interest in one year be on the same deposit at 6% per annum?

(a) Rs. 20.80
(b) Rs. 19.74
(c) Rs.20.24
(d) Rs. 19.5

11. A sum was invested at simple interest at a certain rate for 2 years. Had it been invested at 4% higher rate, it would have fetched Rs. 600 more. The sum is:

(a) Rs. 7,500
(b) Rs. 7,000
(c) Rs.8,200
(d) Rs. 9,400

12. The simple interest at x% for x years will be Rs. x on a sum of:

(a) Rs. x
(b) Rs. x^2
(c) Rs. 100x
(d) Rs. 100x2

13. A sum of money becomes 140% of itself in 4 years at a certain rate of simple interest.

The rate per annum is:

(a) 9%
(b) 10%
(c) 11%
(d) 12%

14. If the simple interest on Rs. 2,000 is less than the Simple Interest on Rs. 3,000 at 5% by Rs. 50, find the time.

(a) 2.5 year
(b) 1 year
(c) 1.5 year
(d) 2 year

15. A sum of Rs. 77,000 is to be divided among three brothers Parkash, Jagdish and Krishan in such a way that simple interest on each part at 5% per annum after 1,2 and 3 years respectively remains equal. The Share of Parkash is more than that of Krishan by?
(a) Rs. 12,000
(b) Rs. 14,000
(c) Rs.22,000
(d) Rs. 28,000

16. The simple interest on Rs. 4,500 at 6% per annum from May 3^{rd} to July 15^{th} in the same year is
(a) Rs. 72
(b) Rs. 54
(c) Rs. 36
(d) Rs 81

17. Find the Simple Interest on Rs. 75,000 at 11% for 2 Years and 5 months.
(a) Rs. 19,942
(b) Rs. 19,912
(c) Rs.19,927
(d) Rs. 19,937

18. Mr Armaan borrowed a sum for 4 years on S.I. at 12%. The total interest paid was Rs. 36,000. Find the Principal.
(a) Rs. 70,000
(b) Rs. 65,000
(c) Rs.80,000
(d) Rs. 75,000

19. The simple interest on Rs. 1,820 from March 9, 2017 to May 21, 2017 at $7\frac{1}{2}$% rate is
(a) Rs. 27.30
(b) Rs. 22.50
(c) Rs. 28.80
(d) Rs. 29

20. At what rate percent of simple interest will a sum of money double itself in 20 years?

(a) 4%
(b) 5%
(c) 6%
(d) 8%

Practice Exercise - 2 Answers with Hints

1 (c)	2 (d)	3 (a)	4 (b)	5 (d)
6 (a)	7 (a)	8 (d)	9 (b)	10 (c)
11 (a)	12 (c)	13 (b)	14 (b)	15 (d)
16 (b)	17 (d)	18 (d)	19 (a)	20 (b)

5. Let x , y and x be his investments at 4%, 6% and 8% respectively

$x \times 4 \times 1/100 = y \times 6 \times 1/100 = z \times 8 \times 1/100 \Rightarrow 4x = 6y = 8z \Rightarrow 2x = 3y = 4z;$

We also know that x + y + z = 26,00,000⇒ x = 12,00,000

i.e., Money invested at 4% = Rs. 12,00,000

11. Think like this 4% of P × 2 = Rs. 60; Hence P is 750.

SIMPLE INTEREST

Practice Exercise - 3

Number of Questions: 33 **Suggested time: 25 minutes**

1. If the simple interest on a certain sum of money for 4 years is 20% of the sum, then the rate of interest being calculated per annum is
(a) 4%
(b) 7%
(c) 6%
(d) 5%

2. A certain sum invested for 'T' years amounts to Rs. 1,200 at 10% per annum. But when invested at 4% per annum, it amounts to Rs. 600. Find the time (T).
(a) 45 years
(b) 60 years
(c) 40 years
(d) 50 Years

3. A sum of money is lent at simple interest for Six years. If the same amount is paid at 4% higher, Shubham would have got Rs. 390 more. Find the principal
(a) Rs. 650
(b) Rs. 1,950
(c) Rs. 1,300
(d) Rs. 1,625

4. Mr. Manav invested an amount of Rs. 12,00,000 at the simple interest rate of 10% per annum and another amount 'P' at simple interest rate of 20% per annum. The total interest earned at the end of one year on the total amount invested yields 14% per annum. Find the total amount invested by Mr Manav.
(a) Rs. 25,00,000
(b) Rs. 15,00,000
(c) Rs. 1,00,00,000

(d) Rs. 20,00,000

5. Mr Naval lent Rs. 40,000 to Mr Ashok for 2 years and Rs. 20,000 to Mr Yatin for 4 years at the same rate of simple interest and received Rs. 22,000 in all from both of them as interest. The rate of interest charged by Mr Naval from them per annum is:

(a) 14%

(b) 15%

(c) 12%

(d) 13.75%

6. NSC certificates, bearing simple rate of interest, were purchased by Mr Deepak Raheja for Rs. 50,000 on 1st Jun 2017 with Rs. 1,00,000 being the maturity value due on 31st May 2021. Find the rate.

(a) 24%

(b) 22%

(c) 16%

(d) 25%

7. If simple interest earned by Mr Ishan on a certain sum of money "P" for 8 years at 4% per annum is same as the simple interest on Rs. 56,000 for 8 years at the rate of 12% per annum then "P" is

(a) Rs. 1,82,000

(b) Rs. 1,04,000

(c) Rs.1,12,000

(d) Rs. 1,68,000

8. If a sum of money trebles itself in 40 years, what is the rate of interest?

(a) 5%

(b) 6%

(c) 4%

(d) None of these

9. What annual instalment will discharge a debt of Rs 1092 due in 3 years at 12% simple interest?

(a) 325

(b) 545

(c) 560

(d) 550

10. In how many years will a sum of Rs. 2,400 at 10% per annum compounded semi annually become Rs. 2,778.30?

(a) 1 year 6 months
(b) 2 years 6 months
(c) 3 years 6 months
(d) 4 years 6 months

11. At what rate percent per annum will a sum of money double itself in 8 years.

(a) 12.5%
(b) 13.5%
(c) 11.5%
(d) 14.5%

12. A sum of money, invested by a wise investor, at simple interest amounts to Rs. 81,500 in Three years and to Rs. 85,400 in Four years. The sum is:

(a) 65,000
(b) 69,000
(c) 69,800
(d) 70,000

13. Mr Vishal invested a sum of money which amounts to Rs 5,040 in 2 years and to Rs 6,820 in 3 ½ years. Find the sum and the rate of interest.

(a) 4,000; 14%
(b) 4,000; 13%
(c) 4,000; 12%
(d) 4,000; 19%

14. Mr Khushwant took loan from a bank at 12% p.a. simple interest. After 3 years he had to pay Rs. 5400 interest. The amount borrowed by him was (Rupees):

(a) 12,000
(b) 10,000
(c) 15,000
(d) 20,000

15. A sum was placed at simple interest at a certain rate for 10 years by Mr Himanshu. Had he put at 5% higher rate, he would have fetched Rs.

6,000 more. Find his investment?

(a) 12,000

(b) 13,000

(c) 15,000

(d) 14,000

16. Simple interest on a certain sum is 64% of the sum. If both Rate (%) and Time (years) are numerically equal, find these.

(a) 7

(b) 8

(c) 6

(d) 5

17. A sum of Rs. 46,500 was lent partly at 8% and at 5% p.a. simple interest. The total interest received after 3 years was Rs. 9,000. The ratio of the money lent at 5% to that lent at 8% is?

(a) 5 : 8

(b) 6 : 7

(c) 16 : 15

(d) 17 : 18

18. Ms Nirmal Goyal, on her retirement, invested an amount of Rs. 13,90,000 divided in two different schemes 'Faayda" and 'Bachat' at simple interest rate of 14% p.a. and 11% p.a. respectively. If the total amount of simple interest earned in 2 years be Rs. 3,50,800, what was the amount invested (Rs) in Scheme 'Bachat'?

(a) 6,40,000

(b) 6,50,000

(c) 7,20,000

(d) 7,50,000

19. Mr Pritam Monga left a will of Rs. 35,00,000 for his two daughters aged 8.5 and 16 such that they may get equal amounts when each of them attains the age of 21 years. It was desired that this amount must be invested at 10% p.a. simple interest. How many lakhs of Rupees did the elder daughter get at the time of the will?

(a) 17.5

(b) 21

(c) 15
(d) 20

20. The simple interest on a sum of money is 1/9 of the principal and the number of years is equal to the rate percent per annum. The rate percent per annum is?
(a) 2 ½
(b) 2 1/3
(c) 3 - 1/3
(d) Data inadequate

21. Mr Bhargava invested a sum of Rs. 72,500 on 1st Jan at a certain rate of interest. On 1st Sept,he invested Rs. 36,250 more - *but at a rate twice the former*. On 31st Dec, he earned Rs. 3,350 as interest from both the investments. Find the original rate of interest (%)?
(a) 3.46
(b) 4.5
(c) 5.0
(d) 6.0

22. Mr Dharam Pal borrows Rs. 50,000 for 2 years at 4% p.a. simple interest. He immediately lends it to Mr Gopal Dass at $\frac{2}{5}x$ p.a for 2 years. Find gain (Rs.)of Mr Dharam Pal in the year.
(a) 1,125
(b) 1,500
(c) 1,675
(d) 1,700

23. A certain amount earns simple interest of Rs. 1750 after 7 years. Had the rate of interest been 2% higher, how much interest (Rupees) would it have earned more?
(a) 35
(b) 245
(c) 350
(d) Data inadequate

24. A sum fetched a total of Rs. 40,162.50 at the rate of 9% p.a. simple interest in 5 years. What is the sum (Rupees)?

(a) 44,625
(b) 80,325
(c) 89,000
(d) 89,250

25. Mr Tarun Kumar invested a sum of money amounts to Rs. 98,000 after 5 years and Rs. 1,20,005 after 8 years at simple interest. The rate of interest (p.a.), same in both cases, is:
(a) 5
(b) 8
(c) 12
(d) 15

26. A sum of money invested for "T" years at 8% p.a. simple interest grows to Rs. 9,000. The same sum of money invested for the same number of years at 4% p.a. simple interest grows to Rs. 6,000 only. Find "T"?
(a) 25
(b) 15
(c) 20
(d) 22

27. How much time will it take for a sum of Rs. 13,500 to yield Rs. 2,430 as interest at 4.5% per annum of simple interest?
(a) 3.5
(b) 4
(c) 4.5
(d) 5

28. How many years will it take Rs. 12,500 to grow to Rs. 1,00,000, if it is invested at 12.5% p.a simple interest?
(a) 65
(b) 56
(c) 45
(d) 57

29. When Mr Akaash invested a sum of Rs. 15,000 he found a difference between the simple interests , from two different sources, after 3 years, as Rs. 135. The difference between their rates of interest (%) is?
(a) 0.1

(b) 0.2
(c) 0.3
(d) 0.4

30. Mr Nishikant borrowed a total of Rs 24,000 from two *"Sahukaars"* of his village Kusumbhi. For one loan, he paid 15% per annum and for the other 18% per annum. At the end of the year, he paid Rs 4,050. How much did he borrow at one of the rates?
(a) 14,000
(b) 12,000
(c) 15,000
(d) 13,000

31. A sum was put at simple interest at a certain rate for 3 years. Had it been put at 2% higher rate, it would have fetched Rs 540 more. Find the sum.
(a) 6,000
(b) 13,500
(c) 7,500
(d) 9,000

32. Find the ratio of simple interest earned by certain amount at the same rate of interest for 6 years and for 9 years?
(a) 1:3
(b) 1:4
(c) 2:3
(d) More data required.

33. Compute simple interest (in Rupees) on Rs 6,250 at 7% per annum for 292 days.
(a) 350
(b) 450
(c) 550
(d) 650

Practice Exercise - 3 Answers with Hints

1 (d)	2 (d)	3 (d)	4 (d)	5 (d)

6 (d)	7 (d)	8 (a)	9 (a)	10 (a)
11 (a)	12 (c)	13 (b)	14 (c)	15 (a)
16 (b)	17 (c)	18 (a)	19 (b)	20 (c)
21 (a)	22 (a)	23 (d)	24 (d)	25 (c)
26 (a)	27 (b)	28 (b)	29 (c)	30 (c)
31 (d)	32 (c)	33 (a)		

19. $x+\frac{5\times10\times x}{100}=(3,500,000-x)+\frac{12.5\times10\times(3,500,000-x)}{100}$

$\Rightarrow x = 2,\ 100,\ 000 = 21\ lakhs$

20. Let the rate be R% per annum and the time be R years then $\frac{x}{9}=\frac{x\times R\times R}{100}$

$\Rightarrow R=\sqrt{\frac{100}{9}}=\frac{10}{3}=3\frac{1}{3}\%$

COMPOUND INTEREST

Practice Exercise - 4

Number of Questions: 15 **Suggested time: 15 minutes**

1. What is the difference between the compound interests on Rs. 5000 borrowed by Ramesh Kumar for $1\frac{1}{2}$ years at 4% per annum compounded yearly and half-yearly?
(a) Rs. 2.04
(b) Rs. 4.80
(c) Rs. 3.06
(d) Rs. 8.30

2. A bank offers 5% compound interest calculated on half-yearly basis. Mr Rohit K Anand deposits Rs. 16,000 each on 1st January and 1st July of a year. At the end of the year, the amount he would have gained by way of interest is:
(a) Rs. 1,200
(b) Rs. 1,210
(c) Rs. 1,230
(d) Rs. 1,220

3. Ms Aniket finds that there is 80% increase in an amount in 8 years at simple interest. What shall be the compound interest of Rs. 14,000 after 3 years at the same rate?
(a) Rs. 3,794
(b) Rs. 3,714
(c) Rs.4,612
(d) Rs. 4,634

4. In how much time (in months), the compound interest, compounded annually, on Rs. 30,000 at 7% per annum is Rs. 4,347?
(a) 12
(b) 24
(c) 36

(d) 42

5. Is there any sum on which the difference between simple and compound interests compounded annually on a certain sum of money for 2 years at 4% per annum is Re. 1? If so find it.
(a) Rs. 600
(b) Rs. 645
(c) Not possible
(d) None of these

6. Mr Hira finds that the difference between compound interest and simple interest on Rs. 15,000 given for 2 years is Rs. 96. What is the rate of interest per annum paid by him?
(a) 9%
(b) 12%
(c) 8%
(d) 6%

7. If the simple interest on a sum of money for 2 years at 5% per annum is Rs. 60, what is the compound interest on the same at the same rate and for the same time?
(a) Rs. 63.5
(b) Rs. 62
(c) Rs. 61.5
(d) Rs. 64

8. The difference between simple interest and compound on Rs. 900 for one year at 10% per annum reckoned half-yearly is:
(a) Rs. 3
(b) Rs. 2.25
(c) Rs. 4.5
(d) Rs. 4

9. What will be the compound interest on a sum of Rs. 40,000 after 3 years at the rate of 11% p.a.?
(a) Rs. 14,705.24
(b) Rs. 14,602.25
(c) Rs. 14,822.26
(d) Rs. 14,322.10

10. At what rate of compound interest per annum will a sum of Rs. 1,400 earn Rs. 173.04 in 2 years?
(a) 4%
(b) 5%
(c) 6%
(d) 8%

11. What will be the minimum number of complete years in which a sum of money put out at 20% compound interest will earn equal to itself?
(a) 5
(b) 4
(c) 4
(d) 2

12. The effective annual rate of interest corresponding to a nominal rate of 6% per annum payable half-yearly is:
(a) 6.07%
(b) 6.08%
(c) 6.06%
(d) 6.09%

13. Anamica invested an amount of Rs. 20,000 in a fixed deposit scheme for 2 years at compound interest rate 4% p.a. How much amount will she earn on maturity of the fixed deposit?
(a) 2,342
(b) 1,632
(c) 2,324
(d) 4,120

14. Simple interest on a certain sum of money "P" for 4 years at 5% per annum is 50% of the compound interest earned on Rs. 3,000 for 2 years at 10% per annum. The sum "P" placed on simple interest is:
(a) Rs. 1,575
(b) Rs. 2,200
(c) Rs. 1,200
(d) Rs. 1,625

15. The compound interest on a certain sum for 2 years at 10% per annum is Rs. 525. Then simple interest on the same principal for double the time

but at half the rate % per annum differs by:
(a) Rs. Nil
(b) Rs. 70
(c) Rs. 50
(d) Rs. 120

Practice Exercise - 4 Answers

1 (a)	2 (b)	3 (d)	4 (b)	5 (d)
6 (c)	7 (c)	8 (b)	9 (a)	10 (c)
11 (b)	12 (d)	13 (b)	14 (a)	15 (a)

COMPOUND INTEREST

Practice Exercise - 5

Number of Questions: 15 **Suggested time: 12 minutes**

1. The difference between simple interest and compound interest on Rs. 2,400 for one year at 10% per annum reckoned half-yearly is Rs. {K% of (18 + 12 – 25 + 15)}. Then K is :
(a) 20
(b) 30
(c) 15
(d) 10

2. The Simple interest on a certain sum for 2 years at 20% per annum is Rs. 80. The corresponding compound interest will exceed by.....
(a) Rs. 6
(b) Rs. 2
(c) Rs. 8
(d) Rs. 9

3. The Compound interest on Rs. 10,240 at $6\frac{1}{2}$% per annum for 2 years 73 days, is:
(a) Rs. 1,465
(b) Rs. 1,110
(c) Rs. 1,511
(d) Rs. 1,525

4. A sum amounts to Rs. 970.20 in 2 years at 5% compound interest. The sum is
(a) Rs. 880
(b) Rs. 904.20
(c) Rs. 924
(d) Rs. 897.60

5. What sum of money at 4% compound interest payable half-yearly amounts to Rs. 13,265.10 in $1\frac{1}{2}$ years?

(a) Rs. 12500

(b) Rs. 11200

(c) Rs. 8840

(d) Rs. 12600

6. Harish had borrowed some money and paid back in two annual instalments of Rs. 8,820 each allowing 5% compound interest. The sum borrowed was:

(a) Rs. 18,200

(b) Rs. 16,400

(c) Rs.12,600

(d) Rs. 14,400

7. Mr Bhola Singh invested certain sum for 2 years at 14% compounded annually which would grow to Rs. 5,458.32. Find his investments.

(a) 4,120

(b) 3,300

(c) 4,200

(d) 4,420

8. Shri Kailash Nath found that the difference between the simple interest and compound interests on some principal amount at 20% for 3 years is Rs. 2,880. Then the principal amount is?

(a) Rs. 21,900

(b) Rs. 19,500

(c) Rs. 23,700

(d) Rs. 22,500

9. Amrit earns an interest of Rs. 1,596 for the third year and Rs. 1,400 for the second year on the same sum Find the rate of interest if he lends at compound interest.

(a) 12%

(b) 13%

(c) 14%

(d) 15%

10. The population of a small village Suchan Kotli, in the year 2015, was 40,000. It consists of males. Due to migration the population decreases by 20 per thousand per year. Find the population after 2 years i.e., in the year 2017.

(a) 38,484

(b) 38,266

(c) 38,416

(d) 38,226

11. If the compound interest on a certain sum for 2 years in Rs. 80.80 and the simple interest is Rs. 0.80 p lesser than it, then the rate of interest per annum is?

(a) 2%

(b) 1%

(c) 3%

(d) 4%

12. The compound interest on a sum for 2 years is Rs. 32 more than the simple interest on the same sum for the same period which is Rs. 800. The difference between the compound and simple interest for 3rd year will be

(a) Rs. 48

(b) Rs. 66.56

(c) None of these

(d) Rs. 98.56

13. On a certain sum of money, the simple interest for 2 years is Rs. 200 at the rate of 7% per annum. Find the difference between Simple and Compound Interest if nothing is changed.

(a) None of these

(b) Rs. 9

(c) Rs. 7

(d) Rs. 11

14. Mrs Kalpana invested Rs. 1,50,000 in the name of her daughter Vridhi in two schemes A and B offering compound interest @ 5% p.a. and 10% p.a. respectively. If the total amount of interest accrued through two schemes together in two years was Rs. 20,750. Find out the amount invested in Scheme A?

(a) Rs. 1,00,000

(b) Rs. 80,000

(c) Rs. 1,20,000

(d) Rs. 1,40,000

15. A sum of money on compound interest amounts to Rs. 82,400 in 2 years and Rs. 98,880 in 3 years. The rate of interest is

(a) 10%

(b) 25%

(c) 20%

(d) 12%

Practice Exercise - 5 Answers with Hints

1 (b)	2 (d)	3 (a)	4 (a)	5 (a)
6 (b)	7 (c)	8 (d)	9 (c)	10 (c)
11 (a)	12 (d)	13 (c)	14 (a)	15 (c)

6. Present worth of Rs. x due T years hence is given by

Present Worth (PW) = $A / (1 + R/100)^T$

i.e., The sum borrowed = Rs. 16,400

9. Interest earned in 3rd year = Rs. 1596 ; Interest earned in 2nd year = Rs. 1400

i.e, in 3rd year, Andrews gets additional interest of (Rs. 1596 - Rs. 1400) = Rs. 196

This means, Rs. 196 is the interest obtained for Rs. 1400 for 1 year

R = 100 × SI/PT = 100 × 196/1400 × 1 = 196/14 = 14%

10. This problem is similar to the problems we saw in compound interest. We can use the formulas of compound interest here as well.

In compound interest, interest (a certain percentage of the principal) will be added to the principal after every year. But in this problem, a certain percentage of the population will be decreased (***Please note that we have to use the -ve sign here instead of the + sign as the population gets***

decreased) from the total population after every year i.e., the formula becomes, $A = P(1-R/100)^T$ where Initial population = P, Rate = R% per annum, Time = T years and A = the population after T years.

$R = 20 \times 100/1000 = 2\%$ ($\therefore$ percentage is calculated for twenty per thousand)

Population after 2 years $= P(1-R/100)^T = 40{,}000\ (1-2/100)^2 = 40{,}000(49/50)^2 = 38{,}416$

(It consists of 48% males–is an irrelevant information and need not be taken note of for any working)

13. Ist year 2^{nd} year

Simple Interest	100	+	100	
Compound Interest	100	+	100 +	Interest on 100@7% for 2^{nd} year
Hence Difference is	=	Rs. 7 only		

15. Interest amounts to Rs. 82,400 in 2 years and Rs. 98,880 in 3

Interest for One viz. 3^{rd} year on Rs. 8,2400 is Rs. 16,480 (Rs.98,880 – 82,400)

Therefore, rate of interest is $(16{,}480 \times 100) / (82{,}400 \times 1) = 20\%$

COMPOUND INTEREST

Practice Exercise - 6

Number of Questions: 25 **Suggested time: 20 minutes**

1. A tree increases annually by $\frac{5}{6}$th of its height. If its height today is 50 cm, what will be the height after 2 years?
(a) 64 cm
(b) 72 cm
(c) 66 cm
(d) 84 cm

2. On a sum of money, the simple interest for 2 years is Rs. 320, while the compound interest is Rs. 340, the rate of interest being the same in both the cases. The rate of interest is:
(a) 15%
(b) 14.25%
(c) 12.5%
(d) 10.5%

3. A bank offers 10% interest rate compounded annually. Mr Roshan Lal deposits Rs. 2,00,000 every year in his account. If he does not withdraw any amount, then how much balance will his account show after four years?
(a) Rs. 10,21,020
(b) Rs. 10,22,200
(c) Rs. 10,42,020
(d) Rs. 10,42,220

4. A sum of money becomes Rs. 2200 after three years and Rs. 4400 after six years on compound interest. The sum is
(a) Rs. 1400
(b) Rs. 1100
(c) Rs. 1000
(d) Rs. 1200

5. What annual payment will discharge a debt of Rs. 1025 due in 2 years at the rate of 5% compound interest?

(a) Rs. 560

(b) Rs. 560.75

(c) Rs. 551.25

(d) Rs. 550

6. A sum of money placed at compound interest doubles itself in 4 years. In how many years will it amount to 8 times?

(a) 10 years

(b) 8 years

(c) 6 years

(d) 12 years

7. On what sum will the compound interest for 2 ½ years at 10% amount to Rs. 31762.5?

(a) Rs. 5000

(b) Rs. 20000

(c) Rs.25000

(d) Rs. 30000

8. The compound interest on Rs. 20,000 at 8% per annum is Rs. 3,328. What is the period (in year)?

(a) 1 year

(b) 4 years

(c) 3 years

(d) 2 years

9. Nitish borrows Rs. 20,000 at 10% compound interest. At the end of every year he pays Rs. 2000 as part repayment. How much does he still owe after three such instalments?

(a) Rs. 24000

(b) Rs. 15000

(c) Rs.20000

(d) Rs. 10000

10. The present worth of Rs. 242 due in 2 years at 10% per annum compound interest is:

(a) Rs. 180

(b) Rs. 240
(c) Rs. 220
(d) Rs. 200

11. The difference between the simple interest on a certain sum at the rate off 10% per annum for 2 years and compound interest which is compounded every 6 months is Rs. 124.05. What is the principal sum?
(a) Rs. 10000
(b) Rs. 12000
(c) Rs. 6000
(d) Rs. 8000

12. A sum of Rs. 6600 was taken as a loan. This is to be repaid in two equal annual instalments. If the rate of interest be 20% compounded annually then the value of each instalment is
(a) Rs. 4320
(b) Rs. 2220
(c) Rs. 4400
(d) Rs. 4420

13. If in a certain number of years Rs. 10000 amount to Rs. 160000 at compound interest in half that time Rs. 10000 will amount to:
(a) Rs. 50000
(b) Rs. 40000
(c) Rs. 80000
(d) Rs. 60000

14. What will be the amount if a sum of Rs. 10000 is placed at compound interest for 3 years while rate of interest for the first, second & third years are 2, 5 and 10 percent, respectively?
(a) Rs. 11781
(b) Rs. 11244
(c) Rs. 11231
(d) Rs. 11658

15. Simple interest on a sum at 5% per annum for 2 years is Rs. 60. The compound interest on the same sum for the same period is
(a) Rs. 62.4
(b) Rs. 61.5

(c) Rs. 62
(d) Rs. 60.5

16. A sum is invested at compounded interest payable annually. The interest in the first two successive years was Rs. 400 and Rs. 420. The sum is
(a) Rs. 8000
(b) Rs. 7500
(c) Rs. 8500
(d) Rs. 8200

17. Alisha borrowed a certain sum from Vikas at a certain rate of simple interest for 2 years. She lent this sum to Sunil at the same rate of interest compounded annually for the same period. At the end of two years, she received Rs. 2400 as compound interest but paid Rs. 2000 only as simple interest to Mr Vikas. Find the calculated rate of interest.
(a) 40%
(b) 30%
(c) 20%
(d) 10%

18. If a sum on compound interest becomes three times in 4 years, then with the same interest rate, the sum will become 81 times in:
(a) 12 years
(b) 18 years
(c) 16 years
(d) 14 years

19. Divide Rs. 3364 between A and B, so that A's Share at the end of 5 years may equal to B's share at the end of 7 years, compound interest being at 5 percent.
(a) Rs. 1764 and Rs. 1600
(b) Rs. 1756 and Rs. 1608
(c) Rs. 1722 and Rs. 1642
(d) None of these

20. A sum is invested for 3 years compounded at 5%, 10% and 20% respectively. In three years, if the sum amounts to Rs. 1386, then find the sum.
(a) Rs. 1500

(b) Rs. 1400
(c) Rs. 1200
(d) Rs. 1000

21. A sum of money lent at compound interest for 2 years at 20% per annum would fetch Rs. 482 more, if the interest was payable half yearly than if it was payable annually. The sum is
(a) Rs. 10,000
(b) Rs. 20,000
(c) Rs. 40,000
(d) Rs. 50,000

22. A sum of money amounts to Rs. 6690 after 3 years and to Rs. 10,035 after 6 years on compound interest. Find the sum.
(a) Rs. 4,360
(b) Rs. 4,460
(c) Rs. 4,560
(d) Rs. 4,660

23. Using which information/s, can we find the rate of interest %.p.a.?
(I) An amount doubles itself in 5 years on simple interest.
(II) Difference between the compound interest and the simple interest earned on a certain amount in 2 years is Rs. 400.
(III) Simple interest earned per annum is Rs. 2000
(a) I only
(b) II and III only
(c) All I, II and III
(d) I only or II and III only

24. Find compound interest on Rs. 8000 at 15% per annum for 2 years 4 months, compounded annually.
(a) Rs. 2,109
(b) Rs. 3,109
(c) Rs. 4,109
(d) Rs. 6,109

25. What annual payment will discharge a debt of Rs. 7620 due in 3years at $16\frac{2}{3}$% per annum interest?

(a) Rs. 5,430
(b) Rs. 4,430
(c) Rs. 3,430
(d) Rs. 2,430

Practice Exercise - 6 Answers with Hints

1 (b)	2 (c)	3 (a)	4 (b)	5 (c)
6 (d)	7 (c)	8 (d)	9 (c)	10 (d)
11 (d)	12 (a)	13 (b)	14 (a)	15 (b)
16 (a)	17 (a)	18 (c)	19 (a)	20 (d)
21 (b)	22 (b)	23 (d)	24 (b)	25 (c)

4. P (1 + R/100)T = 2200 P(1 + R/100)3 = 2200 --- (1)

Amount after 6 years = 4400

P(1 + R/100)T = 4400 P(1 + R/100)6 = 4400 --- (2)

(2)÷(1) ⇒ P(1 + R/100)6 ÷ P(1 + R/100)3 = 4400/2200 = 2 ⇒ (1 + R/100)3 = 2

(Substituting this value in equation 1) ⇒ P × 2 = 2200 P = 1100

5. Let x be the annual payment

Then, present worth of x due 1 year + present worth of x due 2 year hence = 1025

x(1 + 5/100)1 + x(1 + 5/100)2 = 1025 Or x(21/20) + x(21/20)2 = 1025 = Rs. 551.25

9. Balance amount due after three payments :

= [{(20,000 x 1.10 – 2,000) 1.10 – 2,000} × 1.10 – 2,000]

11. P [(1.05)4 – 10 x 2/100] = 124.05 Or P = 8,000

14. A = 10,000 (1.02)1 (1.05)1 (1.10)1 = 11,781

RATIOS & PROPORTIONS

15

Ratio is the relation that exists between two numbers or two quantities of the same units. If **a** and **b** are such quantities, then their ratio is written as **a : b or a/b**. For example the ratio of 5 to 15 will be expressed as

5 : 15 or $\frac{5}{15} = \frac{1}{3}$ or 1 : 3. Here **a is called the antecedent and b is called the consequent.**

The equality of two ratios is called proportion. For example 3 : 5 = 6 : 10.

Here we say that 3, 5, 6 and 10 are in proportion and write as follows:

3 : 5 :: 6 : 10; Here 3, 5, 6, 10 are respectively known as first, second, third and fourth terms.

The first and fourth terms are known as extreme terms.

The second and third terms are known as mean terms.

If four quantities are in proportion, then **Product of means = Product of extremes**

If a **:** x **::** x **:** b **then** ***x*** is called the mean proportion between *a* and *b*. Clearly, x^2 = ab

Thus the mean proportion between any two numbers is equal to the square root of their products.

(A) ***Simple/Direct Ratios*** ***A : B : C : D :: 7 : 5 : 8 : 9***

(Where are units are directly related)

(B) ***Compound/Indirect Ratios*** ***A : B :: 7 : 5; B : C :: 6 : 11***

A		B		C
7		5		
		6		11
42	:	30	:	55

All indirect ratios must be first converted into Direct ratios to find solutions.

Example If 3 : 21 :: x : 63, find the value of x.

Solution

Product of means = Product of extremes ∴ $21 \times x = 3 \times 63$ or $x = \frac{25\times1}{0.128} = 9$

Example Find the fourth proportional of 1, 2, 3.

Solution

Let the fourth proportional be x; ∴ $1 : 3 :: 2 : x$ or $x = 6$.

Example Find the third proportional of 6 and 30.

Solution

Let the third proportional be x

Then $6 : 30 :: 30 : x$ or $x = \frac{30\times30}{6} = 150$

Example Find the mean proportional to .9 and 4.9.

Solution

Let the mean proportional be x.

Then, $x = 14+8\sqrt{3} = \sqrt{4.41} = 2.1$

Example Compare the ratios 5 : 6 and 7 : 8

Solution

$5 : 6 = \frac{1}{4}$; $7 : 8 = \frac{1}{4}$ or $\frac{1}{4}, \frac{1}{4} = \frac{20,21}{24}$

Hence $\frac{1}{4} > \frac{1}{4}$ or $7 : 8 > 5 : 6$

Example If a : b = 5 : 9 and b : c = 18 : 35, then find a : c.

Solution

$a : b = 5 : 9$; or $\frac{1}{4} = \frac{1}{9}$; $b : c = 18 : 35$ or $\frac{1}{4} = \frac{3}{60}$

now, $a : c = \frac{1}{4} = \frac{1}{4} \times \frac{1}{4} = \frac{1}{9} \times \frac{3}{60} = \frac{1}{4} = 2 : 7$

Example Two numbers bear a ratio of 2 : 7. If each of them is increased by 14, then their ratio becomes 4 : 7. Find the numbers.

Solution

Let the number be $2x$ and 70 *(always prefer a multiple of ratios with x)*

Then, $\frac{10\times4\times270}{6}$ or $14x + 98 = 28x + 56$ or $14x = 42$ or $x = 3$

Hence the numbers are 6 and 21.

Practice Exercise - 1

Number of Questions: 25 **Suggested time: 15 minutes**

1. A bag contains 50 P, 25 P and 10 P coins in the ratio 5 : 9 : 4, amounting to Rs. 206. Find the number of coins of each type respectively.
(a) 360, 160, 200
(b) 160, 360, 200
(c) 160, 360, 200
(d) 200, 160, 300

2. A mixture contains alcohol and water in the ratio 4 : 3. If 5 litres of water is added to the mixture, the ratio becomes 4 : 5. Find the quantity of alcohol in the given mixture.
(a) 10
(b) 12
(c) 15
(d) 18

3. A sum of Rs. 312 was divided among 100 boys and girls in such a way that the boy gets Rs. 3.60 and each girl Rs. 2.40 the number of girls is
(a) 35
(b) 40
(c) 45
(d) 50

4. Yash Pal Malhotra has a taste for stamp collections. His collection contains US, Indian and British stamps. If the ratio of US to Indian stamps is 5 to 2 and the ratio of Indian to British stamps is 5 to 1, what is the ratio of US to British stamps?
(a) 10 : 5
(b) 15 : 2
(c) 20 : 2
(d) 25 : 2

5. If Rs. 782 be divided into three parts, proportional to $\frac{1}{2}:\frac{2}{3}:\frac{3}{4}$, then the first part is Rs.?

(a) 182

(b) 190

(c) 192

(d) 204

6. Salaries of Richa and Kanchan are in the ratio 2 : 3. If the salary of each is increased by Rs. 4000, the new ratio becomes 40 : 57. What is Kanchan's salary?

(a) 38,000

(b) 46,800

(c) 36,700

(d) 50,000

7. Number of seats for Mathematics, Physics and Biology in Senior Secondary School Siwani are in the ratio 5 : 7 : 8. There is a proposal to increase these seats by 40%, 50% and 75% respectively. What will be the ratio of increased seats?

(a) 1 : 2 : 3

(b) 2 : 3 : 4

(c) 3 : 4 : 5

(d) 4 : 5 : 6

8. A sum of Rs. 4,27,000 is to be divided among three friends Parbhat, Kuldip and Sandeep such that 3 times Parbhat's share, 4 tunes Kuldip's share and 7 times Sandeep's share are all equal. The share of Sandeep is Rs.:

(a) 84,000

(b) 1,40,000

(c) 1,96,000

(d) 2,40,000

9. In Chhatbir zoo, there are Monkeys and Peacocks. If heads are counted, there are 340 heads and if legs are counted there are 1060 legs. How many Peacocks are there?

(a) 120

(b) 150
(c) 170
(d) 180

10. The Students in three classes are in the ratio 2 : 3 : 5. If 20 students join from new session in each class, the ratio changes to 4 : 5 : 7. The total number of students before the increase was?
(a) 120
(b) 150
(c) 200
(d) 100

11. If (0.75 : x :: 5 : 8), then x is equal to:
(a) 1.12
(b) 1.16
(c) 1.20
(d) 1.30

12. The salaries of Peons, Clericals and Managers in M/s Solvato Consultants (P) Ltd are in the ratio of 1 : 2 : 3. The salary of Clericals and Managers together is Rs. 60,000. By what percent is the salary of Managers more than that of Peons?
(a) 100
(b) 200
(c) 300
(d) 600

13. The compounded ratio of (2 : 3), (6 : 11) and (11 : 2) is:
(a) 1 : 2
(b) 2 : 1
(c) 11 : 24
(d) 36 : 121

14. Rs. 5625 is to be divided among Kailash, Raj and Pushap so that Kailash may receive 1/2 as much as Raj and Pushap together receive. Raj receives 1/4 of what Kailash and Pushap together receive. The share of Kailash is more than that of Raj by:
(a) 1125
(b) 1875

(c) 2625
(d) None

15. A leopard takes 3 leaps for every 5 leaps of a stag. If one leap of the leopard is equal to 3 leaps of the stag, the ratio of the speed of the leopard to that of the stag is:
(a) 9 : 5
(b) 2 : 3
(c) 4 : 7
(d) 5 : 6

16. Two numbers are respectively 20% and 50% more than a third number. The ratio of the two numbers is:
(a) 2 : 5
(b) 3 : 5
(c) 4 : 5
(d) 5 : 4

17. Roshan Lal is a milk man and supplies milk from door to door in his village Kotli. Normally his 20 litres of a mixture contains milk and water in the ratio 3 : 1. He has a habit of mixing water to milk. So how much milk must he add to this mixture to get a mixture containing milk and water in the ratio 4 : 1?
(a) 10
(b) 2
(c) 5
(d) 15

18. The sum of three numbers is 490. If the ratio of the first to second is 2 : 3 and that of the second to the third is 5 : 8, then the second number is:
(a) 50
(b) 100
(c) 150
(d) 200

19. In Government College Chandigarh, the ratio of the number of boys to girls is 8 : 5. If there are 880 girls, the total number of students in the college is
(a) 3399

(b) 2496
(c) 2184
(d) 2288

20. The speed of three cars in the ratio 3 : 4 : 5. The ratio between time taken by them to travel the same distance is
(a) 3 : 4 : 5
(b) 5 : 4 : 3
(c) 20 : 15 : 12
(d) 20 : 12 : 15

21. If 7676 is divided into four parts proportional to 7 : 5 : 3 : 4 then the smallest part is:
(a) 1212
(b) 1515
(c) 1616
(d) 1919

22. The third proportional to $(x^2 - y^2)$ and $(x - y)$ is:
(a) $(x + y)$
(b) $(x - 0)$
(c) $(x + y)/(x - y)$
(d) $(x - y)/(x + y)$

23. What is the third proportional to 16 and 36?
(a) 64
(b) 81
(c) 48
(d) 49

24. A cubical block of metal weighs 6 pounds. How much will another cube of the same metal weigh if its sides are twice as long?
(a) 48
(b) 12
(c) 36
(d) Data Inadequate

25. Find the sub-triplicate ratio of 27 : 125?
(a) 9/25

(b) 3/5
(c) 0.36
(d) 0.6

Practice Exercise 1 - Answers with Hints - need based

1 (c)	2 (a)	3 (b)	4 (d)	5 (d)
6 (a)	7 (b)	8 (a)	9 (b)	10 (d)
11 (c)	12 (b)	13 (b)	14 (d)	15 (a)
16 (c)	17 (b)	18 (c)	19 (d)	20 (b)
21 (a)	22 (d)	23 (b)	24 (a)	25 (b)

5. Given ratio = $\frac{1}{2}:\frac{2}{3}:\frac{3}{4} = 6 : 8 : 9$

6. $(2x + 4000) : (3x + 4000) = 40 : 57$

7. Originally, let the number of seats for Mathematics, Physics and Biology be $5x$, $7x$ and $8x$ respectively

Now (140% of $5x$), (150% of $7x$) and (175% of $8x$) : $\Rightarrow$ The ratio $=7x : 21x/2 : 14x$

$\Rightarrow 2 : 3 : 4$

9. Monkeys, Peacocks = M, 340 – M or 4 M + 2 (340 – M) = 1060

10. $\Rightarrow (2x + 20) : (3x + 20) : (5x + 20) = 4 : 5 : 7$ so, $\frac{2x+20}{4} = \frac{3x+20}{5} = \frac{5x+20}{7}$

14. $B = \frac{1}{4}(A+C) \Rightarrow A+C = 4B$

$B = \frac{1}{4}(A+C) \Rightarrow A+C = 4B$

A + B + C = 5625 using this information in above equations we can get B, A and C

22. Let the third proportional to $(x^2 - y^2)$ and $(x - y)$ be z. Then,

$(x^2 - y^2) : (x - y) :: (x - y) : z$

$$\Rightarrow (x^2 - y^2) \times z = (x - y)^2$$

$$\Rightarrow z = \frac{(x-y)^2}{(x^2-y^2)} = \frac{(x-y)}{(x+y)}$$

MIXTURE & ALLIGATIONS

16

Alligation enables us

- ***To find the average value of a mixture when the prices of two or more varieties are mixed together.***
- ***To find the proportion in which they are mixed and,***
- ***To find the proportion in which these - at given prices - must be mixed to produce a mixture at a given price.***

$$\frac{\text{Amount of Cheap}}{\text{Amount of Costlier}} = \frac{\text{Price of Costlier} - \text{the Average Price}}{\text{Mean Price} - \text{Price of Cheaper}}$$

Example A grocer mixes 18 kg of tea costing Rs. 6.20/kg with 22 kg costing Rs. 8.40/kg. Find the cost of the mixture per kg.

Solution

Cost of 18 kg @ 6.20 = 6.20 × 18 = Rs. 111.60

Cost of 22 kg @ 8.40 = 8.40 × 22 = Rs. 184.80

Cost 40 kg of the mixture = Rs. 111.60 + Rs. 184.80 = Rs. 296.40

∴ Cost of 1 kg of the mixture = Rs. 7.41

Example In what proportion must a grocer mix teas @ 1.22/kg and Rs.1.94/kg so as to make a mixture worth Rs. 1.66/kg.

Solution

Using the formula given above,

$$\frac{194-166}{166-122} = \frac{\text{Amount of Cheap tea}}{\text{Amount of dear tea}} = \frac{7}{11}$$

Example A merchant has 40 kg sugar, part of which he sells at 12% gain and the rest at 28%. He gains 18% on the whole. Find the quantity sold at 12% profit.

Solution

Going by the above Example No. 2 we get the ratio of mixtures as 5 : 3 Therefore the quantity sold at 12% profit is 25 kg.

Example Three glasses each of equal size were filled with a mixture of wine and soda. The proportion of wine to soda in each glass is as follows: **2 : 3; 3 : 4; & 4 : 5.**

The contents of the three glasses are emptied into a single vessel. What is the proportion of spirit and water in it.

Solution

Let each glass be of the size 315 ml (LCM of the sums of the ratio i.e., 5, 7, 9)

		Wine		Soda	
Therefore	1st glass	126		189	
	2nd glass	135		180	
	3rd glass	140		175	
	Total Qty	401		544	
Hence the final quantity is in the ratio		**401**	**:**	**544**	**Answer**

Example From a can of super quality oil containing 96 litres, 12 litres are drawn out, and the can is filled up with inferior quality of oil. The process is repeated thrice, finally what shall the number of litres of super oil left in the can?

Solution

8 Litres = $\frac{3}{60}$ of 96 Litres = $\frac{1}{9}$ of the can.

$\therefore$ The quantity of super oil after 1^{st} operation = $\left(\frac{7}{8}\right)$ of the whole.

$\therefore$ The quantity after the 2^{nd} operation = $\left(\frac{7}{8}\right)^2$ of the whole.

$\therefore$ The quantity after the 3[rd] operation $= \left(\frac{7}{8}\right)^2$ of the whole $= \frac{343}{512}$ of the whole.

$\therefore$ The number of litres left $= \frac{343}{512}$ of $96 = 45\frac{17}{2}$.

A very important result: If a vessel contains 'a' litres wine, and if 'b' litres be withdrawn and replaced then if 'b' litres of the mixture be withdrawn and replaced, and the operation repeated 'n' times in all, then:-

$$\frac{\text{Wine left in vessel after nth operation}}{\text{Whole quantity of wine in vessel}} = \left(\frac{a-b}{a}\right)^n$$

Practice Exercise - 1

Number of Questions: 25 **Suggested time: 25 minutes**

1. Tea worth Rs. 126 per kg are mixed with a third variety in the ratio 1 : 1 : 2. If the mixture is worth Rs. 153 per kg, the price of the third variety per kg will be:
(a) Rs. 169.50
(b) Rs. 170.40
(c) Rs. 175.50
(d) Rs. 180.12

2. A can contains a mixture of two liquids A and B in the ratio 7 : 5. When 9 litres of mixture are drawn off and the can is filled with B, the ratio of A and B becomes 7 : 9. How many litres of liquid A contained by the can initially?
(a) 19
(b) 22
(c) 21
(d) 23

3. 8 litres are drawn from a cask full of wine and is then filled with water. This operation is performed three more times. The ratio of the quantity of wine now left in cask to that of the water is 16 : 65. How many litres of wine the cask hold originally?
(a) 28
(b) 24
(c) 31
(d) 43

4. A vessel is filled with liquid, 3 parts of which are water and 5 parts of syrup. How much of the mixture must be drawn off and replaced with water so that the mixture may be half water and half syrup?
(a) 1/3
(b) 1/4
(c) 1/5
(d) 1/7

5. A merchant has 1000 kg of sugar part of which he sells at 8% profit and the rest at 18% profit. He gains 14% on the whole. The Quantity sold at 18% profit is _____ kg

(a) 402

(b) 569

(c) 600

(d) 644

6. A container contains 40 litres of milk. From this container 4 litres of milk was taken out and replaced by water. This process was repeated further two times. How much milk is now contained by the container (in litres).

(a) 26.43

(b) 27.33

(c) 28.13

(d) 29.16

7. The ratio of expenditure and savings is 3 : 2. If the income increases by 15% and the savings increases by 6%, then by how much percent should his expenditure increases?

(a) 23.23

(b) 21

(c) 12.19

(d) 27

8. How many kilograms of sugar costing Rs. 9 per kg must be mixed with 27 kg of sugar costing Rs. 7 per kg so that there may be a gain of 10% by selling the mixture at Rs. 9.24 per kg?

(a) 46

(b) 49

(c) 51

(d) 63

9. A milk man sells the milk at the cost price but he mixes the water in it and thus he gains 9.09%. The quantity of water in the mixture of 1 litre is:

(a) 83.33

(b) 90.95

(c) 97.09
(d) 112.04

10. The cost of Type 1 rice is Rs. 15 per kg and Type 2 rice is Rs.20 per kg. If both Type 1 and Type 2 are mixed in the ratio of 2 : 3, then the price Rs./kg of the mixed variety of rice is
(a) 19.37
(b) 19.12
(c) 18
(d) 18.30

11. The ratio of petrol and kerosene in the container is 3 : 2 when 10 litres of the mixture is taken out and is replaced by the kerosene, the ratio become 2 : 3. Then total quantity of the mixture in the container is how many litres?
(a) 29
(b) 30
(c) 43
(d) 31.26

12. Find the ratio in which rice at Rs. 7.20 a kg be mixed with rice at Rs. 5.70 a kg to produce a mixture worth Rs. 6.30 a kg.
(a) 5 : 3
(b) 2 : 3
(c) 5 : 4
(d) 4 : 7

13. The diluted wine contains only 8 litres of wine and the rest is water. A new mixture whose concentration is 30%, is to be formed by replacing wine. How many litres of mixture shall be replaced with pure wine if there was initially 32 litres of water in the mixture?
(a) 4.25
(b) 5
(c) 9
(d) 11.56

14. From a container, 6 litres milk was drawn out and was replaced by water. Again 6 litres of mixture was drawn out and was replaced by the water. Thus the quantity of milk and water in the container after these two operations is 9 : 16. The quantity of mixture is:

(a) 15

(b) 19

(c) 35

(d) 36

15. A man travelled a distance of 80 km in 7 hours partly on foot at the rate of 8 km per hour and partly on bicycle at 16 km/hr. Find the distance (km) travelled on foot?

(a) 24

(b) 32

(c) 40

(d) None

16. The average weight of boys in a class is 30 kg and the average weight of girls in the same class is 20 kg. If the average weight of the whole class is 23.25 kg, what could be the possible strength of boys and girls respectively in the same class?

(a) 14, 16

(b) 13, 27

(c) 17, 37

(d) 19 : 27

17. In a mixture of milk and water, there is only 26% water. After replacing the mixture with 7 litres of pure milk, the percentage of milk in the mixture become 76%. The quantity of mixture (litre) is:

(a) 65

(b) 91

(c) 39

(d) 78

18. In what proportion water must be added to pure spirit to gain 20% by selling at the cost price?

(a) 1/4

(b) 1/5

(c) 1/6

(d) Data inadequate

19. From a container of wine, a thief has stolen 15 litres of wine and replaced it with same quantity of water. He repeated the same process and in three attempts the ratio of wine and water became 343 : 169. The initial amount of wine (Litres) in the container was:

(a) 75

(b) 104

(c) 105

(d) 120

20. A mixture of 150 litres of wine and water contains 20% water. How many litres of water should be added so that water becomes 25% of the new mixture?

(a) 10

(b) 25

(c) 35

(d) 45

21. In the 75 litres of mixture of milk and water, the ratio of milk and water is 4 : 1. The quantity of water required to make the ratio of milk and water 3 : 1 is

(a) 4

(b) 3

(c) 8

(d) 5

22. A jar was full with honey. Mrs Sarla used to draw out 20% of the honey from the jar and replaced it with sugar solution. She repeated the same process 4 times and thus there was only 512 gm of honey left in the jar, the rest part of the jar was filled with the sugar solution. The initial amount of honey in the jar was ______ kg?

(a) 1.25

(b) 1.05

(c) 1.5

(d) 1.75

23. The ratio of water and alcohol in two different containers is 2 : 3 and 4 : 5. In what ratio we are required to mix the mixtures of two containers

in order to get the new mixture in which the ratio of alcohol and water be 7 : 5?
(a) 7 : 4
(b) 5 : 8
(c) 8 : 5
(d) 5 : 7

24. From a tank of petrol, which contains 200 litres of petrol, Mr Satpal replaces each time with kerosene when he sells 40 litres of petrol (or mixture). Every time he sells out only 40 litres of petrol (pure or impure). After replacing the petrol with kerosene 4^{th} time, the total amount of kerosene in the mixture is how many litres
(a) 81.92
(b) 96
(c) 118.08
(d) 112.26

25. Four kilograms of a metal contains 1/5 copper and rest in Zinc. Another 5 kg of metal contains 1/6 copper and rest in Zinc. The ratio of Copper and Zinc into the mixture of these two metals:
(a) 49 : 221
(b) 39 : 231
(c) 94 : 181
(d) 231 : 49

Practice Exercise 1 - Answers with Hints - need based

1 (c)	2 (c)	3 (b)	4 (c)	5 (c)
6 (d)	7 (b)	8 (d)	9 (a)	10 (a)
11 (b)	12 (b)	13 (b)	14 (a)	15 (b)
16 (b)	17 (b)	18 (a)	19 (d)	20 (a)
21 (d)	22 (a)	23 (b)	24 (c)	25 (a)

1. Since first second varieties are mixed in equal proportions, so their average price

= Rs.(126 + 135/2) = Rs.130.50

So, the mixture is formed by mixing two varieties, one at Rs. 130.50 per kg and the other at say,

Rs. x per kg in the ratio 2 : 2, i.e., 1 : 1. We have to find x.

$x - 153/22.50 = 1$

$\Rightarrow x - 153 = 22.50 \Rightarrow x = 175.50$

Hence, price of the third variety = Rs.175.50 per kg.

2. Suppose the can initially contains $7x$ and $5x$ litres of mixtures A and B respectively

Quantity of A in mixture left = $\left(7x - \frac{7}{12} \times 9\right) = \left(7x - \frac{21}{4}\right)$ litres.

Quantity of B in mixture left = $\left(5x - \frac{5}{12} \times 9\right) = \left(5x - \frac{15}{4}\right)$ litres.

$$\frac{\left(7x - \frac{21}{4}\right)}{\left(5x - \frac{15}{4} + 9\right)} = \frac{7}{9} \Rightarrow \frac{28x - 21}{20x + 21} = \frac{7}{9} \Rightarrow 252x - 189 = 140x + 147$$

$\Rightarrow x = 3$

So, the can contained 21 litres of A.

3. Let the quantity of the wine in the cask originally be x litres. Then, quantity of wine left in cask after 4 operations = $\left[x\left(1 - \frac{8}{x}\right)^4\right]$ litres

$$\therefore \left[x\left(1 - \frac{8}{x}\right)^4\right] = \frac{16}{81}$$

6. Amount of milk left after 3 operations

$$\left[40\left(1 - \frac{4}{90}\right)^3\right] = \left(40 \times \frac{9}{10} \times \frac{9}{10} \times \frac{9}{10}\right) = 29.16 \text{ litres}$$

9. Profit (%) = 9.09% = $\frac{1}{11}$

Since the ratio of water and milk is 1 : 11,

Therefore the ratio of water is to mixture = 1 : 12

Thus the quantity of water in mixture of 1 liter = $1000\times\frac{1}{12}$ = 83.33 ml

10. Let 2 kg @ Rs.15/kg and 3 kg @ Rs.20/kg is mixed. So total cost of 5 kg = 2 × 15 + 3 × 20 = Rs. 90.

Hence mixed rate is Rs. 18/kg.

14. $\therefore \frac{x\left(1-\frac{6}{x}\right)^2}{x} = \frac{9}{25} \Rightarrow x = 15 \text{ litres}$

16. 'B' boys 'G' girls i.e., 30B + 20G = 23.25 (B + G); Here we can find B/G =?

20. Let 'x' litre of water is added. Therefore, $(30 + x) = 25\% \ (150 + x)$

24. The amount of petrol left after 4 operations = $200\times\left(1-\frac{40}{200}\right)^4 = 200\times\left(\frac{4}{5}\right)^4$

$= 200\times\frac{256}{625} = 81.95 \text{ L}$; Hence the amount of kerosene = 200 – 81.92 = 118.08 litres

PROFIT & LOSS

17

PROFIT

When an article is sold for more than what it costs we say there is a profit or gain.

e.g. If C.P. = Rs. 1200, S.P. = Rs. 1450, then Profit = Rs. 250.

LOSS

When an article is sold for less than what it costs we say there is a Loss

e.g. If C.P. = Rs. 2200, S.P. = Rs. 2080, then Loss = Rs. 120

NET PRICE

The price after deducting the trade discount is called the net price.

Formula

$$\text{Profit} = S.P. - C.P.$$

$$\text{Loss} = C.P. - S.P.$$

$$\text{Profit \%} = \frac{\text{Profit}}{\text{C.P}} \times 100$$

$$\text{Loss \%} = \frac{\text{Loss}}{\text{C.P}} \times 100$$

$$\text{Net Price} = \text{Market Price} - \text{Discount}$$

$$\text{SP} = \frac{\text{CP}(100 + \text{gain \%})}{100} \quad \text{or} \quad = \frac{\text{CP}(100 - \text{loss \%})}{100}$$

TRADE DISCOUNT

The reduction made on the "market price" of an article, is called trade discount.

e.g. If market price = Rs. 800 and discount = 15%, then

$\text{Discount} = \frac{45}{100} \times 800 = \text{Rs. } 120$

Note: Profit or Loss is always calculated on Cost Price i.e., CP and not on Selling Price i.e., SP

Remember

(1) If two items are sold, each at Rs. x, first one at a gain of k% and the other at a loss of k%, there is an overall loss given by $\frac{k^2}{100}\%$

(2) If C. P. of both the items is the same and % loss and gain are equal, then net loss or profit is zero.

Example A shopkeeper uses 960 gram against a weight of One. Find his gain % Gain % = (Margin) × 100/Actual Weight

Solution

$\% \text{ gain} = \frac{39 \times 5 \times 12}{30 \times 6} = 4\frac{1}{4}\% \text{ Ans}$

Example A man purchased some bananas @ 25/ Re and equal # @ 20/ Re. Both together mixed were sold @ 45/ Rs. 2. Gain or loss%?

Solution

Step 1 buy LCM of lots i.e., 25, 20, 45

Step 2 Find total C.P.

Step 3 Find total SP Find Gain/Loss%

$\therefore \text{Loss} = 1\frac{19}{81}\% \quad \text{Ans}$

Example A sells some articles to B at a gain of 20%, while B to C at 25% gain. If C pays Rs. 975/-, How much did A pay for it?

Solution

Let CP of A = 100; SP of A = CP of B = 120

SP of B = CP of C = $1000 \times \frac{1}{12}$ = 150; If C pays Rs. 150, CP of A = 100

If C pays Rs. 975, CP of A = 650 Ans

Example Packets of ball pens were purchased for Rs. 450, One-third of the lot was sold at 10% loss. To gain overall 20% the remaining lot should be sold in what way?

Solution

Cost of one-third of the lot = Rs. 150

The selling price = Rs. 150 × $\frac{45}{100}$ = Rs. 135

The total expected SP = Rs. 450 × $\frac{45}{100}$ = Rs. 540

Hence the selling price of the remaining two-third lot should be = Rs. 405.

But the cost of this lot consisting two-third of the stock is = Rs. 300

∴ Gain = Rs. 105;

∴ Gain % is 35 Ans

Example A trader alloys a discount of 5% for cash payment. How much % above the cost price must be he marks his goods to make a profit of 10%?

Solution

Let the CP be Rs 100; His selling price after profit is = 110

As he charges only 95% of marked price.

95% of the MP = Rs. 110. MP = $\frac{45}{100}$ × 110 = Rs. 115 $\frac{17}{2}$ = Rs. 115.79 Ans

Practice Exercise - 1

Number of Questions: 25 **Suggested Time: 20 minutes**

1. Sumatra buys an old scooter for Rs. 4700 and spends Rs. 800 on its repairs. If he sells the scooter for Rs. 5800 what is his gain percent?
(a) 12
(b) 10
(c) 4 – 4/7
(c) 5 – 5/11

2. The cost price of 20 articles is the same as the selling price of x articles. If the profit is 25%, find the value of *x*
(a) 15
(b) 25
(c) 18
(c) 16

3. If selling price is doubled, the profit triples. What is the profit percent?
(a) 100
(b) 105 1/3
(c) 66 – 2/3
(c) 120

4. In Jindal Crockery Shop, the profit is 320% of the cost. If the cost increases by 25% but the selling price remains constant, find out approximately what percentage of the selling price is the profit?
(a) 250
(b) 100
(c) 70
(c) 30

5. Ram Din bought bananas at 6 for a rupee. How many for a rupee must he sell to gain 20%?
(a) 3
(b) 4

(c) 5

(c) 6

6. The percentage profit earned by selling an item for Rs. 1920 is equal to the percentage loss incurred by selling the same item for Rs. 1280. At what price should the item be sold to make 25% profit?

(a) 2200

(b) 2000

(c) 3000

(c) Data Inadequate

7. Sumit Jindal expects a gain of 22.5% on his cost price. If in a week, his sale was of Rs. 392, what was his profit?

(a) 90

(b) 80

(c) 72

(c) Data inadequate

8. Gulshan buys a scooter for Rs. 1400 and sells it at a loss of 15%. What is the selling price of the scooter?

(a) 1240

(b) 1190

(c) 1090

(c) 1130

9. Murari purchased 20 dozens of toys at the rate of Rs. 375 per dozen. He sold each one of them at the rate of Rs. 33 Find out his profit percentage.

(a) 3.5

(b) 5.6

(c) 4.1

(c) 3.4

10. Some items were bought at 6 items for Rs. 5 and sold at 5 items for Rs. 6 What is the gain percentage?

(a) 44

(b) 33 – 1/3

(c) 30

(c) 28 – 4/7

11. On selling 17 balls at Rs. 720, there is a loss equal to the cost price of 5 balls. What is the cost price of a ball?

(a) 43

(b) 60

(c) 55

(c) 34

12. When an item is sold for Rs. 18,700 by Nirjhar, he loses 15%. At what price should that plot be sold to get a gain of 15%?

(a) 25,100

(b) 24,200

(c) 25,300

(c) 21,200

13. 100 oranges were bought at the rate of Rs. 350 and sold at the rate of Rs. 48 per dozen. What is the percentage of profit or loss?

(a) 25,100

(b) 24,200

(c) 25,300

(c) 21,200

14. Mehta Radios sell one radio for Rs. 840 at a gain of 20% and another for Rs. 960 at a loss of 4%. What total gain or loss percentage?

(a) 5

(b) 6

(c) 6 – 12/17

(c) 5 – 15/17

15. Chawla Sons mix 26 kg of rice at Rs. 20 per kg with 30 kg of rice of other variety at Rs. 36 per kg and sell the mixture at Rs. 30 per kg. What is the profit percentage?

(a) 5

(b) 6

(c) 6 – 12/17

(c) 5 – 15/17

16. If a material is sold for Rs. 34.80, there is a loss of 25%. Find out the cost price of the material?

(a) 46.40

(b) 44

(c) 42

(c) 47.20

17. M/s Mehta Fruits and Vegetables sell apples at the rate of Rs. 9 per kg and thereby lost 20%. At what price (per kg), they should have sold them to make a profit of 5%?

(a) 11.32

(b) 11

(c) 12

(c) 11.81

18. Renu gives 12% additional discount on the discounted price, after giving an initial discount of 20% on the labelled price of an item. The final sale price of the item is Rs. 704. Find out the labelled price?

(a) 1000

(b) 2000

(c) 1200

(c) 920

19. Amritdhari sells two houses at the rate of Rs. 1.995 lakh each. On one he gains 5% and on the other, he loses 5%. What is his gain or loss percent in the whole transaction?

(a) 0.25

(b) 0.3

(c) 0.4

(c) 0.5

20. Milkha Singh purchased a machine for Rs. 80,000. After spending Rs. 5000 on repair and Rs. 1000 on transport he sold it with 25% profit. What price did he sell the machine?

(a) 1,07,000

(b) 1,07,500

(c) 1,08,500

(c) None

21. By selling an item for Rs. 15, a trader loses one sixteenth of what it costs him. The cost price of the item is

(a) 14

(b) 15

(c) 16

(c) 17

22. A shopkeeper sells his goods at cost price but uses a weight of 800 gm instead of kilogram weight. What is his profit percentage?

(a) 18

(b) 40

(c) 25

(c) 20

23. Prasanth bought a car and paid 10% less than the original price. He sold it with 30% profit on the price he had paid. What percentage of profit did he earn on the original price?

(a) 17

(b) 16

(c) 18

(c) 14

24. If a seller reduces the selling price of an item from Rs. 400 to Rs. 380, his loss increases by 2%. What is the cost price of the item?

(a) 17

(b) 16

(c) 18

(c) 14

25. A trader keeps the marked price of an item 35% above its cost price. The percentage of discount allowed to gain 8% is

(a) 30

(b) 25

(c) 20

(d) None

Practice Exercise 1 - Answers with Hints

1 (d)	2 (d)	3 (a)	4 (c)	5 (c)
6 (c)	7 (d)	8 (b)	9 (b)	10 (a)
11 (b)	12 (c)	13 (c)	14 (d)	15 (b)
16 (a)	17 (d)	18 (a)	19 (a)	20 (b)
21 (c)	22 (c)	23 (a)	24 (a)	25 (d)

4. Let cost price = 100; Then, profit = 320; selling price = 100 + 320 = 420

new cost price = 125; profit = 420 − 125=295; Required percentage = 295 × 100/420 ≈ 70%

6. Let CP = x

Therefore, $(1920 - x)/100x = (x - 1280)/100x \Rightarrow 1920 - x = x - 1280 \Rightarrow x = 1600$

Required selling price = 1600 (1.25) = 2000

9. cost price of 12 toys = 375; selling price of 12 toys = 33 × 12 = 396

profit = 396 − 375 = 21; profit percentage =21 × 100/375 = 5.6%

11. loss = (cost price of 17 balls − selling price of 17 balls)

= (cost price of 17 balls − 720)

⇒ (cost price of 17 balls − 720) = cost price of 5 balls

⇒ cost price of 12 balls = 720

⇒ cost price of 1 ball = 60

18. (Labelled price) × 80% × 88% = 704

22. Gain/Loss% = (Margin) × 100/Actual weight

= 200 × 100/800 = 25%

24. 2% of the cost price = 400 − 380 = 20; cost price = 20 × 50 = 1000

Practice Exercise - 2

Number of Questions: 25 **Suggested Time: 20 minutes**

1. Anita bought a computer with 15% discount on the labelled price. She sold the computer for Rs. 2880 with 20% profit on the labelled price. At what price did she buy the computer?
(a) 3000
(b) 2080
(b) 2040
(d) 2000

2. An item was sold for Rs. 27.50 with a profit of 10%. If it was sold for Rs. 25.75, what would have been the percentage of profit or loss?
(a) 3
(b) 2
(b) 4
(d) 5

3. If selling price of an article is Rs. 250, profit percentage is 25%. Find the ratio of the cost price and the selling price.
(a) 5 : 3
(b) 3 : 5
(b) 4 : 5
(d) 5 : 4

4. A material is purchased for Rs. 600. If one fourth of the material is sold at a loss of 20% and the remaining at a gain of 10%, find out the overall gain or loss percentage?
(a) 312
(b) 212
(b) 3
(d) 2

5. Bittu Malhotra buys pencils at 9 for Rs. 16 and sells them at 11 for Rs. 22. Find out his loss or gain **percentage?**

(a) $12\frac{1}{2}$
(b) 12
(b) 14
(d) 11 − 2/3

6. A reduction of 10% in the price of a pen enabled a trader to purchase 9 more for Rs. 540. What is the reduced price of the pen?
(a) 8
(b) 6
(b) 5
(d) 4

7. Anand Parkash purchases two books at Rs. 300 each. He sold one book 10% gain and other at 10% loss. What is the total loss or gain in percentage?
(a) 10
(b) 1
(b) 5
(d) 4

8. Roshan Lal purchases the milk at Rs. x per litre and sells it at Rs. $2x$ per litre still he mixes 2 litres water with every 6 litres of pure milk. What is the profit percentage?
(a) 116
(b) 166.66
(b) 60
(d) 100

9. Naresh Kumar mixes 20 kg of rice at Rs. 39 per kg with 36 kg of rice of other variety at Rs. 45 per kg and sells the mixture at Rs. 45 per kg. His profit percent is:
(a) 5
(b) 8
(b) 10
(d) At par

10. By selling 45 lemons for Rs. 40, a man loses 20%. How many should he sell for Rs. 24 to gain 20% in the transaction?

(a) 16

(b) 18

(b) 20

(d) 22

11. Roor Chand cheats to the extent of 10% while buying and selling, by using false weights. His total gain is.

(a) 20

(b) 21

(b) 22

(d) 23

12. The cost price of 25 articles is the same as the selling price of x articles. If the profit is 33.33%, then the value of x is:

(a) 15

(b) 20

(b) 50

(d) 25

13. The percentage profit earned by selling an article for Rs. 1920 is equal to the percentage loss incurred by selling the same article for Rs. 1280. At what price should the article be sold to make 25% profit?

(a) 2000

(b) 2200

(b) 2400

(d) Data inadequate

14. If books bought at prices ranging from Rs. 200 to Rs. 350 are sold at prices ranging from Rs. 300 to Rs. 425, what is the greatest possible profit that might be made in selling eight books?

(a) 600

(b) 1200

(b) 1800

(d) None

15. If the cost price of 12 pens is equal to the selling price of 8 pens, the gain percent is?

(a) 12

(b) 30

(b) 500

(d) 60

16. Tarun got 30% concession on the labelled price of an article and sold it for Rs. 8750 with 25% profit on the price he bought. What was the labelled price?

(a) 10000

(b) 12000

(b) 13000

(d) 14000

17. A man buys oranges at Rs. 5 a dozen and an equal number at Rs. 4 a dozen. He sells them at Rs. 5.50 a dozen and makes a profit of Rs. 50. How many dozens of oranges does he buy?

(a) 30

(b) 40

(b) 50

(d) 60

18. The marked price of an article is increased by 25% and the selling price is increased by 16.66%, then the amount of profit doubles. If the original marked price be Rs. 400 which is greater than the corresponding cost price by 33.33%, what is the increased selling price?

(a) 240

(b) 360

(b) 420

(d) 600

19. Due to reduction of 25% in price of oranges a customer can purchase 4 oranges more for Rs. 16. What is original price of an orange?

(a) 1.00

(b) 1.33

(b) 1.50
(d) 1.60

20. A man bought an article and sold it at a gain of 5%. If he had bought it at 5% less and sold it for Re 1 less, he would have made a profit of 10%. The C.P. of the article was
(a) 100
(b) 150
(b) 200
(d) 250

21. A trader sold an article at a loss of 5% but when he increased the selling price by Rs. 65 he gained 3.33% on the cost price. If he sells the same article at Rs. 936, what is the profit percentage?
(a) 15
(b) 16.66
(b) 20
(d) data inadequate

22. In what ratio should water and wine be mixed so that after selling the mixture at the cost price a profit of 33.33% is made?
(a) 1 : 4
(b) 1 : 3
(b) 2 : 3
(d) 3 : 4

23. Bharpur Singh purchased 120 reams of paper at Rs. 80 per ream. He spent Rs. 280 on transportation, paid octroi at the rate of 40 paise per ream and paid Rs. 72 to the coolie. If he wants to have a gain of 8%, what must be the selling price per ream?
(a) 90
(b) 89
(b) 87.5
(d) 86.12

24. A dealer sold two of his cattle for Rs. 500 each. On one of them he lost 10% on the other, he gained 10%. His gain or loss percent in the entire transaction was:

(a) 10% loss
(b) 1% loss
(b) 1% gain
(d) At par

25. A person incurs a loss of 5% be selling a watch for Rs. 1140. At what price should the watch be sold to earn 5% profit (Rs.).
(a) 1200
(b) 1230
(b) 1260
(d) 1290

Practice Exercise - 2 Answers with Hints

1 (b)	2 (a)	3 (b)	4 (b)	5 (a)
6 (b)	7 (b)	8 (b)	9 (b)	10 (b)
11 (b)	12 (b)	13 (a)	14 (b)	15 (b)
16 (a)	17 (b)	18 (b)	19 (b)	20 (b)
21 (b)	22 (b)	23 (a)	24 (b)	25 (b)

1. Labelled price = 2880 × 100/120 = 2400; His selling price = 2400 × 85/100 = 2040

2. CP = 27.50 × 100/110 = 25.00

New SP = 25.75

Gain% = 3%

4. SPs = 150 × 0.80 + 450 × 1.10 = 615;

Gain% = $2\frac{1}{2}$%

6. Reduced price of 9 = 10% of 540

7. If CPs are same and the rates are same, it is always NO GAIN NO LOSS.

8. Let the cost price of 1 litre pure milk be Re.1;

8 litre mixture → SP → 8 × 2 = Rs. 16

Profit%= $\left[\frac{16-6}{6}\right]\times 100=\frac{1000}{6}=166.66\,\%$

12. Gain/Loss% = (Difference of items) × 100/SP items.

14. Greatest Selling Price = Rs. (425 ∗ 8) = Rs. 3400.

Greatest Selling Price = Rs. (425 . 8) = Rs. 3400.
Required profit = Rs. (3400 − 1600) = Rs. 1800.

17. Cost price of 2 dozens = Rs. 4 + Rs. 5 = Rs. 9; Selling Price 2 dozens = Rs. 11

Profit of Rs. 2 is made on 2 dozens; Rs. 50 on 50 dozens.

20. Let CP_1 = 100; SP_1 = 105.00; CP_2 = 95; SP_2 = 104.50

Difference is 0.50 then CP = Rs. 100

Difference is 1.00 then CP = Rs. 200

23. Total investment = Rs. (120 ∗ 80 + 280 + (40/100) ∗ 120 + 72)= Rs. (9600 + 280 + 48 + 72) = Rs. 10000.

Sell price of 120 reams = 108% of Rs. 10000 = Rs. 10800.

Sell Price per ream = Rs. [10800/120] = Rs. 90.

PARTNERSHIPS

18

Important Concepts and Formulas - Partnerships

(1) Two or more people can get together to do business by pooling resources. The deal is known as partnership. All the people who have invested money in the partnership are called partners. A partner who manages the business is a working partner. A partner who simply invests the money is a sleeping partner.

(2) If investments of all the partners are for the same time period, the gain or loss is distributed among the partners in the ratio of their investments.

Assume P and Q invest Rs. a and Rs. b respectively for a year in a partnership, then at the end of the year, (P's share of profit) : (Q's share of profit) = a : b

(3) If investments are for different time periods, then equivalent capitals are calculated for a unit of time.

Assume A invests Rs. a for x months and B invests Rs. b for y months then, (A's share of profit) : (B's share of profit)= $ax : by$.

Practice Exercise - 1

Number of Questions: 25 **Suggested Time: 25 minutes**

1. Mr Mohan and Mr Ram invest Rs. 21,000 and Rs. 17,500 respectively in a business. At the end of the year, they make a profit of Rs. 26,400. What is the share of Mohan in the profit?
(a) Rs. 14400
(b) Rs. 26400
(c) Rs. 12000
(c) Rs. 12500

2. Mr Bhushan starts a business with Rs. 45,000. Mr Avinash joins in the business after 3 months with Rs. 30000. What will be the ratio in which they should share the profit at the end of the year?
(a) 1 : 2
(b) 2 : 1
(c) 1 : 3
(c) 3 : 1

3. Suresh started a business with Rs. 20,000. Ramesh joined him after 4 months with Rs. 30,000. After 2 more months, Suresh withdrew Rs. 5,000 of his capital and 2 more months later, Ramesh brought in Rs. 20,000 more. What should be the ratio in which they should share their profits at the end of the year?
(a) 21 : 32
(b) 32 : 21
(c) 12 : 17
(c) 17 : 12

4. Tarun started a business with Rs. 25000 and after 4 months, Kiran joined him with Rs. 60000. Tarun received Rs. 58000 including 10% of profit as commission for managing the business. What amount did Kiran receive?
(a) 75000
(b) 70000

(c) 72000
(c) 78000

5. Atam and Badri started a partnership business investing Rs. 20,000 and Rs. 15,000 respectively. Chetan joined them with Rs. 20,000 after six months. Calculate Badri's share in total profit of Rs. 25,000 earned at the end of 2 years from the starting of the business?
(a) 7500
(b) 8500
(c) 9000
(c) 8000

6. Amrit starts a business with a capital of Rs. 85,000. Banarsi joins in the business with Rs. 42500 after some time. For how much period does Banarsi join, if the profits at the end of the year are divided in the ratio of 3 : 1?
(a) 5 months
(b) 6 months
(c) 7 months
(c) 8 months

7. Akash starts a business with Rs. 40,000. After 2 months, Bhoomi joined him with Rs. 60,000. Chhotu joined them after some more time with Rs. 1,20,000. At the end of the year, out of a total profit of Rs. 3,75,000, Chhotu gets Rs. 1,50,000 as his share. How many months after Bhoomi joined the business, did Chhotu join?
(a) 4 months
(b) 5 months
(c) 6 months
(c) 7 months

8. Ahmed and Bismil invest in a business in the ratio 3 : 2. Assume that 5% of the total profit goes to charity. If Ahmed's share is Rs. 855, what is the total profit?
(a) 1400
(b) 1500
(c) 1600

(c) 1200

9. Amar, Bishnu and Chaitanya invest in a partnership in the ratio: 72 : 43 : 65. After 4 months, Amar increases his share 50%. If the total profit at the end of one year is Rs. 21,600, then what is Bishnu's share in the profit?

(a) Rs. 2000

(b) Rs. 3000

(c) Rs. 4000

(c) Rs. 5000

10. Akbar, Birbal and Chintu jointly thought of engaging themselves in a business venture. It was agreed that Akbar would invest Rs. 6,500 for 6 months, Birbal Rs. 8,400 for 5 months and Chintu Rs. 10,000 for 3 months. Akbar wants to be the working member for which, he was to receive 5% of the profits. The profit earned was Rs. 7,400. What is the share of Birbal in the profit.

(a) 2660

(b) 1000

(c) 2300

(c) 4000

11. Amrita, Balbir and Chintu subscribe Rs. 50,000 for a business. If Amrita subscribes Rs. 4,000 more than Balbir and Balbir Rs. 5,000 more than Chintu, out of a total profit of Rs. 35,000, what will be the amount Amrita receives?

(a) 14,200

(b) 14,700

(c) 14,800

(c) 14,500

12. A, B and C rent a pasture. If A puts 10 oxen for 7 months, B puts 12 oxen for 5 months and C puts 15 oxen for 3 months for grazing and the rent of the pasture is Rs. 1750, then how much amount should C pay as his share of rent more than B?

(a) 450

(b) 350

(c) 550

(c) None of these.

13. A and B entered into a partnership with capitals in the ratio 4 : 5. After 3 months, A withdrew 14 of his capital and B withdrew 15 of his capital. At the end of 10 months, the gain was Rs. 760. What is A's share in the profit?

(a) 310

(b) 330

(c) 370

(c) 350

14. Amrik starts a business with Rs. 3500. After 5 months, Balbir joins with Amrik as his partner. After a year, the profit is divided in the ratio 2 : 3. Balbir's contribution in the capital is

(a) 7000

(b) 8000

(c) 9000

(c) 10000

15. A, B and C decided to share the profit in a business in the ratio 5 : 7 : 8 and partnered for 14, 8 and 7 months respectively. Find the ratio of their investments?

(a) 10 : 12 : 14

(b) 12 : 24 : 28

(c) 20 : 22 : 12

(c) 20 : 49 : 64

16. Ananya and Baarbi started a partnership business investing capital in the ratio of 3 : 5. Casmi joined in the partnership after six months with an amount equal to that of Barbi. At the end of one year, the profit should be distributed among them in --- proportion.

(a) 10 : 5 : 4

(b) 5 : 3 : 4

(c) 3 : 4 : 5

(c) 6 : 10 : 5

17. A and B partner in a business. A contribute 14 of the capital for 15 months and B received 23 of the profit. For how long B's money was used?

(a) 12 months

(b) 10 months

(c) 14 months

(c) 16 months

18. A, B and C started a partnership business by investing Rs. 27,000, Rs. 72,000, Rs. 81,000 respectively. At the end of the year, the profit was distributed among them and C got Rs. 36000 as his share, What is the total profit?

(a) 80000

(b) 90000

(c) 70000

(c) 120000

19. Ankita and Bobby started a partnership business. Ankita's investment was thrice the investment of Bobby and the period of her investment was two times the period of investments of Bobby. If Bobby received Rs 4,000 as profit, find the total profit?

(a) 28000

(b) 30000

(c) 32000

(c) 34000

20. Patil and Qatir invested in a business. The profit earned was divided in the ratio 2 : 3. If Patil invested Rs. 40,000, the amount invested by Qatir is

(a) 40000

(b) 50000

(c) 60000

(c) 70000

21. Anand and Banta started a business jointly. Anand invests Rs 16,000 for 8 month and Banta remains in the business for 4 months. Out of total profit, Banta claims 27 of the profit. How much money was contributed by B?

(a) 11,200

(b) 12,000

(c) 12,400
(c) 12,800

22. A and B starts a business investing Rs. 85,000 and Rs. 15,000 respectively. Find out the ratio in which the profits should be shared.
(a) 10 : 3
(b) 17 : 3
(c) 3 : 10
(c) 3 : 17

23. A, B and C start a business with each investing Rs 20,000. After 5 months A withdraws Rs 5,000, B withdraws Rs 4,000 and C invests Rs 6,000 more. At the end of the year, a total profit of Rs 69,900 was recorded. Find the share of A.
(a) 20,600
(b) 20,700
(c) 20,500
(c) 20,400

24. Ahmed invested Rs 76,000 in a business. After few months, Bosan joined him with Rs 57,000. The total profit was divided between them in the ratio 2 : 1 at the end of the year. After how many months did Bosan join?
(a) 2
(b) 3
(c) 4
(c) 5

25. Three partners A, B and C start a business. B's capital is four times C's capital and twice A's capital is equal to thrice B's capital. If the total profit is Rs 16500 at the end of a year, find out B's share in it.
(a) 4000
(b) 5000
(c) 6000
(c) 7000

Practice Exercise - 1 Answers with Hints

1 (a)	2 (b)	3 (a)	4 (c)	5 (a)
6 (d)	7 (a)	8 (b)	9 (c)	10 (a)
11 (b)	12 (a)	13 (b)	14 (c)	15 (d)
16 (d)	17 (b)	18 (a)	19 (a)	20 (c)
21 (d)	22 (b)	23 (c)	24 (c)	25 (c)

2. (b) = Ratio of the investments multiplied by the time period

= 45000 × 12 : 30000 × 9 = 45 × 12 : 30 × 9 = 3 × 12 : 2 × 9 = 2 : 1

3. (a) = (20000 × 6 + 15000 × 6):(30000 × 4 + 50000 × 4)

= (20 × 6 + 15 × 6) : (30 × 4 + 50 × 4) = (20 × 3 + 15 × 3) : (30 × 2 : 50 × 2) = 105 : 160 = 21 : 32

6. (d) Let B joined for x months. Then

85000 × 12 : 42500 × x = 3 : 1 ⇒ 850 × 12 : 425x = 3 : 1 ⇒ 850 × 4 = 425x ⇒ x = 8

10. (a) A is a working member and for that, he receives 5% of the profit

= 5% of 7400 = 5 × 7400/100 = 370

Remaining amount = 7400 − 370 = 7030

Ratio of their investments

= 6500 × 6 : 8400 × 5 : 10000 × 3 = 65 × 6 : 84 × 5 : 100 × 3 = 13 × 6 : 84 : 20 × 3 = 13 × 2 : 28 : 20

= 13 : 14 : 10

Share of B in the profit = 7030 × 1437 = 190 × 14 =2660

12. (a) A : B : C = 10 × 7 : 12 × 5 : 15 × 3 = 2 × 7 : 12 × 1 : 3 × 3 = 14 : 12 : 9

Amount that C should pay =175×935=5×9=45

23.

(c) A : B : C = (20000 × 5 + 15000 × 7) : (20000 ×5 + 16000 × 7) : (20000 × 5 + 26000 × 7)

= (20 × 5 + 15 × 7) : (20 × 5 + 16 × 7) : (20 × 5 + 26 × 7) = 205 : 212 : 282

A's share = 69900 × 205(205 + 212 + 282) = 69900 × 205/699 = 20500

Practice Exercise - 2

Number of Questions: 25 **Suggested Time: 25 minutes**

1. In a business, A and C invested amounts in the ratio 2 : 1, whereas the ratio between amounts invested by A and B was 3 : 2. If Rs. 157300 was their profit, how much amount did B receive?

(a) 48000
(b) 48200
(c) 48400
(d) 48600

2. P, Q and R started a business by investing Rs. 1,20,000, Rs. 1,35,000 and Rs. 1,50,000 respectively. Find the maximum share of any one of these out of an annual profit of Rs. 56,700.

(a) 21,000
(b) 17,900
(c) 22,000
(d) 18,500

3. If 4 (P's Capital) = 6(Q's Capital) = 10(R's Capital), then out of the total profit of Rs 4,650, R will receive

(a) 600
(b) 700
(c) 800
(d) 900

4. P, Q, R enter into a partnership. P initially invests 25 lakhs and adds another 10 lakhs after one year. Q initially invests 35 lakhs and withdraws 10 lakhs after 2 years. R's investment is Rs 30 lakhs. In what ratio should the profit be divided at the end of 3 years?

(a) 18 : 19 : 19
(b) 18 : 18 : 19
(c) 19 : 19 : 18
(d) 18 : 19 : 18

5. P, Q, R enter into a partnership and their share are in the ratio 12 : 13 : 14 After two months, P withdraws half of his capital and after 10 more months, a profit of Rs 378 is divided among them. What is Q's share?

(a) 114

(b) 120

(c) 134

(d) 144

6. A, B, C started a business with their investments in the ratio 1 : 3 : 5. After 4 months, A invested the same amount as before and B as well as C withdrew half of their investments. The ratio of their profits at the end of the year is :

(a) 1 : 2 : 3

(b) 3 : 4 : 5

(c) 3 : 5 : 10

(d) 5 : 6 : 10

7. A and B invest in a business in the ratio 3 : 2. If 5% of the total profit goes to charity and A's share is Rs. 855, the total profit is:

(a) 500

(b) 1000

(c) 1500

(d) 2000

8. A and B are partners in a business. A contributes 1/4 of the capital for 15 months and B received 2/3 of the profit. For how many months B's money was used?

(a) 3

(b) 6

(c) 10

(d) 12

9. A began a business with Rs. 85,000. He was joined afterwards by B with Rs. 42,500. For how many months did B join, if the profit after an year is divided in the ratio of 3 : 1?

(a) 4

(b) 5

(c) 6
(d) 8

10. A, B and C enter into a partnership and their shares are in the ratio 1/2 : 1/3 : 1/4. After 2 months, A withdraws half of his capital and after 10 months, a profit of Rs. 378 is divided among them. What is B's share?
(a) 144
(b) 169
(c) 225
(d) 339

11. Anand and Deepak started a business investing Rs. 22,500 and Rs. 35,000 respectively. Out of a total profit of Rs. 13,800, Deepak's share is:
(a) 8400
(b) 8200
(c) 8100
(d) 8000

12. A and B started a business jointly A's investment was thrice the investment of B and the period of his investment was two times the period of investment of B. If B received Rs. 4000 as profit, then their total profit is:
(a) 22,000
(b) 28,000
(c) 32,000
(d) 36,000

13. Simran started a software business by investing Rs. 50,000. After six months, Nanda joined her with a capital of Rs. 80,000. After 3 years, they earned a profit of Rs. 24,500. What was Simran's share in the profit?
(a) 10,110
(b) 10,500
(c) 12,000
(d) 13,000

14. P and Q started a business investing Rs, 85,000 and Rs. 15,000 respectively. In what ratio the profit named after 2 years will be divided between P and Q respectively?

(a) 17 : 23

(b) 17 : 3

(c) 17 : 33

(d) 3 : 4

15. A, B, C rent a pasture. A puts 10 oxen for 7 months, B puts 12 oxen for 5 months and C puts 15 oxen for 3 months for grazing. If the rent of the pasture is Rs. 175, how much must C pay as his share of rent?

(a) 45

(b) 50

(c) 60

(d) 55

16. Shekar started a business investing Rs. 25,000 in 1999. In 2000, he invested an additional amount of Rs. 10,000 and Rajeev joined him with an amount of Rs. 35,000. In 2001, Shekar invested another additional amount of Rs. 10,000 and Jatin joined them with an amount of Rs. 35,000. What will be Rajeev's share in the profit of Rs. 1,50,000 earned at the end of 3 years from the start of the business in 1999?

(a) 45,000

(b) 50,000

(c) 70,000

(d) 75,000

17. A, B and C enter into a partnership. They invest Rs. 40,000, Rs. 80,000 and Rs. 1,20,000 respectively. At the end of the first year, B withdraws Rs. 40,000, while at the end of the second year, C withdraws Rs. 80,000. In what ratio will the profit be shared at the end of 3 years?

(a) 2 : 3 : 5

(b) 3 : 4 : 7

(c) 5 : 6 : 4

(d) 1 : 3 : 5

18. A, B and C enter into a partnership with a capital in which A's contribution is Rs. 10,000. If out of a total profit of Rs. 1000, A gets Rs. 500 and B gets Rs. 300, then C's capital is:

(a) 4,000

(b) 5,000

(c) 6,000

(d) 7,000

19. X and Y invested in a business. They earned some profit which they divided in the ratio of 2 : 3. If X invested Rs. 40,000, the amount invested by Y is:

(a) 20,000

(b) 40,000

(c) 60,000

(d) 80,000

20. A, B, C subscribe Rs. 50,000 for a business, A Subscribes Rs. 4000 more than B and B Rs. 5000 more than C. Out of a total profit of Rs. 35,000, A receives:

(a) 14,700

(b) 15,000

(c) 12,000

(d) 13,500

21. Aman started a business investing Rs. 70,000. Rakhi joined him after six months with an amount of Rs. 1,05,000 and Sagar joined them with Rs. 1.4 lakhs after another six months. The amount of profit earned should be distributed in what ratio among Aman, Rakhi and Sagar respectively, 3 years after Aman started the business?

(a) 11 : 13 : 15

(b) 11 : 13 : 17

(c) 12 : 17 : 18

(d) 12 : 15 : 16

22. Reena and Shaloo are partners in a business, Reena invests Rs, 35,000 for 8 months and Shaloo invests Rs. 42,000 for 10 months, out of a profit of Rs. 31,570, Reena's share is:

(a) 12,628

(b) 18,245
(c) 11,235
(d) 10,253

23. In a business, A and C invested amounts in the ratio 2 : 1, whereas the ratio between amounts invested by A and B was 3 : 2. If Rs. 1,57,300 was their profit, how much amount did B receive?
(a) 48,400
(b) 54,200
(c) 64,000
(d) 74,000

24. A starts business with Rs. 3500 and after 5 months, B joins with A as his partner. After a year, the profit is divided in the ratio 2 : 3. What is B's contribution in the capital?
(a) 7,500
(b) 8,000
(c) 8,500
(d) 9,000

25. A and B start a business jointly. A invests Rs. 16,000 for 8 months and B remains in the business for 4 months, Out of total profit, B claims - of the profit. How much money was contributed by B?
(a) 12,000
(b) 12,800
(c) 13,000
(d) 14,500

Practice Exercise - 2 Answers with Explanations

1 (c)	2 (a)	3 (d)	4 (c)	5 (d)
6 (d)	7 (c)	8 (c)	9 (d)	10 (a)
11 (a)	12 (b)	13 (b)	14 (b)	15 (a)
16 (b)	17 (b)	18 (a)	19 C	20 (a)
21 (d)	22 (a)	23 (a)	24 (d)	25 (b)

16. (b) Shekar : Rajeev : Jatin

$= (25,000 \times 12 + 35,000 \times 12 + 45,000 \times 12) : (35,000 \times 24) : (35,000 \times 12)$

$= 12,60,000 : 8,40,000 : 4,20,000 = 3 : 2 : 1$.

Rajeev's Share = Rs. $(1,50,000 \times 2/6)$ = Rs. 50,000.

23. $A : B = 3 : 2 \Rightarrow B : A = 2 : 3 = 4 : 6$ and $A : C = 2 : 1 = 6 : 3$.

So, $B : A : C = 4 : 6 : 3$ or $A : B : C = 6 : 4 : 3$;

B's share = Rs. $(1,57,300 \times 4/13)$ = Rs. 48,400.

TIME & WORK

In finding out the solution of the problems on time and work, it is necessary to determine the work done in a unit time. This unit time is generally taken as an hour or a minute or a day.

The following general rules need the attention

- *If a man can do a piece of work in 10 days, he will finish 1/10 of it in 1 day.*
- *If a man can do 1/8 of a work in an hour, he will complete the whole work in 8 hours.*
- *If Parul can do 1/10 of a work in 1 day and his sister does 1/5 of the same work in 1 day, then both together can do* $\frac{1}{10}+\frac{1}{15}+\frac{3}{10}$ *work in a day.*
- *If A is thrice as good a workman as B, then A's work: B's work = 3 : 1 and time taken by A : time taken by B = 1 : 3 for a certain piece of work.*

Also remember:

- ***If A can do a piece of work in 'n' days, work done by A in 1 day = '1/n'.***
- ***If A does '1/n' work in a day, A can finish the work in 'n' days.***
- ***If M1 men can do W1 work in D1 days working H1 hours per day and M2 men can do W2 work in D2 days working H2 hours per day (where all men work at the same rate), then M1 D1 H1/W1 = M2 D2 H2/W2***
- ***If A can do a piece of work in p days and B can do the same in q days, A and B together can finish it in pq/(p+q) days.***
- ***If A is thrice as good as B in work, then Ratio of work done by A and B = 3 : 1 Ratio of time taken to finish a work by A and B = 1 : 3.***

Example A can do a piece of work in 6 days and B alone can do it in 8 days. How long would it take for both of them to finish.

Solution

A's one day's work = $\frac{1}{4}$ B's one day's work = $\frac{1}{9}$

(A + B)'s one day's work = $\frac{1}{4} + \frac{1}{9} = \frac{22}{11}$ $\therefore$1 work is done in $\frac{22}{11}$ days

Example A and B can do a piece of work in 6 days and A alone can do it in 10 days. In how many days can B alone do it?

Solution

(A + B)'s one day's work = $\frac{1}{4}$; A's one day's work = $\frac{3}{60}$

$\therefore$ B's one day's work = $\frac{1}{4} - \frac{3}{60} = \frac{3}{60}$; $\therefore$ B can complete that work alone in 15 days.

Example A and B can do a piece of work in 40 **days. B and Can do a piece of work in 30 days. C and A can do** a piece of work in 24 days. How long would they all take to do the same work?

Solution

$\therefore$ 2(A + B + C)'s one day's work

$$= \frac{1}{24} + \frac{1}{30} + \frac{1}{40} = \frac{x}{9} = \frac{x \times R \times R}{100}$$

$\therefore$ (A + B + C)'s one day's work = $\frac{17}{2}$

$\therefore$ (A + B + C) together can complete the work in 20 days.

Practice Exercise - 1

Number of Questions: 20 **Suggested time: 20 minutes**

1. Parmod is able to do a piece of work in 15 days and Qasim can do the same work in 20 days. If they work together for 4 days, what fraction of work is now left?
(a) 8/15
(b) 7/15
(c) 11/15
(d) 2/11

2. Tanush's firm can lay railway track between two stations in 16 days. Navansh's firm can do the same job in 12 days. With the help of Vridhi's firm, all can lay the track in 4 days. How many days will it take for Vridhi's firm alone to complete the work?
(a) 9(3/5) days
(b) 9(1/5) days
(c) 9(2/5) days
(d) 10 days

3. Gaurav, Parit and Gaggi can do a work in 20, 30 and 60 days respectively. How many days does it need to complete the work if Gaurav does the work and he is assisted by others on every alternate day?
(a) 10 days
(b) 14 days
(c) 15 days
(d) 9 days

4. Sabir is thrice as good as Anish in work. Sabir is able to finish a job in 60 days less than Anish. Both can finish the work in - days working together.
(a) 18 days
(b) 22 1/2 days
(c) 24 days
(d) 26 days

5. Kalpana can do a particular work in 6 days. Preeti can do the same work in 8 days. Both agreed to do it for Rs. 3200. They completed the work in 3 days with the help of Mini. How much is to be paid to Mini?

(a) Rs. 380

(b) Rs. 600

(c) Rs. 420

(d) Rs. 400

6. A job can be completed by 6 men and 8 women in 10 days. 26 men and 48 women can finish the same work in 2 days. 15 men and 20 women can do the same work in - days.

(a) 4 days

(b) 6 days

(c) 2 days

(d) 8 days

7. Raja can do a piece of work in 4 hours. Raja and Praja together can do it in just 2 hours, while Rani and Praja together need 3 hours to finish the same work. Rani alone can complete the work in _____ hours.

(a) 12 hours

(b) 6 hours

(c) 8 hours

(d) 10 hours

8. Preeti can do a work in the same time in which Kanchan and Sudha together can do it. If Preeti and Kanchan work together, the work can be completed in 10 days. Sudha alone needs 50 days to complete the same work, then Kanchan alone can do it in how many days?

(a) 30 days

(b) 25 days

(c) 20 days

(d) 15 days

9. Narinder completes 80% of a work in 20 days. Then Anju also joins and now together finish the remaining work in 3 days. How long does it need for Anju alone complete the work?

(a) 37 ½ days

(b) 22 days

(c) 31 days
(d) 22 days

10. "SANJAY Book Depot" can print one lakhs books in 8 hours. "DEVINDER & Sons" can print the same number of books in 10 hours while " HARISH & Sons" can print the same in 12 hours. All of them started printing at 0900 hrs. "SANJAY" is stopped at 1100 hrs and the remaining two machines complete the remaining leftover work. Approximately at what time will the printing of one lakhs books be completed?
(a) 1500 hrs
(b) 1400 hrs
(b) 1400 hrs
(d) 1200 hrs

11. A can finish a work in 18 days. B can finish the same work in 15 days. B worked for 10 days and left the job. How many days does A alone need to finish the remaining work?
(a) 8
(b) 5
(c) 4
(d) 6

12. If 3 men and 7 women can complete a work in 10 days and 4 men and 6 women need 8 days to complete the same work. In how many days will 10 women complete the same work?
(a) 50
(b) 40
(c) 30
(d) 20

13. A and B can finish a work 30 days if they work together. They worked together for 20 days and then B left. A finished the remaining work in another 20 days. In how many days A alone can finish the work?
(a) 60
(b) 50
(c) 40
(d) 30

14. A can complete a work in 12 days with a working of 8 hours per day. B can complete the same work in 8 days when working 10 hours a day. If Both start the work together, working 8 hours a day, the work can be completed in _____ days.

(a) 5 5/11

(b) 4 5/11

(c) 6 4/11

(d) 6 5/11

15. P is 30% more efficient than Q. P can complete a work in 23 days. If P and Q work together, how much time will it take to complete the same work?

(a) 9

(b) 11

(c) 13

(d) 15

16. P, Q and R can complete a work in 12, 24 and 6 days respectively. The work will be completed in _____ days if all of them are working together.

(a) 2

(b) 3 3/7

(c) 4 1/4

(d) 4

17. A job can be completed by 10 men in 7 days. While 10 women need 14 days to complete the same work. How many days will 5 men and 10 women need to complete the work?

(a) 5

(b) 6

(c) 7

(d) 8

18. Nirmal can alone complete a work in 20 days. Suresh is 25% more efficient than her. He can complete the work in _____ days.

(a) 14

(b) 16

(c) 18

(d) 20

19. Anil and Rajender are working on a special assignment. Anil needs 6 hours to type 32 pages on a computer and Rajender needs 5 hours to type 40 pages. If both of them work together on two different computers, how much time is needed to type an assignment of 110 pages?

(a) 7 hour 15 minutes

(b) 7 hour 30 minutes

(c) 8 hour 15 minutes

(d) 8 hour 30 minutes

20. Pruthi and Shakil can complete a work in 20 days and 12 days respectively. Pruthi alone started the work and Shakil joined him after 4 days till the completion of the work. How long did the work last?

(a) 5 days

(b) 10 days

(c) 14 days

(d) 22 days

Practice Exercise 1 - Answers with Hints - need based

1 (a)	2 (a)	3 (c)	4 (b)	5 (d)
6 (a)	7 (a)	8 (b)	9 (a)	10 (c)
11 (d)	12 (b)	13 (a)	14 (a)	15 (c)
16 (b)	17 (c)	18 (b)	19 (c)	20 (b)

1. Remaining work = $1 - 4/15 - 4/20$

2. Work completed in every three days = $2 \times (1/20) + (1/20 + 1/30 + 1/60) = 1/5$

So work completed in 15 days = $5 \times 1/5 = 1$

6. $\Rightarrow 52m + 96b = 1$ (2)

$\Rightarrow 60m + 80b = 1$ (1)

from these equations we can find a comparison between a man and a boy's work

9. Work done by A in 20 days = $80/100 = 8/10 = 4/5$

Work done by A in 1 day = (4/5)/20 = 4/100 = 1/25 (1)

Work done by A and B in 3 days = 20/100 = 1/5 (Because remaining 20% is done in 3 days by A and B)

Work done by A and B in 1 day = 1/15 (2)

Work done by B in 1 day = 1/15 – 1/25 = 2/75

⇒ B can complete the work in 75/2 days = 37 1/2 days

13. Amount of work done by A and B in 20 days = 20 × (1/30) = 20/30 = 2/3

Remaining work – 1 – 2/3 = 1/3 which A completes in 20 days

⇒ A can complete the work in 60 days

19. Pages typed by Anil in 1 hour = 32/6 = 16/3

Pages typed by Rajender in 1 hour = 40/5 = 8

Pages typed by Anil and Rajender in 1 hour = 16/3 + 8 = 40/3

Time taken to type 110 pages if Anil and Rajender work together = 110 × 3/40 = 3/4

= 8 1/4 hours = 8 hour 15 minutes

20. If the work lasted for 'x' days then : $x/20 + (x - 4)/12 = 1$

TIME AND WORK

Practice Exercise - 2

Number of Questions: 25 **Suggested time: 20 minutes**

1. Prabhu takes twice as much time as Kanta or thrice as much time as Raghav to finish a piece of work. They can finish the work in 2 days if work together. How much time will Kanta take to do the work alone?
(a) 4
(b) 5
(c) 6
(d) Data inadequate

2. Parshant and Par Brahm could complete a work in 15 days and 10 days respectively. They had started the work together but Par Brahm left after 2 days. Parshant, thereafter, alone completed the remaining work. The work was finished in _____ days.
(a) 12
(b) 16
(c) 20
(d) 24

3. P and Q can do a work in 30 days. Q and R can do the same work in 24 days and R and P in 20 days. They started the work together, but Q and R left after 10 days. How many days more will P take to finish the work?
(a) 10
(b) 15
(c) 18
(d) 22

4. If Q, half as fast as P is, alone can complete a work in 12 days, then both P and Q will finish the work in _____ days
(a) 1
(b) 2
(c) 3

(d) 4

5. A work can be finished in 16 days by 20 girls. The same work can be finished in 15 days by 16 boys. The ratio between the capacity of a boy and a girl is

(a) 1 : 3

(b) 4 : 3

(c) 2 : 3

(d) 2 : 1

6. P and Q need 8 days to complete a work. Q and R need 12 days to complete the same work. But P, Q and R together can finish it in 6 days. How many days will be needed if P and R together do it?

(a) 3

(b) 8

(c) 12

(d) 4

7. A can do a work in 24 days. B can do the same work in 9 days and C can do the same in 12 days. B and C start the work and leave after 3 days. A finishes the remaining work in _____ days.

(a) 7

(b) 8

(c) 9

(d) 10

8. Daily wages of a Supervisor is double to that of a workman. How many Supervisor should work for 25 days to earn Rs. 14,400? Given that wages for 40 workmen for 30 days are Rs. 21,600.

(a) 12

(b) 14

(c) 16

(d) Data inadequate

9. P,Q and R together earn Rs. 1620 in 9 days. P and R can earn Rs. 600 in 5 days. Q and R in 7 days can earn Rs. 910. How much amount does R can earn per day?

(a) Rs. 40

(b) Rs. 70

(c) Rs. 90
(d) Rs. 100

10. Assume that 20 horses and 40 sheep can be grazed for 10 days for Rs. 4,600. If the charges of keeping 5 sheep is the same as the charges of grazing 1 horse, what will be the charges for keeping 50 horses and 30 sheep for 12 days?
(a) Rs. 11,040
(b) Rs. 10,000
(c) Rs. 9,340
(d) Rs. 12,100

11. There is a group of persons each of whom is capable to complete a piece of work in 16 days, when working individually. On the first day one person works, on the second day another person joins him, on the third day one more person joins them and this process continues till the work is completed. How many days are needed to complete the work?
(a) 3 1/4 days
(b) 4 1/3 days
(c) 5 1/6 days
(d) 6 1/5 days

12. A can do a piece of work in 10 days, B in 15 days. They work together for 5 days, the rest of the work is finished by C in two more days. If they get Rs. 18,000 as wages for the whole work, what are the daily wages of B (in Rs):
(a) Rs. 14,000
(b) Rs. 12,000
(c) Rs. 24,000
(d) Rs. 15,000

13. If 12 men can complete a work in 8 days. 16 women can complete the same work in 12 days. 8 men and 8 women started the work and continued for 6 days. How many more men are to be added to complete the remaining work in 1 day?
(a) 8
(b) 12
(c) 16

(d) 24

14. A is twice as good a workman as B and is therefore able to finish a piece of work in 30 days less than B. In how many days they can complete the whole work; working together?

(a) 9
(b) 17
(c) 14
(d) 20

15. A, B and C can do a piece of work in 24 days, 30 days and 40 days respectively. They began the work together but C left 4 days before the completion of the work. In how many days was the work completed?

(a) 11
(b) 12
(c) 13
(d) 14

16. A can finish a certain work in the same time in which B and C together can finish it. If A and B together could do it in 20 days and C alone in 60 days,then B alone could do it in how many days?

(a) 20
(b) 40
(c) 50
(d) 60

17. P can complete a work in 12 days working 8 hours a day. Q can complete the same work in 8 days working 10 hours a day. If both P and Q work together, working 8 hours a day, on which the work will complete?

(a) 6^{th}
(b) 7^{th}
(c) 5^{th}
(d) 8^{th}

18. It is given that $(x - 2)$ men can do a piece of work in 'x' days and $(x + 7)$ men can do 75% of the same work in $(x - 10)$ days. Then in how many days can (x+10) men finish the work?

(a) 27
(b) 12
(c) 25
(d) 18

19. A is thrice efficient as B and C is twice as efficient as B. What is the ratio of number of days taken by A,B and C, when they work individually?
(a) 2 : 6 : 3
(b) 2 : 3 : 6
(c) 1 : 2 : 3
(d) 3 : 1 : 2

20. The ratio of efficiency of A is to C is 5:3. The ratio of number of days taken by B is to C is 2:3. A takes 6 days less than C, when A and C complete the work individually. B and C started the work and left after 2 days. The number of days taken by A to finish the remaining work is:
(a) 4 1/2
(b) 5
(c) 6
(d) 9 – 1/3

21. Pipe A alone can fill the tank in 4 hours, while pipe B alone can fill it in 6 hours. Pipe C can empty the whole tank in 4 hours. Mr Ram Bishan, whose job is to observe the filling of tank, opened the pipe A and B simultaneously to fill the empty tank. He wanted to adjust his alarm so that he could open the pipe C when the tank was half-filled, but he mistakenly adjusted his alarm at a time when his tank would be $3/4^{th}$ filled. Now please find the time difference between both the cases, to fill the tank fully:
(a) 48 minutes
(b) 54 minutes
(c) 30 minutes
(d) Not possible to fill the tank

22. A and B can do a work in 4 hours and 12 hours respectively. Their day starts at 06.00 hrs when A starts the work. If they work alternately for an hour each when will the work be completed?

(a) 12.00 hrs
(b) 13.00 hrs
(c) 11.00 hrs
(d) Not possible to finish

23. When A, B and C are deployed for a task, A and B together do 70% of the work and B and C together do 50% of the work. Who is most efficient?
(a) A
(b) B
(c) C
(d) No comparison

24. A tank has an inlet and outlet pipe. The inlet pipe fills the tank completely in 2 hours when the outlet pipe is plugged. The outlet pipe empties the tank completely in 6 hours when the inlet pipe is plugged. If there is a leakage also which is capable of draining out the liquid from the tank at half of the rate of outlet pipe, then how many hours will it take to fill the empty tank when both the pipes (inlet and leakage) are opened?
(a) 3
(b) 4
(c) 5
(d) Not possible

25. A contractor undertook a project to complete it in 20 days which needed 5 workers to work continuously for all the days estimated. But before the start of the work the client wanted to complete it earlier than the scheduled time, so the contractor decided to increase 5 additional men every 2 days to complete the work in the time as the client desired. Not it happened that the work was further increased by 50% but the contractor continued to increase 5 workers after every 2 days then how many more days are required over the initial time specified by the client.
(a) 1
(b) 2
(c) 5

(d) Not possible

Practice Exercise 2 - Answers with Explanations

1 (c)	2 (a)	3 (c)	4 (d)	5 (b)
6 (b)	7 (d)	8 (c)	9 (b)	10 (a)
11 (c)	12 (b)	13 (b)	14 (d)	15 (a)
16 (d)	17 (a)	18 (b)	19 (a)	20 (c)
21 (b)	22 (a)	23 (a)	24 (b)	25 (b)

2. If P continued for P days, so Q worked for 2 days only

Therefore P/15 + 2/10 = 1.

3. 2p + 2q + 2r = 1/30 + 1/24 + 1/20 = 15/120 = 1/8 ⇒ p + q + r = 1/16

Work done by P, Q and R in 10 days = 10 × (1/16) = 10/16 = 5/8

Remaining work = 1 – 5/8 = 3/8

Work done by P in 1 day = 1/16 – 1/24 = 1/48

Number of days P needs to work to complete the remaining work = (3/8)/(1/48) = 18

5. Such like questions should always be solved through MANDAYS theory

i.e., Women days : Man days or 16 × 20 : 15 × 16

11. Work completed in 1st day = 1/16

Work completed in 2nd day = (1/16) + (1/16) = 2/16

Work completed in 3rd day = (1/16) + (1/16) + (1/16) = 3/16

The work done in 3 days = 1/16 + 2/16 + 3/16 = (1 + 2 + 3)/16 = 6/16

The work done in 4 days = (1 + 2 + 3 + 4)/16 = 10/16

The work done in 5 days = (1 + 2 + 3 + 4 + 5)/16 = 15/16, almost close, isn't it?

The work done in 6 days = (1 + 2 + 3 + 4 + 5 + 6)/16 > 1

Hence the answer is less than 6, but greater than 5. Hence the answer is 5 $^{1/6}$ days.

An easy way to attack such problems is from the choices.

You can see the choices are very close to each other. So just see one by one.

12. A's 5 days work = 50%

B's 5 days work = 33.33%

C's 2 days work = 16.66% [100 – (50 + 33.33)]

Ratio of contribution of work of A, B and C

$=50:33\frac{1}{3}:16\frac{2}{3}=3:2:1$

13. 1 man's 1 day work = $\frac{3}{60}$; 1 woman's 1 day work = $\frac{45}{100}$work done in 6 days $=6\left(\frac{8}{96}+\frac{8}{192}\right)=\left(6\times\frac{1}{8}\right)=\frac{3}{4}$

Remaining work $=\left(1-\frac{3}{4}\right)=\frac{1}{4}$

(8 men + 8 women)'s 1 day work $=1\left(\frac{8}{96}+\frac{8}{192}\right)=\frac{1}{8}$

Remaining work $=\left(\frac{1}{4}-\frac{1}{8}\right)=\frac{1}{8}$

$\frac{3}{60}$ work is done in 1 day by 1 man

$\therefore \frac{5}{6}$ work will be done in 1 day by $\left(96\times\frac{1}{8}\right)=12$ men

17. $\therefore$ P's 1 hour work = 1/96 and Q's 1 hour work = 1/80

(P + Q)'s 1 hour's work $=\left(\frac{1}{96}+\frac{1}{80}\right)=\frac{11}{480}$

so both P and Q will finish the work in $16\frac{2}{3}$ hrs

$\therefore$ Number of days of 8 hours each $=\left(\frac{480}{11}\times\frac{1}{8}\right)=\frac{60}{11}$

18. Working on man days theory, we get:

$\frac{3}{4}\times(x-2)x=(x+7)(x-10)$

$\Rightarrow x = 20$ and $x = -14$ so, the acceptable values is x = 20

Using this information, the desired value can be found.

20. A : B : C

Efficiency → 10 : 9 : 6

No of days → $9x$: $10x$: $15x$

$\Rightarrow 15x - 9x = 6 \therefore x = 1$

work done by B and C in initial 2 days $=\frac{2\times1}{6}=\frac{1}{3}$

Number of days required by A to finish remaining $\frac{1}{2}$ work = $\frac{2/3}{1/9}$ = 6 days

21. In the first case:

Time taken to fill the half tank by A and B = $\frac{20,21}{24}$ = $\frac{5}{6}$ hours

Time taken by A,B and C to fill rest half of the tank = $\frac{50}{16.66}$ = 3 hours

Total time $=\frac{6}{5}+3$ = 4 hours 12 min

In second case:

Time taken to fill $\frac{1}{2}$ tank by A and B = $\frac{39\times5\times12}{30\times6}$ hours

Time taken by A,B and C to fill rest $\frac{1}{2}$ tank = $\frac{25}{16.66}=\frac{3}{2}$ hours

Total time = $\frac{4500}{100}$ = 3 hours 18 min

Therefore, difference in time = 54 minutes

22. Work done by A and B in the first two hours, working alternatively

= First hour A + Second hour B = (1/4) + (1/12) = 1/3.

Thus, the total time required to complete the work = 2 (3) = 6 days

23. A + B = 70%

$$\begin{bmatrix} \therefore (A+B)+(B+C)-(A+B+C)=\mathrm{B} \\ 70+50-100=20 \end{bmatrix}$$

B + C = 50%

⇒ B = 20% A = 50% and C = 30%

Hence A is most efficient

24. Rate of leakage = 8.33% per hour

Net efficiency = 50 – (16.66 + 8.33) = 25%

Time required = 100/25 = 4 hours

25. Total work = 100 + 50 = 150 man-days

In 8 days 100 man-days work has been completed. Now on 9^{th} and 10^{th} day there will be 25 workers. So in 2 days they will complete additional 50 man-days work. Thus the work requires 2 more days.

PIPES & CISTERNS

20

Practice Exercise - 1

Number of Questions: 25 **Suggested time: 25 minutes**

1. Pipes A and B can fill a tank in 5 and 6 hours respectively. Pipe C can empty it in 12 hours. If all the three pipes are opened together, then the tank will be filled in:

(a) 60/17

(b) 113/17

(c) 30/11

(d) 4 1/2

2. Two pipes A and B can fill a cistern in 37 1/2 minutes and 45 minutes respectively. Both pipes are opened. The cistern will be filled in just half an hour, if pipe B is turned off after _____ minutes

(a) 5

(b) 9

(c) 10

(d) 15

3. A pump can fill a tank with water in 2 hours. Because of a leak, it took 2-2/3 hours to fill the tank. The leak can drain all the water of the tank in _____ hours

(a) 6

(b) 8

(c) 9

(d) 10

4. Three pipes A, B and C can fill a tank from empty to full in 30 minutes, 20 minutes, and 10 minutes respectively. When the tank is empty, all the three pipes are opened. A, B and C discharge chemical solutions P,Q and R respectively. What is the proportion of the solution R in the liquid in the tank after 3 minutes?

(a) 6/11
(b) 5/11
(c) 7/11
(d) 8/11

5. A tank is filled by three pipes with uniform flow. The first two pipes operating simultaneously fill the tank in the same time during which the tank is filled by the third pipe alone. The second pipe fills the tank 5 hours faster than the first pipe and 4 hours slower than the third pipe. Time required by the first pipe to fill the tank is _____ hours

(a) 30
(b) 15
(c) 10
(d) 6

6. Two pipes A and B together can fill a cistern in 4 hours. Had they been opened separately, then B would have taken 6 hours more than A to fill the cistern. How much time will be taken by A to fill the cistern separately?

(a) 6 hours
(b) 2 hours
(c) 4 hours
(d) 3 hours

7. Two pipes A and B can fill a tank in 12 and 24 minutes respectively. If both the pipes are used together, then how long will it take to fill the tank?

(a) 9 min
(b) 8 min
(c) 6 min
(d) 4 min

8. Two pipes A and B can fill a tank in 15 minutes and 40 minutes respectively. Both the pipes are opened together but after 4 minutes, pipe A is turned off. What is the total time required to fill the tank?

(a) 10 min 10 sec

(b) 25 min 20 sec
(c) 29 min 20 sec
(d) 20 min 10 sec

9. Two pipes can fill a tank in 25 and 30 minutes respectively and a waste pipe can empty 3 gallons per minute. All the three pipes working together can fill the tank in 15 minutes. The capacity of the tank is:
(a) 250 gallons
(b) 450 gallons
(c) 120 gallons
(d) 150 gallons

10. A tank is filled in 10 hours by three pipes A,B and C. Pipe C is twice as fast as B and B is twice as fast as A. How much time will pipe A alone take to fill the tank?
(a) 70 hours
(b) 30 hours
(c) 35 hours
(d) 50 hours

11. One pipe can fill a tank four times as fast as another pipe. If together the two pipes can fill the tank in 36 minutes, then the slower pipe alone will be able to fill the tank in:
(a) 180 min
(b) 144 min
(c) 126 min
(d) 114 min

12. A tap can fill a tank in 4 hours. After half the tank is filled, three more similar taps are opened. What is the total time taken to fill the tank completely?
(a) 3 hr
(b) 1 hr 30 min
(c) 2 hr 30 min
(d) 2 hr

13. A tap can fill a tank in 4 hours. After half the tank is filled, two more similar taps are opened. What is the total time taken to fill the tank completely?

(a) 1 hr 20 min
(b) 4 hr
(c) 3 hr
(d) 2 hr 40 min

14. Three pipes A,B and C can fill a tank in 6 hours. After working at it together for 2 hours, C is closed and A and B can fill the remaining part in 7 hours. The number of hours taken by C alone to fill the tank is:
(a) 10
(b) 12
(c) 14
(d) 16

15. A large tanker can be filled by two pipes A and B in 60 minutes and 40 minutes respectively. How many minutes will it take to fill the tanker from empty state if B is used for half the time and A and B fill it together for the other half?
(a) 15 min
(b) 20 min
(c) 27.5 min
(d) 30 min

16. Three taps A, B and C can fill a tank in 12, 15 and 20 hours respectively. If A is open all the time and B and C are open for one hour each alternately, the tank will be full in:
(a) 6 – 2/3
(b) 6
(c) 7 1/2
(d) 7

17. A leak in the bottom of a tank can empty the full tank in 6 hours. An inlet pipe fills water at the rate of 4 litres a minute. When the tank is full, the inlet is opened and due to the leak, the tank is empty in 24 hours. How many litres does the tank hold?
(a) 4010 litre
(b) 2220 litre
(c) 1920 litre
(d) 2020 litre

18. A cistern can be filled by a tap in 3 hours while it can be emptied by another tap in 8 hours. If both the taps are opened simultaneously, then after how much time will the cistern get filled?

(a) 4.8 hr

(b) 2.4 hr

(c) 3.6 hr

(d) 1.8 hr

19. Two taps A and B can fill a tank in 5 hours and 20 hours respectively. If both the taps are open then due to a leakage, it took 40 minutes more to fill the tank. If the tank is full, how long will it take for the leakage alone to empty the tank?

(a) 28 hr

(b) 16 hr

(c) 22 hr

(d) 32 hr

20. Bucket P has thrice the capacity as bucket Q. It takes 80 turns for bucket P to fill the empty drum. How many turns it will take for both the buckets P and Q, having each turn together to fill the empty drum?

(a) 30

(b) 45

(b) 45

(d) 80

21. A booster pump can be used for filling as well as for emptying a tank. The capacity of the tank is 2400 m^3. The emptying of the tank is 10 m^3 per minute higher than its filling capacity and the pump needs 8 minutes lesser to empty the tank than it needs to fill it. What is the filling capacity of the pump?

(a) 20 m^3/min.

(b) 40 m^3/min.

(c) 50 m^3/min.

(d) 60 m^3/min.

22. Two pipes A and B can separately fill a cistern in 40 minutes and 30 minutes respectively. There is a third pipe in the bottom of the cistern to empty it. If all the three pipes are simultaneously opened, then the

cistern is full in 20 minutes. In how much time, the third pipe alone can empty the cistern?
(a) 120 min
(b) 100 min
(c) 140 min
(d) 80 min

23. Two pipes A and B can fill a tank in 9 hours and 3 hours respectively. If they are opened on alternate hours and if pipe A is opened first, in how many hours will the tank be full?
(a) 4 hr
(b) 5 hr
(c) 2 hr
(d) 6 hr

24. 13 buckets of water fill a tank when the capacity of each bucket is 51 litres. How many buckets will be needed to fill the same tank, if the capacity of each bucket is 17 litres?
(a) 33
(b) 29
(c) 39
(d) 42

25. Pipe A can fill a tank in 8 hours, pipe B in 4 hours and pipe C in 24 hours. If all the pipes are open, in how many hours will the tank be filled?
(a) 2.4 hr
(b) 3 hr
(c) 4 hr
(d) 4.2 hr

Practice Exercise 1 - Answers with Hints - need based

1 (a)	2 (b)	3 (b)	4 (a)	5 (b)
6 (a)	7 (b)	8 (c)	9 (b)	10 (a)
11 (a)	12 (c)	13 (d)	14 (c)	15 (d)
16 (d)	17 (c)	18 (a)	19 (a)	20 (c)

21 (c)	22 (a)	23 (b)	24 (c)	25 (a)

4. LCM(30,20,10) = 60

Suppose capacity of the tank is 60 litre. Then,

quantity filled by pipe A in 1 minute = 60/30 = 2 litre.

quantity filled by pipe C in 1 minute = 60/20 = 3 litre.

quantity filled by pipe C in 1 minute = 60/10 = 6 litre.

Quantity filled by pipe C in 3 minutes = 3 × 6 = 18 litre.

Quantity filled by pipe A,B,C together in 3 minutes = 3(2 + 3 + 6) = 33 litre.

Required proportion = 18/33 = 6/11

5. Suppose the first pipe alone can fill the tank in x hours. Then, second pipe in $(x - 5)$

third pipe in $(x - 5) - 4 = (x - 9)$ hours.

Part filled by first pipe & second pipe together in 1 hr = Part filled by 3rd pipe in 1 hr

$\Rightarrow 1/x + 1/x - 5 = 1/x - 9$ It is advisable to hit the choices for solution.

6. Suppose pipe A alone can fill the cistern in x hours; Then pipe B in $(x + 6)$ Part filled by pipe A in 1 hr = $1/x$ B in 1 hr = $1/x + 6$; Both in 1 hr = $1/x + 1/x + 6$ or $\Rightarrow 1/x + 1/x + 6 = 1/4$. This point onward, better hit the choices.

8. 4/15 + B/40 = 1

9. Net input in 15 minutes of all (If C empties the tank in C minutes)

= 15/25 + 15/30 – 15/C = 1; Capacity of the tank = 3C litres.

10. Suppose pipe A can fill the tank in x hours. Then B = 2x hours, C = 4x hours.

Therefore, part filled by all pipes together in 1 hour = 1/x + 2/x + 4/x = 7/x

i.e., pipe A, pipe B and pipe C together can fill the tank in x/7 hours. = 10

14. $\Rightarrow$ Part filled by pipes A,B,C together in 2 hr = 2 *1/6 = 1/3

This remaining part of 2/3 is filled by pipes A and B in 7 hours.

Therefore, part filled by pipes A and B in 1 hr = (2/3)7 = 2/21

Part filled by pipe C in 1 hr

= (1/6 − 2/21) = 1/14 i.e., C alone can fill the tank in 14 hours.

15. Part filled by pipe A in 1 minute = 1/60; by pipe B in 1 minute = 1/40

Part filled by both pipes in 1 minute = 1/60 + 1/40 = 5/120 = 1/24

Suppose the tanker is filled in x minutes

Then, to fill the tanker from empty state, B is used for x/2 minutes and A and B together is used for the

rest x/2 minutes.

$x/2 \times 1/40 + x/2 \times 1/24 = 1 \Rightarrow x/2(1/40 + 1/24) = 1 \Rightarrow x = 15 \times 2 = 30$ minutes

18. Net input per hour = 1/3 – 1/8 = 5/24 i.e., the tank can be filled in 24/5 hrs.

21. Let the filling capacity of the pump = x m^3/min.

Then the emptying capacity of the pump = (x + 10) m^3/min.

Time required for filling the tank = 2400/x minutes.

Time required for emptying the tank = 2400/x + 10 minutes.

Pump needs 8 minutes lesser to empty the tank than it needs to fill it

$\Rightarrow 2400/x - 2400/(x + 10) = 8 \Rightarrow 300/(x + 10) - 300/x = x(x + 10) \Rightarrow x^2 + 10x - 3000$

$\Rightarrow (x + 60)(x - 50) = 0$

$\Rightarrow x = 50$ or -60

Since x can not be negative, x = 50 i.e., filling capacity of the pump = 50 m^3/min.

23. Part filled by pipe A in 1hour = 1/9; Part filled by pipe B in 1 hour = 1/3

Pipe A and B are opened alternatively.

Part filled in every 2 hours = 1/9 + 1/3 = 4/9; Part filled in 4 hour = 8/9

Remaining part = 1 − 8/9 = 1/9

Now it is pipe A's turn.

A fills this remaining 1/9 part in next 1 hour

Total time taken = 4 hour + 1 hour = 5 hour.

PIPES AND CISTERNS

Practice Exercise - 2

Number of Questions: 20 **Suggested time: 15 minutes**

1. Pipe K fills a tank in 30 minutes. Pipe L can fill the same tank 5 times as fast as pipe K. If both the pipes were kept open when the tank is empty, It how many minutes, will it take for the tank to overflow?

(a) 3
(b) 2
(c) 4
(d) 5

2. A cistern has a leak which would empty the cistern in 20 minutes. A tap is turned on which admits 4 litres a minute into the cistern, and it is emptied in 24 minutes. How many litres does the cistern hold?

(a) 360
(b) 480
(c) 320
(d) 420

3. Three taps A,B and C can fill a tank in 20,30 and 40 minutes respectively. All the taps are opened simultaneously and after 5 minutes tap A was closed and then after 6 minutes tap B was closed. At that moment a leak developed which could alone empty the full tank in 70 minutes. What is the total time (minutes) taken now for the completely full tank?

(a) 24.315
(b) 26.166
(c) 22.154
(d) 24

4. Two pipes A and B can separately fill a cistern in 60 min and 75 min respectively. There is a third pipe in the bottom of the cistern to empty it. If all the three pipes are simultaneously opened, then the cistern is

full in 50 min. In how many minutes, the third pipe alone can empty the cistern?
(a) 85
(b) 95
(c) 105
(d) 100

5. One pipe can fill a tank three times as fast as another pipe. If together the two pipes can fill tank in 36 min, then the slower pipe alone will be able to fill the tank in?
(a) 85
(b) 92
(c) 187
(d) 144

6. Pipe A can fill a tank in 16 minutes and pipe B cam empty it in 24 minutes. If both the pipes are opened together after how many minutes should pipe B be closed, so that the tank is filled in 30 minutes?
(a) 21
(b) 24
(c) 20
(d) 22

7. Two pipes A and B can separately fill a cistern in 10 and 15 minutes respectively. A person opens both the pipes together when the cistern should have been was full he finds the waste pipe open. He then closes the waste pipe and in another 3 minutes the cistern was full. In how many minutes can the waste pipe empty the cistern when fill?
(a) 8.21
(b) 8
(c) 8.57
(d) 8.49

8. A water tank is two-fifth full. Pipe A can fill a tank in 10 minutes and pipe B can empty it in 6 minutes. If both the pipes are open, how long (minutes) will it take to empty or fill the tank completely?
(a) 6
(b) 6 ½

(c) 7
(d) 8

9. One pipe can fill a tank three times as fast as another pipe. If together the two pipes can fill the tank in 36 min, then the slower alone will be able to fill the tank in:
(a) 181
(b) 128
(c) 144
(d) 172

10. Twelve buckets of water fill a tank when the capacity of each tank is 13.5 litres. How many buckets will be needed to fill the same tank, if the capacity of each bucket is 9 litres?
(a) 28
(b) 25
(c) 26
(d) 18

11. Three taps A,B and C can fill a tank in 12,15 and 20 hours respectively. If A is open all the time and B,C are open for one hour each alternatively, the tank will be full in:
(a) 4.24
(b) 7
(c) $7\frac{1}{2}$
(d) $8\frac{1}{2}$

12. A cistern which could have been filled in 8 hours, takes two hours longer to fill because of a leak in its bottom. If the cistern is full, the leak will empty it in _____ hrs?
(a) 34
(b) 18
(c) 20
(d) $7\frac{1}{2}$

13. A Cistern is filled by pipe A in 8 hrs and the full Cistern can be leaked out by an exhaust pipe B in 12 hrs. If both the pipes are opened in what time the Cistern is full?

(a) 12 hrs

(b) 24 hrs

(c) 16 hrs

(d) 32 hrs

14. Two pipes A and B can fill a tank is 8 minutes and 14 minutes respectively. If both the pipes are opened simultaneously, and the pipe A is closed after 3 minutes, then how much more time will it take to fill the tank by pipe B?

(a) 6 min 15 sec

(b) 5 min 45 sec

(c) 5 min 15 sec

(d) 6 min 30 sec

15. A water tank is two-fifth full. Pipe A can fill a tank in 12 minutes and pipe B can empty it in 6 minutes. If both the pipes are open, how long will it take to empty or fill the tank completely?

(a) 2.8 min

(b) 4.2 min

(c) 4.8 min

(d) 5.6 min

16. Pipes A and B can fill a tank in 8 and 24 hours respectively. Pipe C can empty it in 12 hours. If all the three pipes are opened together, then the tank will be filled in:

(a) 18 hr

(b) 6 hr

(c) 24 hr

(d) 12 hr

17. One pipe can fill a tank 6 times as fast as another pipe. If together the two pipes can fill the tank in 22 minutes, then the slower pipe alone will be able to fill the tank in:

(a) 164 min

(b) 154 min

(c) 134 min
(d) 144 min

18. Two pipes A and B can fill a tank in 2 and 6 minutes respectively. If both the pipes are used together, then how long will it take to fill the tank?
(a) 164 min
(b) 154 minutes
(c) 134 min
(d) 144 minutes

19. Tap 'A' can fill the tank completely in 6 hrs while tap 'B' can empty it by 12 hrs. By mistake, the person forgot to close the tap 'B', As a result, both the taps, remained open. After 4 hrs, the person realized the mistake and immediately closed the tap 'B'. In how much time now onwards, would the tank be full?
(a) 2 hours
(b) 4 hours
(c) 5 hours
(d) 1 hour

20. A Cistern is filled by pipe A in 8 hrs and the full Cistern can be leaked out by an exhaust pipe B in 12 hrs. If both the pipes are opened in what time the Cistern is full?
(a) 12 hrs
(b) 24 hrs
(c) 16 hrs
(d) 32 hrs

Practice Exercise 1 - Answers with Hints - need based

1 (d)	2 (b)	3 (b)	4 (d)	5 (d)
6 (a)	7 (c)	8 (a)	9 (c)	10 (d)
11 (b)	12 (c)	13 (b)	14 (b)	15 (c)
16 (d)	17 (b)	18 (d)	19 (b)	20 (b)

6. Let the pipe B be closed after 'K' minutes.

$30/16 – K/24 = 1 \Rightarrow K/24 = 30/16 – 1 = 14/16$

$\Rightarrow K = 14/16 \times 24 = 21$ min.

12. In 2 minutes (5 litres – 4 litres = 1lit) is filled into the reservoir.

It takes 2 minutes to fill 1lit $\Rightarrow$it takes 50 minutes to fill 25 litres into the reservoir.

In the 51^{st} minute inlet pipe is opened and the reservoir is filled.

14. Let pipes A, B work for 3 minutes, (3 + x) minutes respectively.

$3/8 + (3 + x)/14 = 1$

19. Tap A can fill 1/6 of the tank

Tap B can empty 1/12 of the tank

i.e., In one hour, Tank A and B together can effectively fill

$(1/6 - 1/12)=1/12$ of the tank

$\Rightarrow$ In 4 hours, Tank A and B can effectively fill $1/12 \times 4=1/3$ of the tank.

Time taken to fill the remaining $(1-1/3)=2/3$ of the tank = 4 hours

20. Pipe A can fill 1/8 of the cistern in 1 hour.

Pipe B can empty 1/12 of the cistern in 1 hour

Both Pipe A and B together can effectively fill $1/8-1/12=1/24$ of the cistern in 1 hour

i.e, the cistern will be full in 24 hrs.

STOCKS & SHARES

21

Important Formulas - Stocks and Shares

Introduction

To start a big business or an industry, a large amount of money is required. This may be beyond the capacity of one or two individuals. Hence, a number of individuals join hands to form a company called Joint Stock Company.

Stock Capital

The total amount of money required by the company is called the stock Capital.

Shares or Stock

The whole capital of the company is divided into equal units. Each unit is called a share or a stock.

Shareholder or Stockholder

Each individual who purchases one or more shares is called a shareholder (stockholder) of the company.

The company issues share certificates to each of its shareholders indicating the number of shares allocated and the value of each share.

Face value

Face value of a share is the value printed on the share certificate. It is also called nominal value or par value. ***The face value of a share always remains the same.***

Market value

The stocks of different companies can be traded (bought or sold) in the market through brokers at stock exchanges. The price at which a stock is traded in the market is called its market value.

- *If the market value of a stock is higher than its face value, the stock is traded at a premium or above Par.*

- *If the market value of a stock is same as its face value, the stock is traded at par.*
- *If the market value of a stock is less than its face value, the stock is traded at a discount or below Par.*
- *The market value (trading price) of a share can vary time to time.*

Assume that the face value of stock of company X is Rs. 10 and it is now traded at a premium of Rs. 2. Then its market value now is (Rs. 10 + Rs. 2) = Rs. 12.
Similarly, if the stock of company X having face value of Rs. 10 is now traded at a discount of Rs. 2, it means the market value of X now is (Rs. 10 – Rs. 2) is Rs. 8.

DIVIDEND

- ***The annual profit of a company is distributed among its shareholders. The distributed profit is called the dividend.***
- ***Dividends are declared annually, semi-annually and quarterly as per the company regulations.***
- ***Dividend on a share is normally expressed as a certain percentage of its face value. Sometimes, it is also expressed as a certain amount per share.***

Brokerage

As we have seen earlier, stocks of different companies can be traded (bought or sold) in the market through brokers at stock exchanges. The brokers charge is called brokerage

- ***Brokerage is added to the cost price when the stock is purchased.***
- ***Brokerage is subtracted from the selling price when the stock is sold.***

Number of shares held by a person

= Total Investment of the person/Investment in 1 share

= Total Income/Income from one share

= Total Face Value/Face value of 1 share

What does the statement "Rs. 100, 8% stock at Rs. 110" mean?

It means,

- *The face value of the stock = Rs. 100*
- *Market value (MV) of the stock = Rs. 110*
- *Annual dividend per share = 8% of the face value = 100 × 8/100 = Rs. 8*
- *i.e., an investment of Rs. 110 gives an annual income of Rs. 8*
- *Rate of interest per annum = Annual income from an investment of Rs. 100 = 8 × 100/110 = 7.27%*
- *Please note that generally investors invest in shares not merely because of this annual return. They also will have a capital gain if the market value the share goes higher.*

Example The price of a Rs. 10 share of a company is Rs. 13 and the company declares dividend of 12%, find how many shares can be purchased for a sum of Rs. 2,600?

Also find – (i) the dividend, (ii) the rate of interest on the investment.

Solution

No. of shares = $\frac{4500}{100}$ = 200

Nominal value of the shares = 10 × 200 = Rs. 2,000

Dividend = 12% of Rs. 2,000 = Rs. 240

Rate of interest = $\frac{240}{2600} \times 100 = 9\frac{3}{13}\%$

Example A company declares dividend of 12% on Rs. 100 shares. A man buys such shares and gets 15% on his investment. Find at what price he had bought the shares.

Solution

Dividend on Rs. 100 share = Rs. 12

Let the market value of a share be Rs. x.

$\therefore$ % income on Rs. x = $\frac{17}{2} \times 100 \therefore \frac{1}{100} = 15 \therefore x =$ Rs. 80

Example Which is better investment : (i) 5% shares at Rs. 120 or (ii) 6% shares at Rs. 132? (The nominal value of the share is Rs. 100)

Solution

Income on Rs. 120 = Rs. 5 $\quad \therefore$ Income on Rs. 1 cash = $\frac{5}{120} = \frac{1}{24}$

Also, income on Rs. 132 cash = 6 $\therefore$ Income on Rs. 1 cash

$= \frac{6}{132} = \frac{1}{22}$ Since $\frac{1}{22} > \frac{1}{24}$ $\quad \therefore$ Second investment is better

Example What is the market price of a 3% Rs. 100 stock if it yields 4% after paying an income-tax of 4 P in the rupees?

Solution

Income-tax on Rs. 3 = 3 × 4 P = 12 P

$\therefore$ Now Rs. 4 = interest on Rs. 100

Now Rs. 4 = interest on Rs. 100

$\therefore$ Rs. 2.88 = interest on Rs. $\frac{1}{100} \times 2.88$ = interest on Rs. 72

$\therefore$ The market price of the stock is Rs. 72.

Practice Exercise - 1

Number of Questions: 11 **Suggested time:10 minutes**

1. The cash realized by selling a $5\frac{1}{4}$% stock at 106 $\frac{1}{4}$, brokerage being ($\frac{1}{4}$)%, is:

(a) Rs. 105 $\frac{1}{4}$

(b) Rs. 106 $\frac{1}{4}$

(c) Rs. 106

(d) none of these.

2. By investing in a 6% stock at 96, an income of Rs. 100 is obtained. How much of the investment has been made:

(a) Rs. 1600

(b) Rs. 1504

(c) Rs. 1666.66

(d) Rs. 5760

3. Rs. 2780 are invested partly in 4% stock at 75 and 5% stock at 80 to have equal amount of incomes. The investment is 5% stock is:

(a) Rs. 1500

(b) Rs. 1280

(c) Rs. 1434.84

(d) Rs. 1640

4. By investing Rs. 1100 in a 5 $\frac{1}{4}$% stock one earns Rs. 77. The stock is then quoted at:

(a) Rs. 93

(b) Rs. 107

(c) Rs. $78\frac{1}{4}$

(d) Rs. $97\frac{1}{4}$

5. A man invests in a $4\frac{1}{4}$% stock at 96. The interest obtained by him is:

(a) 4%
(b) 4.5%
(c) 4.69%
(d) $\frac{1}{4}$

6. A invested some money in 4% stock at 96. Now, B wants to invest in an equally good 5% stock. B must purchase a stock worth of:

(a) Rs. 120
(b) Rs. 124
(c) Rs. 76.80
(d) Rs. 80

7. I want to purchase a 6% stock which must yield 5% on my capital. At what price must I buy the stock?

(a) Rs. 111
(b) Rs. 101
(c) Rs. 83.33
(d) Rs. 120

8. A man invests some money partly in 3% stock at 96 and partly in 4% stock at 120. To get equal dividends from both, he must invest the money in the ratio?

(a) 16 : 15
(b) 3 : 4
(c) 4 : 5
(d) 3 : 5

9. A man invested Rs. 388 in a stock at 97 to obtain an income of Rs. 22. The dividend from the stock is:

(a) 12%
(b) 3%
(c) $5\frac{1}{4}$%

(d) 22.68%

10. A man buys Rs. 20 shares paying 9% dividend. The man wants to have an interest of 12% on his money. The market value of each share must be:

(a) Rs. 12
(b) Rs. 15
(c) Rs. 18
(d) Rs. 21

11. A man invested Rs. 4455 in Rs. 10 shares quoted at Rs. 8.25. If the rate of dividend be 6%, his annual income is:

(a) Rs. 267.30
(b) Rs. 327.80
(c) Rs. 324
(d) Rs. 103.70

Practice Exercise - 1 Answers with Hints

1 (c)	2 (a)	3 (b)	4 (c)	5 (c)
6 (d)	7 (a)	8 (c)	9 (b)	10 (c)

2. Income is Rs. 6, when the investment is Rs. 96

Income is Rs. 100, when the investment is Rs.?

4. Value of stock is 5 1/2 : 100 :: 77 : S

S = 1400 (bought by investment of 1100)

6. Rs. 4 is income when stock is = Rs. 96

Rs. 5 is income when stock is Rs. 120

9. The stock is worth Rs. 400 yields Rs. 22 as income.

11. Value of stock is Rs. 5400 (8.25 : 10 :: 4455 : S)

STOCKS AND SHARES

Practice Exercise - 2

Number of Questions: 20 **Suggested time:15 minutes**

1. Mr Rohit Anand buys Rs. 20 shares paying 9% dividend. He wants to have an interest of 12% on his money. What is the market value of each share?
   (a) Rs. 12
   (b) Rs. 18
   (c) Rs. 15
   (d) Rs. 21
2. Mr Nitin Kohli invested Rs. 1552 in a stock at 97 to obtain an income of Rs. 128. What is the dividend from the stock?
   (a) 8%
   (b) 9.7%
   (c) 7.5%
   (d) None
3. The cost price of a Rs. 100 stock at 4 discount, when brokerage is 15% is:
   (a) Rs. 96.25
   (b) Rs. 96.2
   (c) Rs. 97.25
   (d) Rs. 97.5
4. In order to obtain an income of Rs. 650 from 10% stock at Rs. 96, Mr Vikas, a wise investor, must make an investment of:
   (a) Rs. 3100
   (b) Rs. 6500
   (c) Rs. 6240
   (d) Rs. 9600
5. By investing in 16– 2/3% stock at 64, Mr Neeraj earns Rs. 1500. The investment made by Neeraj is _____
   (a) Rs. 9600

(b) Rs. 7500
(c) Rs. 5640
(d) Rs. 5760

6. Mr Subhash Mehta invested Rs. 4940 in Rs. 10 shares quoted at Rs. 9.50. If the rate of dividend be 14%, his annual income is?
(a) Rs. 728
(b) Rs. 648
(c) Rs. 720
(d) Rs. 622

7. Mr Rajender Sharma invests some money partly in 12% stock at 105 and partly in 8% stock at 88. To obtain equal dividends from both, he must invest the money in the ratio:
(a) 31 : 44
(b) 31 : 27
(c) 16 : 15
(d) 35 : 44

8. Mr Rawat bought 40 shares of Rs. 60 at 5 discount, the rate of dividend being 12 ½%. The rate of interest obtained is
(a) 13.64%
(b) 15.5%
(c) 14%
(d) 14.25%

9. By investing Rs. 1800 in 9% stock, Ms Shyama earns Rs. 120. The stock is then quoted at:
(a) Rs. 135
(b) Rs. 96
(c) Rs. 85
(d) Rs. 122

10. The market value of a 10.5% stock, in which an income of Rs. 756 is derived by investing Rs. 9000, brokerage being $\frac{1}{4}$%, is:
(a) Rs. 124.75
(b) Rs. 108.25
(c) Rs. 125.25
(d) Rs. 112.20

11. A 14% stock which yields annually 8% is quoted in the market at:
(a) Rs. 125
(b) Rs. 83.33
(c) Rs. 120
(d) Rs. 175

12. Which is better investment:
(i) 11% stock at 143 or (ii) 9 ¾% stock at 120?
(a) 9 ¾% stock at 120
(b) No comparison, as the total amount of investment is not given.
(c) 11% stock at 143
(d) Both are equally good

13. Mr Suri invests a part of Rs. 12,000 in 12% stock at Rs. 120 and the remainder in 15% stock at Rs. 125. If his total dividend per annum is Rs. 1360, how much does he invest in 12% stock at Rs. 120?
(a) Rs. 6000
(b) Rs. 4500
(c) Rs. 5500
(d) Rs. 4000

14. Rs. 9800 was invested partly in 9% stock at 75 and 10% stock at 80 to have equal amount of incomes. The investment in 9% stock is:
(a) Rs. 4800
(b) Rs. 5400
(c) Rs. 5000
(d) Rs. 5600

15. A 6% stock yields 8%. The market value of the stock is:
(a) Rs. 133.33
(b) Rs. 96
(c) Rs. 75
(d) Rs. 48

16. 12500 shares, of par value Rs. 20 each, are purchased from Mr Sandeep by Mr Vineet at a price of Rs. 25 each. Find the amount required to purchase the shares.
(a) 3,12,500
(b) 3,11,500

(c) 3,13,500
(d) 3,14,500

17. Mr Krishan purchased 12,500 shares, of par value Rs. 20 each, from Mr Mehta at a price of Rs. 25 each and further sells the shares at a premium of Rs. 11 each, find his gain in the transaction.
(a) Rs. 75,000
(b) Rs. 70,000
(c) Rs. 85,000
(d) Rs. 65,000

18. Find the annual dividend received by Nisha Devansh from 1200 preferred shares and 3000 common shares - both of par value Rs. 50 each - if the dividend paid on preferred shares is 10% and semi-annual dividend of 3½% is declared on common shares.
(a) Rs. 18500
(b) Rs. 16500
(c) Rs. 14500
(d) Rs. 19500

19. Mr Bajrang Lal invested Rs. 26,00,000 in 5% stock at 104. He sold the stock when the price rose to **Rs. 120 and invested the sale proceeds in 6% stock. By doing this his income increased by Rs. 2,50,000. At what price did he purchase the second stock?**
(a) Rs. 125
(b) Rs. 48
(c) Rs. 24
(d) None of these

20. Ms Devanshi invested Rs. 32,400 in 8% stock at 90. She sold out Rs. 18,000 stock when the price rose to Rs 95 and the remaining stock at Rs. 98. She then invested the total sale proceeds in 10% stock at 96½. Find the change in her income.
(a) Rs. 2,220
(b) Rs. 120
(c) Rs. 2,720
(d) Rs. 720

Practice Exercise - 2 Answers with Hints - need based

1 (c)	2 (d)	3 (b)	4 (c)	5 (d)
6 (a)	7 (d)	8 (a)	9 (a)	10 (a)
11 (d)	12 (a)	13 (d)	14 (c)	15 (c)
16 (a)	17 (a)	18 (b)	19 (b)	20 (d)

1. Dividend per share = 9% of 20 = X% of 12

2. By investing Rs. 1552, income = Rs. 128

By investing Rs. 97, income = Rs.?

6. Number of shares = 4940/9.50 = 520

Face Value of a share = Rs. 10 (dividend = 14%)

His annual income = 520 × 1.4 = Rs. 728

7. Let he invest A; B

Therefore 12% of A/105 = 8% of B/88

10. Assume that face value = Rs. 100

Dividend per share = Rs. 10.5 (as it is 10.5% stock; Add brokerage while buying)

By investing Rs. 9000, earnings = Rs. 756

To get an earning of Rs. 10.5, investment required = 9000 × 10.5/756 = Rs. 125

i.e, market value of Rs. 100 stock = Rs. (125 – 1/4) = Rs. 124.75 (subtract brokerage while selling)

14. Let the investments be ‘x’ and “9800 – x”

Therefore (9% of x)/75 = (10% of 9800 – x)/80

19. Assuming face value of the first stock = Rs. 100 the dividend per stock = Rs. 5

Market Value of the first stock = Rs. 104

Number of stocks purchases = 26,00,000/104 = 25000

His total income from all these stocks = Rs. 25000 × 5 = Rs. 1,25,000

He sells this stock at Rs. 120 i.e, amount he earns = Rs. 120 × 25000 = Rs. 30,00,000

He invest this Rs. 30,00,000 in 6% stock His new income = Rs. 3,75,000

ie., By Rs. 30,00,000 of investment, he earns an income of Rs. 3,75,000

To get an income of Rs. 6, investment needed = 30,00,000 × 6/3,75,000 = *Rs*. 48

which is the market value of the second stock

Practice Exercise - 3

Number of Questions: 25 **Suggested time: 20 minutes**

1. Mr Ankur Srivastva bought 20 shares of Rs. 50 at 5 discount, the rate of dividend being paid at 13 1/2%. The rate of interest yield is

(a) 12 1/2%
(b) 15%
(c) 15 1/2%
(d) 13 1/2%

2. Find the income of Mr Madhuvrat on 7½% stock of Rs. 2,000 purchased by him at Rs. 80.

(a) Rs. 6
(b) Rs. 160
(c) Rs. 148
(d) Rs. 150

3. Ms Nicky Kataria buys 100 shares, of par value at Rs. 5 each, of a steel company, which pays an annual dividend of 12% at such a price that she gets 10% on her investment. Find the market value of a share paid by her.

(a) Rs. 6
(b) Rs. 12
(c) Rs. 4
(d) Rs. 8

4. Mr Ganga Dhar invested Rs. 5,050 in 5% stock at 99 and sold it when the price rose to Rs. 101. He invested these sale proceeds in 8% stock at 88. Find the change in his income if the Brokerage paid at each transactional time is Rs. 2.

(a) 0
(b) Rs. 160
(c) Rs. 180
(d) Rs. 190

5. To produce an annual income of Rs. 800 from a 8% stock at 90, the amount of stock needed is:

(a) Rs. 10,000

(b) Rs. 14,400

(c) Rs. 10,800

(d) Rs. 16,000

6. Mr Madhukar sells 4000 shares of a Jaguar (each of par value Rs. 10), which paid a dividend of 40%, at Rs. 30 per share. He invests the sale proceeds in shares of Czar (each of par value Rs. 25) that pays a dividend of 15%. If the market value of Czar is Rs. 15, find the number of its shares purchased by Madhukar.

(a) 8000

(b) 12000

(c) 16000

(d) None of these

7. "J K Steels" issued 20,000 shares of par value Rs. 10 each. If the total dividend declared is Rs. 24,000, find the rate of dividend paid by it.

(a) 12%

(b) 12.5%

(c) 14%

(d) 14.5%

8. "Reetha Powders" declared a semi-annual dividend of 12%. Find the annual dividend of Mr Hamraaz who owns 2,000 shares of the company having a par value of Rs. 10 each.

(a) Rs. 4,800

(b) Rs. 2,400

(c) Rs. 3,600

(d) Rs. 3,200

9. "T - N & Vriddhi (P) Ltd" has issued 500 preferred shares and 400 common shares - both of par value Rs. 100 each. The dividend on a preferred share and a common share is 8% and 12%, respectively. The company had a total profit of Rs. 1,50,000. The company keeps some amount in reverse fund and distributed the remaining as dividend. Find the amount kept in reserve fund.

(a) Rs. 1,41,200
(b) Rs. 1,60,000
(c) Rs. 7,200
(d) Rs. 1,82,200

10. Amani invested Rs. 3,33,000 in 5 1/2% stocks at 110. If brokerage is Re.1, what is her annual income from this investment.
(a) Rs. 16,500
(b) Rs. 12,500
(c) Rs. 16,000
(d) Rs. 18,000

11. Find the income on 8% stock of Rs. 1200 purchased at Rs. 120?
(a) Rs. 96
(b) Rs. 66
(c) Rs. 90
(d) Rs. 88

12. The cost price of a Rs. 100 stock at 4 discount, when brokerage is 1/4% is:
(a) Rs. 95.75
(b) Rs. 96
(c) Rs. 96.25
(d) Rs. 104.25

13. A 6% stock yields 8%. The market value of the stock is:
(a) Rs. 48
(b) Rs. 75
(c) Rs. 96
(d) Rs. 133.33

14. A man invested Rs. 14,400 in Rs. 100 shares of a company at 20% premium. If his company declares 5% dividend at the end of the year, then how much does he get?
(a) Rs. 500
(b) Rs. 600
(c) Rs. 650
(d) Rs. 720

15. A man invests in a 16% stock at 128. The interest obtained by him is _____ %:

(a) 8
(b) 12
(c) 12 1/2
(d) 16

16. In order to obtain an income of Rs. 650 from 10% stock at Rs. 96, one must make an investment of:

(a) Rs. 3100
(b) Rs. 6240
(c) Rs. 6500
(d) Rs. 9600

17. A 12% stock yielding 10% is quoted at:

(a) 67
(b) 110
(c) 112
(d) 120

18. By investing in 16 2/3% stock at 64, one earns Rs. 1500. The investment made is:

(a) Rs. 5,000
(b) Rs. 7,500
(c) Rs. 5,760
(d) Rs. 4,768

19. By investing Rs. 1,62,000 in 8% stock, Mirchandani earns Rs. 13,500. The stock is then quoted at:

(a) 80
(b) 96
(c) 106
(d) 108

20. Rs. 9,760. (9.5% stock at 4 discount (brokerage 1/4 %)

(a) 8,000
(b) 11,192
(c) 11,160
(d) 11,156

Practice Exercise - 3 Answers with Hints - need based

1 (b)	2 (d)	3 (a)	4 (d)	5 (a)
6 (a)	7 (a)	8 (a)	9 (a)	10 (a)
11 (a)	12 (c)	13 (b)	14 (b)	15 (c)
16 (b)	17 (d)	18 (c)	19 (b)	20 (b)

4. Purchase price of the first stock = Rs. 99 + Rs. 2 = Rs. 101

Number of stocks purchased in this case = 5050/101 = 50

Income = 50 × 5 = Rs. 250

Sale Price of the stock = 101 – 2 = 99; Sale proceeds by selling the stock = 50 × 99 = 4950

Now he invests this Rs. 4950 in 8% stock at 88

Purchase price of this stock = 88+2 = 90; Stocks purchased = 4950/90 = 55

Income = 55 × 8 = Rs. 440

Change in income = 440 – 250 = Rs. 190

6. The question has been filled with confusing and unwanted information.

Market Value of Company X (his selling price) = Rs. 30

Total shares sold = 4000; Amount he gets = Rs. (4000 × 30)

He invests this amount in ordinary shares of Company Y

His purchasing price = 15; Number of shares he purchases = 4000 × 30/15 = 8000

9. Total dividend in all preferred shares = 500 × 100 × 8/100 = *Rs*. 4,000

Total dividend in all common shares = 400 × 100 × 12/100 = Rs. 4,800

Total dividend = Rs. 4,000 + Rs. 4,800 = Rs. 8,800

amount kept in reserve fund = Rs. 1,50,000 – Rs. 8,800 = Rs. 1,41,200

TIME, SPEED & DISTANCE

22

(i) Speed $=1000\times\frac{1}{12}$

(ii) Time $=\frac{\text{Distance}}{\text{Speed}}$

(iii) Distance = Speed × Time

(iv) x km/hr = (x × $\frac{17}{2}$) m/sec.

(v) y m/sec $=\left(\frac{1}{6}\times 30\right)$ km/hr.

(vi) If a certain distance is covered at x km/hr and the same distance covered at y km/hr, then the average speed during whole journey = $\frac{2xy}{x+y}$ km/hr.

(vii) If a man change his speed in the ratio m:n, then the ratio of times taken become n:m.

Example Express 63 km/hr into meters/sec.

Solution

63 km/hr = $\left(\frac{1}{10}\times 30\right)$m/sec = 17.5 m/sec.

Example Convert 8 m/sec into km/hr.

Solution

8 m/sec $=\left(8\times\frac{18}{5}\right)$km/hr $=28.8$ km/hr.

Example A man covers a certain distance at 75 km/hr and returns back to the starting point at 50 km/hr. Find his average speed during the whole journey:

Solution

This is a question where the D is constant and speed varies. Hence Harmonic Mean of averages will be applicable.

Average speed = $\frac{2xy}{(x+y)} = \left(\frac{2\times 75\times 50}{75+50}\right)$ km/hr = 60 km/hr.

Example A distance is covered in 3 hours 48 min. at 5 kmph. How much time will be taken to cover it at 28.5 kmph?

Solution

Distance = (Speed · Time) = $\left(5\times\frac{19}{5}\right)$ km = 19 km.

Now, distance = 19 km and speed = 28 km/hr

∴ Time Taken = $\frac{Distance}{speed} = \frac{19}{28.5}$ hrs $= \left(\frac{19}{28.5}\times 60\right)$ min. = 40 min.

Example A and B are two stations 330 km apart. A scooter rider starts from A at 6.30 a.m. and travels towards B at 60 kmph. Another scooter rider starts from B at 7.30 a.m. and travels towards A at 75 kmph. At what time do they both meet?

Solution

Suppose they meet x hours after 6.30 a.m. Then, distance moved by 1st in x hours) + [Distance moved by 2nd in $(x - 1)$ hours] = 330

∴ $60x + 75\,(x - 1) = 330 \Leftrightarrow 135x = 405 \Leftrightarrow x = 3$ hrs

So, they shall meet at 09.30 am.

Example A scooter rider travels for 6 hours, the first half of the journey time at 60 kmph and the rest at
48 kmph. Find the total distance traveled by him.

Solution

Let the total distance be 'x' km

$\frac{x}{9} = \frac{x \times R \times R}{100}$ $= 6 \Leftrightarrow 4x + 5x = 480 \times 6 \Leftrightarrow x = 320$ km.

(Problems on Trains)

(i) Time taken by a train x meters long in passing a signal post or a pole or a standing man is the same as the time taken by the train to cross 'x' meters with its own speed.

(ii) Time taken by a train 'x' meters long in passing a stationary object of length 'y' meters is the same as the time taken by the train to cover (x + y) meters with its own speed.

(iii) Suppose two trains or bodies are moving in the same directions at 'u' km/hr and 'v' km/hr such that u > v, then their relative speed = (u – v) km/hr.

(iv) If two trains of lengths 'x' km and 'y' km move in the same direction at 'u' km/hr and 'v' km/hr (where u > v), then time taken by them to cross each other is $\left(\frac{x+y}{u-v}\right)$ ***hrs.***

(v) Suppose two trains or bodies are moving in the opposite directions at 'u' km/hr and 'v' km/hr. Then, relative speed = (u + v) km/hr.

(vi) If two trains of length 'x' km and 'y' km move in the opposite directions at 'u' km/ph and 'v' km/pr, then time taken to cross each other = $\left(\frac{x+y}{u+v}\right)$.

(vii) If two trains start at the same time from stations A and B towards each other and after crossing, they take 'a' & 'b' hours in reaching B and A respectively. Then,

A's speed : B's speed = $(\sqrt{5}-\sqrt{2})$***.***

Example A train 100 m long is running with a speed of 70 km/hr. In what time will it pass a man who is running at 10 km/hr in the same direction in which the train is going?

Solution

Speed of train relative to man = (70 – 10) kmph = $\left(60 \times \frac{3}{50}\right) m/\sec = \left(\frac{50}{3}\right) m/\sec$.

Time taken by the train to cross the man

= Time taken by it to cover 100 m at $\left(\frac{50}{3}\right) m/\sec = \left(100 \times \frac{3}{50}\right)\sec$ = 6 sec.

(Boats & Streams)

Let the speed of a boat in still water be x km/hr and the speed of the stream be y km/hr. Then:

Speed downstream $= (x + y)$ *km/hr*

Speed upstream $= (x - y)$ *km/hr*

Let speed downstream = u km/hr & speed upstream = v km/hr Then :-

Speed in still water $= \frac{1}{2}(v + v)$ *km/hr*

Speed of current $= \frac{1}{2}(u - v)$.

Example A man can row upstream at 8 km/hr and downstream at 10.6 km/hr. Find man's speed in still water and the rate of the current.

Solution

Rate in still water = $\frac{1}{4}$ (8 + 10.6) km/hr = 9.3 km/hr

Rate of current = $\frac{1}{4}$ (10.6 – 8) km/hr = 1.3 km/hr.

Example A man can row 6 kmph in still water. It takes him twice as long as to row up the river as to row down the river. Find the rate of stream.

Solution

Let man's rate upstream = x km/hr Then, man's rate downstream = 2x km/hr

∴ Man's rate in still water = $\frac{1}{4}$ $(x + 2x)$ km/hr ∴ $\frac{3x}{2} = 6 \Leftrightarrow x = 4$

∴ Rate of upstream = 4 km/hr, Rate downstream = 8 km/hr

∴ Rate of current = $\frac{1}{4}(8 - 4)$ km/hr = 2 km/hr

Please remember:

(1) ***Speed & Distance are directly proportional. So if the speed is doubled the distance traveled is also doubled (Time is constant).***

(2) ***Distance and time are directly proportional. If distance to be traveled is tripled, then the time taken would also be tripled (Speed is constant).***

– ***The time is inversely related to the speed.***

– ***If the distance remains same and speed doubles then the time taken to travel that distance becomes 1/2 times the original time taken at the original speed.***

– ***Similarly if the time taken to travel the same distance has become 5 times the original time then we can conclude that the speed must have become $1/5^{th}$ of the original speed.***

Also Remember:

(i) If two objects are moving in opposite directions towards each other or away from each other on a straight line at speeds U & V, then they seem to be moving towards each other or away from each other at a relative speed = (speed of 1^{st}) + (speed of 2^{nd}) = U + V.

(ii) If two men move in the same direction with speeds U and V, then relative speed = difference of their speeds i.e., U - V.

(iii) If a man and train are running simultaneously at 'm' & 't' – may be in the compartment or on the ground – assume the man is stationary and speed of the train is 'm + t' if man is running opposite and 'm-t' if running in the same direction.

(iv) Whether the two trains are moving in the same direction or in the opposite direction the total distance required to be traveled before

they cross each other completely = sum of the lengths of the two trains. And this distance is to be covered at the relative speeds of the train.

(v) Problems in a circular motion make use of both the relative speed and the LCM concept.

(vi) The number of rounds the faster person makes is always one round more than the slow runner whenever and wherever they meet for the first time.

(vii) Moving at X kmph means moving at X.5/18 m/sec.

Practice Exercise - 1

Number of Questions: 15 **Suggested time: 15 minutes**

1. A train has speed of 45 km/ph without stoppages and a speed of 30 km/ph with stoppages. How many minutes per hour does the train stop?
(a) 10
(b) 15
(c) 20
(d) 24

2. A can beat B by 20 m in a 200 m race. B can beat C by 10 min a 250 m race. By how many metres can A beat C in a 100 m race?
(a) 14 m
(b) 12 m
(c) 13.6 m
(d) 14.8 m

3. A train 110 m long is travelling at the speed of 58 km/hr. What is the time in which it will pass a man walking in the same direction at 4 km/hr?
(a) 6 s
(b) 7.5 s
(c) 7.33 s
(d) 7.33 min

4. If a person travels 10.2 km in 3 hours, then the distance covered by him in 5 hours will be
(a) 18 km
(b) 17 km
(c) 16 km
(d) 15 km

5. If I double my speed but reduce the time by one-third, the distance I travel will be what part of my original distance.
(a) four-third
(b) two-third

(c) one-third
(d) same

6. Two trains of equal lengths take 3 seconds and 4 seconds in crossing a telegraph pole. In how much time will these trains cross each other while moving in opposite direction?
(a) 1 sec
(b) 7 sec
(c) $\sqrt{11}$ sec
(d) $3\frac{1}{4}$ sec

7. A car can finish a journey in 10 hours at a speed of 48 kmph. In order to cover the same distance in 8 hours, the speed of the car must increase by
(a) 6 kmph
(b) 7.5 kmph
(c) 12 kmph
(d) 15 kmph

8. X and Y walk around a circular course 35 km in circumference, starting together from the same point. If they walk at the speeds of 4 km/ph and 5 km/ph respectively in the same direction, after how many hours shall they meet next?
(a) 15 hours
(b) 21 hours
(c) 20 hours
(d) 35 hours

9. A man in a train travelling at 30 km/ph notices that a train going in the opposite direction passes him in 9 seconds. If the length of this train is 200 m, find its speed.
(a) 50 kmph
(b) 51 kmph
(c) 52 kmph
(d) 55 kmph

10. I travel a certain distance in X hours at Y km/ph. If I increase the speed by 20 km/ph, I can save an hour. However, on reducing my speed by 20 km/ph, I am on road for 2 hours more. What is X?

(a) 4
(b) 5
(c) 6
(d) can not be determined

11. Ram starts from Inderpuri at 8 a.m. at a speed of 6 km/ph. Shyam starts from the same place in the same direction at 9 a.m. with a speed of 9 km/ph. When will Shyam meet Ram?
(a) 12 noon
(b) 11 a.m
(c) 10 a.m
(d) 1 p.m

12. A student walks from his house at 5 km/ph, reaches school 10 minutes late. If his speed had been 6 km/ph, he would have reached 15 minutes early. The distance of the school from his house is
(a) 5/2 kms
(b) 5/22 kms
(c) 25/2 kms
(d) 25/22 kms

13. A students walks from point A to point B at 12 miles per hour and reaches 5 minutes late. Had he walked at 16 miles per hour, he would have been 5 minutes early. What is his normal rate of walking?
(a) $\sqrt{5}+\sqrt{2}$ kmph
(b) $\frac{39\times5\times12}{30\times6}$
(c) $\frac{30\times30}{6}$ kmph
(d) $\frac{25\times1}{0.128}$ kmph

14. A train 110 m in length is running at a speed of 58 km/ph. The time in which it will cross a passer-by who is walking at 4 km/ph in the same direction is
(a) 6 sec
(b) 7 1/3 sec
(c) 7 1/3 sec

(d) 7 1/3 minute

15. Two trains 200 km away from each other start from opposite directions at the same time. They cross each other at a distance of 110 km from one of the stations. What is the ratio of their speeds?

(a) 11 : 20

(b) 9 : 20

(c) 11 : 9

(d) 9 : 9

Practice Exercise 1 - Answers with Hints - need based

1. (c)	2. (c)	3. (c)	4. (b)	5. (a)
6. (d)	7. (c)	8. (d)	9. (a)	10. (b)
11. (b)	12. (c)	13. (b)	14. (c)	15. (c)

Practice Exercise - 2

Number of Questions: 25 **Suggested time: 25 minutes**

1. Ms Harsimran takes 5 hours 45 min in walking to a certain place and driving back. She would have gained 2 hours by driving both ways. The time (hrs) she would take to walk both ways is.
(a) 11
(b) 8
(c) 7.45
(d) 9.2

2. Mr Jyoti crosses a 600 metre long street in 5 minutes. What is his speed in km per hour?
(a) 8.2
(b) 4.2
(c) 6.1
(d) 7.2

3. When stoppages are not included, the speed of a bus is 54 km/ph and including stoppages, it is 45 km/ph. For how many minutes does the bus stop per hour?
(a) 12
(b) 11
(c) 10
(d) 9

4. Anubhav completes a journey in 10 hours. He travels first half of the journey at the speed of 21 km/hr and second half at the speed of 24 km/hr. Find the total journey in km travelled by him.
(a) 121
(b) 242
(c) 224
(d) 112

5. A train travelling with 5/7 of its actual speed covers 42 km in 6048 seconds. What is the actual speed of the train in km/hr?

(a) 30
(b) 35
(c) 25
(d) 40

6. Mr Sirat covered a certain distance at some speed. Had he moved 3 km/ph faster, he would have taken 2/3 hours less. Had he moved 2 km/ph slower, he would have taken 2/3 hours more. What is the distance covered in km?
(a) 36
(b) 38
(c) 40
(d) 42

7. A and B walk around a circular track. A and B walk at a speed of 2 rounds per hour and 3 rounds per hour respectively. If they start at 10.30 a.m. from the same point in opposite directions, how many times shall they cross each other at noon?
(a) 5
(b) 6
(c) 7
(d) 8

8. Ms Manjit and Ms Harpreet start from the same place walking at the rate of 15 km/ph and 16.5 km/ph respectively in the same direction. In what time (hrs) will they take to be 25.5 km apart?
(a) 17
(b) 14
(c) 12
(d) 19

9. In covering a distance of 30 km, Mr Bhupinder takes 2 hours more than Mr Pritam. If Bhupinder doubles his speed, then he would take 1 hour less than Bhupinder. What is Pritam's speed?
(a) 8
(b) 5
(c) 4
(d) 7

10. A car travels first 160 km at @ 64 km/hr and the next 160 km at @ 80 km/hr. What is the average speed for the first 320 km of the tour?
(a) 70.24
(b) 74.24
(c) 71.11
(d) 72.21

11. Mr Madhukar travelled a distance of 61 km in 9 hours. He travelled partly on foot at 4 km/hr and partly on bicycle at 9 km/hr. What is the distance travelled on bicycle?
(a) 49
(b) 47
(c) 45
(d) 43

12. Walking $6/7^{th}$ of his usual speed, Mr Jatana is 12 minutes too late. What is the usual time (hrs) taken by him to cover that distance?
(a) 1 hr 42 min
(b) 1
(c) 2
(d) 1 hr 12 min

13. Mr Avinash goes to his Bank from his house at a speed of 3 km/hr and returns at a speed of 2 km/hr. If he takes 5 hours in going and coming, find the distance between his house and office (km)?
(a) 3
(b) 4
(c) 5
(d) 6

14. Mr Bharat Bhushan is fond of bicycle. One day he rides his bike 10 km at an average speed of 12 km/hr and again travels 12 km at an average speed of 10 km/hr. Find his average speed (km/ph) for the entire trip approximately?
(a) 11.2
(b) 10
(c) 10.2
(d) 10.8

15. An aeroplane covers a certain distance at a speed of 120 km/ph in 150 minutes. To cover the same distance in 1 – 2/3 hours, it must travel at a speed of:

(a) 660

(b) 680

(c) 700

(d) 720

16. Haryana Express travels at 1 1/2 faster than a car. Both start from Sirsa Railway Station at the same time and reach Bathinda, 75 kms away. On the way, however, the Express lost about 12.5 minutes while stopping at mid-way stations. Find the speed of the car?

(a) 80

(b) 102

(c) 120

(d) 140

17. In a flight of 600m, an aircraft was slowed down due to bad weather. Its average speed for the trip was reduced by 200 km/hr and the time of flight increased by 30 minutes. What is the duration (hrs) of the flight?

(a) 2

(b) 1 1/2

(c) 1/2

(d) 1

18. If Bal Mukand walks at 14 km/hr instead of 10 km/hr, he would have walked 20 km more. What is the actual distance (km) travelled by him?

(a) 80

(b) 70

(c) 60

(d) 50

19. The ratio between the speeds of two trains is 7:8. If the second train covers 250 km in 2 1/2 hours, What is the speed of the first train?

(a) 85

(b) 87.5

(c) 90

(d) 92.5

20. It takes eight hours to cover 600 km journey. If 120 km is covered by train and the rest by car. But it takes 1/3 hrs more, if 80 km is covered more by train and the rest by car. Ratios of speeds are?

(a) 3 : 4

(b) 2 : 3

(c) 1 : 2

(d) 1 : 3

21. Mr Shubham is travelling on his cycle and plans to reach at 1400 hours if he travels at 10 km/ph. He will reach there at 1200 hours if he increases his speed by 5 km/ph. What ideal speed will take him to reach at 1300 hrs.

(a) 8

(b) 10

(c) 12

(d) 14

22. A Volvo Bus travels from Chandigarh to Delhi at an average of 50 miles per hour for 2 1/2 hours and then travels at a speed of 70 miles per hour for 1 1/2 hours. Find the distance travelled.

(a) 210

(b) 230

(c) 250

(d) 260

23. A Bus takes 12 hours from Jaipur to Chandigarh. Its speed increases by 2 km/ph after every one hour. If the distance travelled in the first one hour was 35 km, find the total distance travelled by it?

(a) 422

(b) 552

(c) 502

(d) 492

24. Sound is assumed to travel at about 1100 feet per second. A forest guard hears a sound of an axe striking a tree somewhere, 115 seconds after it strikes the tree. How far is the guard from the wood chopper?

(a) 1800

(b) 2810

(c) 3020
(d) 2420

25. An athlete runs 200 meters race in 24 seconds. What is his speed in km/ph?
(a) 27.5
(b) 20
(c) 25
(d) 30

Practice Exercise 2 - Answers with Hints - need based

1 (c)	2 (d)	3 (c)	4 (c)	5 (b)
6 (c)	7 (c)	8 (a)	9 (b)	10 (c)
11 (c)	12 (d)	13 (d)	14 (d)	15 (d)
16 (c)	17 (d)	18 (d)	19 (b)	20 (a)
21 (c)	22 (b)	23 (b)	24 (d)	25 (d)

7. Relative speed = Speed of A + Speed of B ($\therefore$ they walk in opposite directions)

= 2 + 3 = 5 rounds per hour

Therefore, they cross each other 5 times in 1 hour and 2 times in 12 hour

Time duration from 10.00 a.m. to 12.00 hrs. = 1.5 hour

Hence they cross each other 7 times.

11. $x/9 + (61 - x)/4 = 9$ is the shortest method. Now hit the choices.

13. Let he goes 'x' km. Therefore $x/2 + x/3 = 5$ (hit choices onward)

20. $\Rightarrow 120x + (600 - 120)y = 8 \Rightarrow 15x + 60y = 1$ ... (1)

$\Rightarrow 200x + (600 - 200)y = 25/3 \Rightarrow 24x + 48y = 1$ ... (2)

from (1) and (2) we can find the values of x and y.

Practice Exercise - 3

Number of Questions: 25 **Suggested time: 20 minutes**

1. A train is moving at the speed of 80 km/hr. What is its speed in metres per second?
(a) 2229
(b) 22
(c) 2119
(d) 21

2. The distance between two cities A and B is 330 km. A train starts from A at 8 a.m. and travels towards B at 60 km/hr. Another train starts from B at 9 a.m. and travels towards A at 75 km/hr. At what time will they meet?
(a) 10.30
(b) 10
(c) 12
(d) 11

3. A man walking at the rate of 5 km/hr crosses a bridge in 15 minutes. What is the length of the bridge (in metres)?
(a) 1250
(b) 1280
(c) 1320
(d) 1340

4. A train travelled at an average speed of 100 km/hr, stopping for 3 minutes after every 75 km. How long did it take to reach its destination 600 km from the starting point?
(a) 6 hrs 21 min
(b) 6
(c) 7 hrs 22 min
(d) 7 hrs 14 min

5. A person travels from A to B at a speed of 40 km/hr and returns by increasing his speed by 50%. What is his average speed for both the

trips?
(a) 60
(b) 56
(c) 52
(d) 48

6. A man in a train notices that he can count 21 telephone posts in one minute. If they are known to be 50 metres apart, at what speed is the train travelling?
(a) 61
(b) 56
(c) 63
(d) 60

7. A truck covers a distance of 550 metres in 1 minute whereas a train covers a distance of 33 kms in 45 minutes. What is the ratio of their speed?
(a) 2 : 1
(b) 1 : 2
(c) 4 : 3
(d) 3 : 4

8. A person has to cover a distance of 6 km in 45 minutes. If he covers one-half of the distance in two-thirds of the total time, to cover the remaining distance in the remaining time, what should be his speed in km/hr?
(a) 14
(b) 12
(c) 8
(d) 10

9. The speed of a car increases by 2 kms after every one hour. If the distance travelling in the first one hour was 35 kms. what was the total distance travelled in 12 hours?
(a) 456
(b) 482
(c) 552
(d) 556

10. A thief is noticed by a policeman from a distance of 200 m. The thief starts running and the policeman chases him. The thief and the policeman run at the rate of 10 km and 11 km per hour respectively. What is the distance between them after 6 minutes?

(a) 100

(b) 150

(c) 190

(d) 200

11. A man walking at the rate of 5 km/hr crosses a bridge in 15 minutes. The length of the bridge (in metres) is.

(a) 600

(b) 750

(c) 1000

(d) 1250

12. The number of degrees that the hour hand of a clock moves through between noon and 2.30 in the afternoon of the same day is

(a) 720

(b) 180

(c) 75

(d) 65

13. Two horses start trotting towards each other, one from A to B and another from B to A. They cross each other after one hour and the first horse reaches B, 5/6 hour before the second horse reaches A. If the distance between A and B is 50 km. what is the speed of the slower horse in km/ph?

(a) 30

(b) 15

(c) 25

(d) 20

14. A man takes 6 hours 15 minutes in walking a distance and riding back to the starting place. He could walk both ways in 7 hours 45 minutes. The time taken by him to ride both ways, is.

(a) 4

(b) 4 1/2

(c) 4 3/4
(d) 5

15. Two trains A and B start simultaneously in the opposite direction from two points P and Q and arrive at their destinations 16 and 9 hours respectively after their meeting each other. At what speed (kmph) does the second train B travel if the first train travels at 120 km/h.
(a) 90
(b) 160
(c) 67.5
(d) Cant say

16. A man on tour travels first 160 km at 64 km/hr and the next 160 km at 80 km/hr. The average speed for the first 320 km of the tour is _____ km/ph
(a) 35.55
(b) 36
(c) 71.11
(d) 71

17. The ratio between the speeds of two trains is 7 : 8. If the second train runs 400 kms in 4 hours, then the speed of the first train is km/ph?
(a) 83.5
(b) 84.5
(c) 86.5
(d) 87.5

18. A person crosses a 600 m long street in 5 minutes, What is his speed in km per hour?
(a) 3.6
(b) 7.2
(c) 8.4
(d) 10

19. A man in a train notices that he can count 82 telephone posts in one minute. If they are known to be 25 metres apart, then at what speed (kmph) is the train travelling?
(a) 60
(b) 100
(c) 110

(d) 120

20. A boat sails 15 km of a river towards upstream in 5 hours. How long will it take to cover the same distance downstream, if the speed of current is one-fourth the speed of the boat in still water:

(a) 1.8

(b) 3

(c) 4

(d) 5

21. The distance from town A to town B is five miles. C is six miles from B. Which of the following could be the maximum distance from A to C?

(a) 11

(b) 8

(c) 2

(d) 4

22. Mr Anil Chawla leaves Mumbai at 6 am and reaches Bangalore at 10 am. Ms Sonia leaves Bangalore at 8 am and reaches Mumbai at 11:30 am. At what time do they cross each other?

(a) 10.00

(b) 08.32

(c) 08.56

(d) 09.20

23. A machine puts 'c' caps on bottles in 'm' minutes. How many hours will it take to put all caps on 'b' bottles.

(a) 60 bm/c

(b) bm/60c

(c) bc/60m

(d) 60b/cm

24. A train overtakes two girls, Ms Soumaya and Ms Prerna who are walking in the opposite direction in which the train is going at the rate of 3 km/h and 6km/h and passes them completely in 36 seconds and 30 seconds respectively. The length of the train (m) is:

(a) 120

(b) 150

(c) 125

(d) Data inadequate

25. The distance of the office and home of Mr Miglani is 80 km. One day he was late by 1 hour than the normal time to leave for the office. He increased his speed by 4 km/h and thus reached at the normal time. What is the changed (or increased) speed of Miglani?

(a) 28

(b) 30

(c) 40

(d) 20

Practice Exercise 3 - Answers with Hints - need based

1 (a)	2 (d)	3 (a)	4 (a)	5 (d)
6 (d)	7 (d)	8 (b)	9 (c)	10 (a)
11 (d)	12 (c)	13 (d)	14 (c)	15 (b)
16 (c)	17 (d)	18 (b)	19 (d)	20 (b)
21 (a)	22 (c)	23 (b)	24 (b)	25 (c)

4. Given that the train stops after every 75 km or 600/75 = 8.

It means, the train stops 7 times before 600 km and 1 time just after 600 km. Hence

we need to take only 7 stops into consideration for the 600 km journey.

Hence, total stopping time in the 600 km journey $=7 \cdot 3 = 21$ minutes

Total time needed to reach the destination = 6 hours + 21 minutes

= 6 hours 21 minutes.

12. The hour hand moves from pointing to 12 to pointing to half way between 2 and 3. The angle covered between each hour marking on the clock is 360/12 = 30.

Since the hand has covered 2 1/2 of these divisions the angle moved through is 75°.

13. If the speed of the faster horse be f_s and that of slower horse be S_s then

$$f_s + s_s = \frac{50}{1} = 50$$

$$\frac{50}{s_s} - \frac{50}{f_s} = \frac{5}{6}$$

Now, hit the options

15. $\frac{s1}{s2} = \sqrt{\frac{t2}{t1}} \Rightarrow \frac{120}{s2} = \sqrt{\frac{9}{16}} = \frac{3}{4}$

25. Let the normal speed be *x* km/h, then

$$\frac{80}{x} - \frac{80}{(x+4)} = 1$$

TRAINS

23

(1) Distance = Speed × Time

Speed = Distance/Time

Time = Distance/Speed

(2) Convert kilometres per hour (km/hr) to metres per second (m/s)

X km/hr = X × 5/18 m/s

(3) Convert metres per second (m/s) to kilometres per hour (km/hr)

X m/s = X × 18/5 km/hr

(Problems on Trains)

(1) Time taken by a train x meters long in passing a signal post or a pole or a standing man is the same as the time taken by the train to cross 'x' meters with its own speed.

(2) Time taken by a train 'x' meters long in passing a stationary object of length 'y' meters is the same as the time taken by the train to cover (x + y) meters with its own speed.

(3) Suppose two trains or bodies are moving in the same directions at 'u' km/hr and 'v' km/hr such that u > v, then their relative speed = (u – v) km/hr.

If two trains of lengths 'x' km and 'y' km move in the same direction at 'u' km/hr and 'v' km/hr (where u > v), then time taken by them to cross each other is $\left(\frac{x+y}{u-v}\right)$ ***hrs.***

(4) Suppose two trains or bodies are moving in the opposite directions at 'u' km/hr and 'v' km/hr. Then, relative speed = (u + v) km/hr.

If two trains of length 'x' km and 'y' km move in the opposite directions at 'u' km/ph and 'v' km/pr, then time taken to cross each

***other* = $\left(\dfrac{x+y}{u-v}\right)$**

(5) If two trains start at the same time from stations A and B towards each other and after crossing, they take 'a' & 'b' hours in reaching B and A respectively. Then,

A's speed : B's speed = ($\sqrt{5}-\sqrt{2}$).

Example A train 100 m long is running with a speed of 70 km/hr. In what time will it pass a man who is running at 10 km/hr in the same direction in which the train is going?

Solution

Speed of train relative to man = (70 – 10) kmph = $\left(60\times\dfrac{5}{18}\right)$ m / sec = $\left(\dfrac{50}{3}\right)$ m / sec. Time taken by the train to cross the man = Time taken by it to cover 100 m at $\left(\dfrac{50}{3}\right)$ m/sec. = $\left(100\times\dfrac{3}{50}\right)$ seconds = 6 seconds.

Practice Exercise - 1

Number of Questions: 20 **Suggested Time: 15 minutes**

1. A train is running at a speed of 40 km/hr and it crosses a post in 18 seconds. What is the length of the train?
(a) 190
(b) 160
(c) 200
(d) 120

2. A train, 130 metres long travels at a speed of 45 km/hr crosses a bridge in 30 seconds. The length of the bridge in meter is?
(a) 190
(b) 160
(c) 200
(d) 120

3. A train has a length of 150 metres. It is passing a man who is moving at 2 km/hr in the same direction of the train, in 3 seconds. Find out the speed of the train in km/ph.
(a) 182
(b) 180
(c) 152
(d) 169

4. A train having a length of 240 metre passes a post in 24 seconds. How many seconds will it take to pass a platform having a length of 650 metre?
(a) 120
(b) 99
(c) 89
(d) 80

5. A train 360 metre long runs with a speed of 45 km/hr. What time (in seconds) will it take to pass a platform of 140 metre long?
(a) 38

(b) 35
(c) 44
(d) 40

6. Two trains running in opposite directions cross a man standing on the platform in 27 seconds and 17 seconds respectively. If they cross each other in 23 seconds what is the ratio of their speeds?
(a) 3 : 1
(b) 1 : 3
(c) 3 : 2
(d) insufficient data

7. A jogger is running at 9 kmph alongside a railway track in 240 metres ahead of the engine of a 120 metres long train. The train is running at 45 kmph in the same direction. How many seconds does it take for the train to pass the jogger?
(a) 46
(b) 36
(c) 18
(d) 22

8. Two trains of equal length are running on parallel lines in the same direction at 46 km/hr and 36 km/hr.
If the faster train passes the slower train in 36 seconds, what is the length of each train?
(a) 88
(b) 70
(c) 62
(d) 50

9. Two trains having length of 140 metre and 160 metre long run at the speed of 60 km/hr and 40 km/hr respectively in opposite directions (on parallel tracks). The time (in seconds) which they take to cross each other, is
(a) 10.8
(b) 12
(c) 9.8
(d) 8

10. Two trains are moving in opposite directions with speed of 60 km/hr and 90 km/hr respectively. Their lengths are 1.10 km and 0.9 km respectively. The time (in seconds) taken by the slower train to cross the faster train is

(a) 56

(b) 48

(c) 47

(d) 26

11. A train passes a platform in 36 seconds. The same train passes a man standing on the platform in 20 seconds. If the speed of the train is 54 km/hr, The length of the platform is

(a) 280

(b) 240

(c) 200

(d) None

12. A train moves past a post and a platform 264 metre long in 8 seconds and 20 seconds respectively. What is the speed in km/ph of the train?

(a) 79.2

(b) 69

(c) 74

(d) 61

13. Two trains having equal lengths, take 10 seconds and 15 seconds respectively to cross a post. If the length of each train is 120 metres, in what time (in seconds) will they cross each other when travelling in opposite direction?

(a) 10

(b) 25

(c) 12

(d) 20

14. Two trains, one from P to Q and the other from Q to P, start simultaneously. After they meet,the trains reach their destinations after 9 hours and 16 hours respectively. The ratio of their speeds is

(a) 2 : 3

(b) 2 : 1

(c) 4 : 3
(d) 3 : 2

15. A train having a length of 14 mile, is travelling at a speed of 75 mph. It enters a tunnel 312 miles long. How many seconds does it take the train to pass through the tunnel from the moment the front enters to the moment the rear emerges?
(a) 3
(b) 4.2
(c) 3.4
(d) 5.5

16. A train runs at the speed of 72 kmph and crosses a 250 metre long platform in 26 seconds. What is the length of the train?
(a) 270
(b) 210
(c) 340
(d) 130

17. A train overtakes two persons who are walking in the same direction to that of the train at 2 kmph and 4 kmph and passes them completely in 9 and 10 seconds respectively. What is the length (meter) of the train?
(a) 62
(b) 54
(c) 50
(d) 55

18. A train is travelling at 48 kmph. It crosses another train having half of its length, travelling in opposite direction at 42 kmph, in 12 seconds. It also passes a railway platform in 45 seconds. What is the length of the platform?
(a) 500
(b) 360
(c) 480
(d) 400

19. A train having a length of 270 metre is running at the speed of 120 kmph. It crosses another train running in opposite direction at the speed of 80 kmph in 9 seconds. What is the length of the other train?

(a) 320
(b) 190
(c) 210
(d) 230

20. Two trains, each 100 metre long are moving in opposite directions. They cross each other in 8 seconds. If one is moving twice as fast the other, the speed (km/ph) of the faster train is
(a) 75
(b) 60
(c) 35
(d) 70

Practice Exercise 1 - Answers with Hints - need based

1 (c)	2 (b)	3 (a)	4 (c)	5 (d)
6 (d)	7 (b)	8 (d)	9 (a)	10 (d)
11 (c)	12 (a)	13 (c)	14 (c)	15 (a)
16 (a)	17 (c)	18 (d)	19 (d)	20 (b)

2. Speed = (Distance)/(Time) $\Rightarrow$ 45 × 5/18 = (130 + B)/30

$\Rightarrow$ 130 + B = 375

$\Rightarrow$ B = 245

4. Let required time be ‘t’ seconds.

240 : (240 + 650) = 24 : t

$\Rightarrow$ t = 890/10 = 89.

9. Distance = (140 + 160) = 300 m = 0.3 km

Relative speed = (60 + 40) = 100 km/hr

Required time = 0.3100 hr = 0.3100 × 3600 seconds = 10.8 seconds

13. Speeds of trains are 120/10 = 12 m/s; and 120/15 = 8 m/s

As they travel in opposite direction, relative speed = 12 + 8 = 20 m/s

Distance covered = 120 + 120 = 240 metres

Time = 240/20 = 12 seconds

14. The given ratio, after the trains meet each other, are $S_1 : S_2 :: \sqrt{T_2} : \sqrt{T_1}$

Required Ratio = $\sqrt{16} : \sqrt{9} = 4 : 3$

17. Let speed of the train be 'S' km/ph

(S − 2) : (S − 4) = 10 : 9 ($\because$ speed and time are inversely proportional)

$\Rightarrow$ 9 S − 18 = 10 S − 40 $\Rightarrow$ S = 22; Length of the train = (22 − 2) × 5/18 × 9 = 50 metre

18. 1 ½ times the length of the first train = (48 + 42) × 5/18 × 12 = 300 metre

$\Rightarrow$ Length of the first train = 200 metre

Distance covered by the first train in 45 seconds = 48 × 5/18 × 45 = 600 metre

Therefore, length of the platform = 600 − 200 = 400 metre

19. Let the length of the other train be 'x' meter

Therefore, (270 + x) = (120 + 80) × 5/18 × 9 $\Rightarrow$ x = 230 m

20. Solution 2 Relative Speed = (100 + 100)/8 = 25 m/s

$\Rightarrow$ Sum of their speeds = 25 m/s

Ratio of their speeds = 1 : 2

Therefore, speed of the faster train

= 25 × 2/3 = 50/3 m/s or = 50/3 × 18/5 km/hr = 60 km/hr.

Practice Exercise - 2

Number of Questions: 25 **Suggested Time: 15 minutes**

1. Two stations P and Q are 110 km apart on a straight track. One train starts from P at 7 a.m. and travels towards Q at 20 kmph. Another train starts from Q at 8 a.m. and travels towards P at a speed of 25 kmph. At what time will they meet?
(a) 10:30
(b) 10:00
(c) 9:10
(d) 11:00

2. A train overtakes two persons walking along a railway track. The first person walks at 4.5 km/hr and the other walks at 5.4 km/hr. The train needs 8.4 and 8.5 seconds respectively to overtake them. What is the speed (km/ph) of the train if both the persons are walking in the same direction as the train?
(a) 81
(b) 88
(c) 62
(d) 46

3. A train, having a length of 110 metre is running at a speed of 60 kmph. In what time (seconds), it will pass a man who is running at 6 kmph in the direction opposite to that of the train
(a) 10
(b) 8
(c) 6
(d) 4

4. A 300 metre long train crosses a platform in 39 seconds while it crosses a post in 18 seconds. What is the length of the platform?
(a) 150
(b) 350
(c) 420
(d) 600

5. A train crosses a post in 15 seconds and a platform 100 metre long in 25 seconds. Its length is

(a) 150

(b) 300

(c) 400

(d) 180

6. A train, 800 metre long is running with a speed of 78 km/hr. It crosses a tunnel in 1 minute. What is the length of the tunnel (in metres)?

(a) 440

(b) 500

(c) 260

(d) 430

7. Two train each 500 m long, are running in opposite directions on parallel tracks. If their speeds are 45 km/hr and 30 km/hr respectively, the time taken by the slower train to pass the driver of the faster one is

(a) 50

(b) 58

(c) 24

(d) 22

8. Two trains are running at 40 km/hr and 20 km/hr respectively in the same direction. If the faster train completely passes a man sitting in the slower train in 5 seconds, the length of the faster train (in meter) is:

(a) 19

(b) 27.79

(c) 13.29

(d) 33

9. Two trains are running in opposite directions in the same speed. The length of each train is 120 metre. If they cross each other in 12 seconds, the speed of each train (in km/hr) is

(a) 42

(b) 36

(c) 28

(d) 20

10. A train 108 metre long is moving at a speed of 50 km/hr. It crosses a train 112 metre long coming from opposite direction in 6 seconds. What is the speed of the second train?

(a) 82

(b) 76

(c) 44

(d) 58

11. How many seconds will a 500 metre long train moving with a speed of 63 km/hr, take to cross a man walking with a speed of 3 km/hr in the direction of the train?

(a) 42

(b) 50

(c) 30

(d) 28

12. A man sitting in a train which is traveling at 50 kmph observes that a goods train, traveling in opposite direction, takes 9 seconds to pass him. If the goods train is 280 m long, find its speed?

(a) 60

(b) 62

(c) 64

(d) 65

13. Two cogged wheels of which one has 32 cogs and other 54 cogs, work into each other. If the latter turns 80 times in three quarters of a minute, how often does the other turn in 8 seconds?

(a) 48

(b) 24

(c) 38

(d) 36

14. A train moves past a telegraph post and a bridge 264 m long in 8 seconds and 20 seconds respectively. What is the speed of the train?

(a) 69.5

(b) 70

(c) 79

(d) 79.2

15. A train of length 110 meter is running at a speed of 60 kmph. In what time, it will pass a man who is running at 6 kmph in the direction opposite to that in which the train is going?

(a) 10

(b) 8

(c) 6

(d) 4

16. Two trains having equal lengths, take 10 seconds and 15 seconds respectively to cross a post. If the length of each train is 120 meters, in what time (in seconds) will they cross each other when travelling in opposite direction?

(a) 10

(b) 25

(c) 12

(d) 20

17. A train travelling at a speed of 75 mph enters a tunnel $\frac{22}{11}$ miles long. The train is $\frac{1}{2}$ mile long. How long does it take for the train to pass through the tunnel from the moment the front enters to the moment the rear emerges?

(a) 2.5

(b) 3

(c) 3.2

(d) 3.5

18. Two trains 140 m and 160 m long run at the speed of 60 km/hr and 40 km/hr respectively in opposite directions on parallel tracks. The time (in seconds) which they take to cross each other, is:

(a) 9

(b) 9.6

(c) 10

(d) 10.8

19. Two trains started at the same time, one from A to B and the other from B to A. If they arrived at B and A respectively 4 hours and 9 hours after

they passed each other the ratio of the speeds of the two trains was?
(a) 2 : 1
(b) 3 : 2
(c) 4 : 3
(d) 5 : 4

20. A 270 metres long train running at the speed of 120 kmph crosses another train running in opposite direction at the speed of 80 kmph in 9 seconds. What is the length of the other train?
(a) 230
(b) 240
(c) 260
(d) 320

21. A train moves with a speed of 108 kmph. Its speed in metres per second is:
(a) 10.8
(b) 18
(c) 30
(d) 38.8

22. How many seconds will a 500 meter long train moving with a speed of 63 km/hr take to cross a man walking with a speed of 3 km/hr in the direction of the train?
(a) 42
(b) 50
(c) 30
(d) 28

23. A train speeds past a pole in 15 seconds and a platform 100 m long in 25 seconds. Its length is:
(a) 50
(b) 150
(c) 200
(d) Data inadequate

24. A train 800 metres long is running at a speed of 78 km/hr. If it crosses a tunnel in 1 minute, then the length of the tunnel (in meters) is:
(a) 130

(b) 360
(c) 500
(d) 540

25. A train 110 metres long is running with a speed of 60 kmph. In how many seconds will it pass a man who is running at 6 kmph in the direction opposite to that in which the train is going?
(a) 5
(b) 6
(c) 7
(d) 10

Practice Exercise 2 - Answers with Hints - need based

1 (b)	2 (a)	3 (c)	4 (b)	5 (a)
6 (b)	7 (c)	8 (b)	9 (b)	10 (a)
11 (c)	12 (b)	13 (b)	14 (d)	15 (c)
16 (c)	17 (b)	18 (d)	19 (b)	20 (a)
21 (c)	22 (c)	23 (b)	24 (c)	25 (b)

1. Distance travelled by first train in 1 hour = 20 km

Therefore, at 8 a.m., both trains will be $(110 - 20) = 90$ km apart.

$D = S \times T = 90 = (20 + 25) \times T$ or $T = 2$ i.e., they will meet at 10 a.m.

2. Let the speed of the train be 'S' km/ph

$(S - 4.5) : (S - 5.4) = 8.5 : 8.4$ ($\because$ speed and time are inversely proportional)

$\Rightarrow (S - 4.5) : (S - 5.4) = 85 : 84 \Rightarrow 84(S - 4.5) = 85(S - 5.4) \Rightarrow S = 81$

8. $D = S \times T = (40 – 20) \times (5/18) \times 5 = 27.79$

13. Distance covered by first wheel = Distance covered by second wheel.

Per second turning = $54 \times 80/45 = 32 \times K/8$ (K is number of turns of 2nd wheel)

15. $D = S \times T$ or $110 = (60 + 6) \times 5/18 \times T$; Hence $T = 616$. Speeds of these trains are 120/10, 120/15

i.e., 12 and 8 m/sec

Going in opposite directions @ (12 + 8) m/sec to cover (120 + 120) m they will take 12 seconds.

19. The given ratio, after the trains reach, are $S_1 : S_2 :: \sqrt{T_2} : \sqrt{T_1}$

Required Ratio $= \sqrt{9} : \sqrt{4} = 3 : 2$

20. $(270 + D) = (120 + 80) \times 5/18 \times 9$.

SURDS & INDICES

24

INDICES

The process of multiplying any quantity by itself any number of times is known as involution and the product thus obtained is called power of that quantity. The number which expresses the power is known as index of power or exponent of power. This number is written above a quantity to its right which is known as Base. Index or exponent indicates how many times the base is to be multiplied by itself. For example if a quantity a is multiplied by itself m times, then we write it as follows:

a . a . a . a …….m times = a^m. Here 'a' is the base, 'm' is the index or exponent, a^m is the mth power of 'a'. We read it as "a to the power m".

Thus, it is obvious that,

$(a) \times (a) = a^2$

$(a) \times (a) \times (a) = a^2$

$(a) \times (a) \times (a) \times (a) = a^4$ etc.

The plural of index is indices.

If no index is written on any base, then, the index is regarded as 1. For example, a = a*1*, 4 = 4*1*,
35 = 35*1* etc.

Laws of Indices

(i) $a^m \times a^n = a^{m+n}$

(ii) $a^m \div a^n = a^{m-n}$

(iii) $(a^m)^n = a^{mn}$

(iv) $a^0 = 1$ ***where*** $a \neq 0$

(v) $a^{-m} = \frac{1}{a^m}$

(vi) $\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}$

(vii) $(ab)^m = a^m b^m$

(viii) $m\sqrt{a^m} = a$

(ix) $n\sqrt{5} = a^{1/n}$

(x) $a^{p/q} = (a^p)^{1/q} = ^q\sqrt{a^p} = (a^{1/q})^p$

(xi) $a^m = a^n \Rightarrow m = n$

(xii) $a^m = b^m \Rightarrow a = b$

For solution of such problems always remember:

- ***Reduce all bigger numbers to the lowest shape of factors***
- ***Try to compare the basis or the powers***
- ***Generally avoid dealing with –ve powers***
- ***Generally the answer is 1 where cycle is complete.***

Remember these formulae by heart and apply as and when given a choice:

- $a(b + c) = ab + ac$ (distributive law)
- $(a + b)^2 = a^2 + 2ab + b^2$
- $(a - b)2 = a^2 - 2ab + b^2$
- $(a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$
- $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3 = a^3 + 3ab(a + b) + b^3$
- $(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3 = a^3 - 3ab(a - b) - b^3$
- $a^2 - b^2 = (a - b)(a + b)$
- $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$
- $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$

- $a^n - b^n = (a - b)(a^{n-1} + a^{n-2}b + a^{n-3}b^2 + \cdots + b^{n-1})$
- $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$
- $(a + b)(a + c) = a^2 + (b + c)a + bc$
- $a^n = a.a.a \ldots$ (n times)
- $a^m.a^n \ldots a^p = a^{m+n+p}$
- $a^m.a^n = a^{m+n}$
- $a^m/a^n = a^{m-n}$
- $(am)^n = a^{mn} = (a^n)^m$
- $(ab)^n = a^n b^n$
- $(ab)^n = a^n b^n$
- $a^{-n} = 1/a^n$
- $a^n = 1/a^{-n}$
- $(x + a)^2 = x^2 + 2ax + a^2$
- $(x - a)^2 = x^2 - 2ax + a^2$
- $(x + a)(x + b) = x^2 + (a + b)x + ab$
- $x^2 - a^2 = (x - a)(x + a)$
- $(x^n + 1)$ is completely divisible by $(x + 1)$ when n is odd
- $(x^n + a^n)$ is completely divisible by $(x + a)$ when n is odd
- $(x^n - a^n)$ is completely divisible by $(x - a)$ for every natural number n
- $(x^n - a^n)$ is completely divisible by $(x + a)$ when n is even
- $a^0 = 1$ where $a \in R, a \neq 0$
- if $a^m = a^n$ where $a \neq 0$ and $a \neq \pm 1$, then $m = n$
- if $a^n = b^n$ where $n \neq 0$, then $a = \pm b$

Example Find the value of (i) $(256)^{1/4}$ (ii) 3^{-4}

Solution

(i) $(256)^{1/4} = (4 \times 4 \times 4 \times 4)^{1/4} = (4^4)^{1/4} = 4$

(ii) $3^{-4} = \frac{3x}{2} = 28\sqrt{?} + 1426 = \frac{3}{4}$

Example If $a = b^z$, $b = c^v$ and $c = a^z$ then, prove that $xyz = 1$.

Solution

$a = b^z = (c^y)^z = c^{zy}$ or $(a^z)^{xy} = a^{zyz}$ $\therefore xyz = 1.$

Example Simplify.

$$\left(\frac{a^x}{a^y}\right)^{x+y^2+xy} x \left(\frac{a^y}{a^z}\right)^{y+2^3+yz} x \left(\frac{a^z}{a^x}\right)^{z^2+x^2+yx}$$

Solution

$$\left(\frac{a^x}{a^y}\right)^{x+y^2+xy} x \left(\frac{a^y}{a^z}\right)^{y+2^3+yz} x \left(\frac{a^z}{a^x}\right)^{z^2+x^2+yx}$$

$$= a^{(x-y)(x^2+y^{2+xy})} x a^{(y-z)(y^2+z^2+yz)} x a^{(z-a)(z^2+x^2+zx)}$$

$$= a^{x^2-y^2+y^2-z^2+z^2-x^2} \frac{142}{22}, \frac{264}{192}, \frac{1450}{600}, \frac{375}{625}, \frac{625}{225}, \frac{278}{417} \; a^0 = 1.$$

Example Find the square root of $11 + 6\sqrt{2}$

Solution

Let $\sqrt{1 \ + 6\sqrt{2}} = \sqrt{5} + \sqrt{y}$ Squaring both sides, we get

$11 + 6\sqrt{2} = x + y + 2\sqrt{xy}$ comparing rational and irrational parts we get

$x + y = 11$ & $2\sqrt{xy} = 6 \ \sqrt{2}$ say $\sqrt{xy} = 3\sqrt{2}$ or $xy = 18$

by applying the formula $(x - y)^2 = (x + y)^x - 4\,xy$

we get $x - y = 7$ solving further $x = 9$, $y = 2$

writing in the proper shape $\sqrt{11+6\sqrt{2}} = \sqrt{5} + \sqrt{y} = \sqrt{5} + \sqrt{2} = \mathbf{3} + \sqrt{2}$ **Ans**

Example Simplify $\sqrt{50} - \sqrt{18} + \sqrt{8}$

Solution

$\sqrt{50} - \sqrt{18} + \sqrt{8} = 3\sqrt{2} - \sqrt{3} - 14 + 8\sqrt{3} + \sqrt{2\times2\times2} = 5\sqrt{2} - 3\sqrt{2} + 2\sqrt{2} = 4\sqrt{2}$

Practice Exercise - 1

Number of Questions: 30 **Suggested time: 20 minutes**

1. The value of $8 \times 3\sqrt{26} \times 15 \times 3\ \sqrt{169}$ is
(a) 2880 x $2^{1/3}$
(b) $(480)^{1/3}$
(c) $1480 \times 3^{1/3}$
(d) none

2. Solve $3^{x+2} = 2187$
(a) 5
(b) 6
(c) 3
(d) 4

3. Solve $(\sqrt{5})^{30+3} = 9\,(3)^{30+3} \times 27 \times A$; find A
(a) 3
(b) $-\frac{3}{4}$
(c) can't be solved
(d) none

4. Simplify $(a^x)^{y-z} \times (a^y)^{z-x} \times (a^z)^{x-y}$
(a) 0
(b) 1
(c) 2
(d) $\frac{1}{4}$.

5. Simplify $\frac{2^n.6^{m+1}.10^{m-n}.15^{m+n-2}}{4m.3^{2m+n}.25^{m-1}}$
(a) $\frac{1}{4}$
(b) $\frac{1}{9}$
(c) 1

(d) 2.

6. Find the value of $\sqrt{12-\sqrt{12+\sqrt{12+\sqrt{12+......}}}}$

(a) 1

(b) 2

(c) 3

(d) 4.

7. Find the square root of 7 + 2$3\sqrt{3}$

(a) $\sqrt{5}$ + $\sqrt{2}$

(b) $\sqrt{5}$ − $\sqrt{2}$

(c) $\sqrt{5}$ + $\sqrt{5}$

(d) $\sqrt{5}$ − $\sqrt{5}$.

8. If $\left(\frac{a}{b}\right)^{x-1}=\left(\frac{b}{a}\right)^{x-3}$ then x is equal to

(a) 1

(b) $\frac{1}{4}$

(c) $\frac{1}{4}$

(d) 2.

9. $\frac{5^{m+2}+25^{\frac{m-3}{2}}}{125^{\frac{m+2}{3}}}$

(a) 3125

(b) $\frac{3126}{3125}$

(c) $\frac{3126}{25}$

(d) $\frac{3126}{3125}$.

10. $64^{x-3}=128^{3x-2}$ then x =

(a) $16\frac{2}{3}$

(b) $\frac{17}{2}$

(c) $\frac{2/3}{1/9}$

(d) $\frac{17}{2}$

11. $343^3 - 216^2$ must be ÷ by

(a) 13, 127

(b) 13 only

(c) 559 only

(d) none

12. $4096^2 + 256^3$ is ÷ by

(a) 2^{12}

(b) 21^7

(c) 256^3

(d) all of these.

13. $(19^{26} + 18^{17} - 7^{17} - 8^{26})$ is ÷ by

(a) 11

(b) 22

(c) 33

(d) 44.

14. $\frac{3x\sqrt{5}}{\sqrt{5}-2}$ simplifies to

(a) $11 + 5\sqrt{5}$

(b) $\sqrt{5} + 2$

(c) $3 - \sqrt{5}$

(d) $\sqrt{5} + 1$.

15. $\sqrt{313+\sqrt{115+\sqrt{36}}}$

(a) 24

(b) 16

(c) 11

(d) 18.

16. If $4^m + 3^n = 43$, where m and n are natural numbers find the value of $(m + n)^2 - 4mn$

(a) 1

(b) 24

(c) 37

(d) 25.

17. Convert the mixed surd $(0.1 \times \sqrt{0.1})$ to pure surd.

(a) $14+8\sqrt{3}$

(b) $\sqrt{0.001}$

(c) $\sqrt{0.1}$

(d) $\sqrt{4.41}$

18. If $\sqrt{1-\frac{x}{361}}=\frac{18}{19}$ find the value of x.

(a) 21

(b) 37

(c) 35

(d) 28.

19. $\left(\dfrac{a^{-2/3}b^{4/5}c^{2/7}\left(a^{3/5}b^{5/6}c^{3/7}\right)}{a^{4/5}b^{2/3}c^{1/2}}\right)=a^x b^y c^z$ then the value of x, y, z is

(a) $\frac{-13}{15},\frac{11}{13},\frac{17}{14}$

(b) $\frac{x}{9}=\frac{x\times R\times R}{100}$

(c) $\frac{11}{15},\frac{29}{30},\frac{3}{14}$

(d) $\frac{-11}{15},\frac{29}{30},\frac{-3}{14}$

20. Find the value of $2+\sqrt{2}+\frac{1}{2+\sqrt{2}}+\frac{1}{\sqrt{2}-2}$

(a) $2 + \sqrt{2}$

(b) $\sqrt{2} - 2$

(c) $2\sqrt{2}$

(d) 2.

21. $3^{x+4} - 3^{x+3} = 2$ Find the value of x.

(a) 0

(b) – 3

(c) 2

(d) 2.

22. Express $\frac{1}{1-\sqrt{2}+\sqrt{3}}$ with rational denominator.

(a) $\frac{-(1-\sqrt{2}+\sqrt{3})}{4}$

(b) $\frac{(1-\sqrt{2}-\sqrt{3})}{4}$

(c) $\frac{(1-\sqrt{2}-\sqrt{3})}{4}$

(d) $\frac{-(1-\sqrt{2}+\sqrt{3})\sqrt{2}}{4}$.

23. $\sqrt{17+12\sqrt{2}}$ **simplifies to**

(a) $\sqrt{15+\sqrt{2}}$

(b) $\sqrt{10}+\sqrt{7}$

(c) $\sqrt{5}+\sqrt{2}$

(d) $\sqrt{5}+\sqrt{2}$

24. Find the square root of $11 - 2\sqrt{ab}$

(a) $\sqrt{5} + \sqrt{6}$

(b) $\sqrt{5} - \sqrt{6}$

(c) $\sqrt{7} + 2$

(d) $2 - \sqrt{7}$

25. The value of $(2187)^{-8/7}$ is

(a) $\frac{3}{60}$

(b) $\frac{1}{100}$

(c) $\frac{343}{512}$

(d) none

26. Find the value of $\sqrt{20-\sqrt{20-\sqrt{20-\sqrt{20-......}}}}$

(a) 5

(b) 10

(c) 3

(d) 4.

27. The value of $(a^{\frac{2}{3}}+a^{\frac{3}{4}})(a^{\frac{1}{3}}-a^{\frac{1}{4}})$ is

(a) $a^{1/3}$

(b) $a^{1/3}$

(c) $a^{1/3}$

(d) none

28. The value of $\left[\sqrt{32^{-2/5}x125^{2/3}} \times 8\right]$ is

(a) $\frac{1}{4}$

(b) $\frac{1}{4}$

(c) $\frac{1}{4}$

(d) 10

29. The reverse value of the square of $\left(\frac{64}{125}\right)^{-\frac{1}{3}}$ x is

(a) $\frac{1}{4}$

(b) $\frac{1}{4}$

(c) $\frac{1}{4}$

(d) 25/16

30. If 2 x = $(1024)^{\sqrt{7}}$, what is the value of x?

(a) 7/10

(b) – 7/10
(c) 10/7
(d) None

Practice Exercise - 1 Answers with Hints - need based

1 (d)	2 (a)	3 (d)	4 (b)	5 (a)
6 (d)	7 (a)	8 (d)	9 (b)	10 (d)
11 (d)	12 (c)	13 (a)	14 (b)	15 (d)
16 (a)	17 (b)	18 (b)	19 (c)	20 (d)
21 (b)	22 (b)	23 (c)	24 (b)	25 (d)
26 (d)	27 (d)	28 (d)	29 (d)	30 (b)

6, 26 Always factorise the number into two **consecutive factors**. Select the greater if +ve is used and the smaller if –ve is used.

7, 23 Bifurcate the number under square into two parts that their sum is the first part e.g. $10 = 2 \times 5$, $7 = 2 + 5$

5, 10, 11, 12, 29 Always/Prefer factorise bigger numbers into smaller perfect squares 4, 9, 16, 25, 36 or
factors.

Practice Exercise - 2

Number of Questions: 26 **Suggested time: 15 minutes**

1. $(132)^7 \times (132)^? = (132)^{11.5}$.
(a) 3
(b) 3.5
(c) 4
(d) 4.5

2. $(ab)^{x-2} = (ba)^{x-7}$ then what is the value of x?
(a) 3
(b) 3.5
(c) 4
(d) 4.5

3. If $7^{(x-y)} = 343$ and $7^{(x+y)} = 16807$, what is the value of x?
(a) 4
(b) 3
(c) 2
(d) 1

4. $(0.04)^{-2.5} = A^5$; find the value of A
(a) 0.2
(b) 16
(c) 8
(d) None

5. $(6)^{6.5} \times (36)^{4.5} \div (216)^{4.5} = (6)^{2k}$; then K is?
(a) ½
(b) 1
(c) 2
(d) 3

6. The value of $11 + P^{n-m} + 11 + P^{m-n}$ =?
(a) 2
(b) 11 + P

(c) 1

(d) 1/P

7. If $x = (8 + 2\sqrt{7})$, what is the value of $(\sqrt{x} - 1/\sqrt{x})$?

(a) 12

(b) 4

(c) 2

(d) None

8. Given that $6^m = 46656$, What is the value of 6^{m-2}

(a) 36

(b) 7776

(c) 216

(d) 1296

9. Find the value of $10^{222} \div 10^{220}$ =?

(a) 10

(b) 100

(c) 1000

(d) 10000

10. If m and n are whole numbers and $m^n = 196$, what could be the value of $(m-3)^{(n+1)}$?

(a) 2744

(b) 1331

(c) 121

(d) Cant define

11. $(1024)^{n/5} \times (4)^{2n+1} + 16^n \times 4^{n-1}$ =?

(a) 256

(b) 64

(c) 16

(d) 9

12. If $(ab)^{x-4} = (ba)^{x-8}$, what is the value of x?

(a) 4

(b) 6

(c) 8

(d) 10

13. If $100^{0.20} = x$, $10^{0.60} = y$ and $x^z = y^2$, what is the value of z?
(a) 3
(b) 6
(c) 4.2
(d) 2.2

14. 14 $36 \times 36 \times 36 \times 36 = 6^?$
(a) 10
(b) 8
(c) 4
(d) 6

15. What is the value of $(7^{-14} - 7^{-15}) =$?
(a) 6×7^{-15}
(b) 6×7^{-14}
(c) 7×7^{-15}
(d) 7×7^{-14}

16. If $3^{(n+4)} - 3^{(n+2)} = 8$, What is the value of n?
(a) 0
(b) – 1
(c) – 2
(d) 2

17. What is the value of $(2 \times 4 \times 5)^{5n}$
(a) $2^{5n} + 4^{5n} + 5^{5n}$
(b) $(40^5)^n$
(c) $(40)^{5n}$
(d) $(40n)^5$

18. If $(\sqrt{}\ 3)n = 6561$, then find the value of $(n)^{3/2} =$?
(a) 64
(b) $64\sqrt{3}$
(c) $16\sqrt{3}$
(d) 16

19. $5^x \times 2^3 = 36.\ 5^{(x+1)} =$?
(a) 22

(b) 21
(c) 20.5
(d) 22.5

20. $36^{120} = (36 \times x)^{40}$. What is the value of x?
(a) 4^4
(b) 4^6
(c) 6^2
(d) 6^4

21. $(6561)^{(1/2)} + (6561)^{(1/4)} + (6561)^{(1/8)} =?$
(a) 98
(b) 86
(c) 93
(d) 81

22. Given that $3^{\sqrt{n}} = 6561$ then find the value of $3^{\sqrt{n}} =?$
(a) 81
(b) 9
(c) 16
(d) 25

23. If $5^{(a\ +\ b)} = 5 \times 25 \times 125$, what is $(a + b)^2$
(a) 12
(b) 16
(c) 34
(d) 36

24. Find the value of $(7^{-1} - 11^{-1}) + (7^{-1} + 11^{-1}) =?$
(a) 2×7^{-1}
(b) 2×11^{-1}
(c) 14
(d) 22

25. Find the value of $(7^{-1} - 11^{-1}) + (7^{-1} + 11^{-1}) =?$
(a) 2×7^{-1}
(b) 2×11^{-1}
(c) 14

(d) 22

26. $(5)^{1.25} \times (12)^{0.25} \times (60)^{0.75} = ?$

(a) 420

(b) 260

(c) 200

(d) 300

Practice Exercise - 2 Answers with Hints - need based

1 (d)	2 (d)	3 (a)	4 (c)	5 (b)
6 (c)	7 (d)	8 (d)	9 (b)	10 (d)
11 (c)	12 (b)	13 (a)	14 (c)	15 (b)
16 (b)	17 (b)	18 (a)	19 (d)	20 (d)
21 (c)	22 (a)	23 (d)	24 (a)	25 (d)
26 (b)				

SERIES

25

The series consist of numbers which move in a pattern following some of the rules on a uniform basis.

Some of the factors to be kept in mind while dealing with series.

- ***Generally the first or the last number is not wrong in odd man out.***
- ***Find the pattern to find the odd figure out***
- ***Always check the next number in the series whether it suits the pattern or not.***
- ***Majority of the series consist of Prime numbers, squares, cubes, multiples, product or divisibility tests. Hence these should be tested before finding any pattern.***
- ***If the series contains ups and downs – there are chances that the series contains double series.***
- ***If the numbers are + ve or –ve alternatively, the role of double series or multiplication/division by –ve factors may exist.***
- ***If the series moves up or down slowly, role of addition or subtraction may exist.***
- ***If the series rises or falls role of multiplication/division may exist.***

MISSING NUMBER SERIES

Practice Exercise - 1

Number of Questions: 30 **Suggested time: 20 minutes**

1. Insert the missing number. 12, 25, 49, 99, 197, 395, (...)
(a) 789
(b) 1579
(c) 722
(d) 812

2. Insert the missing number. 34, 7, 37, 14, 40, 28, 43, (...)
(a) 31
(b) 42
(c) 46
(d) 56

3. Find the missing number. 1, 4, 9, 16, 25, 36, 49, (....)
(a) 64
(b) 54
(c) 56
(d) 81

4. Insert the missing number. 2, 7, 10, 22, 18, 37, 26, (?)
(a) 42
(b) 52
(c) 46
(d) 62

5. 4, 12, 48, 240, 1440, (...)
(a) 7620
(b) 10080
(c) 6200
(d) 10020

6. 2, 3, 6, 0, 10, –3, 14, (...)
(a) 6

(b) 2

(c) – 2

(d) – 6

7. 5, 10, 13, 26, 29, 58, 61, (....)

(a) 128

(b) 64

(c) 125

(d) 122

8. 1, 8, 27, 64, 125, 216, (....)

(a) 354

(b) 392

(c) 343

(d) 245

9. 4, 5, 7, 11, 19, (...)

(a) 22

(b) 35

(c) 27

(d) 32

10. 23, 29, 31, 37, 41, 43, (...)

(a) 53

(b) 47

(c) 48

(d) 59

11. 8, 24, 12, 36, 18, 54, (....)

(a) 27

(b) 68

(c) 108

(d) 72

12. 7, 26, 63, 124, 215, 342, (....)

(a) 481

(b) 391

(c) 511

(d) 421

13. 2, – 6, 18, – 54, 162, (...)

(a) 486

(b) –486

(c) 422

(d) –428

14. 2, 6, 12, 20, 30, 42, 56, (...)

(a) 61

(b) 72

(c) 64

(d) 70

15. 8, 12, 18, 27, 40.5, (...)

(a) 60.75

(b) 62

(c) 58

(d) 60.5

16. 196, 144, 100, 64, 36,?

(a) 16

(b) 8

(c) 18

(d) 12

17. 2, 3, 5, 16, 231,?

(a) 62005

(b) 52120

(c) 422

(d) 53105

18. 279936, 46656, 7776, 1296, 216,?

(a) 60

(b) 46

(c) 36

(d) 66

19. 220, 218, 214, 208, 200, 190,?

(a) 166

(b) 178

(c) 182
(d) 190

20. 12, 38, 116, 350, 1052,?
(a) 1800
(b) 2200
(c) 2800
(d) 3158

21. 9, 35, 91, 189, 341,?
(a) 502
(b) 521
(c) 559
(d) 611

22. 78, 64, 48, 30, 10, (...)
(a) –12
(b) –14
(c) 2
(d) 8

23. 16384, 8192, 2048, 256, 16,?
(a) 1
(b) 8
(c) 2
(d) 0.5

24. 46080, 3840, 384, 48, 8, 2,?
(a) 1
(b) $\frac{1}{64}$
(c) $\frac{1}{8}$
(d) None of these

25. 7, 21, 63, 189, 567,?
(a) 1521
(b) 1609
(c) 1307
(d) 1701

26. 1, 0.5, 0.25, 0.125, 0.0625,?

(a) 0.015625
(b) 0.03125
(c) 0.0078125
(d) None of these

27. Complete the series 20, 19, 17,...., 10, 5
(a) 15
(b) 16
(c) 13
(d) 14

28. Which fraction will come next, 12, 34, 58, 716?
(a) 932
(b) 930
(c) 928
(d) 1132

29. 4, 10, (?), 82, 244, 730
(a) 26
(b) 28
(c) 40
(d) 48

30. Complete the series 95, 115.5, 138, ..., 189
(a) 160.5
(b) 162.5
(c) 164.5
(d) 166.5

Practice Exercise - 1 Answers with Hints - need based

1 (a)	2 (d)	3 (a)	4 (b)	5 (b)
6 (d)	7 (d)	8 (c)	9 (b)	10 (b)
11 (a)	12 (c)	13 (b)	14 (b)	15 (a)
16 (a)	17 (d)	18 (c)	19 (b)	20 (d)
21 (c)	22 (a)	23 (d)	24 (a)	25 (d)
26 (b)	27 (d)	28 (a)	29 (b)	30 (b)

1. Each number is twice of previous one with 1 added or subtracted alternatively.

2. We have two series here

34, 37, 40, 43,... (Increase by 3); 7, 14, 28,... (Multiply by 2)

Hence, next term is $28 \times 2 = 56$

3. The series is $1^2, 2^2, 3^2, 4^2, 5^2, 6^2, 7^2$,...

Hence, next term = 82 = 64

4. There are two series here; 2, 10, 18, 26,. (Increase by 8);

7, 22, 37,. (Increase by 15)

5. Go on multiplying the given numbers by 3, 4, 5, 6, 7

7. Numbers are alternatively multiplied by 2 and increased by 3

9. Numbers are multiplied by 2 and decreased by 3

12. The series is $(2^3 - 1), (3^3 - 1), (4^3 - 1), (5^3 - 1), (6^3 - 1), (7^3 - 1)$,...

14. The pattern is $1 \times 2, 2 \times 3, 3 \times 4, 4 \times 5, 5 \times 6, 6 \times 7, 7 \times 8$.

15. $8; (8 \times 3) \div 2 = 12; (12 \times 3) \div 2 = 18$

$(18 \times 3) \div 2 = 27$ $(27 \times 3) \div 2 = 40.5; (40.5 \times 3) \div 2 = 60.75$

17. $2; 3; 3^2 - 2^2 = 5; 5^2 - 3^2 = 16; 16^2 - 5^2 = 231; 231^2 - 16^2 = 53105$

21. $1^3 + 2^3 = 9; 2^3 + 3^3 = 35; 3^3 + 4^3 = 91;$

$4^3 + 5^3 = 189; 5^3 + 6^3 = 341; 6^3 + 7^3 = 559$

22. $78 - 14 = 64; 64 - 16 = 48; 48 - 18 = 30; 30 - 20 = 10; 10 - 22 = -12$

24. $46080 \div 12 = 3840; 3840 \div 10 = 384; 384 \div 8 = 48$

$48 \div 6 = 8; 8 \div 4 = 2; 2 \div 2 = 1$

ODD MAN OUT - SERIES

Practice Exercise - 2

Number of Questions: 30 **Suggested time: 15 minutes**

1. Find the odd man out. 1, 3, 9, 12, 19, 29
a) 12
(b) 9
(c) 1
(d) 3

2. Find the odd man out. 1, 8, 27, 64, 125, 196, 216, 343
a) 64
(b) 196
(c) 216
(d) 1

3. Find the odd man out. 15, 25, 30, 51, 85, 90, 115
a) 15
(b) 25
(c) 51
(d) 90

4. Find the odd man out. 24, 36, 52, 72, 96
a) 72
(b) 52
(c) 36
(d) 24

5. Find the odd man out. 187, 264, 386, 473, 682, 781
a) 386
(b) 187
(c) 781
(d) 682

6. Find the odd man out. 12, 24, 34, 48, 64, 84
a) 48

(b) 34
(c) 24
(d) 12

7. Find the odd man out. 362, 482, 551, 263, 344, 284
a) 362
(b) 482
(c) 551
(d) 344

8. Find the odd man out. 742, 743, 633, 853, 871, 990, 532
a) 532
(b) 990
(c) 633
(d) 742

9. Find the odd man out. 1, 5, 11, 17, 23, 29
a) 29
(b) 17
(c) 11
(d) 1

10. Find the odd man out. 7, 13, 19, 25, 29, 37, 43
a) 19
(b) 29
(c) 25
(d) 43

11. Find the odd man out. 1, 9, 16, 51, 121, 169, 225
a) 169
(b) 51
(c) 16
(d) 1

12. Find the odd man out. 1, 4, 9, 17, 25, 36, 49
a) 1
(b) 9
(c) 17
(d) 49

13. Find the odd man out. 2, 5, 10, 17, 26, 38, 50, 65

a) 50

(b) 38

(c) 26

(d) 65

14. Find the odd man out. 18, 16, 12, 24, 11, 34, 46

a) 16

(b) 46

(c) 16

(d) 11

15. Find the odd man out. 1, 27, 216, 512, 1024, 1331

a) 1024

(b) 512

(c) 27

(d) 1

16. Find the odd man out. 1, 16, 81, 255, 625, 1296

a) 255

(b) 1296

(c) 81

(d) 1

17. Find the odd man out. 6, 13, 18, 25, 30, 37, 40

a) 40

(b) 30

(c) 37

(d) 25

18. Find the odd man out. 445, 221, 109, 46, 25, 11, 4

a) 25

(b) 109

(c) 46

(d) 221

19. Find the odd man out. 1050, 510, 242, 106, 46, 16, 3

a) 46

(b) 106

(c) 510
(d) 1050

20. Find the odd man out. 2, 3, 5, 9, 12, 17, 23
a) 12
(b) 9
(c) 23
(d) 2

21. Find the odd man out. 3, 8, 18, 38, 78, 158, 316
a) 38
(b) 158
(c) 316
(d) 8

22. Find the odd man out. 5, 6, 14, 45, 185, 925, 5556
a) 5556
(b) 925
(c) 185
(d) 6

23. Find the odd man out. 23, 27, 36, 52, 77, 111, 162
a) 162
(b) 111
(c) 52
(d) 27

24. Find the odd man out. 241, 263, 248, 271, 255, 277, 262
a) 277
(b) 271
(c) 263
(d) 241

25. Find the odd man out. 125, 127, 130, 135, 142, 153, 165
a) 165
(b) 142
(c) 153
(d) 130

26. Find the odd man out. 5, 10, 40, 81, 320, 640, 2560

a) 40
(b) 81
(c) 320
(d) 2560

27. Find the odd man out. 12, 21, 32, 45, 60, 77, 95
a) 95
(b) 45
(c) 32
(d) 21

28. Find the odd man out. 3, 5, 15, 75, 1120, 84375
a) 84375
(b) 1120
(c) 15
(d) 3

29. Find the odd man out. 3576, 1784, 888, 440, 216, 105, 48
a) 105
(b) 216
(c) 888
(d) 1784

30. Find the odd man out. 30, -5, -45, -90, -145, -195, -255
a) -5
(b) -145
(c) -255
(d) -195

Practice Exercise - 2 Answers with Hints

1 (a)	2 (b)	3 (c)	4 (b)	5 (a)
6 (b)	7 (d)	8 (d)	9 (d)	10 (c)
11 (b)	12 (c)	13 (b)	14 (d)	15 (a)
16 (a)	7 (a)	18 (c)	19 (b)	20 (b)
21 (c)	22 (c)	23 (b)	24 (b)	25 (a)
26 (b)	27 (a)	28 (b)	29 (a)	30 (b)

1. 12 is an even number. All other given numbers are odd

2. 196 is not a perfect cube

3. All except 51 are multiples of 5

4. All except 52 are multiples of 6

5. In all numbers except 386, the middle digit is the sum of other two digits.

6. All numbers except 34 are multiples of 4

7. In all numbers except 344, the product of first and third digits is the middle digit.

8. In all numbers except 742, the difference of third and first digit is the middle digit.

10. All given numbers except 1 are prime numbers.

One is not a prime number because it does not have two factors. It is divisible by only 1

13. The pattern is $(1 \times 1) + 1$, $(2 \times 2) + 1$, $(3 \times 3) + 1$, $(4 \times 4) + 1$, $(5 \times 5) + 1$, $(6 \times 6) + 1$....

Hence, in place of 38, the right number was $(6 \times 6) + 1 = 37$

17. The difference between two successive terms from the beginning are 7, 5, 7, 5, 7, 5

Hence, in place of 40, right number is $37 + 5 = 42$

14. To obtain next number, subtract 3 from the previous number and divide the result by 2

19. 1050; $(1050 - 30)/2 = 510$; $(510 - 26)/2 = 242$

$(242 - 22)/2 = 110$; $(110 - 18)/2 = 46$; $(46 - 14)/2 = 16$

$(16 - 10)/2 = 3$; Hence, 110 should have come in place of 106

22. $5 \times 1 + 1 = 6$; $6 \times 2 + 2 = 14$; $14 \times 3 + 3 = 45$;

$45 \times 4 + 4 = 184$; Hence, 184 should have come instead of 185

23. $23 + 22 = 27$; $27 + 32 = 36$; $36 + 42 = 52$

$52 + 52 = 77$; $77 + 62 = 113$; $113 + 72 = 162$

Hence, 113 should have come in place of 111

24. Alternatively 22 is added and 15 is subtracted from the terms. Hence, 271 is wrong

25. Prime numbers 2, 3, 5, 7, 11, 13 are added successively. Hence, 165 is wrong

26. Alternatively 2 and 4 are multiplied with the previous terms

28. Pattern : $1^{st} * 2^{nd} = 3^{rd}$, $2^{nd} * 3^{rd} = 4^{th}$, etc.

29. 3576; (3576 – 8)/2 = 1784; (1784 – 8)/2 = 888;

(888 – 8)/2 = 440; (440 – 8)/2 = 216; (216 – 8)/2 = 104

Hence, 105 is wrong. 104 should have come in place of 105

30. 30; 30 – 35 = –5; – 5 – 40 = –45;

–45 – 45 = – 90; –90 – 50 = –140;

Hence, – 145 is wrong. –140 should have come in place of –145

DATA INTERPRETATION

26

Data interpretation is the most scoring and time consuming section in IBPS and other competitive examinations. In Quantitative Aptitude section, generally, you will find at least two data interpretation sets each having 5 questions. In IBPS Probationary Officers and other similar examinations, there are 50 questions in Quantitative Aptitude section. So if you solve those two sets corrected you need to solve 10 questions out of remaining 40 questions to reach a normal cut off of 18–20 marks.

Here are some of the important techniques to make Data Interpretation calculations fast

VISUAL ESTIMATION

It is a well known fact that it is near to impossible to solve 200 questions in 120 minutes accurately. Term "Accurately" is important here because I have seen many candidates attempting 190 + questions but they fail to qualify. The reason behind their failure is the low level of accuracy and many times this accuracy level falls below 40%. Important point to be noted down here is that even if by an attempt of 190 + questions in the examinations, but with low accuracy, you get less time for questions you are definitely sure about and more so there is negative marking in most of the competitive examinations.

So reduce the time of actual working and let us have a look as to how to use VISUAL ESTIMATION technique to solve Data Interpretation questions:

The data interpretation and analysis questions can be divided into four broad categories as under:

TABLE CHARTS assess your capability to analyse data in a tabular formats.

BAR CHARTS show your data in the shapes of "bars" which have to be examined sometimes minutely to understand.

PIE CHARTS test your skills in visual data analysis where the data is presented in a round shape of "pie" by means of angular distribution or actual figures or in fractional divisions.

LINE CHARTS presentation of data related problems is, perhaps, the simplest form and can be understood in the quickest possible manner.

Question Set - 1

Study the following table and answer the questions based on it.

Annual Expenditures of **T & N ASSOCIATES (P) Ltd, Hari Kunj** over the given Years.

(Rs. In lakhs)

Year	Item of Expenditure				
	Taxes	Interest on Loans	Bonus	Fuel and Transport	Net Salary
2008	83	23.4	3.00	98	288
2009	108	32.5	2.52	112	342
2010	74	41.6	3.84	101	324
2011	88	36.4	3.68	133	336
2012	98	49.4	3.96	142	420

1. The total expenditure of the company over these items during the year 2010 is?
(a) 501.21
(b) 544.44
(c) 496.46
(d) 387.78

2. What is the average amount of interest per year which the company had to pay during this period?
(a) 36.66
(b) 39.47
(c) 38.45
(d) 35.28

3. The total amount of bonus paid by the company during the given period is roughly how much percent of the total amount of salary paid during this period?
(a) 0.14
(b) 1.0
(c) 0.28

(d) 0.84

4. Total expenditure on all these items in 2008 was approximately what percent of the total expenditure in 2012?

(a) 63.45

(b) 69.0

(c) 58.45

(d) 73.25

5. The total expenditure of the company over these items during the year 2010 is how much more or less than that of year 2008?

(a) 59.04

(b) 39.04

(c) 58.04

(d) 49.04

6. Total expenditure on all these items in 2008 was approximately what percent of the total expenditure in 2012?

(a) 10 :13

(b) 11 : 23

(c) 9 : 34

(d) 12 : 59

Question Set - 2

The following table shows the number of new employees added to different categories of employees on Pay roll of "VRIDDHI & SMRIDDHI Finance Company (P) Ltd" and also the number of employees from these categories who had left (or are projected to leave) the company every year since the establishment of the Company in 2015.

FY ending March * projected	MANAGERS		TECHNOCRATS		OPERATIONS		ACCOUNTANTS		HELPERS	
	NEW	LEFT	NEW	LEFT	NEW	LEFT	NEW	LEFT	NEW	LEFT
2015	1500	-	2400	-	1760	-	2320	-	1640	-
2016	560	240	544	240	512	208	400	200	368	192
2017	358	184	480	256	480	240	448	208	304	176
2018	296	176	472	192	416	200	496	192	392	160
2019*	320	144	512	200	384	224	544	176	448	240
2020*	386	192	576	224	496	288	500	184	400	208

1. What is the difference between the total number of Technocrats added to the Company and the total number of Accountants added to the Company during the years 2016 to 2020?
(a) 88
(b) 138
(c) 276
(d) 176

2. What will be the total number of Helpers working in the Company in the year 2019?
(a) 2628
(b) 2384
(c) 2186
(d) 1746

3. For which of the following categories the percentage increase in the number of employees working in the Company from 2015 to 2020 will be the maximum?
(a) Managers

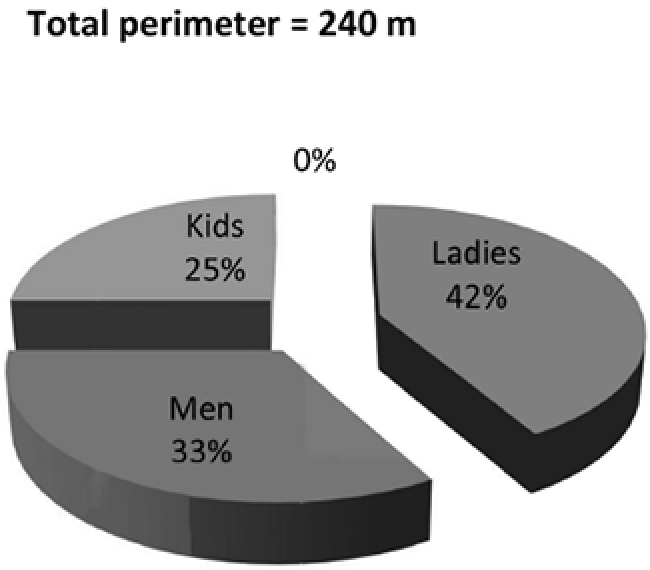
(b) Technocrats
(c) Operations
(d) Helpers

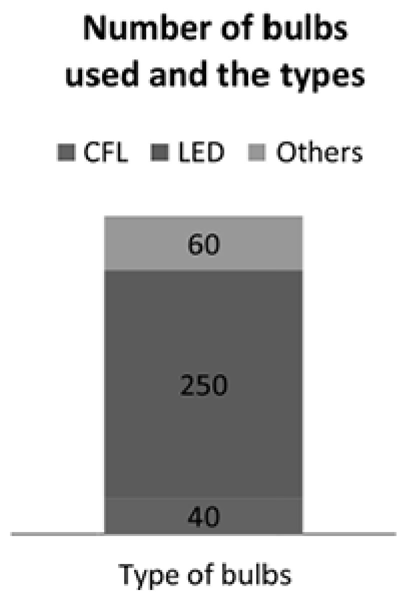
4. What is the pooled average of the total number of employees of all categories in the year 2017?
(a) 2650
(b) 2390
(c) 2530
(d) 2470

5. During the period between 2015 and 2020, find what is the% of the the total number of Operators who had left (or are projected to leave) the Company over the total number of Operators who had joined (or projected to join) the Company?
(a) 29%
(b) 27%
(c) 21%
(d) 24%

Question Set - 3

An exhibition in Art museum in Yamuna Nagar was opened for public featuring traditional cloths consists of men, ladies and kids sections. Each section is allotted a squared arena and the perimeter of all the 3 sections adds up to 240 m with the distribution as follows:





Three types of bulbs were used to light up the entire area and they are distributed in accordance with the following criteria.

Bulb type	LED	CFL	Others
Distribution	1 per 5 m^2	1 per 25 m^2	1 per each 4m of perimeter

1. What is the ratio between the areas of ladies, kids and gents sections?
(a) 42 : 33 : 25
(b) 23 : 16 : 11
(c) 25 : 9 : 16
(d) 26 : 9 : 15

2. If the furnishing of each square meter requires Rs.12.50p, How much more money (Rs.) is required to furnish the gents area than the kids section?
(a) 2487.50

(b) 2877.50
(c) 2187.50
(d) 2297.50

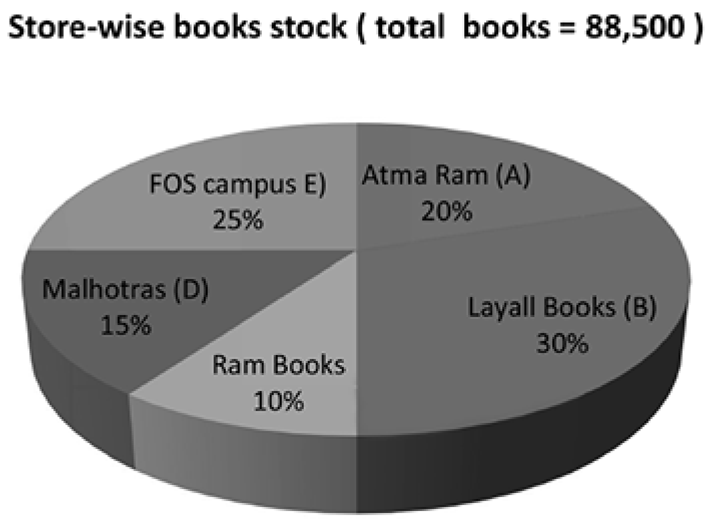
3. What will be cost (Rs.) for lighting up the ladies section provided that each LED costs Rs. 80, CFL Rs. 75 and other bulb Rs. 12?
(a) 12,175
(b) 12,725
(c) 13,715
(d) 12,975

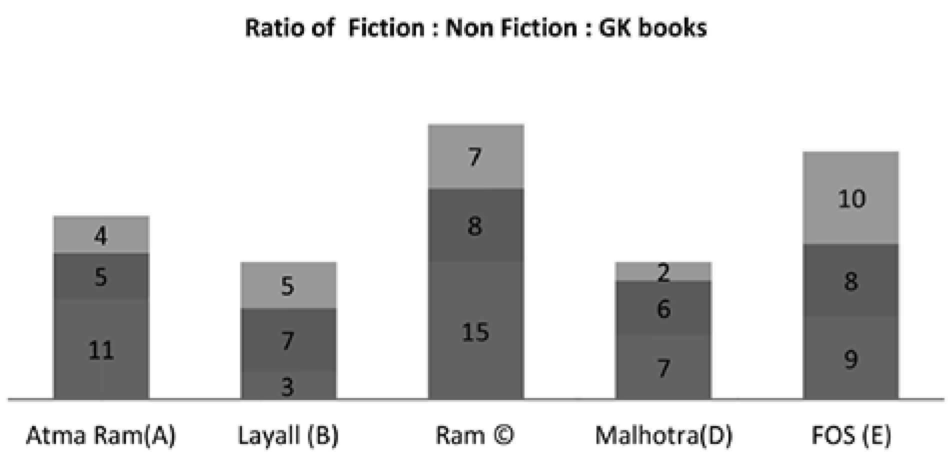
4. What percentage of the LED bulbs is utilized to light up the kids section?
(a) 14
(b) 27
(c) 18
(d) 22

5. What will the total electricity consumption (units) of the ladies and gents section together in a day provided that consumption of each LED is 0.2 units/day, CFL is 0.4units/day, and an ordinary other type of bulb is 1 unit/day respectively?
(a) 102.7
(b) 102.4
(c) 107.79
(d) 114.26

Question Set - 4

Answer the below given questions based upon the information presented in the pie chart and table.





1. The difference between the number of Fiction and GK Books sold by Store A, B, and D together and the number of Non-fiction and GK books sold by Stores C and E is what percentage of the total number of Non-fiction and GK books sold by Stores C and E?

(a) 50

(b) 54

(c) 51

(d) 40

2. Average number of Fiction and GK Books sold by Store A, B, and D together constitutes what percentage of the total number of books sold by Stores C, D, and E?

(a) 27.65
(b) 26.67
(c) 26.16
(d) 25.67

3. How many stores sold more than 5000 copies of non-fiction?

(a) 3
(b) 2
(c) 4
(d) 1

4. If each fiction costs 15.50 rupees and GK costs 12.50 rupees, what is the cost of each non-fiction book provided that total the total business of C is Rs. 120, 360.

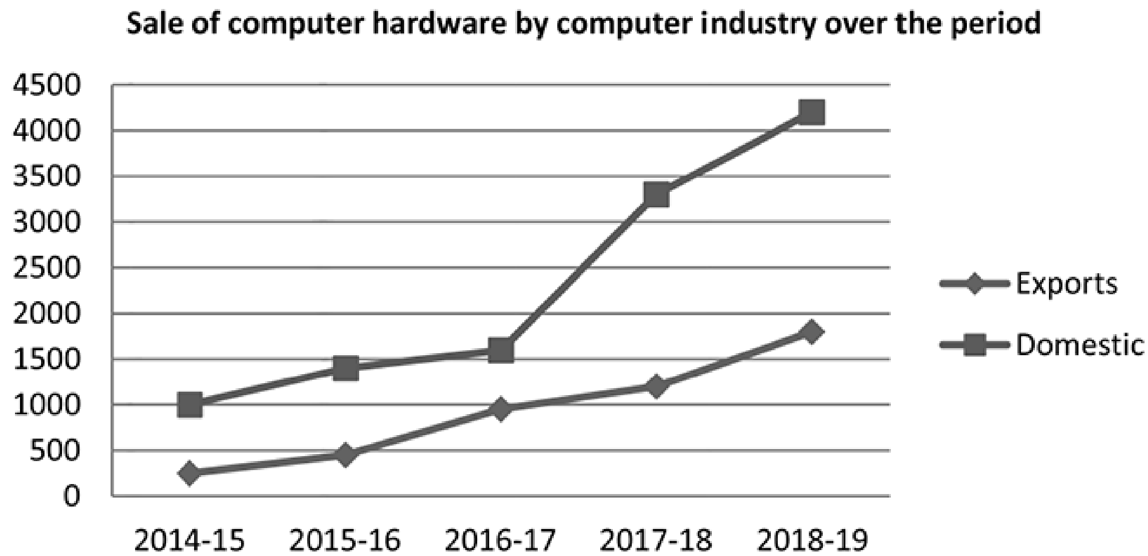
(a) 12
(b) 13
(c) 10
(d) 11

5. What will be the total business of store C (in approximate lakhs Rs.) If the price of fiction and non-fiction is increased by 10% and that of GK is unchanged?

(a) 1.30
(b) 1.35
(c) 1.25
(d) 1.40

Question Set - 5

Study the following line chart and answer the questions given below:



1. What was the difference between the total hardware sale in exports sector in 2015-16 and 2016-17 together and hardware sale in domestic sector in 2016-17?

(a) 150

(b) 200

(c) 300

(d) 400

2. What was the difference (Rs. Crores) in the average sale of hardware between the domestic and exports sector?

(a) 600

(b) 750

(c) 900

(d) None

3. Approximately what was the percentage increase in the sale of hardware in domestic sector from 2016 - 17 to 2018–19?

(a) 25

(b) 30

(c) 35
(d) None

4. What was the difference in the sale of hardware between domestic and export markets in 2016–17?
(a) 500 cr
(b) 700 cr
(c) 1000 cr
(d) 1200 cr

5. In which of the following years was the percentage increase in sale of hardware in domestic sector maximum over the preceding year?
(a) 2014–15
(b) 2016–17
(c) 2017–18
(d) 2018–19

Question Set - 6

(Time: 4 minutes)

Read the following table carefully and answer the questions given below it. Data is related to number of students who got admission and who had left the given five colleges **SUCHAN, DING, MORIWALA, SIKENDERPUR & BAHAWDIN** during the given years.

College/ Year	SUCHAN		DING		MORIWALA		SIKENDERPUR		BAHAWDIN	
	Admin	Left	Admin	Left	Admin	Left	Admin	Left	Admin	Left
2013	161	-	148	-	179	-	116	-	128	-
2014	148	58	172	60	161	90	208	60	191	50
2015	135	69	188	96	143	101	169	45	167	79
2016	112	88	173	59	165	58	142	56	185	82
2017	141	39	151	48	179	66	155	108	142	91

1. What is the average number of students who got admission in Sikenderpur during all the given years taken together?
(a) 156
(b) 164
(c) 166
(d) 158

2. If the respective ratio of number of boys and girls in DING at the end of 2015 was 5:6, what was the number of girls in it at the end of 2015?
(a) 212
(b) 196
(c) 218
(d) 192

3. In which of the colleges the number of students was the highest at the end of 2014?
(a) SUCHAN
(b) DING
(c) BAHAWDIN
(d) MORIWALA

4. What was the total number of students in SUCHAN at the end of 2016?
(a) 385
(b) 341
(c) 387
(d) 339

5. Number of students in BAHAWDIN at the end of 2014 is what % more than the number of students is MORIWALA at the end of 2014?
(a) 9.2
(b) 11.2
(c) 3.8
(d) 7.6

6. What is the respective ratio between total number of students who joined MORIWALA in 2015 and 2016 together and total number of students who left BAHAWDIN in 2015, 2016 and 2017 together?
(a) 11:9
(b) 16:7
(c) 13:8
(d) 15:8

Question Set - 7

(Time: 3 minutes)

The famous township of **Siwani Mandi** has one **Durga Colony** in its MC area - mainly inhabited by women power. A library run by "**Mohan Lal Sahuja Charitable Trust**" has 2800 female members on its roll. Out of which 650 members read only English newspaper, 550 members read only Hindi newspaper and 450 members read only Marathi newspaper. The number of members reading all the three newspapers completes a century. Members reading Hindi as well as English newspaper are 200. A total of 400 members read Hindi as well as Marathi Newspapers and 300 members read English as well as Marathi newspapers.

From the above given information:

1. Find the difference between number of members reading English as well as Marathi newspapers and the number of members reading English as well as Hindi newspaper.
(a) 100
(b) 120
(c) 180
(d) 80

2. How many members read at least 2 newspapers?
(a) 100
(b) 1000
(c) 1180
(d) 980

3. Find the number if members reading Hindi newspaper.
(a) 1100
(b) 950
(c) 1000
(d) 1050

4. How many members read only one newspaper?
(a) 1100

(b) 1206
(c) 1800
(d) 1650

Question Set - 8

(Time: 3 minutes)

A school *"Anand Public School"* run by Mr Ghanshyam Dass in Panjuwana township has 2800 students on its roll. The following information is further given about the school and its pupils:

- *The ratio of boys to girls is 5:9 respectively.*
- *All the students are enrolled in any one or more of the hobby classes, namely, dancing, singing and painting.*
- *12% of the boys learn only singing.*
- *16% of the girls learn only dancing.*
- *The number of students enrolled only in painting is 925.*
- *One-fourth of the boys are enrolled in all the three classes.*
- *Number of girls enrolled only in singing is 250% of the boys enrolled in the same.*
- *The remaining girls are enrolled in all the three classes.*
- *23% of the boys are enrolled only in dancing; and*
- *The remaining are enrolled in only painting.*

From the above given information, you are required to find:

1. What is the respective ratio of the number of boys enrolled only in Dancing to the number of girls enrolled in the same?
   (a) 11:12
   (b) 115:144
   (c) 18 : 17
   (d) 8 : 9
2. What is the number of girls enrolled in all the three classes?
   (a) 1015
   (b) 1774
   (c) 1212
   (d) 980
3. Number of boys enrolled in painting only is what per cent of the girls enrolled in the same?

(a) 86.24
(b) 74.12
(c) 76.19
(d) 79.24

4. How many boys are enrolled in dancing?
(a) 400
(b) 420
(c) 480
(d) 380

Question Set - 9

(Time: 3 minutes)

In a class of 84 students *"in F.O.S. CAMPUS PANIPAT" admitted for Competition Examinations,* the number of boys and girls are in the ratio 5:7 respectively. Among the girls, 7 can speak Hindi and English. 50% of the total students can only speak Hindi. The ratio between the students who speak only Hindi and only English is given as 21:16. The ratio between the numbers of boys and girls speaking English is known as 3 : 5.

From the above given information find:

1. What is the number of boys who speak both the languages?
(a) 3
(b) 5
(c) 6
(d) 4

2. What is the number of girls who only speak English?
(a) 24
(b) 20
(c) 40
(d) 35

3. What is the ratio between the number of boys and girls respectively who speak Hindi only?
(a) 10:3
(b) 17:9
(c) 10:13
(d) 10:11

4. How many girls can speak Hindi?
(a) 10
(b) 12
(c) 22
(d) 19

5. What is the ratio between the number of boys and girls respectively who speak English?

(a) 2 : 5

(b) 1: 4

(c) 5 : 3

(d) 3 : 5

Question Set - 10

(Time: 2 minutes)

M/s **SOLVATO CONSULTANCY (P) Ltd New Delhi** is a famous name in placement of various categories of work force. In the year 2017, It was asked by a firm "**M/s NT & Associates**" to search and find suitable candidates as they wanted to employ Four Managers, Twelve Observers and Seven Assistants for them. **M/S SOLVATO** was given parameters to offer an employment to any candidate as under:

I. Each observer gets Rs. 12,000 per month more than an assistant.

II. An observer and an assistant together get Rs. 32,000 per month.

III. The total salary per month of a manager and an observer is Rs. 57,000

From the above information and the parameters, find:

1. What is the monthly salary of an assistant?
(a) 11500
(b) 13500
(c) 10000
(d) 14000

2. What will be total annual wage bill (approximately Rs. Lakhs) of **M/s NT & Associates** if its required strength is arranged by **M/S SOLVATO** for it?
(a) 53
(b) 50
(c) 48
(d) 57

3. The consultancy firm gets service charges of 10% of the first wage bill for the work force arranged by it. After paying 8% GST on it, what net amount (thousands) will be credited to its account in a month?
(a) 39
(b) 42
(c) 37

(d) 34

Question Set - 11

1. What is the circumference of semi-circle in cm? Given the below information/s for use:

I. The area of semi-circle is half of the areas of parallelogram.

II. The length of parallelogram is 1.5 times the radius of the semi circle.

III. The difference between the length and breadth of parallelogram is 8 cm.

Which option is sufficient to answer the question?

(a) I/II either

(b) II/III either

(c) 1 only

(d) None

Question Set - 12

CHANDIGARH TRICITY (consisting of Chandigarh, Mohali and Panchkula) is surrounded by many small towns and cities like Patiala, Rajpura, Ambala, Derabassi, Naraingarh, Kalka, Pinjore, Ropar, Kurali, Ludhiana and Morinda etc., A good number of commuters come to this **Tricity** for jobs and in search of work from these surrounding towns using one or the other mode of conveyance. To study their behaviour and different modes, a survey was got conducted by **"M/s T & N VRIDDHI - SMRIDDHI EXPLORERS PANCHKULA."** As an outcome of the survey, it was observed that out of the total number of commuters commuting daily into the TRICITY:

- *17,171 commuters commute only by trains;*
- *7,359 commuters commute only by bikes;*
- *22,077 commuters commute by bus;*
- *14,718 commuters commute by cars;*
- *4,906 commute only by autos;*
- *7,359 commuters commute only by taxis;*
- *26,983 commuters commute by autos or by train;*
- *12,265 commuters commute by bus and cars both;*

From the above given information, find:

1. The total number of commuters commuting by trains forms nearly what percent of the total number of commuters commuting daily?
(a) 20
(b) 30
(c) 35
(d) 15

2. The total number of commuters commuting by bikes and taxis together forms nearly what percent of the total number of commuters commuting daily?
(a) 40
(b) 11

(c) 25
(d) 20

3. The total number of commuters commuting by autos forms what percent of the total number of commuters commuting daily?
(a) 12
(b) 30
(c) 25
(d) 6

4. The total number of commuters commuting by cars only forms what percent of the total number of commuters commuting daily?
(a) 13
(b) 21
(c) 30
(d) 36

5. On Toll plazas, surrounding Tricity, each bus commuter is charged @ Rs. 0.45 p, what will be the revenue earned (Rs.) through Tolls daily?
(a) 7,000
(b) 8,000
(c) 10,000
(d) 13,000

6. It is assumed that each car is, on an average, charged Rs. 18 at these Tolls on which 5% GST has to be deposited by the Toll authorities with the Government. How much amount (Rs. in lakhs) will the revenue deposited by them in Government Treasury in the month of February 2018?
(a) 3
(b) 4
(c) 5
(d) 6

Question Set - 13

M/s Chawla & Chawla, a partnership firm, is one of the known rice merchants of Haryana State and have a big fleet of transport vehicles to run its business. Study the following table chart and answer questions based on it.

(Annual Expenditures (in Lakhs Rupees) over the given Years).

Year	Salary	Fuel and Transport	Bonus	Interest on Loans	Taxes
2008	288	98	3.00	23.4	83
2009	342	112	2.52	32.5	108
2010	324	101	3.84	41.6	74
2011	336	133	3.68	36.4	88
2012	420	142	3.96	49.4	98

1. What is the average amount of interest (in Lakhs Rs.) per year which the firm had to pay during this period?
(a) Rs. 36.66
(b) Rs. 36.36
(c) Rs. 36.26
(d) Rs. 36.06

2. The total amount of bonus paid by the firm during the given period is approximately what percent of the total amount of salary paid during this period?
(a) 0.5%
(b) 1%
(c) 1.5%
(d) 2%

3. Total expenditure on all these items in 2008 was approximately what percent of the total expenditure in 2012?
(a) 71%
(b) 37%
(c) 56%

(d) 69%

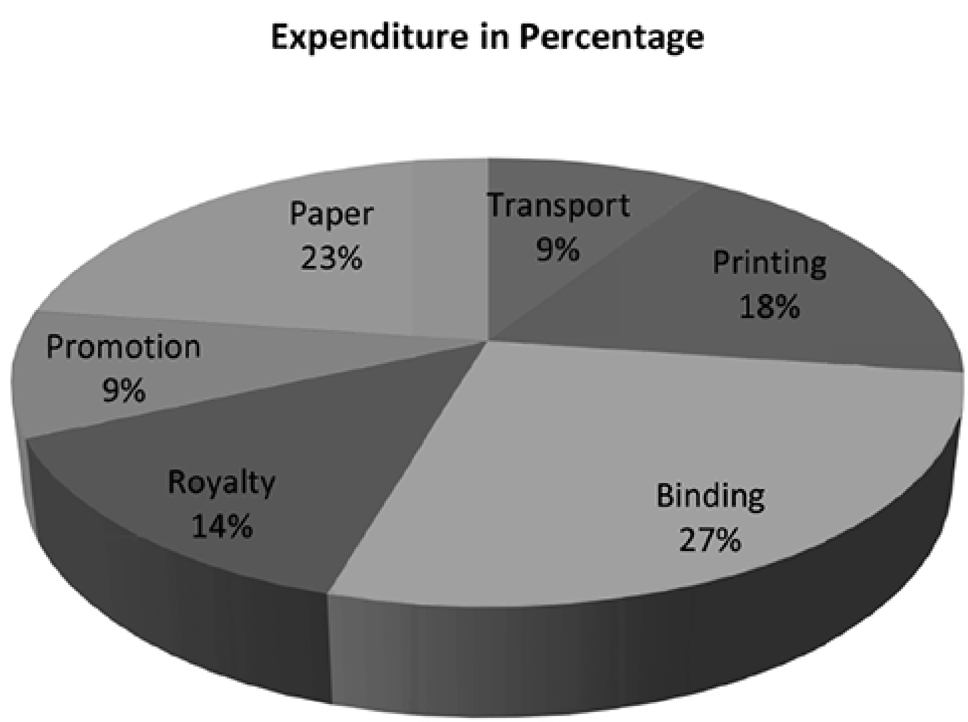
4. Calculate the total expenditure (in Lakhs Rs.) of the firm over these items during the year 2010 from the table chart given.

(a) 543.44
(b) 544.44
(c) 545.44
(d) 546.44

5. The ratio between the total expenditure on Taxes for all the years and the total expenditure on Fuel and Transport for all the years respectively is approximately?

(a) 4:13
(b) 7:13
(c) 10:13
(d) 11:13

Question Set - 14



1. What is the central angle (in degrees)of the sector corresponding to the expenditure incurred on Royalty?
(a) 54
(b) 48
(c) 45
(d) 40

2. Which two expenditures together have a central angle of 108^2 in pie chart?
(a) Binding, Transportation
(b) Binding, Royalty
(c) Printing,Transportation
(d) None

3. If the difference between the two expenditures are represented by 18 degree in the pie chart, then which option of following can be correct?
(a) Binding/Royalty
(b) Paper/Printing
(c) Paper/Royalty
(d) Royalty/Promotion

4. If for a edition of a book, the cost of paper is Rs. 56, 250 then find the promotion cost (Rs.) for this edition?
(a) Rs. 21,000
(b) Rs. 23,000
(c) Rs. 22,000
(d) Rs. 24,000

5. If for the certain quantity of books, the publisher has to pay Rs. 30,600 as printing cost, then what will be the amount of royalty to be paid for these books?
(a) Rs. 22,650
(b) Rs. 27,750
(c) Rs. 26,850
(d) Rs. 23,800

6. If 5,500 copies are published and the transportation cost on them amounts to Rs. 82, 500 then what should be the selling price of the book so that the publisher can earn a profit of 25% overall?

(a) Rs. 180.50
(b) Rs. 182.50
(c) Rs. 183.50
(d) Rs. 187.50

7. Royalty on the book is less than the printing cost by?
(a) 20%
(b) 25%
(c) 30%
(d) 35%

Question Set - 15

(Q. 1 – 5) Study the chart carefully to answer the following questions.

	Temperature				
	Dehradun	**Queta**	**Kolkata**	**Lucknow**	**Ranchi**
January	20°C	15°C	20°C	22°C	35°C
February	21°C	16°C	18°C	20°C	30°C
March	22°C	18°C	16°C	22°C	32°C
April	25°C	20°C	15°C	25°C	36°C
May	28°C	22°C	14°C	18°C	38°C

1. What is the difference between the average temperature of Dehradun and that of Queta?
(a) 8°C
(b) 5°C
(c) 9°C
(d) 7°C

2. What is the difference between the average temperature of all cities in May and that if February?
(a) 14°C
(b) 18°C
(c) 3°C
(d) 12°C

3. The average temperature of Ranchi is approximately what percent more than that of Kolkata?
(a) 101%
(b) 106%
(c) 103.5%
(d) 97.21

4. What is the ratio of the average temperature of Lucknow to that of Queta?
(a) 91 : 99
(b) 107 : 91

(c) 97 : 87
(d) 103 : 95

5. The average temperature in May is what percent of the average temperature in March of the given five cities?
(a) 119.91%
(b) 113.51%
(c) 110%
(d) 109.09%

Answers with Explanations

Question Set 1

1 (b)	2 (a)	3 (b)	4 (b)	5 (a)

1. Total expenditure during 2010 = (324 + 101 + 3.84 + 41.6 + 74) = 544.44

2. Average amount of interest paid during the period = (49.4 + 36.4 + 41.6 + 32.5 + 23.4)/5 = 36.66

3. Explanation: Required percentage

$$\left[\frac{(3.00+2.52+3.84+3.68+3.96)}{(288+342+324+336+420)}*100\right]\%$$

$$\left[\frac{17}{1710}*100\right]\%$$

$$\approx 1\%$$

6. Required ratio (83 + 108 + 74 + 88 + 98) : (142 + 133 + 101 + 112 + 98)

Question Set 2

1 (d)	2 (b)	3 (a)	4 (b)	5 (a)

3. Number of Managers working in the Company:

In 2015 = 1520;

In 2020 = 1520 + 560 + 358 + 296 + 320 + 386 − 240 − 184 − 176 − 144 − 192 = 2504

4. Managers 2014; Technocrats 2928; Operations 2304; Accounts 2766; Helpers 1944;

Therefore, Average = 2390

5. Total number of Operators who left the Company during 2015–2020

= (208 + 240 + 200 + 224 + 288) = 1160.

Total number of Operators who joined the Company during 2015–2020

= (1760 + 512 + 480 + 416 + 384 + 496 = 4048

Therefore Required % = (1160/4048) × 100 = 29%.

Question Set 3

1 (c)	2 (c)	3 (a)	4 (c)	5 (b)

Perimeter of a square = 4 × length of its one side

Ladies: 42% of 240 = 100 m (each side is 25m and area is 625 $m^{2)}$

Gents: 33% of 240 = 80 m (each side is 20m and area is 400 $m^{2)}$

Kids: 25% of 240 = 60m (each side is 15m and area is 225 m^2)

From the above data we can arrive create the following table.

Bulb Type	Distribution	Ladies	Gents	Kids	Total
LED	1 per 5 m^2	125	80	45	250
CFL	1 per 25 m^2	25	16	9	50
Others	1 per each 4m of perimeter	25	20	15	60

1. Ratio between the areas of ladies, kids and gents sections is 625 : 225 : 400 = 25 : 9 : 16

2. Area of gents section- Area of kids section = 400–225 = 175m^2

Furnishing charge is Rs. 125/m^2

Expense for furnishing extra 175 m^2 = 175*125 = 21,875

3. Cost for lighting up the ladies section = 125*80 + 25*75 + 25*12 = Rs. 12,175

4. %age of the LED bulbs utilised to light up the kids section = (45/250)*100 = 18%

5. Total number of LED, CFL, and other bulbs in ladies and gents section together is 205, 41, and 45 respectively.

Therefore the total electricity consumption in a day = 205*0.2 + 41*0.4 + 45*1 = 102.4

Question Set 4

1 (a)	2 (b)	3 (a)	4 (d)	5 (a)

1. Total number of Non-fiction and GK books sold by Store C = 88500*(20/100)*15/30 = 17700*15/30

= 8850 + Stores E = 88500*(25/100) = 14750*18/27 = 14750 **Together = 23,600**

Number of Fiction and GK Books sold by Store A, B, and D

A = 17700*15/20 = 13275 + B = 26550*8/15 = 14160 + D = 13275*9/15 = 7965;

Total = 13275 + 14160 + 7965 = 35,400

Difference = 11,800. **Percentage = 11800/23600 * 100 = 50%**

2. Average number of Fiction and GK Books sold by Store A, B, and D = 11,800

Total number of books sold by Stores C, D, and E = 44,250

Percentage = (11800/44250)*100 = 26.67%

3. Non-fiction sold by

A = 4425; B = 12390; C = 2360; D = 5310

E = 6556; **Answer is B,D and E = 3 stores**

4. Total number of non-fictions sold = 2360; Total cost = Rs. 25,960

(Total Cost of non-fiction/Number of non-fiction) = Price per copy = 25960/2360 = Rs. 11

5. Price of fiction and non-fiction is increased by 10%

Now fiction will cost Rs. 17.05, non-fiction Rs. 12.1, and GK Rs. 12.5 (Unchanged)

Total business of C will be equal to (4425*17.05 + 2360*12.1 + 2065*12.5)

= Rs. 129,814.75

Question Set 5

1 (c)	2 (d)	3 (d)	4 (b)	5 (c)

Question Set 6

1 (d)	2 (d)	3 (c)	4 (b)	5 (d)
6 (a)				

1. Average = (116 + 208 + 169 + 142 + 155)/5 = 158

2. Total admitted students till 2015 = 148 + 172 + 188 = 508; Total who left till 2015 = 60 + 96 = 156; Difference = 352; Number of girls = 6/11 * 352 = 192

3. SUCHAN = 161 + 148 – 58 = 251; DING = 148 + 172 – 60 = 260

MORIWALA = 179 + 161 – 90 = 250; SIKENDERPUR = 116 + 208 – 60 = 264

BAHAWDIN = 128 + 191 – 50 = 269

4. (116 + 148 + 135 + 112) – (58 + 69 + 88) = 341

5. Students of BAHAWDIN at the end of 2014 = 128 + 191 – 50 = 269

Students of MORIWALA at the end of 2014 = 179 + 161 – 90 = 250

Percentage = (269 – 250)/250 * 100 = 38/5 = 7.6

6. Ratio = (143 + 165) : (79 + 82 + 91) or 11 : 9

Question Set 7

1 (a)	2 (b)	3 (d)	4 (d)

Question Set 8

1 (b)	2 (b)	3 (c)	4 (c)

Question Set 9

1 (a)	2 (b)	3 (d)	4 (c)	5 (d)

Question Set 10

1 (c)	2 (d)	3 (b)

Question Set 11

1. (d) : None of the options can, not even all the three together can answer the question.

Question Set 12

1 (b)	2 (d)	3 (a)	4 (c)	5 (b)

Commuters are :

Bikes	7,359	Autos only	4,906	Taxis	7,359
Cars only	2,453	Bus and Cars both	12,265	Bus only	9,812
Auto & Taxis both	4,906	Train	22,077	Total commuters	81,149

(for the convenience of calculations, you are advised to round all these figures to either upper or lower thousands as also the choices have been given with wide ranges)

Question Set 13

1 (a)	2 (b)	3 (c)	4 (b)	5 (c)

Question Set 14

1 (a)	2 (d)	3 (b)	4 (c)	5 (d)
6 (d)	7 (b)			

1. Central angle corresponding to the royalty will be = 15% of $360^\circ = 54^\circ$

2. Such questions can be solved, first by getting 108° is how much% of 360, and thereafter picking up the correct option. 108 is 30% of 360, So from the options given, it is clear that Printing cost (20%) + Transportation cost(10%) = 30%

3. Move as per question 2 above technique

4. Apply this method 23% : 56,250 :: 9% :?

5. 18% : 30,600 :: 14% :?

6. Selling price of each book = 1.25 × 82,500/5,500 = 187.50

7. Instead of calculating by any other method, we should adopt this method viz. Difference%

= (20% – 15%)/20% * 100 (without actually calculating the costs).

Question Set 15

1 (b)	2 (c)	3 (b)	4 (b)	5 (d)

1. Average temperature of Dehradun = 23.2; Queta = 18.2, Difference 5° C

2. Average temperature in May = 24, Feb = 21; Difference = 3° C

3. Average temperature in Ranchi = 34.2; Kolkata = 16.6; Difference% = 106.2%

4. Average temperature of Lucknow = 21.4; Queta = 18.2; Ratio: 21.4 : 18.2 = 107:91

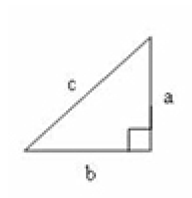
5. Average temperature in May = 24, March = 22; Required% = 109.09%

(For more questions on DATA INTERPRETATION – please refer to the Author's book "DATA INTERPRETATION Simplified by JAGGAN SANEJA" – available on all e-commerce stores like Amazon.com, Bookadda.com, Infibeam.com, Flipkart.com and NOTION PRESS.com).

AREAS & VOLUMES

27

1. PYTHAGOREAN THEOREM (PYTHAGORAS' THEOREM)



In a right angled triangle, the square of the hypotenuse is equal to the sum of the squares of the other two sides. $c^2 = a^2 + b^2$ where c is the length of the hypotenuse and a and b are the lengths of the other two sides.

Length of the longest rod that can be placed in a box of length l, breadth b and height h = $\sqrt{l^2 + b^2 + h^2}$.

Pi is a mathematical constant which is the ratio of a circle's circumference to its diameter. It is denoted by π (π ≈ 3.14 ≈ 22/7)

Sum of Interior Angles of a polygon = 1800 (n − 2); where n = number of sides

Example Number of sides of a triangle = 3.
Hence, sum of the interior angles of a triangle = 180(3 − 2) = 180 × 1 = 180°

Example Number of sides of a quadrilateral = 4
Hence, sum of the interior angles of any quadrilateral = 180(4 − 2) = 180 × 2 = 360°

- ***If each of side of a rectangle or any two dimensional shape is increased by x%, its area is increased by (x^2/100 +***

2x)%

- ***If radius of a circle is increased by x%, its area is increased by (x^2/100 + 2x)%***

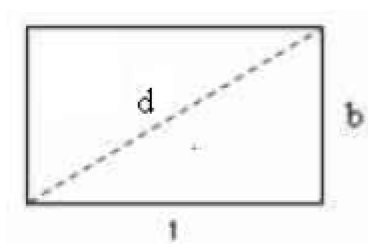
Example If each side of a triangle is doubled, what is the percentage increase in its area?

Here, each side is increased by 100% (because each side is doubled) percentage increase in its area = (100^2/100 + 2 × 100)% = 300%

Example If circumference of a circle is increased by 50%, what is the percentage increase in its area?

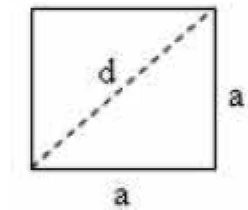
- *Circumference of the circle is increased by 50%. Since circumference of the circle is 2πr, radius, r is increased by 50%*
- *Percentage increase in its area = (502/100 + 2 × 50)% = 125%*

RECTANGLE



L = Length;	B = Breadth;	D = Length of diagonal
Area = LB	Perimeter = 2(L + B)	d = $\sqrt{L^2 + B^2}$

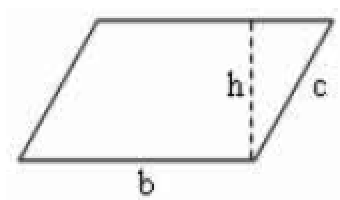
SQUARE



a = Length of a side; d = Length of diagonal

Area = $a^2 = 12d^2$; Perimeter = $4ad = \sqrt{2}a$

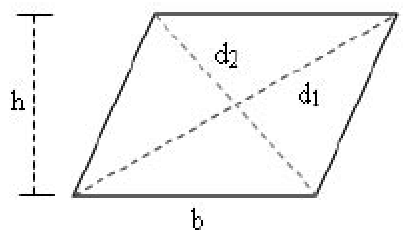
PARALLELOGRAM



b and c are sides; b = base; h = height;

Area = $b \times h$; Perimeter = $2(b + c)$

RHOMBUS



length of each side = a

a = length of each side; b = base; h = height

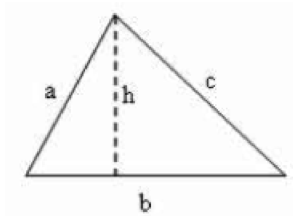
d_1, d_2 are the diagonals

Area = bh (Formula 1 for area)

Area = $12d^1d^2$ (Formula 2 for area)

Perimeter = $4a$

TRIANGLE



a,b and c are sides; b = base; h = height;

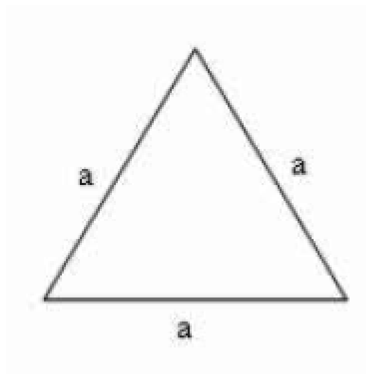
Area = half bh .... (Formula 1 for area)

Area = √s(s − a)(s − b)(s − c) where s is the semi perimeter = (a + b + c)/2 .... (Formula 2 for area)

Perimeter = a + b + c

Radius of in circle of a triangle of area A = A/s; 's' is the semi perimeter = (a + b + c)/2

EQUILATERAL TRIANGLE

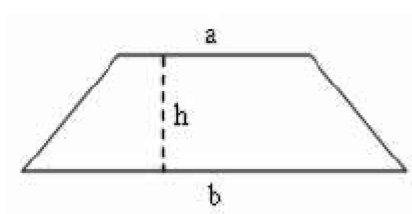


a = side; Area = $\sqrt{3}a^2/4$; Perimeter = 3a

Radius of in circle of an equilateral triangle of side a = $a/2\sqrt{3}$

Radius of circum circle of an equilateral triangle of side a = $a/\sqrt{3}$

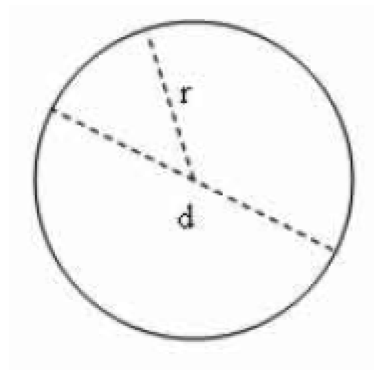
TRAPEZIUM (TRAPEZOID)



Base 'a' is parallel to base 'b'; h = height

Area = (a + b) h/2

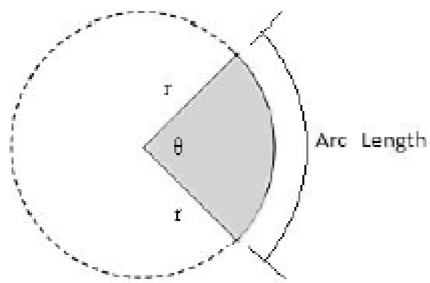
CIRCLE



R = radius; d = diameter = 2r

Area = $\pi r^2 = 1/4\ \pi d^2$; Circumference = $2\pi r = \pi d$

SECTOR OF CIRCLE



r = radius; θ = central angle = l/r radians = Arc/radius

$180° = 200^g = \pi$ radians (here $\pi = 22/7$ or 3.14)

Area, A = $\theta/360\ \pi r^2$ (if angle is in degrees) = $1/2\ r^2\theta$ (if angle is in radians)

Arc Length, s = $\theta/180\ \pi\ r$ (if angle in degrees) = $r\ \theta$ (if angle in radians)

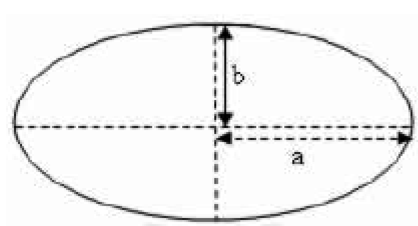
In the radian system for angular measurement,

2π radians = 360° $\Rightarrow$ 1 radian = 180°π $\Rightarrow$ 1° = $\pi/180$ radians

Hence, Angle in Degrees = Angle in Radians × 180°π

Angle in Radians = Angle in Degrees × π180°

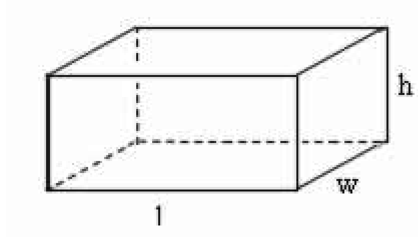
ELLIPSE



Major axis length = 2a; Minor axis length = 2b

Area = πab; Perimeter ≈ $2\pi\ [\sqrt{}a^2 + b^2]/2$

RECTANGULAR SOLID

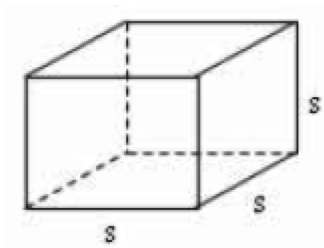


L = Length; B = w = width; h = height

Total Surface Area = 2 LB + 2BH + 2HL = 2 (LB + BH + HL)

Volume = LBH

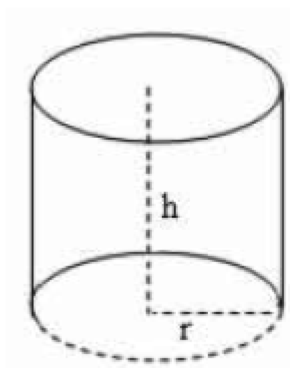
CUBE



S = edge = a; Total Surface Area = $6s^2 = 6a^2$

Volume = $s^3 = a^3$

RIGHT CIRCULAR CYLINDER



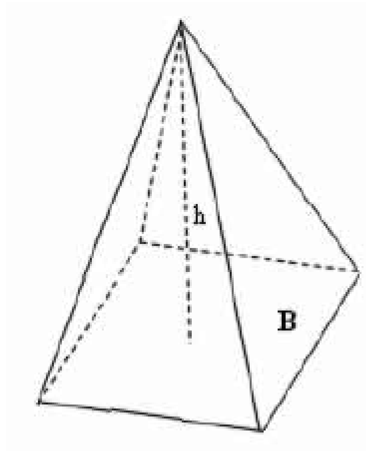
H = height r = radius of base

Lateral Surface Area = $(2\pi r)h$

Total Surface Area = $(2\pi r)h + 2(\pi r^2)$

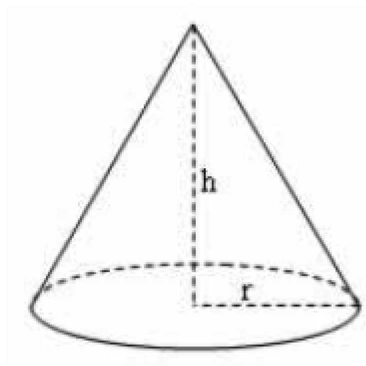
Volume = $(\pi r^2)h$

PYRAMID



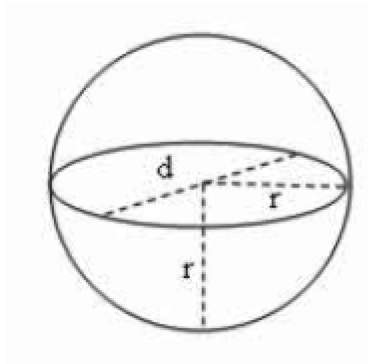
H = height; B = area of the base
Total Surface Area = B + Sum of the areas of the triangular sides
Volume = 1/3 Bh

RIGHT CIRCULAR CONE



h = height; r = radius of base
Lateral Surface Area = $\pi r (\sqrt{r^2 + h^2}) = \pi rs$
where s is the slant height = $\sqrt{r^2 + h^2}$
Total Surface Area = $\pi r (\sqrt{r^2 + h^2}) + \pi r^2 = \pi rs + \pi r^2$

SPHERE



r = radius; d = diameter; d = 2r
Surface Area = $4\pi r^2 = \pi d^2$
Volume = $4/3\ \pi r^3 = 1/6\ \pi d^3$

Important properties of Geometric Shapes Properties of Triangle

Sum of the angles of a triangle = 180°

- ***Sum of any two sides of a triangle is greater than the third side.***
- ***The line joining the midpoint of a side of a triangle to the + ve vertex is called median.***
- ***The median of a triangle divides the triangle into two triangles with equal areas.***
- ***Centroid is the point where the three medians of a triangle meet.***
- ***Centroid divides each median into segments with a 2:1 · ratio***
- ***Area of a triangle formed by joining the midpoints of the sides of a triangle is ¼ of the area of the given triangle.***
- ***An equilateral triangle is a triangle in which all three sides are equal.***
- ***In an equilateral triangle, all three internal angles are congruent to each other.***
  - » ***In an equilateral triangle, all three internal angles are each 60°***
  - » ***An isosceles triangle is a triangle with (at least) two equal sides.***
  - » ***In isosceles triangle, altitude from vertex bisects the base.***
  - » ***The diagonals of a rectangle are equal and bisect each other.***
  - » ***Opposite sides of a rectangle are parallel.***

- ***Opposite sides of a rectangle are congruent.***
- ***Opposite angles of a rectangle are congruent.***
- ***All four angles of a rectangle are right angles.***
- ***The diagonals of a rectangle are congruent.***
- ***All four sides of a square are congruent.***
- ***Opposite sides of a square are parallel.***
- ***The diagonals of a square are equal.***
- ***The diagonals of a square bisect each other at right angles.***
- ***All angles of a square are 900***
- ***A square is a special kind of rectangle where all the sides have equal length.***
- ***The opposite sides of a parallelogram are equal in length.***
- ***The opposite angles of a parallelogram are congruent (equal measure).***
- ***The diagonals of a parallelogram bisect each other.***
- ***Each diagonal of a parallelogram divides it into 2 triangles of the same area.***
- ***All the sides of a rhombus are congruent.***
- ***Opposite sides of a rhombus are parallel.***
- ***The diagonals of a rhombus bisect each other at right angles.***
- ***Opposite internal angles of a rhombus are congruent (equal in size).***
- ***Any two consecutive internal angles of a rhombus are supplementary; i.e., the sum of their angles = 180°***
- ***Sum of the interior angles of a quadrilateral is 3600***
- ***If a square and a rhombus lie on the same base, area of the square will be greater than area of the rhombus (In the special case when each angle of the rhombus is 90°***
- ***rhombus is also a square and therefore areas will be equal).***
- ***A parallelogram and a rectangle on the same base and between the same parallels are equal in area.***
- ***Of all the parallelogram of given sides, the parallelogram which is a rectangle has the greatest area.***
- ***Each diagonal of a parallelogram divides it into two triangles of the same area.***
- ***A square is a rhombus and a rectangle.***

AREAS & VOLUMES

Practice Exercise - 1

Number of Questions: 20 **Suggested time: 15 minutes**

1. An error 2% in excess is made while measuring the side of a square. What is the percentage of error in the calculated area of the square?
(a) 4.04
(b) 2.02
(c) 4
(d) 2

2. A rectangular park 60 m long and 40 m wide has two concrete crossroads running in the middle of the park and rest of the park has been used as a lawn. The area of the lawn is 2109 sq.m. what is the width of the road?
(a) 5
(b) 4
(c) 2
(d) 3

3. A towel, when bleached, lost 20% of its length and 10% of its breadth. What is the percentage decrease in area?
(a) 30
(b) 28
(c) 32
(d) 26

4. If the length of a rectangle is halved and its breadth is tripled, what is the percentage change in its area?
(a) 25 up
(b) 25 down
(c) 50 down
(d) 50 up

5. A person walked diagonally across a square plot. Approximately, what was the percent saved by not walking along the edges?

(a) 35
(b) 30
(c) 20
(d) 25

6. A rectangular field has to be fenced on three sides leaving a side of 20 feet uncovered. If the area of the field is 68 sq. feet, how many feet of fencing will be required?
(a) 95
(b) 92
(c) 88
(d) 82

7. A rectangular parking space is marked out by painting three of its sides. If the length of the unpainted side is 9 feet, and the sum of the lengths of the painted sides is 37 feet, find out the area of the parking space in square feet?
(a) 126
(b) 64
(c) 100
(d) 102

8. The area of a rectangular plot is 460 square metres. If the length is 15% square metres. If the length is 15%
(a) 14
(b) 20
(c) 18
(d) 12

9. A large field of 700 hectares is divided into two parts. The difference of the areas of the two parts is one-fifth of the average of the two areas. What is the area of the smaller part in hectares?
(a) 400
(b) 365
(c) 385
(d) 315

10. The length of a room is 5.5 m and width is 3.75 m. What is the cost of paying the floor by slabs at the rate of Rs. 800 per sq. metre.

(a) 12000
(b) 19500
(c) 18000
(d) 16500

11. The length of a rectangle is twice its breadth. If its length is decreased by 5 cm and breadth is increased by 5 cm, the area of the rectangle is increased by 75 sq.cm. What is the length of the rectangle?
(a) 18
(b) 16
(c) 40
(d) 20

12. If a square and a rhombus stand on the same base, then what is the ratio of the areas of the square and the rhombus?
(a) 1:2
(b) 3:4
(c) >1
(d) 1

13. The breadth of a rectangular field is 60% of its length. If the perimeter of the field is 800 m, find out the area of the field
(a) 37,500
(b) 30,500
(c) 32,500
(d) 40,000

14. A room 5 m 44 cm long and 3 m 74 cm broad needs to be paved with square tiles. What will be the least number of square tiles required to cover the floor?
(a) 176
(b) 124
(c) 224
(d) 186

15. The length of a rectangular plot is 20 metres more than its breadth. If the cost of fencing the plot @ Rs. 26.50 per metre is Rs. 5300, what is the length of the plot in metres?
(a) 60

(b) 100
(c) 75
(d) 50

16. The ratio between the length and the breadth of a rectangular park is 3:2 If a man cycling along the boundary of the park at the speed of 12 km/hr completes one round in 8 minutes, then what is the area of the park (in sq. m)?
(a) 1,42,000
(b) 1,12,800
(c) 1,42,500
(d) 1,53,600

17. What is the percentage increase in the area of a rectangle, if each of its sides is increased by 20%?
(a) 45
(b) 44
(c) 40
(d) 42

18. If the difference between the length and breadth of a rectangle is 23 m and its perimeter is 206 m, what is its area?
(a) 2800
(b) 2740
(c) 2520
(d) 2200

19. The ratio between the perimeter and the breadth of a rectangle is 5:1. If the area of the rectangle is 216 sq. cm, what is the length of the rectangle?
(a) 16
(b) 18
(c) 14
(d) 20

20. What is the least number of square tiles required to pave the floor of a room 15 m 17 cm long and 9 m 2 cm broad?
(a) 814
(b) 802

(c) 836
(d) 900

Practice Exercise 1 - Answers with Hints - need based

9. Average of the two areas = 700/2 = 350

one-fifth of the average of the two areas = 350/5 = 70

Let area of the smaller part = x hectares.

Then, area of the larger part = x + 70 hectares.

$x + (x + 70) = 700 \Rightarrow 2x = 630 \Rightarrow x = 315$

11. Let breadth = x cm; Then, length = 2x cm

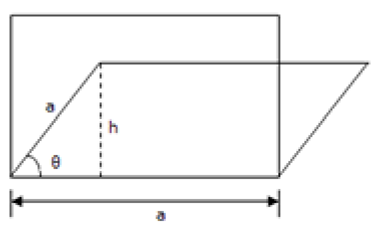
Area = $x \times 2x = 2x^2$ sq.cm. New length = (2x − 5) cm; New breadth = (x + 5) cm

New area = (2x − 5)(x + 5) sq.cm.

Given that, new area = initial area + 75 sq.cm.

$\Rightarrow (2x - 5)(x + 5) = 2x^2 + 75 \Rightarrow x = 100/5 = 20$ cm

12. If a square and a rhombus lie on the same base, area of the square will be greater than area of the rhombus (In the special case when each angle of the rhombus is 90° rhombus is also a square and therefore areas will be equal)



13. breadth = 60% of length; perimeter = 800 m

⇒ 2(length + breadth) = 800

⇒ 2(length + 60% of length) = 800

⇒ 2(160% of length) = 800 ⇒ length = 400 × 100/160 = 250 m

breadth = 250 × 60100 = 150 m

Area = $250 \times 150 = 37500$ m^2

14. In all such cases, the side of the square is always HCF of sides.

Now we need to find out HCF(Highest Common Factor) of 544 and 374 = 34

Area of each square tile = 34×34 cm^2

Number of tiles required = $544 \times 374/34 \times 34 = 16 \times 11 = 176$

15. Hence, let's take the length as L metre and breadth as (L − 20) metre.

Length of the fence = perimeter = 2(length + breadth) = 2(2L − 20) metre

Total cost = $2(2L - 20) \times 26.50$ = Rs. 5300 (given)

L = 60 metre

AREAS & VOLUMES

Practice Exercise - 2

Number of Questions: 20 **Suggested time: 15 minutes**

1. The diagonal of the floor of a rectangular room is 712 feet. The shorter side of the room is 412 feet. What is the area of the room?
(a) 27
(b) 22
(c) 24
(d) 20

2. The diagonal of a rectangle is $\sqrt{41}$ cm and its area is 20 sq. cm. What is the perimeter of the rectangle?
(a) 16
(b) 10
(c) 12
(d) 18

3. A tank is 25 m long, 12 m wide and 6 m deep. What is the cost (Rupees) of plastering its walls and bottom at the rate of 75 paise per sq.m?
(a) 558
(b) 502
(c) 516
(d) 612

4. It is decided to construct a 2 metre broad pathway around a rectangular plot on the inside. If the area of the plot is 96 sq.m. and the rate of construction is Rs. 50 per square metre, what will be the total cost of construction?
(a) 3500
(b) 4200
(c) 4400
(d) Insufficient Data

5. The area of a parallelogram is 72 cm^2 and its altitude is twice the corresponding base. What is the length of the base

(a) 6
(b) 7
(c) 8
(d) 12

6. Two diagonals of a rhombus are 72 cm and 30 cm respectively. What is its perimeter?
(a) 166
(b) 156
(c) 244
(d) 212

7. The base of a parallelogram is (p + 4), altitude to the base is (p − 3) and the area is (p2 − 4) find out its actual area
(a) 46
(b) 56
(c) 49
(d) 60

8. A circle is inscribed in an equilateral triangle of side 24 cm, touching its sides. What is the area of the remaining portion of the triangle?
(a) $144\sqrt{3} - 48\pi$
(b) $21\sqrt{3} - 6\pi$
(c) $44\sqrt{3} - 3\pi$
(d) $21\sqrt{3} - 8\pi$

9. A rectangular plot measuring 90 metres by 50 metres needs to be enclosed by wire fencing such that poles of the fence will be kept 5 metres apart. How many poles will be needed?
(a) 36
(b) 54
(c) 56
(d) 65

10. If the diagonals of a rhombus are 24 cm and 10 cm, what will be its perimeter?
(a) 46
(b) 56

(c) 66
(d) 52

11. What will be the length of the longest rod which can be placed in a box of 80 cm length, 40 cm breadth and 60 cm height?
(a) $\sqrt{11600}$
(b) $\sqrt{14100}$
(c) $\sqrt{21500}$
(d) None

12. A spherical ball is covered with two colors blue and orange. The ratio of area covered by blue and orange colors on the whole ball is 2:3. The ratio of area for blue and orange colours on the upper part of the ball is 1:2. Find the ratio of area covered by blue and orange colours on the lower half
(a) 4 : 5
(b) 3 : 7
(c) 5 : 8
(d) 7 : 8

13. A circle is inscribed in a square whose length of the diagonal is $12\sqrt{2}$ cm. An equilateral triangle is inscribed in that circle. The length of the side of the triangle is
(a) $4\sqrt{3}$ cm
(b) $8\sqrt{3}$ cm
(c) $6\sqrt{3}$ cm
(d) $11\sqrt{3}$ cm

14. The whole surface of a rectangle solid is 448 sq.cm. The length of the solid is twice its breadth and the height is half its breadth. Find the volume of the solid.
(a) 448
(b) 512
(c) 336
(d) 496

15. There is a cylinder circumscribing the hemisphere such that their bases are common. Find the ratio of their volumes
(a) 5/2

(b) 7/2
(c) 3/2
(d) 9/2

16. The whole surface of a rectangle solid is 448 sq.cm. The length of the solid is twice its breadth and the height is half its breadth. Find the volume of the solid.
(a) 480
(b) 524
(c) 624
(d) 512

17. The difference between the sides of two squares is 18 cm. The perimeter of the smaller square is 56 cm. What is the area of the bigger square?
(a) 1480
(b) 1024
(c) 1624
(d) 512

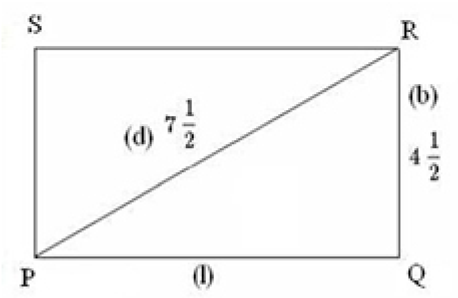
18. If each edge of a cube is increased by 50%, find the percentage increase in Its surface area
(a) 135
(b) 145
(c) 66
(d) 125

19. How many cubes of 3 cm edge can be cut out of a cube of 18 cm edge
(a) 216
(b) 245
(c) 166
(d) 125

20. A rectangular block 6 cm by 12 cm by 15 cm is cut up into an exact number of equal cubes. Find the least possible number of cubes.
(a) 15
(b) 25
(c) 30
(d) None

Practice Exercise 2 - Answers with Hints - need based

1 (a)	2 (d)	3 (a)	4 (d)	5 (a)
6 (b)	7 (d)	8 (a)	9 (c)	10 (d)
11 (a)	12 (d)	13 (a)	14 (b)	15 (c)
16 (d)	17 (b)	18 (d)	19 (a)	20 (d)

1. 

3. Total Surface area of the rectangular solid,

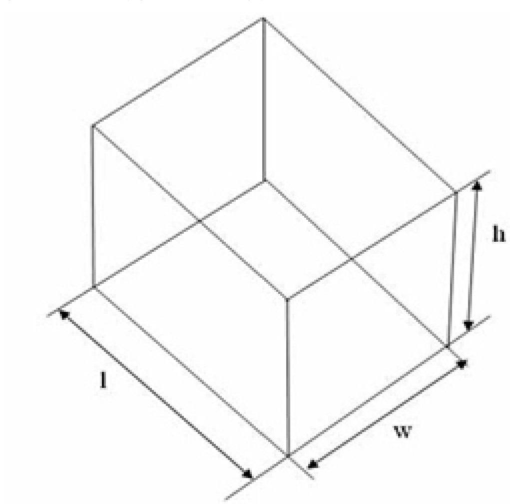
S = 2lw + 2lh + 2wh = 2(lw + lh + wh)

In this case, l = 25 m; w = 12 m, h = 6 m (all surface to be plastered except top)

Hence, total area to be plastered = Total surface area - Area of the top face

= (2lw + 2lh + 2wh) − lw = lw + 2lh + 2wh = (25 × 12) + (2 × 25 × 6) + (2 × 12 × 6) = 744

Cost of plastering = 744 × 0.75 = Rs. 558

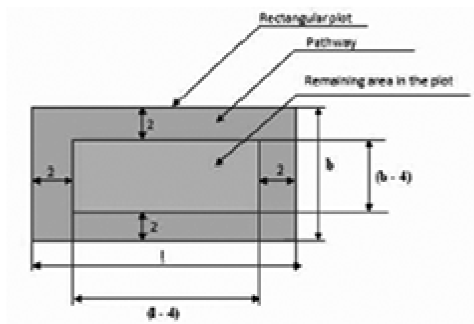


4. Area of the rectangular plot = L B = 96 sq.m,; width of pathway is 2 m

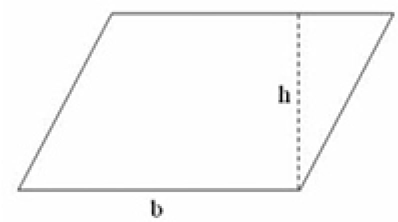
Remaining area in the plot = (L − 4)(B − 4)

Area of the pathway = Area of the rectangular plot - Remaining area in the plot

= 96 − (L − 4)(B − 4) = 4L + 4B − 16 ***(We do not know the values of L and B and hence area of the pathway cannot be found out. So we cannot determine total cost of construction).***



5.

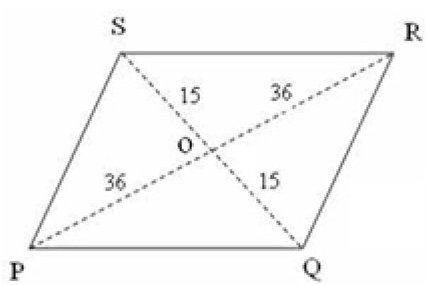


Area of a parallelogram, A = bh

Let base = x cm; Then, height = 2x cm(∵ altitude is twice the base)

Area = $2x^2 = 72\ cm^2 \Rightarrow x = 6$ cm

6.



Remember the following two properties of a rhombus which will be useful in solving this question

1. All the sides of a rhombus are congruent.

2. The diagonals of a rhombus bisect each other at right angles.

Let the diagonals be PR and SQ

PO = OR = 72/2 = 36 cm

SO = OQ = 30/2 = 15 cm

PQ = QR = RS = SP

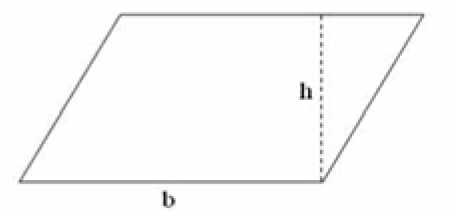
$= \sqrt{(36^2 + 15^2)} = 39$ cm; Perimeter $= 4 \times 39 = 156$ cm

7. Area of a parallelogram, as given below is, A = bh

$P^2 - 4 = (P + 4)(P - 3) \Rightarrow P^2 + P - 12$

$\Rightarrow -4 = P - 12 \Rightarrow P = 8$

Hence, actual area $= (P^2 - 4) = 64 - 4 = 60$ sq. units



8. Radius of in circle of an equilateral triangle (as below) of side $a = a^2 \sqrt{3}$

Radius of the circle inscribed $= 12/\sqrt{3}$ cm

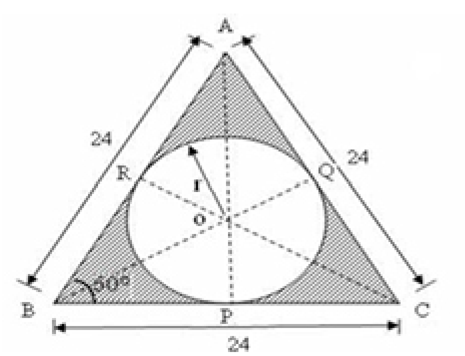
Area of the circle inscribed $= \pi(12/\sqrt{3})^2 = \pi \times 144/3 = 48\pi$ cm2

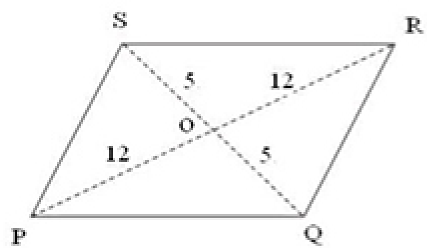
Area of the equilateral triangle $= \sqrt{3}/4\, a^2 = 144\sqrt{3}$ cm2

Area of the remaining portion of the triangle

= Area of the equilateral triangle – Area of inscribed circle

$= 144\sqrt{3} - 48\pi$ cm2



10. 

Use Pythagoras to find PQ = 13, Perimeter = 52 cm

11. Length of the longest rod that can be placed in a box = $\sqrt{L^2 + B^2 + C^2}$

$= \sqrt{80^2 + 40^2 + 60^2} = \sqrt{11600}$ cm

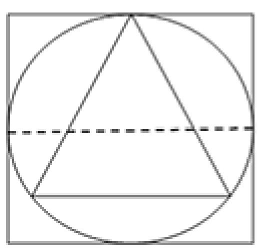
12. Take total area = 30x

$\Rightarrow$ Total blue = 12x and total orange = 18x;

Half area = 15x

According to ratio, blue = 5x and orange = 10x

Ratio of another half = (12x − 5x):(18x − 10x) = 7 : 8

13. 

14. Side of the square = $12\sqrt{2}\sqrt{2}$ = 12 cm

Therefore, diameter of the circle = 12 cm; r = 6 cm

Side of the equilateral triangle = $6 \times \sqrt{3} = 6\sqrt{3}$ cm

AREAS & VOLUMES

Practice Exercise - 3

Number of Questions: 20 **Suggested time: 15 minutes**

1. Three solid cubes of sides 1 cm, 6 cm and 8 cm are melted to form a new cube. Find the surface area of the cube so formed

(a) 486
(b) 445
(c) 566
(d) 125

2. A right triangle with sides 3 cm, 4 cm and 5 cm is rotated the side of 3 cm to form a cone. The volume of the cone so formed is:

(a) 12π
(b) 15π
(c) 16π
(d) 20π

3. A hall is 15 m long and 12 m broad. If the sum of the areas of the floor and the ceiling is equal to the sum of the areas of four walls, the volume of the hall is:

(a) 720
(b) 900
(c) 1200
(d) 1800

4. A cistern 6m long and 4 m wide contains water up to a depth of 1 m 25 cm. The total area of the wet surface is:

(a) 49
(b) 50
(c) 53
(d) 55

5. The volume of a wall, 5 times as high as it is broad and 8 times as long as it is high, is 12.8 cu. meters. Find the breadth of the wall.

(a) 40

(b) 30
(c) 20
(d) 10

6. The diagonal of a rectangle is 41 cm and its area is 20 sq. cm. The perimeter of the rectangle must be:
(a) 9
(b) 18
(c) 20
(d) 41

7. Find the length of the longest pole that can be placed in a room 12 m long 8m broad and 9m high.
(a) 15
(b) 14
(c) 19
(d) 17

8. A cistern of capacity 8000 litres measures externally 3.3 m by 2.6 m by 1.1 m and its walls are 5 cm thick. The thickness of the bottom is _____ meter:
(a) 1.1 m
(b) 1
(c) 9.90
(d) 2.1

9. The curved surface area of a cylindrical pillar is 264 sq.m and its volume is 924 cub.m. Find the ratio of its diameter to its height?
(a) 3 : 7
(b) 7 : 3
(c) 4 : 7
(d) 7 : 6

10. Find the number trucks of bricks (one truck carries thousand bricks), each measuring 24 cm × 12 cm × 8 cm, required to construct a wall 24 m long, 8m high and 60 cm thick, if 10% of the wall is filled with mortar?
(a) 36
(b) 45

(c) 75
(d) 95

11. How many cubes of 10cm edge can be put in a cubical box of 1m edge
(a) 10
(b) 100
(c) 1000
(d) 10000

12. The dimensions of an open box are 50 cm, 40 cm and 23 cm. Its thickness is 2 cm. If 1 cubic cm of metal used in the box weighs 0.5 gms, find the weight of the box in kilograms.
(a) 8.04
(b) 9.04
(c) 8.34
(d) 8.74

13. Fifty took a dip in a water tank 40 m long and 20 m broad on a religious day. If the average displacement of water by a man is 4m3, then the rise in the water level in the tank will be:
(a) 26
(b) 25
(c) 95
(d) 65

14. Two cones have their heights in the ratio 1:3 and the radii of their bases in the ratio 3:1. Volume of the bigger is what% more than the smaller.
(a) 200
(b) 700
(c) 400
(d) 400

15. How many hundreds of bricks, each measuring 25 cm × 11.25 cm × 6 cm, will be needed to build a wall of 8 m × 6 m × 22.5 cm?
(a) 71
(b) 69
(c) 64
(d) 78

16. In a shower, 5 cm of rain falls. The volume of water that falls on 1.5 hectares of ground is:

(a) 75

(b) 750

(c) 75000

(d) 7500

17. The surface area of a cube is 1734 sq. cm. Find its volume

(a) 3356

(b) 2394

(c) 4913

(d) 3778

18. Find the surface area of a10 cm × 4 cm × 3 cm brick

(a) 174

(b) 24

(c) 164

(d) 84

19. A wall is 8 m × 6 m × 22.5 m. Find the approximate total cost of white wash on this if it costs Rs. 24 per square meter.

(a) 20000

(b) 16000

(c) 64000

(d) 72000

20. The capacity of a tank of dimensions (8 m × 6 m × 2.5 m) in litres is

(a) 1200

(b) 120

(c) 12000

(d) 1.20 lakhs

Practice Exercise 3 - Answers with Hints - need based

1 (a)	2 (a)	3 (c)	4 (a)	5 (a)
6 (b)	7 (d)	8 (b)	9 (b)	10 (b)
11 (c)	12 (a)	13 (b)	14 (a)	15 (c)
16 (b)	17 (c)	18 (c)	19 (a)	20 (d)

3. Given relationship is = 2 (15 + 12) h = 2 (15 × 12)

h = 20/3 m

V = (15 × 12 × 20/3) m^3 = 1200 cubic meter

5. Let the breadth of the wall be x metres.

Then, Height = 5x metres and Length = 40x metres.

x × 5x × 40x = 12.8

$$\Rightarrow x^3 = \frac{12.8}{200} = \frac{128}{2000} = \frac{64}{1000} \Rightarrow x = \frac{4}{10} m$$

$$\Rightarrow x = \frac{4}{10} \times 100 = 40cm$$

6. $\sqrt{l^2 + b^2} = \sqrt{41}$

Also, l × b = 20

= (l + b) = 9.

Perimeter = 2(l + b) = 18 cm.

7. Length of longest pole = Length of the diagonal of the room

= $(L^2 + B^2 + H^2)^{½}$

= $(12^2 + 8^2 + 9^2)^{½}$

= 17 m

8. Let the thickness of the bottom be 'x' cm

⇒ (330 – 10) × (260 – 10) × (110 – x) = 8000 × 1000

⇒ 110 – x = (8000 × 1000)/(320 × 250) = 100

Therefore x = 10 cm

9. Number of bricks = [(90/100) × (2400 × 800 × 60)]/(24 × 12 × 8) = 45000

12. Volume of the metal used in the box = External Volume – Internal Volume

= [(50 × 40 × 23) – (44 × 34 × 20)]cm^3 = 16080 cm^3

Weight of the metal = [(16080 × 0.5)/1000] kg = 8.04 kg.

17. Given $6a^2 = 1734 \Rightarrow a = 17$ cm $\Rightarrow$ Volume $= a^3 = 17^3 = 4193$ cubic cm

PERMUTATIONS & COMBINATIONS

28

Assuming that I teach three students namely a,b,c. It so happens with them they come in a group of two each day.

Hence, they can keep me busy by coming in a group of two each day say (a,b),(b,c), (c,a) i.e., only three ways.

But if I keep a watch on their movements in my class and also observe the order they enter my class then it would amount to 6 ways viz. (ab), (ba), (ca), (ac), (bc),(cb).

Combinations: If the order does not matter in an arrangement i.e., we would have three combinations of groups consisting of two each i.e., (a,b), (b,c), (c,a)

Here the order within each set selected does not matter.

Permutations: If three items a,b,c, were to be arranged in groups of two, there would be six arrangements i.e., (a,b), (b,a), (c,a), (a,c), (b,c),(c,b).

Notice that an order within each combination does not matter while in Permutation it does.

Some Rules covering Permutations & Combinations

- *If there are 'm' ways of one arrangements and 'n' ways of another*
- *then there are total of 'm.n' ways*
- *the number of ways of arranging n different things taken all of them at a time is n! or [n] ways*
- *the number of ways of arranging 'n' different things taken only 'r' at a time is n P r or = (n) (n – 1) (n – 2) …. (n – r + 1) ways*
- *The number of permutations of 'n' dissimilar things taken 'r' at a time is n.n.n.n…….r times*
- *If we consider 'n' items in a Circular arrangements then this can be done in [n-1] ways*

Practice Exercise – 1

Number of Questions: 30 **Suggested time: 25 minutes**

1. Out of 7 consonants and 4 numerals, how many codes of 3 consonants and 2 numerals can be formed?
(a) 24400
(b) 21300
(c) 210
(d) 25200

2. In a group of 6 Men and 4 Women, four member team is to be selected. In how many different ways can they be selected such that at least one Man should be there?
(a) 159
(b) 209
(c) 201
(d) 212

3. From a group of 7 Gentlemen and 6 Ladies, five member committee is to be selected so that at least 3 Gentlemen are included in the committee. In how many ways can it be done?
(a) 624
(b) 702
(c) 756
(d) 812

4. In how many different ways can the letters of the word 'COUNCIL' be arranged so that the vowels always come together?
(a) 610
(b) 720
(c) 825
(d) 920

5. In how many different ways can the letters of the word "CORPORATION' be arranged so that the vowels always come together?

(a) 47200
(b) 48000
(c) 42000
(d) 50400

6. In how many ways can a group of 5 boys and 2 girls be made out of a total of 7 boys and 3 girls?
(a) 1
(b) 126
(c) 63
(d) 64

7. In how many different ways can the letters of the word 'MATHEMATICS' be arranged such that the vowels must always come together?
(a) 9800
(b) 100020
(c) 120960
(d) 140020

8. Out of 8 executives and 10 work force we need to form a committee of 5 executives and 6 work force. In how many ways can the committee be formed?
(a) 10420
(b) 11
(c) 11760
(d) None of these.

9. How many three-letter words, can be formed using letters of 'LOGARITHMS', if repetition is not allowed?
(a) 720
(b) 420
(c) None of these
(d) 5040

10. In how many different ways can the letters of the word 'ROADING' be arranged such that thevowels always come together?
(a) 122
(b) 720

(c) 420

(d) None of these

11. A coin is tossed 3 times. How many possible outcomes will be there?.

(a) None of these

(b) 8

(c) 2

(d) 1

12. In how many different ways can the letters of the word 'RETAIL' be arranged such that the vowels do not occupy any EVEN position?

(a) 36

(b) 64

(c) 120

(d) None

13. A bag contains 2 Pink, 3 black and 4 Green balls. In how many ways can 3 balls be drawn from the bag, picking up at least one black ball included in the draw?

(a) 64

(b) 128

(c) 32

(d) None of these

14. In how many different ways can the letters of the word 'BUDGE' be arranged to have the vowels always together?

(a) None of these

(b) 48

(c) 32

(d) 64

15. In how many ways can the letters of the word 'SEATER' be arranged?

(a) 480

(b) 120

(c) 360

(d) 720

16. How many words can be formed by using all letters of the word 'TIHAR'?

(a) 720

(b) 24
(c) 120
(d) 60

17. How many lakhs of arrangements can be made using the letters of the word 'ENGINEERING'?
(a) 9.24
(b) 2.77
(c) 4.79
(d) 1.82

18. How many 3 digit numbers can be formed from the digits 1, 4, 5, 8, 3 and 9 which are divisible by 5 and none of the digits is repeated?
(a) 20
(b) 16
(c) 8
(d) 24

19. How many words with or without meaning, can be formed by using all the letters of the word, 'TEHRI' using each letter exactly once?
(a) 720
(b) 24
(c) 365
(d) 120

20. What is the value of ${}^{100}P_2$?
(a) 9801
(b) 12000
(c) 5600
(d) 9900

21. In how many different ways can the letters of the word 'HONOUR' be arranged?
(a) 315
(b) 128
(c) 360
(d) 180

22. There are 6 class sessions on each working day of a Rania school. In how many ways can one organize 5 subjects such that each subject is allotted at least one period?

(a) 3200

(b) None of these

(c) 1800

(d) 3600

23. How many Six-digit telephone numbers can be formed if each number starts with 27 and no repetition of digits is allowed?

(a) 720

(b) 360

(c) 1420

(d) 1680

24. Mr Chinu is a event manager and has ten patterns of Sofas and eight patterns of Tables.In how many ways can he make a pair of Sofa and Table?

(a) 100

(b) 80

(c) 110

(d) 64

25. Between Patiala and Mansa 25 buses ply during the day. In how many ways can Mr Bhushango from Patiala to Mansa and return by a different bus?

(a) 448

(b) 600

(c) 576

(d) 625

26. Navansh has a box which contains 4 Pink, 3 Orange and 2 blue balls. He draws three balls at random. Find out the number of ways so that Navansh gets balls of different colours?

(a) 62

(b) 48

(c) 12

(d) 24

27. In an examination, Vridhi got a question paper which has two parts A and B, each with 10 questions. If she selects 8 questions from part A and 4 from part B, in how many ways can she attempt the examination ?

(a) 5820

(b) 7020

(c) 8200

(d) None

28. In how many different ways can 5 lions and 5 lambs form a circle in a circus such that the lions and lambs stand alternate?

(a) 2880

(b) 1400

(c) 1200

(d) 3212

29. Find out the number of ways in which 6 shirts of different colours can be worn in 3 parties ?

(a) 120

(b) 720

(c) 125

(d) 729

30. In how many ways can 5 men use 5 different stairs if no stairs can be used more than once?

(a) 360

(b) 720

(c) 60

(d) 120

Practice Exercise 1 - Answers with Explanations - need based

1 (d)	2 (b)	3 (c)	4 (b)	5 (d)
6 (c)	7 (c)	8 (c)	9 (a)	10 (b)
11 (b)	12 (d)	13 (a)	14 (b)	15 (c)
16 (c)	17 (b)	18 (a)	19 (d)	20 (d)
	22 (c)		24 (b)	

21 (d)		23 (d)		25 (b)
26 (d)	27 (d)	28 (a)	29 (d)	30 (d)

1. $= {}^7C_3 \cdot {}^4C_2$

3. Total number of ways

$= {}^7C_5 + ({}^7C_4 \cdot {}^6C_1) + ({}^7C_3 \cdot {}^6C_2)$

$= {}^7C_2 + ({}^7C_3 \cdot {}^6C_1) + ({}^7C_3 \cdot {}^6C_2)[\because {}^nC_r = {}^nC_{(n-r)}]$

9. Hence, the number of 3-letter words (with or without meaning) formed by using these letters $= {}^{10}P_3 = 10 \times 9 \times 8 = 720$

11. When a coin is tossed once, there are two possible outcomes: Head (H) and Tale(T)

Hence, when a coin is tossed 3 times, the number of possible outcomes $= 2 \times 2 \times 2 = 8$

(The possible outcomes are HHH, HHT, HTH, HTT, THH, THT, TTH, TTT)

22. 5 subjects can be selected in ${}^5C_5$ ways; 1 subject can be selected in ${}^5C_1$ ways.

These 6 subjects can be arranged themselves in 6! ways.

Since two subjects are same, we need to divide by 2!

Therefore, total number of arrangements $= {}^5C_5 \times {}^5C_1 \times 6!2! = 1800$

24. He has 10 patterns and 8 patterns

A sofa can be selected in 10 ways

A table can be selected in 8 ways.

Hence one sofa and one table can be selected in 10×8 ways = 80 ways

25. He can go in any of the 25 buses (25 ways).

Since he cannot come back in the same bus, he can return in 24 ways.

Total number of ways $= 25 \times 24 = 600$

26. Total number of ways $= {}^4C_1 \times {}^3C_1 \times {}^2C_1 = 4 \times 3 \times 2 = 24$

29. The first ring can be worn in any of the 3 fingers (3 ways).

Similarly each of the remaining 5 rings also can be worn in 3 ways.

Hence total number of ways = $3 \times 3 \times 3 \times 3 \times 3 \times 3 = 36 = 729$

PERMUTATIONS AND COMBINATIONS

Practice Exercise - 2

Number of Questions: 20 **Suggested time: 20 minutes**

1. In how many ways can a team of 5 persons be formed out of a total of 10 persons such that two particular persons should not be included in any team?

(a) 56
(b) 112
(c) 28
(d) 128

2. How many triangles can be formed by joining the vertices of an regular octagon?

(a) 56
(b) 28
(c) 112
(d) 120

3. If there are 9 horizontal lines and 9 vertical lines in a chess board, how many regular rectangles can be formed in the chess board?

(a) 920
(b) 1024
(c) 64
(d) 1296

4. Find the number of diagonals of a regular decagon?

(a) 16
(b) 28
(c) 35
(d) 12

5. Find the number of triangles that can be formed using 14 points in a plane when 4 points are collinear?

(a) 480
(b) 360

(c) 240
(d) 120

6. What is the sum of all 4 digit numbers formed using the digits 2, 3,4 and 5 without repetition?
(a) 93324
(b) 92314
(c) 93024
(d) 91242

7. In a birthday party, every person shakes hand with every other person. If there was a total of 28 handshakes in the party, how many persons were present in the party?
(a) 9
(b) 8
(c) 7
(d) 6

8. There are 8 points in a plane out of which 3 are collinear. How many straight lines can be formed by joining them?
(a) 16
(b) 26
(c) 22
(d) 18

9. How many quadrilaterals can be formed by joining the vertices of an regular octagon?
(a) 60
(b) 70
(c) 65
(d) 74

10. How many straight lines can be formed by joining 12 points on a plane out of which no points are collinear?
(a) 72
(b) 66
(c) 58
(d) 62

11. If ${}^{n}C_8 = {}^{n}C_{27}$, what is the value of n?

(a) 35

(b) 22

(c) 28

(d) 41

12. In how many ways can 10 students can be arranged in a straight line ?

(a) 9!

(b) 6!

(c) 8!

(d) 10!

13. Find the number of triangles which can be drawn out of 'n' given points on a circle?

(a) ${}^{(n+1)}C_1$

(b) ${}^{n}C_1$

(c) ${}^{(n+1)}C_3$

(d) ${}^{n}C_3$

14. In how many ways can 10 books be arranged on a shelf such that a particular pair of books is always together?

(a) 9! × 2!

(b) 9!

(c) 10! × 2!

(d) 10!

15. In how many ways can 10 books be arranged on a shelf such that a particular pair of books will never be together?

(a) 9! × 8

(b) 9!

(c) 9! × 2!

(d) None

16. Amanat wants to send invitation letter to her 7 friends. In how many ways can she send the invitation letter through 4 servants to carry the invitation letters?

(a) 16384

(b) 10801
(c) 14152
(d) 12308

17. How many three digit numbers divisible by 5 can be formed using any of the digits from 0 to 9 such that none of the digits is repeated?
(a) 108
(b) 112
(c) 124
(d) None

18. In how many ways can 7 beads be arranged to form a necklace?
(a) 720
(b) 360
(c) 120
(d) 60

19. A telegraph has 10 arms and each arm can take 5 distinct positions (including position of the rest). How many signals can be made by the telegraph?
(a) ${}^{10}P_5$
(b) $5^{10} - 1$
(c) 5^{10}
(d) ${}^{10}P_5 - 1$

20. There are two books each of 5 volumes and two books each of two volumes. In how many ways can these books be arranged in a shelf so that the volumes of the same book should remain together?
(a) $4! \times 5! \times 2!$
(b) $4! \times 14!$
(c) $14!$
(d) $4! \times 5! \times 5! \times 2! \times 2!$

Practice Exercise 2 - Answers with Explanations - need based.

1 (a)	2 (a)	3 (c)	4 (c)	5 (b)
6 (a)	7 (b)	8 (b)	9 (b)	10 (b)

11 (a)	12 (d)	13 (d)	14 (a)	15 (d)
16 (a)	17 (d)	18 (b)	19 (b)	20 (d)

2. Number of triangles that can be formed by joining the vertices of a polygon of n sides = $^nC_3$

7. Suppose there are n persons present in a party and every person shakes hand with every other person. Then, total number of handshakes = $^nC_2$ = n(n−1)2

9. Number of quadrilaterals that can be formed by joining the vertices of a polygon of n sides = $^nC_4$

13. Suppose there are n points in a plane out of which no three points are collinear. Number of triangles that can be formed by joining these n points = $^nC_3$

19. The 1st arm can take any of the 5 distinct positions. Similarly, each of the remaining 9 arms can take any of the 5 distinct positions. Hence, total number of signals = 510 But there is one arrangement when all the arms are in rest. In this case there will not be any signal. Hence required number of signals = 510 − 1

20. 1 book: 5 volume; 1 book: 5 volume 1 book: 2 volume 1 book: 2 volume

Given that volumes of the same book should remain together. Hence, just tie the same volume books together and consider as a single book. Hence we can take total number of books as 4. These 4 books can be arranged in $^4P_4$ = 4! Ways.

5 volumes of the 1st book can be arranged among themselves in $^5P_5$ = 5! ways.

5 volumes of the 2nd book can be arranged among themselves in $^5P_5$ = 5! ways.

2 volumes of the 3rd book can be arranged among themselves in $^2P_2$ = 2! Ways

2 volumes of the 4th book can be arranged among themselves in $^2P_2$ = 2! ways.

Hence total number of ways = $4! \times 5! \times 5! \times 2! \times 2!$

Practice Exercise - 3

Number of Questions: 18 **Suggested time: 15 minutes**

1. Find the total number of words that can be made by using all the letters from the word "MEHNATI" using only once.
(a) 7320
(b) 5040
(c) 720
(d) 12340

2. What is the total number of words that can be made by using all the letters of the word, KHERA, using each letter only once?
(a) 60
(b) 180
(c) 120
(d) None

3. How many different 4-digit numbers can be made from the first 4 natural numbers, using each digit only once?
(a) 24
(b) 18
(c) 12
(d) 120

4. How many different 4-digitk numbers can be made from the first 4 whole numbers, using each digit only once?
(a) 24
(b) 18
(c) 12
(d) 120

5. How many different 4-digit numbers can be made from the first 4 natural numbers, if repetition is allowed?
(a) 224
(b) 218
(c) 576

(d) 256

6. How many six-lettered words starting with the letter T can be made from all the letters of the word TAREVL?
(a) 720
(b) 360
(c) 120
(d) None

7. The number of handshakes that can be exchanged among a party of 12 students, if every student shakes hand once with every student is?
(a) 66
(b) 132
(c) 144
(d) 91

8. How many different 4-lettered words can be made from the word MEHA so that H and M are always together?
(a) 24
(b) 32
(c) 12
(d) 84

9. 66 games were played in a tournament, where each player played one against the rest. The number of players are
(a) 33
(b) 12
(c) 13
(d) 11

10. The number of ways, in which can a student choose 5 courses out of 9 courses, if 2 courses are compulsory is:
(a) 35
(b) 25
(c) 45
(d) 9^5

11. The number of 2-digit even numbers formed from the digits 1, 2, 3, 4, 5 and 6, if repetition of digits is not allowed.
(a) 6

(b) 5
(c) 15
(d) 30

12. How many different words can be formed by taking any three letters from the word "CHEMISTRY"
(a) 504
(b) 720
(c) 492
(d) None

13. In how many ways the letters of the word "NUCKLOW" be arranged in such a way that all the vowels used in it are at continuous position?
(a) 720
(b) 1440
(c) 5040
(d) None

14. What are the different numbers which can be made by taking any 4 digit out of the number 9427586 without repetition?
(a) 1680
(b) 720
(c) 240
(d) 840

15. 7 persons P, Q,R,S,T,U,V are to speak in a meeting. In how many ways can they do this without P speaking before Q?
(a) 5040
(b) 7520
(c) 120
(d) 360

16. How many numbers between 4000 and 8000 can be formed by using the digits 1, 3, 6, 7, 4, 5; if each digit can be repeated?
(a) 648
(b) 720
(c) 324
(d) None

17. How many words can be made from the word PRINT so that all the four consonants are always together?

(a) 24

(b) 12

(c) 48

(d) None

18. In how many ways a dancing pair can be made in a dance party, in which there are 10 boys and 8 girls?

(a) 80

(b) 64

(c) 100

(d) 81

Practice Exercise 1 - Answers

1 (b)	2 (c)	3 (a)	4 (b)	5 (d)
6 (c)	7 (a)	8 (c)	9 (b)	10 (a)
11 (c)	12 (a)	13 (b)	14 (d)	15 (b)
16 (a)	17 (c)	18 (a)		

SEQUENCES & FUNCTIONS

29

FUNCTIONS

In the study of natural phenomena and the solution of technical and mathematical problems, we find it
necessary to consider the variation of one quantity as dependent on the movement of another.

For example expenditure is directly or indirectly related with the income or the price of goods prevalent in the market. Also we can say a distance covered by a moving particle is dependent upon the speed of the vehicle. Hence we can say the expenditure or the path traversed is a function of the price/demand or the time. In other words word if we say that the area of a circle 'A' depends upon the size of the radius 'r' here we mean that the area is a function of 'r'

Definition: If to each value of a variable 'x' (within a certain range) there corresponds one definite value of another variable 'y', then 'y' is called the *function of* 'x' or **y = f(x)**

According to modern concept, a function is a relation between two sets A and B in which each element of set A is matched to one and only one element of set B (such a relation is called single-valued and everywhere defined). In other words it reads 'f' is a function of A into B.

In a function each element of A must be matched to one and only one element of B.

Types of functions

- **Power function**
- **General exponential function**
- **Logarithmic function**
- **Trigonometric functions**
- **The rational integral function**
- **Inverse function**
- **Constant function**

- **Composite function**

Inverse Function

$f^{-1}(x)$ is known as <u>Inverse function.</u>

e.g. If $f(x) = x^2$ then $y(say) = x^2$, $x^2 = y$, $x = \sqrt{y}$

$\therefore f^{-1}(x) =$

$f^{-1}(16). = \pm 4,$

$f^{-1}(9). = \pm 3,$

$f^{-1}(1). = \pm 1,$

$f^{-1}(25). = \pm 5,$

Constant Function

If $f(x) = 12$, then $f(a) = 12$, $f(0) = 12$, $f(5) = 12$

All these are examples of Constant Function where the value of $f(x) = y$ does not change with a change in the value of 'x'.

Composite Function

(g.f.) is called a composite of 'f' and 'g'. 'g .f ' is real $\sqrt{x}$ 'g circle f' OR 'g operation f' OR 'g composite f' are other terminologies for(g . f.). [Note that function f is applied first, and then g] $(g.f)(a) = g[f(a)]$

e.g.	$f(x)$	=	$2x,$		$g(x)$	=	x^2
then	$f(5)$	=	$10,$		$g(5)$	=	25
					$g(a)$	=	a^2 etc.,
$(g . f)$	(x)	=	$g[f(x)]$	=	$g(2x)$	=	$4x^2.$

Domain and Range of a function: The set of values of 'x' for which the values of the function 'y' are determined by the rule f(x) is called the domain of function. The set of values of 'y' determined at various possible values of 'x' is called the range of the function.

So Domain ***is the set of values of x***
Range ***is the set of values of y***

A function is summarily a mapping or transformation of 'x' into 'y' or f(x). The variable 'x' presents elements of the **<u>domain</u>** and is called the independent variable. The variable ‘y’ representing the elements of the **range** is called the dependent variable.

Value of a Function at a Point

f(a) is the value of the function f(x) when x takes the value ‘a’ i.e., when ‘x’ is replaced by ‘a’.

Example 1 $y = f(x) = \sqrt{x+1}$ then

$$f(0) = \sqrt{1} = 1$$

$$f(a) = \sqrt{a+1}$$

$$f(a-1) = \sqrt{a-1+1} = \sqrt{a}$$

$$f(-1) = \sqrt{-1+1} = 0$$

$$f(a^2+2a) = \sqrt{a^2+2a+1} = \pm(a+1)$$

Example 2 $f(x) = |x-3|$ then

$$f(0) = 3$$

$$f(3) = 0$$

$$f(-3) = 6$$

$$f(a+3) = |a+3-3| = |a|$$

$$f(3-x) = |3-x-3| = |x|$$

$$f\left(\frac{1}{x}\right) = \left|\frac{1}{x}-3\right|.$$

Example 3 If $f(x) = x^2 + x + 7$, then

$$f(a+b) = (a+b)^2 + (a+b) + 7.$$

$$= a^2 + 2ab + b^2 + a + b + 7.$$

Example 4 If $g(x) = x^{\frac{1}{2}}$ then $g(0) = 0$; $g(2) = \sqrt{2}$; $g(4) = \pm 2$.

SOLVED EXAMPLES

1. The function $f(x) = 3x^2 - 2x + 5$; then find
What is the value of:- $f(3)$, $f(-2)$, $f(½)$, Ans: [26, 21, 19/4]

2. $f(x) = 3x + 14$ where $x \in N$
Find $f(2)$, $f(4)$, $f(0)$, $f(3)$ Ans: [20, 26, 14, 23]

Practice Exercise - 1

1. Find f(4), if $f(x) = x^4 - 3x^3 + 6x^2 - 10x + 16$.

2. Given that $f(x) = 2^{z-2}$, find f(-1.5)

3. If $f(t) = 2t^2 + \frac{2}{t^2} + \frac{5}{6} + 5t$, find $f\left(\frac{1}{t}\right)$.

4. To make a box from sheet metal having the shape of a square 30 X 30 cm. It is necessary to cut small squares from its corners. How long must the side x of the cut-out squares be to bend a box of maximum capacity.

5. Find the nth term of the sequence when n tends to infinity 0.5, 0.55, 0.555, 0.5555, ..

6. Find the value of Tn of the sequence when 'n'$\Rightarrow$ to ∞. $\sqrt{5}, \sqrt{5+\sqrt{5}}, \sqrt{5+\sqrt{5+\sqrt{5}}}, ...$

7. Find in each cases the least positive integer n such that

(a) $n^2 + 2n > 999$

(b) $\frac{\sqrt{192} \times \sqrt{729} \times \sqrt{45}}{\sqrt{125} \times \sqrt{567}} =$

8. If $f(x, y) = 3x^2 - 2xy - y^2 + 4$, find f(1, -1).

9. If $f(x) = x^2 - x - 2$, find f(ax + b).

10. Find for each of the following, a possible nth term of the sequences whose first 5 terms are indicated and find the 6th term.

(a) $-\frac{1}{5}, \frac{3}{8}, -\frac{5}{11}, \frac{7}{14}, -\frac{9}{17}, ...$

(b) 1, 0, 1, 0, 1, ……

(c) $\{(6\frac{4}{5})^2 - (3\frac{1}{5})^2\}$

11. If f(x) = 2x and g(x) = x + 1, find a formula for (f . g) (x) and (g . f) (x).

12. If f: $X \rightarrow Y$ such that Y = x + 1 and g: $Y \rightarrow Z$ such that $z = y^2 + 3$.

what is (g . f) (x) when x = -2, 0, 1, 3.

Practice Exercise 1 - Answers

1. 136

2. 0.088

3. f(t)

4. 4. 5 cm

5. $\frac{5}{6}$

6. 2.79

7. (a) 31 (b) 10

8. 8

9. ax (1 + ax) – b(1 – b) – 2

10. (a) $\frac{(-1)^n(2n-1)}{(3n+2)}, \frac{11}{20}$

(b) $7500 \times \frac{80}{100}$

(c) $\left(\frac{n+3}{n+5}\right), \left(\frac{1-(-1)^n}{2}\right), 0$

11. 2x + 2, 2x + 1.

12. 4, 4, 7, 19.

AVERAGES

The following are the important types of averages generally used in business.

- Arithmetic mean (AM)
- Median (M)
- Mode (Z)
- Geometric mean (GM)
- Harmonic mean (HM)

ARITHMETIC MEAN

The most used measure for representing the entire data by one value is called an 'average' and what the statisticians call the arithmetic mean.

Before we actually talk of data or its applications, we must know what a data is. Data is any information rather statistical about any study or applications. It may be presented in Ungrouped or Raw or Primary Data which is not tilted or defected from its original shape. It directly belongs to the study group and is an individual's study. A grouped information or Final or Secondary Data is a data presentation system in our liking or as per our requirement. It is also called Indirect Data where by the direct source is not available.

Ungrouped or Raw or Primary Data

Marks of 15 students 45, 48, 59 , 95, 86, 79, 87, 98, 57, 48, 39, 85, 74, 82, 73

This is an example of Primary or Raw data where we can study each individual

Grouped or Final or Secondary Data

Marks obtained by 15 students in a class are given as under:

Marks Range	(f)	(cf)
30 – 40	1	1
40 – 50	3	4
50 – 60	2	6
60 – 70	0	6
70 – 80	3	9
80 – 90	4	13
90 – 100	2	15

Here it is not possible to study which individual student scored how many marks. The information, in its raw shape, is available only with the Enumerator and he has presented the information for us in the required manner (in a grouped shape) only.

ARITHMETIC MEAN (AM) – UNGROUPED DATA

The process of computing mean in case of ungrouped data (i.e., where frequencies are not given) is very simple. Add together the various values of the variable and divide the total by the number of observations. The A.M. or $\bar{X}$, read as X bar. It is defined as:

$$\bar{X} = \frac{X_1 + X_2 + \dots\dots\dots X_n}{N}$$

Where X1 + X2 + …. XN *are different observations of the data N is the number of observations.*

Example 1 The following table gives the monthly income of 12 employees of a firm:

On rolls	1	2	3	4	5	6	7	8	9	10	11	12
Monthly Income	280	180	96	98	104	75	80	94	100	75	600	200

Calculate the average monthly income

$$\bar{X} = \frac{\Sigma X}{N}$$

Σ (X) = 1,982, N = 12

$\therefore \frac{240}{2600} \times 100 = 9\frac{3}{13}\%$; Thus the average monthly income is Rs. 165.17.

ARITHMETIC MEAN (AM) – GROUPED DATA

For grouped data, arithmetic mean may be computed by applying any of the following methods:

(i) Direct method

(ii) Indirect method

(i) **Direct Method.** When direct method is used

$$\bar{X} = \frac{\Sigma fm}{N}$$

where 'm' = mid-point of the various classes i.e., 1/2 ($L_0 + L_1$) mean of Lower & Upper limits

f = the frequency of each class

N = the total frequency

Example 2 Here given is the salary of 140 employees: Given that

Income (Rs)	200-400	400-600	600-800	800-1000	1000-1200	1200-1400	1400-1600
Employees	50	30	28	12	10	8	2

Calculate the average salary drawn.

Solution

CALCULATION OF AVERAGE SALARY

Salary Drawn (Rs.)	# of Employees	m	fm
200 – 400	50	300	15000
400 – 600	30	500	15000
600 – 800	28	700	19600
800 – 1000	12	900	10800
1000 – 1200	10	1100	11000
1200 – 1400	8	1300	10400
1400 – 1600	2	1500	3000
	N = 140		$\Sigma fm = 84800$

$\bar{X} = \frac{\Sigma fm}{N} = x = \frac{36 \times 25}{15} = 60$. Thus the average salary is Rs.605.70

Example 3 (a) The average income of a group of 50 persons was calculated to be Rs.39.20. Later it was discovered that a salary of Rs.96 was misread as Rs.69. Calculate the correct average income.

Solution

$\Sigma X = \mathbf{39.2} \times 50 = 1{,}960$

Less incorrect item – 69

Add correct item + 96

$\therefore$ Correct $\Sigma X = 1{,}987$

Hence correct average = Rs. 39.74 p

MEDIAN (M) – UNGROUPED DATA

Remember the following steps:

1. Arrange the data in ascending or descending order of magnitude.

(Both arrangements would give the same answer.)

2. Find the Mid – term

- *if # of terms N is odd the mid – term is* $\frac{1}{2}:\frac{2}{3}:\frac{3}{4}$ **item.**
- *If # of terms N is Even there will be two mid terms Viz. (N/2) and (N + 1)/2*

  M = Size of $\frac{1}{2}:\frac{2}{3}:\frac{3}{4}$ **item.**

Example 4 From the following data of wages of 7 workers compute the median wages:

Wages (in Rs.)	100	150	80	90	160	200	140.

Solution

CALCULATION OF MEDIAN

Sr No.	Wages (arranged)
1	80
2	90
3	100
4	140
5	150
6	160
7	200
8 *	210 *

Case I - Being considered when N is = 7 (odd)

Median = Size of $\left(\frac{N+1}{2}\right)th$ item

$\therefore$ Median = Size of $\frac{7+1}{2}$ = 4^{th} item

Size of 4^{th} item = 140. Hence median wage is Rs.140.

In the above illustration the number of items was odd, and therefore, it was possible to determine the value of 4^{th} item.

Case II - Being considered when N is = 8 (even)

But in case the number of items is even say - *8 the median would be the value of $\frac{2/3}{1/9}$, i.e., 4.5^{th} item. For finding out the value of 4 $.5^{th}$ item we shall take the average of 4^{th} and 5^{th} items i.e., mean of sizes of these two terms which is 140 + 150 = 290/2 = 145 Ans.

MEDIAN (M) – GROUPED DATA

Never re-arrange the frequency in ascending or descending orders.

(i) Determine the particular class in which the value of median lies.
(ii) Find the class in which the mid-term lies
(iii) Apply the following formula

$$\text{Median} = L + \frac{\frac{N}{2} - c.f.}{f} \times i$$

Where

L = Lower limit of the median class, i.e., class in which the middle term lies.

c.f. = Cumulative frequency of the class ***preceding*** *the median class.*

f = Frequency of the median class.

i = The class interval of the median class.

Example 5 147 workers are working in an industrial unit with the following age groups:

Age Group	18-22	22-26	26-30	30-34	34-38
Number of workers	12	12	28	26	15
Age Group	38-42	42-46	46-50	50-54	54-58
Number of workers	18	16	8	7	5

Calculate the median age.

Solution

CALCULATION OF MEDIAN AGE

Age group	(f)	(cf)
18-22	12	12
22-26	12	24
26-30	28	52
30-34	26	78
34-38	15	93
38-42	18	111
42-46	16	127
46-50	8	135
50-54	7	142
54-58	5	147

Median = size of $\frac{N}{2}$ th item

= size of $\frac{45}{100}$ = 73.5th term say 74th term

Hence median lies in the class 30 – 34

Med = L + $\frac{\frac{N}{2} - c.f.}{f} \times i$

L = 30, $\frac{N}{2}$ = 73.5, c f. 52, f = 26, i = 4

Median = 30 + $\frac{73.5 - 52}{26} \times 4$ = 30 + 3.4 = 33.4

Hence the median age is 33.4 years.

MODE (Z)

Mode is a data or frequency with maximum occurrences.
These methods have been discussed for grouped and ungrouped data.

Calculation of Mode – Ungrouped Data

For determining mode count the number of times and the value which occurs the maximum number of times is the modal value. The more often the modal value appears relatively, the more valuable the measure is an average to represent data.

Example 6 In a survey of people having preference for sizes of screen of T.V. sets a group of 30 persons were selected at random. Find the modal size of the T.V. screen.

16	19	16	23	25	23
19	16	19	23	23	19
23	19	16	19	23	25
25	23	23	19	23	23
23	19	23	23	16	23

Solution

CALCULATION OF MODAL SIZE (Z)

Size	Tally Bars	Frequency
16	~~1111~~	5
19	~~1111~~ 111	8
23	~~1111~~ ~~1111~~ ~~1111~~	15
25	11	2
	Total:	30

CALCULATION OF MODE (Z) – GROUPED DATA

In the case of grouped data the following formula may be used for calculating mode.

$$Z = L + \frac{f_1 - f_0}{2f_2 - f_0 - f_2} \times i$$

L = Lower limit of the modal class

f_1 = frequency of the modal class

f_0 = frequency of the class **preceding** the modal class

f_2 = frequency of the class **succeeding** the modal class.

Take the case of Example 5 above

$f_1 = 28$; $f_0 = 12$; $f_2 = 26$ L = 26 i = 4

By applying the above formula we get Z = 29. 6 **Answer**

Example 7 In a moderately asymmetrical distribution the Mode and Mean are 32: 1 and 35.4 respectively. Calculate the Median.

Solution

Mode = 3 Median – 2 Mean

Mode = 32.1 and Mean = 35.4

Substituting the values

32.1 = 3 Median – 2(35.4) = 3 Median –70.8

–3 Median = –32.1 – 70.8 = –102.9 Therefore, Median = 34.3

GEOMETRIC MEAN (G.M.)

Geometric mean is defined as the nth root of the product of N items of a series. If there are two items, we take the square root; if there are three items the cube root, and so on. Symbolically,

$$\text{G.M.} = \sqrt[n]{((X_1)\times(X_2)\times(X_3)\times.......X_n)}$$

Where X_1, X_2, X_3, etc., refer to the various items of the series.

RELATIONSHIP AMONG THE AVERAGES

If all the numbers **X***1*, **X***2* **,…… X** *n* 'n' are **not identical** , the values of A.M., G.M. and H.M. would also differ and will be: ***AM > GM > HM***

if all the numbers X*1*, X*2*,…… X*n* are identical then $A.M. \geq G.M. \geq H.M.$

RESULTS ON DIAGRAMS

BAR CHARTS: Presentation method of Ungrouped Data

HISTOGRAM: It is graphical representation of a frequency distribution in the form of rectangle with class intervals as bases and the corresponding frequencies as heights, there being no gap between any two successive rectangles.

FREQUENCY POLYGON: When a grouped frequency distribution with equal class intervals is given, we represent the class-marks on x-axis and the frequencies on y-axis.

OGIVE (CUMULATIVE FREQUENCY CURVE): We mark the upper class limits along x-axis and cumulative frequencies along y-axis. Plot these points and join them to get a curve, called an ogive.

Example 8 Find the median of the frequency distribution:

Marks Obtained	Number of Students	Cumulative Frequency
15 – 20	6	6
20 – 25	7	13
25 – 30	9	22
30 – 35	8	30
35 – 40	10	40

Solution

Here n = 40, So, $\frac{n}{2} = 20$

Now, 20 lies in the class 25 – 30: So, Median Class is 25 – 30.

$\therefore$ L1 = 25, L2 = 30, n = 40, C = 13 & f = 9.

$\therefore$ Median = $L_1 + \frac{(L_2 - L_1)}{f} \times \left(\frac{n}{2} - C\right) = 25 + \frac{(30-25)}{9} \times (20-13)$

$= \left(25 + \frac{35}{9}\right) = 28.88$

Practice Exercise - 1

1. The average value of the median of 2, 8, 3, 7, 4, 6, 7 and the mode of 2, 9, 3, 4, 9, 6, 9 is:
(a) 9
(b) 8
(c) 7.5
(d) 6

2. Find the mean of 3,6,9,12,15,18,21
(a) 15
(b) 12
(c) 18
(d) 24

3. A candidate obtains the marks as given below: English: 60, Hindi: 75, Mathematics: 63, Physics: 59 and Chemistry: 55. If the weights of 1, 2, 1, 3, 3 respectively are allotted to these subjects, the candidate's weighted mean is:
(a) 62.4
(b) 61.5
(c) 61
(d) 60.5

4. The wages of 5 workers in a factory on 3rd July 2016 were Rs.26, Rs.40, Rs.42, Rs.45 and Rs.30. Find the mean wages
(a) 36.40
(b) 33.45
(c) 33.90
(d) 35.72

5. The average weight of a group of 20 boys was calculated to be 89.4 kg and it was later discovered that one weight was misread as 78 kg instead of the correct one of 87 kg, then the correct average weight is:
(a) 88.95 kg
(b) 89.25 kg
(c) 89.55 kg

(d) 89.85 kg

6. Find the mean of first eight prime numbers

(a) 8.265

(b) 11,625

(c) 8.625

(d) 7.425

7. The median of the following incomplete frequency distribution is 4.

x	1	2	3	4	5	6	7	8
Frequency	2	3	4	5	6	7	8	–

The frequency of 8 is: ?

(a) 1

(b) 2

(c) 3

(d) 4

8. Calculate arithmetic mean of the scores of 3.8, 4.2, 3.3, 3.7, 4.0, 3.7, 4.6, 3.9, 4.4, 4.4

(a) 3

(b) 5

(c) 6

(d) 4

9. A pie chart is drawn to show the areas in millions of square kms of several continents. The area 11.7 sq. km. of Africa is shown by several subtending an angle of 820. If the subtended angle corresponding to North America be 660, its area is:

(a) 10.8 km2

(b) 9.4 km2

(c) 8.8 km2

(d) 5.6 km2

10. The mean heights of 10 students was 153 cm. But later it was observed that 151 was read as 141. Corrected mean is ?

(a) 145

(b) 149

(c) 154

(d) 162

11. The following is the data of wages per day 5, 4, 7, 5, 8, 8, 8, 5, 7, 9, 5, 7, 9, 10, 8. Find the mode of the data:

(a) 7
(b) 5
(c) 8
(d) 10

12. The mean of 5 numbers is 27. If one number is excluded, their mean is 25. Find the number excluded.

(a) 36
(b) 30
(c) 35
(d) 40

13. The mode of the given distribution is:

Weight (in kg)	40	43	46	49	52	55
Number of Children	5	8	16	9	7	3

(a) 40
(b) 46
(c) 55
(d) None of these.

14. Find median of 8 , 7, 3 , 5, 10, 12, 1

(a) 8
(b) 7
(c) 5
(d) 4

15. If the geometric mean of x, 16, 50, be 20, then the value of x is:

(a) 40
(b) 20
(c) 10
(d) 4

16. Find Mode of

Class	(f)
6.5 – 7.5	5
7.5 – 8.5	12
8.5 – 9.5	25
9.5 – 10.5	48
10.5 – 11.5	32
11.5 – 12.5	6
12.5 – 13.5	1

(a) 8
(b) 10
(c) 12
(d) 15

17. If the arithmetic mean of two numbers is 10 and their geometric mean is 8, the numbers are:
(a) 20, 5
(b) 16, 4
(c) 15, 5
(d) 12, 8

18. Given that AM of 6, 8, 5, 7, x and 4 is 7. Find the value of x
(a) 10
(b) 15
(c) 12
(d) 14

19. The median of 0, 2, 2, 2, –3, 5, –1, 5, 5, –3, 6, 6, 5, 6 is:
(a) 0
(b) – 1.5
(c) 2
(d) 3.5

20. Find the mean of 894, 896 , 898 , 900 , 902
(a) 898
(b) 897
(c) 895
(d) 899

21. The median of the following distribution is:

Class-interval	35 – 45	45 – 55	55 – 65	65 – 70
Frequency	8	12	20	10

(a) 56.5
(b) 57.5
(c) 58.7
(d) 59

22. Find the mean if

x	19	21	23	25	27	29	31
f	13	15	16	18	16	15	13

(a) 25
(b) 27
(c) 23.88
(d) 26.19

23. Consider the following table:

Diameter of heart (in mm)	120	121	122	123	124	125
Number of person	5	9	14	8	5	9

The median of the above frequency distribution is:
(a) 122 mm
(b) 123 mm
(c) 122.5 mm
(d) 122.75 mm

24. Find the mean of first 100 odd numbers
(a) 95
(b) 98
(c) 99
(d) none

25. The mode of the following frequency distribution is:

Class interval	3 – 6	6 – 9	9 – 12	12 – 15	15 – 18	18 – 21	21 – 24
Frequency	2	5	21	23	10	12	3

(a) 11.5
(b) 12.4
(c) 12
(d) 11.8

Practice Exercise - 1 Answers with Explanations

1. A frequency polygon is constructed by plotting the frequency of the class and the mid value of the class.

2. b

3. Clearly, (b) describes best.

4. a

5. Total actual weight = (20 × 89.4 – 78 + 87) kg = 1797 kg.

6. c

7. Let the frequency of 8 be f.

x	Frequency (f)	Cumulative frequency
1	2	2
2	3	5
3	4	9
4	1	10
5	2	12
6	4	16
7	2	18
8	f	$18 + f$

8. d

9. 82° represents 11.7 km^2.

10. c

11. Since 8 occurs most often, mode = 8.

12. c

13. Clearly, 46 occurs most often. So, mode = 46.

14. b

15. $(x \times 16 \times 50)1/3 = 20 \Rightarrow x \times 16 \times 50 = (20)3 \Rightarrow x = \left(\frac{20\times 20\times 20}{16\times 50}\right) = 10.$

16. b

17. Let the numbers be a and b. Then,

$$\left[\frac{a+b}{2}=10 \Rightarrow (a+b)=20\right] \text{ and } \left\lfloor \sqrt{ab}=10 \Rightarrow ab=64 \right\rfloor$$

$$\therefore a-b=\sqrt{(a+b)^2-4ab}=\sqrt{400-256}=\sqrt{144}=12.$$

18. c

19. Observations in ascending order are: –3, –3, –1, 0, 2, 2, 2, 5, 5, 5, 5, 6, 6, 6.

Number of observations is 14, which is even.

$\therefore$Median = $\frac{1}{2}$ [7^{th} term + 8^{th} term] = $\frac{1}{2}$ (2 + 5) = 3.5.

20. a

21.

Class-interval	Frequency	Cumulative Frequency
35 – 45	8	8
45 – 55	12	20
55 – 65	20	40
65 – 70	10	50

Here n = 50. So, $\frac{1}{2}$ = 25 ; 25 lies in the class interval 55 – 65.

$\therefore L_1 = 55$, $L_2 = 65$, n = 50, C = 20 & f = 20.

$$\therefore \text{ Median} = L_1+\frac{(L_1-L_2)}{f}\times\left(\frac{n}{2}-c\right)=55+\frac{(65-55)}{20}\times(25\times20)=57.5$$

22. a

23. The Given Table may be represented as:

Diameter of heart (in mm)	Number of persons	Cumulative frequency
120	5	5
121	9	14

122	14	28
123	8	36
124	5	41
125	9	50

Here n = 50. So, $\frac{1}{2}$ = 25 & $\frac{1}{2}$ + 1 = 26.

Median = $\frac{1}{2}$ (25^{th} term + 26^{th} term) = $\frac{122+122}{2}$ = 122.

[∴ *Both lie in that column whose c.f. is* 28]

24. d

25. Maximum frequency is 23. So, model class is 12 – 15.

∴ $L_1 = 12$, $L_2 = 15$, f = 23, $f_1 = 21$ and $f_2 = 10$.

$$\therefore Mode = L_1 + \frac{f_1 - f_2}{2f - f_1 - f_2}(L_2 - L_1) = 12 + \frac{(15-12)}{(46-21-10)}(15-12) = 12.4$$

Practice Exercise - 2

(For convenience of students, all answers are given in brackets alongside the questions)

1. Out of a total population of a town in South Africa 65% belonged to black race. Their mean incomes were 1500 pounds and that of remaining 4000 pounds. Find the average income of the town.

(Answer: 2375 pounds)

2. The mean monthly salary paid to 77 employees in a company was Rs.78. The mean salary of 32 of them was Rs.75 and that of other 25 was Rs. 82. What was the mean salary of the remaining?

(Answer: 77.8)

3. Find the value of X given that arithmetic average of the following distribution is 28.

Profit per shop	0-10	10-20	20-30	30-40	40-50	50-60
No. of shops	12	18	27	x	17	6

(Answer: 20)

4. The numbers 3.2 , 5.8, 7.9 and 4.5 have frequencies x , x + 2 , x–3, x + 6 respectively. If the AM is 4.876, find the value of x

(Answer: 5)

5. The following table gives the weekly wages in rupees in a certain commercial organization:

Weekly wages (Rs)	30-32	32-34	34-36	36-38	38-40
Frequency	2	9	25	30	49
Weekly wages (Rs)	40-42	42-44	44-46	46-48	48-50
Frequency	62	39	20	11	3

Calculate from the above data:

(i) the median and the third quartile wages.

(ii) the number of wage-earners receiving between Rs. 37 and Rs. 45 per week.

(Answer: (i) Median 40.32, Q3 = 422.54, (ii) 175)

6. A train runs 25 miles @ 30 mph. another 50 miles @ 40 mph. then due to the repair of tracks for 6 minutes @ 10 mph and finally covers the remaining distance of 24 miles @ 24 mph. What is the average speed in mph?

(Answer: 31.41 mph)

7. A man gets three annual rises in salary. At the end of the first year he gets an increase of 40% at the end of second an increase of 6% on his salary and it was at the end of the first year and at the end of third year an increase of 9% on his salary as it was at the end of second year. What is the average percentage increase?

(Answer: 6.3%)

8. In a batch of 15 students, five failed. The marks of 10 students who passed were 9,6,7,8,8,9,6,5,4,7. what was the median of all 15 students

***(Answer:* 6)**

(hint -Arrange the marks of pass students in ascending order evidently who failed were lower than the lowest marks thus find the mid - term and subsequently median marks)

9. A motor car covered a distance of 50 miles four times. The first time at 50 m.p.h., the second at 20 m.p.h., the third at 40 m.p.h., and the fourth at 25 m.p.h. Calculate the average speed and explain the choice of the average.

(Answer: 29.63)

10. Compute mode of the following data:-

Marks obtained	>10	>20	>30	>40	>50	>60
No. of students	100	97	92	85	75	63
Marks obtained	>70	>80	>90	>100	>110	>120
No. of students	48	36	30	28	20	10

(Hint: make classes of 10-20, 20-30, …… 120-130 and solve Answer: 65)

11. Mr. A spends Rs.100 for apples costing Rs.5 per kilogram and another Rs.100 for apples costing Rs.4 per kilogram. What is the average price of apples per kilogram?

(Answer: 4.44)

12. For the following frequency table calculate mean, median and mode:

Monthly Rent (Rs.)	No. of firms
200-400	6
400-600	9
600-800	11
800-1000	20
1000-1200	15
1200-1400	10
1400-1600	8

(Answer: Median 1100, Mode 1109.)

13. Three men take 12, 8, 6 hours respectively to husk an acre of corn. Determine the average number of hours to husk an acre.

(Answer: 8.6)

14. Find

(a) Median	when	A.M. = 20;	Z = 18	*(Answer:19.33)*
(b) Median	when	A.M. = 32.1 ;	Z = 35.4	*(Answer: 34.3)*
(c) Mode	when	A.M. = 15.6 ;	M= 15.76	*(Answer: 15.99)*
(d) A.M.	when	Z = 22;	M= 21.4	*(Answer: 21.1)*

15. From the following table showing the wage distribution in a certain factory determine:

(a) The mean wages,

(b) The median wage

(c) The modal wage,

(d) Wage limits for the mid- half of the wage earners,

(e) The percentage of the workers who earned between 75 and 125

(f) The percentage who earned more than Rs.150 per week, and

(g) The percentage who earned less than Rs.100 per week.

Weekly wages (Rs.)	No. of workers	Weekly wages (Rs.)	No. of workers
20-40	8	120-140	35
40-60	12	140-160	18
60-80	20	160-180	7
80-100	30	180-200	5
100-120	40		

(Answer: Mean = 108.5, Median = 108.75,

(c) Mode = 113.3, (d) = 81.25 – 129.3, (e) = 48, (f) 12, (g) 40.)

16. The following distribution represents the number of minutes spent by a group of teenagers in going to movies. What is the median?

Minutes per week	**Number of teens**
0-99	27
100-199	32
200-299	65
300-399	78
400-499	58
500-599	32
600---	9

(Answer: 333.5)

17. A machine is assumed to depreciate 40% in value in the first year. 25% in the second year and 10% per annum in the next three years, each percentage being calculated on the diminishing value. What is the average percentage depreciation for the five years?

(Answer: 20%)

PROBABILTY

31

If an event can happen in ‘a’ ways and fall in ‘b’ ways, and each of these ways is equally likely, then probability or the chance, of its happening is $\frac{3126}{25}$, and that of its failing is $\frac{3126}{25}$. E.g., If in a lottery there are 7 prizes and 25 blanks, the chance that a person holding 1 ticket will win a prize is $\frac{3}{60}$, and his chance of winning is $\frac{3}{60}$.

If p is the probability of happening of an event, the probability of its not happening is 1 - p.

Instead of saying that the chance of the happening of an event is $\frac{3126}{25}$, it is sometimes stated that the odds are ‘a’ to ‘b’ in favour of the event, or ‘b’ to ‘a’ against the event.

If c is the total no. of cases, each being equally likely to occur, and of these ‘a’ are favourable to the event, then the probability that the event will happen is $\frac{1}{2}$, and the probability that it will not happen is $1 - \frac{1}{2}$.

Example What is the chance of throwing a number greater than 4 with an ordinary dice whose faces are numbered from 1 to 6.

Solution:

There are 6 possible ways in which the dice can fall, and of these two are favourable to the event required.

$\therefore$ required chances = $\frac{5}{15} = \frac{1}{3}$

Example From a bag containing 4 white and 5 black balls a man draws 3 at random, what are the odds against these being all black?

Solution:

The total no. of ways in which 3 balls can be drawn is $^9C_3$ and the no. of ways of drawing 3 black balls is $^5C_3$;
therefore the chance of drawing 3 black balls is

$$\frac{^5C_3}{^9C_3} = \frac{5.4.3}{9.8.7} = \frac{5}{42}$$

Thus the odds against the event are 37 to 5.

Example Find the chance of throwing at least one ace in a simple throw with two dice.

Solution:

The possible no. of cases is 6×6 or 36.

An ace on one die may be associated with any of the 6 numbers on the other die, and the remaining 5 numbers on the first die may be associated with the ace on the second die, thus the number of favourable cases is 11.

$\therefore$ required chance $= \frac{3}{60}$.

Practice Exercise - 1

Number of Questions: 20 **Suggested time: 20 minutes**

1. A bag contains 2 yellow, 3 green and 2 blue balls. Two balls are drawn at random. What is the probability that none of the balls drawn is blue?
(a) 1/2
(b) 10/21
(c) 9/11
(d) 7/11

2. A dice is rolled twice. What is the probability of getting a sum equal to 9?
(a) 2/3
(b) 2/9
(c) 1/3
(d) 1/9

3. Three coins are tossed. What is the probability of getting at most two tails?
(a) 7/8
(b) 1/8
(c) 1/2
(d) 1/7

4. When tossing two coins once, what is the probability of heads on both the coins?
(a) 1/4
(b) 1/2
(c) 3/4
(d) None of these

5. What is the probability of getting a number less than 4 when a die is rolled?
(a) 1/2
(b) 1/6
(c) 1/3

(d) 1/4

6. A bag contains 4 black, 5 yellow and 6 green balls. Three balls are drawn at random from the bag. What is the probability that all of them are yellow?

(a) 2/91
(b) 1/81
(c) 1/8
(d) 2/81

7. One card is randomly drawn from a pack of 52 cards. What is the probability that the card drawn is a face card(Jack, Queen or King)

(a) 1/13
(b) 2/13
(c) 3/13
(d) 4/13

8. A dice is thrown. What is the probability that the number shown in the dice is divisible by 3?

(a) 1/6
(b) 1/3
(c) 1/4
(d) 1/2

9. John draws a card from a pack of cards. What is the probability that the card drawn is a card of black suit?

(a) 1/2
(b) 1/4
(c) 1/3
(d) 1/13

10. There are 15 boys and 10 girls in a class. If three students are selected at random, what is the probability that 1 girl and 2 boys are selected?

(a) 1/40
(b) 1/2
(c) 21/46
(d) 7/42

11. What is the probability of selecting a prime number from 1,2,3,... 10?

(a) 2/5

(b) 1/5
(c) 3/5
(d) 1/7

12. 3 balls are drawn randomly from a bag contains 3 black, 5 red and 4 blue balls. What is the probability that the balls drawn contain balls of different colors?
(a) 3/11
(b) 1/3
(c) 1/2
(d) 2/11

13. 5 coins are tossed together. What is the probability of getting exactly 2 heads?
(a) 1/2
(b) 5/16
(c) 4/11
(d) 7/16

14. What is the probability of drawing a "Queen" from a deck of 52 cards?
(a) 1/2
(b) 1/13
(c) 1/6
(d) 1/3

15. A card is randomly drawn from a deck of 52 cards. What is the probability getting an Ace or King or Queen?
(a) 3/13
(b) 2/13
(c) 1/13
(d) 1/2

16. A card is randomly drawn from a deck of 52 cards. What is the probability getting a five of Spade or Club?
(a) 1/52
(b) 1/13
(c) 1/26
(d) 1/12

17. When two dice are rolled, what is the probability that the sum is either 7 or 11?

(a) 1/4

(b) 2/5

(c) 1/9

(d) 2/9

18. A card is randomly drawn from a deck of 52 cards. What is the probability getting either a King or a Diamond?

(a) 4/13

(b) 2/13

(c) 1/3

(d) 1/2

19. John and Dani go for an interview for two vacancies. The probability for the selection of John is 1/3 and whereas the probability for the selection of Dani is 1/5. What is the probability that none of them are selected?

(a) 3/5

(b) 7/12

(c) 8/15

(d) 1/5

20. John and Dani go for an interview for two vacancies. The probability for the selection of John is 1/3 and whereas the probability for the selection of Dani is 1/5. What is the probability that only one of them is selected?

(a) 3/5

(b) None of these

(c) 2/5

(d) 1/5

Practice Exercise 1 - Answers with Hints - need based

1 (b)	2 (d)	3 (a)	4 (a)	5 (a)
6 (a)	7 (c)	8 (b)	9 (a)	10 (c)
11 (a)	12 (a)	13 (b)	14 (b)	15 (a)

16 (c)	17 (d)	18 (a)	19 (c)	20 (c)

1. Total number of balls = 2 + 3 + 2 = 7

n(S) = Total number of ways of drawing 2 balls out of 7 = $^7C_2$

n(E) = Number of ways of drawing 2 balls, none of them is blue;

= Number of ways of drawing 2 balls from the total 5 (= 7 – 2) balls = $^5C_2$

($\because$ There are two blue balls in the total 7 balls. Total number of non-blue balls = 7 – 2 = 5)

P(E) = n(E) / n(S)= $^5C_2$ / $^7C_2$ = 10/21

2. Total number of possible when a die is rolled twice, n(S) = 6 × 6 = 36

n(E) = Getting a sum of 9 when the two dice fall = {(3, 6), {4, 5}, {5, 4}, (6, 3) = 4

P(E) = n(E) / n(S) = 4/36 = 1/9

3. Total number of outcomes possible when a coin is tossed = 2 ($\because$ Head or Tail)

Hence, total number of outcomes possible when 3 coins are tossed, n(S) = 2 × 2 × 2 = 8

($\because$ i.e., S = {TTT, TTH, THT, HTT, THH, HTH, HHT, HHH}

E = event of getting at most two Tails = {TTH, THT, HTT, THH, HTH, HHT, HHH}

Hence, n(E) = 7 P(E) = n(E) / n(S) = 7/8

5. Total number of outcomes possible when a die is rolled = 6 ($\because$ any one face out of the 6 faces)

i.e., n(S) = 6

E = Getting a number less than 4 = {1, 2, 3} Hence, n(E) = 3

P(E) = n(E) / n(S) = 3/6 = 1/2

6. Try to understand this method: P(E) = n(E) / n(S)

= $^5C_3$ / $^{15}C_3$ = $^5C_2$ / $^{15}C_3$ [$\because$ $^nC_r = {}^nC_{(n-r)}$. So $^5C_3 = {}^5C_2$.] = 291

7. Total number of cards, $n(S) = 52$

Total number of face cards, $n(E) = 12$

$P(E) = n(E) / n(S) = 3/13$

9. $P(E) = n(E) / n(S) = 26/52 = 1/2$

10. The best method is: $P(E) = n(E) / n(S) = {}^{15}C_2 \times {}^{10}C_1 / {}^{25}C_3$

12. $P(E) = n(E)n(S) = {}^{3}C_1 \times {}^{5}C_1 \times {}^{4}C_1 / {}^{12}C_3$

13. Hence, total number of outcomes possible when 5 coins are tossed, $n(S) = 2^5$

E = Event of getting exactly 2 heads when 5 coins are tossed

$n(E)$ = Number of ways of getting exactly 2 heads when 5 coins are tossed $= {}^{5}C_2$

$P(E) = n(E)n(S) = {}^{5}C_2 / 2^5 = 5/16$

15. P(Ace or King or Queen) = P (Ace) + P (King)+ P(Queen) = TRUE

$= 1/13 + 1/13 + 1/13 = 3/13$

16. P(Spade Cards of Number 5) + P(Club Cards of Number 5)

$1/52 + 1/52 = 1/26$

17. To get a sum of 7, (1, 6), (2, 5), {3, 4}, (4, 3), (5, 2), (6,1) = 6 ways

To get a sum of 11, (5, 6), (6, 5) = 2 ways FALSE

$= 6/36 + 2/36 = 8/36 = 2/9$

19. Probability that none of them is selected $=[1 - P(A)][1 - P(B)] = (1 - 1/3)(1 - 1/5) = 2/3 \times 4/5 = 8/15$

20. $=P(A)[1-P(B)]+P(B)[1-P(A)]$

$=1/3\,(1 - 1/5) + 1/5\,(1 - 1/3) = 1/3 \times 4/5 + 1/5 \times 2/3 = 4/15 + 2/15 = 2/5$

Practice Exercise - 2

Number of Questions: 25 **Suggested time: 20 minutes**

1. A letter is randomly taken from English alphabets. What is the probability that the letter selected is not a vowel?
(a) 5/25
(b) 2/25
(c) 5/26
(d) 21/26

2. The probability A getting a job is 1/5 and that of B is 1/7. What is the probability that only one of them gets a job?
(a) 11/35
(b) 12/35
(c) 2/7
(d) 1/7

3. A letter is chosen at random from the word 'ASSASSINATION'. What is the probability that it is a *vowel?*
(a) 4/13
(b) 8/13
(c) 7/13
(d) 6/13

4. A letter is chosen at random from the word 'ASSASSINATION'. What is the probability that it is a consonant?
(a) 4/13
(b) 8/13
(c) 7/13
(d) 6/13

5. Tickets numbered 1 to 20 are mixed up and then a ticket is drawn at random. What is the probability that the ticket drawn has a number which is a multiple of 3 or 5?
(a) 1/20
(b) 9/20

(c) 4/20
(d) 1/4

6. One ball is picked up randomly from a bag containing 8 yellow, 7 blue and 6 black balls. What is the probability that it is neither yellow nor black?
(a) 1/3
(b) 1/4
(c) 1/2
(d) 3/4

7. Two cards are drawn together from a pack of 52 cards. The probability that one is a club and one is a diamond?
(a) 13/51
(b) 1/52
(c) 13/102
(d) 1/26

8. Two cards are drawn together at random from a pack of 52 cards. What is the probability of both the cards being Queens?
(a) 1/52
(b) 12/21
(c) 22/21
(d) 1/26

9. Two dice are rolled together. What is the probability of getting two numbers whose product is even?
(a) 17/36
(b) 1/3
(c) 3/4
(d) 11/25

10. When two dice are tossed, what is the probability that the total score is a prime number?
(a) 1
(b) 13
(c) 4
(d) 512

11. A bag contains 6 white and 4 black balls. 2 balls are drawn at random. Find the probability that they are of same colour.

(a) 1/2

(b) 7/15

(c) 8/15

(d) 1/9

12. A problem is given to three students whose chances of solving it are 1/2, 1/3 and 1/4 respectively. What is the probability that the problem will be solved?

(a) 1/4

(b) 1/2

(c) 3/4

(d) 7/12

13. Two cards are drawn at random from a pack of 52 cards. what is the probability that either both are black or both are queen?

(a) 52/221

(b) 55/191

(c) 55/221

(d) 19/221

14. A man and his wife appear in an interview for two vacancies in the same post. The probability of husband's selection is (1/7) and the probability of wife's selection is (1/5). What is the probability that only one of them is selected?

(a) 2/7

(b) 1/7

(c) 3/4

(d) 4/5

15. In a lottery, there are 10 prizes and 25 blanks. A lottery is drawn at random. What is the probability of getting a prize?

(a) 2/7

(b) 5/7

(c) 1/5

(d) 1/2

16. What is the probability of getting 53 Mondays in a leap year?

(a) 1/7
(b) 3/7
(c) 2/7
(d) 1

17. Two dice are thrown together. What is the probability that the sum of the number on the two faces is divided by 4 or 6

(a) 7/18
(b) 14/33
(c) 8/18
(d) 7/35

18. Two cards are drawn together from a pack of 52 cards. The probability that one is a spade and one is a heart, is:

(a) 3/20
(b) 29/34
(c) 47/100
(d) 13/102

19. A bag contains 6 black and 8 white balls. One ball is drawn at random. What is the probability that the ball drawn is white?

(a) 3/7
(b) 4/7
(c) 1/8
(d) 3/4

20. In a class, 30% of the students offered English, 20% offered Hindi and 10% offered both. If a student is selected at random, what is the probability that he has offered English or Hindi?

(a) 1/2
(b) 3/4
(c) 4/5
(d) 2/5

21. If two letters are taken at random from the word HOME, what is the probability that none of the letters would be vowels?

(a) 1/6
(b) 1/2
(c) 1/4

(d) 1/3

22. In a simultaneous throw of pair of dice. Find the probability of getting the total more than 7.

(a) 1/2
(b) 5/12
(c) 7/15
(d) 13/12

23. A word consists of 9 letters; 5 consonants and 4 vowels. Three letters are picked at random. What is the probability that more than one vowel will be selected?

(a) 13/42
(b) 17/42
(c) 5/42
(d) 3/14

24. A bag contains 2 red, 3 green and 2 blue balls. Two balls are drawn at random. What is the probability that none of the balls drawn is blue?

(a) 10/21
(b) 11/21
(c) 1/2
(d) 2/7

25. A bag contains 5 blue and 4 black balls. Three balls are drawn at random. What is the probability that 2 are black and 1 is blue ball?

(a) 5/13
(b) 4/9
(c) 5/12
(d) 2/7

Practice Exercise 2 - Answers with Hints - need based

1 (d)	2 (c)	3 (d)	4 (c)	5 (b)
6 (a)	7 (c)	8 (b)	9 (c)	10 (d)
11 (b)	12 (c)	13 (c)	14 (a)	15 (a)
16 (c)	17 (a)	18 (d)	19 (b)	20 (d)
21 (a)	22 (b)	23 (b)	24 (a)	25 (c)

4. Total Number of letters in the word ASSASSINATION, n(S) = 13

Total number of consonants in the word ASSASSINATION = 7

P (consonant) = 7/13

5. Multiples required are: 3,6,9,12,15,18, 5,10,20 = 9 out of 20 = 9/20

6. Means pick up a Blue only: ${}^{7}C_1 \,/\, {}^{21}C_1$

- The product of even numbers is always even

- The product of odd numbers is always odd

- If there is at least one even number multiplied by any number of odd numbers, the product is always even

Let E = the event of getting two numbers whose product is even

= {(1,2), (1,4), (1,6), (2,1), (2,2), (2,3), (2,4), (2,5), (2,6), (3,2), (3,4), (3,6), (4,1), (4,2), (4,3), (4,4), (4,5), (4,6), (5,2),(5,4), (5,6), (6,1), (6,2), (6,3), (6,4), (6,5), (6,6)}

Hence, n(E) = 27

P(E) = n(E) / n(S) = 27/36 = 3/4

10. Prime numbers are 2, 3, 5, 7, 11, 13, 17, 19, 23, 29 ... etc.,

Let E = the event that the total is a prime number

= {(1,1), (1,2), (1,4), (1,6), (2,1), (2,3), (2,5), (3,2), (3,4), (4,1), (4,3), (5,2), (5,6), (6,1), (6,5)}

Hence, n(E) = 15

P(E) = n(E) / n(S)=15/36 = 5/12

11. Then, n(E) = no of ways (2 balls out of six) or (2 balls out of 4)

$$= {}^{6}C_2 - {}^{4}C_2 = \frac{10\times4\times270}{6} = 15 + 6 = 21$$

Therefore, P(E) = n(E) / n(S) = 21/45 = 7/15

12. P(none solves the problem) = P(not A) and (not B) and (not C)

$$= P(\overline{A} \cap \overline{B} \cap \overline{C})$$

$$x = \sqrt{3+1}/\sqrt{3-1}$$

$$= \frac{1}{2} + \frac{2}{3} + \frac{3}{4} = \frac{1}{4}$$

P (the problem will be solved) = 1 - P(none solves the problem) $7500 \times \frac{80}{100}$

14. Total chances = (Husband selected, wife not) OR (Husband not, wife selected)

= 1/7 × 4/5 + 6/7 × 1/5 = 10 / 35 = 2/7

16. In a leap year there will be 52 Sundays and 2 days will be left.

These days can be: (Sunday, Monday); (Monday, Tuesday); (Tuesday, Wednesday)

(Wednesday, Thursday); (Thursday, Friday); (Friday, Saturday); (Saturday, Sunday)

Of these 7 outcomes, favourable are 2; Hence the probability of getting 53 days = 2/7

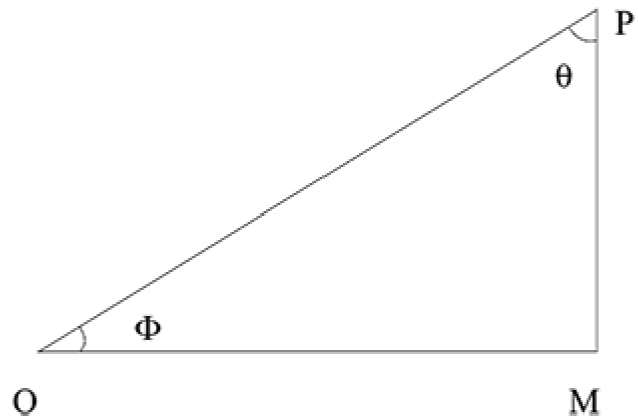
$$P(E) = \frac{30}{100} = \frac{3}{10}, \ P(H) = \frac{20}{100} = \frac{1}{5} \text{ and } P(E \cap H) = \frac{10}{100} = \frac{1}{10}$$

20.
$$\begin{aligned} P(E \text{ or } H) &= P(E \cup H) \\ &= P(E) + P(H) - P(E \cap H) \\ &= \left(\frac{3}{10} + \frac{1}{5} - \frac{1}{10}\right) = \frac{4}{10} = \frac{2}{5} \end{aligned}$$

25. The desired result can be available in $({}^4C_2\ {}^5C_1)\ /\ {}^9C_3$ ways

HEIGHT AND DISTANCE 32

Thus it is obvious that the three sides of a right angle triangle OPM i.e., OM, PM and OP are respectively the base, perpendicular and hypotenuse for angle POM (= Θ). From these we can find out 6 ratios which are known as Trigonometric ratios or functions or circular functions of angle Θ.



They are defined as follows:

(i) Sine θ = $\frac{PM}{OP} = \sin\theta$

(ii) Cosine θ = $\frac{OM}{OP} = \cos\theta$

(iii) Tangent θ = $\frac{PM}{OP} = \tan\theta$

(iv) Cotangent θ = $\frac{OM}{PM} = \cot\theta$

(v) Cosecant θ = $\frac{OP}{PM} = \text{cosec}\,\theta$

(vi) Secant θ = $\frac{OP}{OM} = \sec\theta$

Trigonometric Ratios of some Specific Angles

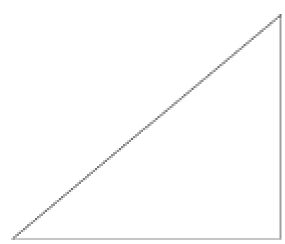
Angles T-Ratios	0°	30°	45°	60°	90°
Sine	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
Cosine	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
Tangent	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	∞
Cotangent	∞	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0
Secant	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	∞
cosecant	∞	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1

Example An arc of 3.5 cm length of a circle of radius 14 cm subtends on angle Θ at the centre of the circle. Find Θ.

Solution:

Angle = $\frac{Arc}{Radius}$ $x = \frac{(200 \times 60)}{250}$ radian.

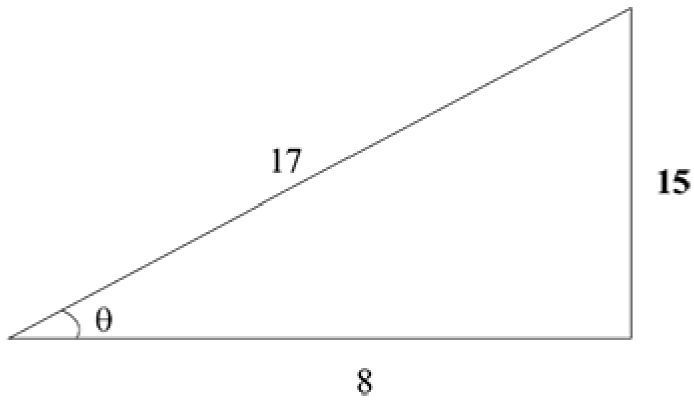
Example If sin Θ = $\frac{17}{2}$, find the value of



Solution:

$\tan \Theta = \frac{17}{2}$

Example If $\cos \Theta = \frac{17}{2}$, find cosec Θ



Solution:

$\text{cosec } \Theta = \frac{17}{2}$

Practice Exercise - 1

Number of Questions: 10 **Suggested time: 10 minutes**

1. Two ships are sailing in the sea on the two sides of a lighthouse. The angles of elevation of the top of the lighthouse observed from the ships are 30º and 45º respectively. If the lighthouse is 100 m high, the distance between the two ships is:
(a) 300 m
(b) 173 m
(c) 273 m
(d) 200 m

2. A man standing at a point P is watching the top of a tower, which makes an angle of elevation of 30º with the man's eye. The man walks some distance towards the tower to watch its top and the angle of the elevation becomes 45º. What is the distance between the base of the tower and the point P?
(a) 9 units
(b) 3√3 units
(c) 12 units
(d) Data inadequate

3. From a point P on a level ground, the angle of elevation of the top tower is 30º. If the tower is 200 m high, the distance of point P from the foot of the tower is:
(a) 346 m
(b) 400 m
(c) 312 m
(d) 298 m

4. The angle of elevation of the sun, when the length of the shadow of a tree is equal to the height of the tree, is:
(a) 70°
(b) 60°
(c) 45°
(d) 30°

5. An observer 2 m tall is $10\sqrt{3}$ m away from a tower. The angle of elevation from his eye to the top of the tower is 30º. The height of the tower is
(a) 16 m
(b) 12 m
(c) 14 m
(d) 10 m

6. The angle of elevation of a ladder leaning against a wall is 60º and the foot of the ladder is 12.4 m away from the wall. The length of the ladder is:
(a) 14.8 m
(b) 6.2 m
(c) 12.4 m
(d) 24.8 m

7. A man on the top of a vertical observation tower observers a car moving at a uniform speed coming directly towards it. If it takes 8 minutes for the angle of depression to change from 30° to 45°, how soon after this will the car reach the observation tower?
(a) 8 min 17 second
(b) 10 min 57 second
(c) 14 min 34 second
(d) 12 min 23 second

8. A man is watching from the top of a tower a boat speeding away from the tower. The boat makes an angle of depression of 45° with the man's eye when at a distance of 100 metres from the tower. After 10 seconds, the angle of depression becomes 30°. What is the approximate speed of the boat, assuming that it is running in still water?
(a) 26.28 km/hr
(b) 32.42 km/hr
(c) 24.22 km/hr
(d) 31.25 km/hr

9. The top of a 15 metre high tower makes an angle of elevation of 60° with the bottom of a pole and angle of elevation of 30° with the top of the pole. Find the height of the pole?

(a) 5 metres
(b) 8 metres
(c) 10 metres
(d) 12 metres

10. The angle of elevation of the top of a tower from a certain point is 30°. If the observer moves 40 m towards the tower, the angle of elevation of the top of the tower increases by 15°. The height of the tower is:
(a) 64.2 m
(b) 62.2 m
(c) 52.2 m
(d) 54.6 m

Practice Exercise 1 - Answers with Explanation

1. (c) Let BD be the lighthouse and A and C be the positions of the ships.

Then, BD = 100 m, ∠BAD = 30° , ∠BCD = 45°
tan 30° = BD/BA ⇒ 1/√3=100/BA ⇒ BA = 100√3
tan 45° = BD/BC ⇒ 1 = 100/BC ⇒ BC = 100 Distance between the two ships
= AC = BA + BC = 100√3 + 100 = 100(√3 + 1) = 273 m

2. (d) tan 45° = SR/QR tan 30° = SR/PR = SR (PQ + QR)

Here we find that there are two equations BUT 3 variables. Hence we cannot find the required value from this combination.

3. (a) Refer to the figure given below.

tan 30° = RQ/PQ = 1/√3 = 200/PQ or PQ = 200√3 = 346 m

4. (c) Consider the diagram shown above where QR represents the tree and PQ represents its shadow. We have, QR = PQ

Let angle QPR = θ; tan θ = QR/PQ = 1 *(since QR = PQ)* ⇒ θ = 45°

5. (b) SR = PQ = 2 m; PS = QR = 10√3 m

tan 30° = TS/PS = 1√3

TS = 10√3/√3 = 10 m or TR = TS + SR = 10 + 2 = 12 m

6. (d) Consider the diagram shown above where PR represents the ladder and RQ represents the wall. Here cos 60° = PQ/PR or 1/2=12.4/PR;

PR= 24.8 m

7. (b) Let AB be the tower and D and C be the positions of the car.

$\angle ADC = 30°$, $\angle ACB = 45°$ Given that Tan45°=AB/BC = $h/x \Rightarrow 1 \Rightarrow h = x(1)$

tan30°=AB/BD = AB/(BC + CD) = $h/x + y \Rightarrow 1/\sqrt{3} = h/x + y \Rightarrow y = \sqrt{3}h - x$

$\Rightarrow y = \sqrt{3}h - h$ ($\because$ from equation 1) $\Rightarrow y = h(\sqrt{3} - 1)$

i.e., Given that distance $y = h(\sqrt{3}-1)$ is covered in 8 minutes.

= Time to travel distance h ($\because$ Since $x = h$ as per equation 1).

Let distance h is covered in t minutes.

$\Rightarrow h(\sqrt{3}-1)/h = 8 / t \Rightarrow (\sqrt{3} - 1) = 8/t \Rightarrow t = 10$ minutes 57 seconds

8. (a) (Use method of Q7 above)

9. (c) AC represents the tower and DE represents the pole.

Given that AC = 15 m, $\angle ADB = 30°$, $\angle AEC = 60°$

Let DE = h; Then, BC = DE = h, or AB = (15 – h) ($\because$ AC=15 ,BC = h) & BD = CE

tan 60° = AC/CE $\Rightarrow \sqrt{3} = 15/CE \Rightarrow CE = 15\sqrt{3} = BD$ …(1)

tan 30° = AB/BD $\Rightarrow 1/\sqrt{3}=15-h/BD \Rightarrow 1/\sqrt{3} = 15 - h/(15\sqrt{3})$ i.e., height of the electric pole = 10 m

10. (d) Let DC be the tower & A, B the positions of observer such that AB = 40 m

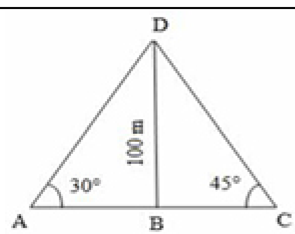
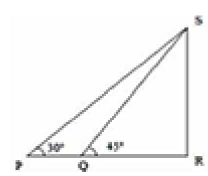
We have $\angle DAC = 30°$, $\angle DBC = 45°$

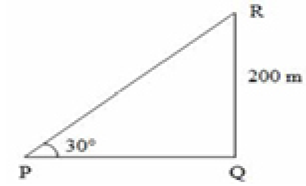
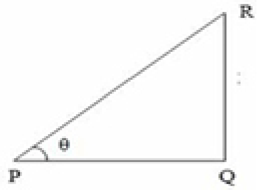
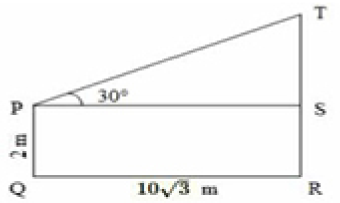
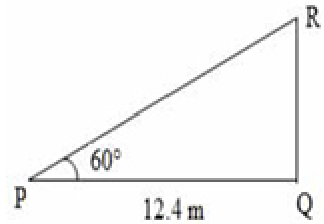
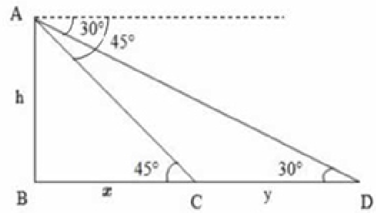
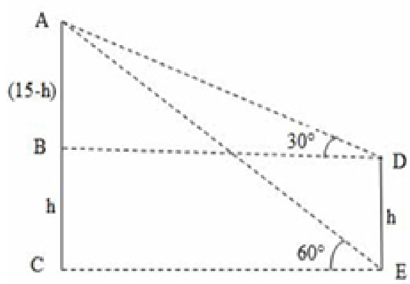
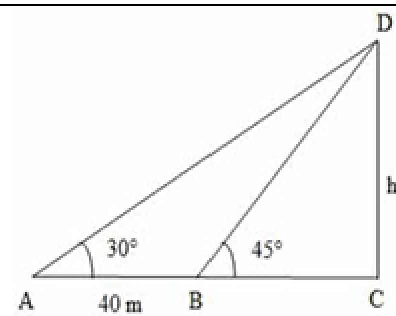
tan 30° = DC/AC $\Rightarrow 1/\sqrt{3} = h/AC \Rightarrow AC = h\sqrt{3}$ …(1)

tan 45° = DC/BC $\Rightarrow 1 = h/BC \Rightarrow BC = h$ …(2)

Also AB = (AC − BC) $\Rightarrow$ 40 = (AC - BC) $\Rightarrow 40 = (h\sqrt{3} - h)$ [$\because$ from (1) & (2)]

$\Rightarrow 40 = h(\sqrt{3} - 1) \Rightarrow h = 40(\sqrt{3} - 1) \times (\sqrt{3} + 1)(\sqrt{3} + 1) = 40(\sqrt{3} + 1)/2 = 54.6$ m

1		2	

3		4	
5		6	
7		8	Attempt this figure as similar to that of Q. 7
9		10	

Practice Exercise - 2

Number of Questions: 15 **Suggested time: 12 minutes**

1. On the same side of a tower, two objects are located. Observed from the top of the tower, their angles of depression are 45° and 60°. If the height of the tower is 600 m, the distance between the objects is approximately equal to:
(a) 272 m
(b) 284 m
(c) 288 m
(d) 254 m

2. A ladder 10 m long just reaches the top of a wall and makes an angle of 60° with the wall. Find the distance of the foot of the ladder from the wall ($\sqrt{3} = 1.73$)
(a) 4.32 m
(b) 17.3 m
(c) 5 m
(d) 8.65 m

3. From a tower of 80 m high, the angle of depression of a bus is 30°. How far is the bus from the tower?
(a) 40 m
(b) 138.4 m
(c) 46.24 m
(d) 160 m

4. The angle of elevation of the top of a lighthouse 60 m high, from two points on the ground on its opposite sides are 45° and 60°. What is the distance between these two points?
(a) 45 m
(b) 30 m
(c) 103.8 m
(d) 94.6 m

5. From the top of a hill 100 m high, the angles of depression of the top and bottom of a pole are 30° and 60° respectively. What is the height of the pole?

(a) 52 m

(b) 50 m

(c) 66.67 m

(d) 33.33 m

6. A vertical tower stands on ground and is surmounted by a vertical flagpole of height 18 m. At a point on the ground, the angle of elevation of the bottom and the top of the flagpole are 30° and 60° respectively. What is the height of the tower?

(a) 9 m

(b) 10.40 m

(c) 15.57 m

(d) 12 m

7. A balloon leaves the earth at a point A and rises vertically at uniform speed. At the end of 2 minutes, John finds the angular elevation of the balloon as 60°. If the point at which John is standing is 150 m away from point A, what is the speed of the balloon?

(a) 0.63 meter/sec

(b) 2.16 meter/sec

(c) 3.87 meter/sec

(d) 0.72 meter/sec

8. The angles of depression and elevation of the top of a wall 11 m high from top and bottom of a tree are 60° and 30° respectively. What is the height of the tree?

(a) 22 m

(b) 44 m

(c) 33 m

(d) None of these

9. Two vertical poles are 200 m apart and the height of one is double that of the other. From themiddle point of the line joining their feet, an observer finds the angular elevations of their topsto be complementary. Find the heights of the poles.

(a) 141 m and 282 m
(b) 70.5 m and 141 m
(c) 65 m and 130 m
(d) 130 m and 260 m

10. To a man standing outside his house, the angles of elevation of the top and bottom of awindow are 60° and 45° respectively. If the height of the man is 180 cm and he is 5 m awayfrom the wall, what is the length of the window?
(a) 8.65 m
(b) 2 m
(c) 2.5 m
(d) 3.65 m

11. The elevation of the summit of a mountain from its foot is 45°. After ascending 2 km towards the mountain upon an incline of 30°, the elevation changes to 60°. What is the approximate height of the mountain?
(a) 1.2 km
(b) 0.6 km
(c) 1.4 km
(d) 2.7 km

12. Two persons are on either sides of a tower of height 50 m. The persons observers the top of the tower at an angle of elevation of 30° and 60°. If a car crosses these two persons in 10 seconds, what is the speed of the car?
(a) $24\sqrt{3}$ km/hr
(b) $24\sqrt{3}$ km/hr
(c) $20\sqrt{33}$ km/hr
(d) None of these

13. Find the angle of elevation of the sun when the shadow of a pole of 18 m height is $6\sqrt{3}$m long?
(a) 30°
(b) 60°
(c) 45°
(d) None of these

14. The angle of elevation of top of the tower from a point on the ground is sin-1 A = (3/5). If the point of observation is 20 meters away from the foot of the tower, what is the height of the tower?

(a) 9 m

(b) 18 m

(c) 15 m

(d) 12 m

15. A person, standing exactly midway between two towers, observes the top of the two towers at angle of elevation of 22.5° and 67.5°. What is the ratio of the height of the taller tower to the height of the shorter tower? (Given that tan 22.5° = $\sqrt{2} - 1$)

(a) $1 - 2\sqrt{2} : 1$

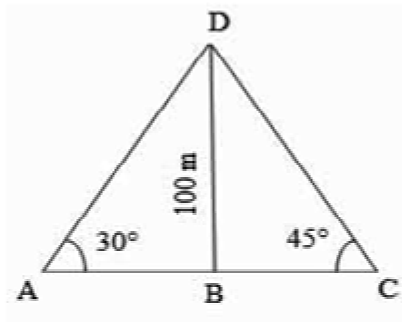
(b) $1 + 2\sqrt{2} : 1$

(c) $3 + 2\sqrt{2} : 1$

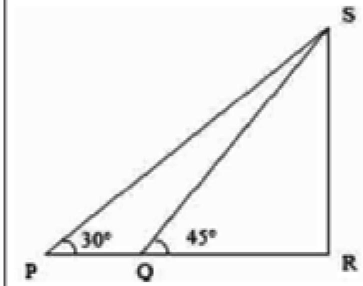
(d) $3 - 2\sqrt{2} : 1$

Practice Exercise 2 - Answers with Hints

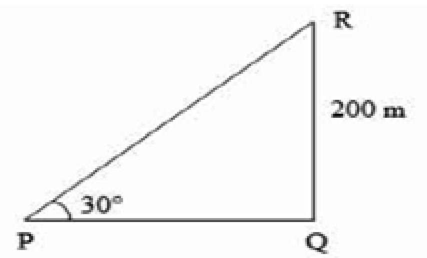
<table>
<tr>
<td>1</td>
<td>



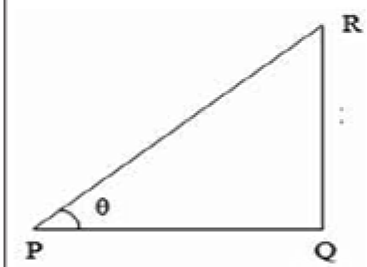
</td>
<td>2</td>
<td>



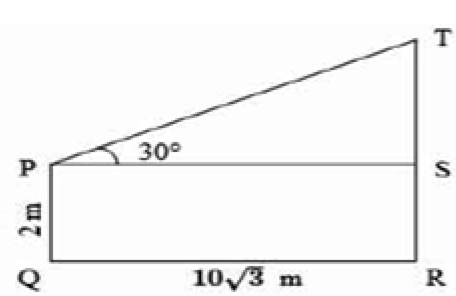
</td>
</tr>
<tr>
<td>3</td>
<td>



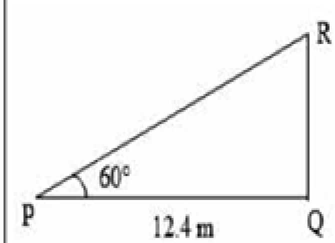
</td>
<td>4</td>
<td>



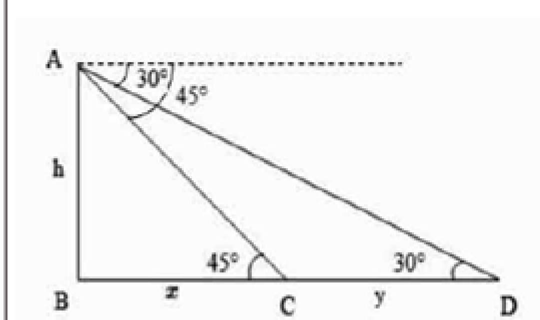
</td>
</tr>
<tr>
<td>5</td>
<td>



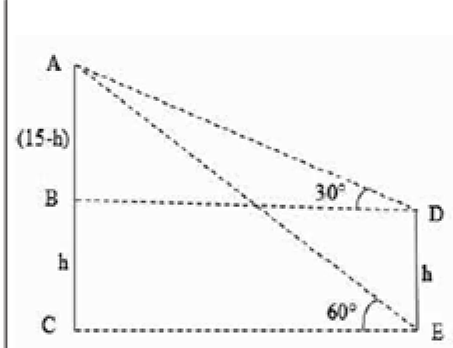
</td>
<td>6</td>
<td>



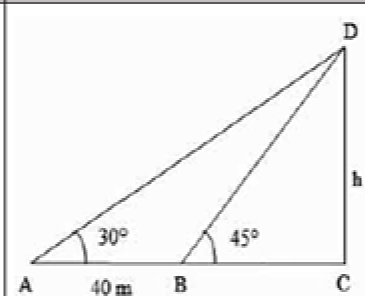
</td>
</tr>
<tr>
<td>7</td>
<td>



</td>
<td>8</td>
<td>Attempt this figure as similar to that of Q. 7</td>
</tr>
<tr>
<td>9</td>
<td>

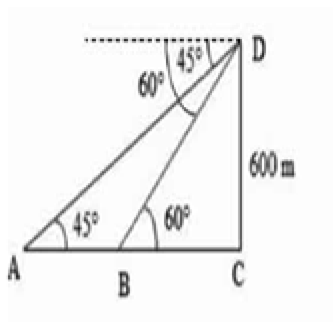
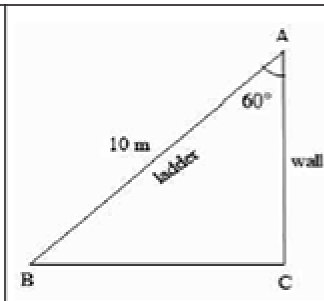
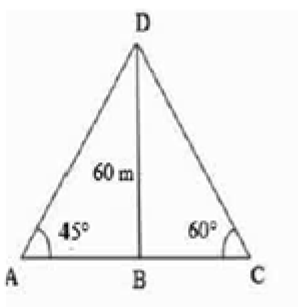
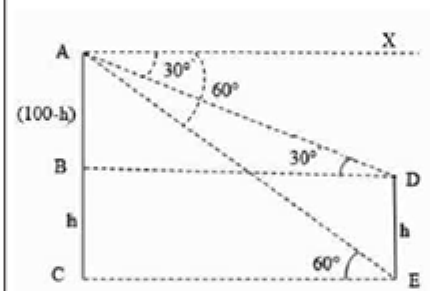
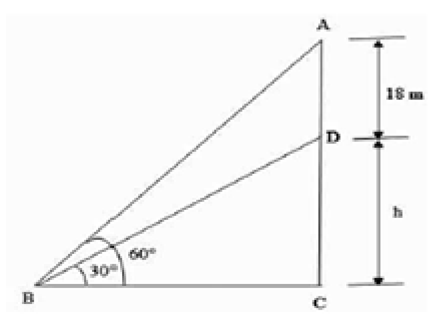
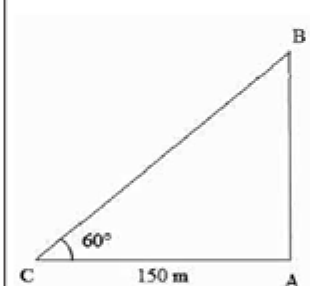
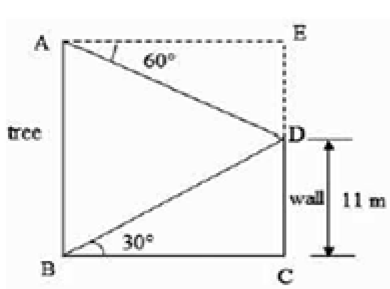
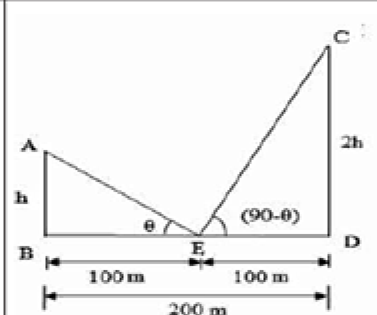


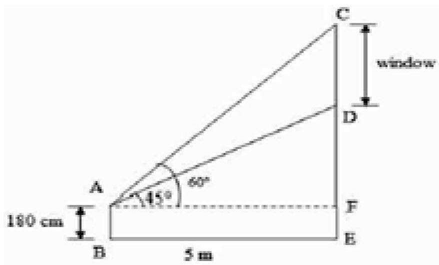
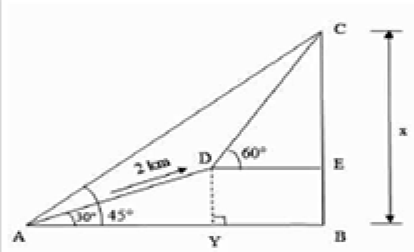
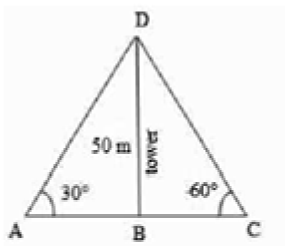
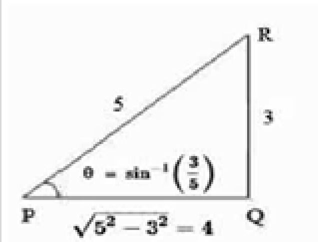
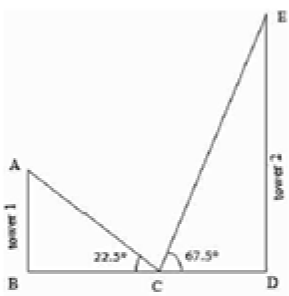
</td>
<td>10</td>
<td>



</td>
</tr>
</table>

11 (d)	12 (d)	13 (b)	14 (d)	15 (c)
16 (a)	17 (b)	18 (b)	19 (b)	20 (d)
21 (d)	22 (a)	23 (b)	24 (c)	25 (c)

11		12	
14		15	
16		17	
18		19	

<table>
<tr>
<td>20</td>
<td>
</td>
<td>.21</td>
<td>
</td>
</tr>
<tr>
<td>.22</td>
<td>
</td>
<td>24</td>
<td>
</td>
</tr>
<tr>
<td>25</td>
<td>
</td>
<td></td>
<td></td>
</tr>
</table>

(***For more questions on TRIGONOMETRY – please refer to the Author's book "MATHEMATICS SIMPLIFIED by JAGGAN SANEJA" – available on all e-commerce stores like Amazon.com, Bookadda.com, Infibeam.com, Flipkart.com and NOTION PRESS.com.***)

TEST SERIES

33

(These tests have been specially created and framed for quick revision of questions. A majority of these questions have appeared in various examinations conducted by different organisations. Even a few of these complete tests have been asked in some competition examinations.)

- *You are advised to strictly adhere to the time limit given for these tests.*
- *Don't go by judgements as these are the real time tests for you*
- *By now you have had enough of practice from the book.*
- *There may be negative markings in some of the examinations, so please be careful in picking up your choices.*
- *Evaluate yourself at the end of each test and correlate your performance viz-a-viz your preparation chart.*
- *In case of any difficulty you can mail your problems to the author at jnsaneja@yahoo.co.in*
- *Wishing you a good luck in the examinations.*

Five Minute Test - 1

In a survey conducted on 400 people in a part of Hyderabad City, 200 were found to read Deccan Herald, 150 for Express and 180 for The Hindu. 40 people read Express & The Hindu, 80 Deccan Herald & Express and 100 read Deccan Herald & The Hindu while 20 read all the papers. Now you are to find the following:-

1. How many people did not read any paper?
(a) 130
(b) 80
(c) 100
(d) 70

2. How many people read Deccan Herald & The Hindu only?
(a) 90
(b) 180
(c) 160
(d) 110

3. How many people read at least two papers?
(a) 200
(b) 180
(c) 160
(d) 140

4. How many people read Maximum One paper?
(a) 25
(b) 150
(c) 80
(d) none

5. How many people read Express or Hindu or both only?
(a) 200
(b) 180
(c) 240
(d) none

Five Minute Test - 2

1. The difference b/w the largest 4 digit number and the smallest 3 digit number is?
(a) 9899
(b) 8999
(c) 9989
(d) 9889

2. If (0.50) is divided by (0.01) the result is equal to?
(a) 50
(b) 5
(c) 500
(d) 1/50

3. If three-fourth of an estate is worth Rs. 90,000 then the value of its two-third is Rs.
(a) 60,000
(b) 65,000
(c) 70,000
(d) 80,000

4. A train running at 36 km/ph takes 2 min 20 sec to cross a bridge then the length of the bridge is __k.m.
(a) 1.4
(b) 2.2
(c) 3.6
(d) None

5. If 9 girls prepare 135 garlands in 3 hours, how many girls are required to prepare 270 garlands in an hour?
(a) 27
(b) 18
(c) 36
(d) 6

6. Given that 42% of a number is 126, then 37% of half of it will exceed 15% of it by __

(a) 9.5
(b) 10.5
(c) 12.5
(d) 13.5

Five Minute Test - 3

1. The measure of an angle is thrice its supplementary; its measure is __ degrees.

(a) 30

(b) 60

(c) 45

(d) None

2. Of the two statements given below, which is true?

(i) three angles of a triangle are equal to those of another

(ii) areas of two similar triangles are equal - then the triangles are congruent

(a) Only i

(b) Only ii

(c) both

(d) neither

3. Given that the length of a side of a rhombus is 13 and one of its diagonals is 24, then its two-third area is .... square units.

(a) 40

(b) 80

(c) 120

(d) 240

4. The volume of a right circular cylinder is 1100 cc and the base radius is 5 cm. Its total curved surface is ___approximately sq. cm

(a) 440

(b) 540

(c) 600

(d) 450

5. A wire was used to make a square and an equilateral triangle. It is given that the area of the latter is $16\sqrt{3}$ square cm, side of square is __cm

(a) 4

(b) $4\sqrt{2}$

(c) $6\sqrt{2}$

(d) 6

6. An equilateral triangle of 7 cm side had a semi circle, a square and an equilateral triangle drawn on its three sides. Sum of their area is __sq. cm nearest to

(a) 70

(b) 110

(c) 90

(d) 100

Five Minute Test - 4

For Questions: **1-3**: A race consists of 3 stretches A, B and C each of 2 km length in which a Car, a Motor Cycle and a Bicycle moving at Minimum speed of 40,30,10 kmph and Maximum speed of 60,50,20 kmph respectively are expected to take part. Previous record has been maintained for six minutes to complete the race. Please answer the following questions based on it:-

1. Anshul travelled at Minimum speed by car at Stretch 'A' and at Maximum speed over the stretch B. What should be his Minimum speed over stretch C to beat the record?.
(a) 15
(b) 20
(c) 17
(d) not possible

2. Mr Hare travelled at the slowest speed in stretch A and took the same time for stretch B also. If he took 15 minutes to complete the stretch his speed in stretch C is? kmph
(a) 10
(b) 15
(c) 20
(d) Data Inadequate

3. Mr Tortoise travelled at an average of 20 kmph. His average speed over the first two stretches was four times that over the last stretch. His speed during the last stretch i.e., C was
(a) 15
(b) 10
(c) 20
(d) can't be determined

Question 4-5: In Leisure Departmental Stores an employee gets off on those days whose first letter begins with his name (e.g. Mr Satpal shall get an off on Saturday or Sunday). Known this fact you are to find:-

4. Raja started a project on Sunday, 25th Feb 2016 and finished on Mar 2, 2016. If Mr Tom started his project on 25th Feb' 2016, he will finish on

(a) 26th Feb

(b) Fourth Mar

(c) 28th Feb

(d) he can't

5. If Raja had actually completed on Apr 2, 2016 and Tom & Shyam were assigned the same task on the day Raja began, then they will finish on?

(a) 15th Mar

(b) 22nd Mar

(c) 29 Mar

(d) None of These

6. How many values of natural number 'n' are possible for the expression (16.n*2* + 7n + 6)/n to assume a value of a natural number?

(a) Infinite

(b) 3

(c) 4

(d) 2 only

7. Nuts were bought at the rate of Rs. 100, 80, 60 per kg and were mixed in the ratio 3:4:5 by weight. If these were sold at 50% profit. The selling price per kg was Rs.?

(a) 110

(b) 80

(c) 70

(d) 115

Five Minute Test - 5

1. It is given that the cost price of 25 oranges + selling price of 15 oranges is the same as the selling price of 35 oranges minus the cost price of 5 oranges. Then it is a case of?
(a) gain
(b) loss
(c) at par
(d) data inadequate

2. Three numbers are such that their sum is zero, then sum of their cubes is surely ...
(a) + ve
(b) - ve
(c) zero
(d) can't say

3. Sixteen was divided in two parts in such a way that sum of their squares added to their product gave a result as 208. Then the smaller part is given by?
(a) 6
(b) 4
(c) 8
(d) 10

4. Find the number of sides of a polygon if sum of its angles is twice that of an octagon.
(a) 16
(b) 20
(c) 8
(d) 14

5. A conical vessel has a hemispherical lid. Total height when closed is 30 cm and the maximum girth is 44 cm. Its total volume given in cubic centimetres is?
(a) 1913
(b) 1200

(c) 900

(d) 1300

6. Chances of hitting a target by 3 shots at it is 1/2, 1/3, 1/4. Find the Probability that at least one shot hits the target.

(a) 3/4

(b) 1/4

(c) 1/24

(d) 13/12

7. Given that $f(x) = 1/x + 1$, $g(x) = 1/x - x$, then $f[g(2)]$ is?

(a) 1/2

(b) - 1/2

(c) - 2

(d) 1/3

Five Minute Test - 6

1. Sixteen balls could tightly be fitted in a rectangular box in two rows - that too with one above the other. What % volume of box could not be occupied?

(a) > **20**

(b) > 30
(c) > 40
(d) > 50
(e) none

2. In writing 1/3rd as (0.3), the %age error involved is
(a) 10
(b) 3
(c) 9.1
(d) 11.1
(e) none

3. A tank could be filled by 5 taps in two hours. One pump could not be made functional from the beginning itself. Extra time required to fill this tank is __ minutes?
(a) 15
(b) 20
(c) 25
(d) 30
(e) none

4. Which of the following is divisible by 33 and 36 both?
(a) 1854724
(b) 745350
(c) 875380
(d) 3569841
(e) none

5. Between the two Natural numbers 748 and 1396 find how many are multiples of 4?

(a) 159

(b) 160

(c) 161

(d) 162

(e) none

6. A monkey scales a greased pole 72 m high. It is going 8 meter up but slipping down 3 meter in every alternate minute. In which attempt it will be on the top the pole.?

(a) 24

(b) 25

(c) 26

(d) 27

(e) none

Five Minute Test - 7

Question (1-4): On her sister's birthday, Amani wants to present an art piece or an electrical gadget or a sports article or a garment. These cost Rs. 20, 25, 30 and Rs. 35 and are available in store A, B, C, D (not in order). Further to her knowledge: -

i) Article available at Store D costs Rs. 20;

ii) Sports article costs Rs. 35;

iii) Garments cost Rs. 30 - but are not available at Store C;

iv) Art piece is available at D;

From the above given information, solve the following problems:-

1. The article that costs Rs. 25 is available at store __
(a) A - and is art piece
(b) B - and is sports article
(c) A - and is electrical
(d) C - and is sports article

2. Store B sells ---
(a) An article costing Rs. 20
(b) the electrical article
(c) the garments
(d) the art piece

3. Amani buys an article costing less than Rs. 25. She does not buy
(a) the electric gadget
(b) the garment
(c) the sports article
(d) from C

4. If (i) statements were not given, we can deduce that:-
(1) the Store D sold electrical gadgets or art piece;
(2) electrical or art piece costs Rs. 20;
(3) The article worth Rs. 30 is sold at D or B;
(a) 2 only
(b) 1 only

(c) 3 only
(d) 2 & 3 only

5. Find the value of (16 + 12 + 88 + 34 + 66 - 21 - 79 + 11 + 89) when divided by 25.
(a) 7.04
(b) 7.44
(c) 8.24
(d) 8.64
(e) none

6. Two successive discounts of 20% and 12% are equivalent to one single discount of %?
(a) 16.0
(b) 29.6
(c) 31.4
(d) 32.0
(e) None

7. Average of 24.5, 31.5, 52.6, 47.4, x is calculated as 39.8. Find the value of x.
(a) 29.8
(b) 39.0
(c) 43
(d) 37.8
(e) 38.2

Five Minute Test - 8

Question 1-5: Refer to the data given below and answer

NATURAL GAS PRODUCTION (million cubic mtrs)

Year	1991-92	1992-93	1993-94	1994-95
Off shore	14,394	13,352	13,356	14,213
On shore	4,251	4,708	4,979	5,165

1. In which year is the ratio of off-shore to on-shore the highest?
(a) 91 - 92
(b) 92 - 93
(c) 93 - 94
(d) 94 - 95
(e) none

2. In which year is the ratio of off-shore to on-shore the lowest?.
(a) 91 - 92
(b) 92 - 93
(c) 93 - 94
(d) 94 - 95
(e) none

3. In which year is the % growth of on-shore over the previous year the lowest?
(a) 91 - 92
(b) 92 - 93
(c) 93 - 94
(d) 94 - 95
(e) none

4. What is the total net production in 1991 - 92? (million cubic mtrs)
(a) 1390
(b) 13947
(c) 46008
(d) 14,970

(e) None

5. Increase in the total net production over the previous year is the highest for which year?

(a) 91 - 92

(b) 92 - 93

(c) 93 - 94

(d) 94 - 95

(e) All same

Five Minute Test - 9

1. How much does 1/2 added to 1/3 and 1/8 and subtracted by 1/15 gives?
(a) 9/8
(b) 16/15
(c) 41/40
(d) 37/35
(e) none

2. Product of 3/100 with 15/49 and by 7/9 is?
(a) 215/44100
(b) 1/140
(c) 1/196
(d) 25/158
(e) none

3. Talking of a cone, its radius is doubled but height reduced to one-third. New volume will be what part of the old volume?
(a) 2/3
(b) 3/2
(c) 4/3
(d) 3/4
(e) none

4. What is the Average of 3.2 , 47/12 and 10/3 is?
(a) 3.55
(b) 3.333
(c) 103/30
(d) 209.20
(e) none

5. What is the value of 1.2% of 5.7 % of 2.25?
(a) 1.539
(b) 15.39
(c) .1539
(d) .01539
(e) none

6. In a class 40% are girls. If 20% of the girls wear glasses. What % of the strength of boys are the girls who wear glasses?

(a) 13.33
(b) 17.25
(c) .75
(d) 0.75
(e) none

7. Goods were marked 30% above CP. If 15% sale-discount was offered and 7% tax was imposed, what is the tax % paid on goods costing Rs. 480?

(a) 7.74
(b) 5.24
(c) 9.34
(d) 5.89
(e) none

8. If in a camp 10 men can survive for 24 days on 15 cans of rations, how many cans are required for 8 men to survive for 36 days?

(a) 15
(b) 16
(c) 17
(d) 18
(e) none

Five Minute Test - 10

1. Two dices are simultaneously thrown. What are the chances that their sum is 5?

(a) 1/10

(b) 1/9

(c) 1/36

(d) 4/11

(e) None

2. Liquid of a cylindrical can of 6 cm height and radius 2 cm are poured into a conical pot of radius 8 cm and 9 cm high. Height of level in the conical pot will be....cm?

(a) 5/8

(b) 4

(c) 6

(d) 9/8

(e) None

3. Four small balls were fitted in a big ball of radius 4 cm and the remaining portion was filled with water. How much (π cc) water is required?

(a) 20

(b) 40

(c) 60

(d) 80

(e) Data inadequate

4. Chord AB of a circle with radius 15cm is a tangent to a smaller concentric circle with radius 12 cm & centre O. Area of triangle ABO in (π cm*2*)?

(a) 36

(b) 48

(c) 27

(d) 6 pie

(e) None

5. Area of a square formed by intersections of four lines $x + y = 1$, $x + y = -1$, $x - y = 1$ and $x - y = -1$ is how many square units?

(a) 1
(b) 2
(c) 4
(d) 2.25
(e) None

6. Find the sum that amounts to Rs. 3000 at 10% in 5 yrs given at SI is

(a) 2000
(b) 2220
(c) 2180
(d) 2400
(e) None

Five Minute Test - 11

1. When half of the sweets of a box were digested by a group of his friends, Mr Kanwar is now left with 4.2 kg. Half of the remaining stock was consumed by another group of friends and now Kanwar is left with 2.7 kg. What was the total weight of the filled box in kg?
(a) 7.2
(b) 4.9
(c) 5.0
(d) none

2. From the above question weight of the box is ..kg
(a) 1.2
(b) .2
(c) .3
(d) .4

3. Which one is a perfect square amongst the given below?
(a) 8452664
(b) 6461764
(c) 21373138
(d) 8472563

4. A number when divided by 7,5,11 leaves 2,3,5 as remainder. If the division order is reversed the last remainder will be divisible by?
(a) 2
(b) 3
(c) 4
(d) Data inadequate

5. An item costing 125.00 was sold at a gain of 20% even when half of the marked price was discounted in sale. Marked price was Rs.?
(a) Rs. 212.50
(b) 300
(c) 247.50
(d) 271.50

6. From Question (5) given above, if no discount were offered the gain would have been..%?
(a) 20
(b) 120
(c) 140
(d) 250

7. A solid sphere of 8 cm radius is melted to form 8 small balls. Without wastage at any stage, find the radius of each small balls (..cm).
(a) 1.25
(b) 2.5
(c) 3.75
(d) 4

8. Water flows @ 5 meter in a minute through a pipe. A conical vessel with 40 cm base and 24 cm depth will take how many seconds to fill in by this inflow?
(a) 55
(b) 5.2
(c) 51/12
(d) Data Inadequate

9. What will be time taken in seconds to fill another spherical tank of 15 m radius by this inlet pipe?
(a) $55 \times 75 \times 75$
(b) $5.2 \times 75 \times 75$
(c) $4 \times 75 \times 75$
(d) Data Inadequate

Five Minute Test - 12

1. A circle is in a square. Find the ratio of their areas.
(a) 14:11
(b) 11:14
(c) 7:11
(d) 11:7

2. A square is in a circle. Find the ratio of their areas.
(a) 14:11
(b) 11:14
(c) 7:11
(d) 11:7

3. A circle, an equilateral triangle and a square are made of 264 cm wire. Which figure has the maximum area covered under it?
(a) Circle
(b) Traingle
(c) Square
(d) All equal

4. A field 45m × 36m has two cross roads each of 5m width. Area of the remaining field in square meter is?.
(a) 1200
(b) 1220
(c) 1230
(d) 1240

5. A circular field of 15400 square meter has roads each 20 m wide around and inside it. Difference between the areas of the roads is how many Pie square meter?.
(a) 700
(b) 5750
(c) 5600
(d) 900

6. A table 80 cm by 75 cm has tumblers 7 cm wide placed on it. No. of tumblers is?.

(a) 110
(b) 120
(c) 116
(d) 125

Five Minute Test - 13

Read the following carefully and answer the following questions.

In a class of 120 students in Anand Vihar School, every student has to study at least one of the subjects viz. History, English or Mathematics. It is known that 59 students study History, 67 go for English and 73 have opted for Mathematics. Also that 34 study Mathematics & History, while 26 like English & Mathematics combinations and finally 33 went for the subjects of History and English.

Based on the above, you are to find:-

1. How many students study exactly two subjects?
(a) 54
(b) 51
(c) 48
(d) 46

2. How many students study all the three subjects?
(a) 12
(b) 15
(c) 13
(d) 14

3. How many students study more than 1 subjects?
(a) 63
(b) 65
(c) 62
(d) 66

4. How many students took English & Mathematics but Not History?
(a) 12
(b) 13
(c) 14
(d) 11

5. How many students study English & Maths or English & History only?
(a) 41

(b) 43
(c) 47
(d) 31

6. How many students took History & English but NOT mathematics?
(a) 19
(b) 18
(c) 17
(d) 20

7. If 70% of students failed in one or more subjects, then the maximum number of failures are in the subject of:-
(a) Mathematics
(b) English
(c) History
(d) None of these

8. If conditions "that all study at least one subject" is relaxed, what is the maximum number of students who study all the subjects?
(a) 26
(b) 33
(c) 45
(d) None of these

9. In above questions Minimum number of students is
(a) 8
(b) 9
(c) 17
(d) None of these

Five Minute Test - 14

In a town of Jodhka inhabited with 1000 people, a survey was got conducted which found that 25% of the inhabitants played football or hockey or cricket or more than one of these games. 85 residents played hockey, 160 played football, 200 played cricket and 45 played all the three games. While 20 played only football and 70 played only cricket, 20 persons played both hockey and football but not cricket. Based on the above information, please find:-

1. How many played only football and cricket?
(a) 110
(b) 75
(c) 70
(d) 120

2. What % of people played only hockey?
(a) 10
(b) 0.85
(c) 8.5
(d) 1.0

3. How many people played hockey & football?
(a) 65
(b) 20
(c) 40
(d) 45

4. What % of people played at least two games?
(a) 150
(b) 20
(c) 15
(d) 24

5. If 30% of the people who play only hockey and cricket are females, how many women play hockey & cricket?
(a) 3
(b) 8

(c) 2
(d) Cannot say

Five Minute Test - 15

Ram Singh Sihag of Sihag Farms maintain a big farm land. The farm is given on lease to many farmers for cultivation. The terms of farming include certain conditions as under:-

- *Every year a farmer can plant 3 out of the 5 crops made available to him which are Apples(a), Grapes (G), Papayas (P), Oranges (O) and Mangoes (M). However on deciding what to plant, he has to keep some other factors in mind-*
  - (a) Two new crops have to be planted each year;
  - (b) If grapes are planted, apples have to be planted;
  - (c) Grapes and oranges cannot be planted together;
  - (d) If papayas are planted in a year, the crop cannot be repeated next year;

1. Given the above, which combination over a period of two years is acceptable?
(a) G,A,P; O,M,P
(b) G,M,P; O,A,G
(c) O,M,P; G,A,M
(d) A,M,P; M,A,G

2. In a year G & M were grown. Which combination may be planted in succeeding year?
(a) O,A,P
(b) M,O,A
(c) P,O,M
(d) (a) and (c)

3. If a farmer plants G,A,P in the first year, he can plant ___ in the third year?
(a) G,A,P
(b) M,G,A
(c) A,O,M
(d) can’t be said

4. If a farmer plants A,O,M in a year, how many combinations of crops are possible in following year?

(a) 1

(b) 2

(c) 3

(d) can't say

5. In how many different ways can a farmer plant his crops in a year so as to grow apples every year?

(a) 1

(b) 2

(c) 3

(d) none

ANSWER KEY - FIVE MINUTE TESTS

TEST / QUESTION NO.	1	2	3	4	5	6	7	8	9
FIVE MINUTE TEST NO. 1	(d)	(b)	(b)	(d)	(d)	NOT APPLICABLE			
FIVE MINUTE TEST NO. 2	(a)	(a)	(d)	(d)	(d)	(b)	NOT APPLICABLE		
FIVE MINUTE TEST NO. 3	(d)	(b)	(b)	(a)	(d)	(b)	NOT APPLICABLE		
FIVE MINUTE TEST NO. 4	(d)	(d)	(d)	(b)	(c)	(c)	(d)	NOT APPLICABLE	
FIVE MINUTE TEST NO. 5	(a)	(b)	(b)	(d)	(d)	(a)	(d)	NOT APPLICABLE	
FIVE MINUTE TEST NO. 6	(c)	(c)	(c)	(d)	(c)	(d)	NOT APPLICABLE		
FIVE MINUTE TEST NO. 7	(c)	(c)	(d)	(d)	(d)	(b)	(b)	NOT APPLICABLE	
FIVE MINUTE TEST NO. 8	(a)	(c)	(d)	(e)	(d)	NOT APPLICABLE			
FIVE MINUTE TEST NO. 9	(e)	(b)	(b)	(c)	(d)	(e)	(a)	(d)	n/a
FIVE MINUTE TEST NO. 10	(b)	(d)	(b)	(a)	(b)	(a)	NOT APPLICABLE		
FIVE MINUTE TEST NO. 11	(b)	(a)	(b)	(b)	(b)	(c)	(d)	(d)	(d)**
FIVE MINUTE TEST NO. 12	(b)	(c)	(a)	(d)	(c)	(a)	** Size of the pipe?		
FIVE MINUTE TEST NO. 13	(b)	(d)	(b)	(a)	(d)	(a)	(a)	(c)	(d)
FIVE MINUTE TEST NO. 14	(b)	(d)	(a)	(c)	(d)	NOT APPLICABLE			
FIVE MINUTE TEST NO. 15	(c)	(d)	(a)	(a)	(b)	NOT APPLICABLE			

Ten Minute Test - 1

1. Each angle of a regular pentagon will be:
(a) 72°
(b) 90°
(c) 108°
(d) 120

2. If one of the interior angles of a regular polygon is found to be equal to (9/8) times of one of the interior angles of a regular hexagon, then the number of sides of the polygon is:
(a) 4
(b) 5
(c) 7
(d) 8

3. The ratio of the measure of an angle of a regular octagon to the measure of its exterior angle is:
(a) 1:2
(b) 1:3
(c) 2:3
(d) 3:1

4. The angles of a pentagon are in the ratio 1: 2: 3: 5: 9. The smallest angle is:
(a) 72°
(b) 45°
(c) 54°
(d) 27

5. If each interior angle of a regular polygon is 11 times its exterior angle, the number of sides of the polygon is:
(a) 22
(b) 24
(c) 18
(d) 11

6. One angle of a hexagon is 100**o** and all the other five angles are equal. What is the measure of each one of the equal angles?

(a) 36°

(b) 144°

(c) 124°

(d) 72

7. The average salary of 50 workers in a factory was Rs. 350. 5 new workers with salaries 250, 300, 320, 350 and 400 were employed. What would be the new average salary?

(a) 346.64

(b) 347.64

(c) 348.64

(d) 349.36

8. Typist A can type a sheet in 5 minutes, typist B in 6 minutes and typist C in 8 minutes. The average number of sheets typed per hour per typist is……sheets.

(a) 8.83

(b) 7.83

(c) 7

(d) 9.83

9. A car runs its first 5000 miles at 40 miles to the gallon of petrol and its next 5000 miles at 36 miles to the gallon. What is the average petrol consumption for the whole 10000, miles (gallon/mile)?

(a) .0263

(b) .0264

(c) .0265

(d) .0266

10. The sum of the interior angles of a hexagon is:

(a) 360°

(b) 540°

(c) 720°

(d) 900° 45.

11. The sum of the interior angles of a polygon is 1080°. The number of sides of the polygon is:

(a) 8
(b) 6
(c) 10
(d) 9

12. Find out the average rate of motion in the case of a person who rides the first mile @ 10 m.p.h., the next mile @ 8 m. p. h and the third mile @ 6 m. p. h.
(a) 6
(b) 7.33
(c) 7.66
(d) 8.33

13. The population of a town increased from 100000 to 150000 in a decade. What is the average % increase per year
(a) 2
(b) 3
(c) 4
(d) 5

14. The average monthly sales for the first eleven months of the year in respect of a certain salesman were Rs. 12,000 but due to his illness during the last month, the average monthly sales for the whole year came down to Rs. 11,375. What was the value of sales during the last month?
(a) 4100
(b) 3900
(c) 4300
(d) 4500

15. My income for 15 days averaged Rs. 7, the average for the first 5 days was Rs. 6 and the average for the last 9 days was Rs. 8. What was the income on the 6^{th} day?
(a) Rs. 2
(b) Rs. 2.25
(c) Rs. 2.50
(d) 3.00

Ten Minute Test - 2

1. Find the sum on which the difference between the simple interest and compound interest for 3 years at 5 per cent per annum is Rs. 73.20

(a) 8100
(b) 500
(c) 9600
(d) 9800

2. If the compound interest on a certain sum for 3 years at 5 p.c. be Rs. 50.44, what would be the simple interest?

(a) 36
(b) 48
(c) 52
(d) 56

3. A sum of Rs. 2550 is borrowed to be paid back in two years by two equal annual payments, 4 p.c. compound interest being allowed. What is the annual payment?

(a) 1350
(b) 1352
(c) 1296
(d) 1560

4. A builder borrows Rs. 1261 from the bank to be paid back with compound interest at the rate of 5 per;cent per annum by the end of 3 years in 3 equal yearly instalments. Find the value of each instalment?

(a) 464
(b) 492
(c) 470
(d) 502

5. If the discount on Rs. 170.04 at 4% be Rs. 14.04, the sum is due after how many years?

(a) 7
(b) 2
(c) 5

(d) 4

6. Find the present worth of Rs. 132 due in 2 years at 5 p.c. simple interest per annum.
(a) 140
(b) 145
(c) 125
(d) none

7. Find sum will discharge a debit of Rs. 5300 due a year and a half hence at 4 % per annum simple interest?
(a) 5000
(b) 6500
(c) 6200
(d) 4800

8. Find the difference between the compound interests on Rs. 5,000 for 1 ½ years at 4 p.c. per annum, according as the interest is paid yearly or half-yearly.
(a) 2.04
(b) 2.97
(c) 3.52
(d) 3.92

9. What is the cash payment equivalent to Rs. 297.54;due in 3 years 6 months, simple interest being reckoned at 4 % per annum?
(a) 265
(b) 256
(c) 2610
(d) 261

10. I buy a horse for Rs. 400 and sell it for Rs. 510 at a credit of 6 months. What do I gain per cent reckoning money worth 4% per annum?
(a) 25%
(b) 33%
(c) 40%
(d) none

11. In a sequence of seven consecutive integers, the average of the first 5 is 'n'. The average of all the seven numbers is (k is a function of 'n')

(a) n
(b) n + 1
(c) k.n
(d) n + 2/7

12. If x > 2, y > - 1, then which of the followings is necessarily true?
(a) xy > 2
(b) - x < 2y
(c) xy < - 2
(d) - x > 2y

13. In how many ways can 6 flags be arranged consisting a red, 3 white and 2 blue flags in a line so that-no two adjacent flags are of same color
(a) 6
(b) 4
(c) 10
(d) 2

14. What is the number of different triangles with integral sides and perimeter 14?
(a) 6
(b) 5
(c) 4
(d) 3

15. A three-digit integer ‘n’ leaves same remainder on dividing 34041 and 32506. ‘n’ is
(a) 289
(b) 367
(c) 453
(d) 307

Ten Minute Test - 3

1. What is the smallest number which when increased by 7 is divisible by 27, 21, 35 & 77

(a) 10402

(b) 10388

(c) 10397

(d) 20783

2. In twelve hours how many times are the hands of clock make supplementary angles

(a) 22

(b) 24

(c) 12

(d) 11

3. Which one is a prime number?

(a) 467

(b) 351

(c) 171

(d) 117

4. What is the remainder if (100000000 – 6) is divided by 3

(a) 0

(b) 1

(c) 2

(d) 3

5. The smallest whole number which should be multiplied to make 576 a perfect cube is?

(a) 8

(b) 24

(c) 4

(d) 3

6. If six-digit number (93p25q) is divisible by 88, the values of 'p' and 'q' are

(a) 2,8

(b) 8,2
(c) 8,6
(d) 6,8

7. The bisectors of any two adjacent angles of a parallelogram intersect at:
(a) 30°
(b) 45°
(c) 60°
(d) 90

8. The bisectors of the angles of ll gm enclose a:
(a) Parallelogram
(b) Rhombus
(c) Rectangles
(d) Square

9. The figure formed by joining the mid points of the adjacent sides of a quadrilateral is a:
(a) Parallelogram
(b) Rectangles
(c) Square
(d) Rhombus

10. A number when divided by 255 leaves 58, find the remainder if it is divided by 19
(a) 2
(b) 3
(c) 1
(d) none

11. What is the value of 39856 × 139
(a) 5839945
(b) 5539964
(c) 5439981
(d) 5539984

12. The greatest number that will divide 964, 1238 and 1400 leaving remainders of 41,31,51 respectively is given by
(a) 64
(b) 71

(c) 69
(d) 72

13. If sum of two numbers is 1000, their LCM is 8919. Find their difference
(a) 986
(b) 978
(c) 982
(d) 974

14. If a man receives on a fourth of his capital 3 p.c. , on two-third 5 p.c. and on the remainder 11 p.c., what percentage does he receives on the whole?
(a) 2%
(b) 3%
(c) 5%
(d) 7%

15. The average weight of 10 men is increased by 1 ½ kg when one of the men who weights 68 kg is replaced by a new man. Find the weight of the new man.
(a) 73 Kg
(b) 83 Kg
(c) 88 Kg
(d) 78 Kg

16. Each interior angle of a regular hexagon is:
(a) 60°
(b) 120°
(c) 108°
(d) 90

Ten Minute Test - 4

1. 50,000; 49,990; 49,890; 48,890 ... the next number in this sequence is
(a) 48790
(b) 37790
(c) 38890
(d) 47790

2. Given that ‘pq’ is - ve, then which one cannot be + ve?
(a) p.q.q
(b) p + q
(c) 4p/q
(d) 5q/(p.p)

Question No. 3-4: A post card costs 15 p, an inland costs 35p, and a cover costs 0.70 p

3. Ravi bought a total of 15 numbers of these items. He paid Rs. 6.65 He must have got __ covers at least?
(a) 4
(b) 2
(c) 3
(d) 5

4. Shekhar bought 20 items in Rs. 5.20 p. He must at least get __ cards?
(a) 2
(b) 4
(c) 13
(d) 9

5. ‘pq’ is odd, then which of the following statements shall hold good?
(I) p is even (II) q=0 (III) p is odd, q is even?
(a) I true
(b) II true
(c) III true
(d) none

6. The sum of two numbers is 50 while sum of their reciprocals is 1/12. The numbers are in the ratio of?
(a) 14/11
(b) 2/3
(c) 12/13
(d) 18/7

7. The denominator of a fraction is 3 more than numerator. If 4 is added to each, the result is a fraction which differs by 3/35 from the original. Denominator of the fraction is?
(a) 8
(b) 10
(c) 11
(d) 13

8. How many numbers between 1006 and 2999 are divisible by 6 but not by 9
(a) 109
(b) 110
(c) 111
(d) none

9. Difference of S.I. and C.I. on a certain sum @ 5% after 2 year is Rs. 3. The sum is Rs. ..?
(a) 1050
(b) 1100
(c) 1200
(d) 1250.

10. Population of a village is 15,000. It increases @ 100 per thousand per year. At the end of 3rd year the population shall be
(a) 18750
(b) 19965
(c) 19675
(d) 19000.

11. Selling an article for Rs. 700 yields some profit. Selling this for Rs. 500 will put me to a loss of 3/5th of the profit earned earlier. Then the cost price of the article is Rs. ...?

(a) 560
(b) 575
(c) 600
(d) 650

12. A sum doubles itself in 25 years, it will earn 4 times of itself in __decades
(a) 100
(b) 10
(c) 50
(d) 75

13. Given that a:b=3:4, b:c=5:6, c:d=11:9, then a:d=?
(a) 50:60
(b) 55:72
(c) 60:70
(d) 65:75

14. A piece of 10 meter cloth 80 cm wide costs Rs. 50. The cost of 15 meter long and 40 cm wide will cost Rs. ..?
(a) 75.00
(b) 62.50
(c) 37.50
(d) 25.00

15. Of the two statements that(a) the sum of two prime numbers is prime and (b) the product of two prime numbers is prime, we can conclude that __is true
(a) a only
(b) b only
(c) neither
(d) both

16. Known that a number 'x' if divided by 907, the remainder is 423. Then the statements that(a) 2x divided by 907- remainder is 846, (b) 2x divided by 1814 shall leave 846, we can say that __ is correct
(a) only 'a'
(b) only 'b'
(c) 'a & b' both

(d) neither

Ten Minute Test - 5

For questions **(Q 1 - Q 6),** use the following instructions and answer:-

(a) - if the question can be answered with the help of ANYONE statement only.

(b) - if the question can be answered with the help of EITHER of the statements.

(c) - if the question can be answered with the help of BOTH statements.

(d) - if the question CANNOT be answered even with the help of both the statements

1. What is the value of ‘y’, given that x , y , z all are positive real
   (a) $y2 = x z$
   (b) $x = z$,

2. What is the value of $T10$?
   (a) $T6 = 192$
   (b) $T11 = 6144$

3. What is the 3rd number of 8 consecutive real numbers
   (a) Their product is 34459425
   (b) Their sum is 84

4. Is $0 < x < 1$?
   (a) $x2 < x$
   (b) $x3 > 0$

5. Is M a perfect square ($M > 0$ & integer)?
   (a) M is divisible by 16
   (b) M + 1 is perfect square

6. What is the value of ‘x’?
   (a) x is 2-digit number
   (b) x = Twice the sum of digits + 4

Q 7–Q. 10 Shown below are the colour-mix patterns used by Indo-Gulf Paints for the factories, to produce the specific shades:

CODE COLOUR   MIXTURE PATTERNS

A.	Royal blue:	Blue 60% + White 30% + Grey 10%
B.	Sky blue:	Royal blue 70% + White 30%
C.	Aqua blue:	Blue 55% + Green 40% + Red 5%
D.	Midnight blue:	Blue 40% + Black 30% + Royal blue 30%
E.	Black blue:	Blue 50% + Black 50%
F.	Ice blue:	Blue 70% + Grey 10% + White 20%

7. Which shade of blue has the largest %age of the colour blue?
(a) A
(b) B
(c) C
(d) D
(e) E

8. What is the %age of the colour blue in midnight blue?
(a) 40
(b) 58
(c) 70
(d) 60
(e) 82

9. What is the highest percentage of white amongst the above 6 shades?
(a) 51
(b) 30
(c) 24
(d) 61
(e) 69

10. If we mixed equal quantities of aqua blue and midnight blue, how much blue would be in the resulting mix?
(a) 56.59
(b) 47.5
(c) 48.2
(d) 54
(e) 35.9

11. $\sqrt{\dfrac{9.5x0.0085x18.9}{0.0017x1.9x2.1}}$

(a) 0.15

(b) 0.5

(c) 15

(d) 250

12. If $\dfrac{5+2\sqrt{3}}{7+4\sqrt{3}} = a + b\sqrt{3}$,, then the value of x is:

(a) 1

(b) 3

(c) 5

(d) 7

13. If $\sqrt{3}$ = 1.732, then the value of $\dfrac{1}{\sqrt{3}}$ is:

(a) 0.617

(b) 0.313

(c) 0.577

(d) 0.173

14. Population of a town in 1980 was 50,000. In 1981 it increased by 2,500. In 1982 it decreased by 1,700. At the end of 1982 the population was _______% of that of 1980.

(a) 102.4

(b) 98.5

(c) 103.4

(d) 101.6.

15. If population of a country increased by 5% for three consecutive years. The over all increase at the end of 3^{rd} year will be ____%

(a) 1.1576

(b) 1.1475

(c) 1.1352

(d) 2.4.

16. Population of a town 'A' increased by 8% and 9% for two years. Population of another town 'B' increased by 7% and 10% which town

had more increase.
(a) A
(b) B
(c) Both equal
(d) Data incomplete.

ANSWER KEY - TEN MINUTE TESTS

QUESTION NO.	TEN MINUTE TEST No. 1	TEN MINUTE TEST No. 2	TEN MINUTE TEST No. 3	TEN MINUTE TEST No. 4	TEN MINUTE TEST No. 5
1	(c)	(c)	(b)	(c)	(d)
2	(d)	(b)	(d)	(c)	(d)
3	(d)	(b)	(a)	(a)	(d)
4	(d)	(a)	(b)	(d)	(d)
5	(b)	(b)	(d)	(c)	(d)
6	(c)	(d)	(c)	(b)	(d)
7	(b)	(a)	(d)	(b)	(a)
8	(d)	(a)	(c)	(b)	(b)
9	(b)	(d)	(a)	(c)	(a)
10	(c)	(a)	(d)	(b)	(a)
11	(a)	(b)	(d)	(b)	(c)
12	(c)	(b)	(b)	(b)	(a)
13	(c)	(a)	(c)	(b)	(c)
14	(d)	(d)	(c)	(c)	(d)
15	(d)	(d)	(b)	(c)	(a)
16	NOT APPLICABLE		(b)	(c)	(a)

Twenty Minute Test - 1

1. Atul and Babita enter into a business contributing Rs. 2,000 for 9 months and Rs. 5000 for 7 months. Annual profit of Rs. 1100 will be divided between them in the ratio:
(a) 6:11
(b) 2:5
(c) 7:9
(d) 18:35

2. If 3: 21:: x: 35, then the value of 'x' is
(a) 5
(b) 4
(c) 7
(d) 8

3. An alloy contains copper and zinc as 5:3 and another alloy contains copper and tin as 8:5. If equal weights of both alloys are melted then weight (kg) of tin in new alloy of 1 kg will be
(a) 26/5
(b) 5/26
(c) 7/31
(d) 31/7

4. LCM & HCF of two numbers are 84 and 21. If the numbers are in the ratio 1:4, then the larger of the two numbers is?
(a) 12
(b) 48
(c) 84
(d) 108

5. A plane flies @ 320 kmph. Wind blows at 40 kmph. Time of journey against the wind is 135 minutes. Time taken for the return journey will be … minutes?
(a) 94
(b) 105
(c) 108

(d) 120

6. Going at 30 kmph a man reaches 10 minutes late but going at 42 kmph he is 10 minutes early. The distance he goes is … km

(a) 36
(b) 35
(c) 40
(d) 48

7. A certain distance is covered by a train at a certain speed. If half of the distance is covered in double of the time, ratio of the new speed to the original one is?

(a) ¼
(b) 4
(c) 2
(d) ½

8. A train running between two stations A & B arrives 10 minutes late if going at 50 kmph. It is 50 minutes late if it moves at 30 kmph. Distance between the stations is …km

(a) 40
(b) 50
(c) 60
(d) 70

9. A man makes upward journey in a river @ 16kmph and downward @ 28 kmph. His average speed is …kmph

(a) 32
(b) 56
(c) 22
(d) 10

10. A student rides @ 8 kmph reaching 25 minutes late. Going @ 10 kmph he is early by 5 minutes. What correct time should he take to reach in time?

(a) 30
(b) 20
(c) 25
(d) 15

11. From a horizontal distance of 50 meters, the angles of elevation of top and bottom of a vertical cliff face are 45 & 30 degrees. Height of cliff is approximately ..meter?

(a) 30

(b) 35

(c) 45

(d) 17

12. A tank 30 x 20 × 12 m is dug in a field 500 m × 30 m. By what height (in cm) the level of the remaining field will rise if the earth thus dug is spread over it?

(a) 33

(b) 5

(c) 25

(d) 4

13. One side of a right angled triangle is double the other, hypotenuse is 10 cm. The area of the triangle in sq cm will be …?

(a) 40

(b) 50

(c) 33

(d) 20

14. If the length of a rectangle(a) is cut by 4 cm and the width increased by 3 cm, a square with the same area equal to that of A would result. The perimeter of A is .. cms?

(a) 44

(b) 46

(c) 48

(d) 50

15. In a circle of radius 10 cm, a chord is drawn 6 cm away from its centre. If a chord half in length is to be drawn, its distance from centre would be …cm?

(a) 9.3

(b) 9

(c) 8

(d) 9.42

16. When the price of a radio was reduced by 20% its sales increased by 80%. Net effect on the sale is …%

(a) 44

(b) - 44

(c) 66

(d) 75

Q. 17-21: There is a group of 5 persons – A,B,C,D,E consisting Professor of each Philosophy, Economics and Psychology. A & D are ladies having no specialisation and are also unmarried. No lady is a philosopher or an economist. There is a married couple with E as husband. B is brother of C and is neither a psychologist nor an economist.

From the above information, please answer to the following questions:-

	(a)	(b)	(c)	(d)
17. Who is the wife of E?	A	B	C	D
18. Which group consists of all males	ABC	BCD	BC	BE
19. Who is the Professor of Philosophy?	A	B	C	D
20. Who is the Professor of Economics?	A	B	C	E
21. Who is the Professor of Psychology?	A	B	C	D

Q. 22-26: There are four friends A,B,C,D. One of them is a cricketer and studies Chemistry and Biology. A & B play football. Both football players study Maths. D is a boxer. One football player also studies Physics. The boxer studies Maths and Accounts. Each of the friends studies two subjects and plays only one game

	(a)	(b)	(c)	(d)
22. Who studies Accounts and plays football?	A	B	A or B	D
23. Who studies Physics?	D	A or B	B	A
24. Who does not study Maths?	A	B	C	D
25. Who is a cricketer?	A	B	C	D

ANSWER KEY - TWENTY MINUTE TESTS - 1

1 (d)	2 (a)	3 (b)	4 (c)	5 (b)

6 (b)	7 (b)	8 (b)	9 (c)	10 (c)
11 (d)	12 (b)	13 (d)	14 (d)	15 (a)
16 (a)	17 (c)	18 (d)	19 (b)	20 (d)
21 (c)	22 (c)	23 (b)	24 (c)	25 (c)

Twenty Minute Test - 2

1. Ram spent $\frac{1}{2}$ of the money that he had at the village fair. What percentage of the amount that he had spent remains with him now?
(a) 66.67%
(b) 33.33%
(c) 16.67%
(d) 50%.

2. Rekha gave 8 marbles to Shamma and is now left with 42 marbles. What % of the original number of marbles remains with her?
(a) 8%
(b) 16%
(c) 84%
(d) 42%.

3. Lakhan sold 8 loaves (out of the 15 he had) in his bakery. What % did he sell?
(a) 53.33%
(b) 52.75%
(c) 56.67%
(d) 51.67%.

4. Mary gave her younger sister $\frac{5}{6}$ of number of her frocks and donated $\frac{1}{2}$ of her stock. What % of frocks is she left with?
(a) 62.5%
(b) 37.5%
(c) 66.67%
(d) 33.33%.

5. Lila has 3 sisters and 5 brothers. What % of her siblings is comprised of sisters?
(a) 60%
(b) 37.5%
(c) 33.3%

(d) 44.4%.

6. Ramu has 20% of all marbles in Myopia. If there are 180 marbles in Myopia, how many marbles does Ramu have?
(a) 144
(b) 216
(c) 36
(d) 900.

7. Rahul earned a 30% commission on the revenues he had generated. If he had generated a revenue of Rs. 79. How much did he earn?
(a) 24.3
(b) 23.7
(c) 22.85
(d) None of these.

8. Bonny has 30% red shirts in his wardrobe. How many red shirts does he have if he had 70 shirts?
(a) 30
(b) 21
(c) 49
(d) 59.

9. 30% of a ft is how many inches?
(a) 3
(b) 4
(c) 3.6
(d) 4.5.

10. 35% of a kilometer is how many metres?
(a) 350
(b) 490
(c) 560
(d) 1040.

11. 33% of a litre is how many ml?
(a) 333
(b) 330
(c) 300
(d) 33.33.

12. 30% of a tonne is how many kilograms?
(a) 30
(b) 300
(c) 3000
(d) 300×10^3.

13. Shankar produces 7 units, which was 20% of the target set for him. What is Shankar's target?
(a) 28
(b) 35
(c) 14
(d) None of these.

14. Sudhir ate 20% of the bananas and 30% of the apples in his stock. If there were 40 bananas and 20 apples, how many fruits did he eat?
(a) 18
(b) 15
(c) 16
(d) 14.

15. Arvind bought 7 out of 18 eggs and 8 out of 17 breads. What % of bread and eggs did he buy?
(a) 42.85%
(b) 43.28%
(c) 87.23%
(d) 39.33%.

16. Keya read $\frac{1}{2}$ of a 49-page business newspaper. It also carried a 21-page supplement which she did not read. What % of the newspaper and supplement did she read?
(a) 16.67%
(b) 10%
(c) 20%
(d) 33.33%.

17. Leela gave 25% of her red beads to Sheela. Sheela gave her 4 yellow beads in return. If Leela originally had 8 red beads and 7 yellow beads then what % of her current collection comprises of yellow beads?

(a) 41.18%
(b) 64.70%
(c) 57.90%
(d) 36.85%.

18. Manoj sells 20% diesel and 12 litres of petrol every day. If he has 9 litres of diesel and 20 litres of petrol what % of his stock in litres does he totally sell?
(a) 52%
(b) 50.72%
(c) 44.35%
(d) 47.6%.

19. Madonna sold 7000 copies of her new book at a sale. This amounts to 11% of her total sales. How many books did she sell?
(a) 77000
(b) 63636
(c) 63333
(d) None of these.

20. Pravin had 9 pairs of shoes but lost laces for 4 of them. What % of the total pairs can he wear now?
(a) 55.55%
(b) 44.44%
(c) 22.22%
(d) None of these.

21. Sunil gave 30% of his earnings to his father, 20% of the remainder to his his mother. What % does he retain with himself?
(a) 50%
(b) 56%
(c) 66%
(d) None of these.

22. In an army of 30,000 jawans 28% jawans are below the age of 30 years. How many new recruits should be taken to make the figure 40%?
(a) 14,400
(b) 12,000
(c) 3,600

(d) 6,000.

23. Suzy had 3 children A, B and C. A's age was 20% more than C but 10% less than B. If Malati was thrice as old as C at 45 years, what is then B's age?

(a) 20

(b) 16

(c) 18

(d) Data inadequate

24. Rohan had scored 70% more in maths than in science. However, the science paper was for 50 marks and maths for 150. If Rohan got 81 marks in both subjects, what was his % in science?

(a) 34%

(b) 36%

(c) 54%

(d) 60%

25. Latif got late on 60% of all his school days. On the remaining days he remained absent on 35% of the days. If school opened for 325 days, how many days was he on time?

(a) 85

(b) 16

(c) 62

(d) 45.

ANSWER KEY - TWENTY MINUTE TESTS - 2

1 (d)	2 (c)	3 (a)	4 (a)	5 (d)
6 (c)	7 (b)	8 (b)	9 (c)	10 (a)
11 (b)	12 (b)	13 (b)	14 (d)	15 (a)
16 (c)	17 (b)	18 (d)	19 (b)	20 (a)
21 (b)	22 (d)	23 (d)	24 (d)	25 (a)

Twenty Minute Test - 3

1. Nitin sold 3 goats and now he has 12 goats left. In % terms how many did he sell?
(a) 25%
(b) 20%
(c) 33.33%
(d) 30%.

2. Sunita has 18 marbles and Anita has 24. What % of the total marbles belong to Anita?
(a) 42.84%
(b) 57.12%
(c) 75%
(d) 52.28%.

3. Priya lost all, but 3, out of 11 of her dolls. What % of the dolls did she not lose?
(a) 33.33%
(b) 30%
(c) 27.27%
(d) 28.28%.

4. Susan's watch loses 12 seconds every hour. In % terms how much does it lose daily?
(a) 3.33%
(b) 0.33%
(c) 0.0033%
(d) 8%.

5. Kedar has 7 servants. If he removes 2 of them, what % of servants he has removed?
(a) 28.57%
(b) 40%
(c) 22.22%
(d) 71.42%.

6. Sohan has 20 servants. 30% are male. How many female servants does he have?

(a) 17

(b) 14

(c) 13

(d) 7.

7. Raghu ate 15% of the mangoes. If there were 24 mangoes how many did he eat?

(a) 160

(b) 28

(c) 24

(d) 3.6

8. In a question paper 3.5% of the questions are not solvable. In a test paper of 180 questions how many are solvable?

(a) 173.7

(b) 170.3

(c) 6.3

(d) 166.3.

9. 14.5% of a kilo meter is how many metres?

(a) 232

(b) 1450

(c) 145

(d) 2320.

10. 18% of all red roses and 30% of white roses have a disease called the Chacha syndrome. If there are 50 roses of each colour, how many have the Chacha syndrome?

(a) 48

(b) 24

(c) 12

(d) 36.

11. 60% of a quintal is how many grams?

(a) 60

(b) 600

(c) 6000

(d) 60000.

12. 7% of half a quintal is how many kilograms?
(a) 7
(b) 14
(c) 3.5
(d) 140.

13. "Rip van Winkle" sleeps after every 3rd awake month for 3 years. What % of the months does he not sleep?
(a) 25%
(b) 75%
(c) 8.33%
(d) 33.33%.

14. Of 80 Barbie dolls, $\frac{1}{2}$ have one eye bigger than the other and $\frac{1}{2}$ have one arm bigger than the other. If no doll has both the defects, then how many dolls are defective?
(a) 32
(b) 52
(c) 28
(d) 42.

15. Alroy gave his sister 40% of his pocket money and was left with Rs. 8. What is his pocket money (Rs.)?
(a) 20
(b) 13.33
(c) 11.20
(d) 12.80.

16. Robin scored 70 runs of the team's total of 250 runs. If the opener scored another 70 runs. What % did the rest of the team score (assume no extras)?
(a) 44%
(b) 56%
(c) 46%
(d) Insufficient data.

17. What % of days February contributed to the year in the year 1996?

(a) 10%
(b) 8.33%
(c) 7.9%
(d) 7.2%.

18. Ajay scored 70% of his runs in Delhi. If he scored 560 runs in Delhi, how many runs did he totally score?
(a) 700
(b) 728
(c) 800
(d) 720.

19. In a family of 9 males and 11 females, a pair of twin sisters are born. What is the percentage of females in the family now?
(a) 50%
(b) 55%
(c) 45%
(d) 59%.

20. 125% of 80 is ______ % of 150.
(a) 50%
(b) 66.67%
(c) 33.37%
(d) 87.5%.

21. Ravi bought a crate of mangoes, 20% of the mangoes was damaged. Another 10% were raw. Of the remainder, only 80% were fit for consumption. If 168 mangoes were fit for eating, how many were there in the crate?
(a) 210
(b) 300
(c) 312
(d) 240.

22. 15% of Malati's sales comprised strawberry cones, 20% mango cones, 12% candies and 159 vanilla cups. How many candies did she sell?
(a) 300
(b) 141
(c) 36

(d) 60.

23. Raj had three notebooks X, Y, Z. If X had 120 pages, Y had 10% more and Z 10% less. Her son Kaushtabh tore 5,10,15% of pages respectively. What % of pages were torn?

(a) 8%

(b) 10%

(c) 7%

(d) 13%.

24. Sid and Amit had an agreement that every time Sid got a blue ball he would get 10% of Amit's money and Amit would get 20% of Sid's if he found a red ball, Sid and Amit both had Rs. 100 each. Sid found 3 blue and Amit 2 red balls. How much money did Sid gain?

(a) 4

(b) - 22

(c) 12

(d) Cannot say.

25. Roy drove at a speed 30% greater than usual speed when he is outside the city limits. If he normally drove at 60 kmph in the city, then how long did he take to do a 624 km stretch on the highway?

(a) 20.8hrs

(b) 8hrs

(c) 10.4hrs

(d) 7.3hrs.

ANSWER KEY - TWENTY MINUTE TESTS - 3

1 (b)	2 (b)	3 (c)	4 (b)	5 (a)
6 (b)	7 (d)	8 (a)	9 (c)	10 (b)
11 (d)	12 (c)	13 (c)	14 (b)	15 (b)
16 (a)	17 (c)	18 (c)	19 (d)	20 (b)
21 (b)	22 (c)	23 (b)	24 (d)	25 (b)

Twenty Minute Test - 4

1. Chandru sanctioned only 85% of the amount on a travel voucher and 75% on food bills. If Ram filled a claim for Rs. 720, 40% being travel and 60% being food, then how much will Chandru give him?
(a) 583.2
(b) 558.8
(c) 568.8
(d) 562.6.

2. A's income is 30% more than B's and their expenses are 80% and 70% of their incomes. By what % is A's savings more/less than B's?
(a) 13.33% more
(b) 13.33% less
(c) 16.67%more
(d) 20% less.

3. In an election half of the electorate voted for BJP, a third of the remaining voted for Congress, a fourth of the remaining voted for JDU and a fifth of the remaining voted for the independent. What % of the total electorate vote in that election?
(a) 20%
(b) 22.5%
(c) 80%
(d) 77.5%.

4. A buys a property which appreciates for the first two years at 10 and 11% respectively. In the year 3 and 4, it depreciates by 11 and 10% respectively. How much is the property worth now?
(a) 100%
(b) 97.8%
(c) 102.2%
(d) 96.4%.

5. A runs 40% of the mile race before dropping out. B runs 60% of a km race before dropping out. What % more or less than B did A run?
(a) 6.25% more

(b) 6.67% more
(c) 6.67%less
(d) 6.25%less.

6. Ramesh filled 30% more gas than Suresh in his bike. Suresh's fuel tank however had 20% lesser capacity. Whose bike contained more gas % terms?
(a) Ramesh
(b) Suresh
(c) Equal
(d) Cannot say.

7. A ran 20% distance more than B and 12% less than C. If B dropped out after 28% of the race, how much of the race did A and C run between them?
(a) 33.6%
(b) 38.18%
(c) 68.24%
(d) 71.78%.

8. Rohit eats 30% of chapatis that Raghu eats, but 8% of the rice that Raghu eats. If there are 72 kgs of rice and 39 chapatis, then what % of the total food does Raghu consume?
(a) 50.4
(b) 14.4
(c) Insufficient data
(d) None .

9. A increases prices of his items by 80% and sees a drop of 80% in sales. How much does his revenue increase or decrease by?
(a) 90% decrease
(b) 36% decrease
(c) 10% decrease
(d) 64% decrease.

10. The population of a town increases from 50,375 to 72,412 in 10 years. What is the average simple annual increase per cent in the population?
(a) 1.4%
(b) 4.4%

(c) 3.04%
(d) 3.74%.

11. The length of a line 3.5 cm was found by measurement to be 3.4 cm. What is the error per cent?
(a) 2.73%
(b) 2.94%
(c) 3.34%
(d) 2.86%.

12. 12.5% of the salary is deducted in a firm for PF. An employee receives Rs. 420 per month after his contribution for the provident fund has been deducted, What is his salary?
(a) 960
(b) 472.5
(c) 480
(d) 787.5.

13. A house agent's commission for letting property is 7.5% on each year's rental. If his annual earnings on letting house is Rs. 112.5 lakhs, how much (in lakhs) does the owner of the house receive after paying the agent's charge?
(a) 1500
(b) 121.62
(c) 1387.5
(d) 1250.

14. 720 buses operate from a depot, but 168 are called back for repairs. What percentage of the total are in service of the roadways?
(a) 76.67%
(b) 23.33%
(c) 80%
(d) 20%.

15. A contractor makes an estimate of Rs. 2,87,500 for a cavity wall insulation, but then offers an 8% discount for immediate payment. The discount offered is Rs.
(a) 26450
(b) 22880

(c) 23000
(d) 25622.

16. 132 trains arrive at a railway station and it is found on an average that 99 of them are on time. What % of the trains are late?
(a) 33.33%
(b) 25%
(c) 75%
(d) 66.67%.

17. A small town's industries have a total turnover of Rs. 25,00,000. If the town council plans to raise Rs. 170,000 through octroi levy, what % levy will it charge on the industries?
(a) 8.5%
(b) 6.8%
(c) 4%
(d) 1.7%.

18. The sales of a company in the year 2014 - 15 was 25% less than that in the year 2015 - 16. If the sales in 2014 - 15 were Rs. 225 crores, what was the sales in Rs. Crores in 2015 - 16?
(a) 300
(b) 281.5
(c) 270
(d) None of these.

19. If the salary of Priya is 4 times more than that of Kavita, then how much is Kavita's salary in terms of Priya's salary?
(a) 80%
(b) 75%
(c) 25%
(d) None of these.

20. The total cost of 4 kg sugar and 6 kg of tea leaves = Rs. 660. If the price of sugar rises by 20%, then the total expenses on sugar and tea works out to Rs. 702. The original price of 1 kg of sugar (Rs.) before the increase is?
(a) 52.50
(b) 63

(c) 75
(d) Data inadequate

21. If $2\sqrt{2}$ = 4.899, then the value of $\sqrt{\frac{8}{3}}$ is:

(a) 544
(b) 2.666
(c) 1.633
(d) 1.333

22. If $\sqrt{x}$ = 2.448, then the value of $\frac{3\sqrt{2}}{2\sqrt{3}}$ is

(a) 1.223
(b) 1.224
(c) 8.816
(d) 0.612

23. If the price of an article with a 7% import duty is Rs. 321, what is its price (Rs.) before the duty?
(a) 343.5
(b) 298.5
(c) 300
(d) 310.

24. A earns 20% more than B and spends 10% more than B. If B saves Rs. 600 from his income of Rs. 6000 how much does A save?
(a) 1840
(b) 1260
(c) 144
(d) 108.

25. A manufacturer increases the price of an article by 24%. Till what % reduction in volume sales, will the manufacturer total sales remain greater than or equal to the original sales value?
(a) 24%
(b) 20%
(c) 19.35%
(d) 19.65%.

ANSWER KEY - TWENTY MINUTE TESTS - 4

1 (c)	2 (b)	3 (c)	4 (b)	5 (b)
6 (a)	7 (a)	8 (d)	9 (d)	10 (b)
11 (d)	12 (c)	13 (b)	14 (a)	15 (c)
16 (b)	17 (b)	18 (a)	19 (d)	20 (a)
21 (a)	22 (b)	23 (d)	24 (b)	25 (c)

Twenty Minute Test - 5

1. Sita and Gita together have Rs. 3500. Sita has Rs. 1500 more than Gita and she requires Rs. 500 more to purchase a T.V. set . The cost of TV set is Rs.?
(a) Rs. 2000
(b) Rs. 2500
(c) Rs. 3000
(d) None of these.

2. The L.C.M. and H.C.F. of two numbers are 720 and 60 respectively. If one of the numbers is 180, what is the second number?
(a) 120
(b) 60
(c) 480
(d) 240.

3. 67952 × 997 is equal to –
(a) 67748144
(b) 67758144
(c) 67648144
(d) none of these.

4. Find the smallest number which when divided by 14, 21, 24, 28 and 36 leaves a remainder of 2.
(a) 254
(b) 250
(c) 502
(d) 506.

5. The average score of a batsman for a number of innings was 42.5 runs. In the next 4 innings he scored 20, 75, 15, 0 runs and his average for all the innings fell by 5. How many innings in all did he play?
(a) 10
(b) 8
(c) 15
(d) 12.

6. The greatest number of 4 digits which is a perfect square is:
(a) 9990
(b) 9996
(c) 9836
(d) 9801.

7. Find the smallest number which when divided by 4,5, 6 and 8 leaves a remainder 3, but when divided by 9 leaves no remainder.
(a) 363
(b) 723
(c) 243
(d) 1089.

8. Find the least number of 4 digits which when divided by 8, 12, 16, 36 leaves remainders 6, 10, 14 and 32 respectively.
(a) 1008
(b) 1010
(c) 1152
(d) 1006.

9. The sum of all the prime numbers between 30 and 50 is?
(a) 156
(b) 199
(c) 166
(d) 162.

10. Which number K should replace both the question marks in K/49 = 784/K
(a) 7
(b) 196
(c) 14
(d) 28.

11. The average age of 32 students is 10 years. If the teacher's age is also included, the average age increases by one year. What is teacher's age in years?
(a) 33
(b) 21
(c) 43

(d) 53.

12. The value of $(101)^3$

(a) 303

(b) 1000003

(c) 1033001

(d) 1030301.

13. The smallest number which is a perfect square and contains 7936 as a factor is

(a) 12008

(b) 246016

(c) 61504

(d) 240616.

14. In a box, there were a dozen cakes of soap in each horizontal and vertical row. If there were 12 such layers in the box, find the total number of soap cakes in the box?

(a) 462

(b) 144

(c) 36

(d) None of these.

15. ¼ of marks obtained by Mr Boy in English & one third of marks in Hindi are equal. Total number of marks secured in both the subjects are 140. Marks obtained in English are?

(a) 60

(b) 80

(c) 75

(d) None of these.

16. L.C.M. of two numbers (a, b) is equal to twice the first number. Value of ‘b’is?

(a) > a

(b) < a

(c) = a or < a

(d) < 0

17. My friend Mr Yash Pal Malhotra and me meet every Sunday. The first time we met at 12.30, the next time at 1.20, then at 2.30, then at 4.00:

when did we meet after that?

(a) 5.20

(b) 6.20

(c) 6.00

(d) 5.50.

18. The square of a number consists of 8 digits. How many digits would there be in the number itself?

(a) 2

(b) 5

(c) 3

(d) 4.

19. What should come in place of (*) in the expression 38 *4 + 98 = 3942?

(a) 4

(b) 8

(c) 3

(d) 5.

20. A woman who was found shopping was asked, "How many number of pieces of fruits did you buy?" Her reply was, "they are all oranges but eight, all bananas but eight and all apples but eight !" How many fruits did she altogether buy?.

(a) Eight

(b) Twelve

(c) Sixteen

(d) Twenty four.

21. The difference between the squares of any two consecutive numbers (a,b) is equal to:

(a) ab

(b) ½ ab

(c) a + b

(d) (a + b)(a - b)

22. Bucket P has thrice the capacity as bucket Q has. It takes 60 turns for Bucket P to fill an empty drum. How many turns will it take for both the buckets P and Q, having each turn together, to fill the empty drum?

(a) 90

(b) 30
(c) 20
(d) 5.

23. One litre of crude oil yields about 20 ml of good quality fuel. 950 litres of crude oil shall yield how many litres of fuel?
(a) 1.9
(b) 19
(c) 190
(d) 19,000

24. The sum of two numbers is 24. If their product is maximum, then the numbers are?
(a) 8, 16
(b) 9,15
(c) 10, 14
(d) 12, 12.

25. If a fraction has its numerator increased by 1, it becomes equal 1/3, but if its denominator is increased by 1, it becomes equal to 1/4., what is the fraction?
(a) 1/2
(b) 1/3
(c) 2/9
(d) 4/15.

ANSWER KEY - TWENTY MINUTE TESTS - 5

1 (c)	2 (d)	3 (a)	4 (d)	5 (d)
6 (d)	7 (c)	8 (d)	9 (b)	10 (b)
11 (c)	12 (d)	13 (b)	14 (d)	15 (b)
16 (c)	17 (d)	18 (d)	19 (a)	20 (b)
21 (c)	22 (d)	23 (b)	24 (d)	25 (d)

DATA SUFFICIENCY

34

This section is designed to test your reasoning and logical ability. Like the Problem Solving section, it requires a basic knowledge of the principles of arithmetic, algebra, and geometry and other topics. Each Data Sufficiency question consists of a mathematical problem and two statements containing information relating to it. You must decide whether the problem can be solved by using information from:

(A) the first statement alone, but not the second statement alone;

(B) the second statement alone, but not the first statement alone;

(C) both statements together, but neither alone; or

(D) either of the statements alone.

(E) if the problem cannot be solved, even by using both statements together.

Approaching Data Sufficiency problems properly will help you use this time wisely.

CARE: LEGENDS / options may vary from examination to examination. You are advised to go through the instructions carefully to select the right options.

Always keep in mind the fact that you are never asked to supply an answer for the problem; you need only determine if there is sufficient data available to find the answer. Therefore, don't waste time figuring out the exact answer. Once you know whether or not it is possible to find the answer with the given information you are through. If you spend too much time doing unnecessary work on one question you may not be able to finish the entire section.

STRATEGY FOR DATA SUFFICIENCY QUESTIONS

A systematic analysis can improve your score on Data Sufficiency sections. By answering three questions, you will always arrive at the correct place. In addition, if you can answer any one of the three questions, you can eliminate at least one of the possible choices so that you can make an intelligent guess.

The three questions are:

(i) Is the first statement alone sufficient to solve the problem?

(ii) Is the second statement alone sufficient to solve the problem?

(iii) Are both statements together sufficient to solve the problem?

As a general rule try to answer the questions in the order I, II, III, since in many cases you will not have to answer all three to get the correct choice.

Here is how to use the three questions:

(i) If the answer to I is YES, then the only possible choices are (A) OR (D). Now, if the answer to II is YES, the choice must be (D), and if the answer to II is NO, the choice must be (A)

(ii) If the answer to I is NO then the only possible choices are (B), (C), or (E). Now, if the answer to II is YES, then the choice must be (B), and if the answer to II is NO, the only possible choices are (C) or (E)

(iii) So, finally, if the answer to III is YES, the choice is ©, and if the answer to III is NO, the choice is (E)

A better way to see this is to use a decision tree.

If first statement sufficient? (I)

}{

}{

YES NO

}{ }{

Choice is (A) or (D) Choice is (B), (C) or (E)

}{ }{

If second statement sufficient? (II) Is second statement sufficient? (II)

}{ }{

}{ }{

------------------ ---------------------

YES NO YES NO

}{ }{ }{ }{

Choice is (D) Choice is (A) Choice is (B) Choice is (C) or (E)

Are both statements together Sufficient ? (III)

}{

---------}{ --------

YES NO

}{ }{

Choice is (C) Choice is (E)

To use the tree simply start at the top and by answering YES or NO move down the tree until you arrive at the correct choice. For example, if the answer to I is YES and the answer to II is NO, then the correct choice is (A). (Notice that in this case you don't need to answer III to find the correct choice.)

The decision tree can also help you make intelligent guesses. If you can only answer one of the three questions, then you can eliminate the choices that follow from the wrong answer to the question.

BUT

- *Please don't conclude immediately;*
- *Check all the possibilities before you conclude;*
- *Never conclude midway until you are sure of results.*

Advice 1: You know the answer to I is YES. You can eliminate choices (B), (c) and (E)

Advice 2: You know the answer to II is NO. You can eliminate choices (D) and (B) since they follow from YES for II

Advice 3 You know the answer to III is YES. You can eliminate choice (E) since it follows from NO for III

Advice 4: You know the answer to I is NO and the answer to III is YES. You can eliminate (E) since it follows from NO to III. You also can eliminate (A)k and (D) since they follow from YES to I

Since you get one raw score point for each correct choice and lose only one quarter of a point for an incorrect choice, you should guess whenever you can answer one of the three questions.

Read the following directions carefully and then try the sample Data Sufficiency questions below. Allow yourself 8 minutes total time. All numbers used are real numbers. A

ILLUSTRATIONS

DIRECTIONS: Each of the following problems have a question and two statements which are labeled (I) and (2). Use the data given in (I) and (2) together with other available information (such as the number of hours in a day, the definition of clockwise, mathematical facts, (etc.) to decide whether the statements are sufficient to answer the question. Then choose.

(A) if you can get the answer from (1) alone but not from (2) alone;

(B) if you can get the answer from (2) alone but not from (1) alone;

(C) if you can get the answer from (1) and (2) together, although neither statement by itself suffices;

(D) if statement (1) alone suffices and statement (2) alone suffices;

(F) if you cannot get the answer from statements (1) and (2) together, but need even more data.

All numbers used are real numbers. A figure given for a problem is intended to provide information consistent with that in the question, but not necessarily consistent with the additional information contained in the statements.

1. A rectangular field is 40 yards long. Find the area of the field

(1) A fence around the entire boundary of the field is 140 yards long

(2) The width of the field is more than 20 yards.

2. Is X a number greater than zero?

(1) X2 − 1 = 0

(2) X3 + 1 = 0

3. An industrial plant produces bottles. In 1961 the number of bottles produced by the plant was twice the number' produced in 1960. How many bottles were produced altogether in the years 1960, 1961 and 1962?

(1) In 1962 the number of bottles produced was 3 times the Number produced in 1960

(2) In 1963 the number of bottles produced was one half The total produced in the year 1960, 1961 and 1962

4. A man 6 feet tall is standing near a tube light fitted on the top of a Pole . What is the length of the shadow cast by the man?

(1) The pole is 18 feet high.

(2) The man is 12 feet from the pole.

5. Find the length of RS if z is 90 degree and PS = 6

(1) PR = 6

(2) X = 45 degree

6. Working at a constant rate it takes 'U' worker in all 3 hours to fill up a ditch with sand. How long would it take for 'V' workers to fill up the same ditch working without any changes.

(1) Working together but at the same time U and V can fill in the ditch in 1 hour 52 ½ minutes.

(2) In any length of time worker V files in only 60% as much as worker U does in the same time.

7. Did John go to the beach yesterday?

(1) If John goes to the beach, he will be sunburned the next day.

(2) John is sunburned today.

Analysis:

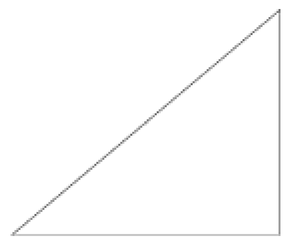
(1) (A) The area of a rectangle is the length multiplied by the width. Since you know the length is 40 yards, you must find out the width in order to solve the problem. Since statement (2) simply says the width is greater than 20 yards you cannot find out the exact width using (2). So (2) alone is not sufficient. Statement (1) says the length of a fence around the entire boundary of the field is 140 yards. The length of this fence is the perimeter of the rectangle, the sum of twice the length and twice the width. If we replace the length by 40 in P = 2L + 2W we have 140 = 2 (40) + 2W and solving for W yields 2W − 60 or W = 30 yards. Hence the area is (40)(30) = 1200 square yards. **Thus (1) alone is sufficient but (2) alone is not.**

(2) (B) Statement (1) means $X^2 = 1$, but there are two possible solutions to this equation, X = 1, X = -1. Thus using (1) alone you can not deduce whether X is positive or negative. Statement (2) means $X^3 = -1$ but there is only one possible (real) solution to this, X = -1. Thus X is not greater than zero which answers the question. **And (2) alone is sufficient.**

(3) (E) T, the total produced $P_0 + P_1 + P_2$, where P_0 is the number produced in 1960,P1 the number produced in 1961, andP2 the number produced in 1962. You are given that $P_1 = 2P_0$.Thus $2 = P_0 + P_1 + P_2 = 2P_0 + = 3P_0 + P_2$. So we must find out P_0 and P_2 to answer the question. Statement (1) says $P_2 = 3P_0$; thus by using (1) if we can find the value of P_0 we can find T. But (1) gives us no further information about P_0 . Statement (2) says T equals the number produced in 1963, but it does not say what this number is. Since there are no relations given between production in 1963 and production in the individual year 1960, 1961, or 1962 you cannot use (2) to find out what P_0 is. **Thus even (1) and (2) together are not sufficient.**

(4) (C) Sometimes it may help to draw a picture. By proportions or by similar triangles the height of the pole, h, is to 6 feet as the length of shadow, s, + the distance to the pole, x, is to s. So h/6 = (s + x)/s. Thus hs = 6s + 6x by cross multiplication. Solving for s gives hs − 6s = 6x. or s(h − 6) = 6x, finally we have s = 6x/(h − 6). Statement (1) says h =

18; thus s = 6x/12 = x/2, but using (1) alone we cannot deduce the value x. Thus (1) alone is not sufficient. Statement (2) says x equals 12; thus, using (1) and (2) together we deduce s = 6, but using (2) alone all we can deduce is that s = 72 / (h − 6), which cannot be solved for s unless we know h. **Thus using (1) and (2) together we can deduce the answer but (1) alone is not sufficient nor is (2) alone.**



(5) (D) Since z is a right angle,, $(RS)^2 = (PS)^2 + (PR)^2$. Sp $(RS)^2 = (6)^2 + (PR)^2$, and RS will be the positive square root of $36 + (PR)^2$. Thus if you can find the length of PR the problem is solved. Statement (1) says PR = 6, thus $(RS)^2 = 36 + 36$, so RS = $6\sqrt{2}$. Thus (1) alone is sufficient. Statement (2) says x = 45º but since the sum of the angles in a triangle is 180º and z is 90º them y = 45º. So x and y are equal angles and that means the sides opposite x and opposite y must be equal or PS = PR. Thus PR = 6 and RS = $6\sqrt{2}$ so (2) alone is also sufficient.

(6) (D) (1) says U and V together can fill in the ditch in 1 7/8 hours. U would fill in (1/3)(15/8) = 5/8 of the ditch. So V fills in 3/8 of the ditch in 1 7/8 hours. Thus V would take (8/3)(15/8) = 5 hours to fill in the ditch working by himself. Therefore, statement (1) alone is sufficient. According to statement (2) since U fills the ditch in 3 hours, V will fill 3/5 of the ditch in 3 hours. Thus V will take 5 hours to fill in the ditch working by himself.

(7) (E) Obviously, neither statement alone is sufficient. John could have gotten sunburned at the beach, but he might have gotten sunburned somewhere else. Therefore (1) and (2) together are not sufficient. This problem tests your grasp of an elementary rule of logic rather than your mathematical knowledge.

Data Sufficiency Test - 1

(A) **i**f the question can be answered with the help of ANYONE statement only.

(B) if the question can be answered with the help of EITHER of the statements.

(C) if the question can be answered with the help of BOTH statements.

(D) if the question CANNOT be answered even with the help of both the statements.

1. How many apples did Victor pick?
I. He ate half the apples plus half an apple and had 7 still left
II Picking @6 apples every half hour, it took him 1 hr 15 min to pick up.

2. Is Ram elder than Shyam?
I Shyam is elder than Rohan
II Rohan is elder than Ram

3. Can pipe A fill the tank in 2 hours (with both inlet pipes)
I The tank has only two piles A& B
II Together they fill the tank in 4 hours

4. Range of a group of # is Max-Min. If mean of 5 #s is 60, range of #s is
I The mean of 4 largest # is 72
II The mean of 4 smallest # is 44

5. If x & y are integers (Not = to 1) What is the value of y ?
I Mod $(x^2.y) = 36$
II $2\ xy = 24$

6. Is $z > 0$
I $x/y < 0$ and $z/y > 0$
II $x < 0$

7. Has the work done in the office daily increased by 10%?
I The %age of people absent is constant each day.
II The amount of work done by each employee daily has gone up by 10%

8. Is (a.b.c) > 0 ?

I ab < bc

II ac = _/ (ba)

9. Is the truck heavier than the car ?

I The car moves easily

II The truck faces a lot of friction

10. Is x between 1 and 2 ?

I square of x is > than 1 and < than 4

II square of x is > than x and < than 2x

11. From among 18 persons, One-third being men, a person is selected. The probability that the person selected is a man who is a college graduate?

I The probability that a person selected is a college graduate is 5/18

II The probability that a woman selected is 2/3

12. Is x > y ?

I x = 26.5

II Numerical value of y is always 40

13. Average of 3 quotations for a share is Rs.118. Is the highest quotation > than Rs.129 ?

I The lowest quotation is Rs.100

II One of the quotations is Rs.115

14. Should the Company produce a product Q?

I R is four times more profitable than Q

II A unit of Q requires 5 units of labour, A unit of R requires 2 units of labour and available labour hours is 100 with no other constrains

15. During a scheme if a shopper saved Rs.3 on the price of a soap, then the original price of the soap is?

I The original price was 25% higher than the price the shopper had paid.

II Other brands in the same segment cost Rs.12 per cake on an average

Data Sufficiency Test - 1 Answers with Hints

1 (b)	2 (c)	3 (a)	4 (c)	5 (d)

6 (c)	7 (c)	8 (c)	9 (d)	10 (a)
11 (d)	12 (d)	13 (c)	14 (c)	15 (a)

5. Values for x can be = 1,−1, 2,−2, 3,−3, 6,−6, for which respectively y = 36, 9, 4, 1

9. Neither of the statements gives weights of the vehicles.

11. First statement tells about the selection of a **person** not of a man

Second refers to selection of **women** and not of **men**

12. The value of Y could be 40 or - 40 ; hence no concrete results of be obtained.

Data Sufficiency Test - 2

(Time 15 Minutes)

(A) if the question can be answered with the help of ANYONE statement only.

(B) if the question can be answered with the help of EITHER of the statements.

(C) if the question can be answered with the help of BOTH statements.

(D) if the question CANNOT be answered even with the help of both the statements.

1. If a, b c are consecutive integers in the ascending order, what is the value of a?

I $b = (c + a)/2$

II $ac = -1$

2. In a class, some study Algebra and some Geometry while some study both. What fraction reads both subjects ?

I 3/4 of the students study Algebra

II 2/3 of these read Geometry

3. Saurabh walked from his home to school and back again. His average speed was ?

I He walked to his school at an average of 4 mph

II He walked to his home at an average of 5 mph

4. Train X leaves town A for B while train Y leaves B for A (both at constant speeds). Which train is faster (towns A,B,C are in line)

I Train Y arrives at town C before train X

II C is closer to B than to A

5. What is the radius of the circle?

I Ratio of its area to circumference is > 7

II Diameter of the circle is 32 units

6. How many people read the Hindu and the HT

I Out of 200 readers, 100 read Hindu. 120 HT

II Out of 200 readers, 100 read Hindu, 120 HT and 50 neither

7. What is the value of prime number X ?

I X(X+1) is a two digit number > 50

II Cube of X is a 3 digit number

8. How old is the teacher ?

I The average age of 10 students and their teacher is 35 years.

II The teacher is 10 years older than the oldest student in the class.

9. In an election 3 candidates A,B,C were contesting. How many votes did each receive?

I A received 1005 votes more than B and 1215 more than C

II Total votes cast were 15415 and there were 10% invalid votes.

10. What is the time difference b/w NY and London?

I The departure time at NY is 0900 hrs and arrival at London is 1000 hrs − same day

II The flight time is 5 hours

11. A says to B "I am 3 times as old as you were 3 years ago" How old is A?

I B's age 17 years from now is same as A's present age

II A's age nine years from now is 3 times B's present age

12. What is the area of the triangle ABC ?

I ABC is equilateral triangle side of 7 cms

II ABC is a right-angled triangle circumscribed by a circle of radius 5 cms

13. Is $x > y$

I $16x - 3y = 0$

II $2x = 4y + 4$

14. If a rope of 16m is cut into three pieces of unequal length, the shortest length is?

I The combined length of 2 pieces of rope is 12 meters

II The combined length of the 2 other pieces of rope is 11 meters

15. If X is divided by 2, the remainder is 1. What is remainder if it is divided by 4

I $31 < x < 35$

II x is a multiple of 3

Data Sufficiency Test - 2 Answers with Hints

1 (a)	2 (c)	3 (c)	4 (d)	5 (b)
6 (a)	7 (d)	8 (d)	9 (c)	10 (c)
11 (a)	12 (a)	13 (d)	14 (c)	15 (a)

4. It is not mentioned that C is how much closer

otherwise also Y is bound to reach C earlier than X because of lesser distance)

7. (1) possible values for x could be 7,8 or 9

(2) Possible values for x could be > 11

13. Neither statement holds water for − ve values of x, y

Data Sufficiency Test - 3

(Time 10 Minutes)

(A) if the question can be answered with the help of I (but not II) statement only.

(B) if the question can be answered with the help of II(but not I) statement.

(C) if the question can be answered with the help of BOTH statements.

(D) if each statement alone is sufficient to answer the question.

(E) if the question CANNOT be answered even with the help of both the statements.

1. Exactly how many of the 20 people are data processors in the office

I There are more than 10 data processors in the office

II The number of people in the office who are NOT data processors is 3.

2. Which of the towns A or B is farther from town C?

I The distance from town A to town B is 30 miles

II The distance from town B to town C is 20 miles.

3. What percent of X is Y ?

I Y/X = 2/5

II X is 250 % of Y

4. Container X holds how many more ltrs of liquid than container Z ?

I if Z were full and were emptied into empty X, X would be half filled.

II X and Z together hold 6 ltrs

5. If a salesman received a commission of $400 in January, what was his total amount of sales in that month?

I He received a commission of 10% on his first $1000 sales each month

II He received a commission of 20% on his 2nd $ 1000 sales each month.

6. Plane X is flying at 25,000 ft while Plane Y at 10,000 ft of altitude. If X starts descending just as Y starts to ascend, in what time both will reach same altitude?

I Plane X descends 9 times as fast as plane Y does

II Plane Y ascends @80 ft / sec and plane X descends @720 ft/sec

7. How long will a stenographer take to type − without break − at uniform speed?

I He types 10 pages per hour

II At the end of an hour, he has typed $2/5^{th}$ of his matter.

8. The distance b/w Centre of a circle and a point P on its circumference is__

I The circumference of the circle is 6 pie

II Area of the circle is 9 pie

9. What is the value of integer X ?

I X is an integer multiple of 2, 3 and 5

II $20 < x < 50$

10. Does line *"m"* contain the point P with coordinates (x,y)?

I m contains the origin

II $x = 0$

11. What is the total # of students in the 4 grades of the high school?

I There is an average of 200 students in each grade

II The 9^{th} grade has same # as the 10^{th} & each has 10 more than 11^{th} & 12^{th} grades

12. Is 'r' an integer ?

I $r > 0$

II $3(3) + 4(4) = r (r)$

13. What are the chances that a coin will come up heads on the 8^{th} toss ?

I the coin is unbiased and always comes up either tail or head

II The first seven tosses came up heads

14. How many cents are saved by buying 10 gallons of gasoline in State Y than State X?

I Gas costs 5 cents more per gallon in State X than in State Y

II Gas costs 25% more per gallon in State X than in State Y.

15. Rectangle ABCD is inscribed in a circle. What is the area of the Circle?

I The length of the arc CD is $1/3^{rd}$ of the circumference.

II The length of the rectangle is 8/3 cm and the width is 8 cm

Data Sufficiency Test - 3 Answers

1 (b)	2 (e)	3 (d)	4 (c)	5 (e)
6 (b)	7 (b)	8 (d)	9 (c)	10 (c)
11 (a)	12 (b)	13 (a)	14 (a)	15 (b)

2. Geographical location of the towns A,B & C is not given.
5. Rates of commission beyond sales of $ 2000 is not given

Data Sufficiency Test - 4

(Time 7 Minutes)

(A) if the question can be answered with the help of ONLY ONE of the statements.

(B) if the question can be answered with the help of EITHER statement.

(C) if the question can be answered with the help of BOTH statements.

(D) if the question CANNOT be answered even with the help of both the statements

1. Consider three real numbers X,Y & Z. Is Z the smallest of these numbers?

I X is greater than at least one of Y and Z

II Y is greater than at least one of X and Z.

2. Let X be a real number. Is the Modulus of X necessarily less than 3 ?

I $X(X + 3) < 0$

II $X(X - 3) > 0$

3. How many people are watching TV Programme P ?

I # of people watching Programme Q is 1000 and both P & Q is 100

II # of people watching either P or Q or both is 1500

4. Triangle PQR has angle PRQ equal to 90 degrees. What is value of PR + RQ?

I Diameter of the inscribed circle of the triangle PQR is equal to 10 cm

II Diameter of the circumscribed circle of triangle PQR is 18 cm

5. Harshad bought and sold these shares very next day each time paying 1% brokerage. His profit earned per rupee spent on these shares is ?

I The sales price per share was 1.05 times that of purchase price.

II The number of shares purchased was 100

6. For any two real numbers

a @ b = 1 if both a and b are +ve or both −ve

= −1 if one of the two numbers is +ve and the other −ve

What is (2 @ 0) @ (-5 @ -6)?

I a @ b is zero if a is zero.

II a @ b = b @ a

7. There are two straight lines in the x, y plane with equations

$ax + by = c$ $dx + ey = f$

Do these two lines intersect?

I a,b,c,d,e and f are real numbers

II c & f are non-zero

8. O is the centre of two concentric circles, AE is a chord of the outer circle and it intersects the inner circle at points b & d. C is a point on the chord in b/w b & d. What is the value of ac/ce?

I bc / cd = 1

II A third circle intersects the inner circle at b and d and the point c is on the line joining the centres of the third circle and the inner circle.

9. Ghosh Babu has decided to take a non-stop flight from Mumbai to No-man's-land in South America. He is scheduled to leave Mumbai at 0500 hrs on 23rd November 2016. What is the local time when he reaches there?

I The average speed of the plane is 700 kmph

II The flight distance is 10,500 km

10. What are the ages of two individuals X & Y?

I The age difference b/w them is 6 years.

II The product of their ages is divisible by 6

Data Sufficiency Test - 4 Answers with Hints

1 (c)	2 (a)	3 (c)	4 (c)	5 (a)
6 (c)	7 (d)	8 (b)	9 (d)	10 (d)

7. It is no where clear that a,b,c,d,e,f are distinct from each other

9. The stoppages in between the journey are not given, nor the time difference between the two locations has been given.

10. There could be 'n' number of age combinations like (12,18), (30,36), 54,60) and so on.

Data Sufficiency Test - 5

(Time 7 Minutes)

(A) if the question can be answered with the help of I (but not II) statement only.

(B) if the question can be answered with the help of II(but not I) statement.

(C) if the question can be answered with the help of BOTH statements.

(D) if the question CANNOT be answered even with the help of both the statements.

1. How many boys attended the class today?

I The class of 108, has a ratio of 1:8 b/w girls and boys

II There were five absentees including 3 boys.

2. Did the Gitanjali Express arrive at its scheduled time at 0550 hrs ?

I The Gitanjali Express left at 0607 hrs after 11 minutes of wait

II The Howrah Mail (arrival time 0612 hrs) arrived before Gitanjali Express

3. In a full deck of cards, I removed two cards. Do I have at least one heart?

I I have one red card

II One card is red and the other black

4. Can Mr Mehta post a profit on today's sales of 1,80,000 lakh shares of IMS Ltd at a price of Rs.154 per share

I He had purchased 95,000 shares at Rs.132 per share

II He had purchased 85,000 shares at Rs.186 per share

5. Is the angle B of a Triangle ABC an acute angle, if it has at least one obtuse angle?

I Angle A = 102*

II Angle C = 42*

6. A triangle is inscribed in a circle. Is one of the angles 90*

I The triangle is isosceles

II One of the sides of the triangle is equal to the diameter of the circle.

7. A four side figure is inscribed in a circle. Is it a square?

I The internal angles are supplementary

II One of the diagonals is equal to twice the radius

8. There are 3 red, 2 blue , 3 white balls in a bag. I remove 2 balls at random, without replacement. Do I have at least 1 blue ball?

I I pick 5 more balls and there is at least one of each colour.

II I pick 5 more balls and there are 2 red and 2 white among them.

9. I sell goods for Rs.45,000 to Harsh. Do I make a profit?

I. Harsh resells it at a profit.

II Harsh resells it for Rs.48,000/-

10. A,B,C stay in a building. Does A stay above B?

I B stays on the same floor as C.

II C stays on the ground floor.

Data Sufficiency Test - 5 Answers with Hints

1 (c)	2 (c)	3 (d)	4 (b)	5 (a)
6 (b)	7 (d)	8 (d)	9 (d)	10 (d)

3. I There could be other red cards also other than heart

II It is but natural that the card has to be red or black.

7. I Any quadrilateral inscribed in a circle has this result.

II It could be a square or a rectangle.

8. I Five more balls picked up could have different combinations even if one ball is of each colour like (1,1,3), (2,1,2), (3,1,1) and so on.

II Five picked up in the second leave 3 more balls (1+1+1) which does not ensure the favourable result.

9. In neither of the statements my cost price can be derived.

10. A could stay on Ground floor, higher floors, basement etc

Questions-Bank Test No. 101

Percentagia - 01 **Maximum Time : 10 Minutes**

- *Compute orally as much as you can.*
- *Use standard fractions wherever approximation exists (e.g. 16.666 be read as 16-2/3 and so).*
- *Table memorization will be of immense help.*

(1) 3 % of 30 =____________________

(2) 55.555% of 11700 =____________

(3) 87.5% of 1.04 =_________________

(4) 25% of 30100 =_________________

(5) 77.77% of 8100 =________________

(6) 12.5% of 9200 =_________________

(7) 33.33% of 25800 =______________

(8) 37.5% of 2000 =__________________

(9) 44.444% of .63 =_________________

(10) 62.5% of 5200 =_________________

(11) 88.88% of 18.09 =________________

(12) 83.33% of 276 =_________________

(13) 75% OF 7.5_____________________

(14) 58.333% of 16.8 =________________

(15) 25% of 55 =_____________________

(16) 1/5% of 2500 =__________________

(17) 11 1/9% of 67.5 =________________

(18) 50% of .003 =___________________

(19) 5.555% of 804600 =______________

(20) 35% is 35000 =__________________

(21) 75 is __ % of 375________________

(22) 80% of 12.5 =___________________

(23) 30% of 540 =___________________

(24) 3% of 9________________________

(25) 62.5% of __is 800_______________

(26) 25% of 350 =___________________

(27) 15% of 125 is___________________

(28) 6% of 7600 =___________________

(29) 62½%of __ is 369.6______________

(30) 2.5 is __% of 95_________________

Now take a calculator to compute these and find:

how many were right?

how many were wrong?

why & why???????

Questions-Bank Test No. 102

Number of Questions: 26 ***Suggested time: 15 minutes***

(***For All Questions Bank Tests: Mark 'e' as answer for none of the choices)***

	[A]	[B]	[C]	[D]
Part A :find square root of				
(1) 7056	84	94	74	86
(2) 170569	412	413	433	437
(3) 7921	79	99	81	83
(4) 9.3636	3.6	3.06	3.03	3.04
(5) 17.64	4.8	4.2	5.2	5.8
(6) 204304	457	462	442	452
(7) 7586.41	86.1	87.1	97.9	77.1
(8) 5776	75	76	86	87
(9) 8464	98	92	93	88
(10) 10. 28.09	6.3	5.7	5.3	5.4
(11) 519841	722	731	721	729
(12) .144 × .0169 × .0256 =	.2496	.02496	.002496	2.496
(13) 36 × 49 × 64 × 25	168	1680	1680	0 168000
(14) 72 × 48 × 216 864	1728 3456	432		
(15) 240 × 360 × 8 × 27 36	216	1296	728	
(16) 490 × 360 × 2500	2100	21000	210000	210
(17) 35 × 45 × 65 × 63 × 2197	2197	2197 × 169	2197 × 13	none
(18) 6724 =	82	87	92	97
(19) 596225 =	795	815	725	890
(20) 169744 =	387	392	387	492
(21) 389376 =	696	624	724	596
(22) 15825.64 =	125	312	.8125	131.5
(23) (.00000576)	.24	.024	.0024	.00024
(24) (0.00002025)	.45	.045	.0045	.00045
(25) (.9 X.016 X 0.25)	.6	.06	.006	.0006
(26) (.08 × 0.009 × .0016) =	.0108	.00108	.108	.0011

how many did you do? ***how many could you do?*** ***how many were wrong?***

Questions-Bank Test No. 103 **(Total 15 Minutes)**

Q. 1 Complete the following block of tables ***(in 5 minutes)***

	9	7	8	12	11	6	10	5	14	15	13
15	135										
17											221
12									168		
8						48					
14								70			

Q. 2 Replace *Y* by appropriate multiple/factors if x.y is = ***(2 minutes)***

find y	?	?	?	?	?	?
x = 15,xy =	195	240	315	495	5580	825
find y	?	?	?	?	?	?
x = 17,xy =	289	238	374	867	1989	2601

Q. 3 fill in the blank: ***(in 3 minutes)***

X^2	X^3	X
16	?	?
?	64	?
?	?	27
?	125	?
?	?	1.5
?	?	$5\sqrt{5}$

Q. 4 fill in the blanks ***(in 5 minutes)***

X^2	Y	Z	X/Y	XYZ	Y^2
45	9	?	?	?	?
?	12	?	15	?	?
18	?	12	6/7	?	?
?	?	?	7/3	?	24
?	?	2	3	?	121

Questions-Bank Test No. 104

Number of Questions: 26 NUMBERS GAME Suggested time: 15 minutes

1. To make 78514k6 divisible by 8 and also by 3, “k” should be
(a) 2
(b) 8
(c) 5
(d) 6

2. A number divisible by 9 but not by 8 is
(a) 785464
(b) 856872
(c) 758462
(d) 548766

3. Product of 4 consecutive even numbers must have __as a factor.
(a) 8
(b) 16
(c) 32
(d) 64

4. Sum of any 10 consecutive numbers must have .. as a factor.
(a) 11
(b) 4
(c) 8
(d) 6

5. The relation $(98 \times 98 - 18 \times 18)$ must have __as a factor
(a) 15
(b) 80
(c) 32
(d) 64

6. Last digit in 28 to the power 97 is
(a) 8
(b) 4

(c) 2
(d) 6

7. The last digit in $(7 \times 8 \times 9 \times 4 \times 6 \times 3 \times 7)$ is
(a) 2
(b) 3
(c) 4
(d) 6

8. The number (89715938*) is divisible by 4. Non-zero (*) is
(a) 2
(b) 3
(c) 4
(d) 6

9. $R = gS - 4$, when $S = 8$, $R = 16$, find g
(a) 2
(b) 2.5
(c) 3
(d) 3.5

10. The number 798k463 must be divided by 11, k can be
(a) 3
(b) 7
(c) 18
(d) 13

11. What should be multiplied to 13 to have all nines
(a) 75923
(b) 76933
(c) 76924
(d) 76923

12. What should be multiplied to 7 to have all fives
(a) 79366
(b) 79364
(c) 78365
(d) 79365

13. Sum of all primes b/w 120 & 130 must be more near to

(a) 260
(b) 250
(c) 130
(d) 135

14. How many prime numbers are there b/w 70 and 90
(a) 4
(b) 5
(c) 6
(d) 7

15. How many factors of 72 are there other than 1 and itself
(a) 8
(b) 9
(c) 10
(d) 11

16. The numbers 78914, 856405 and 384540 are all divisible by
(a) 9
(b) 11
(c) 17
(d) 15

17. The difference of squares of 145 & 55 must be divisible by
(a) 18
(b) 1800
(c) 18000
(d) 180

18. The sum of cubes of 87 and 34 must always be divisible by
(a) 11
(b) 121
(c) 143
(d) 22

19. Two numbers whose sum or difference have 11 as a factor should be such that --- may be multiples of 11
(a) one
(b) neither

(c) both
(d) no value

20. An odd # and an even # operated with +,−,×,division must be
(a) odd
(b) even
(c) any #
(d) not certain

21. The number of primes b/w 200 & 250 must be b/w __ & ___
(a) 4 , 6
(b) 6, 8
(c) 8 , 10
(d) 2 , 4

22. The sum of squares of all $N < 46$ must have one factor as
(a) 15
(b) 315
(c) 161
(d) all of them

23. Sum of all numbers possible using only 3,5 or 7 is
(a) 3520
(b) 3130
(c) 3330
(d) 3100

24. The difference of a three-digit number and the number when digits interchanged with all possibilities must be divisible by..
(a) 9
(b) 99
(c) 11
(d) 999

25. The left digit of a two-digit number when multiplied to the number results as 280, difference of the digits is
(a) 2
(b) 3
(c) 1

(d) can’t say

26. A two digit number whose sum of digits is 14 when decreased by 54, the digits are reversed. Square of the number is

(a) 3576

(b) 2916

(c) 3116

(d) none

Questions-Bank Test No. 105

Number of Questions: 25 QUANTITATIVE APTITUDE **Suggested time: 15 minutes**

1. The average of 4 , 5 , 3.5 , 7.5 , 9.5 and 5.0 is
(a) 5.0
(b) 5.2
(c) 5.5
(d) 7.5

2. The length of a rectangle is 1 cm more than the width,while perimeter is 14 cm. Find its area _____ sq. cm
(a) 16
(b) 14
(c) 12
(d) 10

3. Two nos. differ by 215, greater is 827, smaller number is
(a) 621
(b) 612
(c) 522
(d) 521

4. The sides of two squares are 3:4, their areas are in ratio
(a) 3/4
(b) 6/7
(c) 9/16
(d) 4/3

5. The smallest among .5 , .499 , .049 , .0599 is
(a) .05
(b) .499
(c) .049
(d) .0599

6. square root of (36 + 64) is
(a) _/36 + _/64

(b) _/36 × _/64
(c) _/64 / _/36
(d) 10

7. The price gone down by 10% can be restored by raising it by
(a) 10
(b) 11.11
(c) 9.9
(d) 11

8. By selling a fan for Rs.475 a person loses 5%. To gain 5% he should sell it for
(a) 500
(b) 525
(c) 535
(d) 575

9. In an exam 900 were boys and 1100 girls. 32% boys and 38% girls passed. What % of the strength failed
(a) 35.3
(b) 62.0
(c) 64.7
(d) 68.0

10. Sita buys for 70 and sells at 20% loss. Selling price will be
(a) 60
(b) 58
(c) 56
(d) 55

11. The length is five-fourths of breadth. Perimeter is 243m then the area of rectangle is ___sq. m
(a) 364.5
(b) 3645
(c) 1458
(d) 14580

12. .0072 × .0405 =
(a) .0027

(b) .0029
(c) .00028
(d) .00029

13. CI exceeds SI for 2 years @ 15%p.a. by Rs.144/ =. Sum is Rs.
(a) 6000
(b) 6200
(c) 6400
(d) 6300

14. The smallest perfect square divisible by 2,3,4,5 and 6 is
(a) 784
(b) 900
(c) 3600
(d) 2500

15. 10101.0101 × 100 is equal to
(a) 101.10101
(b) 1010101.1
(c) 101.010101
(d) 1010101.01

16. 0.0000000192 is also = K × 10 to the power -7 where K is
(a) .192
(b) 1.92
(c) .0192
(d) 192

17. (6.359 × 6.359 − 2.641 × 2.641) divided by (6.359 − 2.641) =
(a) 3.718
(b) 9.718
(c) 9.000
(d) 0

18. The rope with which a donkey can be tathered to cover 2464 sq. m of an area is __m radius
(a) 28
(b) 56
(c) 168

(d) 88

19. The largest N by which product of 3 consecutive N is divisible by is
(a) 2
(b) 3
(c) 6
(d) 12

20. Difference b/w SI & CI on Rs. 1625 for 3 yrs @ 4%p.a. will be
(a) 7.95
(b) 7.9
(c) 7.2
(d) 7.75

21. Six men working 8 hrs earn Rs.720 per week. 8 men working 6 hrs will earn how much in two weeks
(a) 1080
(b) 1440
(c) 1920
(d) 2160

22. The sides of a rectangle are 7:4 and area 252 sq.m Cost of fencing @ Rs.5 per meter is
(a) 66
(b) 264
(c) 330
(d) 462

23. The largest 4 digit number & the smallest 5 digit differ by
(a) 1
(b) 99
(c) 999
(d) 9999

24. Price gone up by 25% shall compel a cut in consumption by
(a) 25
(b) 20
(c) 33
(d) 16

25. A room is 6m × 5 m × 4m. Its diagonal is __m

(a) 7.9

(b) 8.1

(c) 8.7

(d) 9.4

Questions-Bank Test No. 106

Number of Questions: 20 QUANTITATIVE APTITUDE **Suggested time: 20 minutes**

1. A book contains 428 pages. Total number of digits used in printing all these page numbers of this book must be
(a) 850
(b) 1175
(c) 1176
(d) 1284

2. The greatest no. that will divide 2930 & 3250 leaving 7,11 is
(a) 37
(b) 53
(c) 59
(d) 73

3. The least perfect square divisible by 2,3,4,6,8 is
(a) 1024
(b) 1296
(c) 900
(d) 784

4. The value of (.728 sq − .272 sq) divided by (.456) is
(a) 4
(b) 3
(c) 2
(d) 1

5. A garrison had 2000 men ration for 54 days. After 15 days x men join them and it will now last for 20 days. X is
(a) 3900
(b) 1900
(c) 2000
(d) 1000

6. Wages of 45 women are Rs. 15,525/- in 48 days. A man gets twice of a woman. No. of men to receive Rs.5,750/- in 16 days is
(a) 20
(b) 25
(c) 30
(d) 15

7. 3 successive discounts of 20% on the marked price of a commodity are equivalent to a single discount of___%
(a) 60
(b) 55.8
(c) 50.2
(d) 48.8

8. In how many years Rs.4100 will earh Rs.1394 @17% p.a
(a) 4
(b) 3
(c) 2.5
(d) 2

9. At what compounded rate Rs.1200 will become Rs.1323 in 2 years
(a) 8
(b) 7
(c) 6
(d) 5

10. Four vessels contain Milk to Water as 3:7,4:7,5:9,7:13,Water is max in ___vessel
(a) 4^{th}
(b) 3
(c) 2
(d) 1

11. Sixteen was divided in two parts in such a way that sum of their squares added to their product gave 208, then the smaller part is
(a) 6
(b) 4
(c) 8
(d) 10

12. The smallest Natural number but a perfect square and also ending in 3 identical digits must lie between

(a) 1000 − 2000

(b) 2000 − 3000

(c) 3000 − 4000

(d) > 4000

13. The sum of all prime numbers from 1 to 20 is

(a) 75

(b) 76

(c) 77

(d) 78

14. The number 89715938* is divisible by 4. then * can be

(a) 2

(b) 3

(c) 4

(d) 6

15. The HCF OF 1134 , 1344 , 1512 is

(a) 64

(b) 44

(c) 24

(d) 42

16. Length and breadth of rectangle are 3:2,Perimeter 30m,area is

(a) 34

(b) 50

(c) 54

(d) 60

17. The point (8,4) divides the line joining (5, − 2),(9,6) in

(a) 2:1

(b) 1:3

(c) 3:1

(d) 1:2

18. The lines $x + q = 0$, $y-2 = 0$, $3x + 2y + 5 = 0$ are concurrent, then q is

(a) 1

(b) 2
(c) 3
(d) 5

19. The radius of a cylinder is doubled and height halved. The ratios of the new and previous volumes will be
(a) 1/2
(b) 2/1
(c) 2/3
(d) 3/2

20. The height of a cylinder is 6 m. Thrice the sum of the areas of its two circular faces is twice the curved surface, then radius is
(a) 4 m
(b) 3
(c) 2
(d) 1

Questions-Bank Test No. 107

Number of Questions: 20 **Suggested time: 20 minutes**

1. A chemist preparing a solution should have included 35 mm of a chemical but actually used 36.4 mm, then %age error is
(a) .04
(b) .05
(c) 1.4
(d) 3.85

2. A retailer buys a radio for Rs.75.but marks up the price by 1/3 and sells it at a discount of 20% .His Gain (Rs.) is ...
(a) 5
(b) 6.67
(c) 7.5
(d) 13.33

3. Radius of a circle is increased by 50%, increase(%) in area?
(a) 25
(b) 50
(c) 100
(d) 125

4. Which fraction is the highest?
(a) 403/134
(b) 79/26
(c) 527/176
(d) 99/34

5. Eight men can chop down 28 trees in a day, 20 men shall chop how many?
(a) 28
(b) 160
(c) 70
(d) 100

6. product of 3/100 , 15/49 , 7/9 is

(a) 25/44100
(b) 1/140
(c) 1/196
(d) 25/158

7. 1440 dozen pens bought @ Rs. 2.50 per dozen were sold at the rate of 25p each . Profit is Rs.
(a) 60
(b) 72
(c) 720
(d) 874

8. Radius of a cone is doubled and height made $1/3^{rd}$. The new and the old Volumes are in the ratio
(a) 2:3
(b) 3:2
(c) 4:3
(d) equal

9. Difference b/w CI & SI on Rs.6000 @12% p.a. for 2 years is
(a) 86.4
(b) 84.4
(c) 82.4
(d) 144

10. The sum of the squares of any three consecutive odd numbers when increased by 1 is always divisible by
(a) 6
(b) 12
(c) 24
(d) 18

11. A man wants to have a rectangular field but has only 30 m of fence with him. Using a wall of his house as one side, the field can have a maximum perimeter of __m
(a) 25
(b) 30
(c) 35
(d) 53

12. Two numbers whose difference, sum and product are in the ratio of 1:7:24 can be

(a) 24,30

(b) 8,6

(c) 12,18

(d) 4,3

13. Find sum of first 13 terms of an A.P., whose first term is one and the 2^{nd}, 10^{th} & 34^{th} terms form G.P. series.

(a) 45

(b) 39

(c) 52

(d) 65

14. A train after 30 minutes of starts, meets with an accidents is detained for an hour and then moves at $3/5^{th}$ of its speed reaching ‘x’ miles away 3 hours late. Had the accident taken place 50 miles farther, it would have arrived 1.5 hrs earlier. Then ‘9x’= ___ miles

(a) 400

(b) 600

(c) 800

(d) 200

15. Two such numbers whose sum is 9 and the sum of their fourth powers is 2417 are

(a) 7,1

(b) 6,3

(c) 5,4

(d) 7,2

16. Two groups of birds, sitting on different trees spoke thus, “if two of you flew to join us, we would be double of what you remain” said one group. “But why should not three of you join us to make us equal” retorted the other group. They were ......

(a) 18,20

(b) 12,18

(c) 20,13

(d) 24,15

17. The mean of five consecutive even numbers all > or = to 3 is
(a) 4
(b) 5
(c) 6
(d) 7

18. Sides of a square figure are up by 30% & 20%. Area is up by %
(a) 20
(b) 30
(c) 156
(d) 56

19. A soap cake measures 5cm × 4 cm × 1.5 cm. How many such cakes can a box 59cm × 49 cm × 16 cm contain.
(a) 660
(b) 1320
(c) 1980
(d) 2640

20. Some men took a job for 20 days. Eight of them not turning up now they took 24 days . How many were originally expected?
(a) 47
(b) 48
(c) 49
(d) 50

Questions-Bank Test No. 108

Number of Questions: 24 **Maximum Time:15 minutes**

1. _/72 + _/128 − _/98 + _/50 = _/X ; then X is
(a) 12_/2
(b) 8_/2
(c) 7_/2
(d) 5_/2

2. Which is the greatest _/5651 , 2561/33, square of 8.67
(a) _/5651
(b) 2561/33
(c) 8.76
(d) ALL EQUAL

3. _/.000036 × _/.049 × _/2.5 × _/.0081 = 189 × (how many tens)
(a) − 7
(b) − 6
(c) − 9
(d) −10

4. .36 × .28 × .0025 × .004 = 100.8 × 10 to the power
(a) − 7
(b) − 10
(c) − 9
(d) −8

5. (.009 × .002 × .00004 × .00096) /(.3 × .00032 × .0048)is = 1.5×
(a) 1/10000
(b) 1/1000000
(c) 1/100000
(d) 1/1000

6. 18% of 35% of x = 27% of 14% of 24.5
(a) 1.47
(b) 14.7
(c) 147

(d) 1470

7. Spending 35% is saving p/qth part of it. Then p/q is
(a) .35
(b) .65
(c) 13/20
(d) 7/20

8. 28% share of 15% of two-third of 180 is = 14% of x,then x?
(a) 900
(b) 3600
(c) 360
(d) 4500

9. What is 2/7th of a number whose 3/5th exceeds its 5/9th by 98.
(a) 630
(b) 1260
(c) 63
(d) 12600

10. A loss of 15% is made by selling an article for Rs.345.To gain 2% it should be sold for Rs. ... more
(a) 412
(b) 414
(c) 69
(d) 57

11. Selling for Rs.240 makes twice gain than at Rs.180.CP is
(a) 1230
(b) 120
(c) 240
(d) 60

12. Two houses were sold for Rs.2 lakhs each gaining/losing 15% respectively. Minimum loss is around Rs. ..
(a) 5000
(b) 6500
(c) 9000
(d) 12000 (e) 14000

13-14. Three successive discounts of 35%,40%,65% are offerred on goods marked 20% above cost.

13. They are = to....% 14)Net discount offerred is?

(a) 16

(b) 84

(c) 15

(d) 83 (e) 17

14. Rs. 45 is pocketed more in selling goods @15% gain instead of at 3% loss. The CP is Rs.

(a) 250

(b) 260

(c) 450

(d) 360 (e) 320

16-17. A gets 20% more than B, who gets 25%more than C. If C gets 40% less than D, then

15. D gets __%of A

16. A gets __%of D

(a) 9−1/11

(b) 111−1/11

(c) 90

(d) 10

17. 40% goods sold at 40%gain, 30% goods at 30%loss. Remaining be sold at what gain/loss % to be at par.

(a) +23

(b) −23

(c) at par

(d) 30

18. A General put his men in a solid square. Even if three-fourth of the men were not present - still in square, but then the men in front row would be 64 less. He had how many men with him?

(a) 128

(b) 256

(c) 128 × 128

(d) 256×256 (e) 16

19. X is a square. 65 added to _/X is again a square. 40 added further is again a square. The number can be

(a) 3136

(b) 256

(c) 1225

(d) 6241

20. 'A' securing 48% votes defeated his only rival by 2448 votes, though 4% votes were invalid and 17% were not cast.Total votes?

(a) 1440

(b) 14400

(c) 144

(d) 144000

21. LCM of 10,15,25,30 & 35 is how times their HCF

(a) 21

(b) 1050

(c) 5

(d) 210

22. HCF of two Nos. is 15,their product is 5400.How many pairs are there? Difference of the Greatest & the smallest such no.

(a) 2/345

(b) 3/345

(c) 2/315

(d) 3/375

23. A perfect square also divisible by 6,12,18,24,30 & 36 is?

(a) 3600

(b) 900

(c) 2025

(d) 14400

(e) 8100

Questions-Bank Test No. 109

Number of Questions: 20 **Maximum Time:12 minutes**

1. LCM of 12, 34 and 48 is
(a) 408
(b) 816
(c) 576
(d) 1632

2. Product of 7183 and 34881 is
(a) 250550232
(b) 250550243
(c) 350550123
(d) 250505223

3. 70% of 80% is
(a) 56
(b) 506%
(c) .056
(d) .56

4. The value of square root of .8 is nearest to
(a) .8671
(b) .9
(c) .8
(d) .8374

5. A boy was asked to find 5/16 of a certain number. He found 5/6 of it and got 125 more. The correct answer is
(a) 150
(b) 160
(c) 200
(d) 180

6. 6.Fifty men eat 100 kg in 4 days.200 men will eat 50 kg in
(a) 2.5
(b) 10

(c) 1
(d) 2

7. What percent of Rs. 22 is Rs.1.10p
(a) 50
(b) 5
(c) .5
(d) 20

8. In a village of 4500 voters 11/18 are males,40% of whom are married. Number of females not married are
(a) 650
(b) 700
(c) 750
(d) 1100

9. Ages of a father and son are in the ratio 12:5 The father is 28 yrs older than son. He is
(a) 24
(b) 48
(c) 52
(d) 36

10. He sold a car for Rs.36,000 with a discount of 8% on 20000 and 5% on the remaining. The car fetched net Rs.
(a) 30,000
(b) 33,480
(c) 33,600
(d) 32,480

11. The largest fraction is
(a) 7/13
(b) 9/16
(c) 8/15
(d) 23/40

12. Rs. 227.15 is divided by 59 men:each will get Rs.
(a) 3.75
(b) 3.85
(c) 3.95

(d) 3.70

13. A train @ 60 kmph passes two poles 100 m apart in 18 seconds. The length of the train is __m

(a) 100

(b) 200

(c) 150

(d) 300

14. A ship has enough to survive food for 50 men for 72 days. 30 days later they pick up "x" survivors and have enough for 35 days only. Then "x" is

(a) 60

(b) 15

(c) 10

(d) 12

15. A mixture has A & B in the ratio 2:5.If the price of A is thrice that of B and the total cost, including 0.80p labour, is Rs.5.20. Then the cost of A is Rs... per ltr

(a) 50

(b) 5

(c) .5

(d) 20

16. the largest fraction among the following is

(a) 5/7

(b) 7/11

(c) 8/13

(d) 9/14

17. A # whose 7^{th} part exceeds its 9^{th} part by 4 is ___

(a) 252

(b) 339

(c) 126

(d) 63

18. 50 match sticks are to be used to make 12 figures of triangles or pentagons, How many are pentagons

(a) 5

(b) 6
(c) 7
(d) 8

19. In a party there were 91 handshakes done. If no one shook hand twice with any of the guests, total guests are
(a) 13
(b) 12
(c) 14
(d) 15

20. In how many other ways can the letters of the word "MIDDLE" be arranged begining M.
(a) 120
(b) 60
(c) 720
(d) 48

Questions-Bank Test No. 110

1. What is the number which when increased by 7 is divisible by 27,21,35 & 77
(a) 10402
(b) 10388
(c) 10397
(d) 20783

2. In twelve hours how many times are the hands of clock make supplementary angles
(a) 22
(b) 24
(c) 12
(d) 11

3. Which one is prime number
(a) 467
(b) 351
(c) 171
(d) 117

4. What is the remainder if (100000000 – 6) is divided by 3
(a) 0
(b) 1
(c) 2
(d) 3

5. The smallest whole number which should be multiplied to make 576 a perfect cube
(a) 8
(b) 24
(c) 4
(d) 3

6. A field 30 × 20 m has a path of uniform width inside it. Area excluding the path is 375, the width of the path is..m
(a) 2.5

(b) 1.5
(c) 3
(d) 1.25

7. Five kg tea @ Rs.75 mixed with 10 kg @ Rs.180. What % is profit if sold @ Rs.200/ = per kg
(a) 36
(b) 38
(c) 42
(d) 51

8. A train 400m which overtook a man walking at 3.6 kmph in 25 seconds, reached the station in 10 minutes. How long will the man take (minutes)
(a) 50
(b) 170
(c) 90
(d) 145

9. If 25 men take 24 days to build a wall 120 m long, then 15 men will make 75 m wall in __days
(a) 25
(b) 40
(c) 24
(d) 15

10. At what rate (%) will a sum become Rs.6050 in two years but Rs.6655 in three years
(a) 15
(b) 12
(c) 10
(d) 20

11. How many four digit numbers are there whose decimal notation contains not more than two distinct digits?
(a) 495
(b) 576
(c) 565
(d) none

12. A,B,C can do a piece of work in 15 days. B alone in 30 days, C alone in 40 days, A & C together will take _days

(a) 120

(b) 24

(c) 30

(d) 40

13. A merchant allows 10% discount on the list price. What should be the list price if he wants to make 20% on his cost price of Rs.600

(a) 720

(b) 792

(c) 800

(d) 666.67

14. Find Ist and 4th term of a GP if the first number exceeds the second by 36 and 3rd is 4 more than the 4th

(a) $-27,-1$

(b) $27,-1$

(c) $54,-18$

(d) $-54,18$

15. The radius of a circle inscribed in a triangle of sides 5,12,13 cm is __ cm

(a) 2

(b) 2.5

(c) 3

(d) 3.5

16. The shortest and the greatest altitudes differ by __ cm when drawn in a triangle of 3.5, 12, 12.5 cm sides

(a) 3.36

(b) 4.24

(c) 5.32

(d) 2.18

17. A train 500 m long moves around a square track 8 km long at 24 km/ph. When (minutes) will the engine reach its starting point after a complete round of the track

(a) 17.25

(b) 18.75
(c) 19.5
(d) 20.0

18. A plane flies around a equi-triangular field at 200,400,600 kmph, its average speed km/ph
(a) 100
(b) 109
(c) 119
(d) 129

19. Nuts at Rs.100,80,60/kg were mixed in the ratio 3:4:5 by weight and sold at 50% profit. The rate of sale was Rs.../kg
(a) 110
(b) 90
(c) 70
(d) 80

20. A candidate when asked to find $7/8^{th}$ of a number instead found $7/18^{th}$ and got his answer lesser by 770. The number talked is
(a) 1260
(b) 6160
(c) 1584
(d) 1644

Questions-Bank Test No. 104 - Answers

1 (c)	2 (d)	3 (d)	4 (a)	5 (b)
6 (a)	7 (d)	8 (c)	9 (b)	10 (b)
11 (d)	12 (d)	13 (a)	14 (b)	15 (c)
16 (c)	17 (c)	18 (b)	19 (b)	20 (d)
21 (b)	22 (d)	23 (c)	24 (a)	25 (c)
26 (d)				

Questions-Bank Test No. 105 - Answers

1 (d)	2 (c)	3 (b)	4 (c)	5 (c)
6 (d)	7 (d)	8 (b)	9 (c)	10 (c)
11 (b)	12 (d)	13 (c)	14 (b)	15 (d)
16 (b)	17 (c)	18 (a)	19 (b)	20 (c)
21 (b)	22 (c)	23 (a)	24 (b)	25 (c)

Questions-Bank Test No. 106 - Answers

1 (c)	2 (d)	3 (d)	4 (d)	5 (b)
6 (b)	7 (d)	8 (d)	9 (d)	10 (c)
11 (b)	12 (a)	13 (c)	14 (c)	15 (d)
16 (c)	17 (c)	18 (c)	19 (b)	20 (c)

Questions-Bank Test No. 107 - Answers

1 (e)	2 (a)	3 (d)	4 (b)	5 (c)
6 (b)	7 (c)	8 (a)	9 (a)	10 (a)
11 (d)	12 (b)	13 (b)	14 (d)	15 (d)
16 (b)	17 (c)	18 (d)	19 (b)	20 (b)

Questions-Bank Test No. 108 - Answers

1(e)	2 (b)	3(e)	4 (c)	5 (b)
6 (b)	7 (c)	8 (b)	9 (a)	10 (b)
11 (b)	12 (c)	13 (a)	14 (b)	15 (a)
16 (b)	17 (a)	18 (b)	19 (c)	20 (b)
21 (b)	22 (d)	23 (a)	24 (a)	

Questions-Bank Test No. 109 - Answers

1 (d)	2 (a)	3 (a)	4 (c)	5 (b)
6 (a)	7 (d)	8 (b)	9 (c)	10 (a)
11 (b)	12 (d)	13 (b)	14 (d)	15 (e)
16 (c)	17 (c)	18 (c)	19 (c)	20 (e)

Questions-Bank Test No. 110 - Answers

1 (b)	2 (d)	3 (a)	4 (b)	5 (d)
6 (a)	7 (b)	8 (b)	9 (a)	10 (c)
11 (a)	12 (c)	13 (c)	14 (b)	15 (a)
16 (c)	17 (b)	18 (b)	19 (a)	20 (c)

PROFICIENCY TESTS

Proficiency Test - 1001

No. of Questions: 10 **Suggested time: 7 minutes**

Directions for Q 1 to 5: Refer the following table about required monthly payments needed to satisfy a loan disbursed at the rate of 10.0%.p.a.

Loans Amount (Rs.)/ YEARS	13	14	15	16	17	18	19
50	0.58	0.56	0.54	0.53	0.52	0.50	0.50
100	1.15	1.11	1.08	1.05	1.03	1.00	0.99
200	2.30	2.22	2.15	2.10	2.05	2.00	1.97
300	3.45	3.33	3.23	3.14	3.07	3.00	2.95
400	4.60	4.44	4.30	4.19	4.09	4.00	3.93
500	5.74	5.55	5.38	5.23	5.11	5.00	4.91
600	6.89	6.65	6.45	6.28	6.13	6.00	5.89
700	8.04	7.76	7.53	7.33	7.15	7.00	6.87
800	9.19	8.87	8.60	8.37	8.17	8.00	7.86
900	10.34	9.98	9.68	9.42	9.20	9.00	8.84
1000	11.48	11.09	10.75	10.46	10.22	10.00	9.82
2000	22.96	22.17	21.50	20.92	20.43	20.00	19.63
3000	34.44	33.25	32.24	31.38	30.64	30.00	29.44
4000	45.92	44.33	42.99	41.84	40.85	40.00	39.26
5000	57.40	55.42	53.74	52.30	51.07	50.00	49.07
6000	68.88	66.50	64.48	62.76	61.28	60.00	58.88
7000	80.35	77.58	75.23	73.22	71.49	69.99	68.69
8000	91.83	88.66	85.97	83.68	81.70	79.99	78.51
9000	103.31	99.74	96.72	94.14	91.91	89.99	88.32
10000	114.79	110.83	107.47	104.60	102.13	99.99	98.13

1. At the 10% rate what would be the monthly payment needed to pay off a Rs.12,000 loan in 15 years?
(a) Rs.107.47
(b) Rs. 128.95
(c) Rs. 128.97
(d) Rs.110.83.

2. Paying an EMI of Rs. 50 in what time a loan of Rs.5,000/- shall be paid back given that rate of interest is 10%?

(a) 8

(b) 12

(c) 1

(d) 18.

3. What is the difference in each monthly payment between a 17 year loan of Rs.13000 at the 10% rate and an 18-year loan for the same amount at the same rate?

(a) Rs. 2.85

(b) Rs.2.01

(c) Rs. 2.78

(d) Rs. 2.25.

4. According to this table for an 18-year loan the monthly payment for a Rs.1000 loan is what percent of the amount of the loan?

(a) 0.1%

(b) 1.0%

(c) 1.5%

(d) 10.0%.

5. How much money will be paid in interest on a Rs.8000 loan at 10% over a period of 18 years?

(a) Rs.9,390

(b) Rs.9,000

(c) Rs.9,280

(d) Rs.9,290

DIRECTIONS for Q 6 to 10 Refer the following table:

Income (INR)	**Tax (INR)**
0–4,000	1% of income
4,000–6,000	40 + 2% of income over 4,000
6,000–8,000	80 + 3% of income over 6,000
8,000–10,000	140 + 4% of income over 8,000

10,000–15,000	220 + 5% of income over 10,000
15,000–25,000	470 + 6% of income over 15,000
25,000–50,000	1,070 +7% of income over 25,000

6. How much tax is due on an income of Rs. 11,500?
(a) Rs.75
(b) Rs.80
(c) Rs.295
(d) Rs.150.

7. Your income for a year is Rs. 24,000. You receive a raise so that next year your income will be 29,000. How much more will you pay in taxes next year if the tax rate remains the same?
(a) Rs. 370
(b) RS. 380
(c) Rs.300
(d) Rs.340

8. Shyama paid Rs.740as tax. If x was her income, which of the following statements is true?
(a) 10000< x <14,000
(b) 16,000 < x <19,000
(c) 19,000 < x <23,000
(d) 24,000 < x <26,000

9. The town of Ding has a population of 50,000. The average income of a person who lives in Ding is Rs. 3,700 per year. What is the total amount paid in taxes by the people of Ding?
(a) Rs. 37
(b) Rs. 3,700
(c) Rs. 50000
(d) Rs. 1,850,000.

10. A person who has an income of Rs.13,600 pays what percent (to the nearest percent) of his or her income in taxes?
(a) 1.8
(b) 2.2
(c) 2.9

(d) 3.6

Proficiency Test – 1001 : Answer Key

1 (c)	2 (d)	3 (c)	4 (b)	5 (c)
6 (c)	7 (d)	8 (c)	9 (d)	10 (c)

Proficiency Test - 1002

Total Questions = 9 **Suggested time: 6 minutes**

Directions for Q 1 to 5 Refer the following table:

Relative Sweetness Of Different Substances	
Lactose	0.16
Maltose	0.32
Glucose	0.74
Sucrose	1.00
Fructose	1.70
Saccharin	675.0

1. (Fructose & Glucose) together are how many time sweeter than (Lactose& Maltose)?
(a) 5.09
(b) 5.1
(c) 5.8
(d) 5.08

2. If equal amounts of maltose is substituted for lactose, what % increase in sweetness is obtained?
(a) 16
(b) 50
(c) 100
(d) Data inadequate

3. How many grams of sucrose be added to one gram of saccharin to make a mixture 100 times as sweet as glucose?
(a) 7
(b) 8
(c) 9
(d) 10.

4. What is approximate ratio of Fructose to Lactose in a mixture equal in sweetness as Maltose is ?

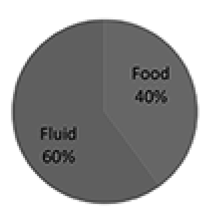
(a) 2/17
(b) 1/5
(c) 9/2
(d) 4/43.

5. Approximately how many times sweeter than sucrose is a mixture of glucose, sucrose, and fructose in the ratio of 1 : 2 : 3?
(a) 0.6
(b) 1
(c) 1.3
(d) 2.9.

Directions for Q 6 to 9 Refer the following table and graph.

Weight distribution average adult (Total body weight 63.00 kg)	
Organ	**Weight (grams)**
Lungs	900
Liver	1530
Brain	1350
Gastro	1800
Blood	4500
Skelton	9000
Muscles	27000

Total 2500 cc



6. Which of the following can be inferred from the table and graphs?
I. Half of the total body weight is the skeleton and blood.
II. About 2.4% of the total body weight is the lever.

III. The weight of the blood in the average adult is twice the weight of the skeleton.

(a) I only
(b) II only
(c) III only
(d) II and III

7. If the weight of the liver is represented as 'g' grams, the total body weight is ?

(a) 41.17 g
(b) 61.470 + g
(c) 63−g
(d) 24g.

8. What is the angle of the sector in Graph I representing daily water intake through fluids?

(a) 144°
(b) 72°
(c) 120°
(d) 216°.

9. How many 'cc' of the output through the kidneys is the daily intake in the shape of fluids?

(a) 800
(b) 1200
(c) 1500
(d) 1800

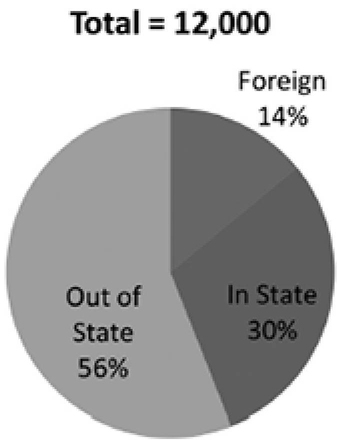
Proficiency Test – 1002 : Answer Key

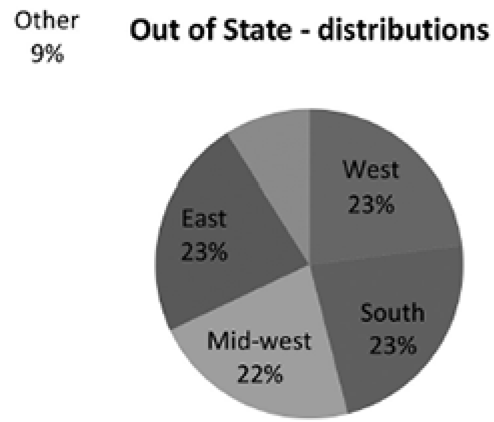
1 (d)	2 (d) (Total quantity is not given)	3 (b)	4 (a)	5 (c)
6 (c)	7 (d)	8 (c)	9 (d)	

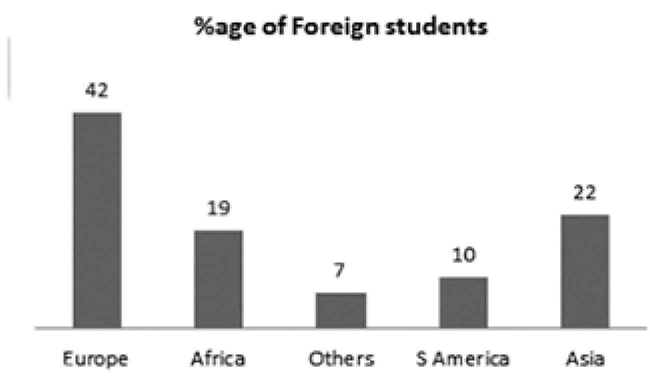
No. of Questions: 9 **Suggested time: 6 minutes**

Directions for Q 1 to 5 refer the following charts

Residence Breakdown of Students at University of Utrakhand (October 1, 2000)







1. Which of the following statements can be inferred from the graphs?

I. More students registered at University Utrakhand on October, 1, 2000, were from Europe than from the west.

II. Over twice as many students from Africa were registered at University Utrakhand on October 1, 2000, as students from South America.

III. The Midwest accounted for approximately 13% of the students registered at University Utrakhand on October 1, 2000.

(a) III only

(b) I and II only
(c) II and III only
(d) I, II and III

2. If half of the students from the East were from New York State, then approximately how many of the students who were registered at University Utrakhand on October 1,2000 were from New York State?
(a) 810
(b) 640
(c) 724
(d) 864.

3. How many of the students who were registered at University Utrakhand on October 1, 2000, were from the South?
(a) 1,600
(b) 1,680
(c) 1,800
(d) 1,750.

4. On October 1, 2000, how many more of the students registered at University Utrakhand were from Europe than from Asia?
(a) 360
(b) 380
(c) 336
(d) 540.

5. How many more students were there from "South" than from South America?
(a) 1,800 (b) 504 3 1,512 (d) None

Directions for Questions 6 to 9 refer the following table

		Size of Portion		Calories	Protein (grms)
Milk	whole	1 glass	8 oz	160	9
	skimmed/buttermilk			90	9
	chocolate drink			190	8
Cheese	American or Swiss	1 cube	1 oz	110	8
	Foods – cheddar type	2 tbsp		90	6
	cottage, creamed			30	4
	cream			110	2
Butter		1 tbsp	½ oz	100	..
Butter		1 tbsp or small pat		35	..
Cream	light table	2 tbsp		60	1
	heavy whipped	1 heaping tbsp		50	..
	Half-and-half	¼ cup		80	2
Ice cream	Vanilla	½ cup		145	3
	a la mode	medium scoop		115	2
Sherbet		½ cup		130	1

6. How many tablespoons of light cream have the same number of calories as 8 ounces of buttermilk?

(a) 2

(b) 3

(c) 4

(d) 5.

7. How many calories of cream would there be in the amount needed to furnish the same number of grams of protein as there are in 4 ounces of chocolate milk.

(a) 55

(b) 95

(c) 110

(d) 220.

8. Which of the following has the greatest number of calories per pound?

(a) Cottage cheese

(b) Cheddar-type cheese

(c) Swiss cheese

(d) Whole milk.

9. Which of the following furnishes the greatest number of grams of protein per unit weight?

(a) Whole milk
(b) Cottage cheese
(c) American cheese
(d) Chocolate milk.

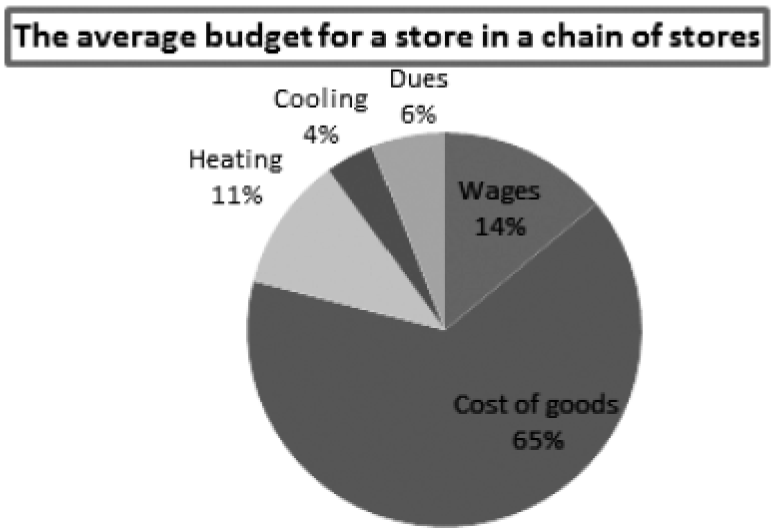
Proficiency Test – 1003 : Answer Key

1 (a)	2 (b)	3 (b)	4 (c)	5 (d)
6 (c)	7 (d)	8 (d)	9 (c)	

PROFICIENCY TEST - 1004

Total Questions: 10 **Suggested time: 6 minutes**

Directions for Q 1 to 5: Refer the following graphs / data to answer the given questions: -



The budget of M/s Buddy's Stores

Description	**Budget (INR)**
Cost of Goods	2,00,000
Wages	50,000
Cooling Energy	20,000
Heating	20,000
Dues	10,000

1. An average store in the chain has a budget of Rs.400,000. How much should be budgeted for heating and cooling?
(a) Rs.1700
(b) Rs.24,00
(c) Rs. 60,000
(d) Rs.17,000

2. What is the ratio of wages paid by M/s Buddy's Stores to its total non wage expenditure?

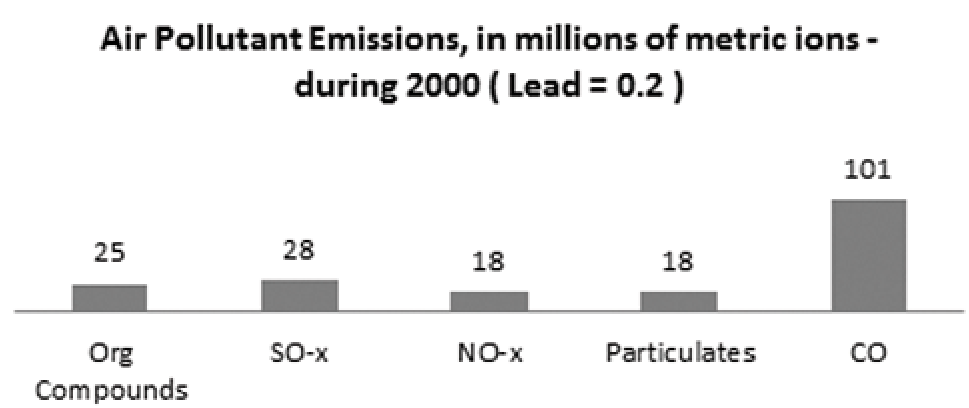
(a) 1/3
(b) ¼
(c) 1/7
(d) 1/6.

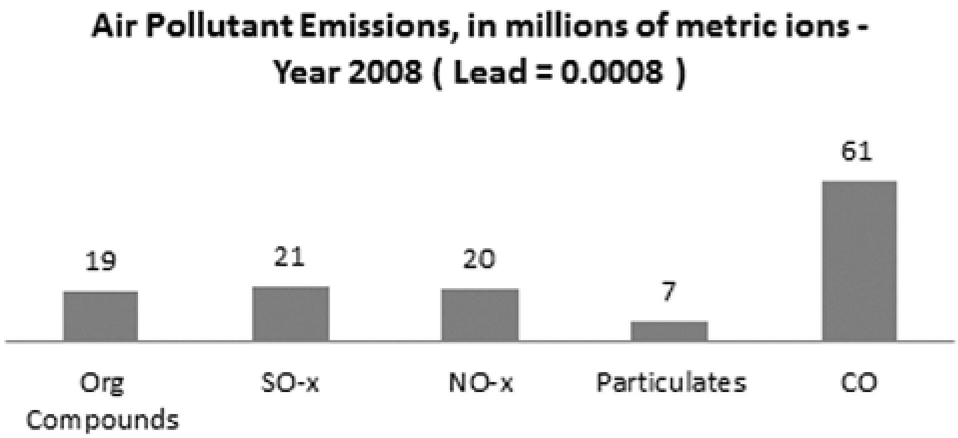
3. What % more/less does M/s Buddy's Stores spend for heating than the average store in the chain?
(a) 7%
(b) 40%
(c) 4%
(d) 18%.

4. Calculate the ratio of the percentage M/s Buddy's Stores spends on wages to the percentage the average store spends on wages.
(a) 8/10
(b) 4/3
(c) 5/4
(d) 25/21.

5. If there are 15 stores in the chain and the average budget for each store is Rs.4,00,000, how much should be budgeted for the total chain for cooling energy?
(a) Rs.1,10,000
(b) Rs. 2,20,000
(c) Rs. 2,40,000
(d) Rs. 8,20,000.

Directions for Q 6 to 10 : Refer the following graphs and answer the given questions : -





6. Select the emission that increased in tonnage from 2000 to 2008
(a) NOx
(b) SOx
(c) Organic Compounds
(d) Lead.

7. Calculate the percentage decrease in lead emissions between 2000 and 2008.
(a) 4%
(b) 22%
(c) 48%
(d) 96%.

8. Estimate the percentage the percentage contribution of CO to the emissions listed in the charts in 2000.
(a) 47%
(b) 49%
(c) 51%
(d) 53%.

9. Vehicle emissions contributed 65, 9 and 6 million metric tons of CO, organic compounds and NOx, respectively, in 2000. Estimate their percentage contribution to the total emissions.
(a) 42%
(b) 44%
(c) 46%
(d) 48%.

10. If the circle graph of 2000 were drawn to scale, approximate the angle that would represent lead.

(a) 0.04°

(b) 0.4°

(c) 0.2°

(d) 0.02°

Proficiency Test – 1004 : Answer Key

1 (c)	2 (d)	3 (b)	4 (d)	5 (c)
6 (a)	7 (d)	8 (d)	9 (a)	10 (b)

CHHOTTE HAIN BUT KHOTTE HAIN 37

(Chhote hain but khotte hain – hum ko aisa vaisa no samjho, hum bade kaam ki chees)

Given here are some small cases which would help you a lot in understanding the smallest of the tricks. They seem to be of little importance but will shake your head while dealing with them. So, try your hands on these.

Case Let - I

Time : 5 Minutes

Read the following carefully and answer the following questions.

In a class of 125 students, each student studies at least one of the subjects viz. History, English or Mathematics except a group of six students who have been exempted to attend any class. 59 students study History, 67 English and 73 Mathematics. 34 study Mathematics& History, 26 English & Mathematics and 33 History and English.

1. How many study exactly two subjects?
(a) 54
(b) 51
(c) 48
(d) 46

2. How many study all the three subjects?
(a) 12
(b) 15
(c) 13
(d) 14

3. How many study more than 1 subjects?
(a) 63
(b) 67

(c) 62
(d) 66

4. How many English& Mathematics but Not History?
(a) 12
(b) 13
(c) 14
(d) 11

5. How many study English&(Mathematics or History) ?
(a) 41
(b) 43
(c) 47
(d) 57

6. How many History & English but NOT maths?
(a) 19
(b) 18
(c) 17
(d) 20

7. If 70% of students failed in one/ more subjects, then the maximum number of failures are in
(a) Mathematics
(b) English
(c) History
(d) None

8. What is the number of students who study all the subjects?
(a) 12
(b) 16
(c) 15
(d) 13

9. How many students study Maximum One subjects?
(a) 42
(b) 49
(c) 47
(d) 44

Case Let - II

Time : 5 Minutes

Study the following data and answer the below given questions. (Sales unit in '00'000)

Year/Company	Alpha	Beta	Galaxy	Drum	Elegy	Fox
Sales in 2015	1020	4500	3200	1800	1830	2575
Sales in 2016	1150	4200	3500	1750	1775	2625

1. Which company given had the least %age change from the year 2015 to 2016 is?
(a) Alpha
(b) Galaxy
(c) Elegy
(d) Drum

2. The net change in sales of all companies together is __% in the year 2016 over 2015?
(a) .076
(b) .9
(c) 0.5
(d) .76

3. Average sales per company per month forthe year 2016 were ?
(a) 2400
(b) 208
(c) 2500
(d) 200

4. 4. The maximum %age change in sales between the two years anywhere is ?
(a) 12.25
(b) 10.525
(c) 10.75
(d) 9.375

5. If the sales of year 2016 for Elegy is actually 2275, then the answer for Q 1 given above would be ?
(a) Beta

(b) Elegy
(c) Alpha
(d) Galaxy

6. What is the market % share of Beta in value terms in the year 2015? (assume market is captured by these only)
(a) 28
(b) 30
(c) 32
(d) can't say

7. If the average cost price of a product sold by Beta to that of Alpha's is 28:29, what is the ratio of volume of sales of company Alpha :Beta for the given two years period is ?
(a) 21:34
(b) 31:120
(c) 13:23
(d) 31:13

Case Let - III

Time : 3 Minutes

A person moves 4 KMs north and reaches B. From B he moves 3 km East to reach C. From C he proceeds 5 km N/E to reach D, then 5 km towards North to reach E. Now further 10 km West to reach F. Ultimately he decides otherwise and heads back straight to A from F.

1. What could be the possible distance AF in km?
(a) 11.8
(b) 12.7
(c) 13.7
(d) 16.2

2. What could be the possible distance in kilometer of segment CF?
(a) 11.4
(b) 8.8
(c) 15.2
(d) 14.2

3. If the time of total journey taken by him is 50 minutes. Find his average speed km/hr.
   (a) <45
   (b) >51
   (c) >55
   (d) <40

4. Moving at the speed of 36 km/hr had he returned back along F⇒E⇒D⇒C⇒B⇒A. how much time (minutes) would he have taken?
   (a) 54
   (b) 45
   (c) 60
   (d) 36

Case Let - IV

Time: 2 Minutes

We are talking of a manufacturing unit which manufactures parts for Tractor. All its production parts are sold at a profit of 20%. One day when the unit was functioning normal, suddenly after a production of 3000 units, the production line developed a snag and thereafter 25% of the products produced were found defective. These defective product shall have to be sold at 80% of the cost price. However, all its produced units are sold on the same day.

In view of the changed scenario, answer the following questions: -

1. What is the loss (%) in a production of 5000 units?
   (a) 2
   (b) 3
   (c) 3.6
   (d) 100

2. At what production level of units given below the profit will be Minimum
   (a) 7000
   (b) 3000
   (c) 5000
   (d) 6000

3. What should be the production to have maximum profits?
(a) 9000
(b) 12000
(c) 3000
(d) 2000

Case Let - V

Time : 3 Minutes

A company sold 12000 trucks in 2013 (which was 120% of the sale of 2012). The total sales for the year 2012 were Rs.290 crores. Its market share, however, declined to 60% in 2013 – a decline of 2% over the market share in 2012. In 2014, the company expected the market to grow by 25% with its market share to remain steady at 60%.

1. The total market in the year 2012 was approximately worth Rs…. crores
(a) 460
(b) 425
(c) 435
(d) 402

2. The average price of a truck (lakhs) in 2012 was
(a) 2.70
(b) 2.90
(c) 3.30
(d) 1.90

3. The total market in 2014 is expected to be Rs crores.
(a) 924
(b) 945
(c) 950
(d) 962

4. The price of a tuck in 2013has gone down by how much Rs…. than in 2012 is
(a) 47000
(b) 57000
(c) 51000

(d) 71000

Case Let - VI

Time : 3 Minutes

"Dhani" Bank is a nationalized bank in the Land of Swarg Lok with a credit and deposit history as under:-

Year	Credit Rs Crores	Deposits Rs. Crores
1985	124	148
1986	123	156
1987	139	159
1988	139	163
1989	141	175
1990	148	179

It is further known that 45% of its 1400 branches are in rural areas and remaining distributed equally amongst semi-urban and urban areas. Of the urban areas branches 20% are in Metro. Lending in 1990 was done to the extent of 75% towards priority sectors, of which 1/3 is for transportation. 25% deposits in 1990 were long term, 40% in medium term and rest were in short term

1. In which of the following years was the Credit/Deposit ratio minimum?
(a) 1985
(b) 1986
(c) 1987
(d) 1988

2. In 1985, if 40% of its lending was through rural branches, the bank gave through a rural branch on an average an amount of Rs…. (lakhs). ?
(a) 7.12
(b) 7.94
(c) 8.36
(d) 9.14

3. Highest growth rate in deposits in any year is approximately how much %.
(a) 5.5
(b) 6
(c) 6.5

(d) 7

4. If the long-term deposits grew @10% p.a., what % ofthe bank's deposit was long-term deposits in 1988?

(a) 42

(b) 45

(c) 36

(d) 28

Case Let - VII

Time : 3 minutes

Below are given some information. You are advised to go through these and answer the questions given below:-

1. Advertising expenditure of USA in 1974 was 76.780 million US$. This was about 2% of the GNP of USA;

2. GNP of UK, Japan and West Germany in 1974 was 175,413 and 3510 US$ respectively;

3. Per capita advertising expenditure of Japan and USA was 38 and 125 US$ respectively;

4. Advertisement expenditure of W Germany was 14% more than that of UK but 40% lower than that of Japan;

5. Per capita GNP of Japan is US$ while its population is only 0.109 billion.

(1 billion = 1000 million = 109; 1 million = 10,00,000 =106).

1. What % of UK's GNP is the GNP of Japan?

(a) 235

(b) 242

(c) 254

(d) 225

2. What is the advertising Expenditure of Japan as % of its GNP

(a) 0.5

(b) 1.0

(c) 5

(d) 3.8

3. Advertising expenditure of W Germany is how much billion ?

(a) 3.89

(b) 2 2.485

(c) 1.972

(d) 3.854

4. If advertising Expenditure as a % of GNP for Germany is equal to that of UK, by how many billions US $ must GNP of Germany decrease if expenditure remains unchanged ?

(a) 490

(b) 480

(c) 510

(d) 520

Case Let - VIII

Time : 5 minutes

Harish bought 260 kg of wheat @Rs. 20/ per kg. He then sold half of it to Arif @21/- kg and sold 110 kg in the open market at @ Rs. 23/kg. He distributed the remaining wheat stock at cost price to the poor.

Further known also that Arif sold 40% of the wheat to Sharif @ Rs.27/kg and the rest of his stock @ Rs.22/kg to some traders.

From the above information, answer the following questions.

1. How much profit (Rs) would Harish have made had he preferred not to donate to the poor and instead sold in the open market ?

(a) 60

(b) 560

(c) 400

(d) 460

2. How many more kg should Harish sell in the open market (instead of to Arif)in order to make a total profit of Rs.512?

(a) 12

(b) 8

(c) 6

(d) Not possible

3. If Arif had bargained for a price at Rs.20.50/kg with Harish, by how many Rupees would the latter's profits have dropped?
(a) 65
(b) 130
(c) 260
(d) 195

4. What % of Arif's profits came due to Sharif's purchase made from him?
(a) 40
(b) 60
(c) 80
(d) 90

5. If Arif had bought 20 kg more from Harish in order to sell to Sharif, then his total profits would have gone up by % ?
(a) 12.6
(b) 30
(c) 520
(d) 16.66

Case Let - IX

Time : 5 minutes

The film "Mera Kanoon" ran for a total of 27 weeks at Metro theatre and 17 weeks at Novelty theatre in Mumbai. At Metro it earned an average of Rs.2.17 lakhs per show for the first 9 weeks when there were 3 shows per day. In the next 5 weeks, it earned Rs.1.68 lakhs per show per day. In the next 6 weeks the earnings stayed at Rs.1.54 lakhs. But after 20 weeks the number of shows dropped to 2 per day and the earnings stayed at Rs.1.08 lakhs per show till the end.Novelty ran only 1 show per day but its earnings per show were the same as those of Metro.

From the above information, answer the questions given below:-

1. How many lakhs of Rupees did the movie earn in 4 days in the city in its 15^{th} week?
(a) 24.64
(b) 18.48
(c) 13.86

(d) 30.8

2. By how much %age did the daily earnings drop from 11th week to the 21st week ?

(a) 52

(b) 57

(c) 68

(d) 34

3. In the 16th week Metro had 280 seats which are priced at Rs.42/- per seat. If Novelty's rate is Rs.36 per seat then how many seats does Novelty have?

(a) 280

(b) 206

(c) 240

(d) No relation.

4. Each theatre pays 45% entertainment tax on each ticket, then how much tax (Rs Lakhs) Metro paid for first 10 weeks?

(a) 84.84

(b) 63.63

(c) 38.18

(d) 28.63

5. If the movie showing costs Rs. 55 lakhs and yet another Rs.9 lakhsis spent on marketing and advertisement, will it have some profits after paying taxes during the entire run?

(a) It's loss

(b) Yes

(c) At par

(d) data inadequate

Case Let - X

Time : 4 minutes

Mumbai Metro information is given. Please read the same and answer the questions given :-

A. A train starts from Boriwali at 10:32 hrs and reaches Andheri by 10:46, Bandra by 10:52, Dadar by 11:00 and Mumbai Central by 11;06

;

B. It halts for a minute at each of the station on its way, except Dadar and Mumbai Central where it halts for 2 minutes and 3 minutes respectively;

1. If Mahim is the only station, halfway, between Bandra and Dadar, what time should I reach Mahim to catch this train?

(a) 10:54

(b) 10:55

(c) 10:56

(d) 10:57

2. If it takes just two minutes of travel from Borivali to Kandivli, at what time will the train reach Malad which is further 2 minute away.

(a) 10:36

(b) 10:37

(c) 10:38

(d) can't say

3. If there are two stops between Andheri and Bandra, and the halt time is not changed, then what is the average running time between each station?

(a) 1 minute

(b) 1 ½ min

(c) 2 minutes

(d) 1 min 40 sec

4. If Churchgate is 7 minute journey from Mumbai Central, what is the departure time from here if the train halts for 6 minutes ?

(a) 11:19

(b) 11:20

(c) 11:22

(d) 11:21

Case Let - XI

Time : 4 minutes

Given below are the colour-mix patterns used by **M/s NKT Paints (Pvt.) Limited**, for its factories in Kurukshetra, to produce the specific shades:-

Colour Code	Description	Contents
RB	Royal Blue	Blue 60% + White 30% + Grey 10%
SB	Sky Blue	RB 70% + White 30%
AB	Aqua Blue	Blue 55% + Green 40% + Red 5%
MB	Midnight Blue	Blue 40% + Black 30% + RB 30%
BB	Black Blue	Blue 50% + Black 50%
IB	Ice Blue	Blue 70% + Grey 10% + White 20%

1. Which colour code shade has the minimum %age of the colour blue?
(a) RB
(b) BB
(c) AB
(d) MB

2. What is the %age of the colour White in Sky blue?
(a) 55
(b) 51
(c) 30
(d) 60

3. What is the minimum percentage of Blue amongst the above 6 shades?
(a) 42
(b) 55
(c) 58
(d) 60

4. If we mixed equal quantities of aqua blue and midnight blue, how much % blue would be in the resulting mix?
(a) 56.5
(b) 57.5
(c) 48.2
(d) 51.5

Case Let - XII

Time : 5 minutes

A company had a capacity of 1000 units of production, but could only produce 800 items in the year 2010. The capacity to produce was increased by 20% in 2012 and again by 25% in 2014.

The actual production went up by 100 units only each year for next four years. However, sales (which were 700 in 2010) were not consistent. Sales went up by 40% in 2011 butwent down by 150 in 2012. However, after 2012sales went 100 units up every year.

Any surplus production over sales was added to stocks at the end of the calendar year. However, the excess of sales over production in any year, if any,was to be drawn from the stocks. Stock in 2010 beginning is taken as Zero.

Sales price was Rs.10 per unit in 2010but was reduced by Re1 per unit next year. Thereafter no change in sales price till 2014.

(Capacity utilization = Production / Capacity)

1. In which given year were the sales proceed the maximum ?
(a) 2010
(b) 2011
(c) 2012
(d) Can't say

2. When was the Production over Capacity ratio at its maximum?
(a) 2010
(b) 2011
(c) 2013
(d) 2014

3. Which year-end amongst the given below saw the stocks next to maximum ?
(a) 2010
(b) 2011
(c) 2013
(d) 2014

4. What was the totallevel of stocks during the given time period?
(a) 700
(b) 800
(c) 820
(d) 780

5. What was the average sales figure over the 5 years period?

(a) 920
(b) 936
(c) 840
(d) 790

Test Let - XIII

Time : 5 minutes

A town has population of 1,000 people. To study the taste of its inhabitants about various games, a sample survey was conducted on 25% population who played either football or hockey or cricket or at least one of these games. 85 persons played hockey, 160 had a taste for football, while 200 played cricket. Only 45 people played all the three games. 20 persons played only football while 70 played only cricket. 20 people played both hockey and football but not cricket.

1. How many played only football and cricket?
(a) 110
(b) 75
(c) 70
(d) 120

2. What % of the population played only hockey?
(a) 10
(b) 85
(c) 8.5
(d) 1

3. How many people played hockey & football?
(a) 65
(b) 20
(c) 40
(d) 45

4. played atleast two games?
(a) 150
(b) 15
(c) 60
(d) 24

5. If 30% of the people who play hockey and/or cricket are females, how many women play hockey & cricket only?

(a) 3
(b) 8
(c) 2
(d) can't say

6. How many people played exactly two games ?

(a) 105
(b) 124
(c) 135
(d) 150

7. How many played only Football or only Cricket or both games ?

(a) 155
(b) 160
(c) 165
(d) 170

8. How many played Football and Hockey but not Cricket?

(a) 20
(b) 30
(c) 65
(d) 75

Case Let - XIV

Time : 3 minutes

Kiran, a housewife, inherited some money after the death of her aunty who had already advised her to use the funds carefully. After much pondering and consultations, Kiran invested the whole amount in stock market. But due to instability in the market, after 3 months of investments, she was advised to liquidate her portfolios immediately. She made a profit of 25% on her investments. She kept hold of ¼ of this profit and invested the remaining funds in a Fixed Deposit with a Financer earning 20% p.a. After a year of the death of her aunty, she decided to again try her luck and invest in stocks and withdrew the total amount in Fixed Deposits. However, this time, she lost Rs.10,000 in the first 3 months which compelled her in immediately selling of her holdings at par. Out of the proceeds received, she paid back the loss suffered and invested half of the remaining funds in a 1-year bank deposit earning 10% p.a.. On maturity of deposit she got Rs.53,900/=.

1. What amount did Kiran inherit (Rs.) from her aunty?
(a) 1,12,000
(b) 1,50,000
(c) 1,28,000
(d) 1,15,000

2. Overall gain/loss by Kiran from her investments was to the extent of Rs. ?
(a) 42,700
(b) 35,800
(c) 40,000
(d) 44,900

3. Total interest earned on her deposits was Rs. ?
(a) 22,900
(b) 15,000
(c) 27,500
(d) 2,500

4. Over the entire period her total income was Rs. ?
(a) 37,500
(b) 27,500
(c) 10,000
(d) 25,000

Case Let - XV

Time : 4 minutes

Four friends Manthon, Rahul, Sunny & Victor competed in a contest where maximum possible score was 300. All friends scored more than 60 and none scored fractional scores;

I - Manthon scored less than 200 and his score was a perfect square but 70 less than double of Victor's scores;

II - Rahul& Sunny together scored exactly 270, and score of one of the two was one less than a perfect square (like X^2-1 shape);

III - The score of neither Rahul nor Victor was a multiple of 11; but score of one of them was a multiple of 17.

1. Scores of Sunny and Manthon were?
(a) 150,100
(b) 120,150
(c) 150,121
(d) 85,120

2. Ratio of scores of (Victor &Rahul) to that of (Manthon & Sunny) is
(a) 41:50
(b) 4:5
(c) 50:41
(d) 41:25

3. Score of Sunny is __ % of that of Victor ?
(a) 20
(b) 253
(c) 178
(d) 182

4. Difference between the scores of __was double of a square
(a) Manthon &Sunny
(b) Manthon &Victor
(c) Sunny &Victor
(d) Sunny &Rahul

5. The sum total score of all the four was ?
(a) 557
(b) 455
(c) 579
(d) 540

Answers - Case Lets

CASE LET - I:	1 a	2 c	3 b	4 b	5 d	6 d	7 a	8 d	9 a
CASE LET - II :	1 d	2 c	3 b	4 d	5 a	6 d	7 b		

6 d (value cannot be found due to obvious reasons)

CASE LET - III : 1 c 2 a 3 b 4 b

(*hints: AB = 4, BC = 3 , CD= 5, ED = 5 If CD,ED meet at K (such that BCK and KDE make a right angle at K), then either (CK = 3 & KD =4) or (CK = 4 & KD = 3) and move further.*

CASE LET - IV : 1 a 2 d 3 c

CASE LET - V : 1 c 2 b 3 d 4 b

CASE LET - VI : 1 b 2 b 3 a 4 c

CASE LET - VII : 1 a 2 b 3 b 4 a

CASE LET - VIII : 1 b 2 c 3 a 4 c 5 b

CASE LET - IX : 1 a 2 c 3 c 4 d 5 b

CASE LET - X : 1 c 2 b 3 b 4 d

CASELET - XI : 1 d 2 b 3 a 4 a

CASE LET - XII : 1 b 2 c 3 c 4 d 5 b

CASE LET - XIII : 1 b 2 d 3 a 4 c 5 a 6 a 7 c 8 a

CASE LET - XIV : 1 c 2 c 3 a 4 b

(Let she inherited = 400 x ; On liquidation profit 25% = 100 x ; Total funds 500 x Now 375 x invested in Fixed Deposit @ 20% which remained for 9 months only; Total amount due 375 x + 56 ¼ x = 1725x/4 or ½ (1725x/4 – 10000) x 1.10 = 53,900)

CASE LET - XV: 1 c 2 a 3 c 4 a 5 b

(*hint: better select the choices by rejection (e.g. Q 1: 'd' choice is rejected because of 'III' condition and choice 'b' is rejected because of condition laid down in condition 'II')*

For more questions please also refer to the books:
(i) MATHEMATICS Simplified by JAGGAN SANEJA
(ii) DATA INTERPRETATION Simplified by JAGGAN SANEJA

These books are available at all leading e-commerce stores like Amazon.com, Book adda.com, Flipkart.com, Infibeam.com and NOTIONPRESS.com

For any queries , please refer to the author at : jnsaneja@yahoo.co.in